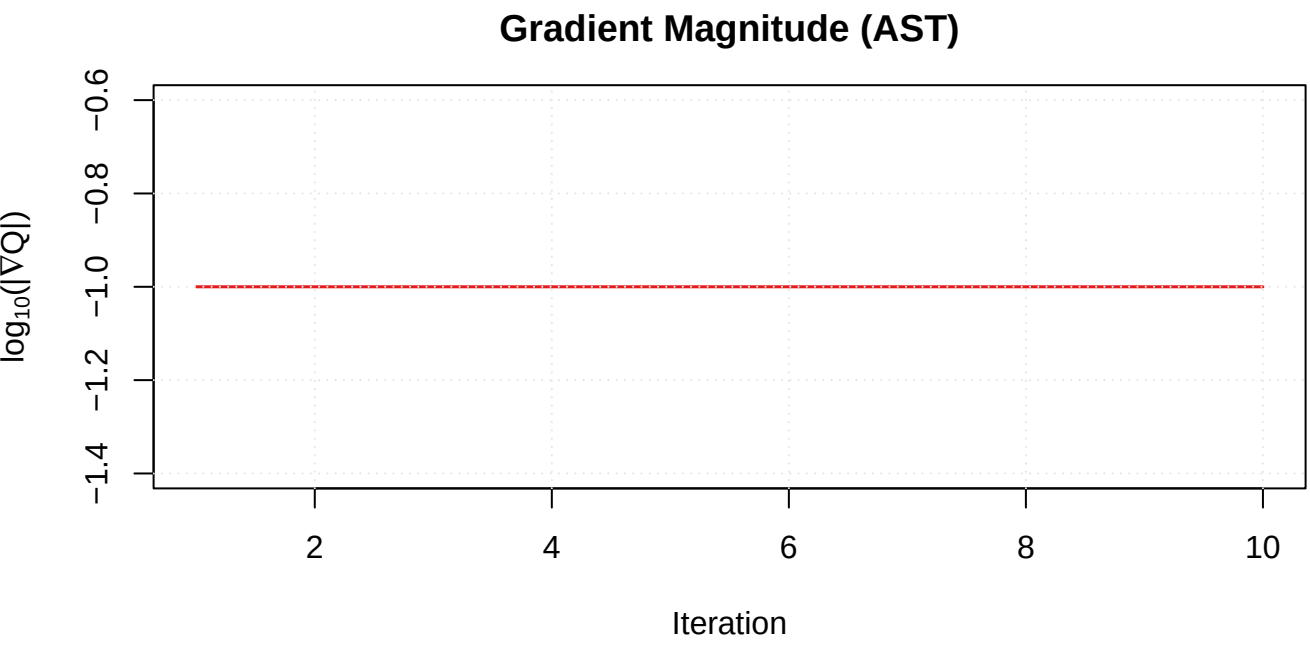
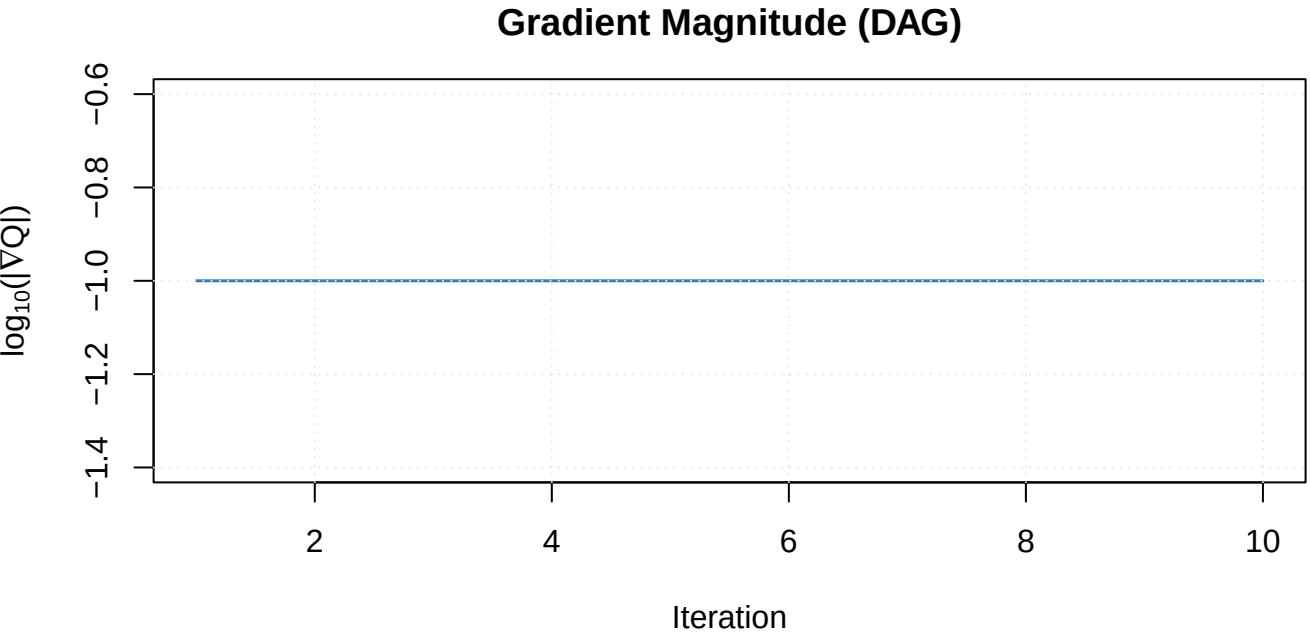
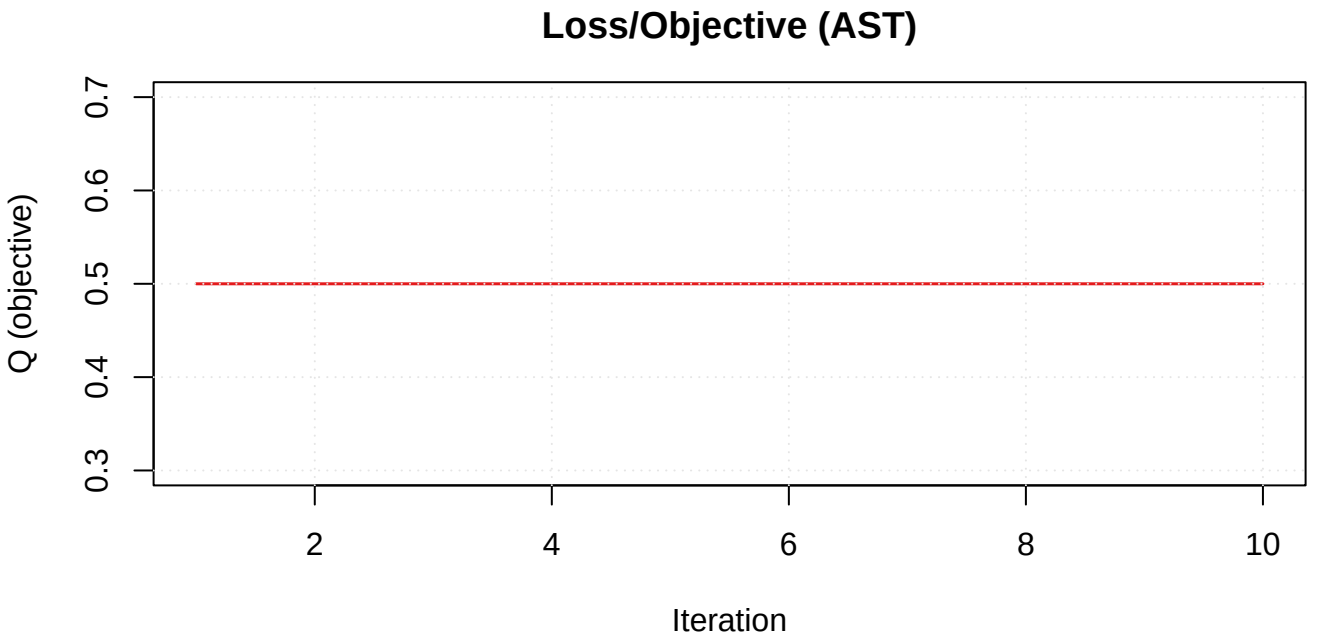
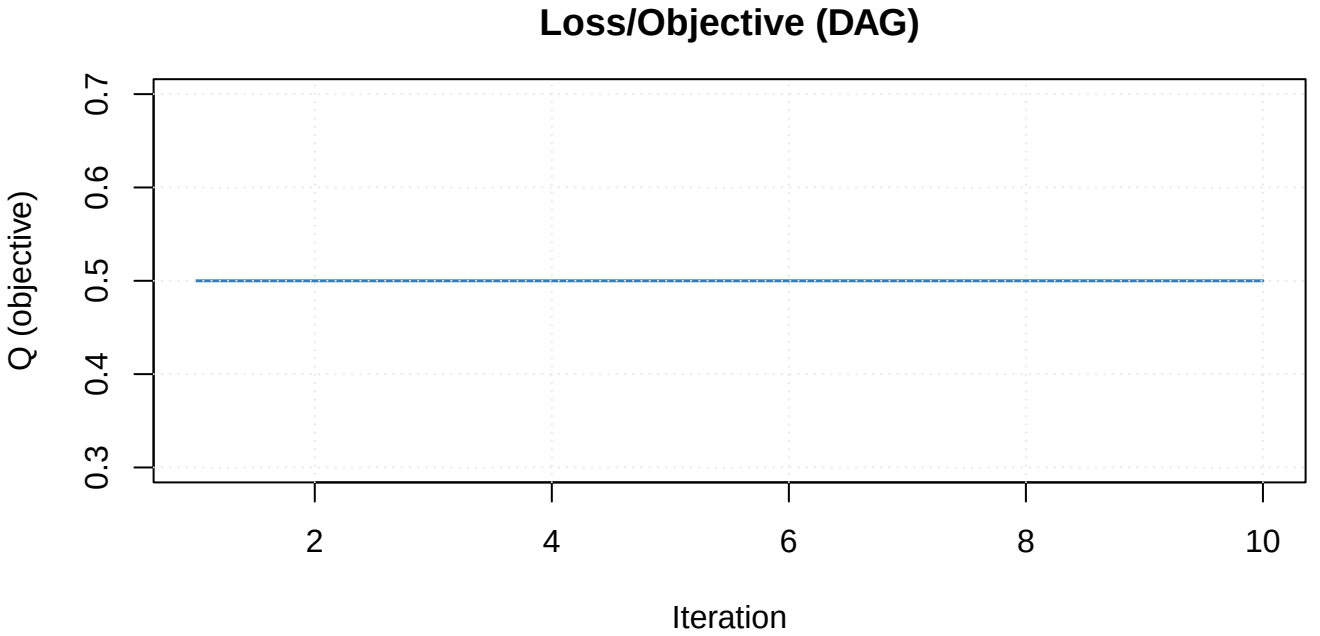
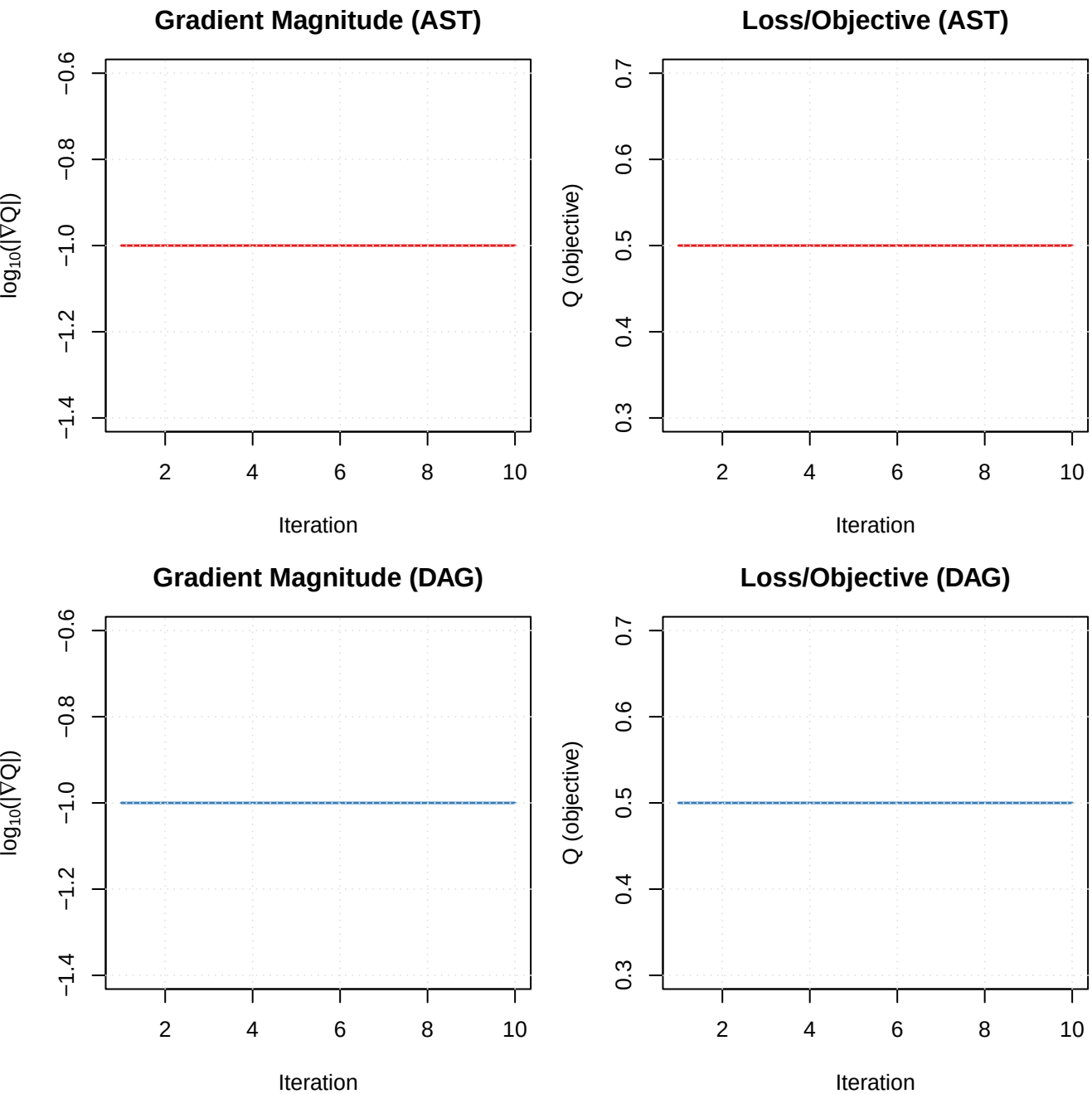
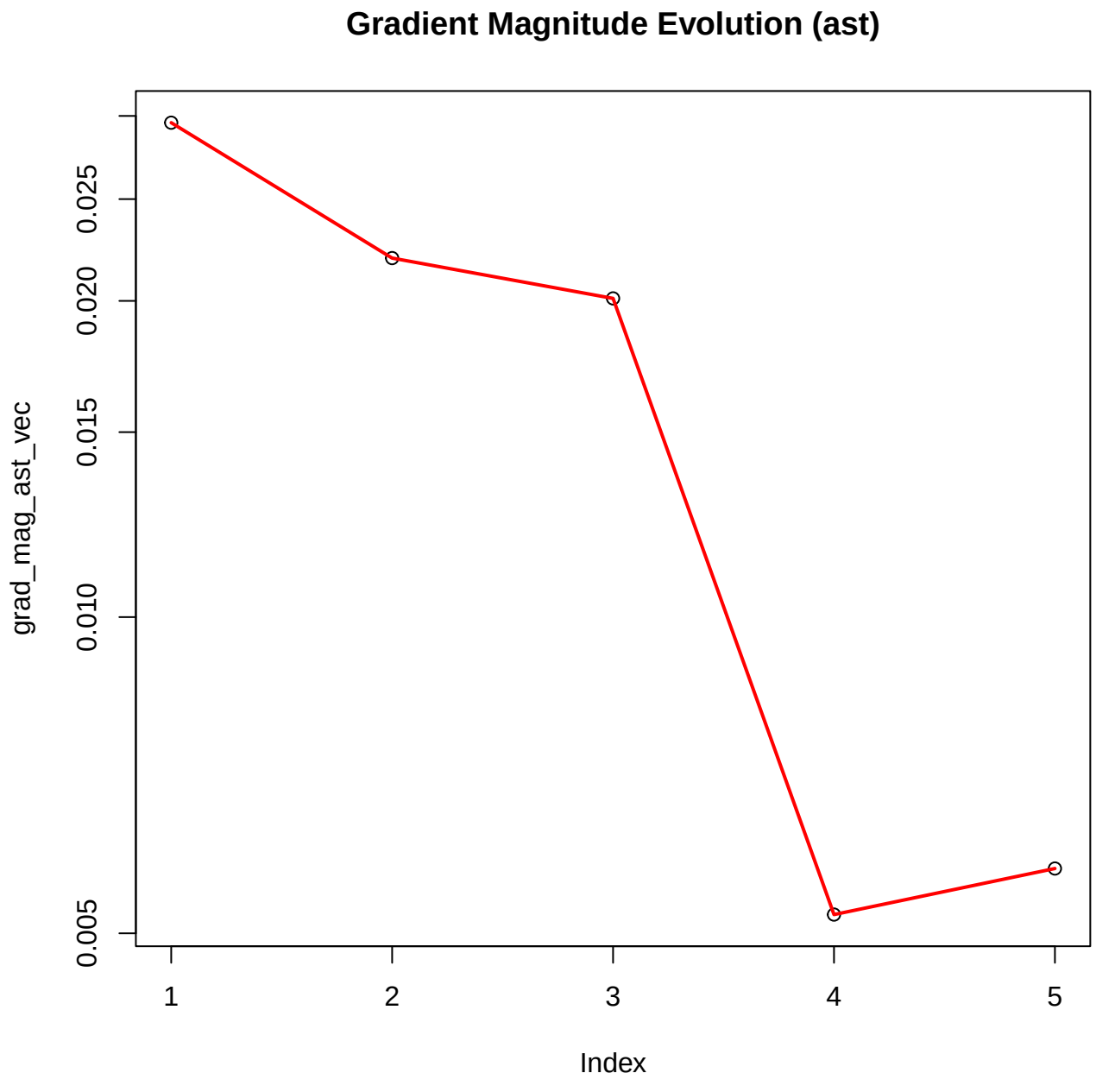

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

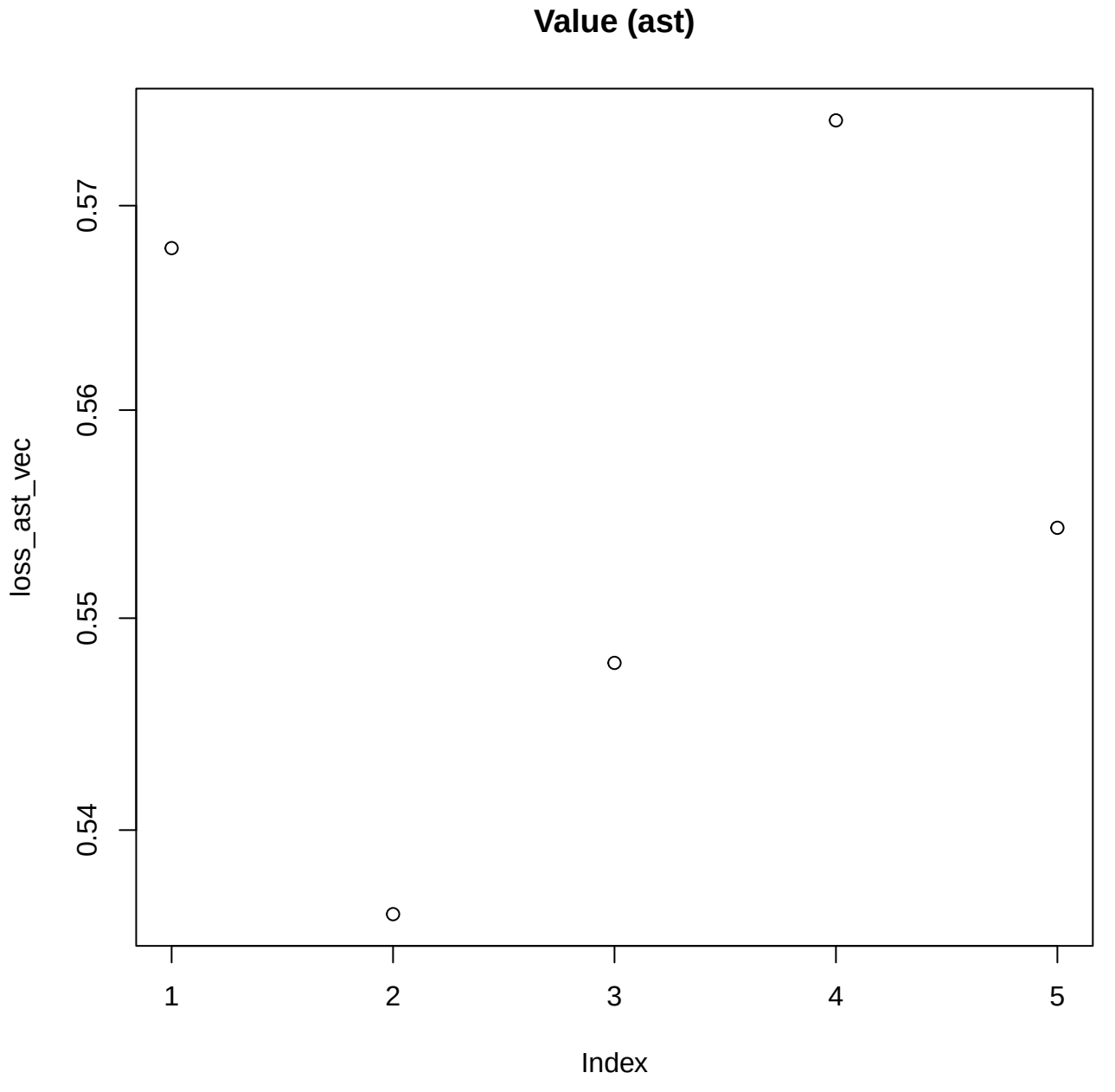

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

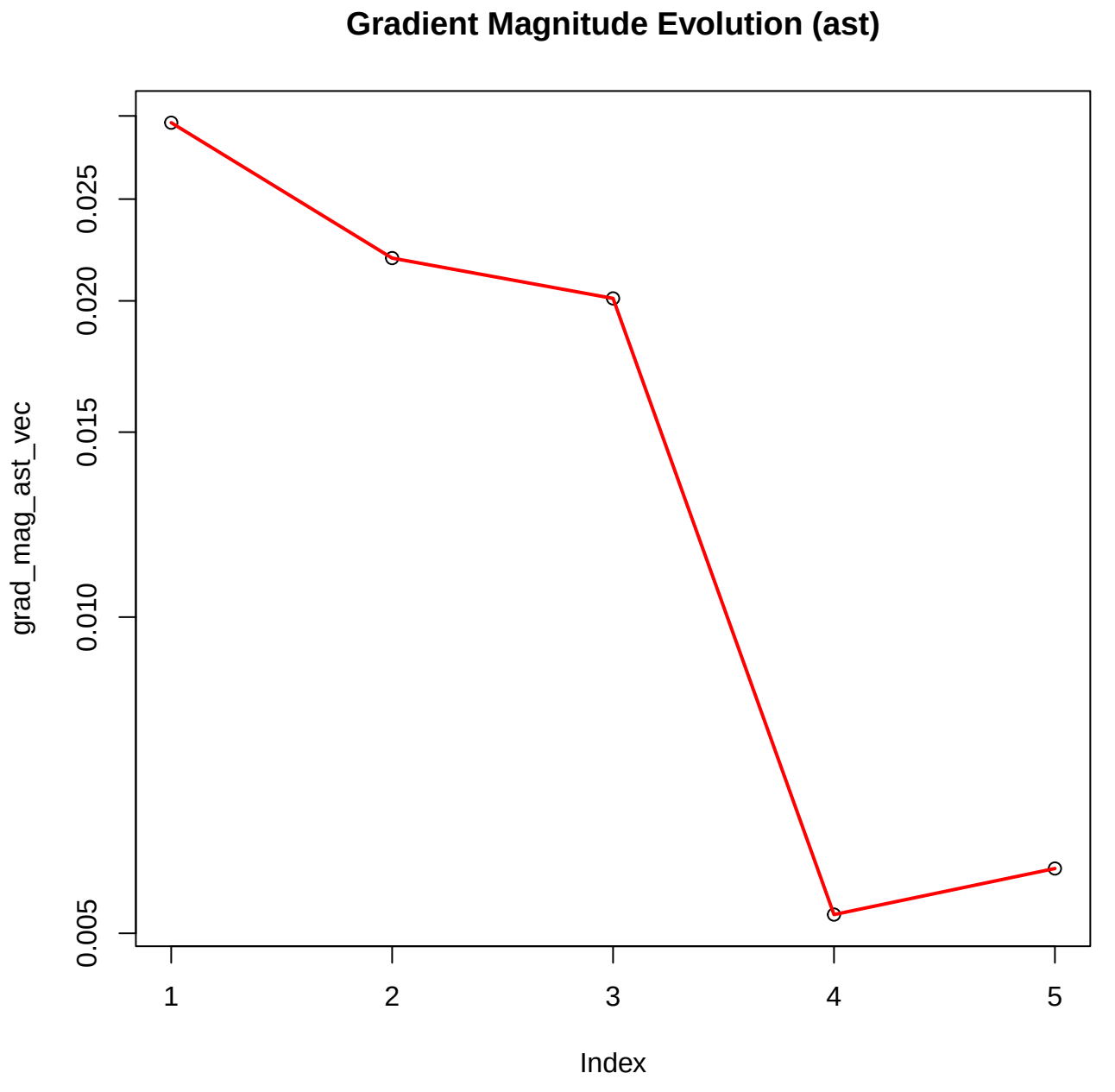

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

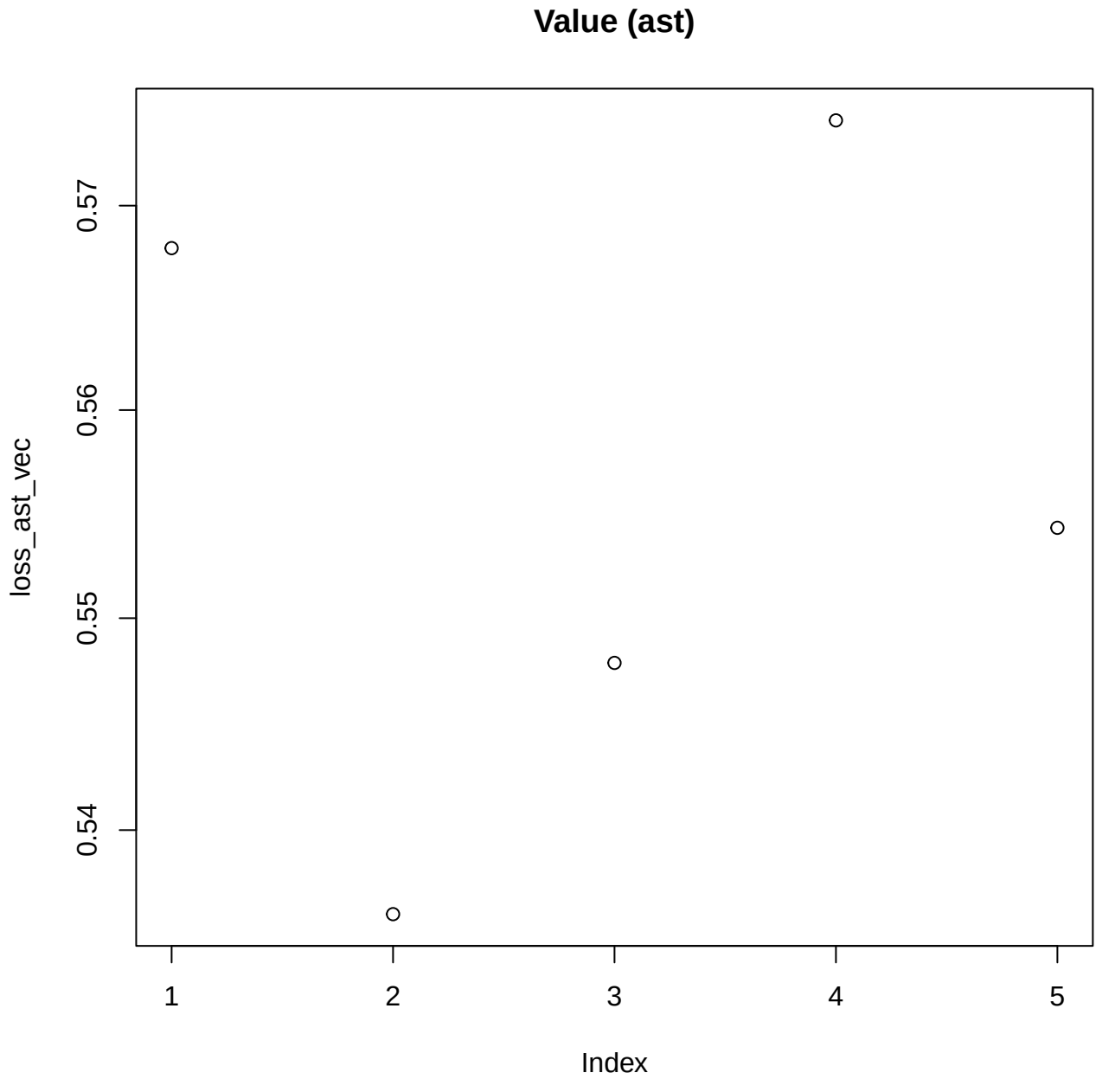

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

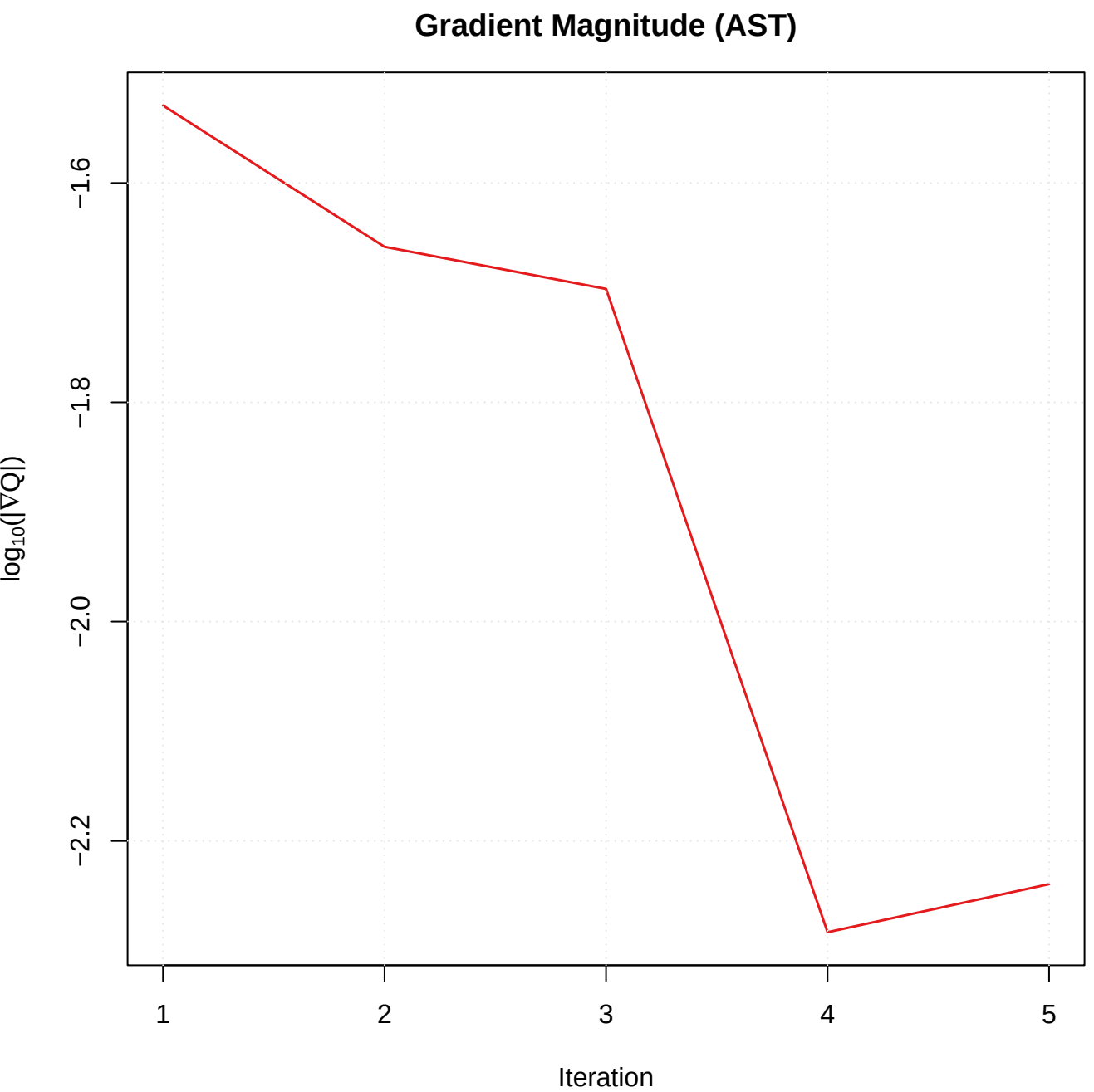

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

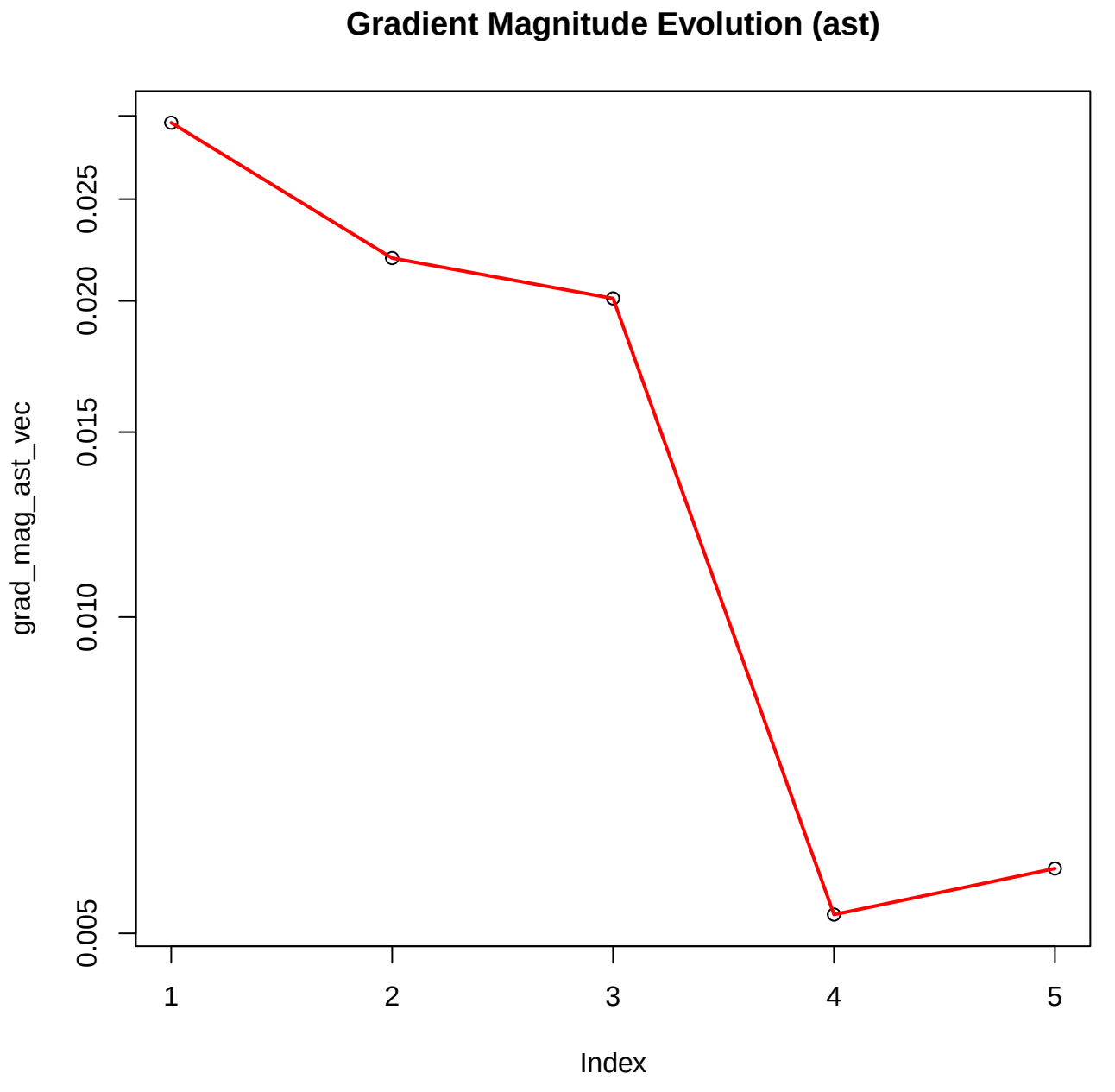

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

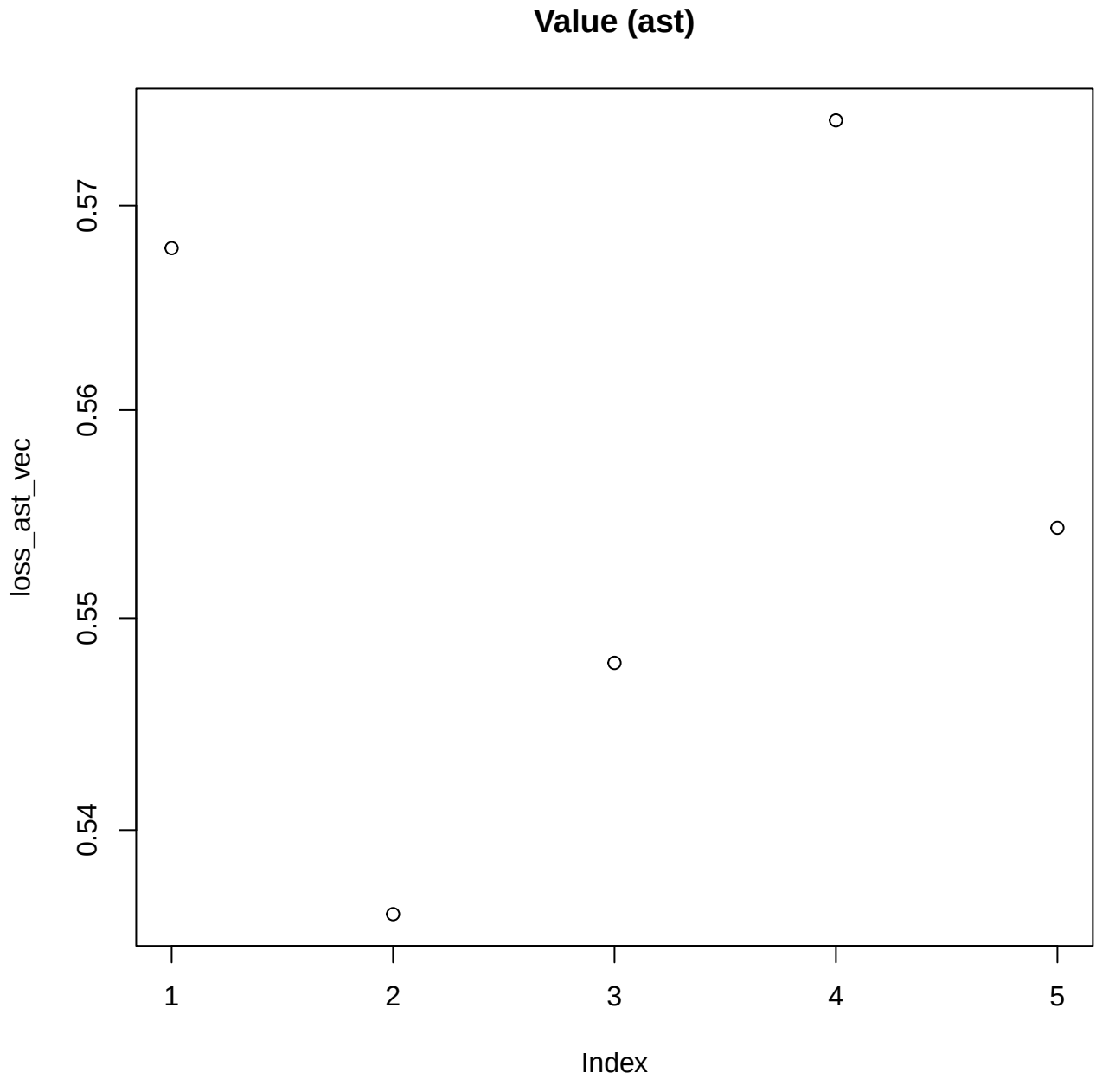

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

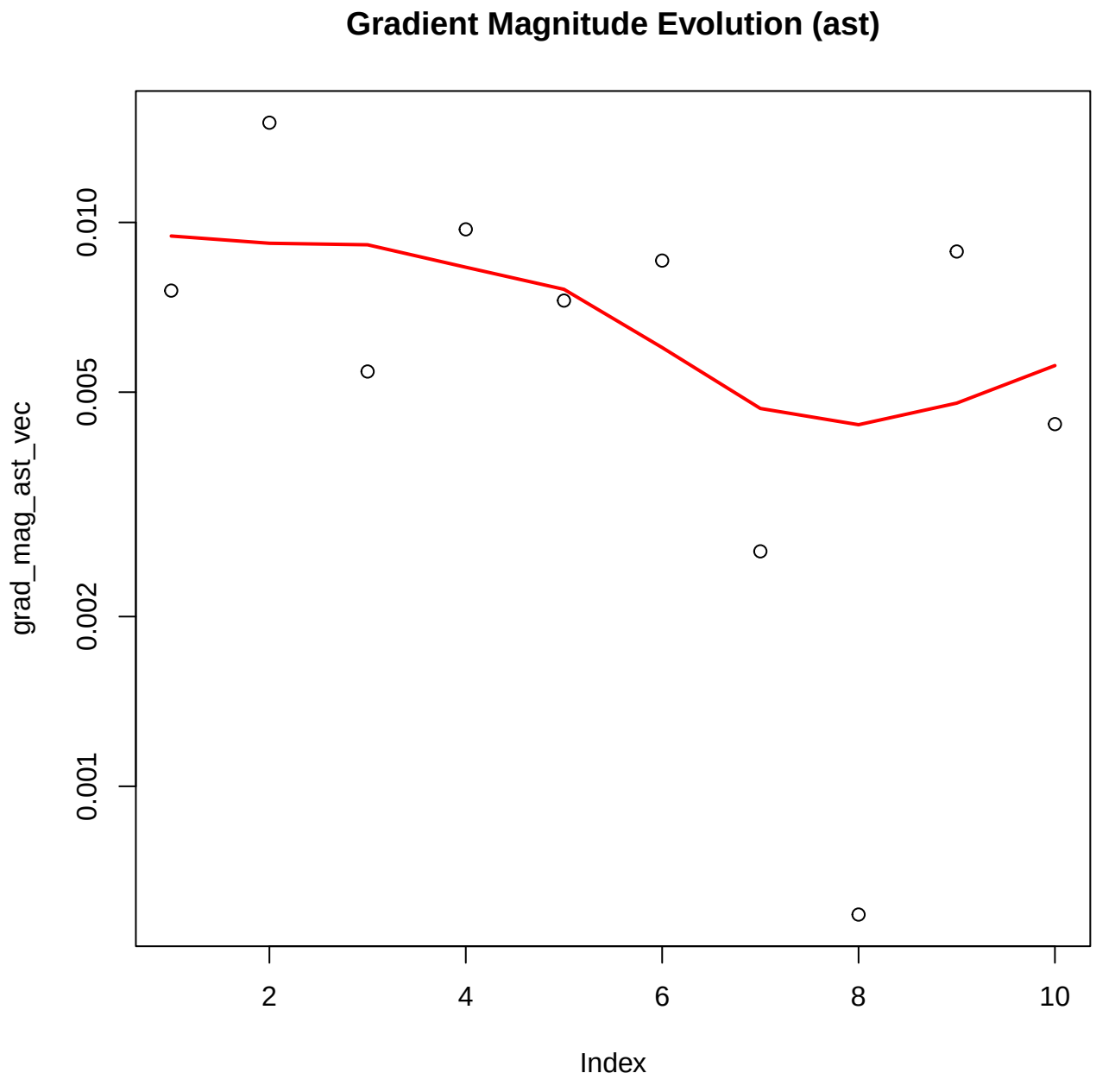

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

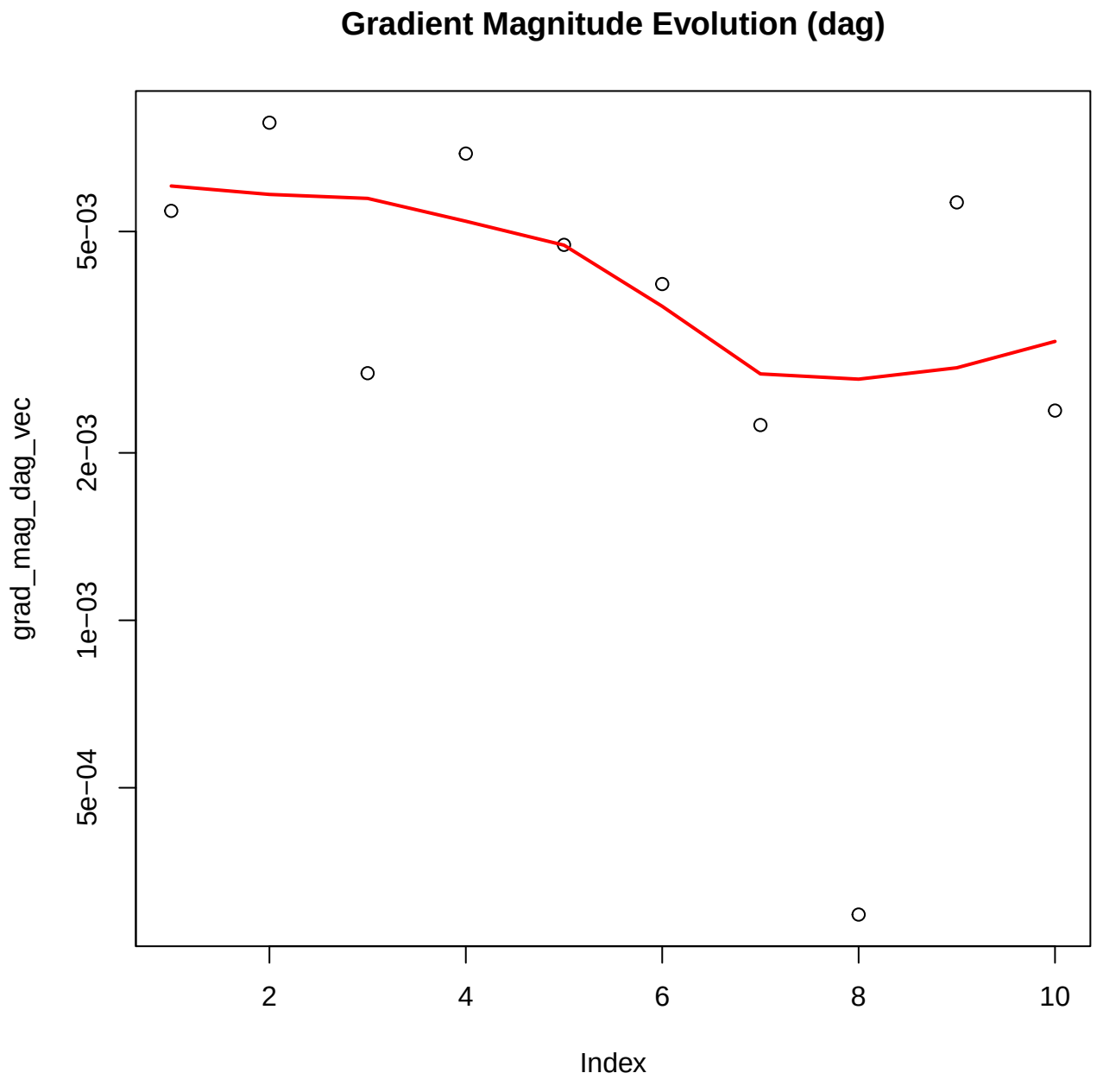

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

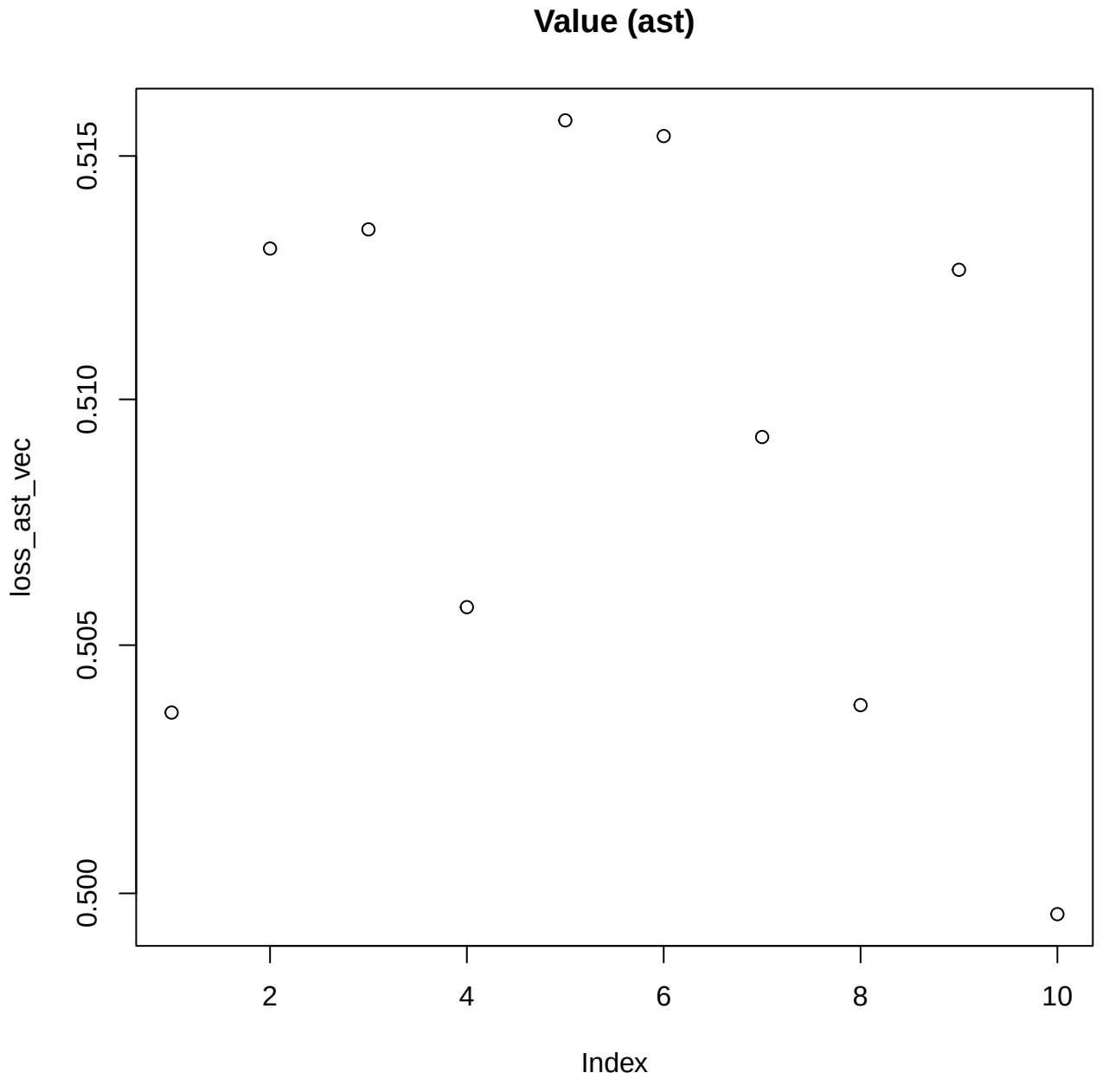

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

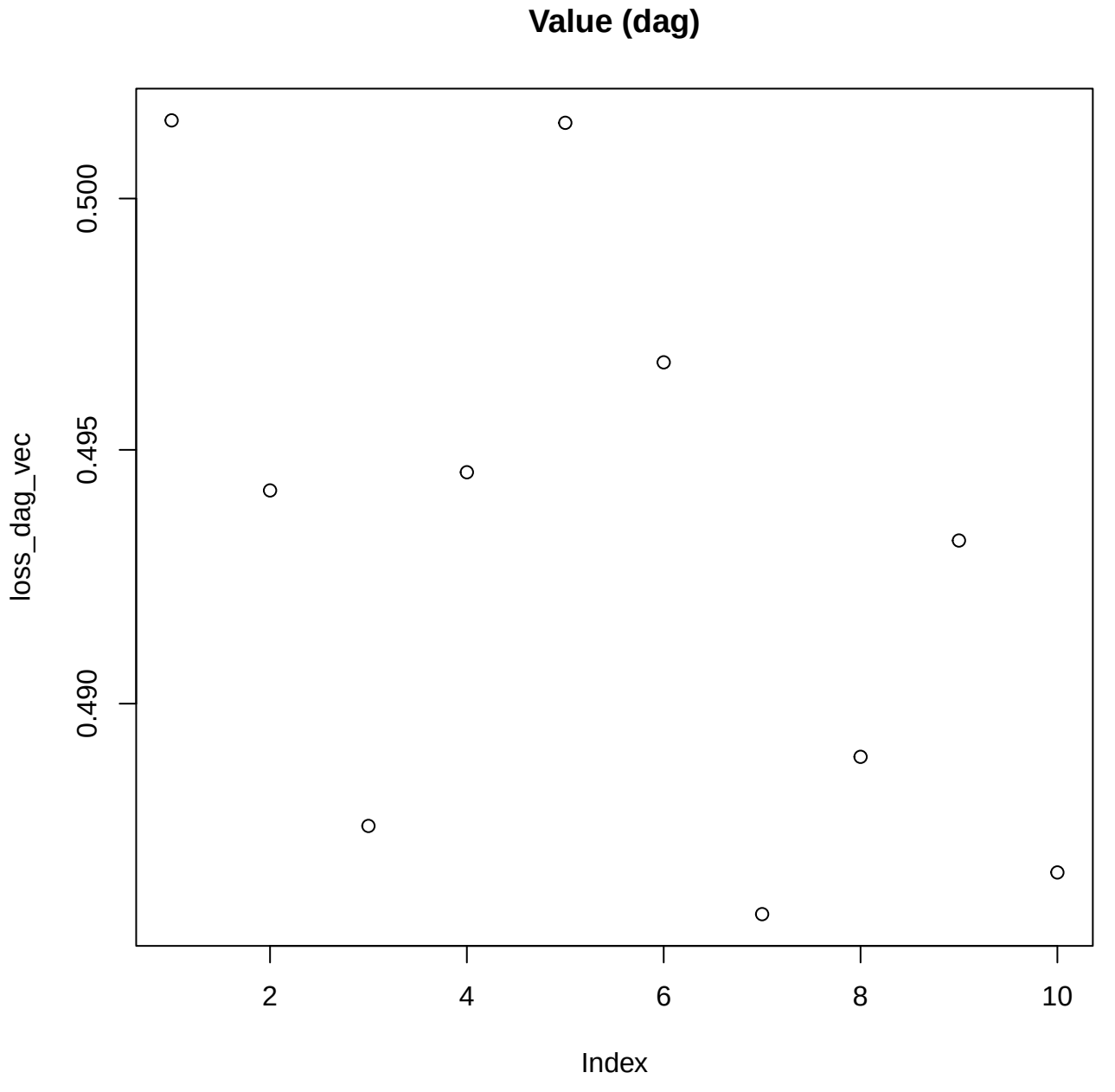

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

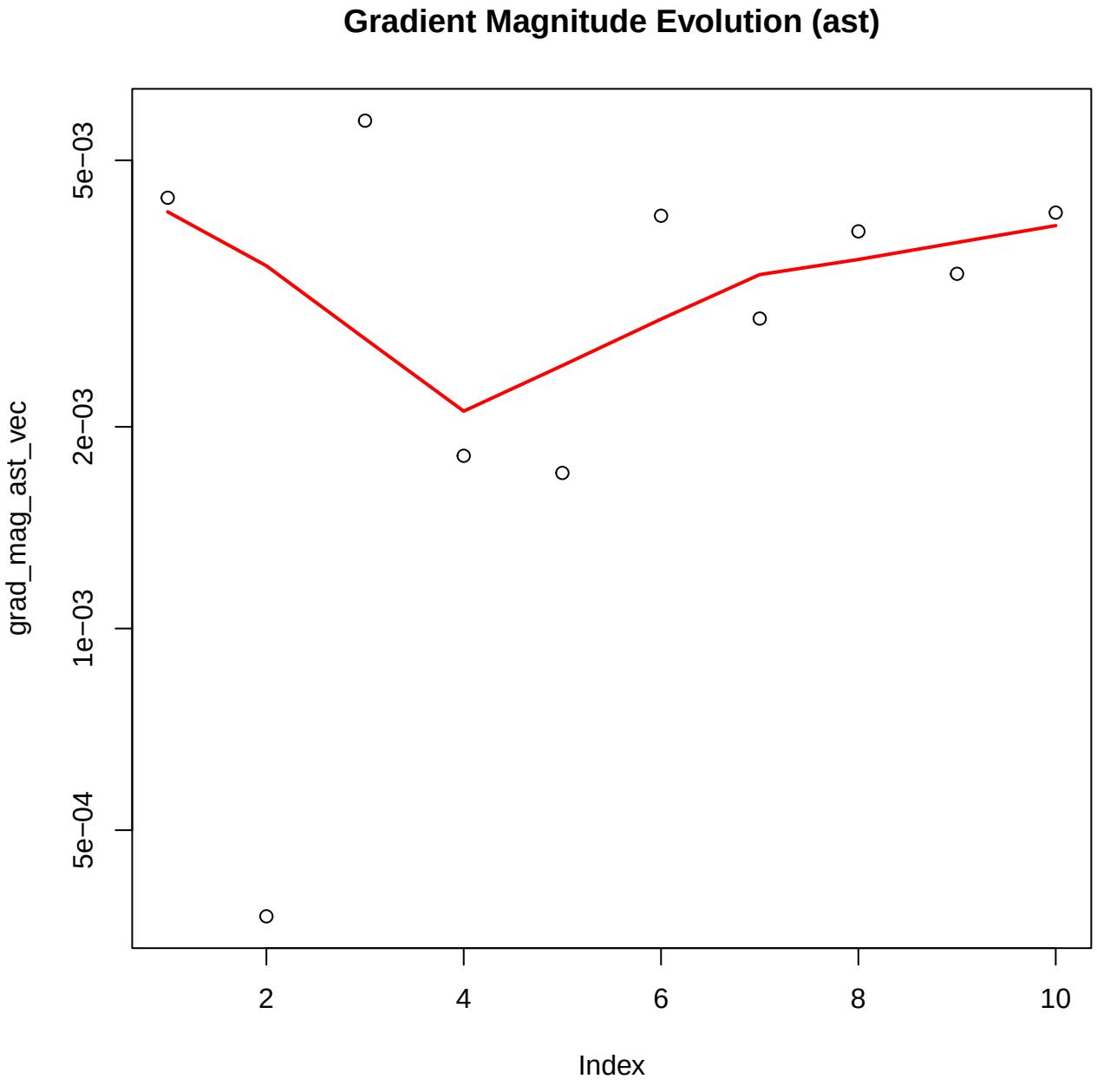

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

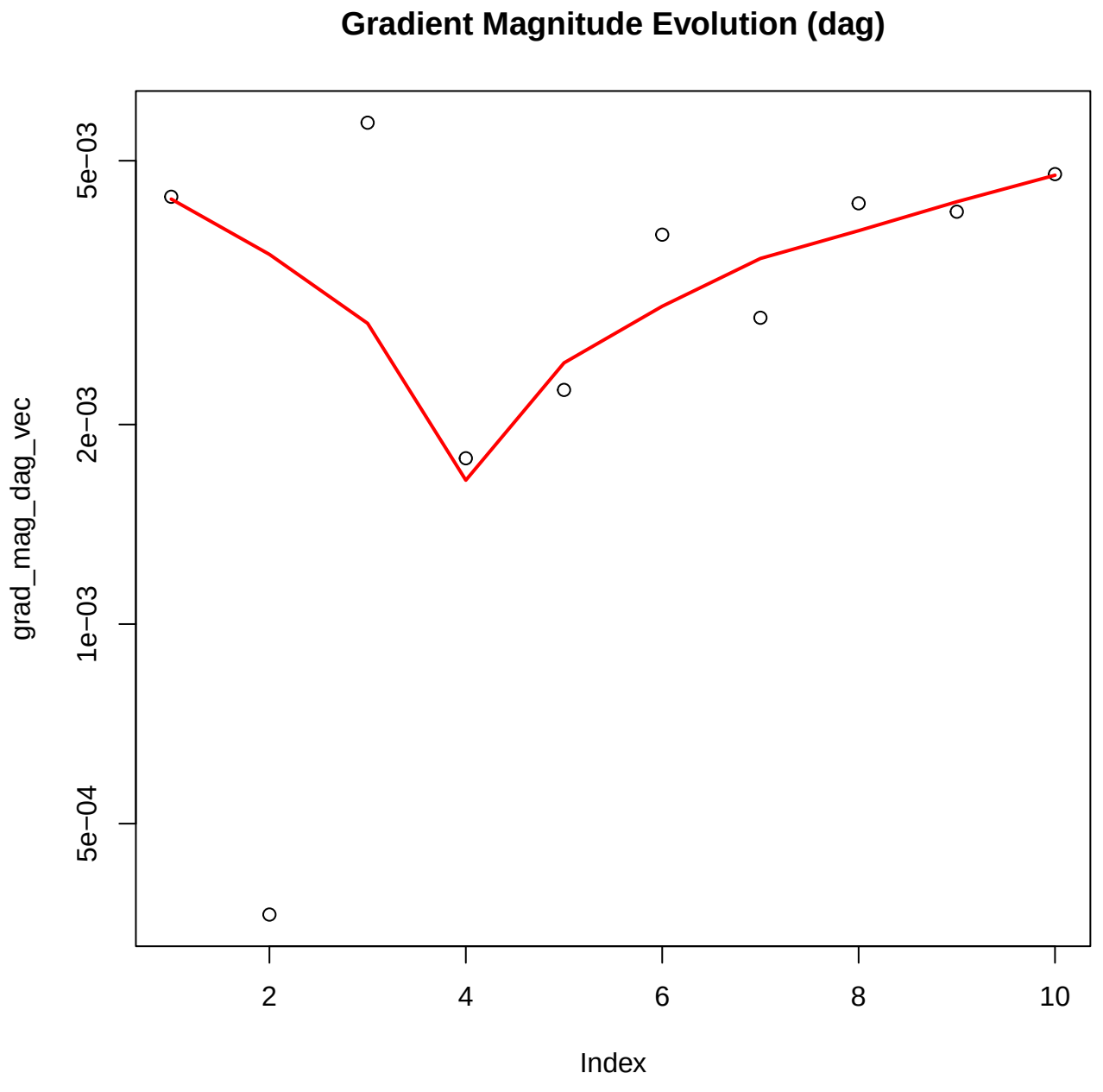

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

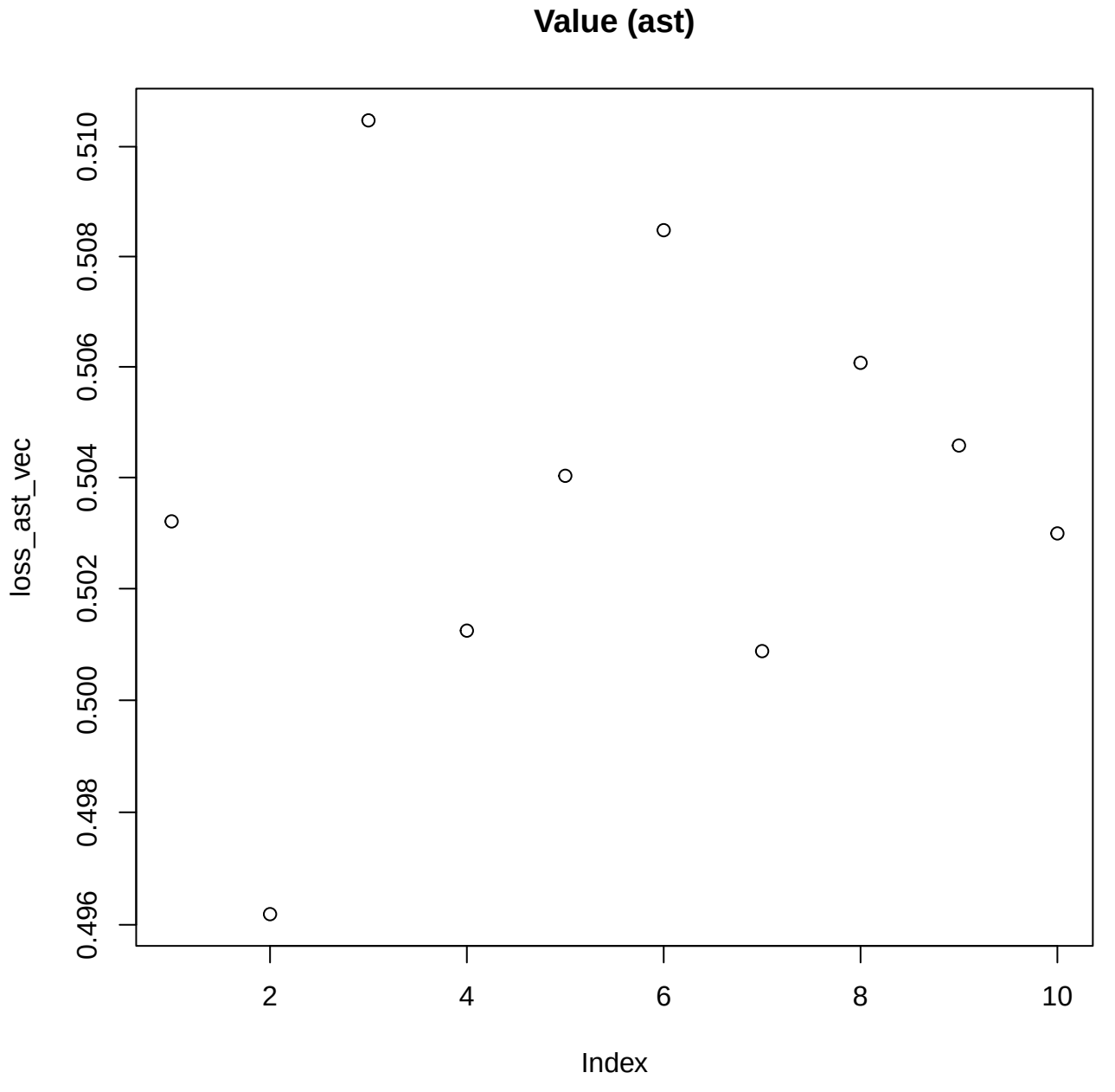

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

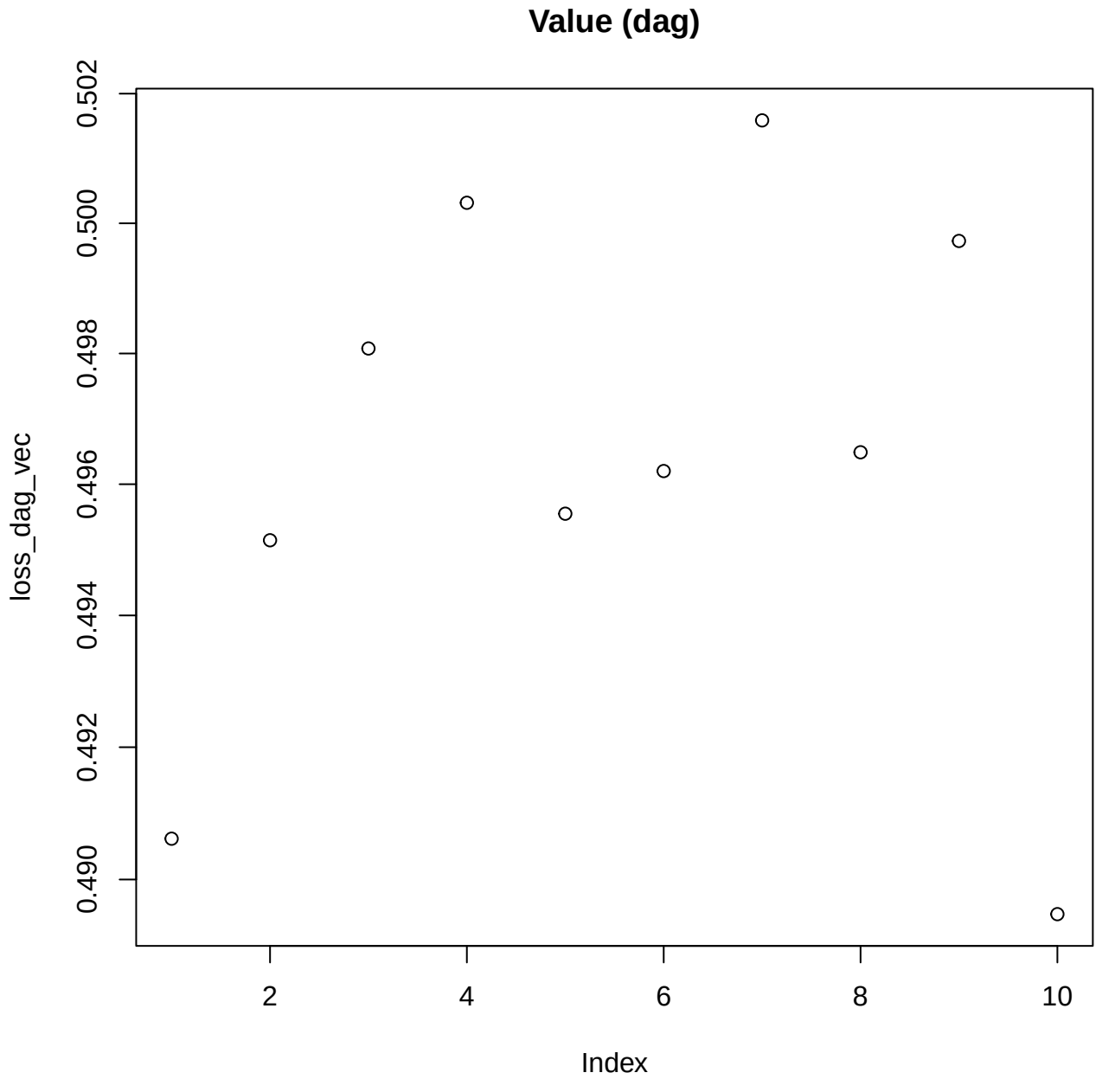

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

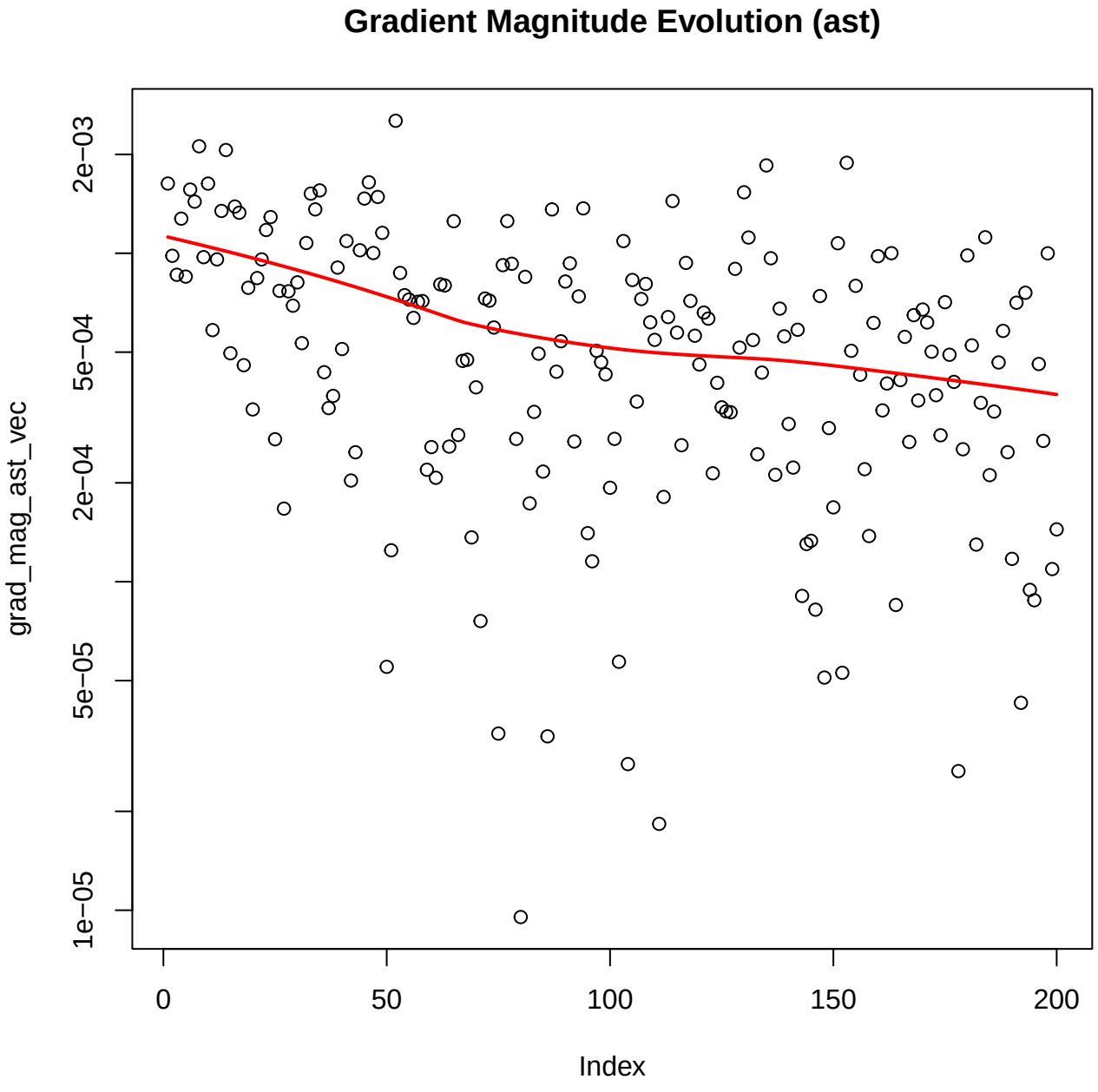

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

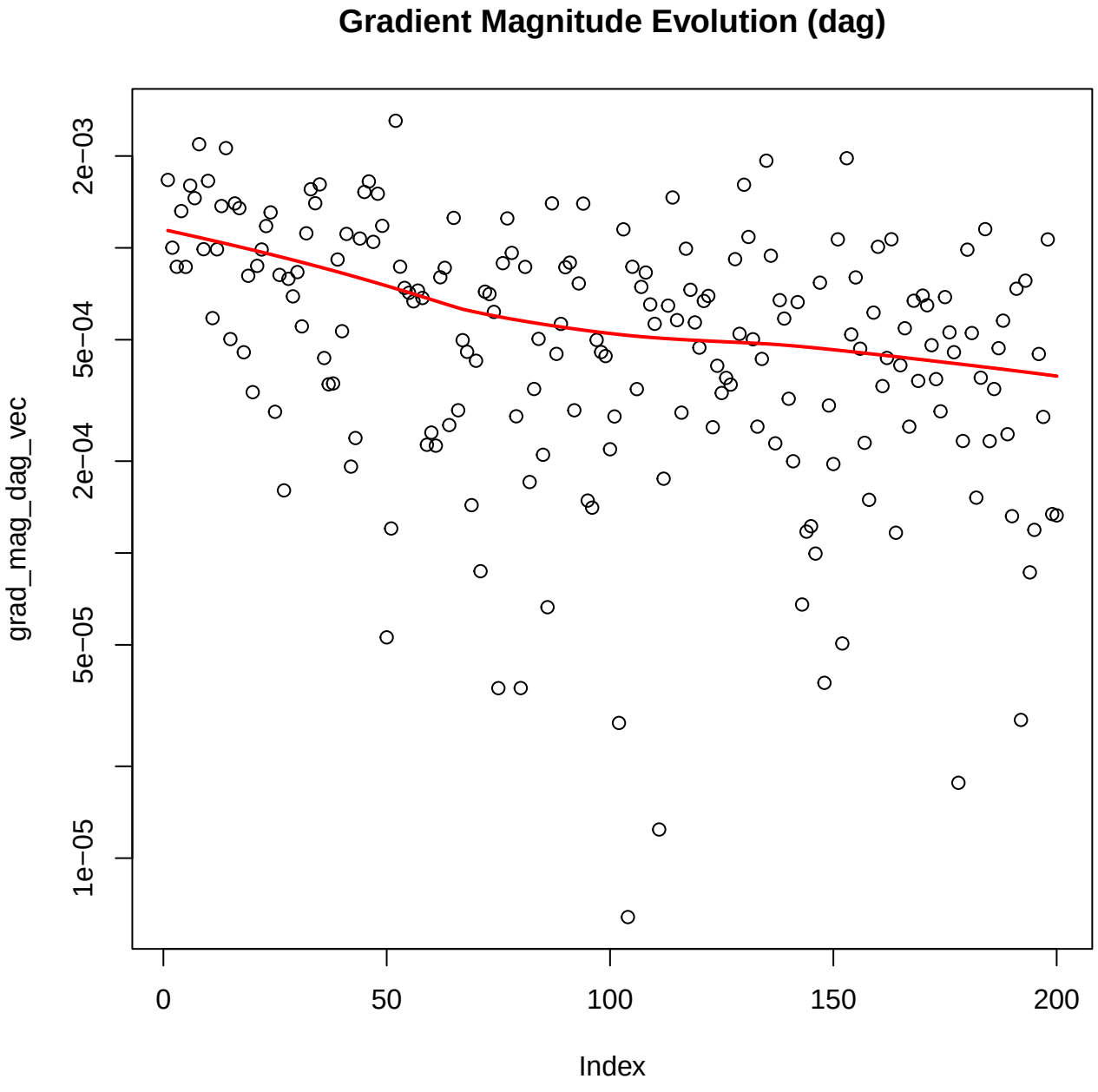

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

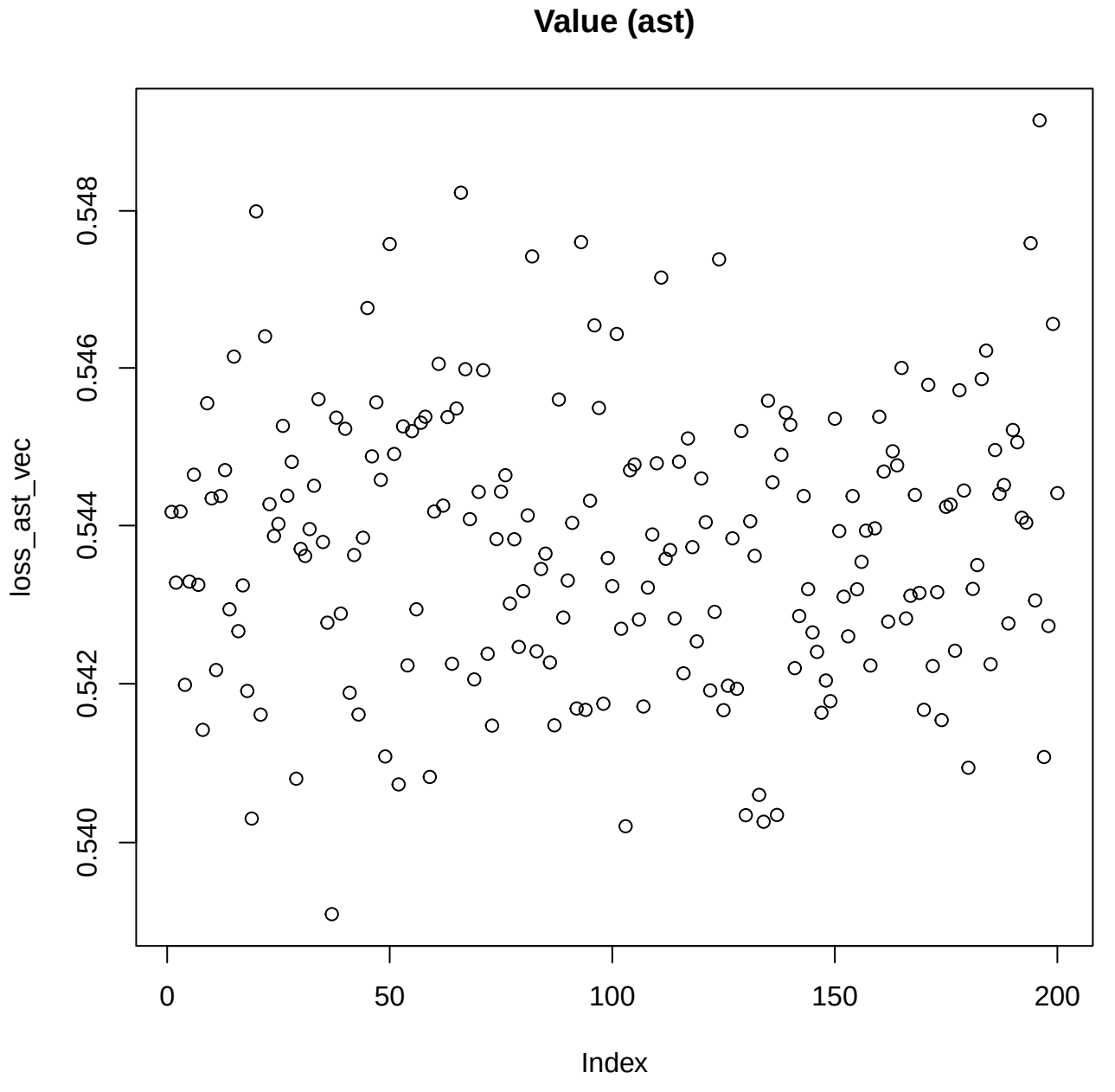

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

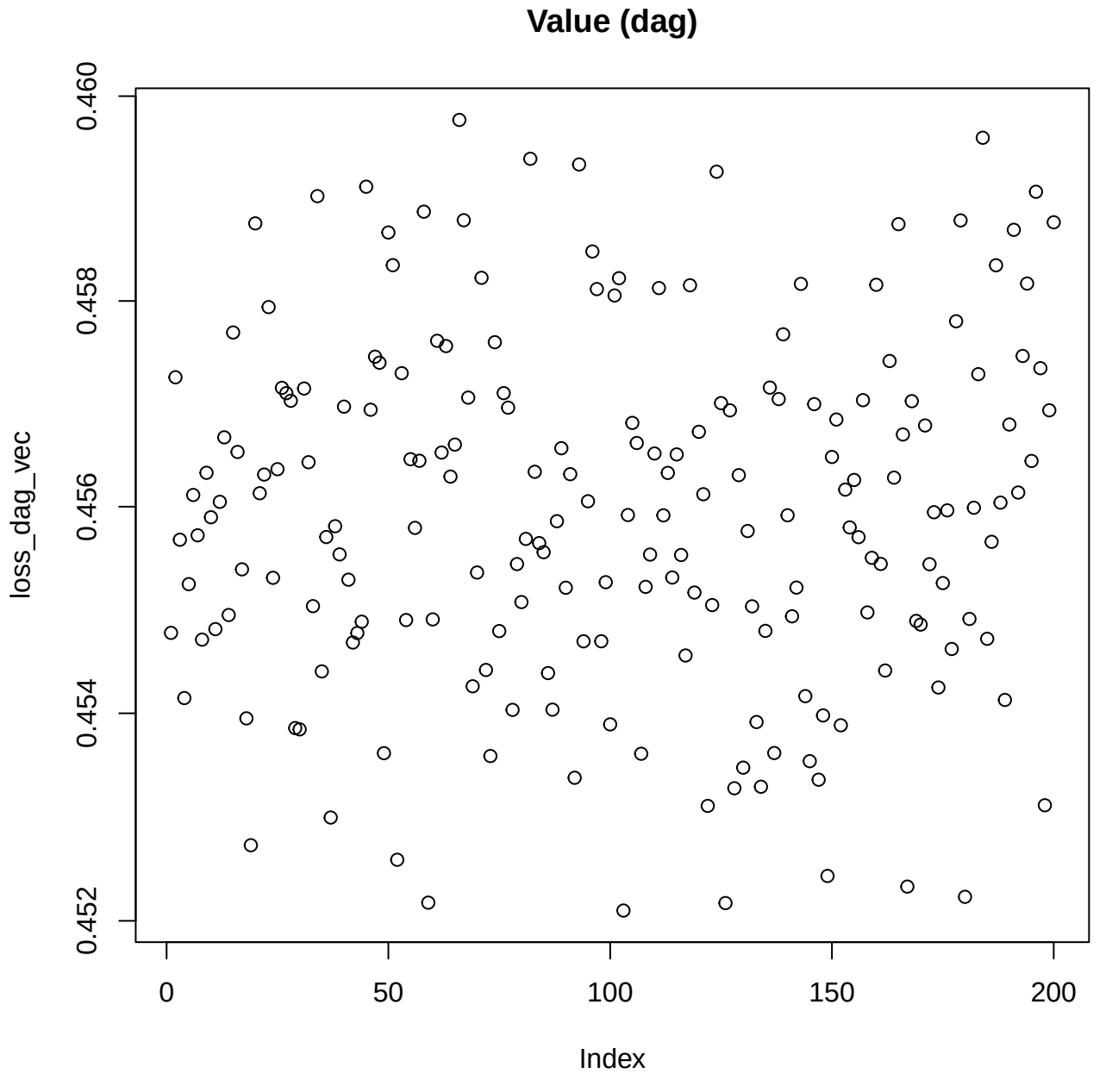

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

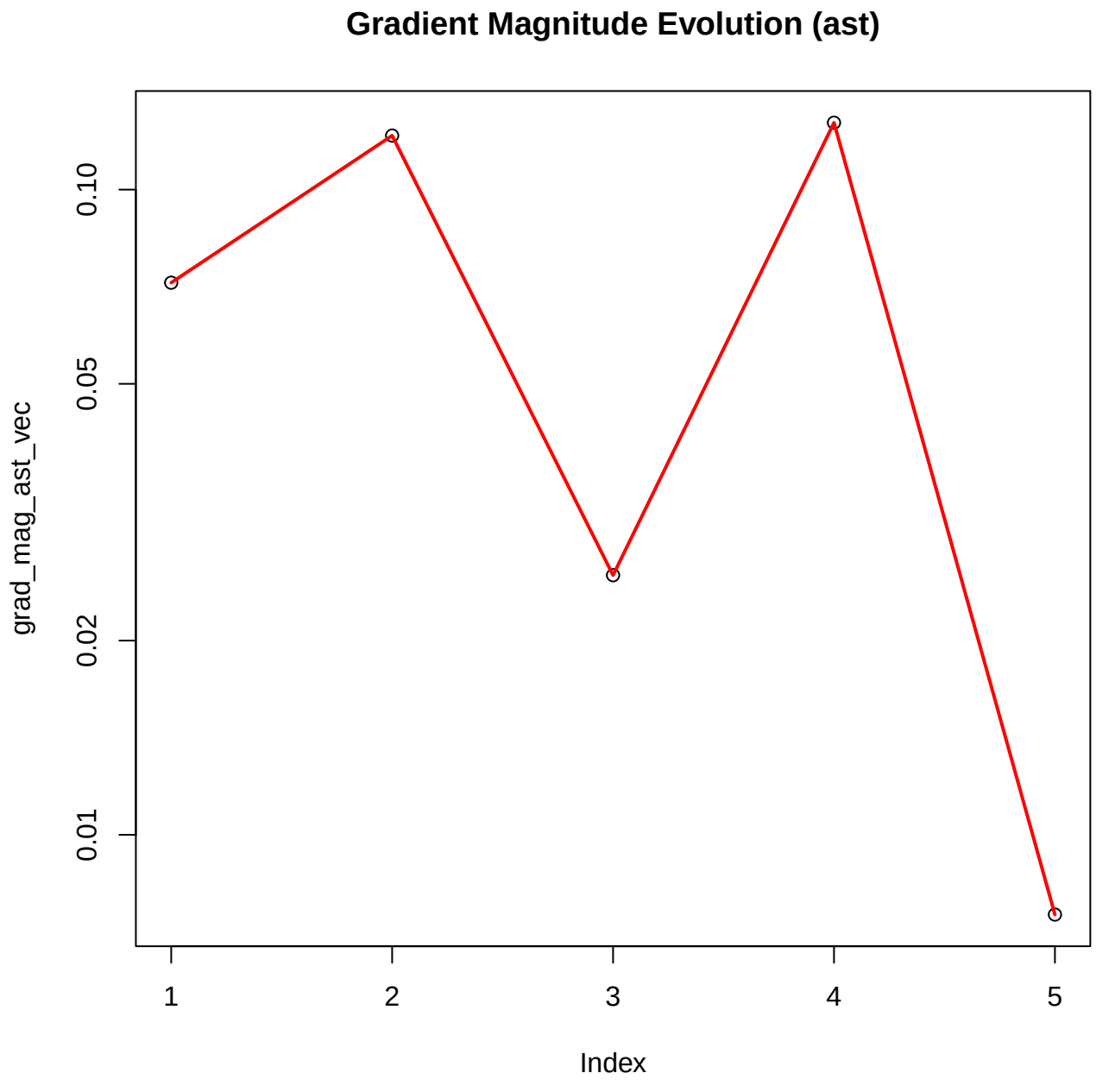

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

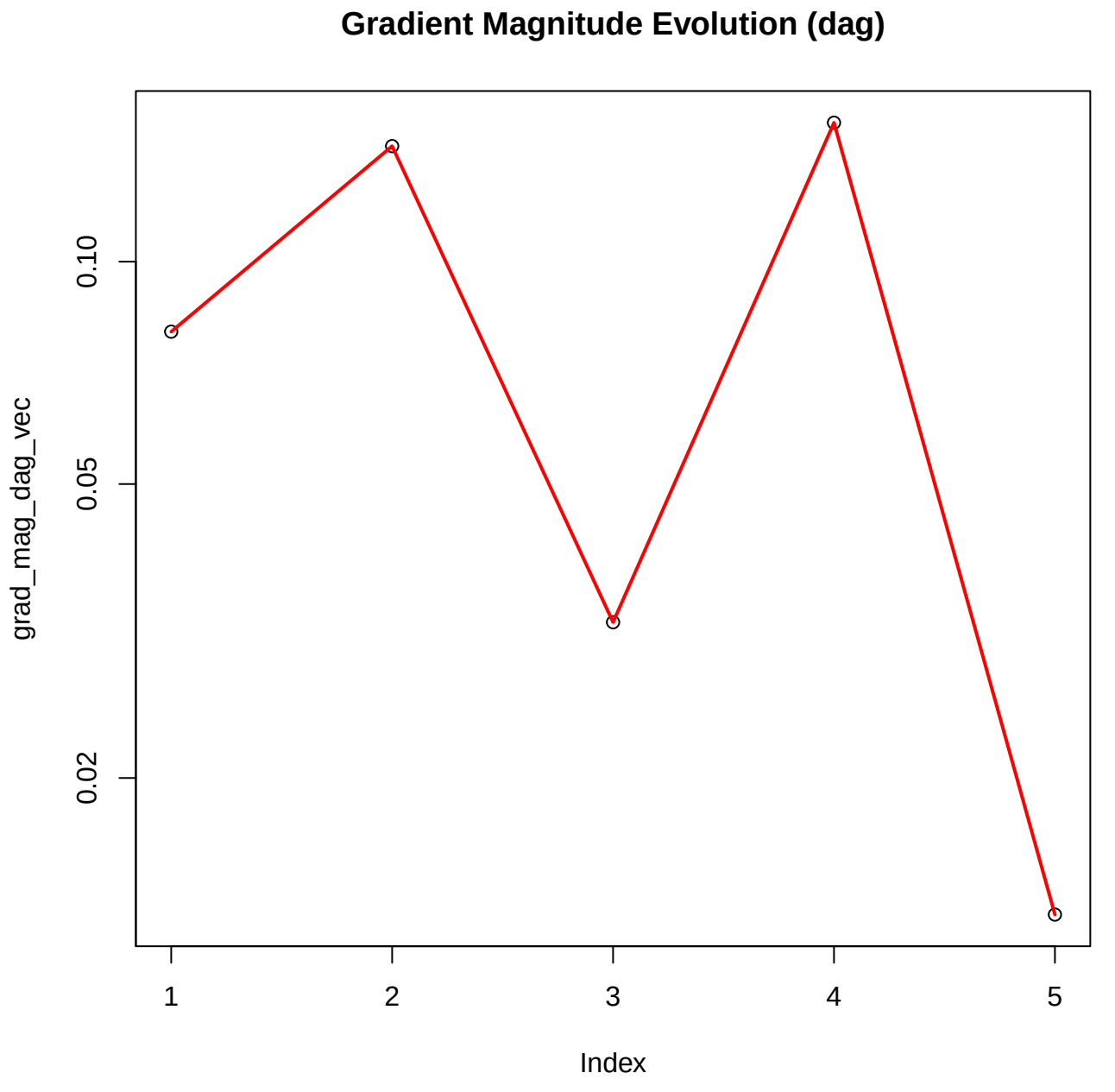

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

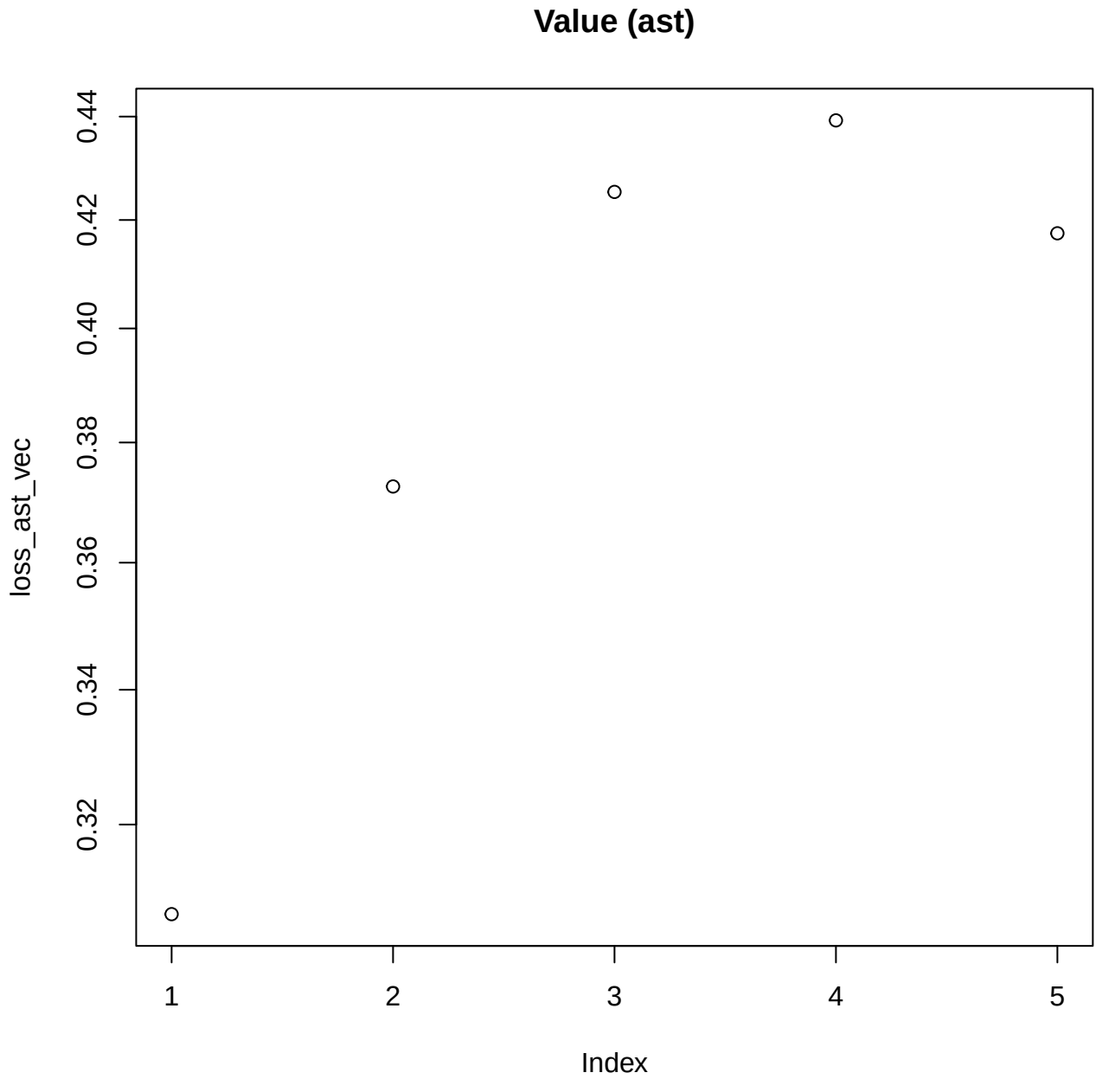

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

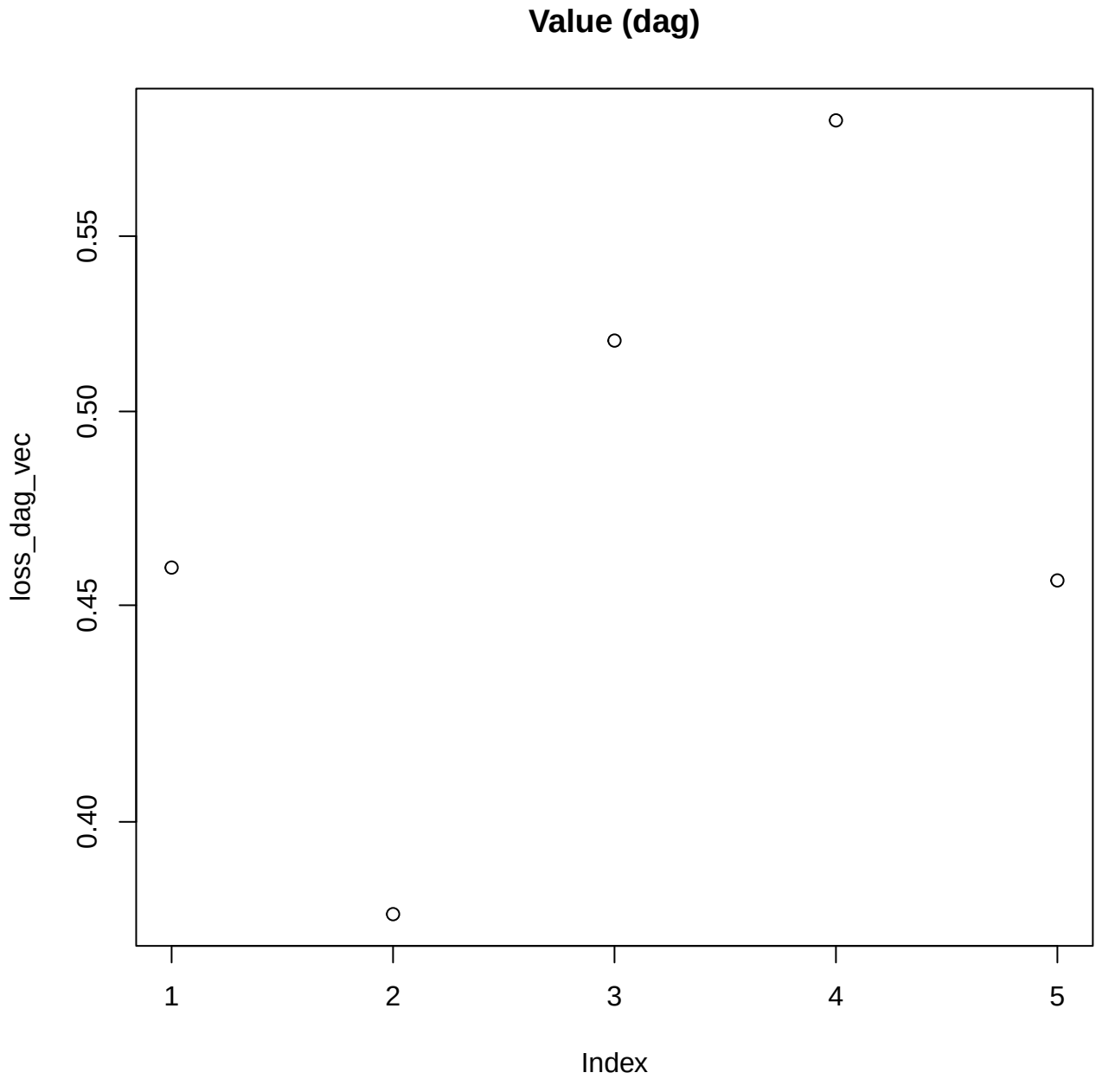

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

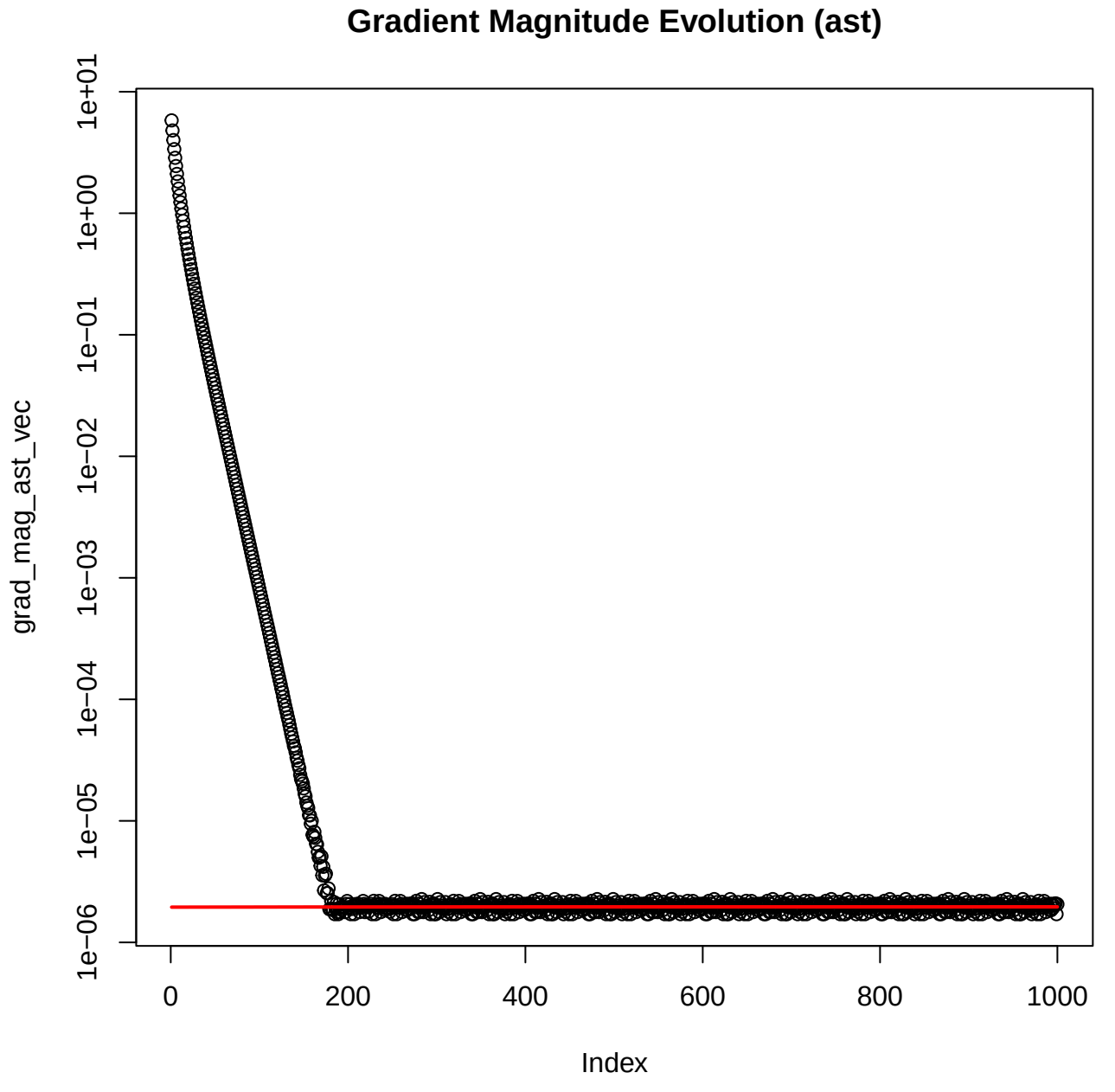

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

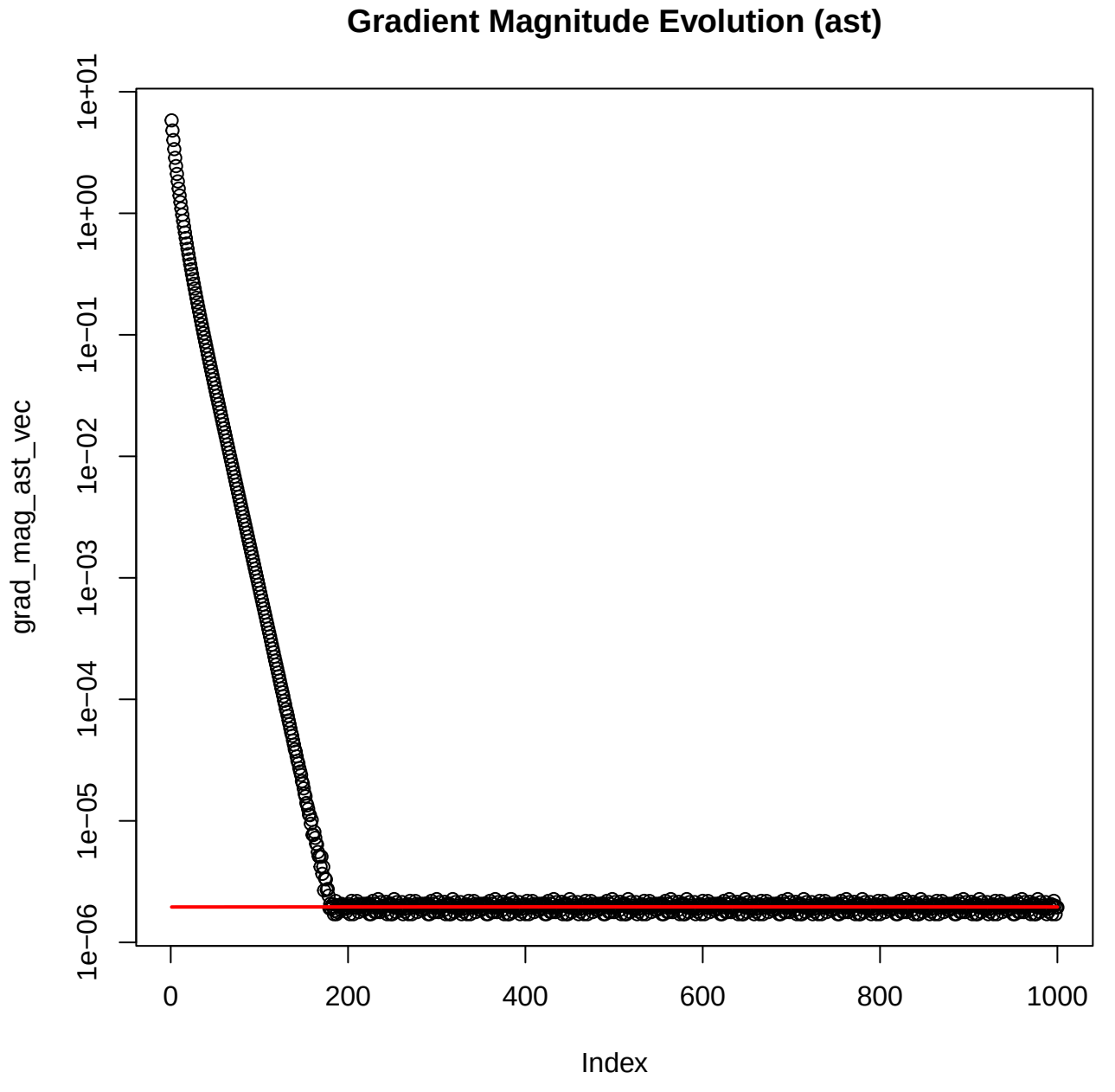

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

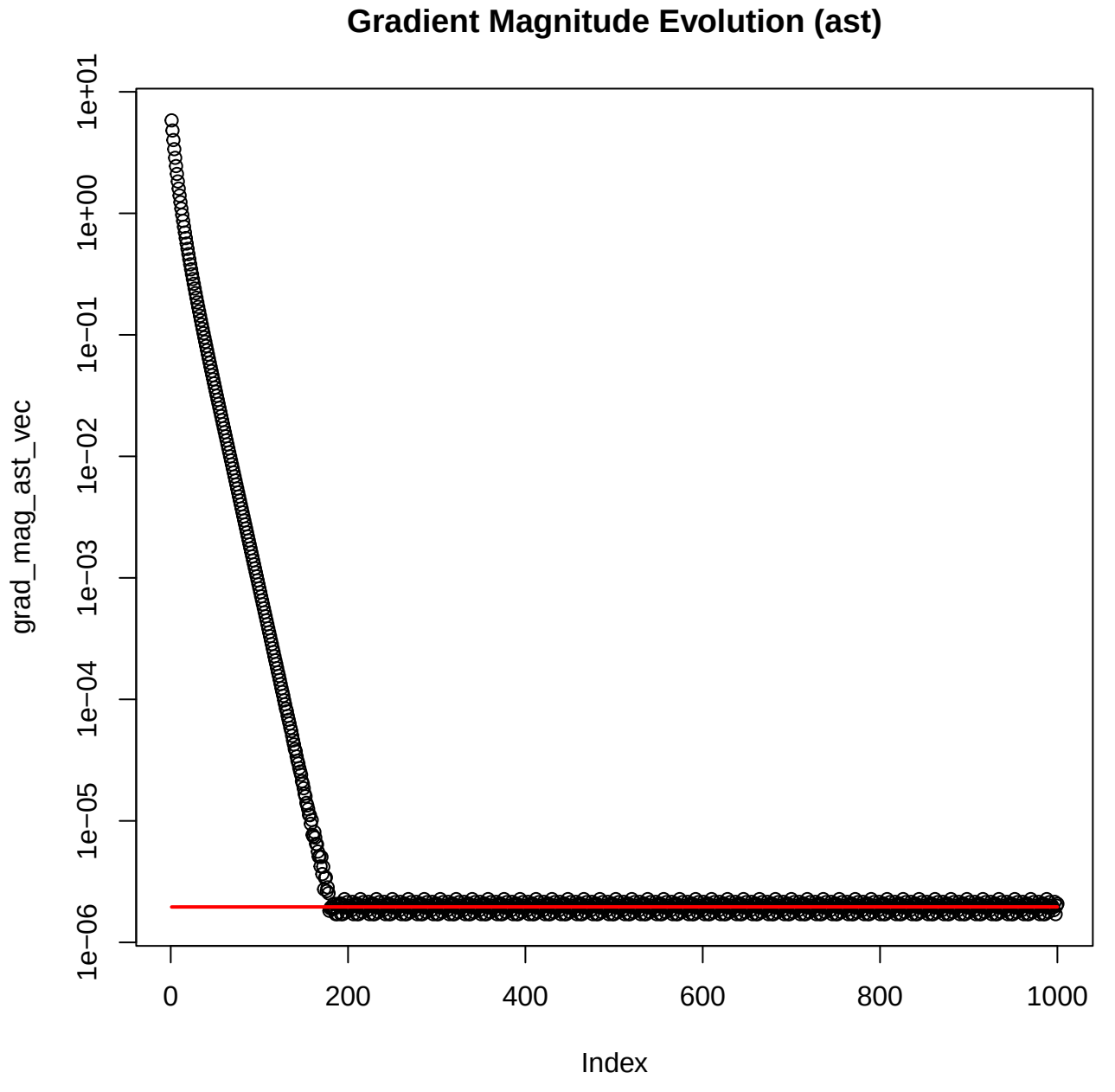

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

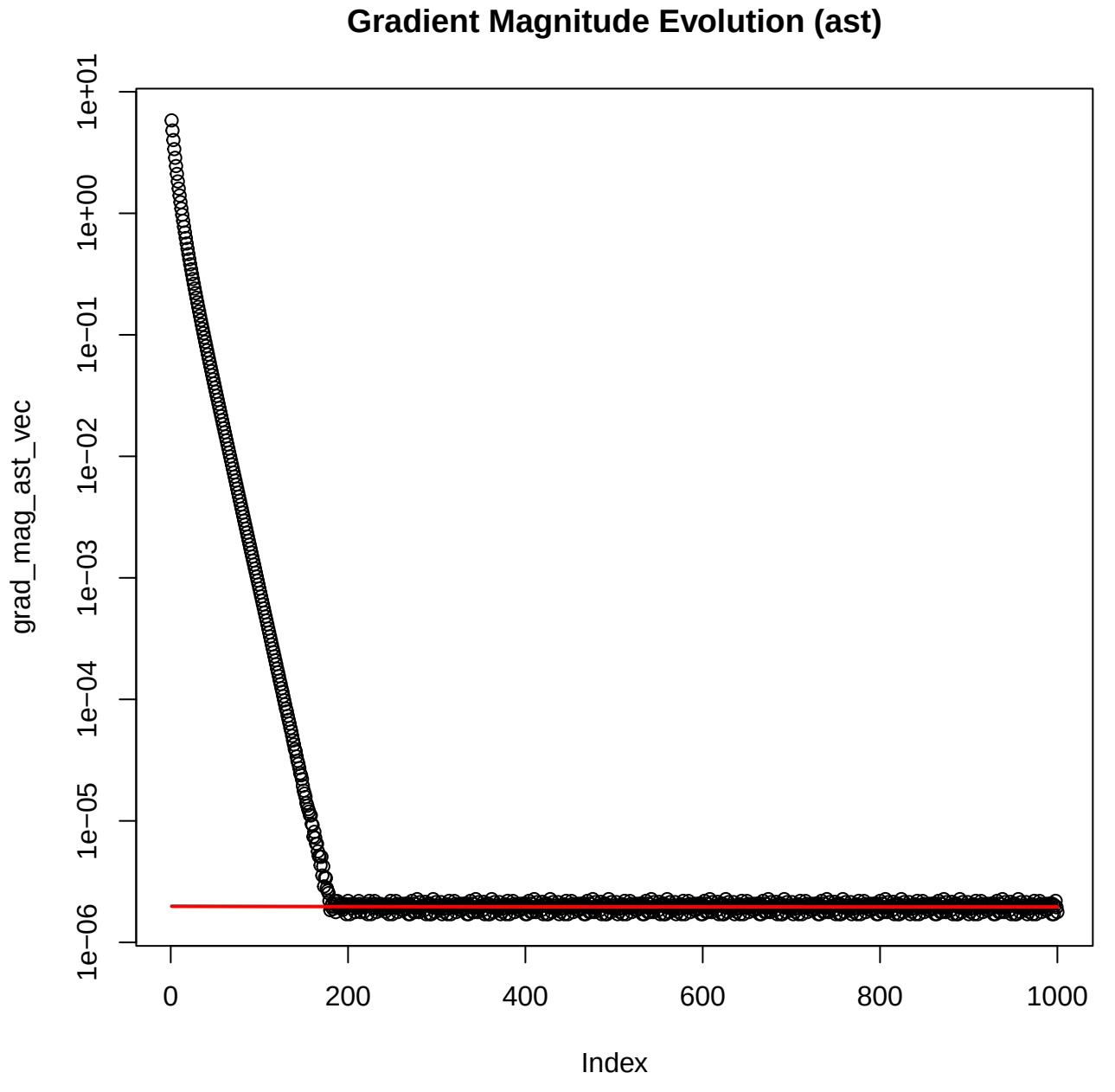

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

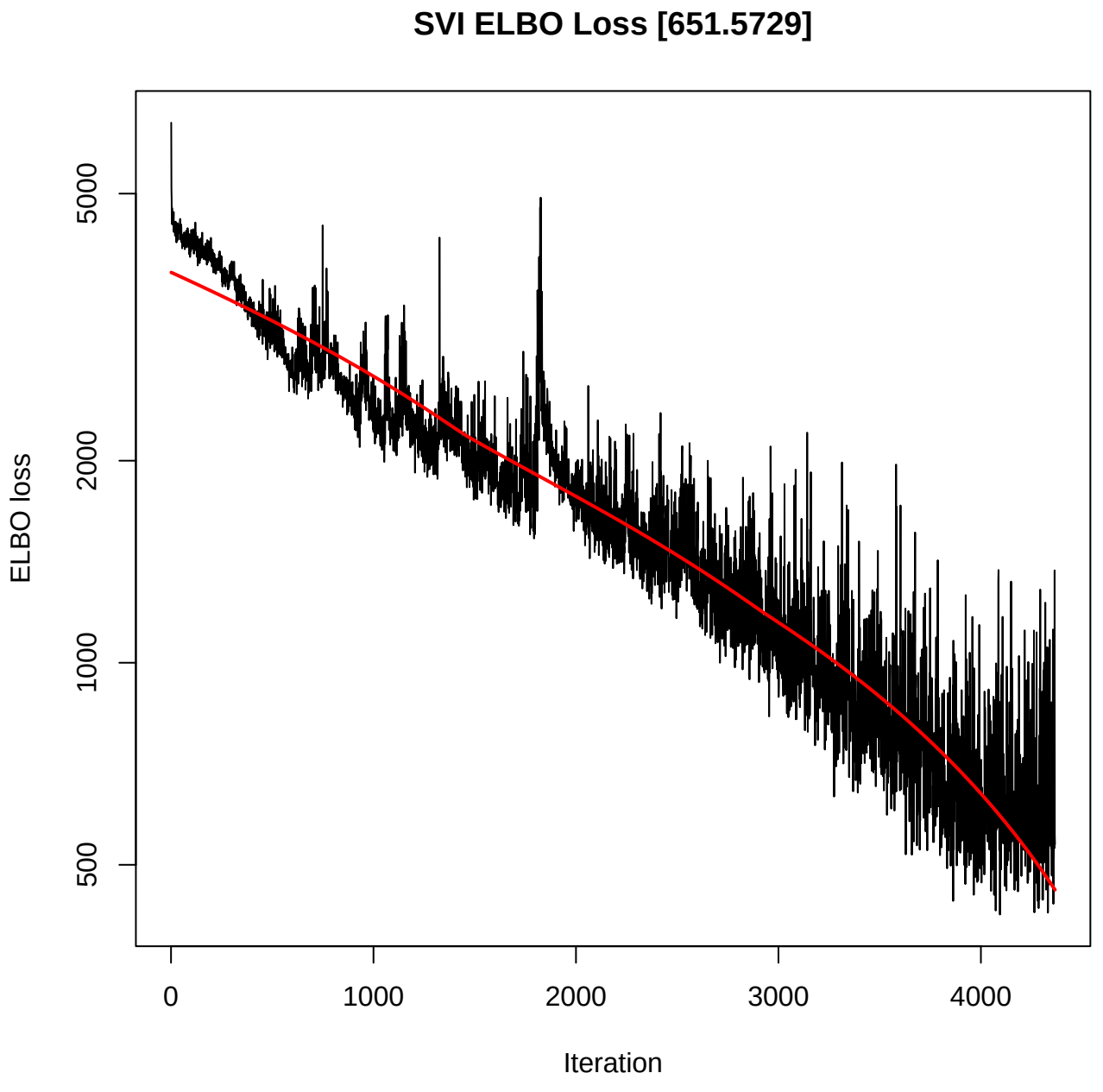

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

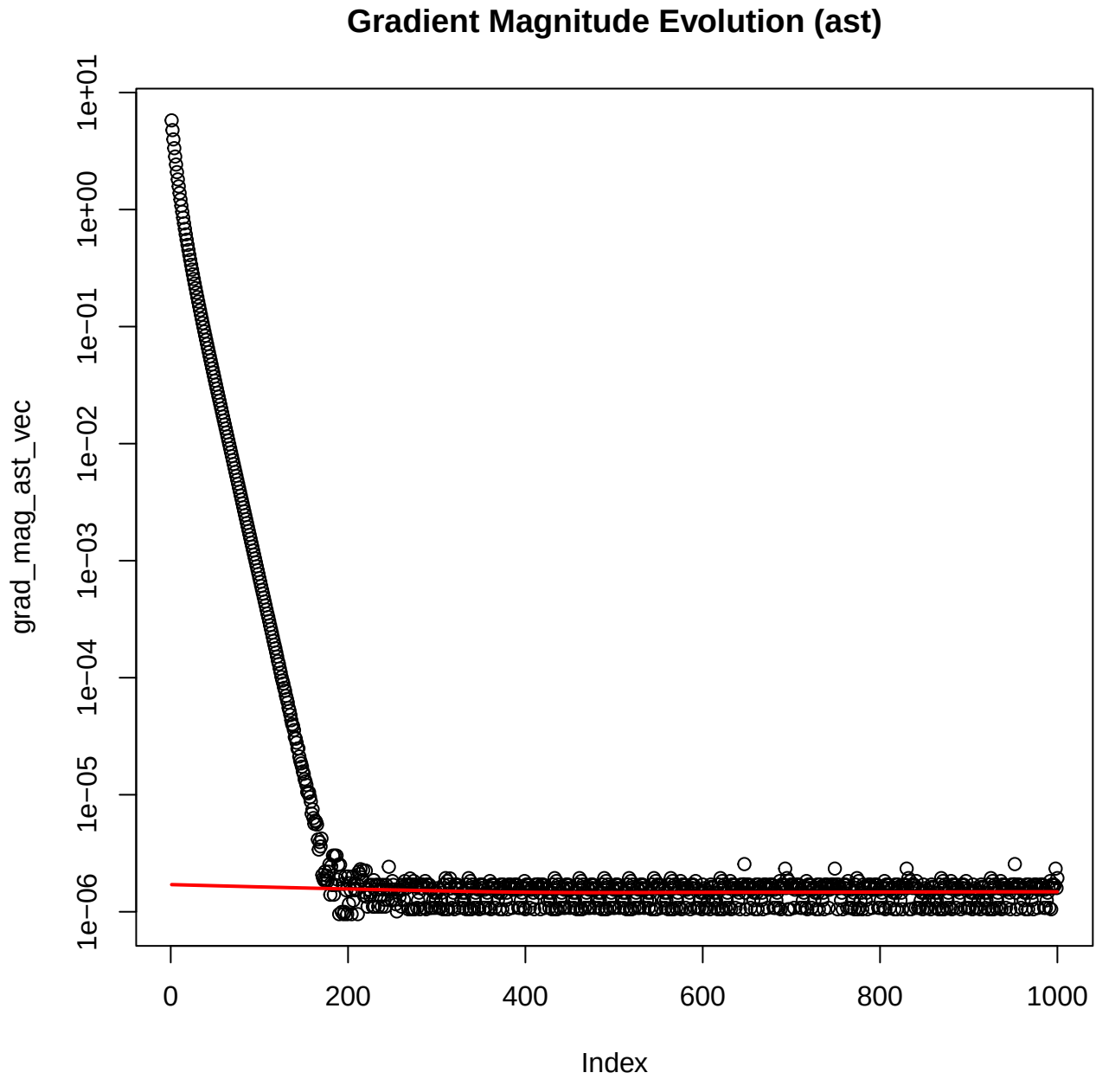

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

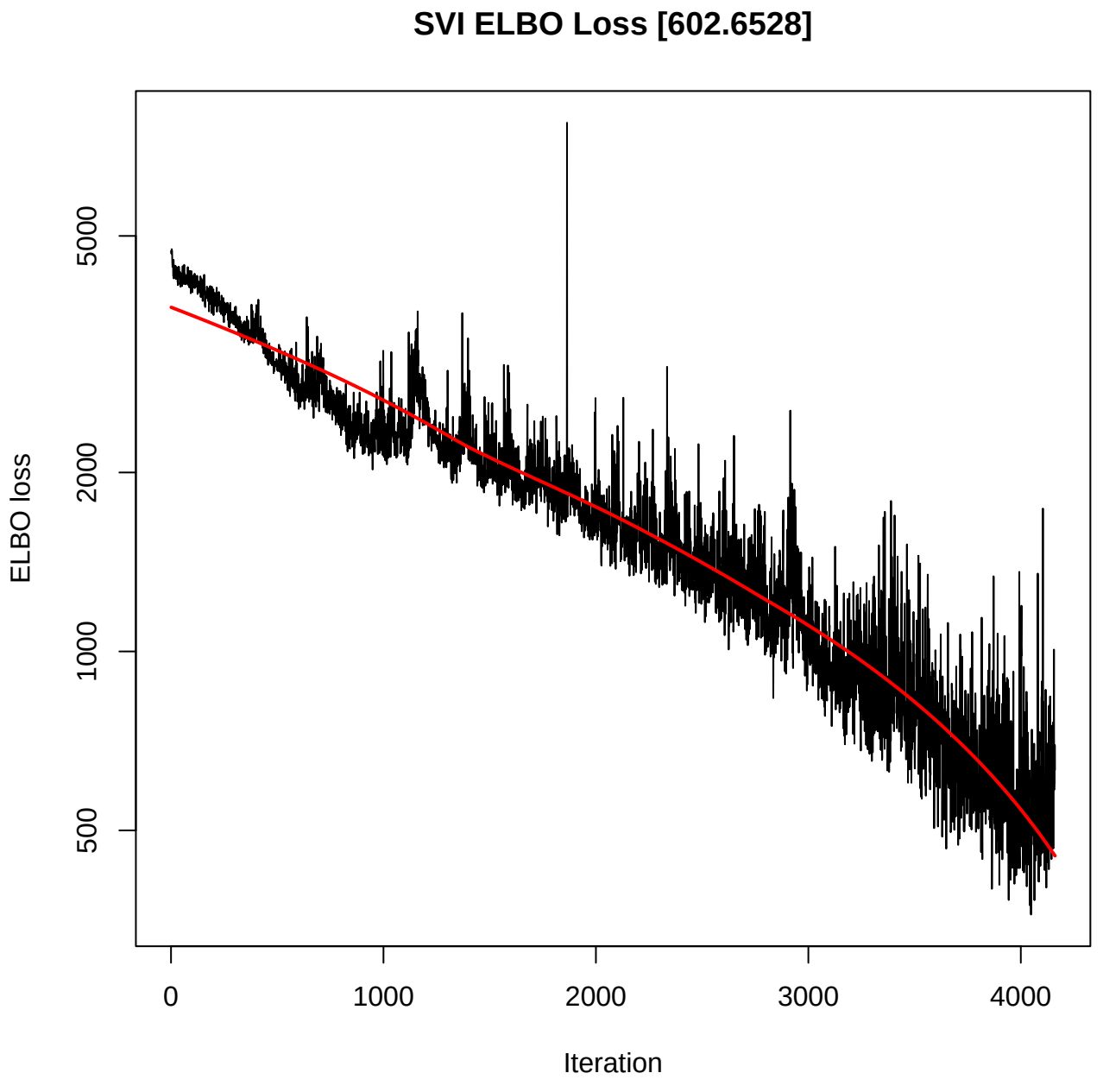

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

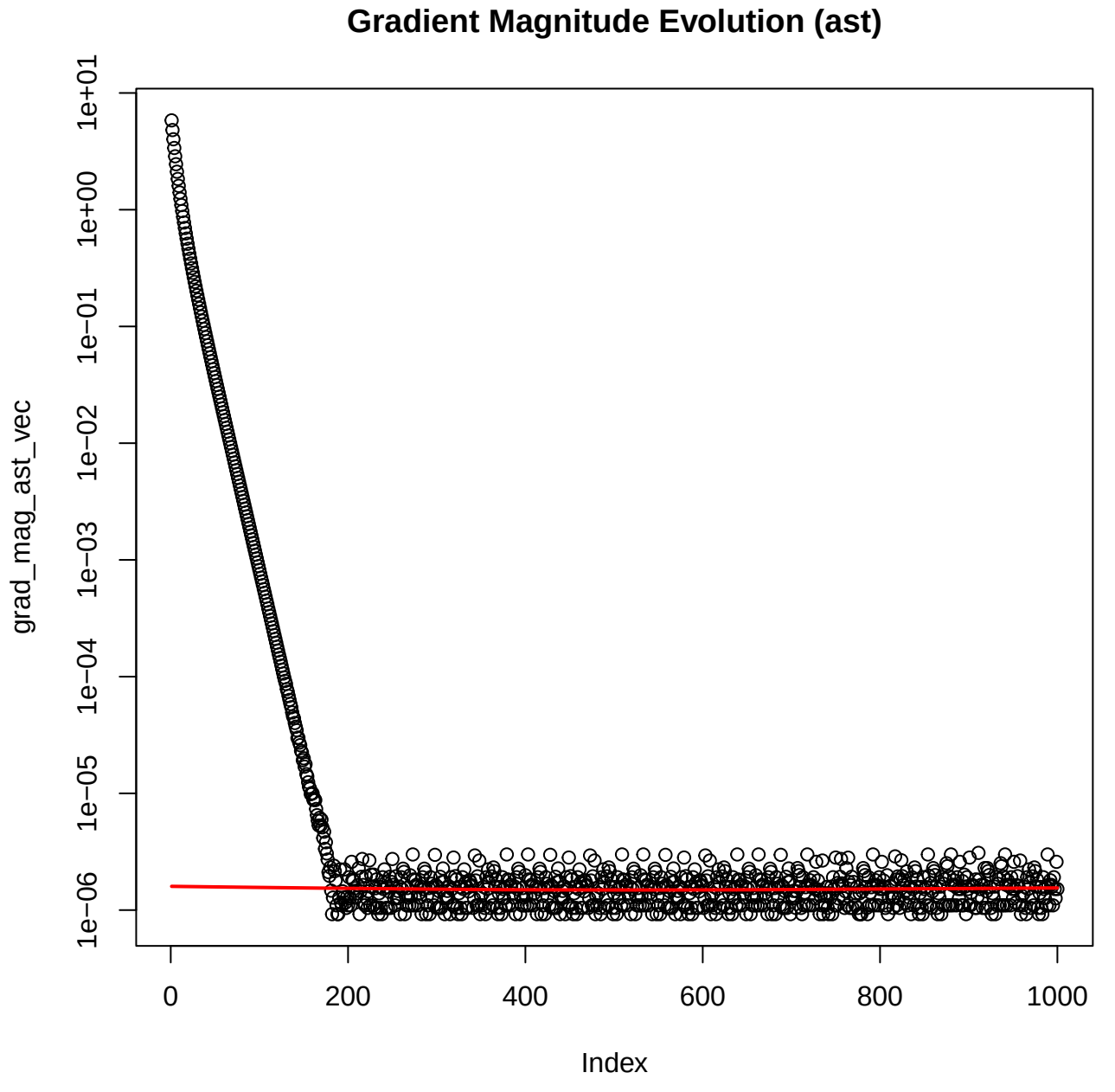

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

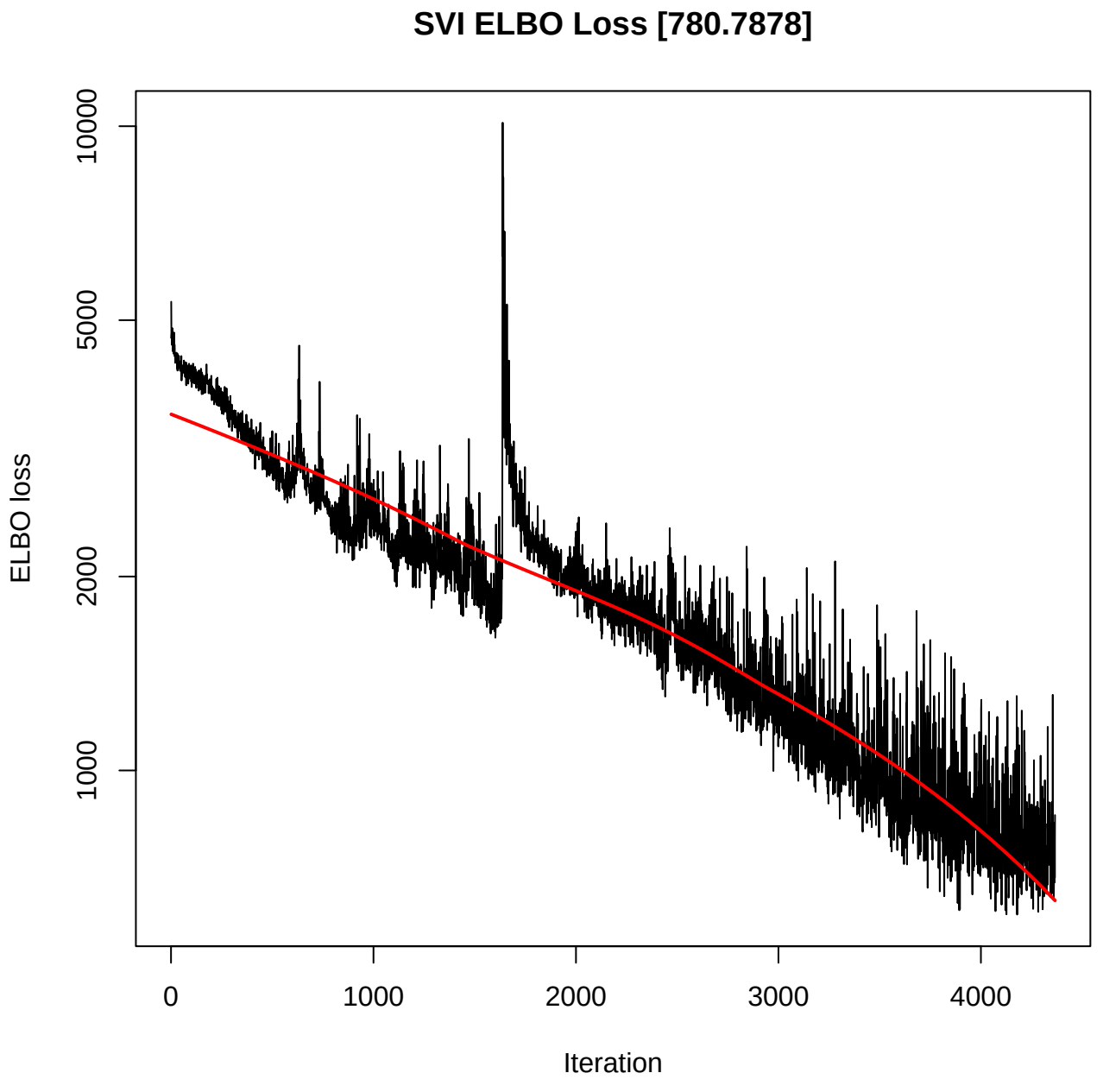

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

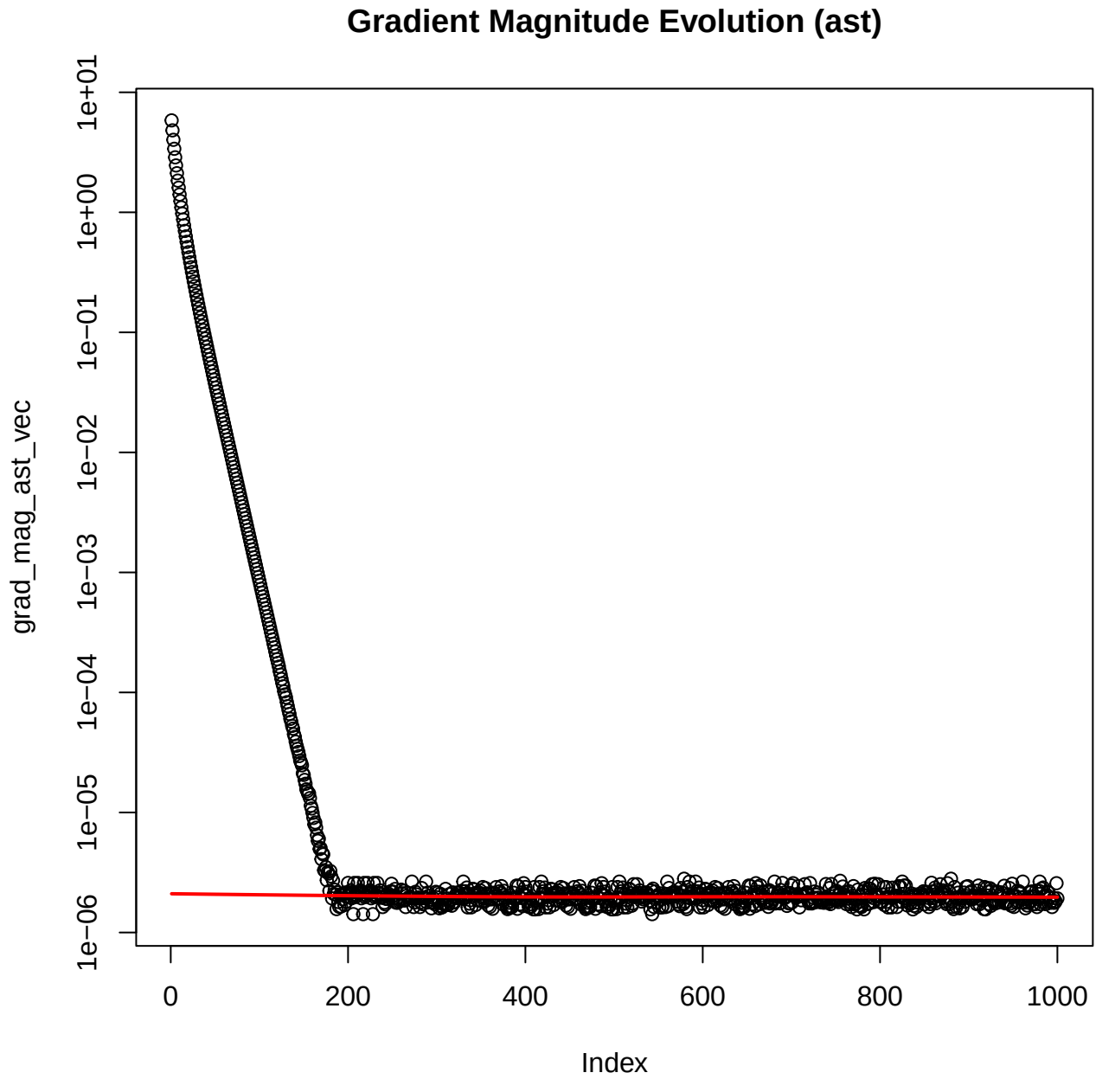

SVI ELBO Loss [651.5729]
ELBO loss
500
1000
2000
5000
0
1000
2000
3000
4000
Iteration

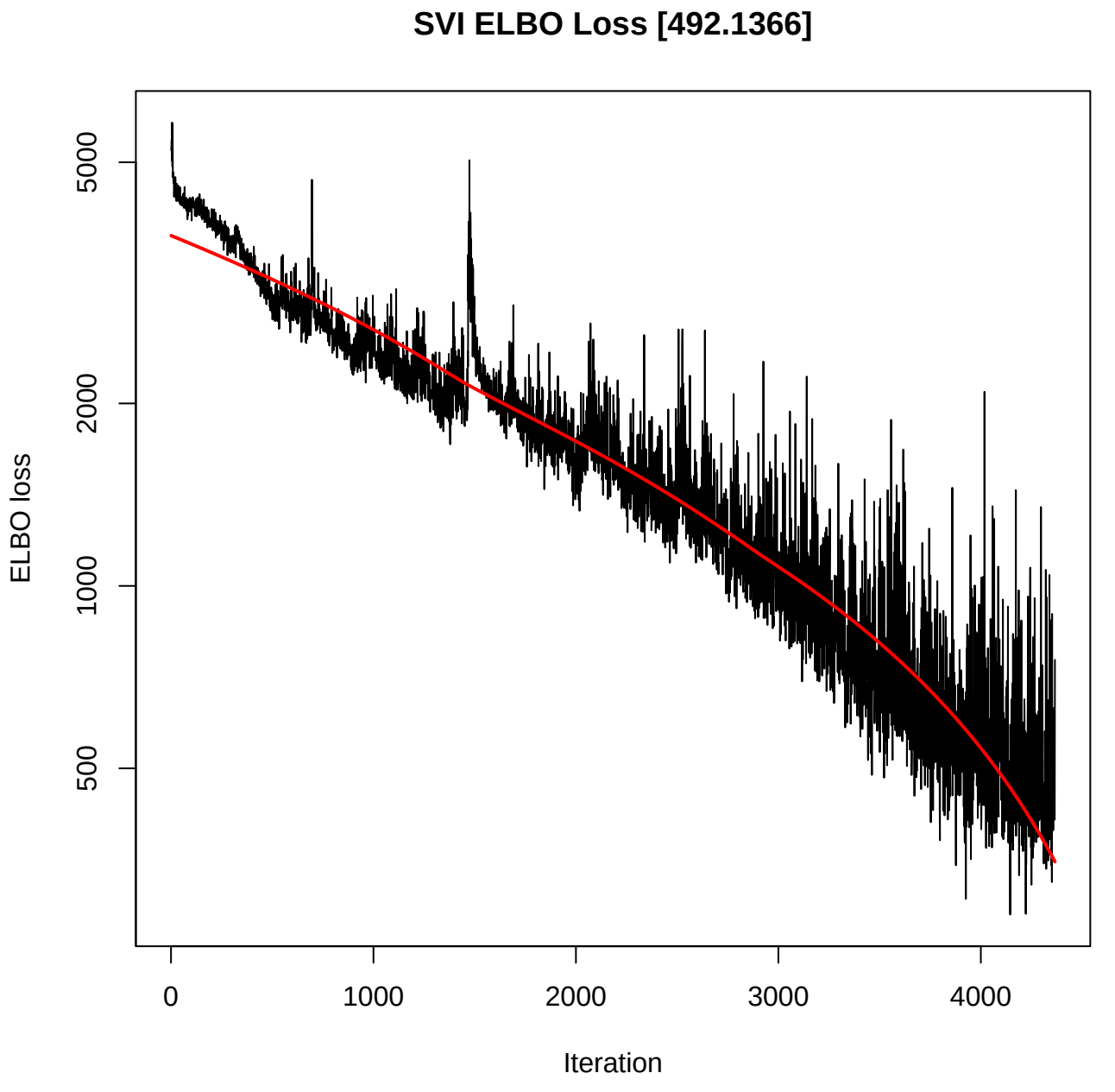

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

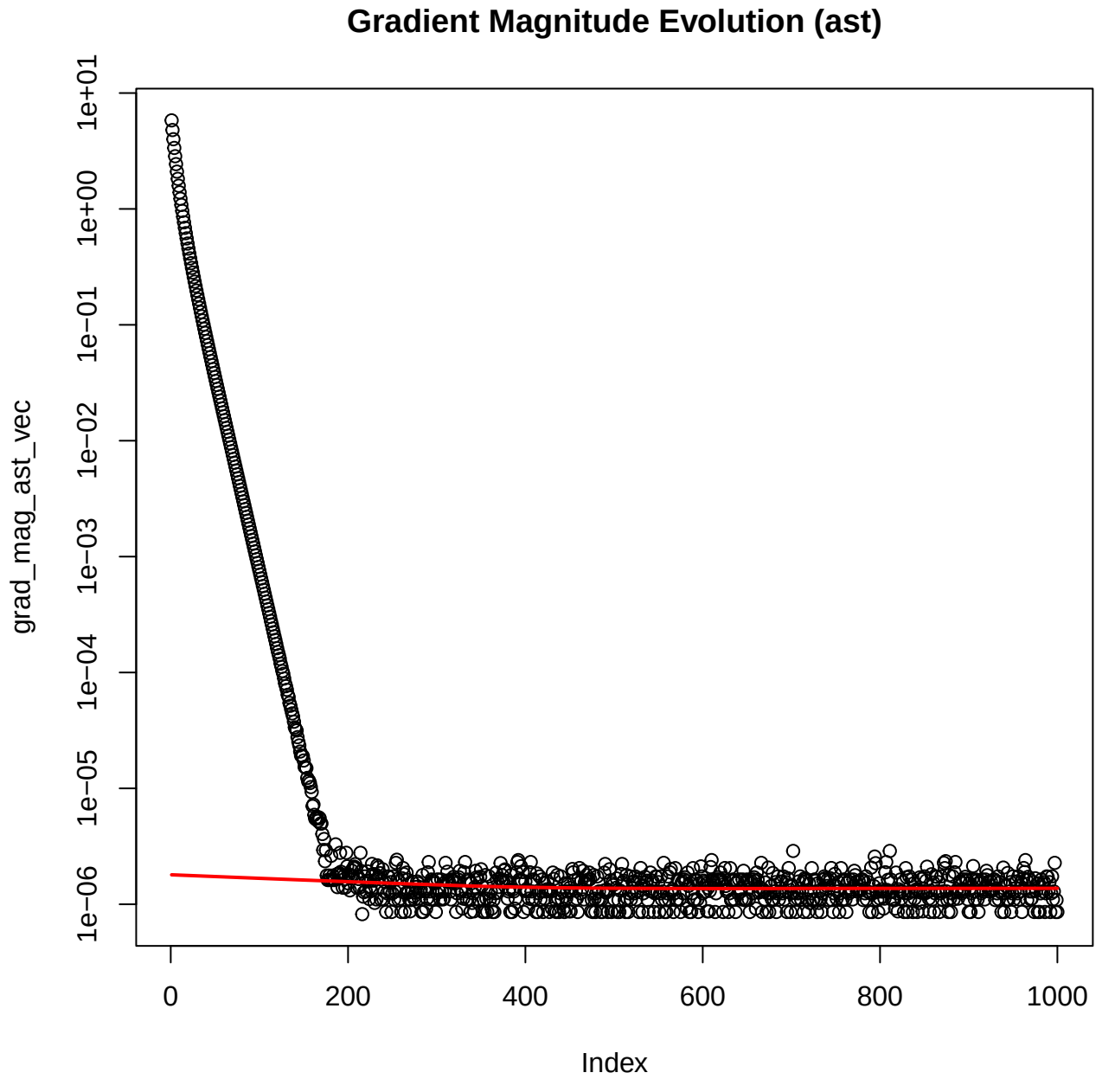

SVI ELBO Loss [602.6528]
ELBO loss
500
1000
2000
5000
0
1000
2000
3000
4000
Iteration

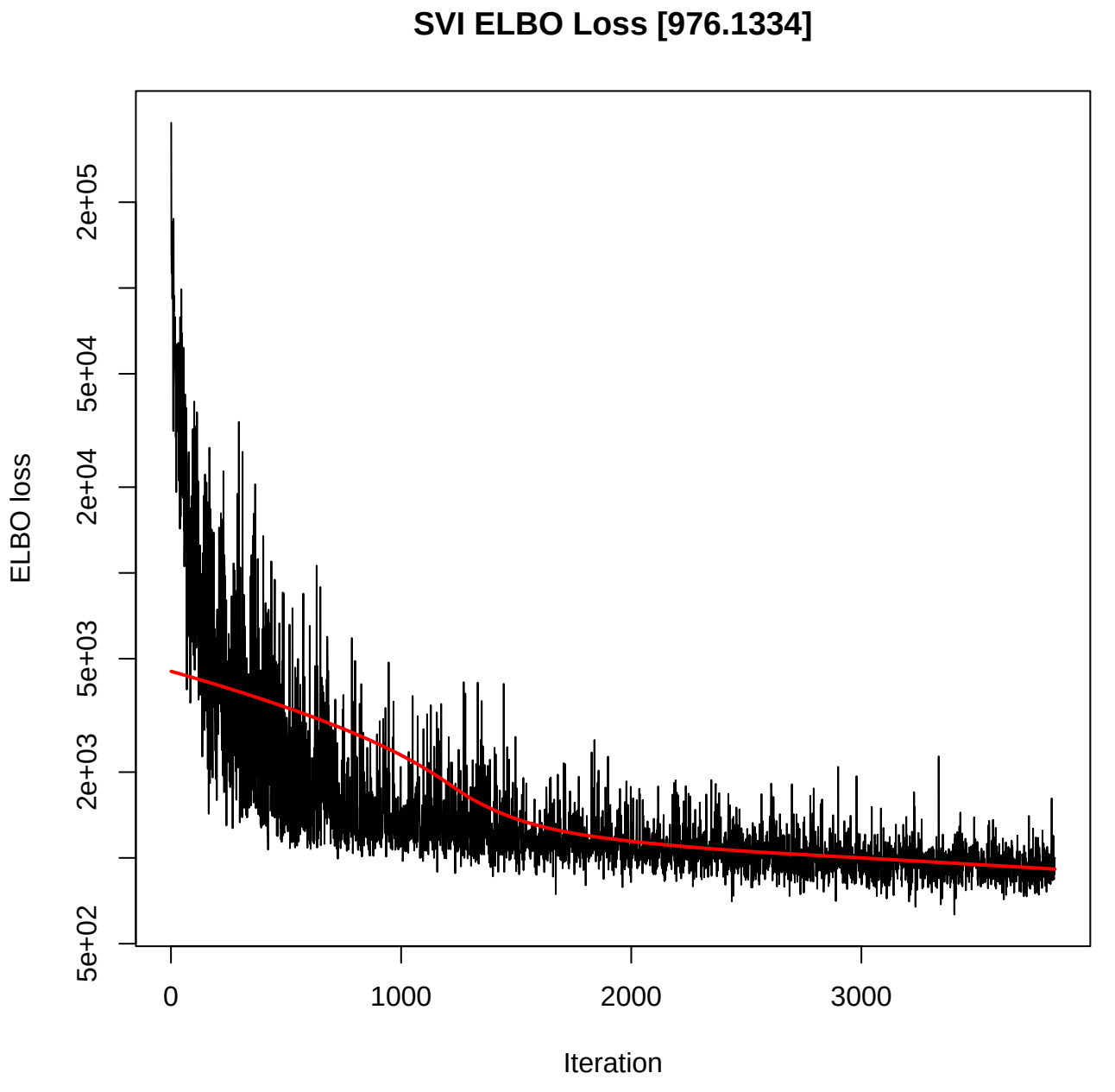

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

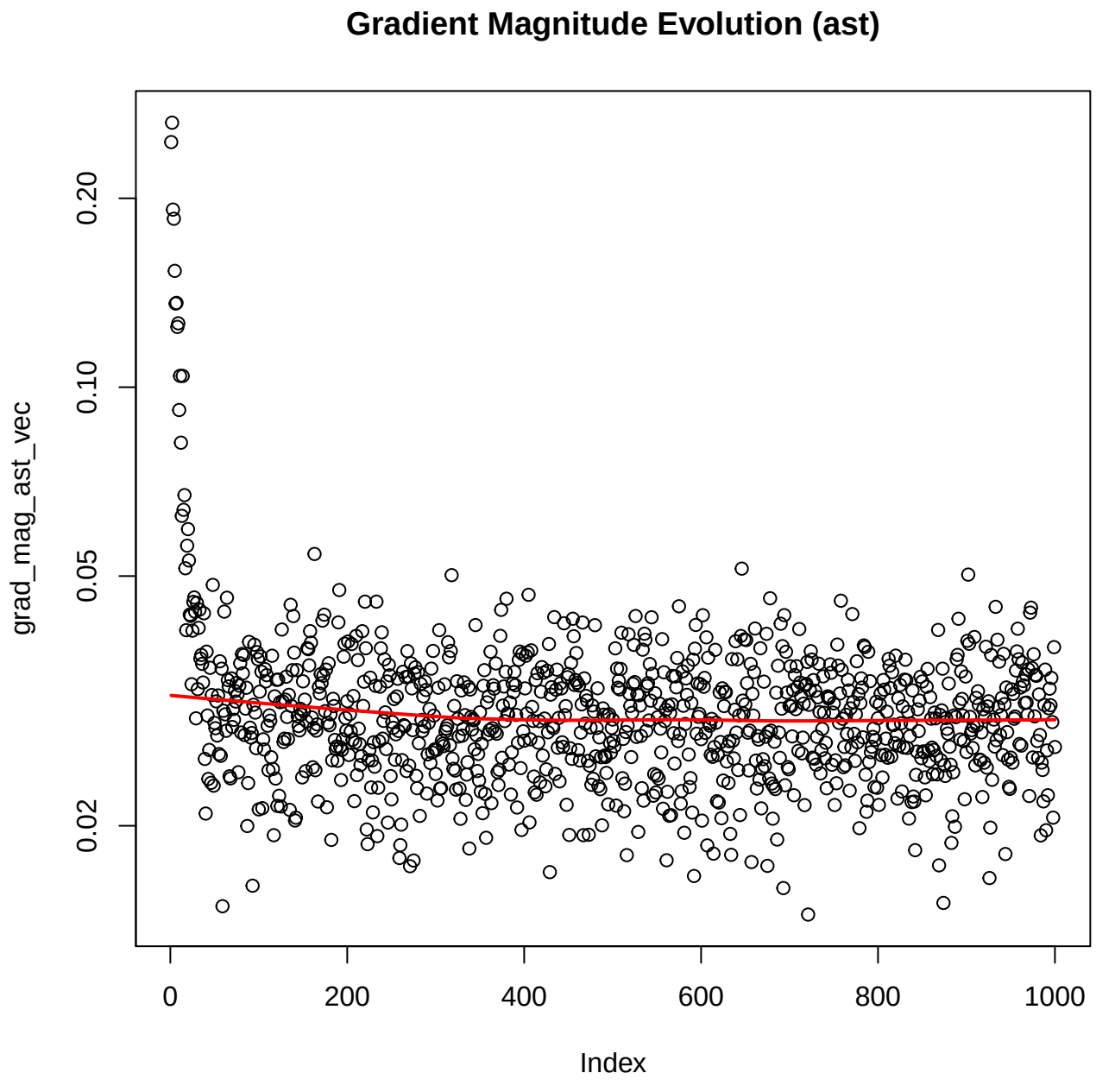

SVI ELBO Loss [780.7878]
ELBO loss
10000
5000
2000
1000
0
1000
2000
3000
4000
Iteration

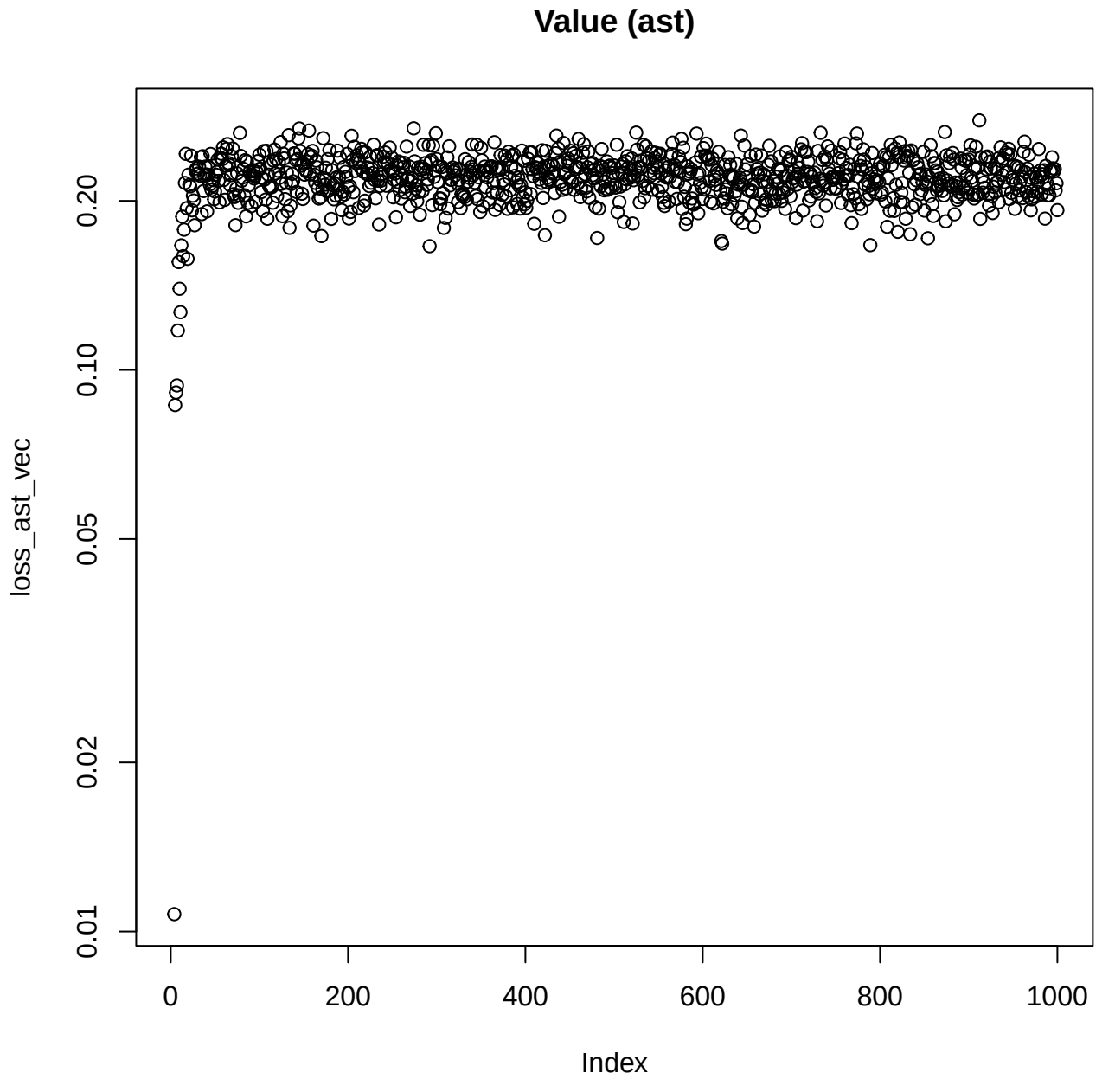

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

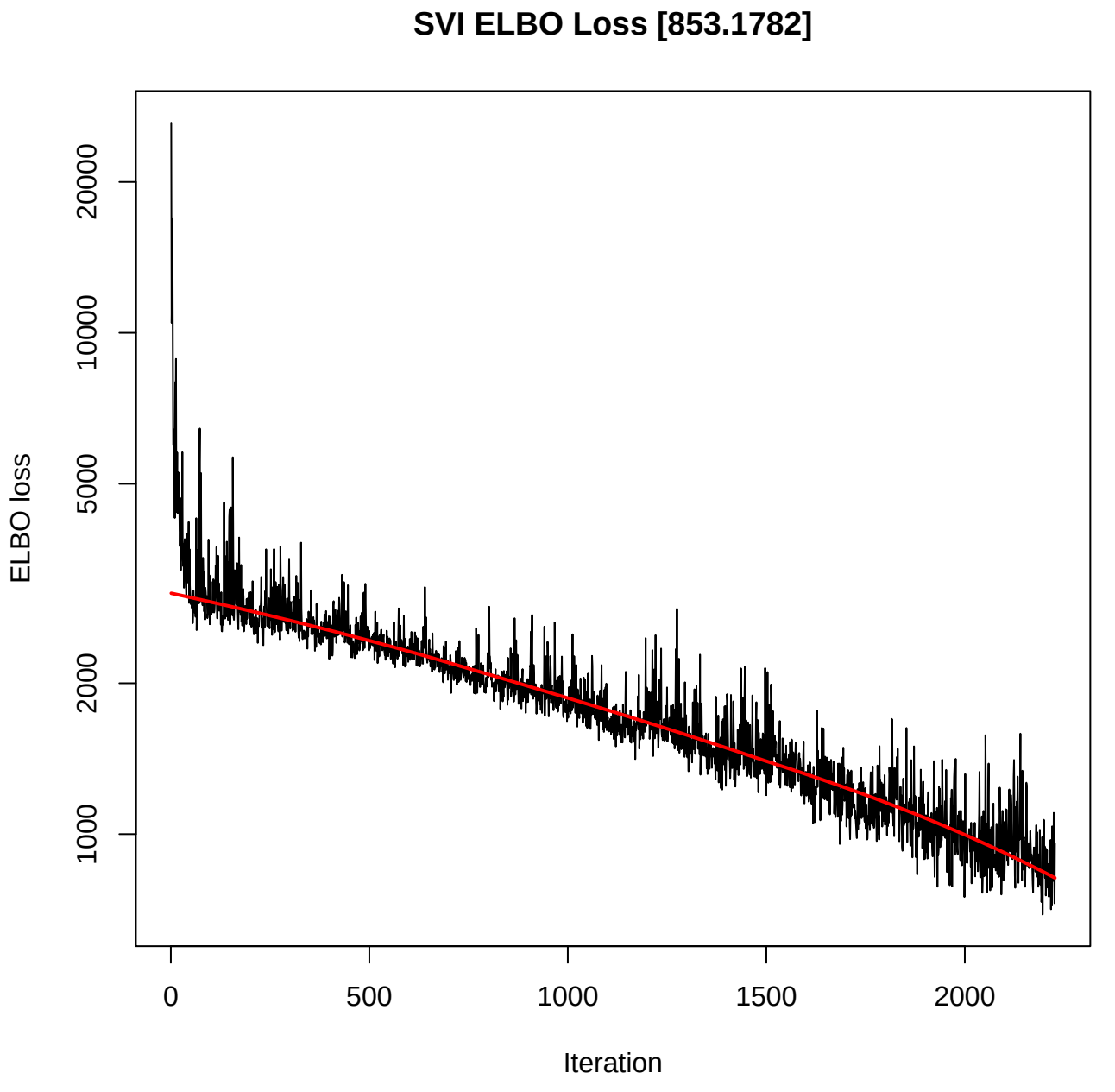

SVI ELBO Loss [492.1366]
ELBO loss
5000
2000
1000
500
0
1000
2000
3000
4000
Iteration

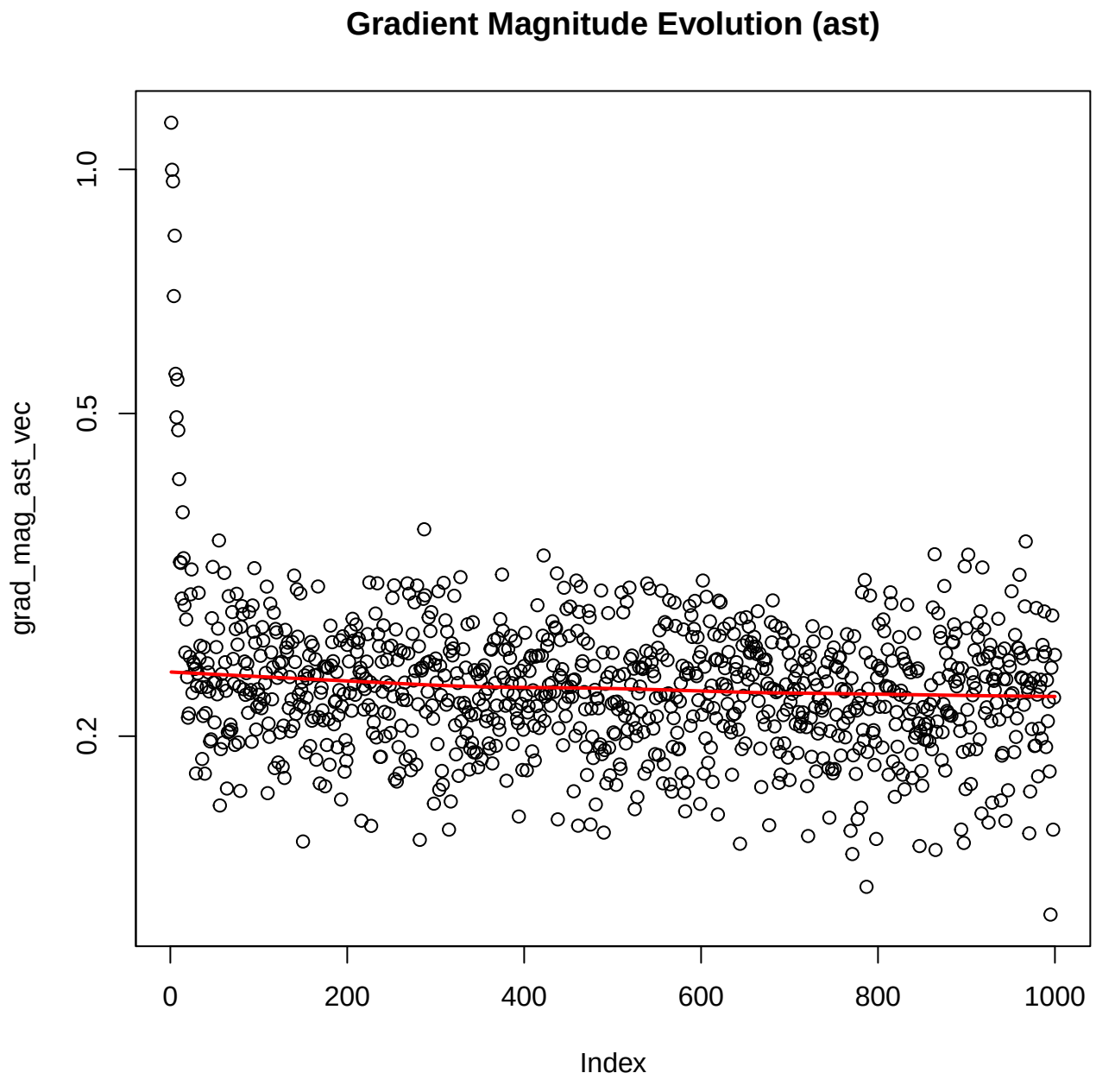

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

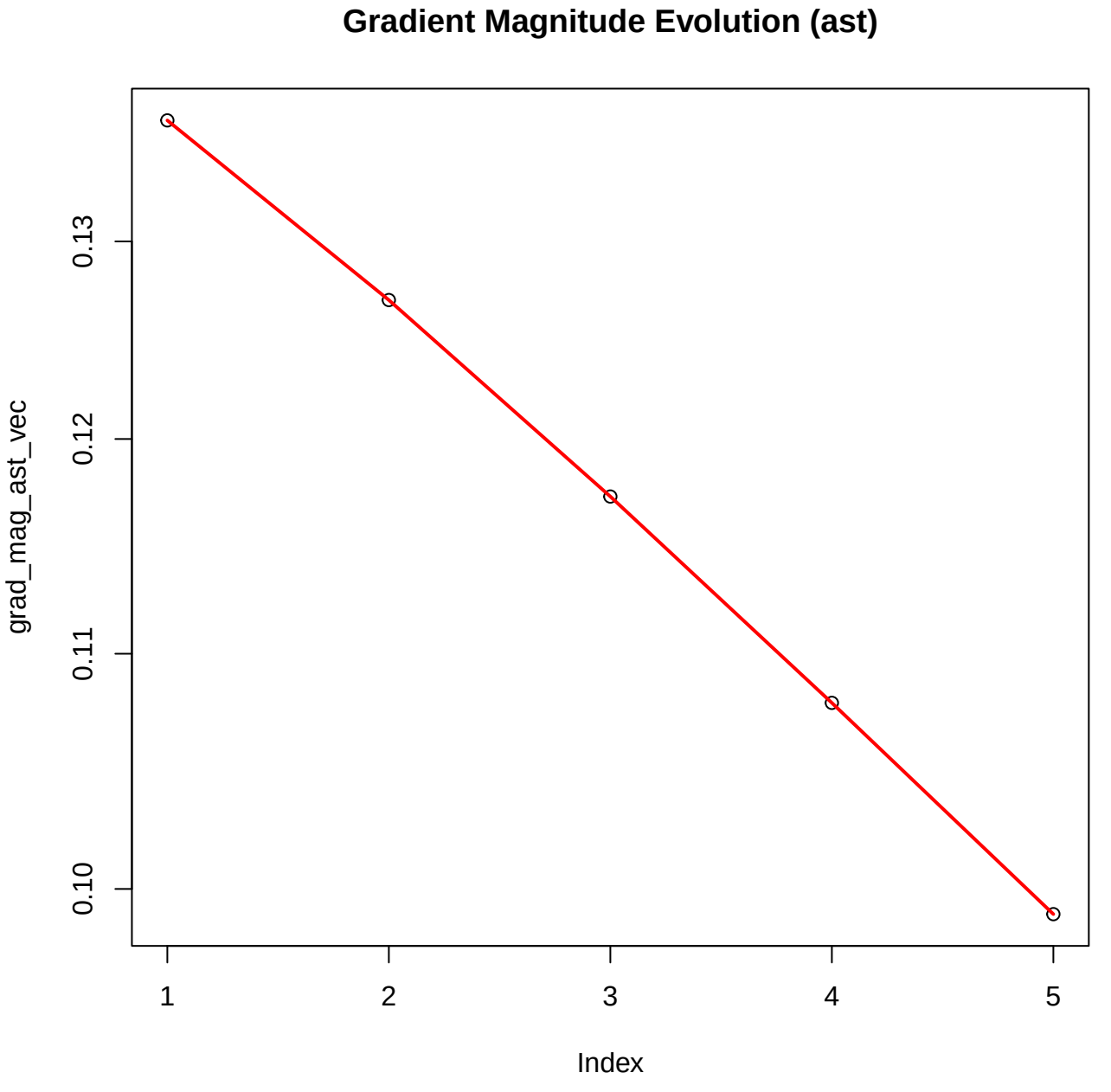

SVI ELBO Loss [976.1334]
ELBO loss
2e+05
5e+04
2e+04
5e+03
2e+03
5e+02
0
1000
2000
3000
Iteration

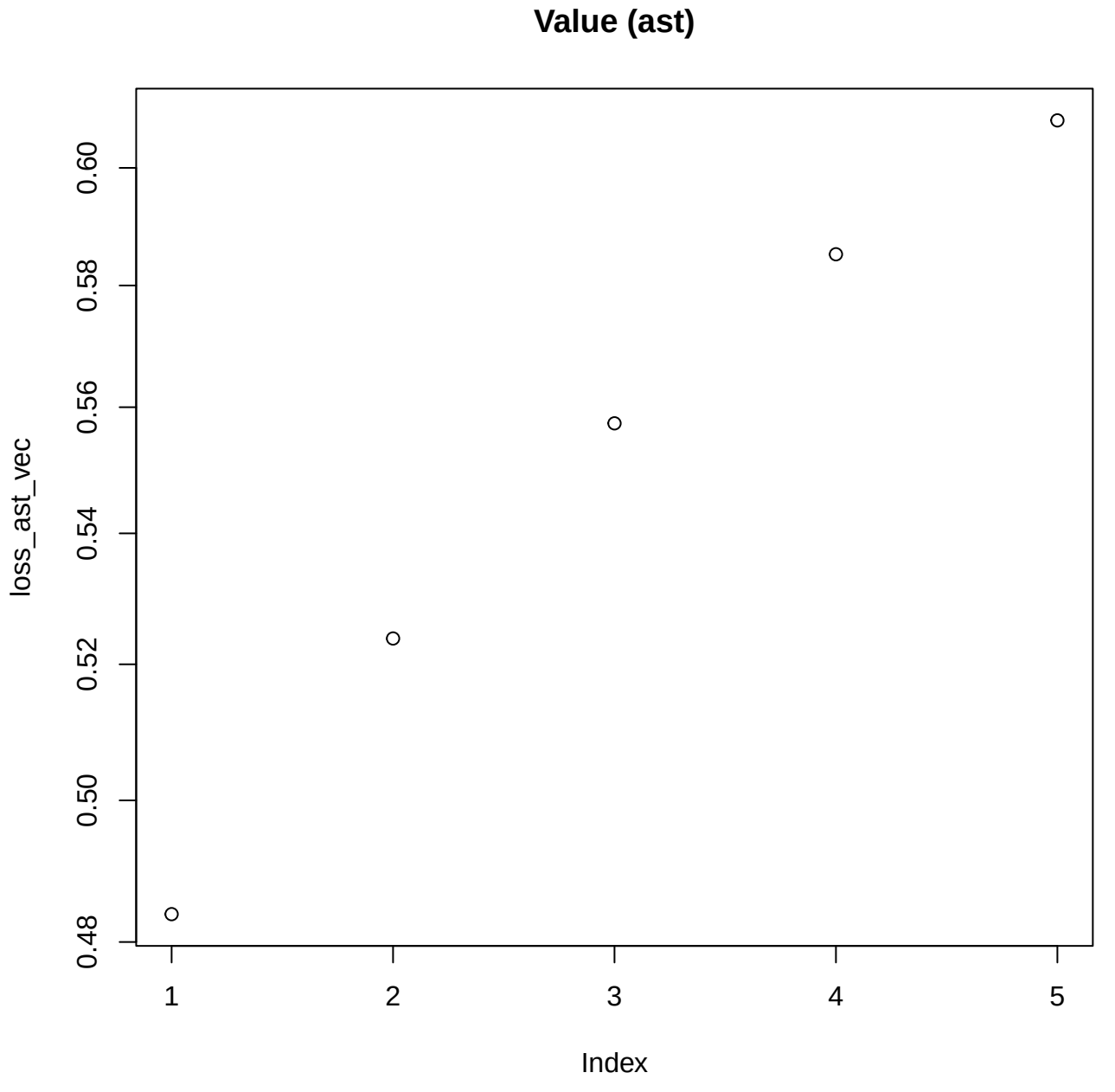

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index

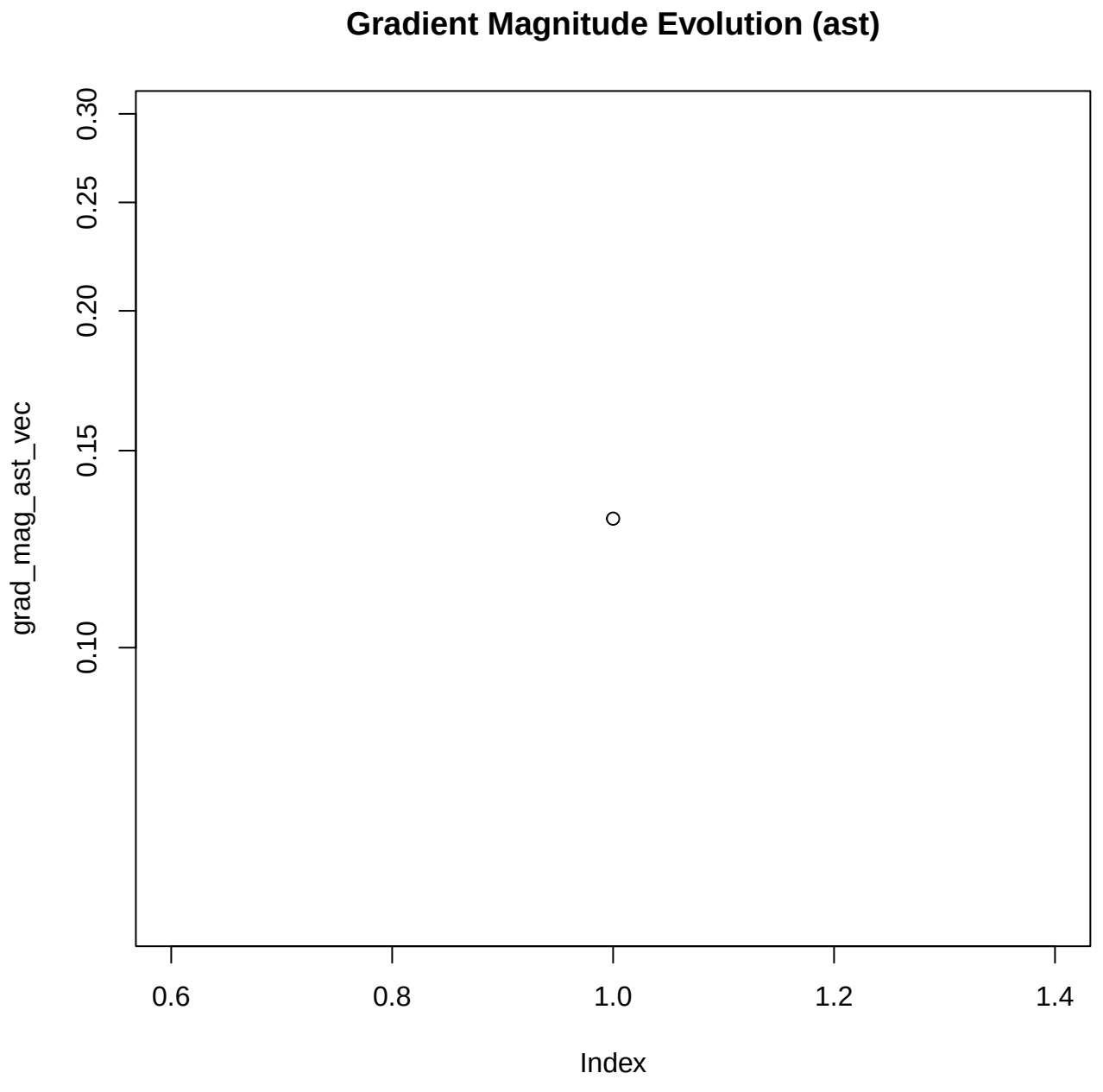

Value (ast)
loss_ast_vec
0.01
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index

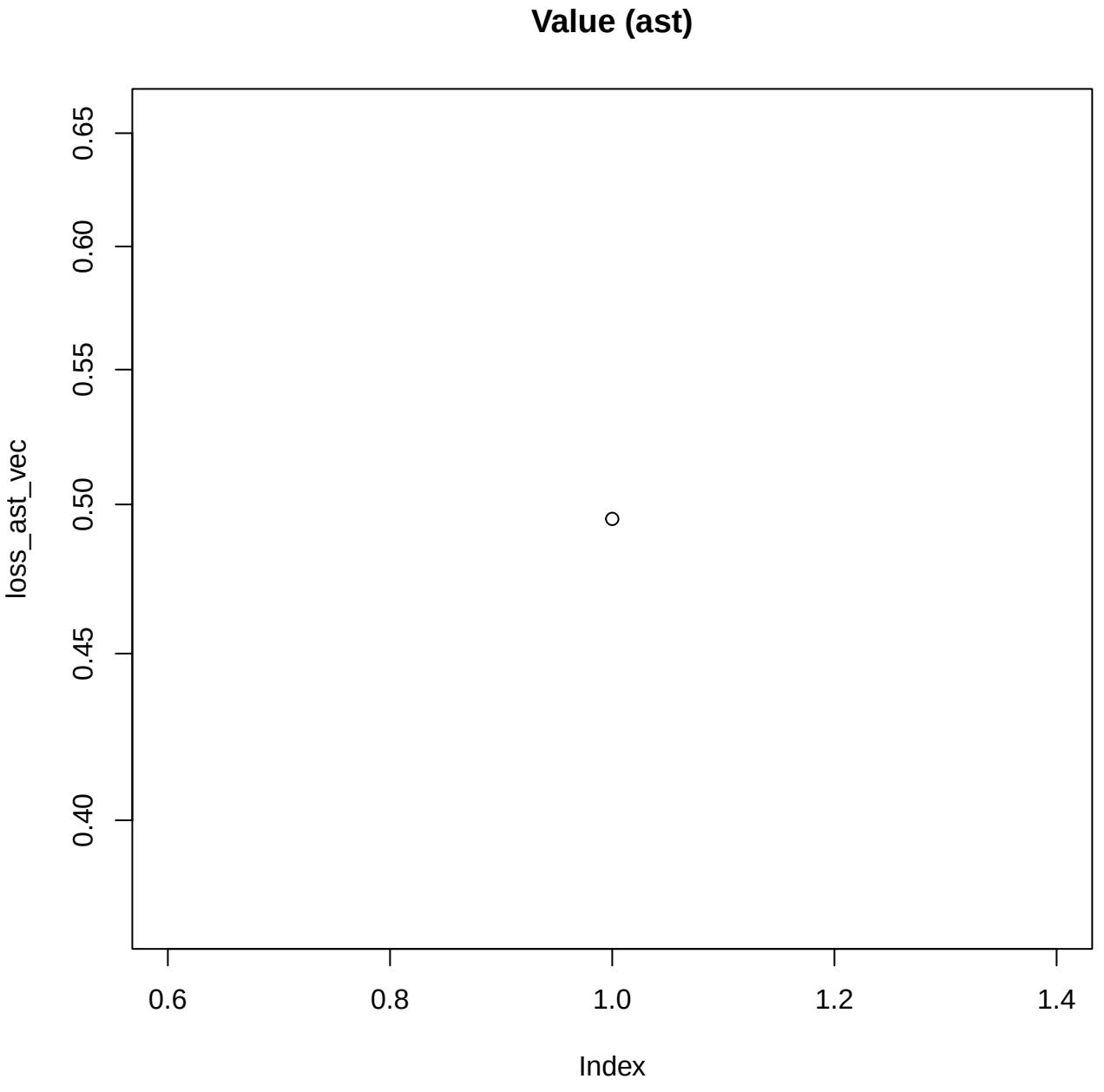

SVI ELBO Loss [853.1782]
ELBO loss
20000
10000
5000
2000
1000
0
500
1000
1500
2000
Iteration

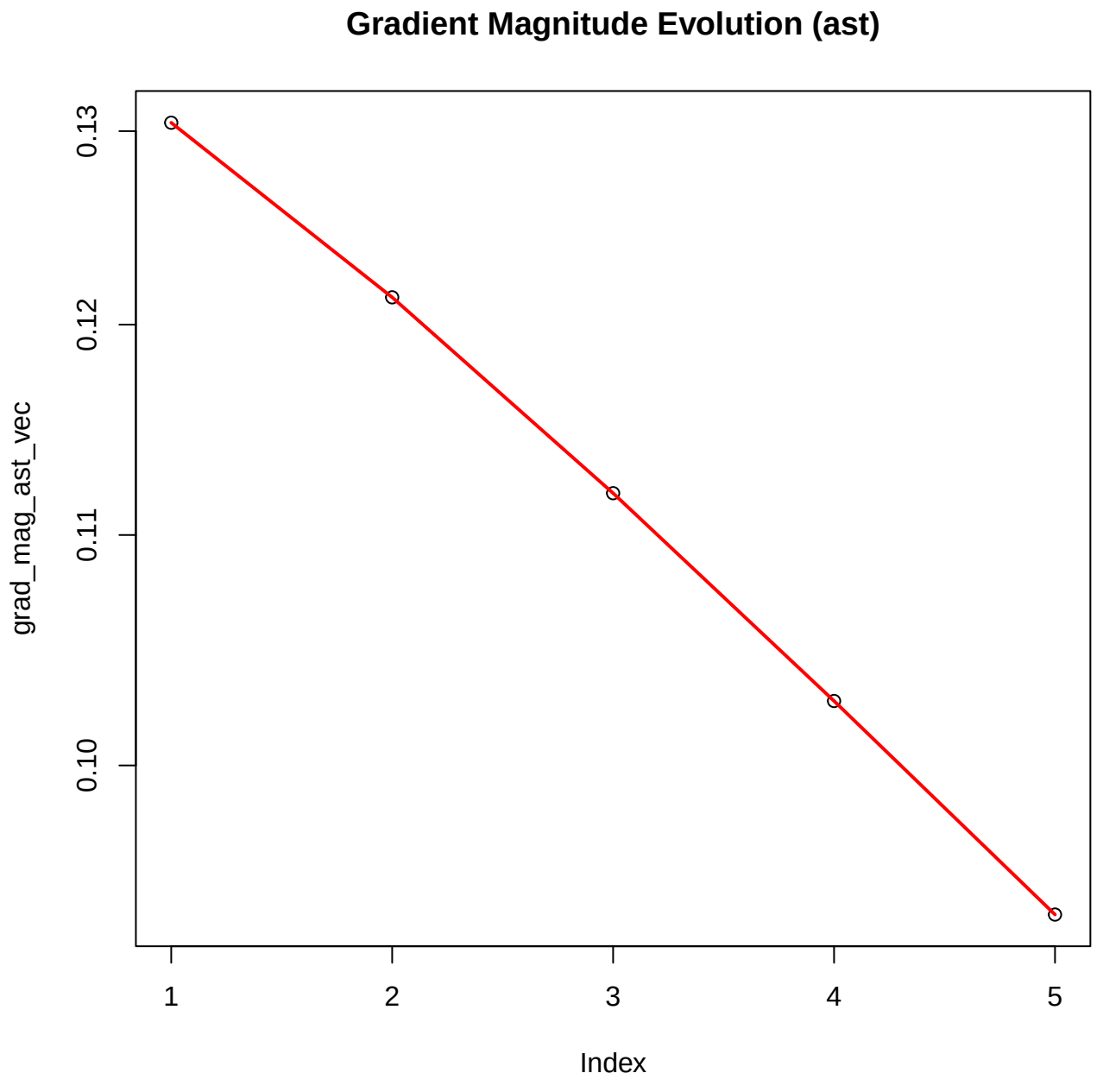

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1.0
0.5
0.2
0
200
400
600
800
1000
Index

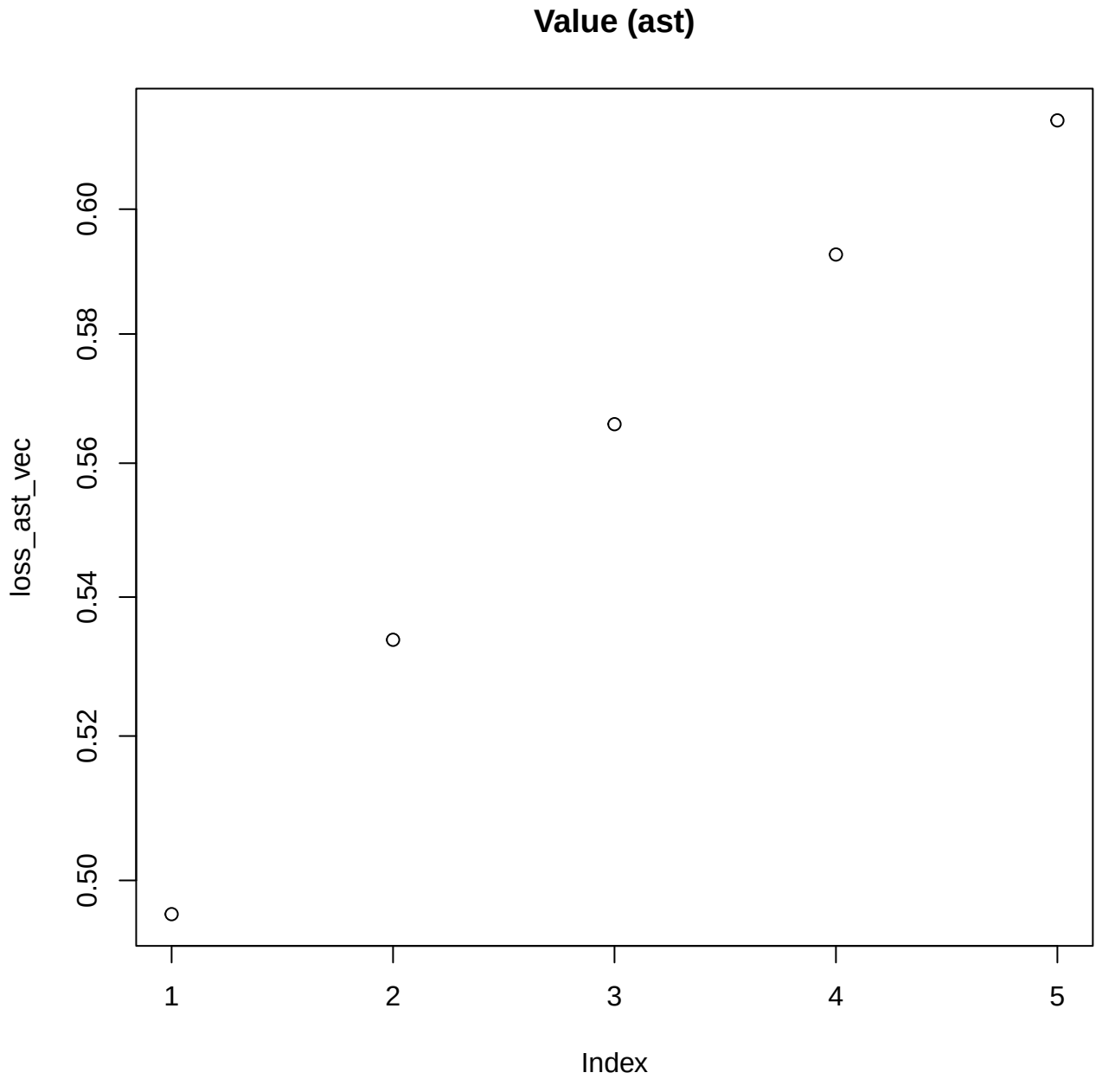

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

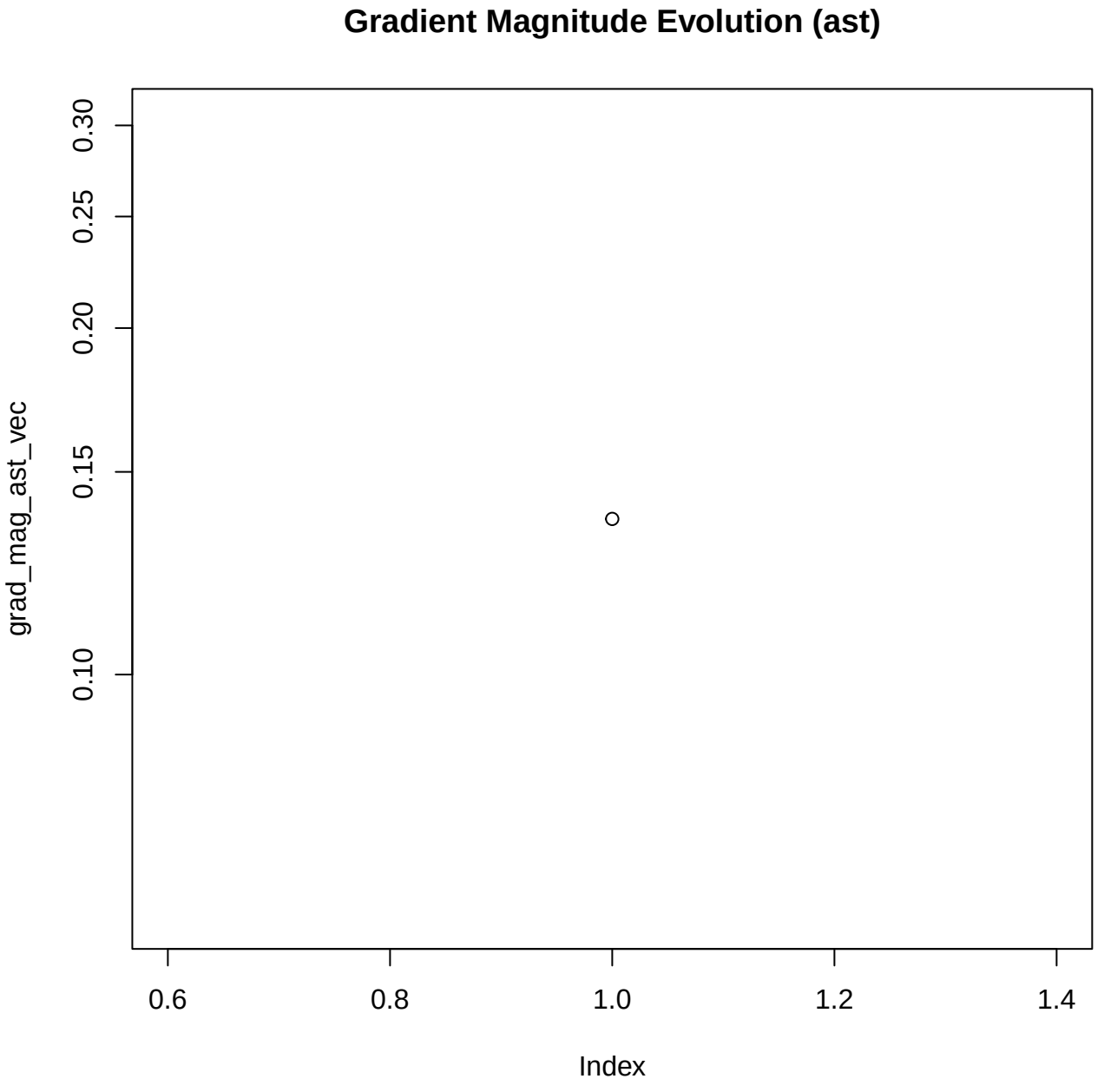

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

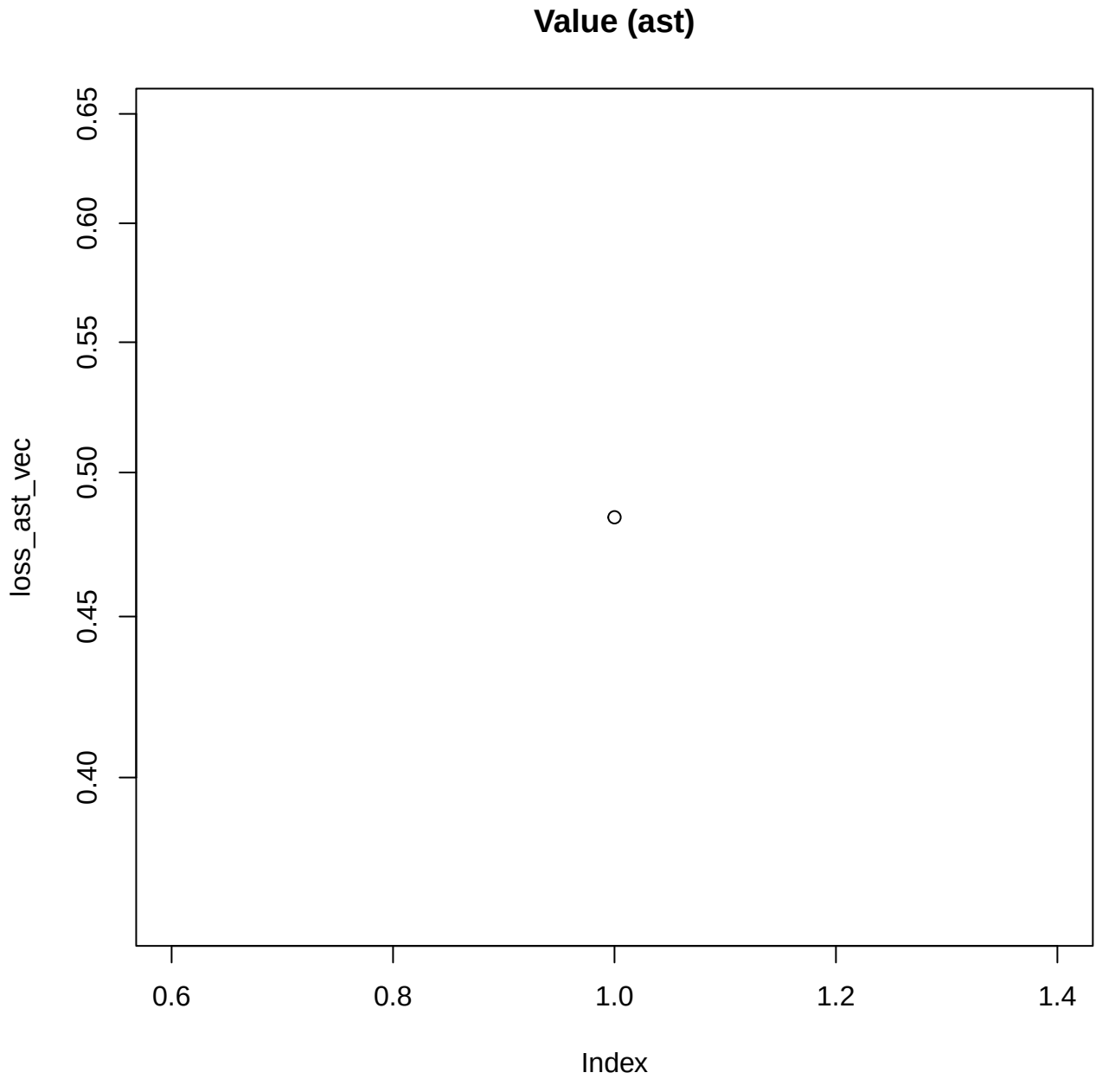

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

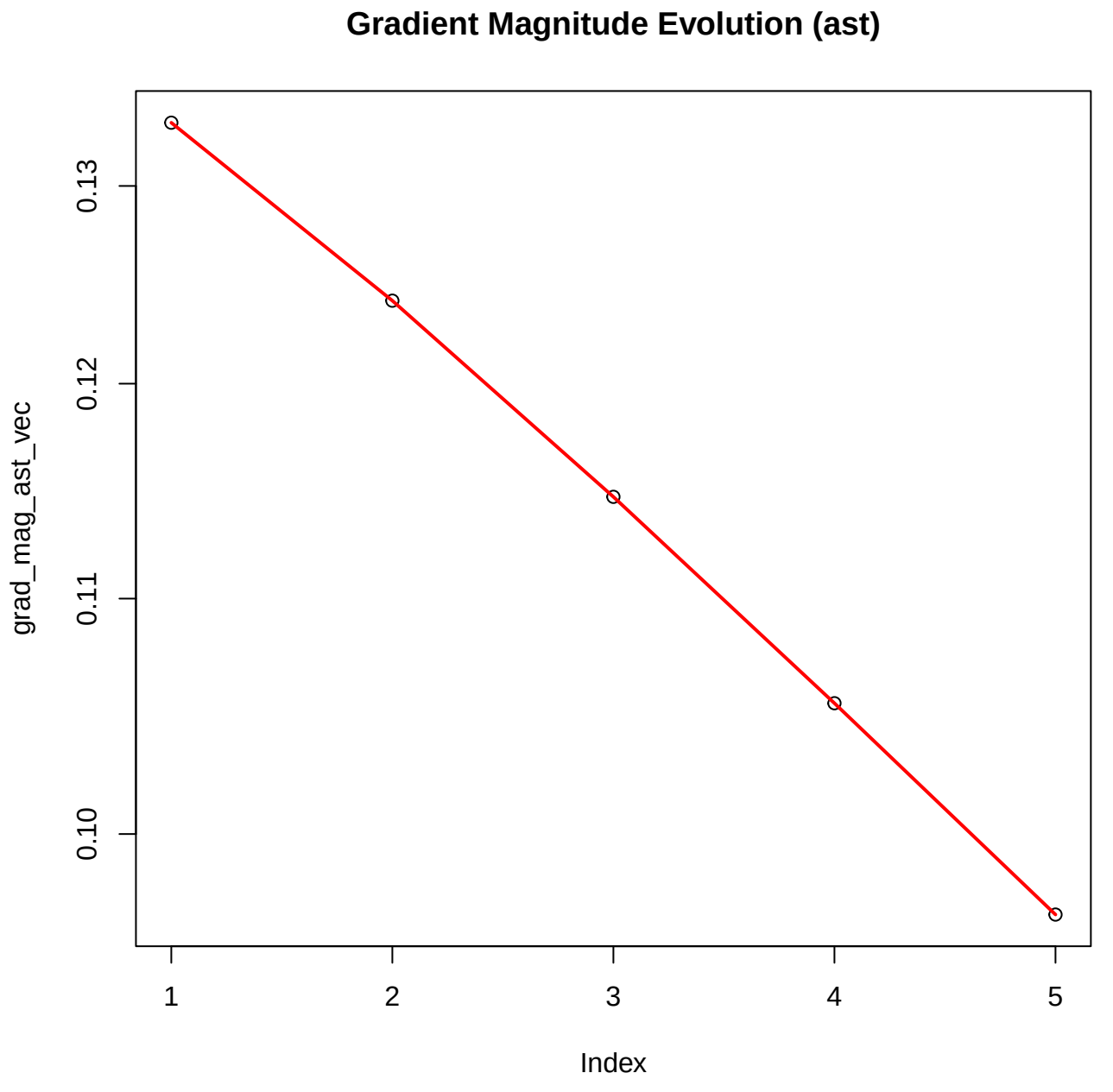

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

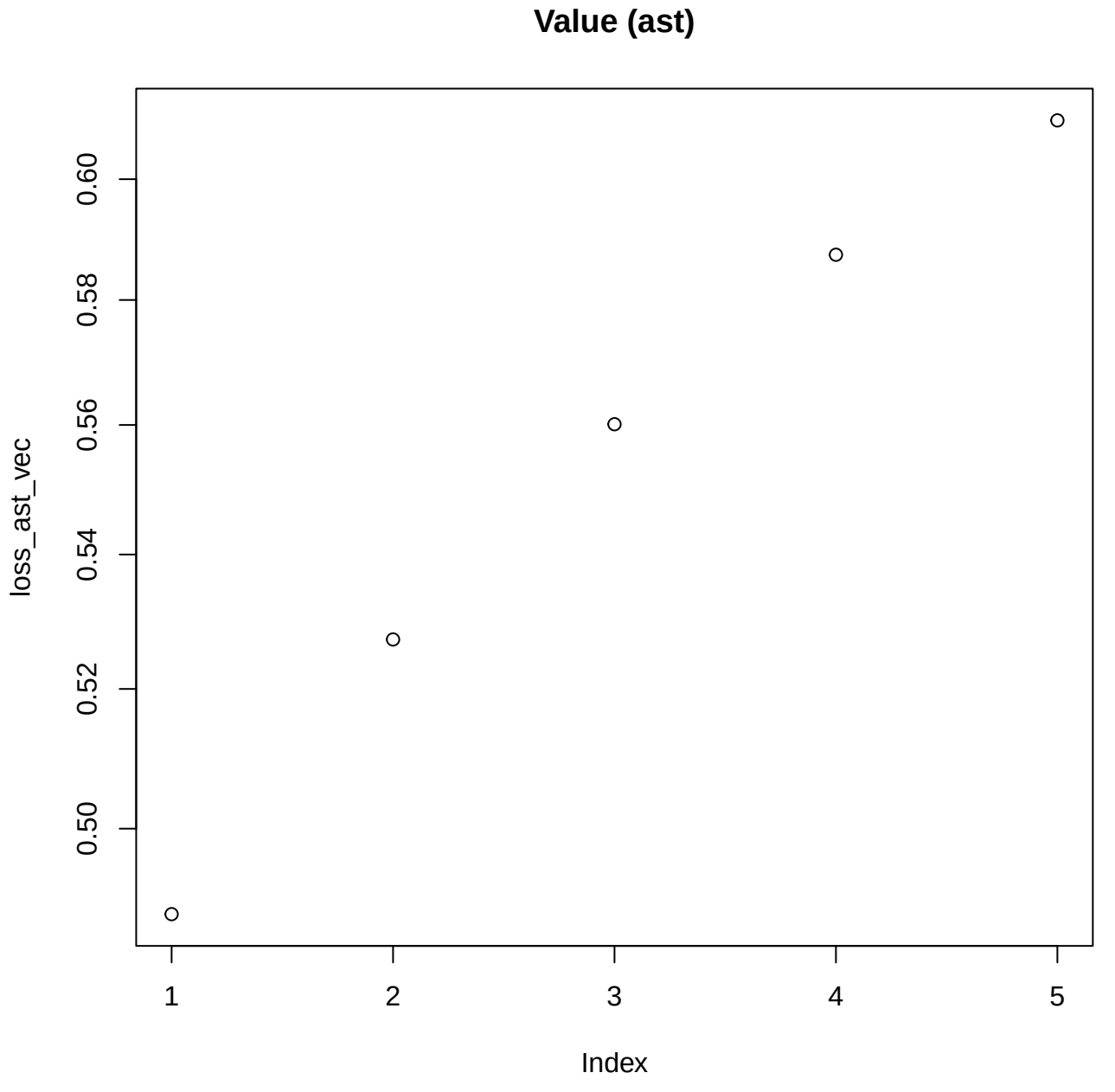

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

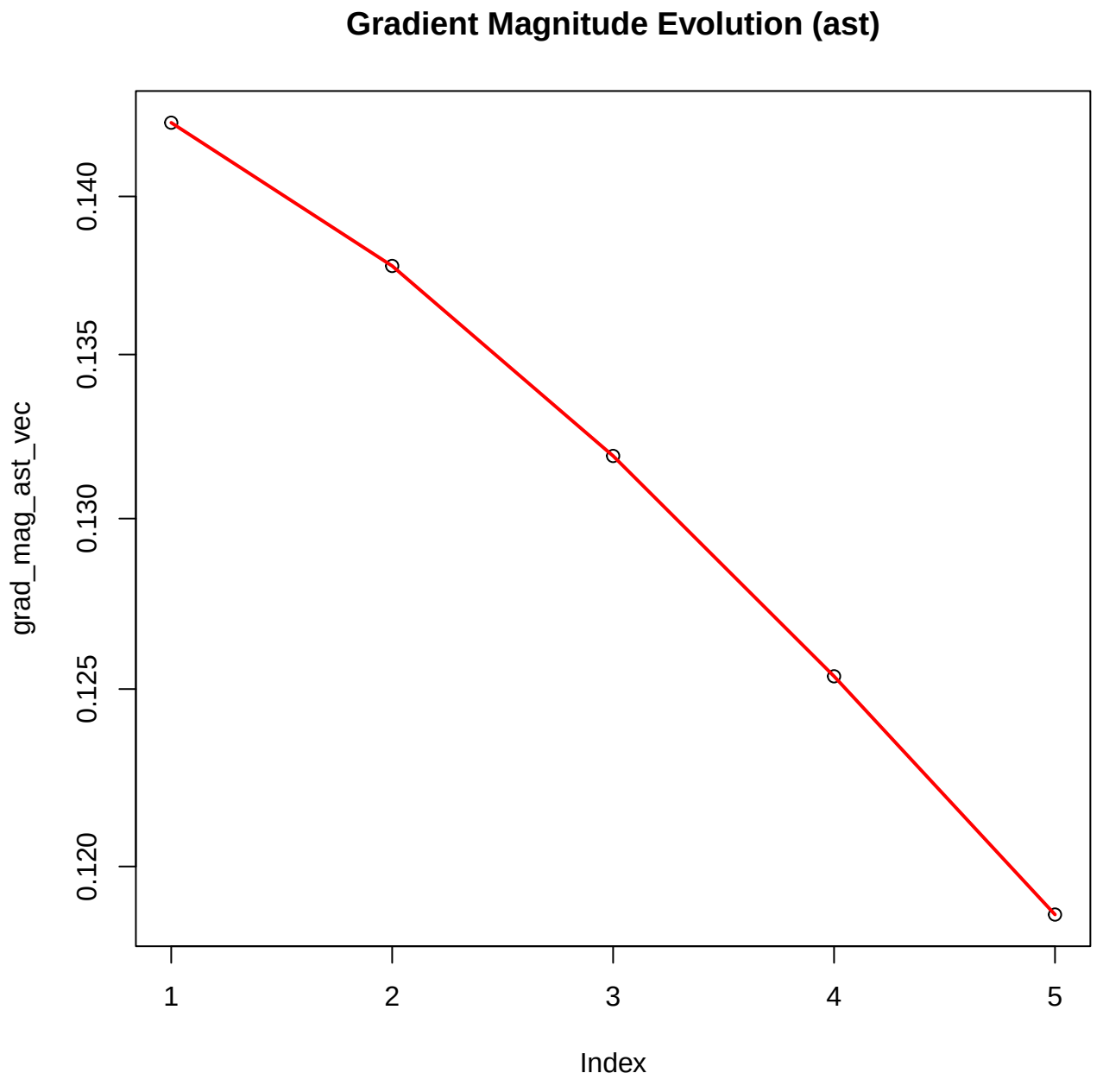

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

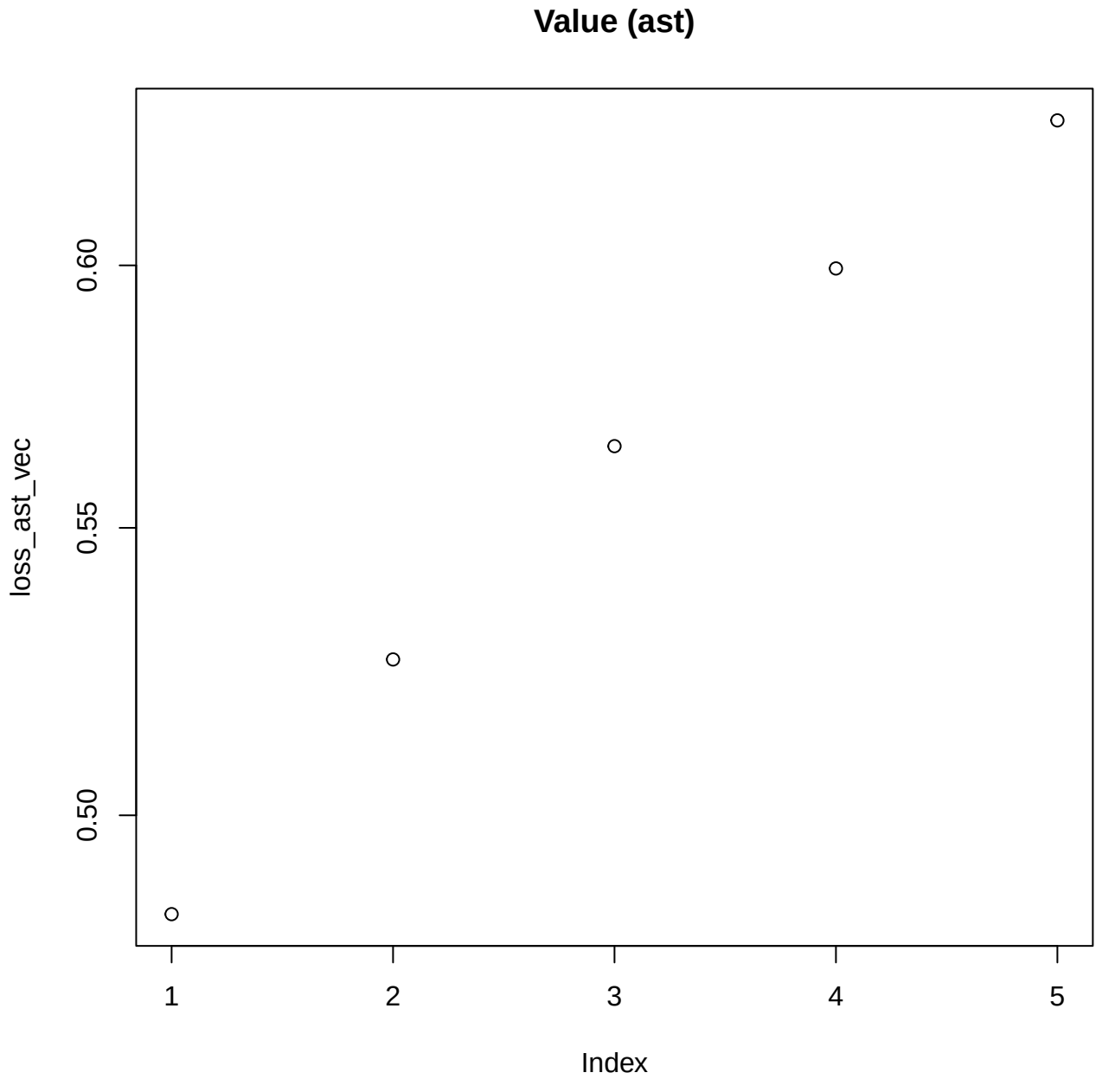

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

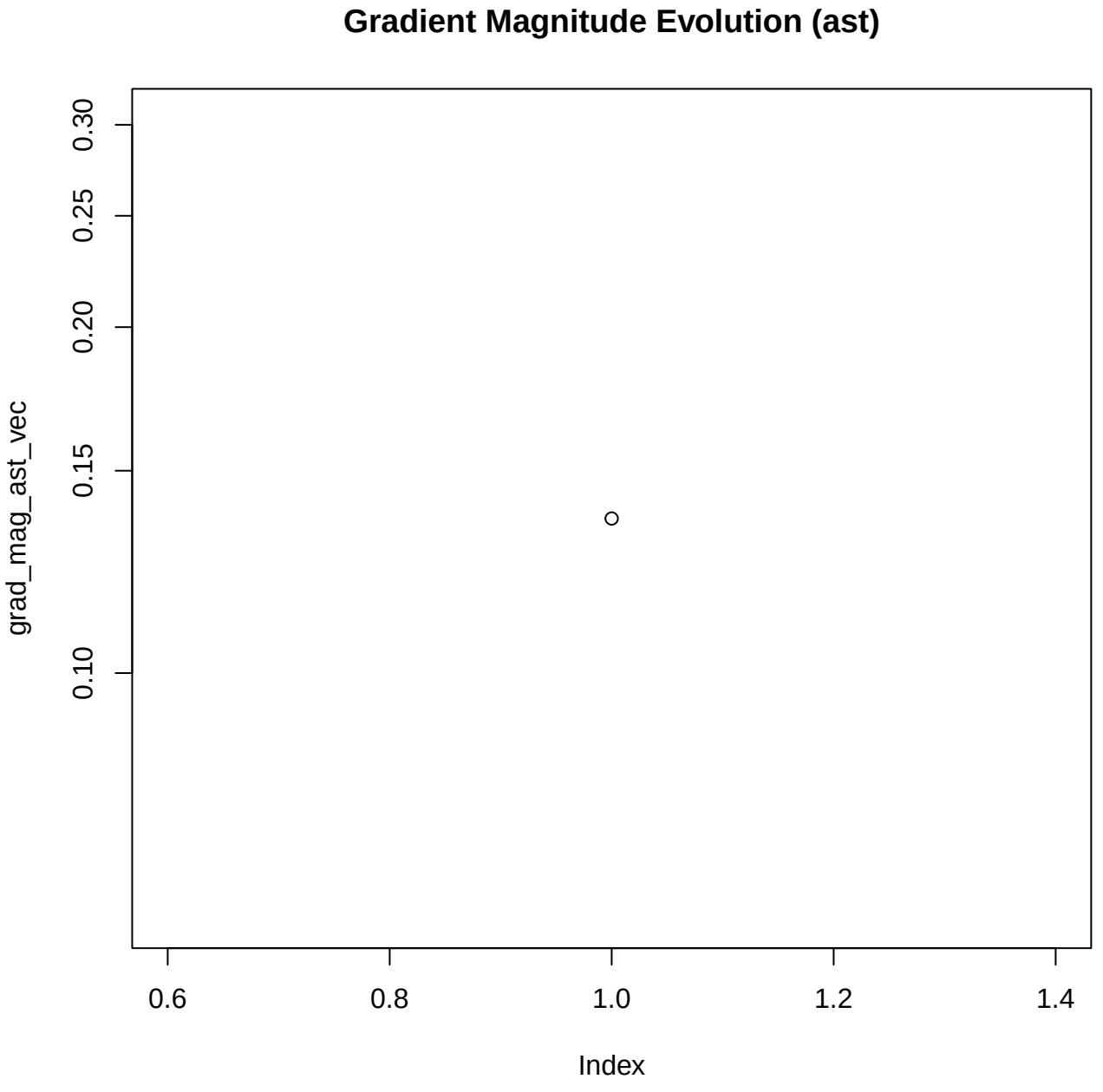

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

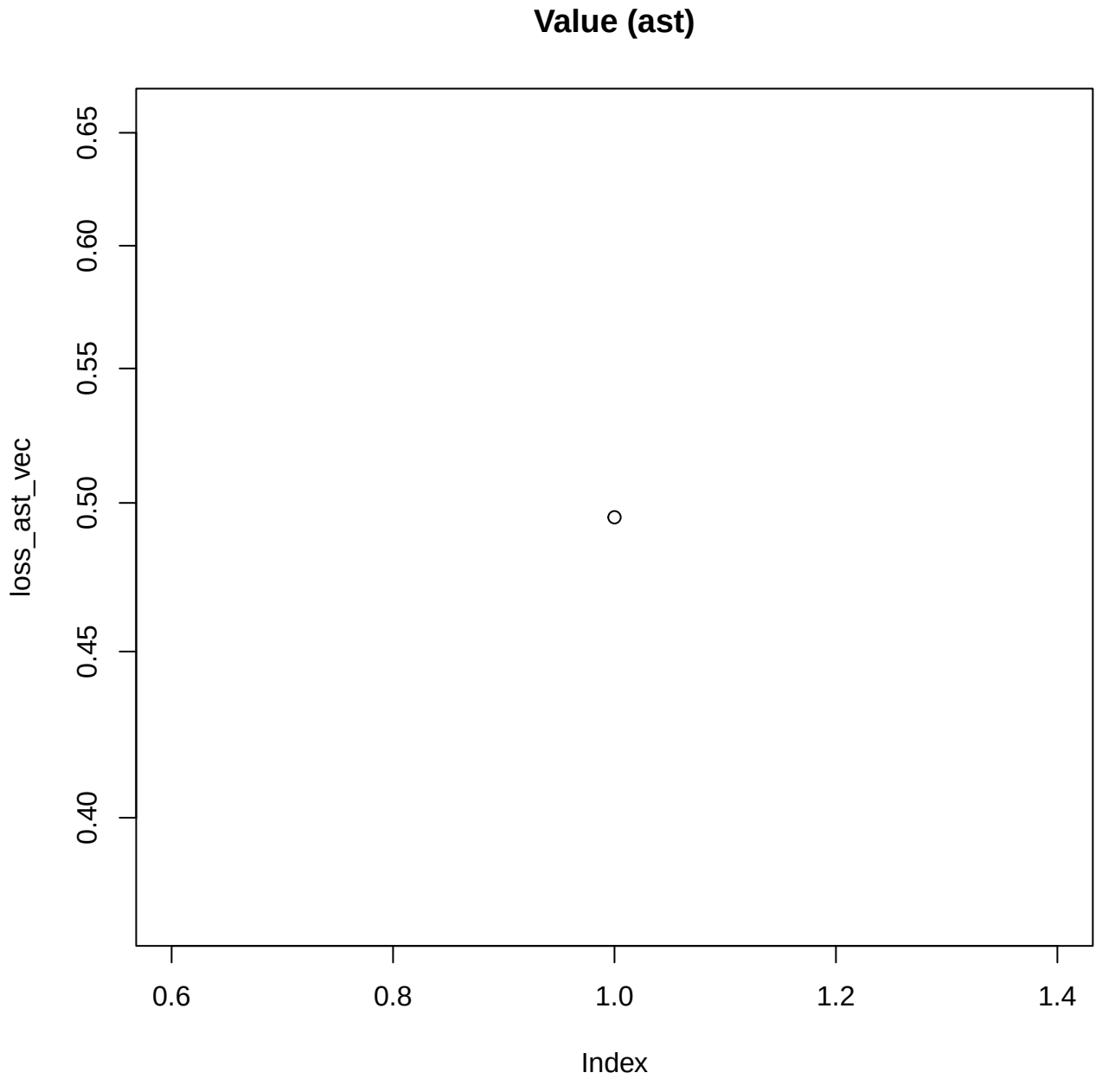

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

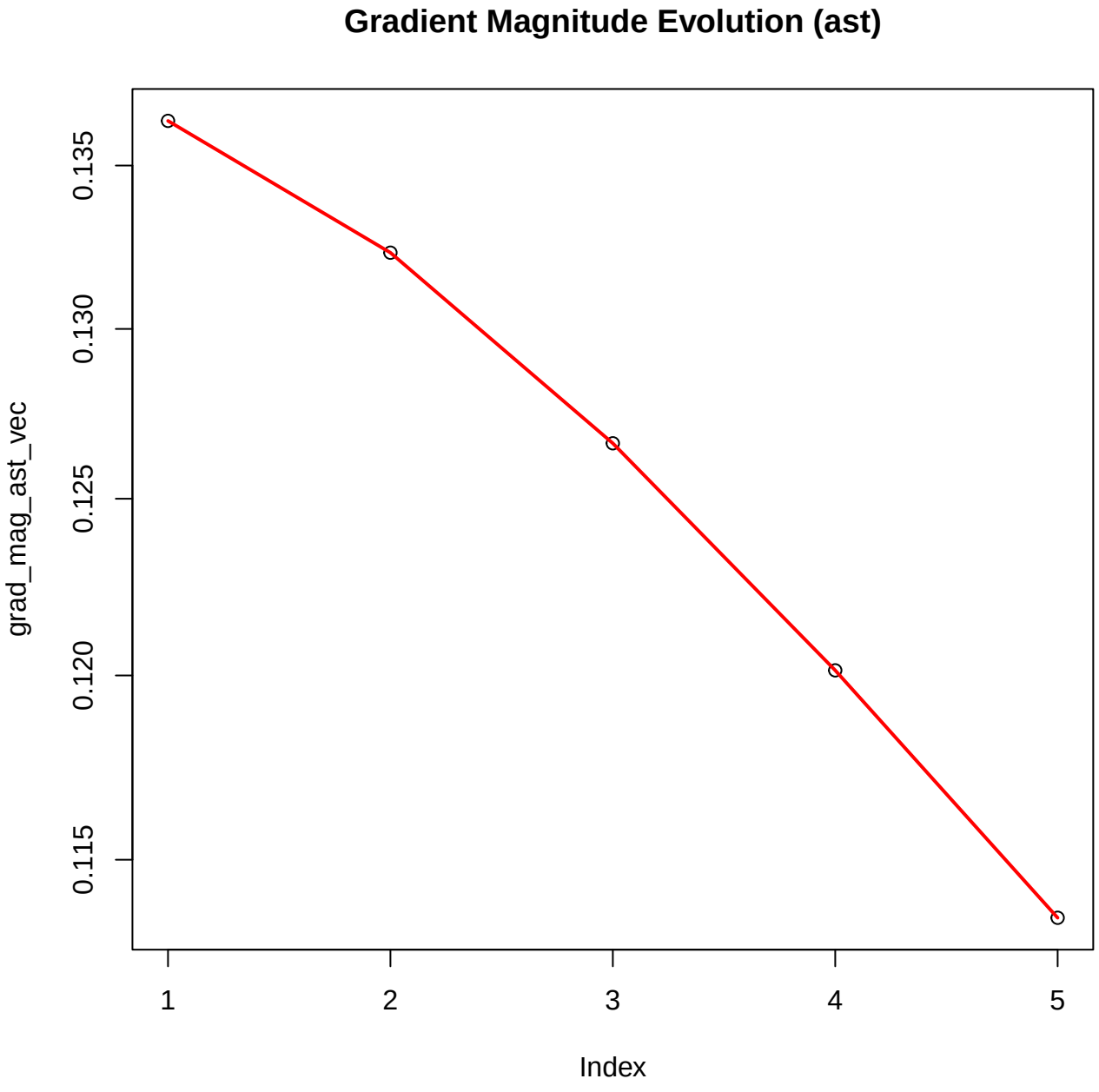

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

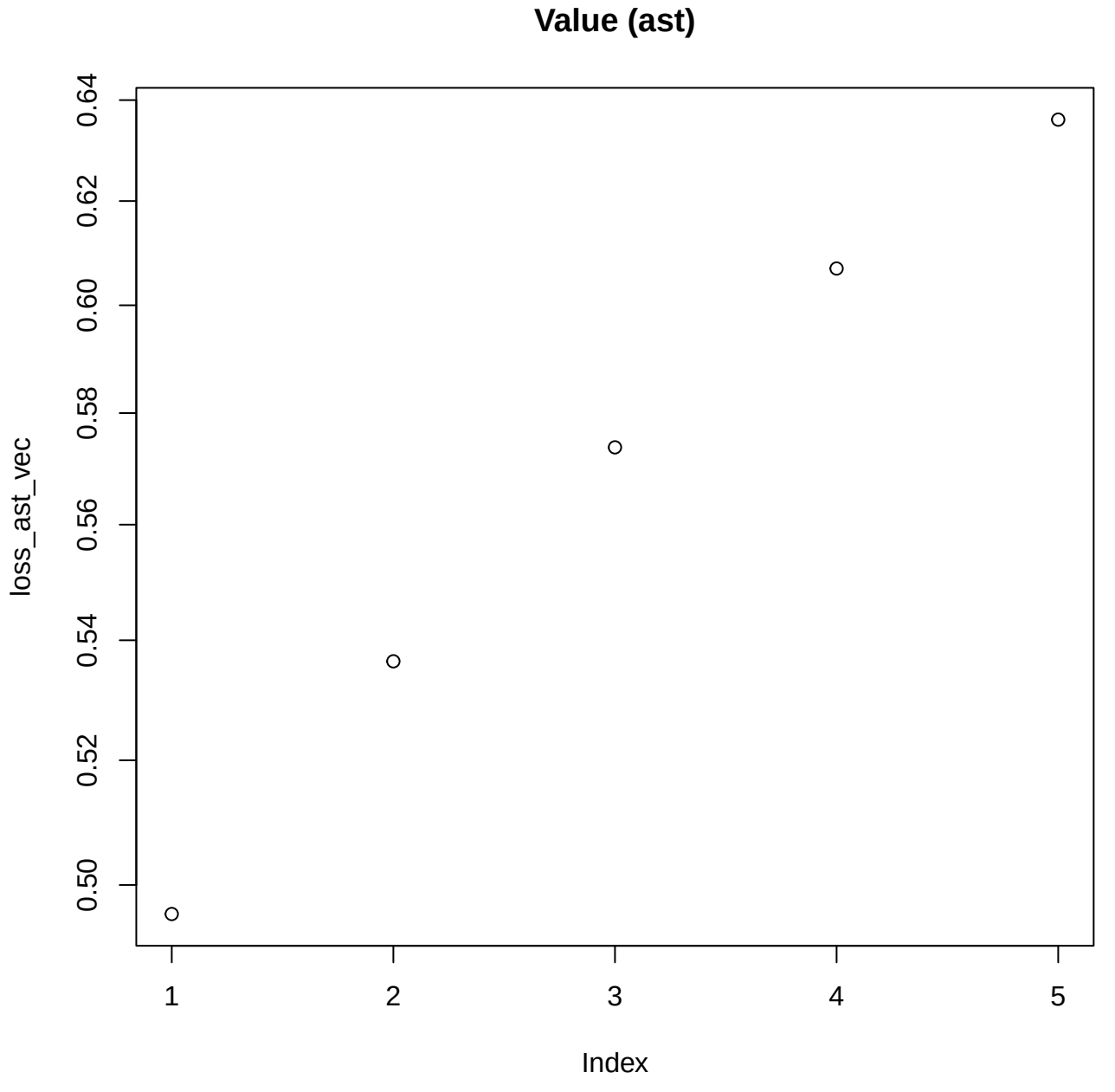

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.120
0.125
0.130
0.135
0.140
1
2
3
4
5
Index

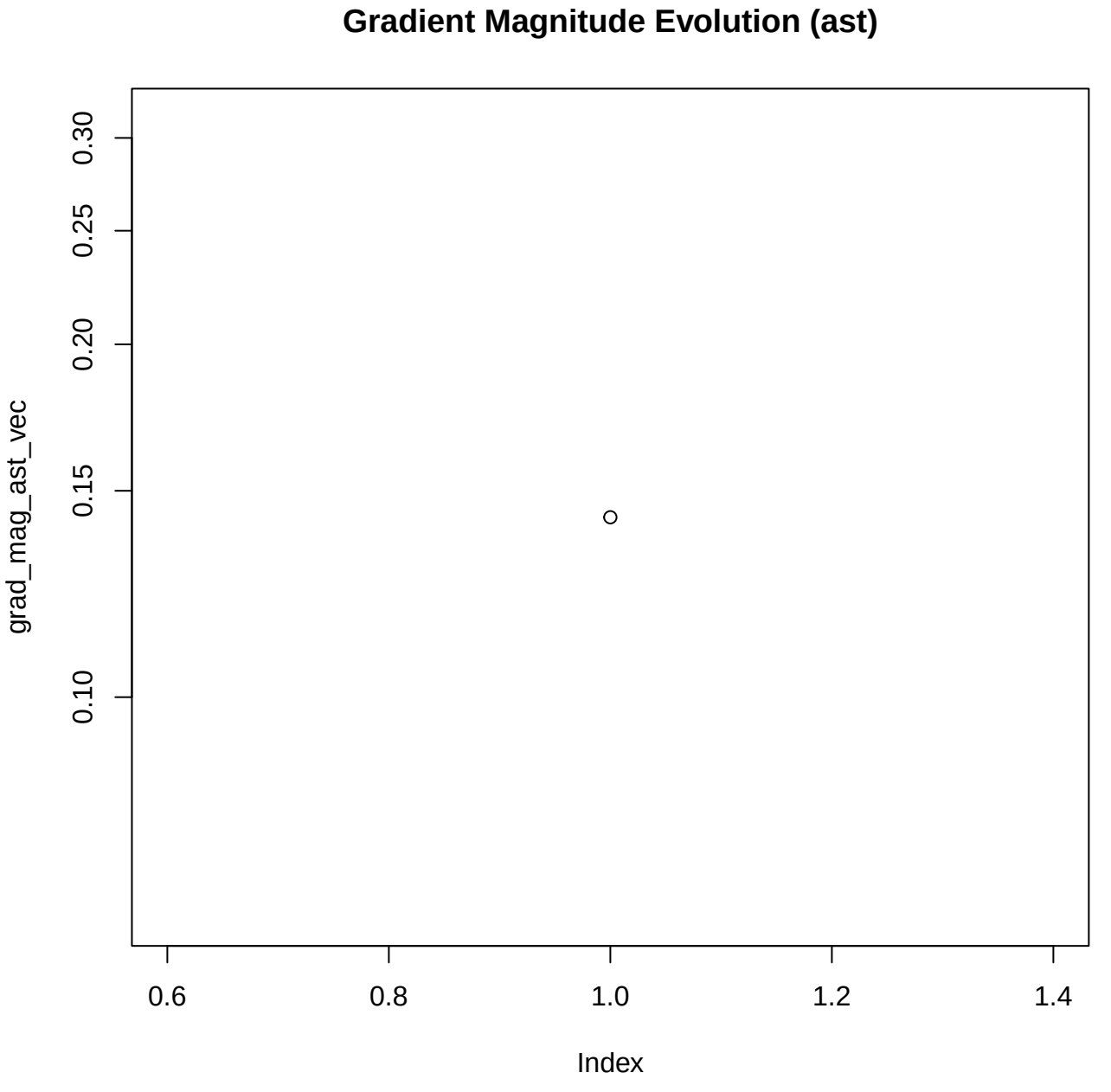

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

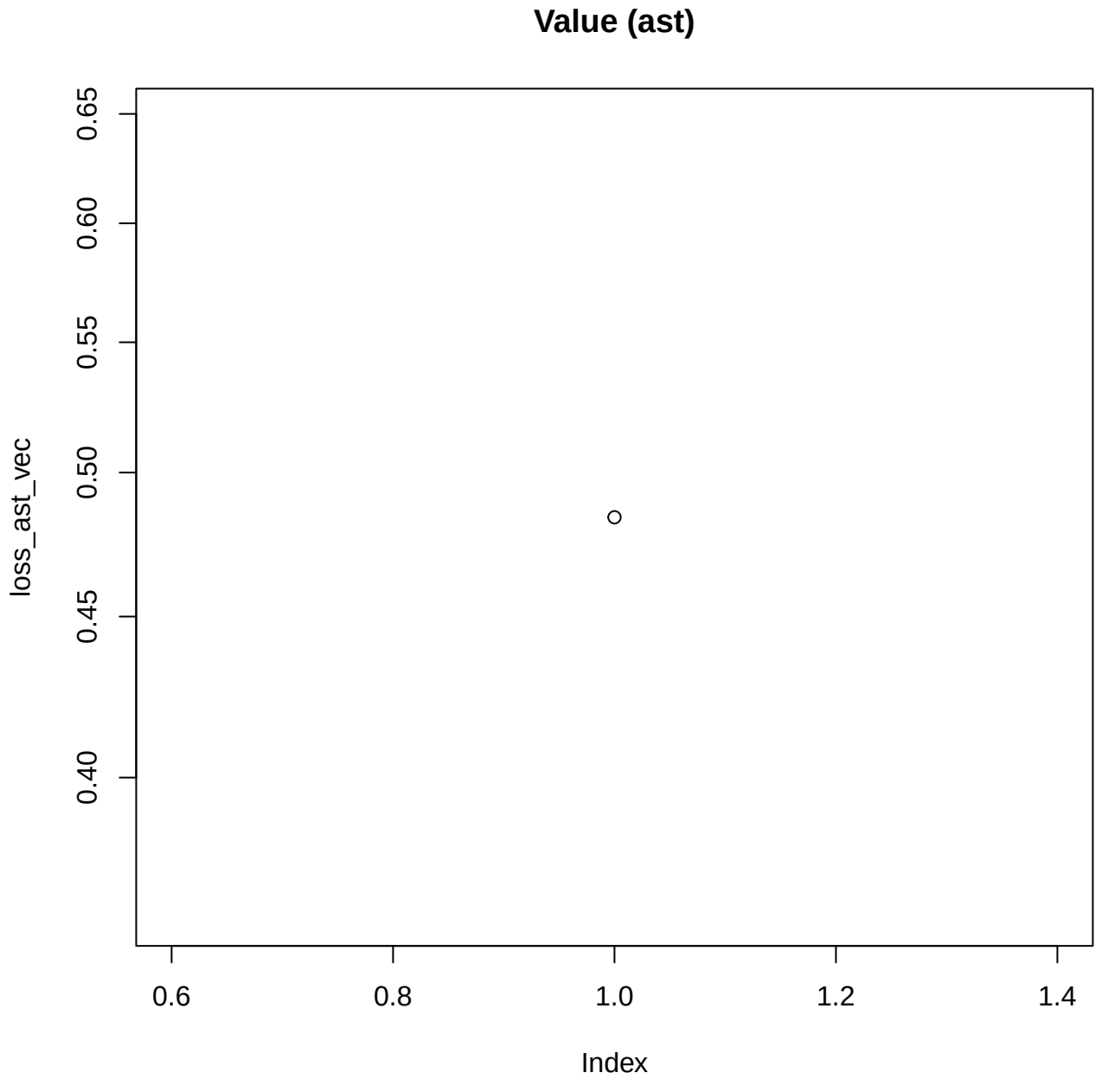

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

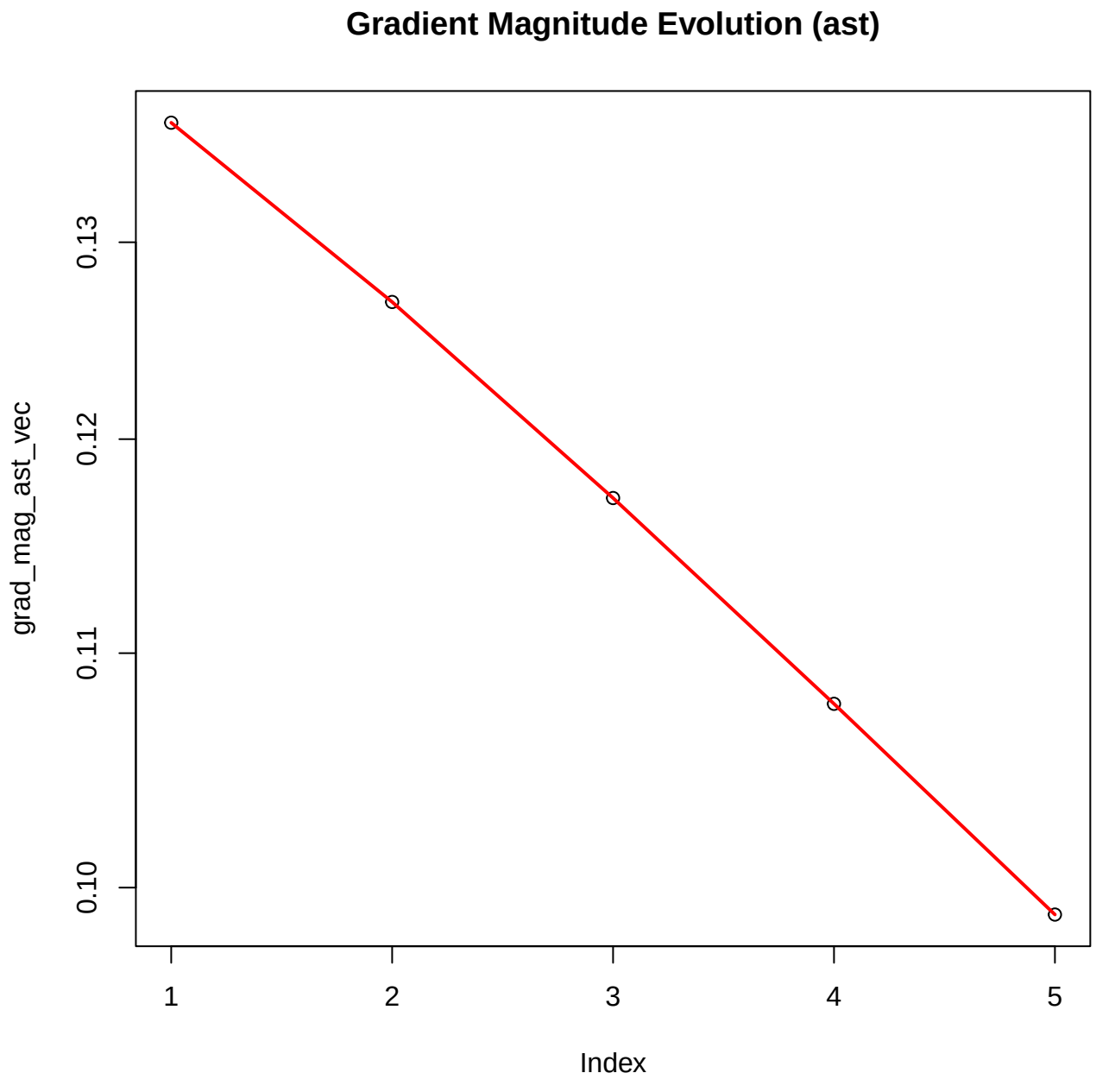

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

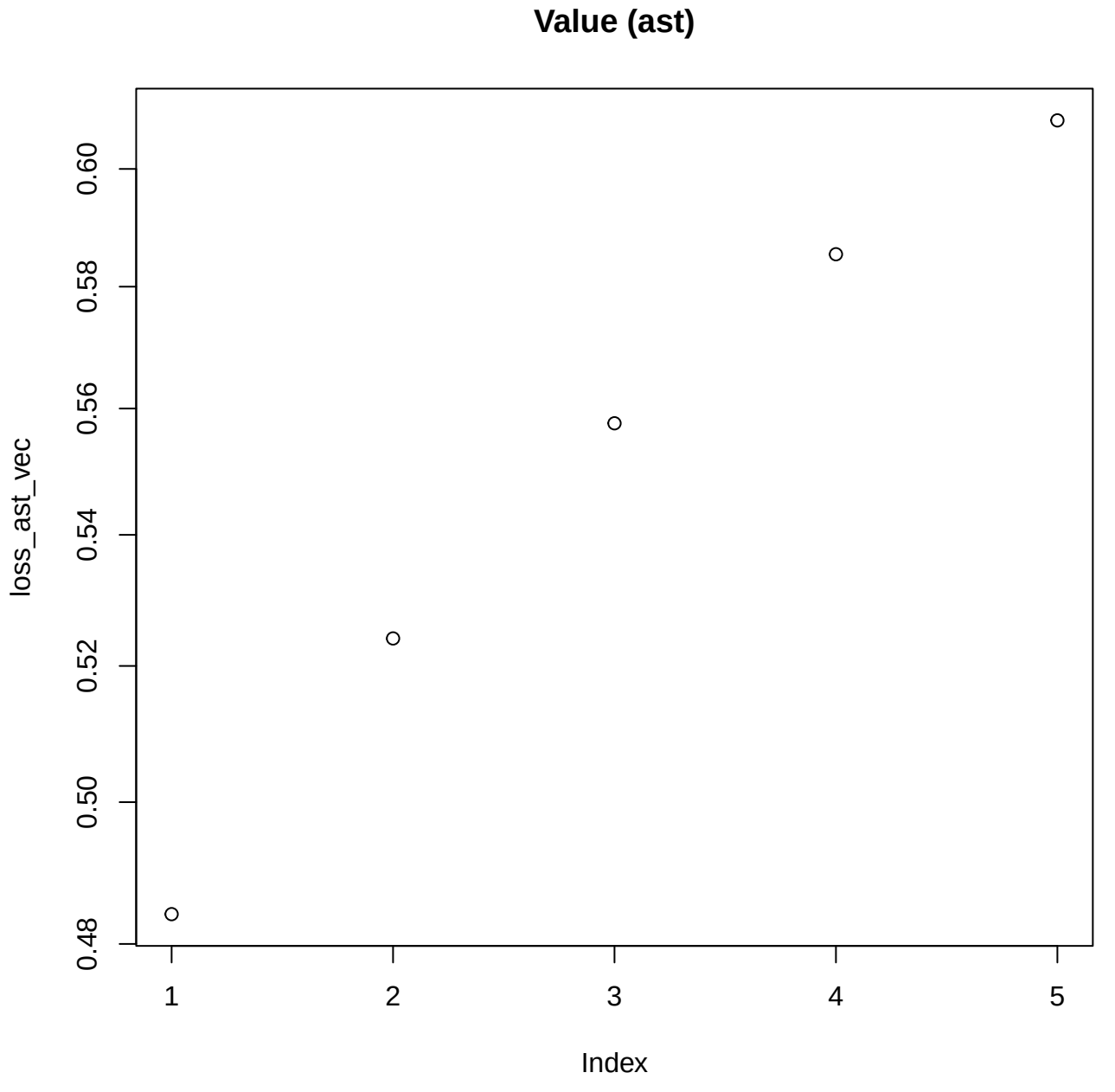

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.115
0.120
0.125
0.130
0.135
1
2
3
4
5
Index

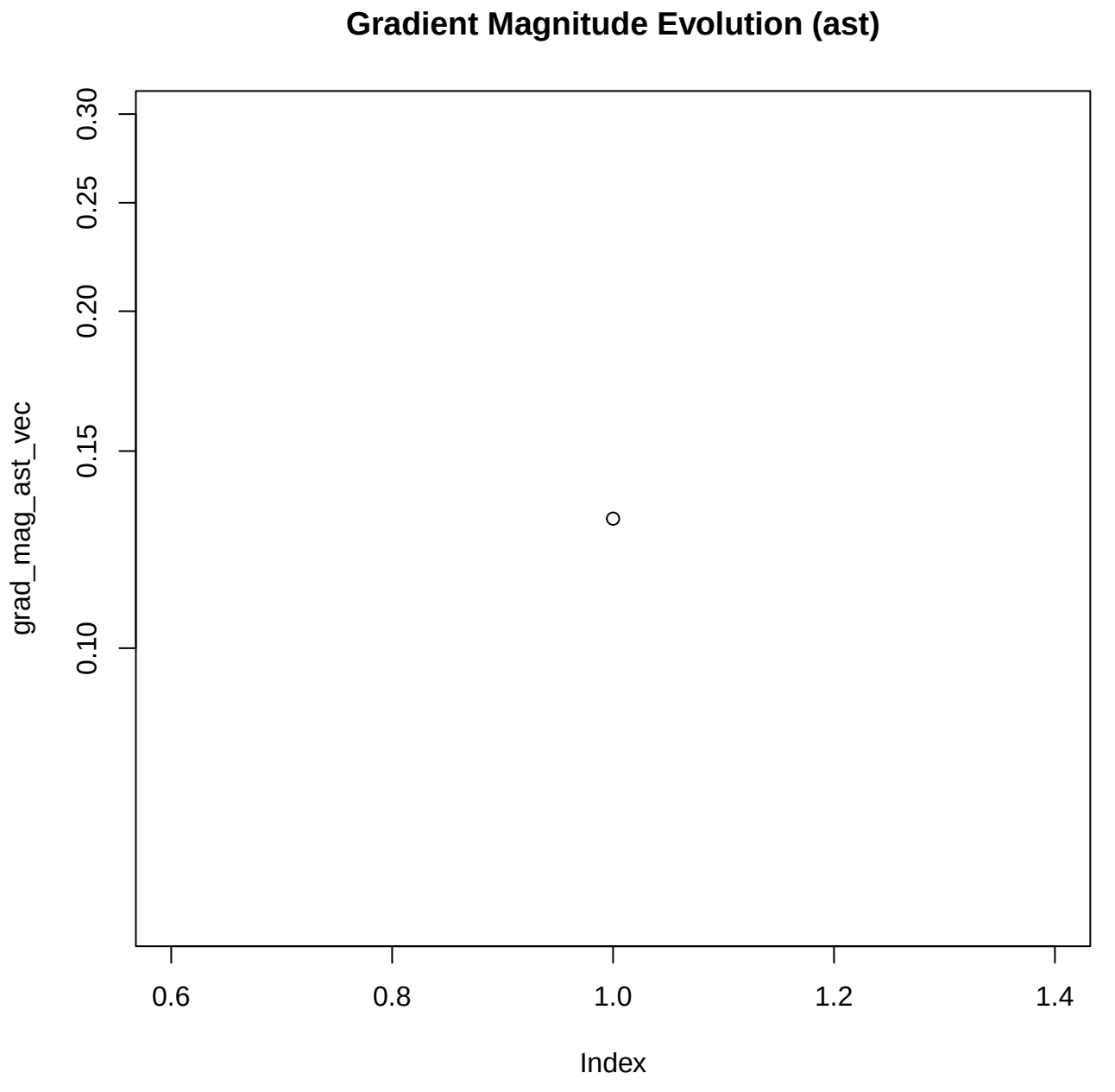

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
0.62
0.64
1
2
3
4
5
Index

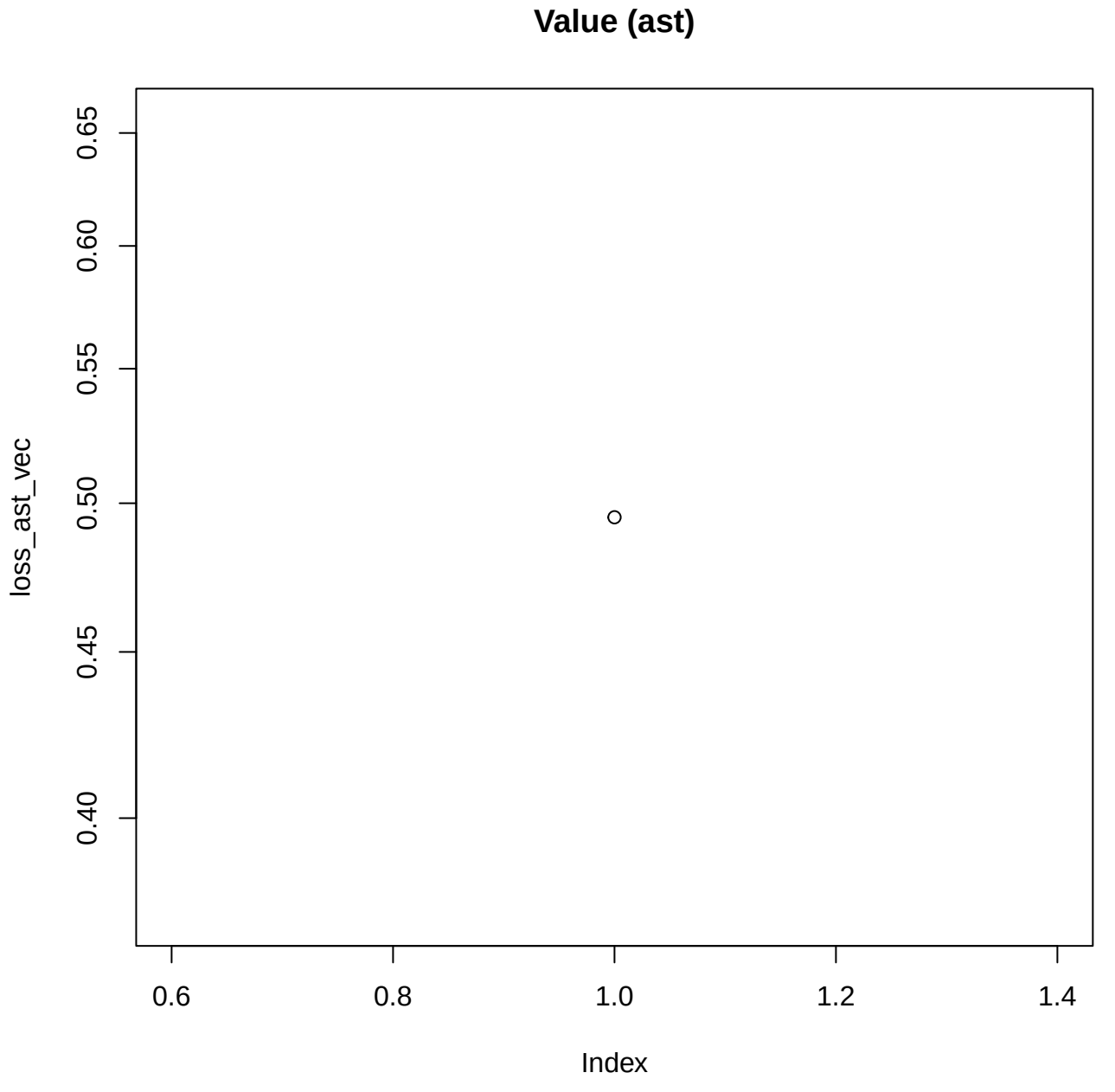

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

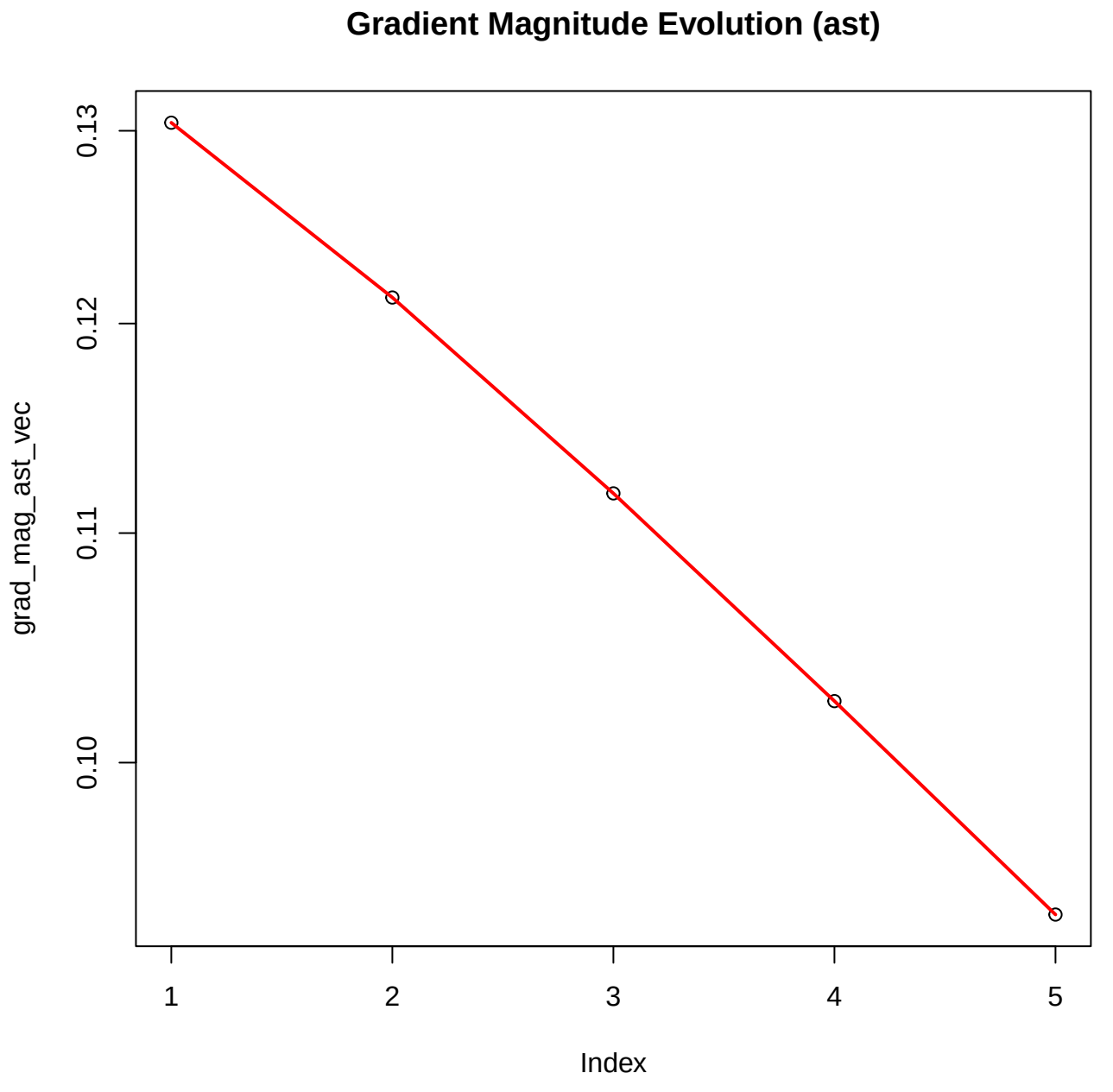

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

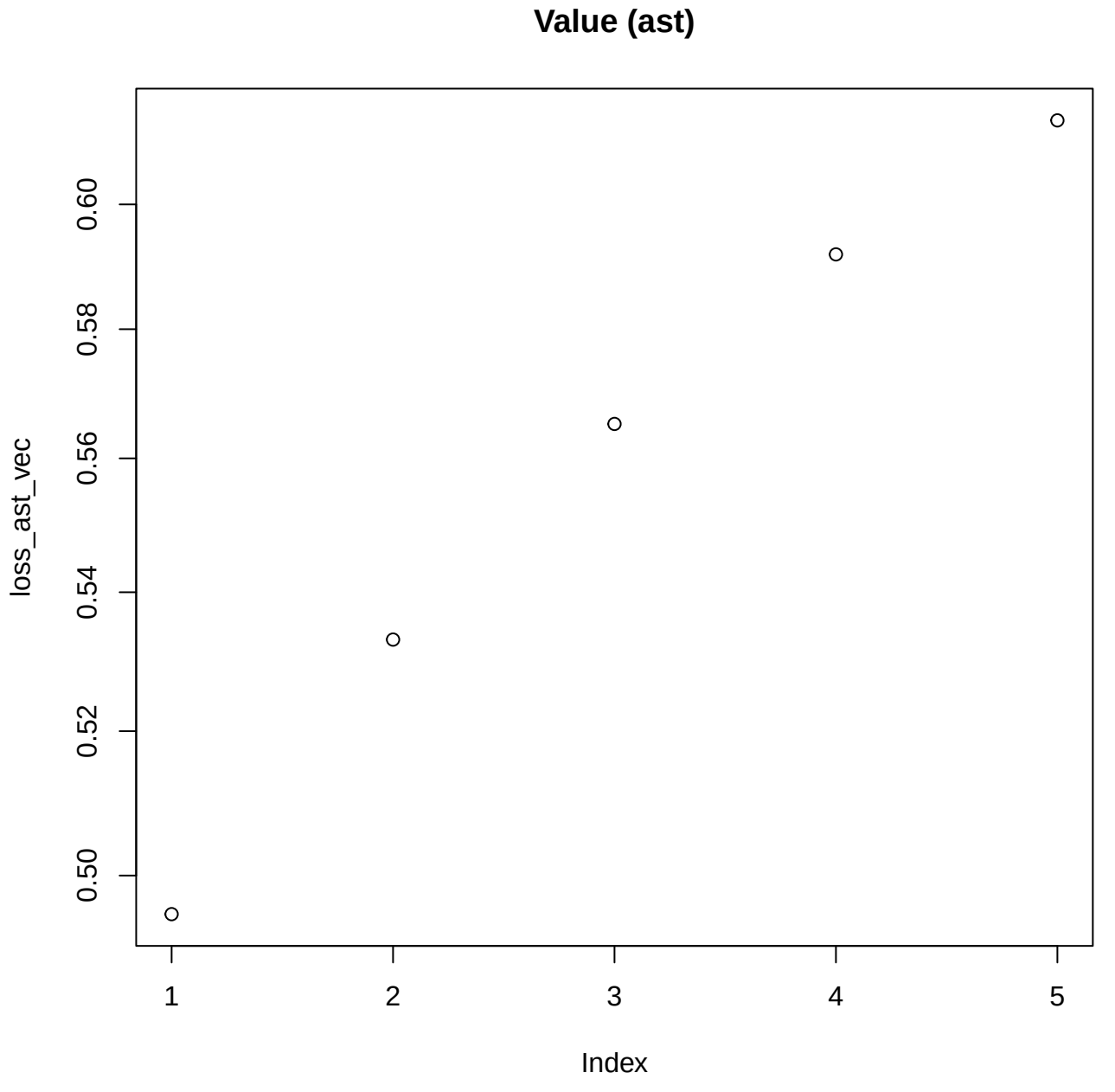

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

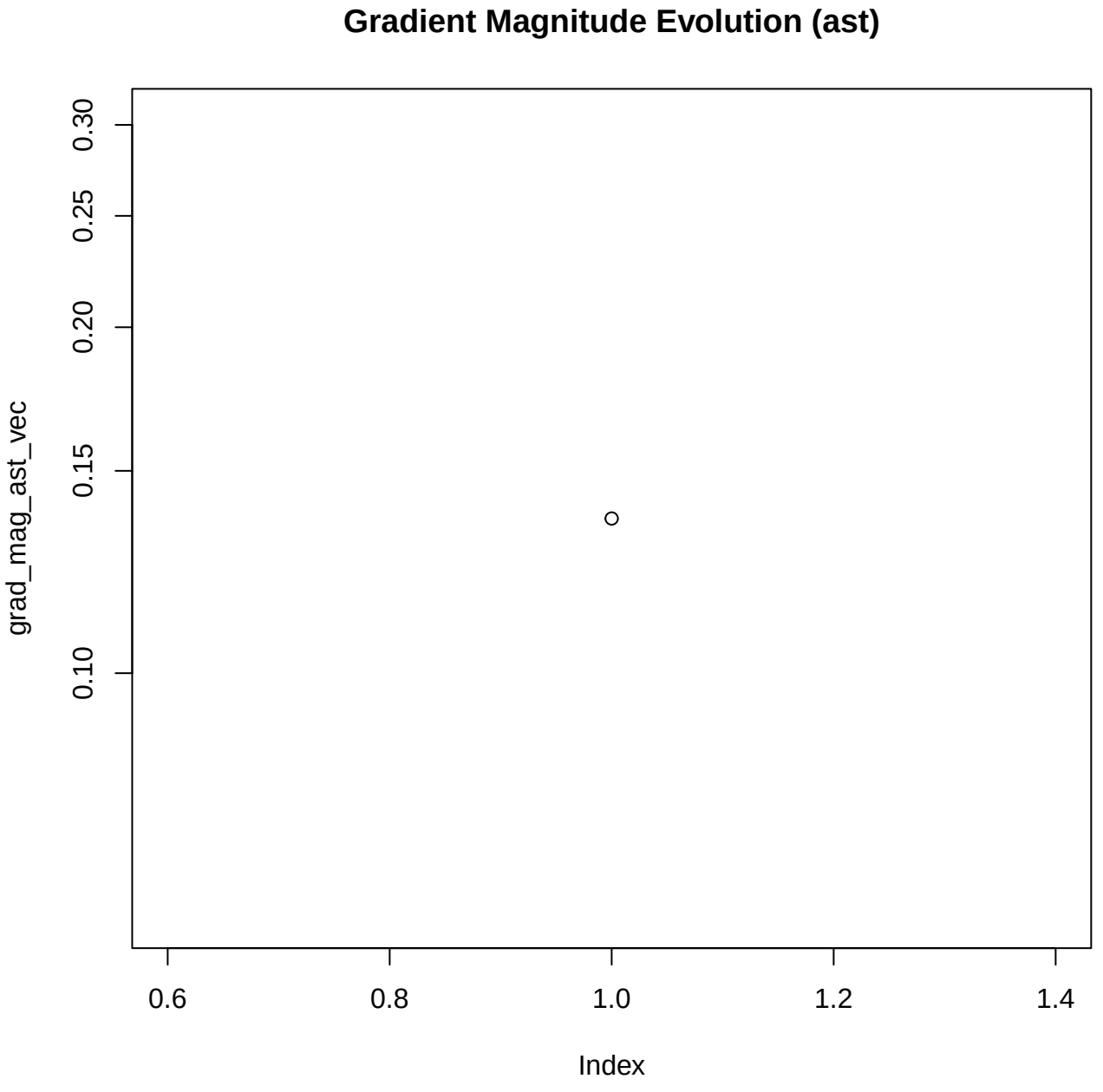

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

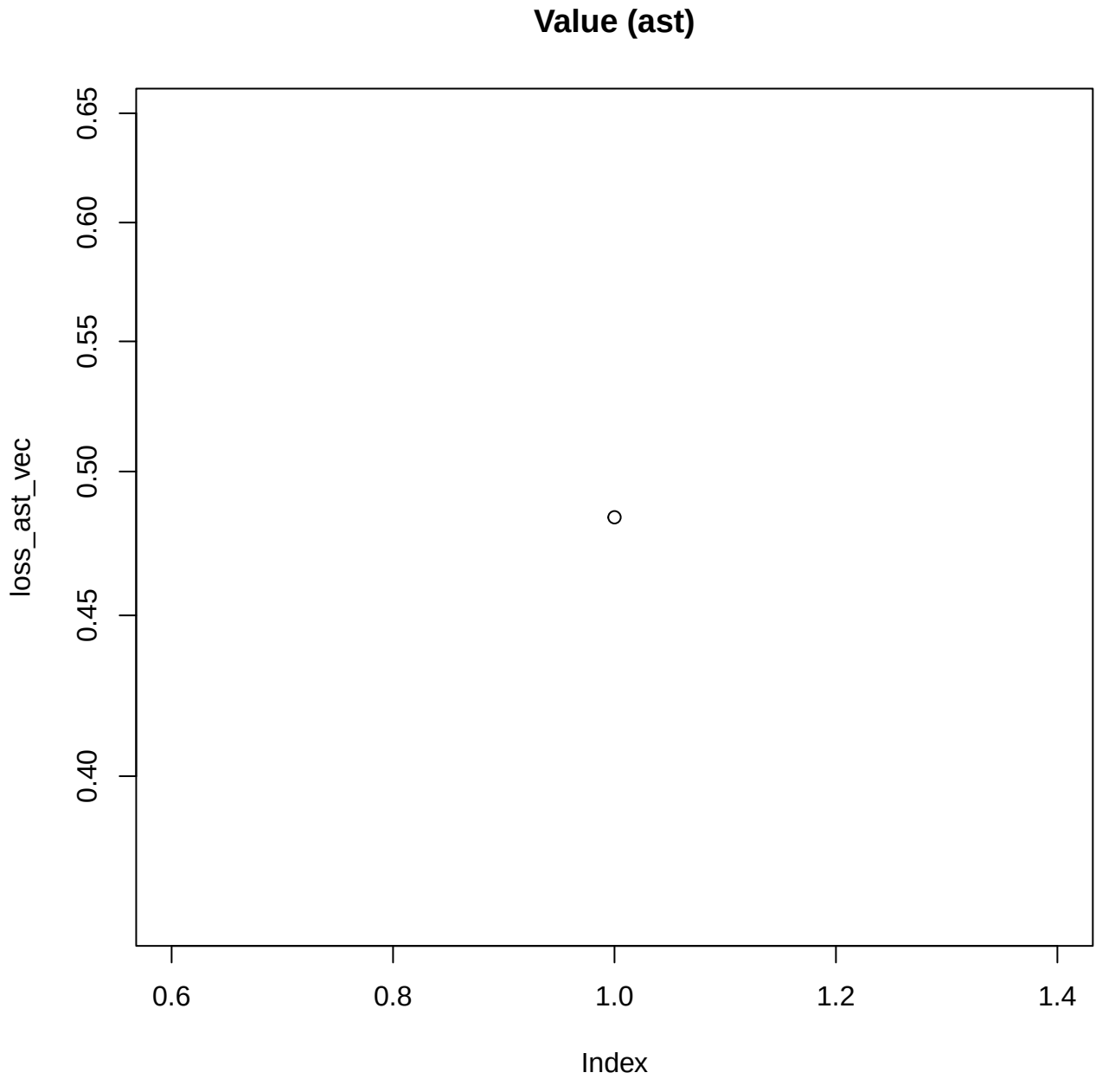

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

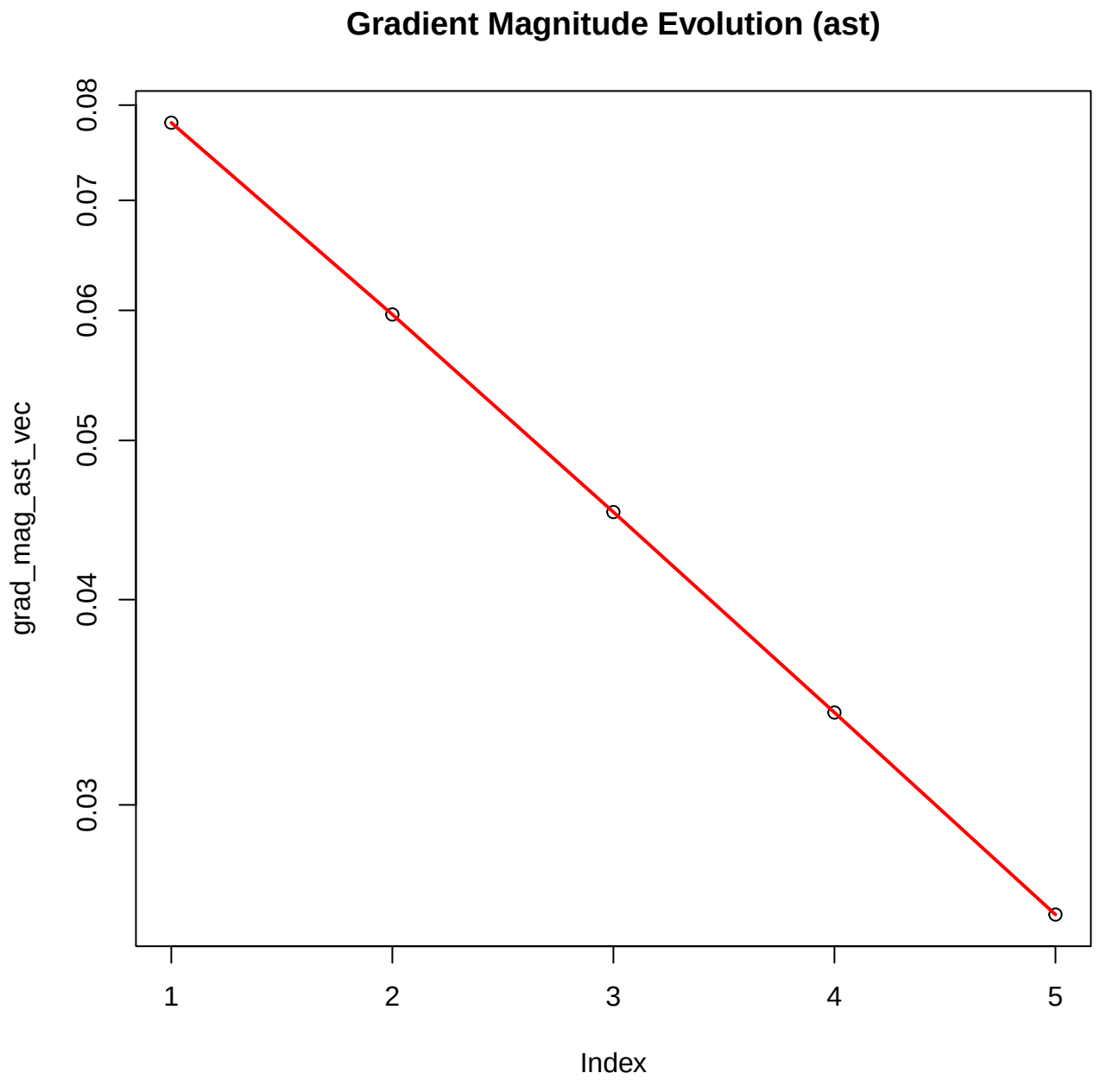

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.6
0.8
1.0
1.2
1.4
Index

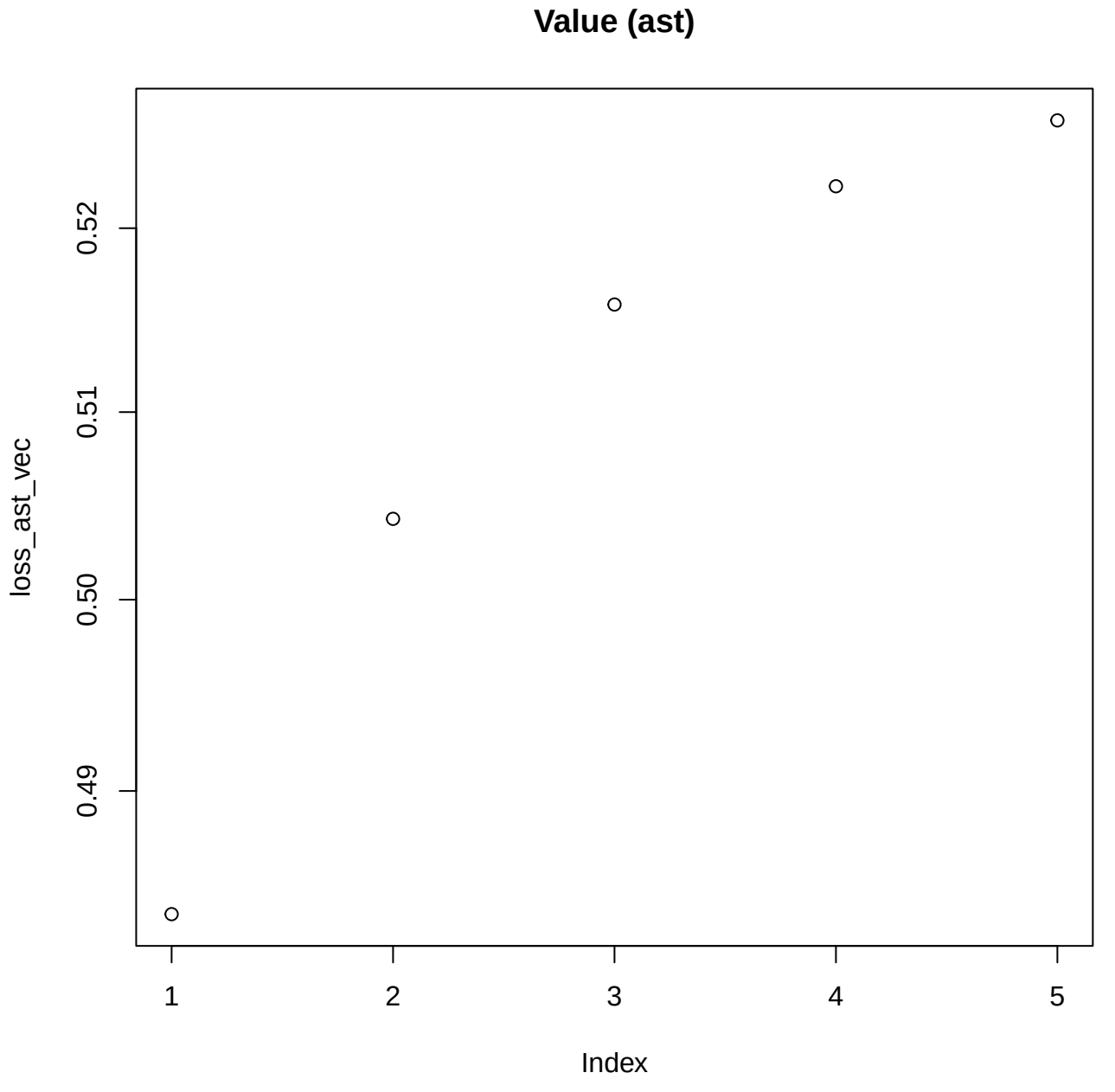

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

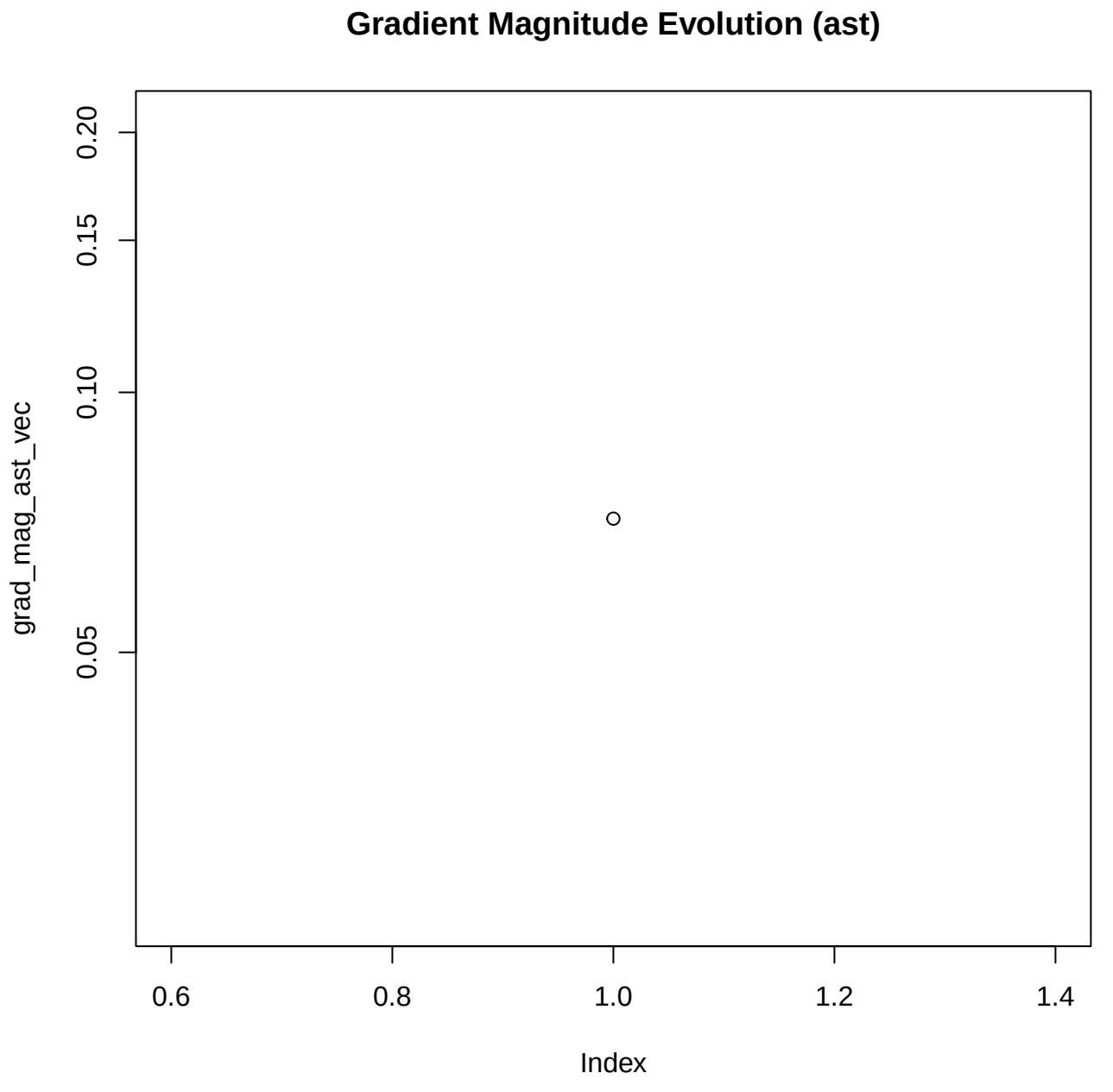

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

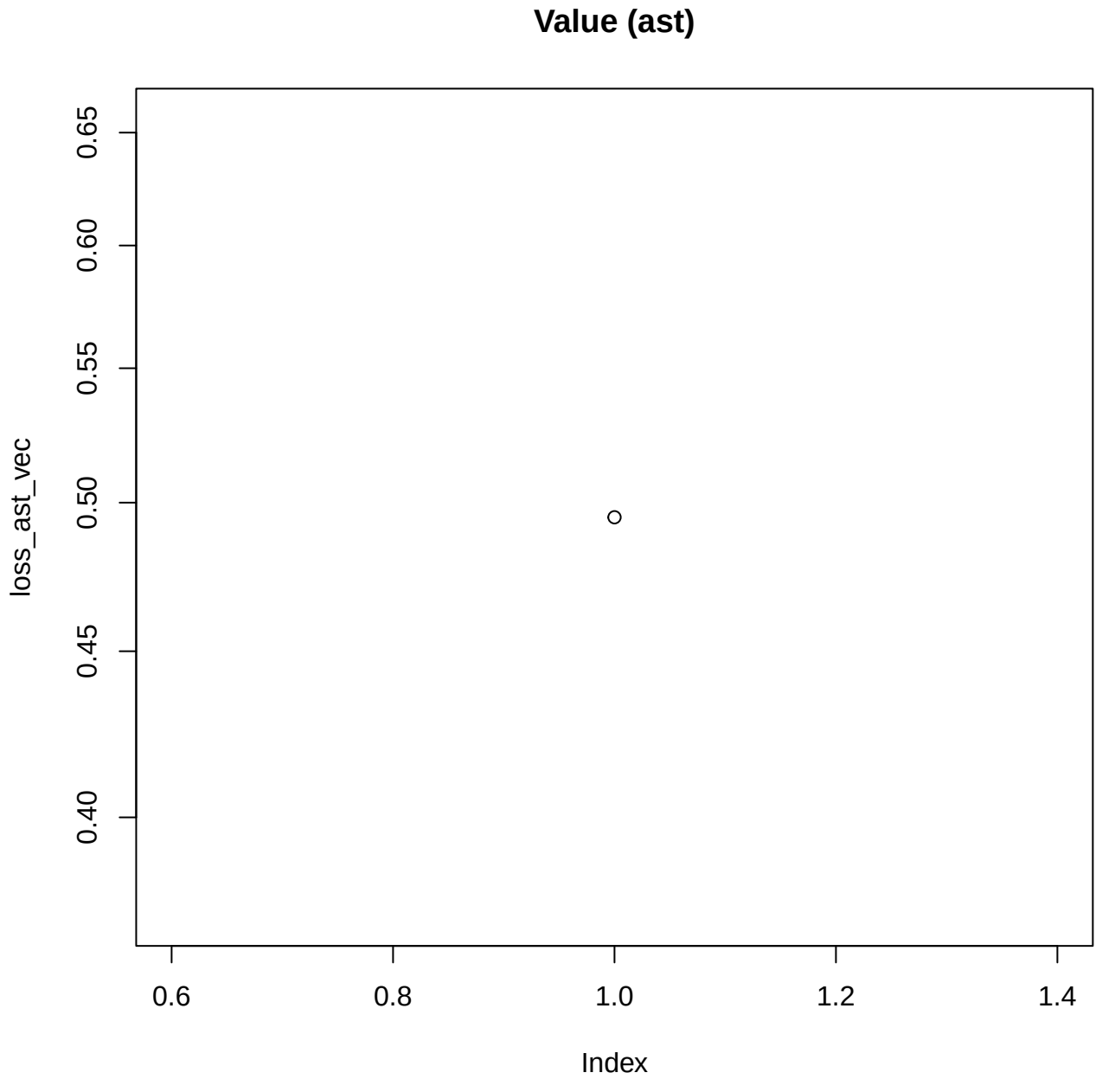

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

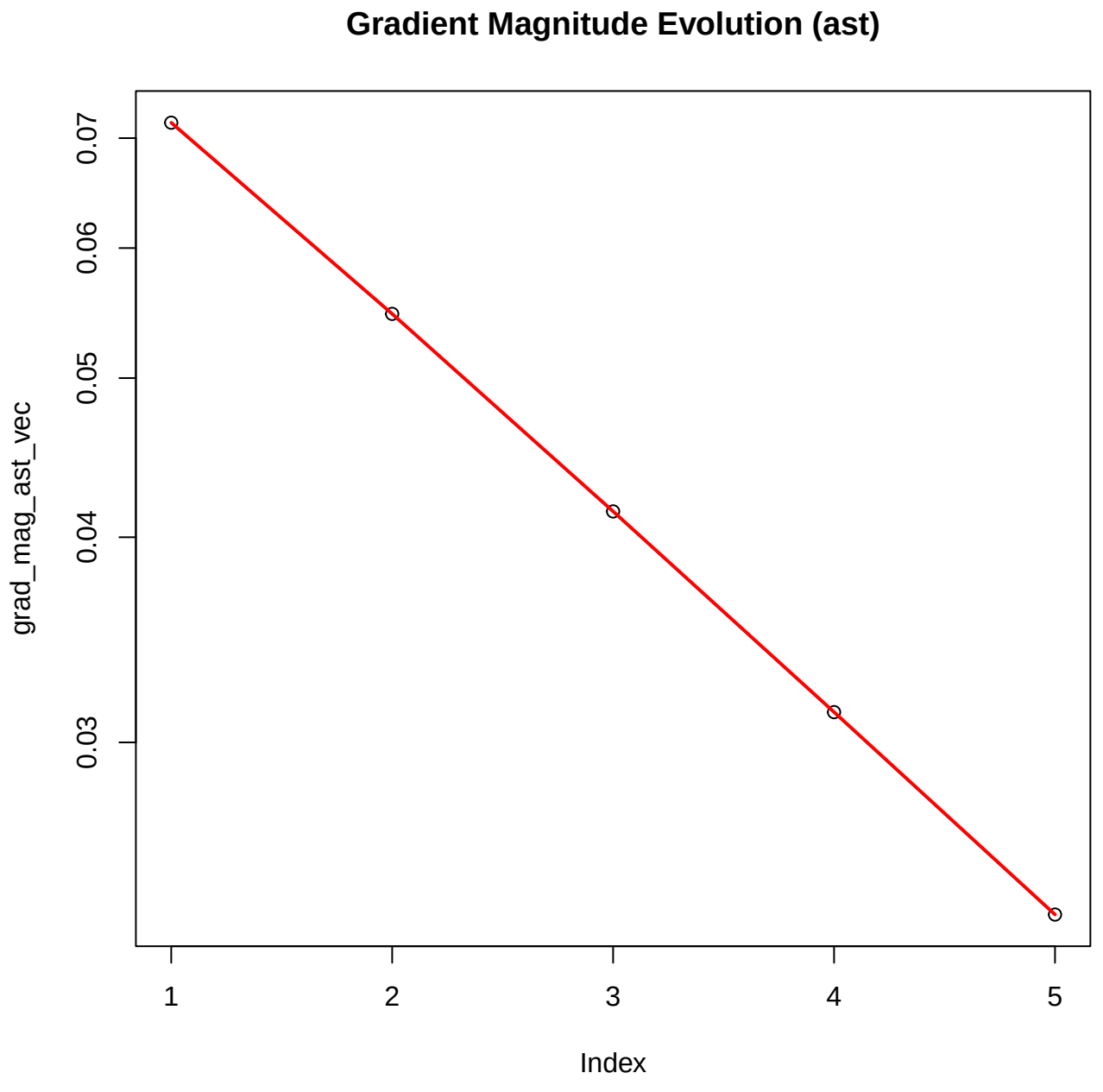

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

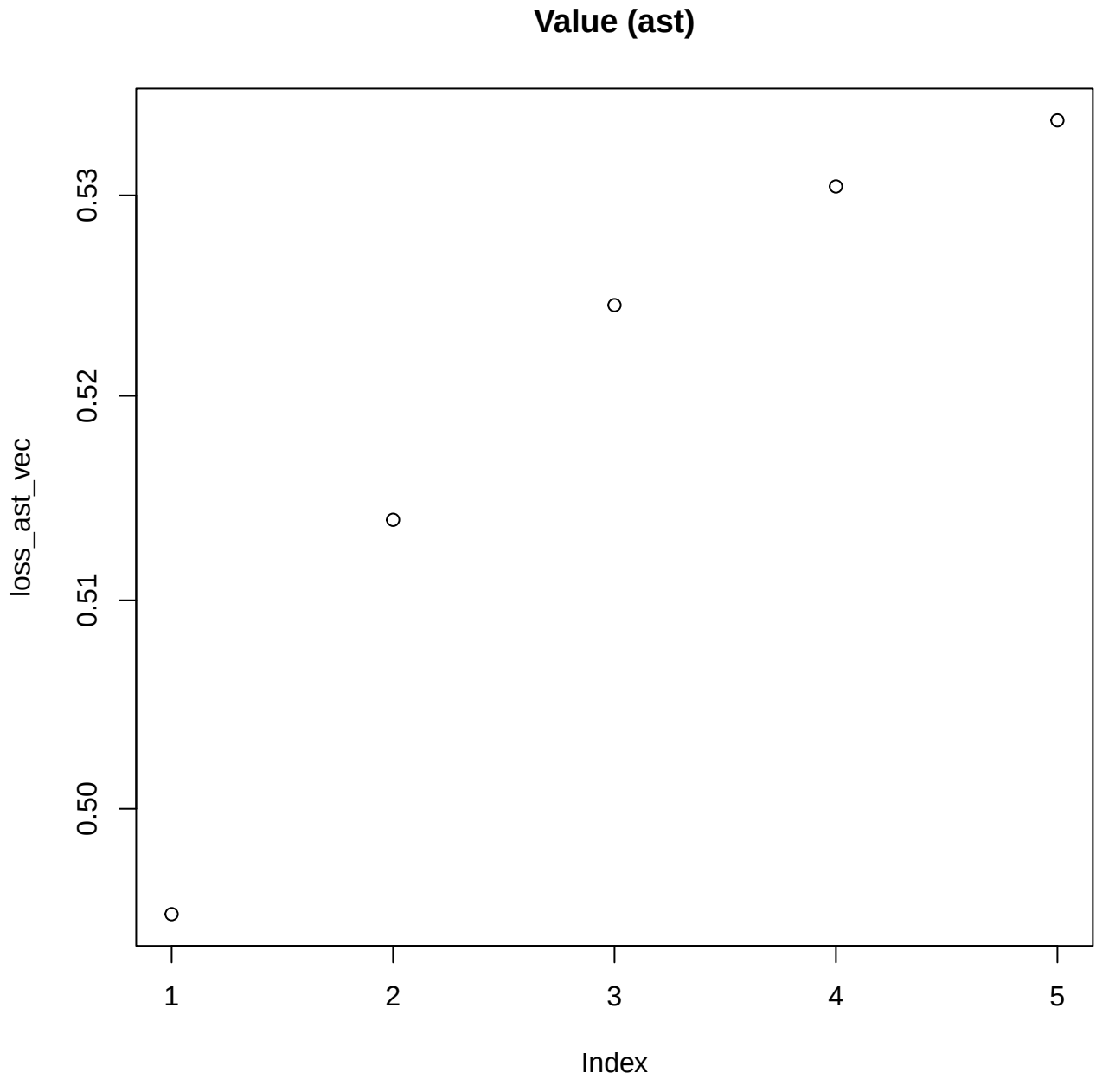

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.07
0.06
0.05
0.04
0.03
1
2
3
4
5
Index

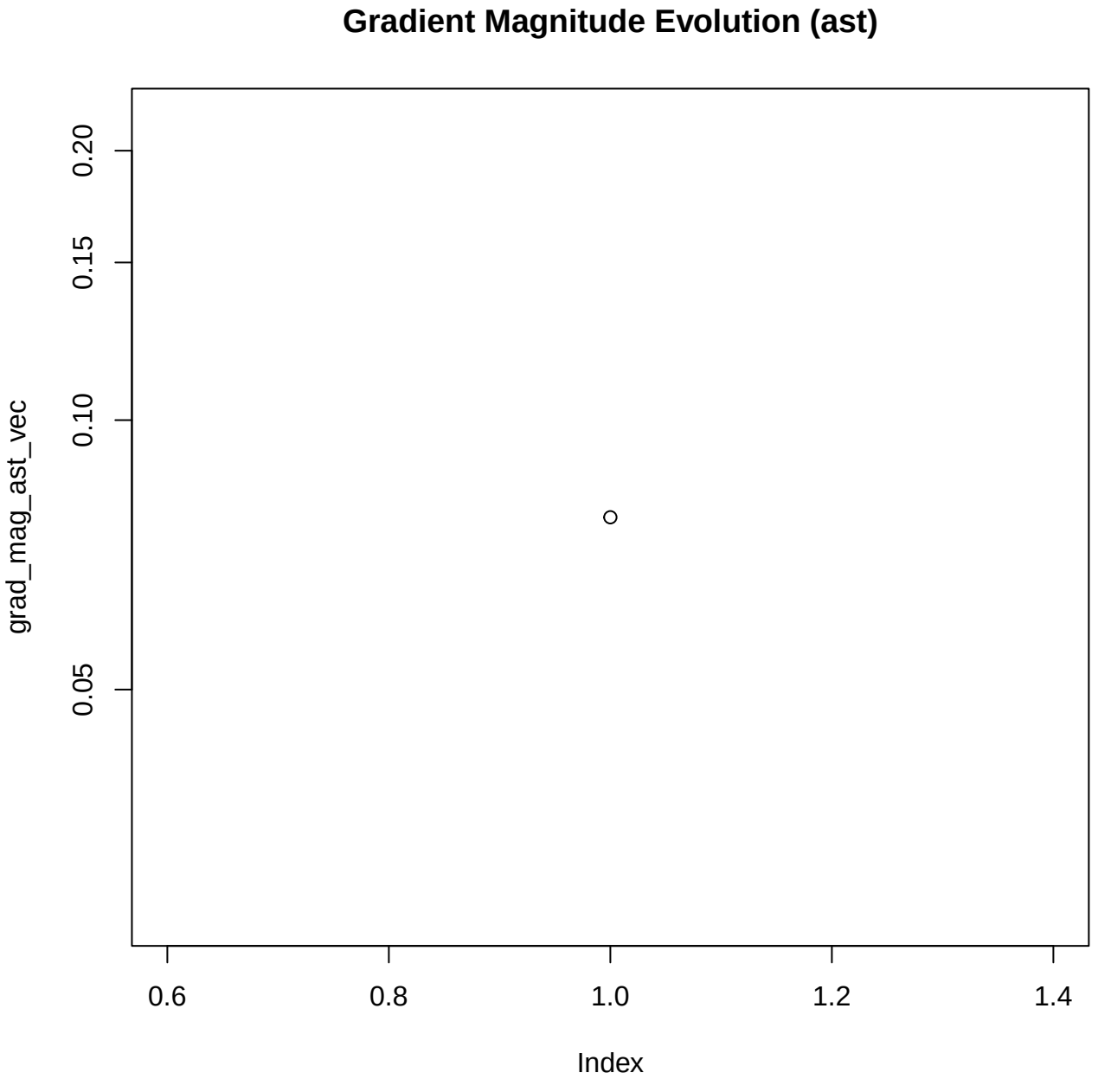

Value (ast)
loss_ast_vec
0.49
0.50
0.51
0.52
1
2
3
4
5
Index

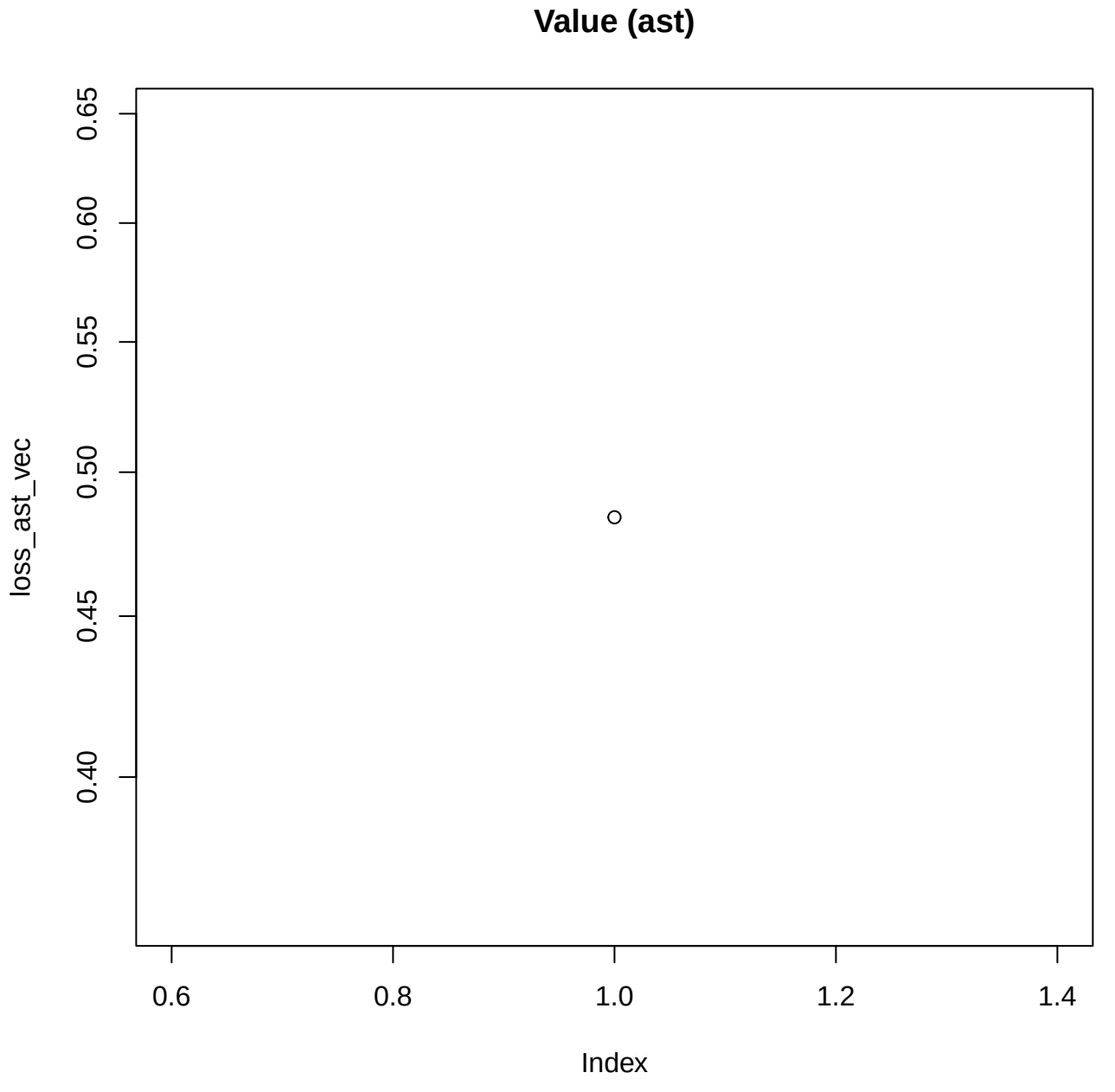

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.05
0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
Index

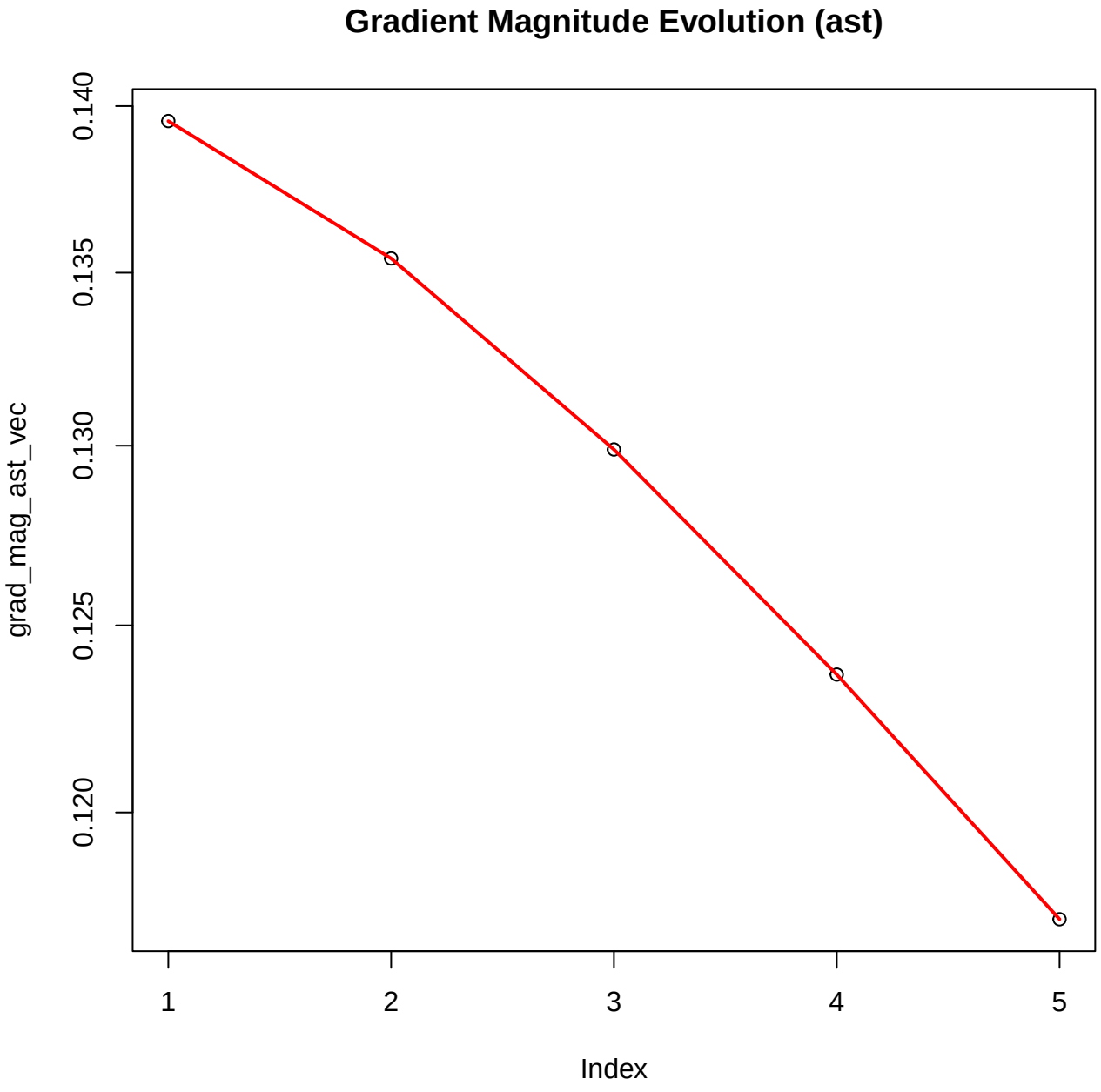

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

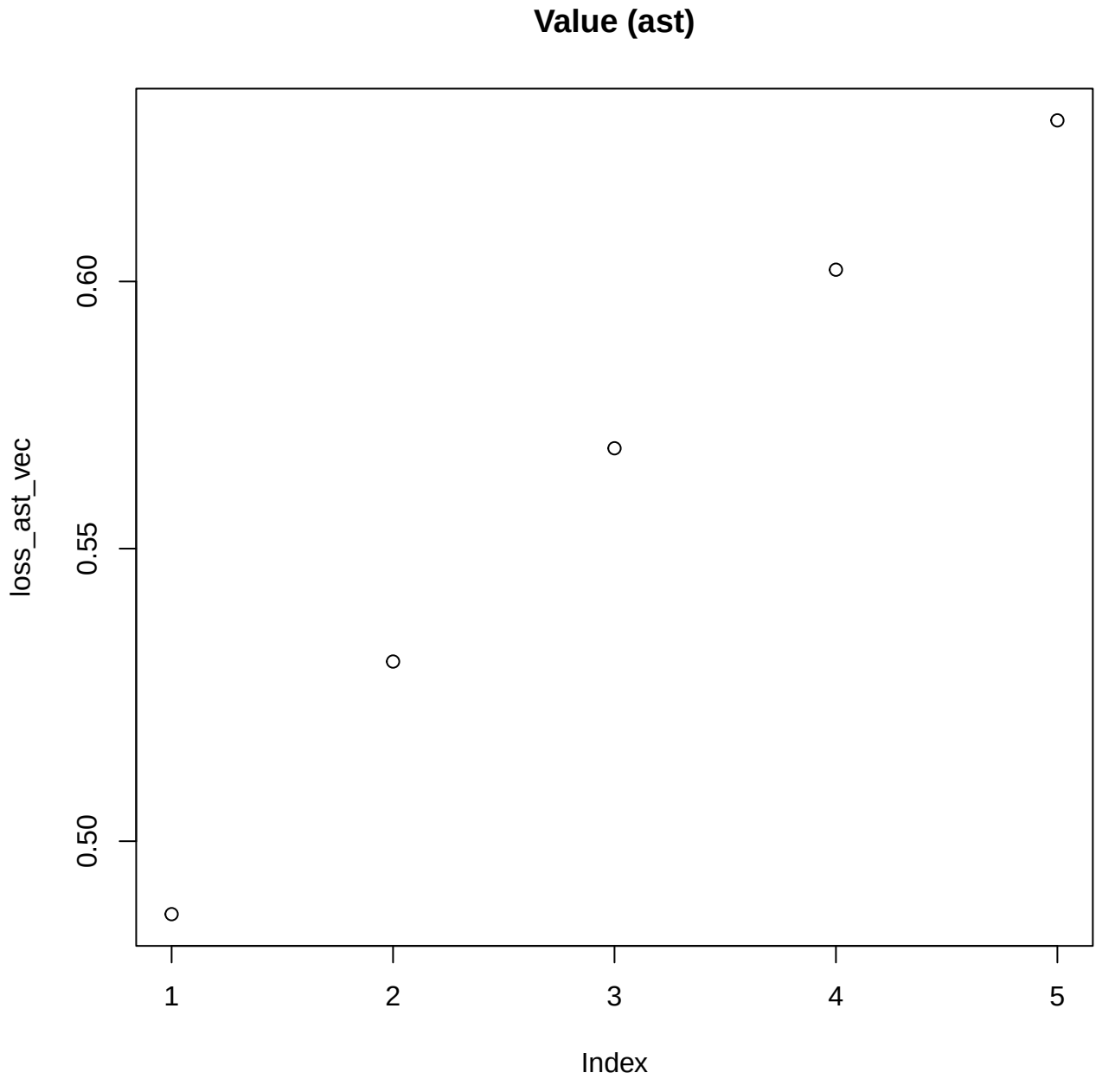

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index

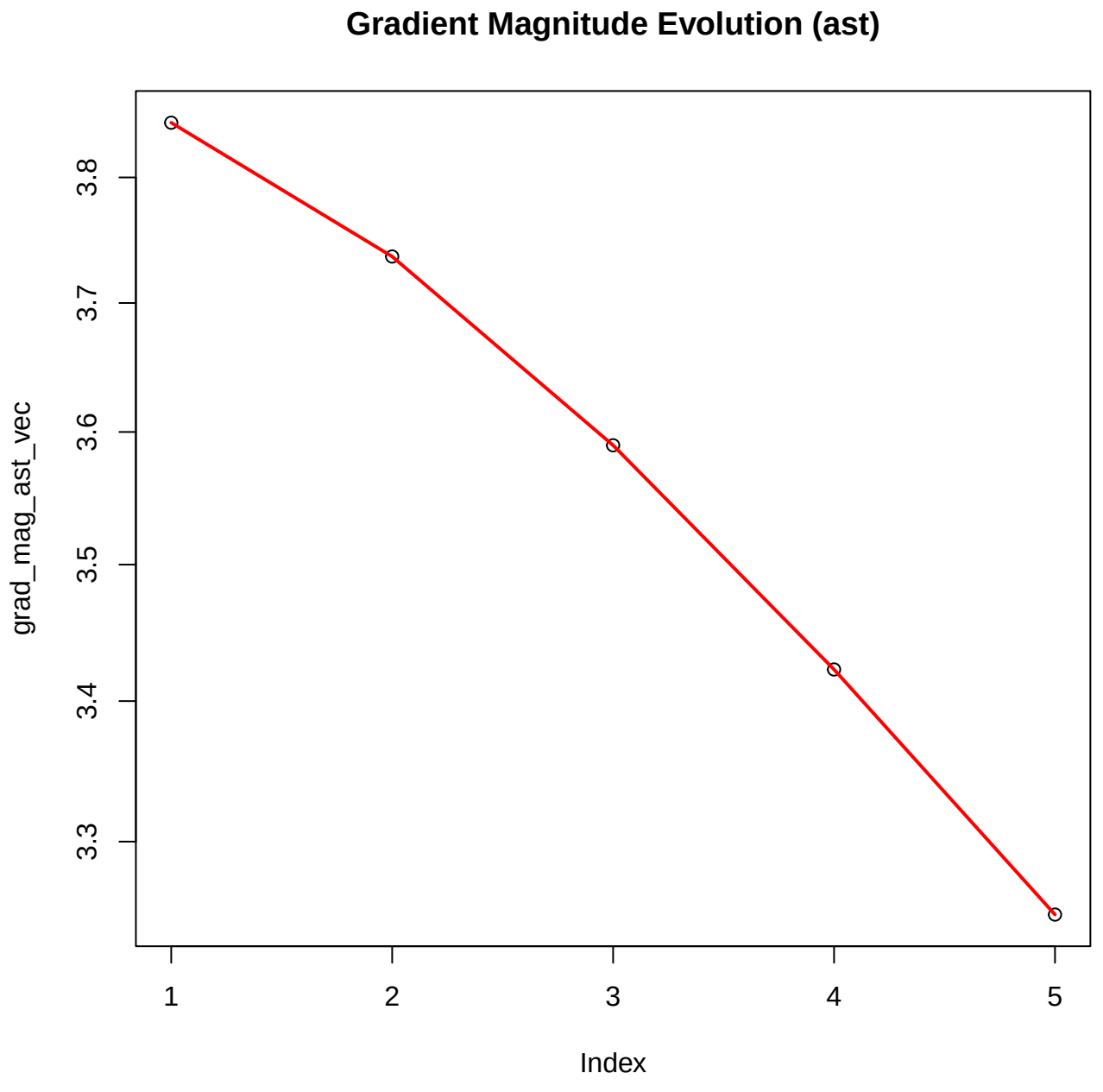

Value (ast)
loss_ast_vec
0.50
0.51
0.52
0.53
1
2
3
4
5
Index

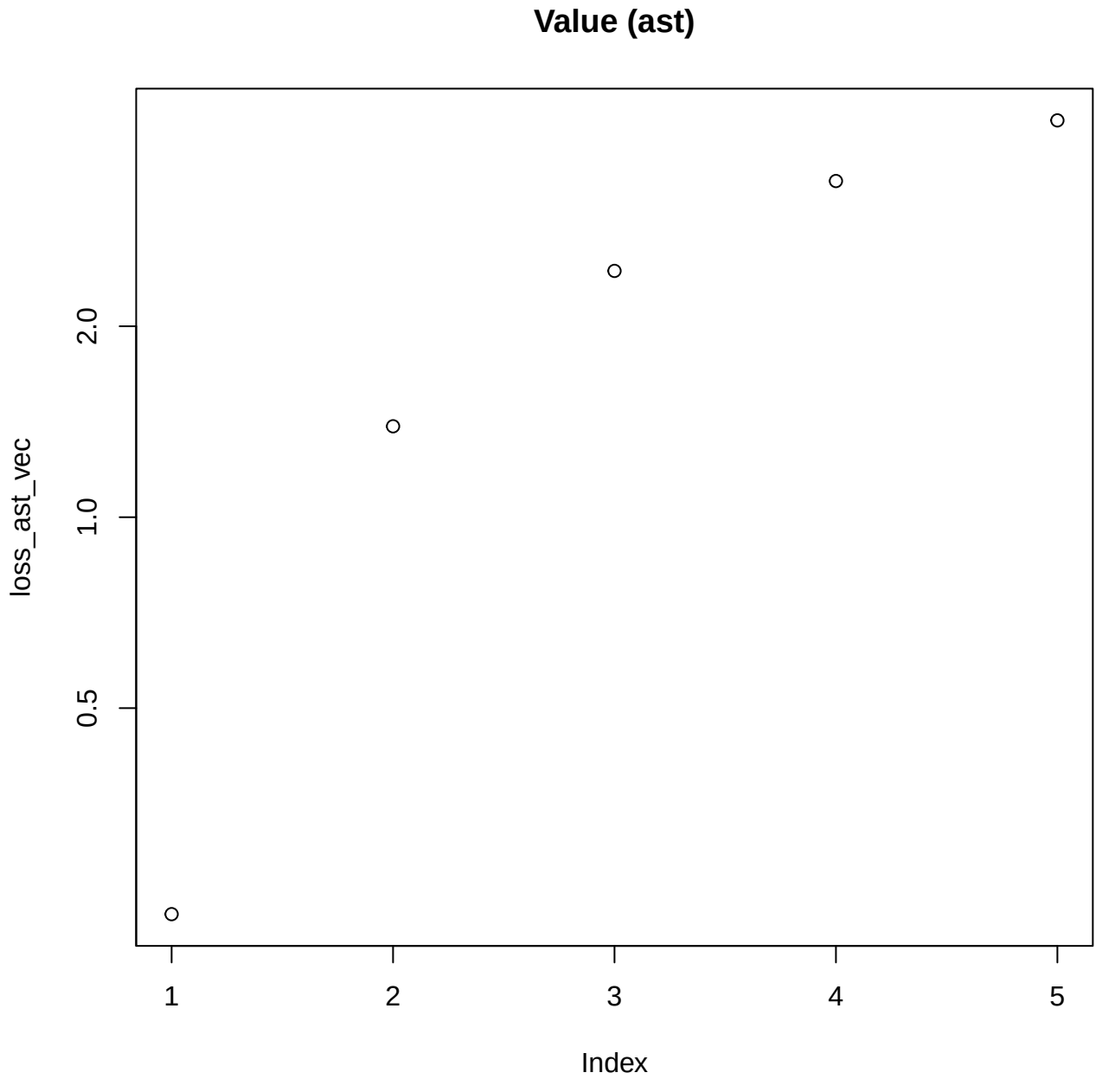

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.05
0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
Index

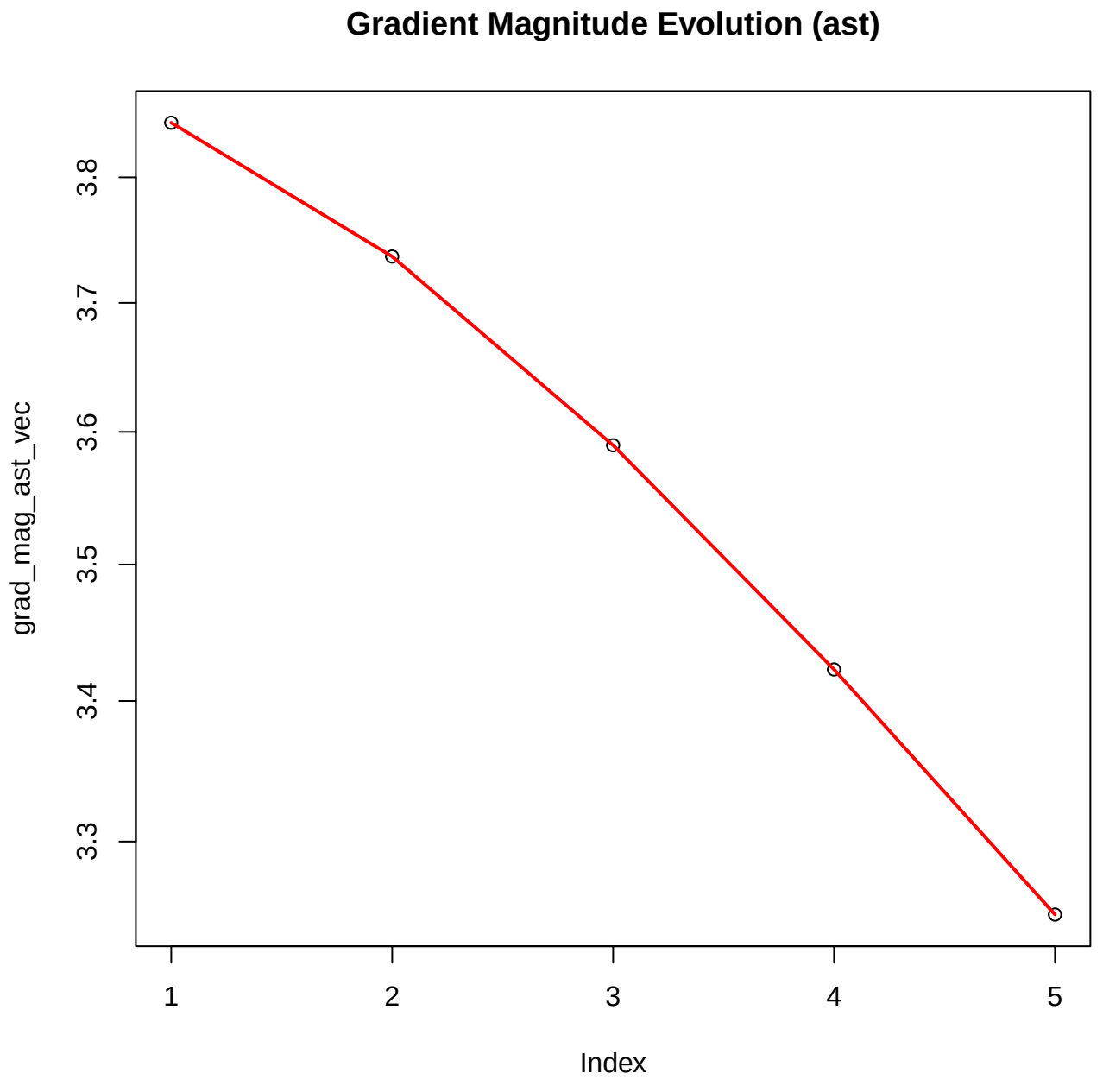

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

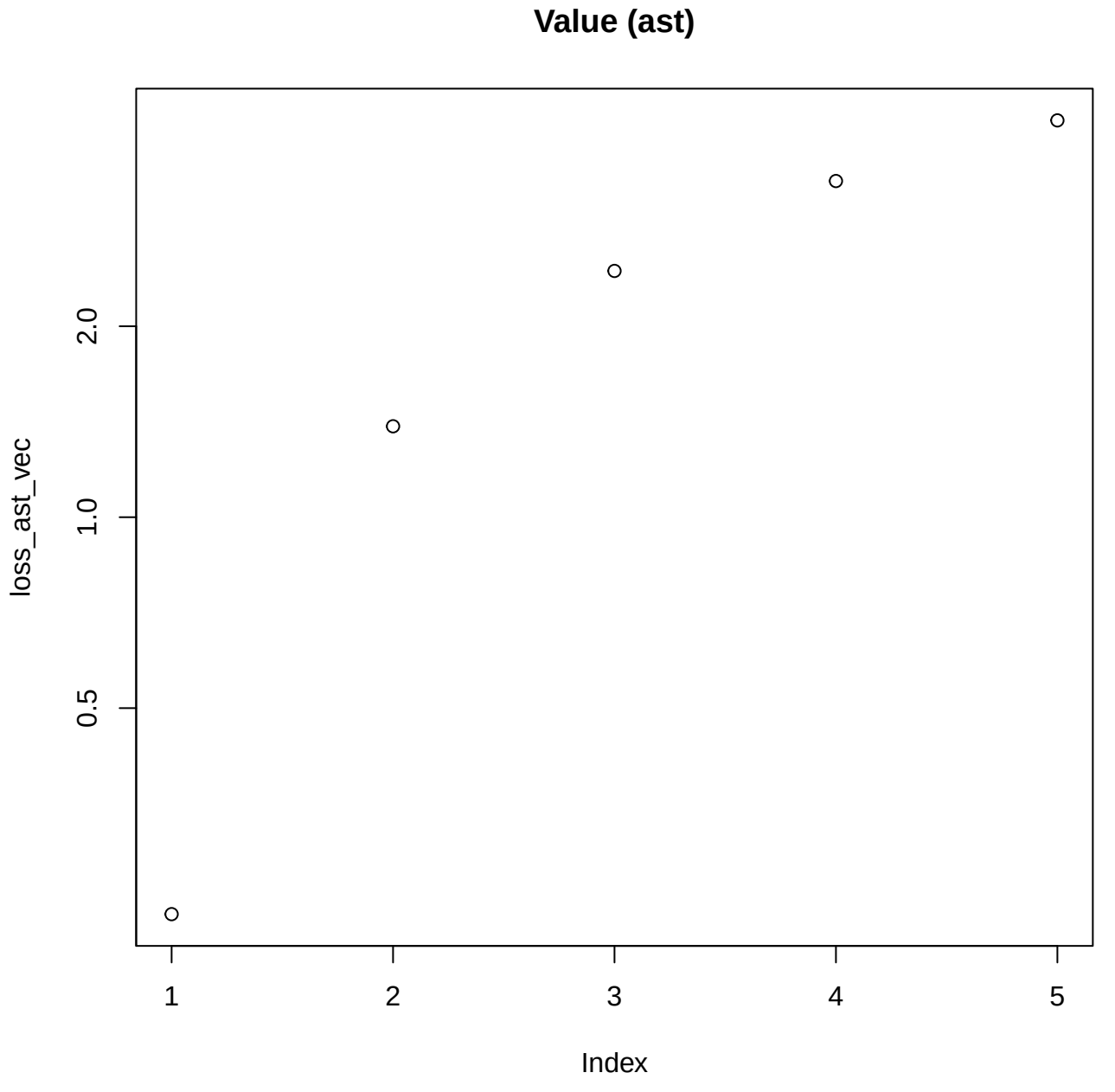

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.140
0.135
0.130
0.125
0.120
1
2
3
4
5
Index

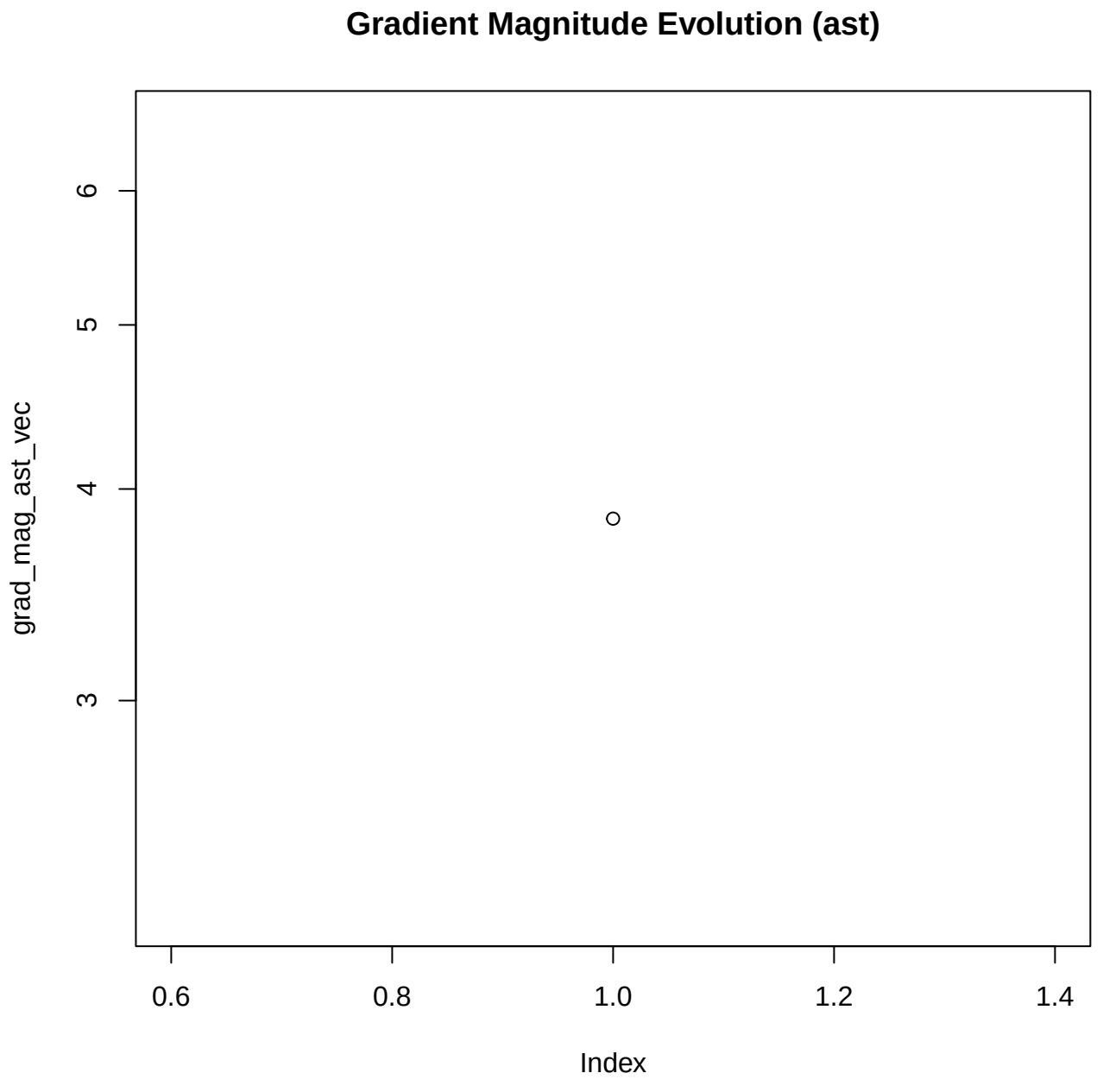

Value (ast)
loss_ast_vec
0.50
0.55
0.60
1
2
3
4
5
Index

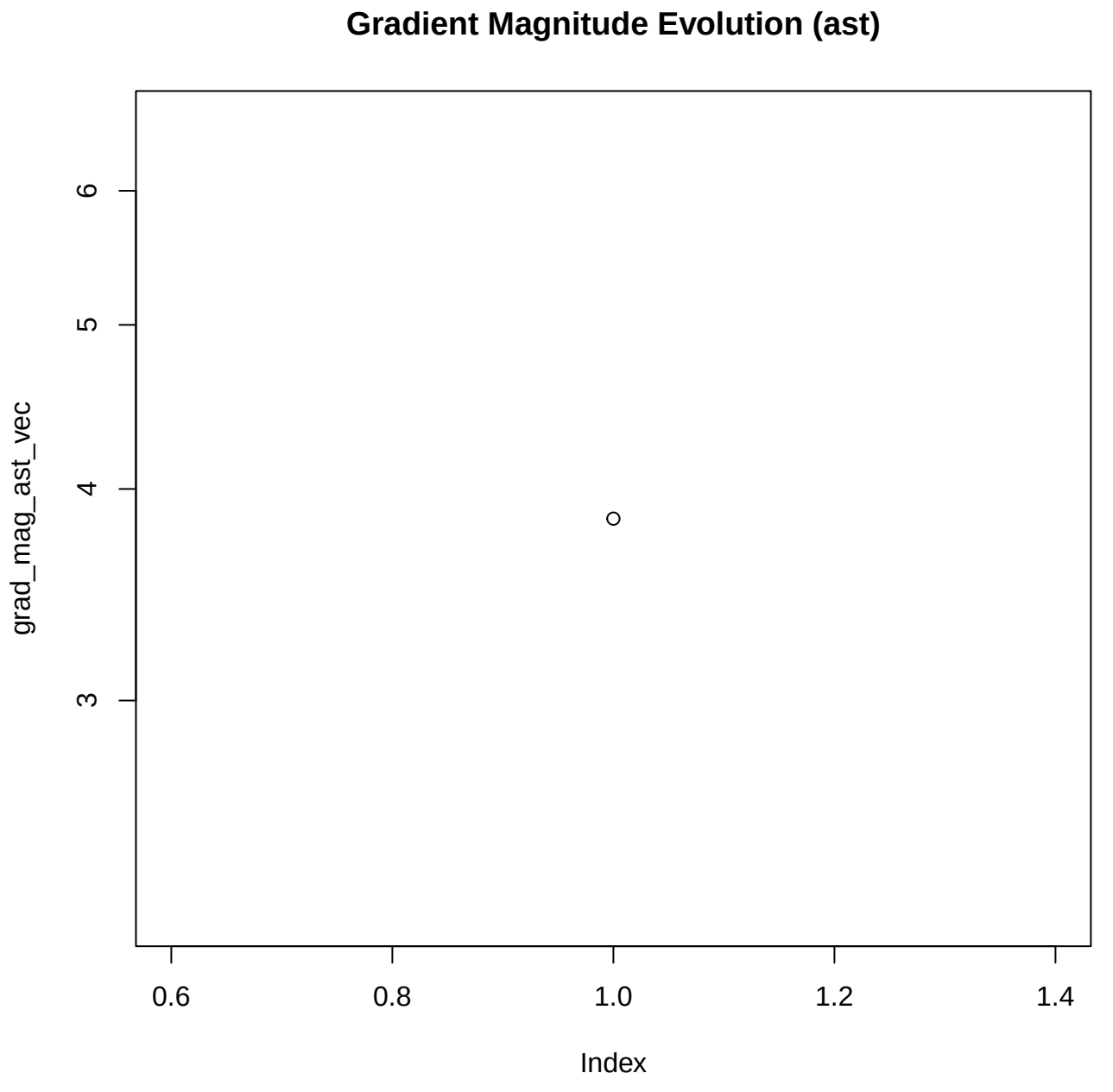

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

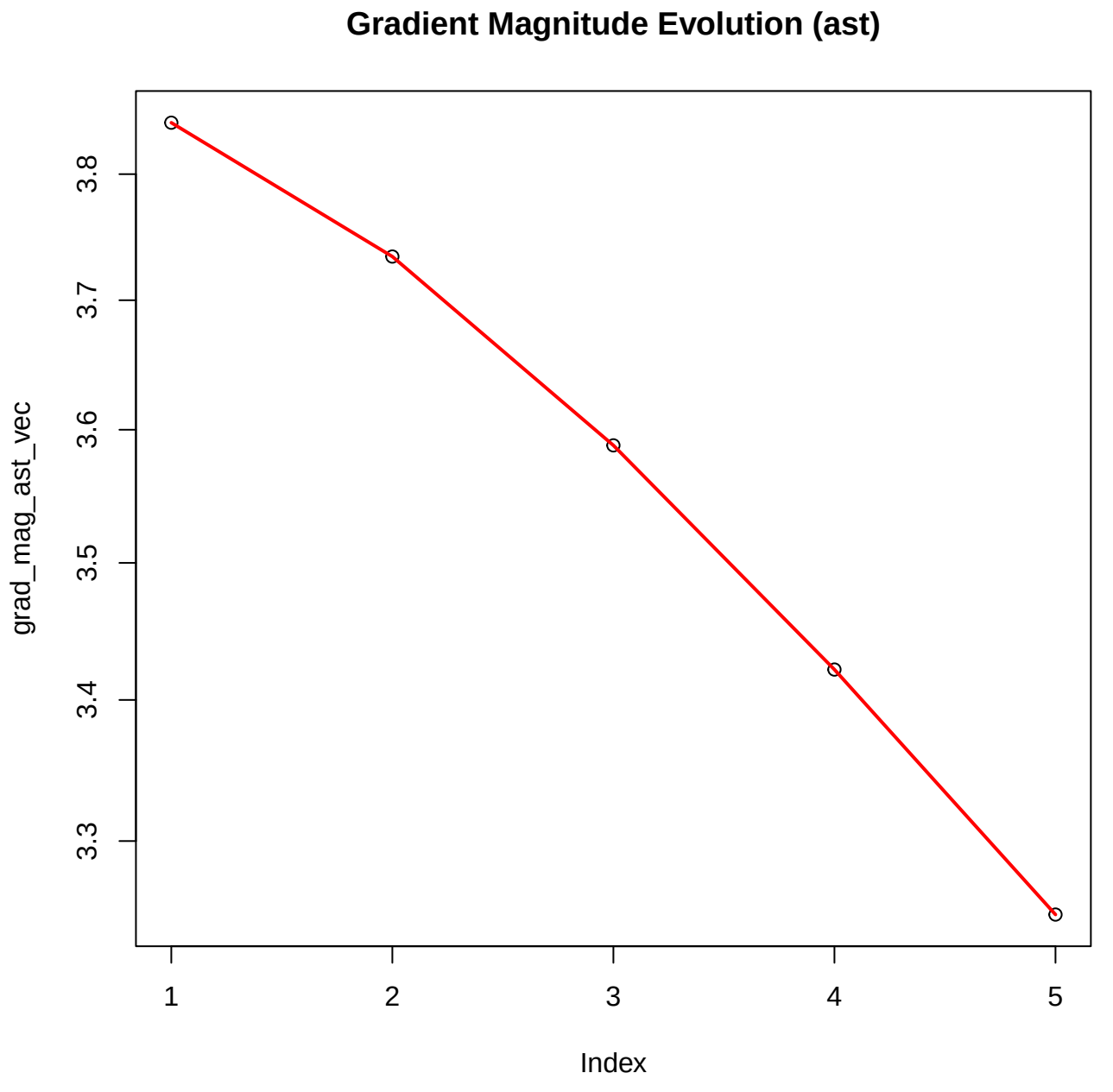

Value (ast)
loss_ast_vec
0.5
1.0
2.0
1
2
3
4
5
Index

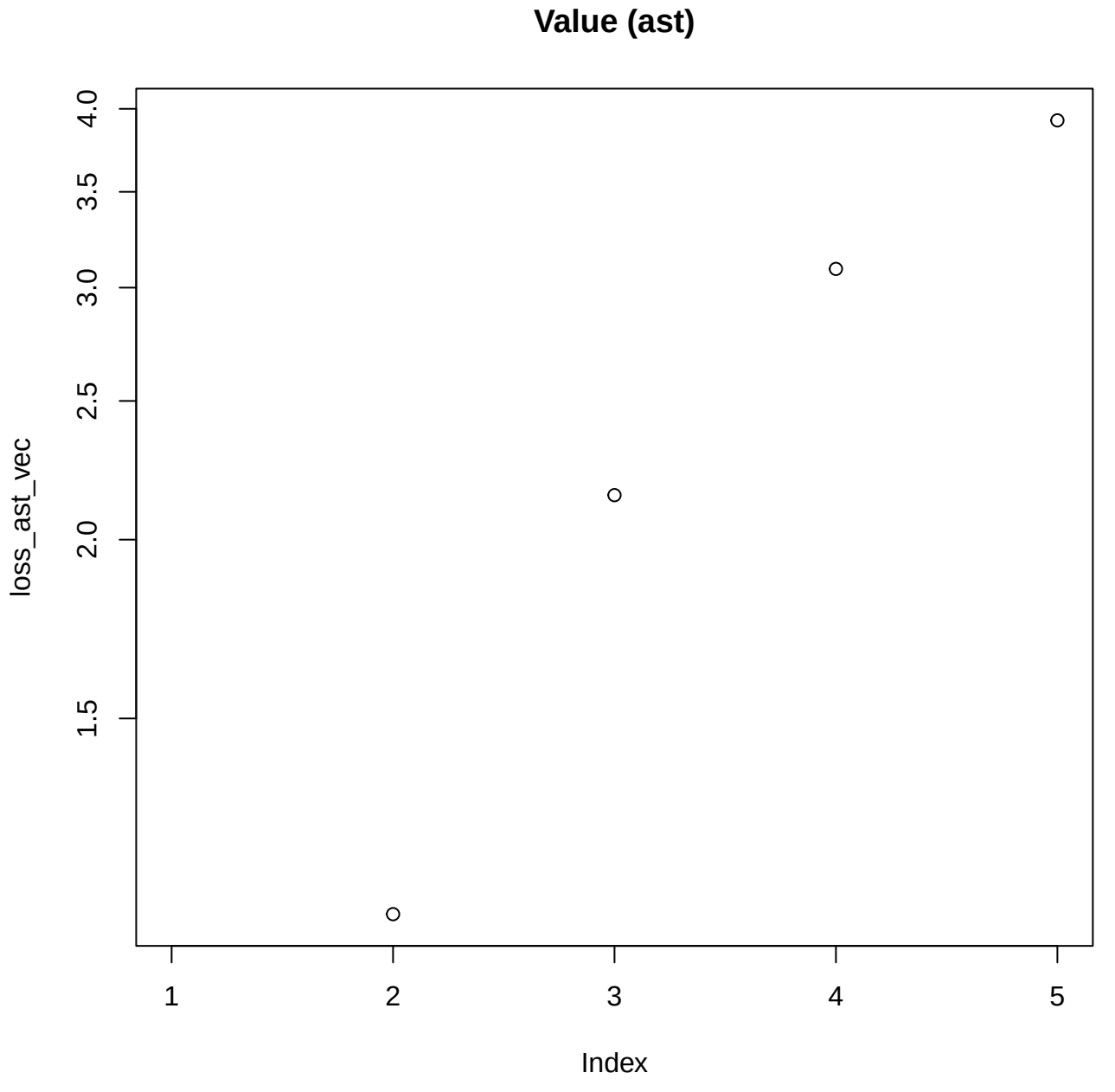

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

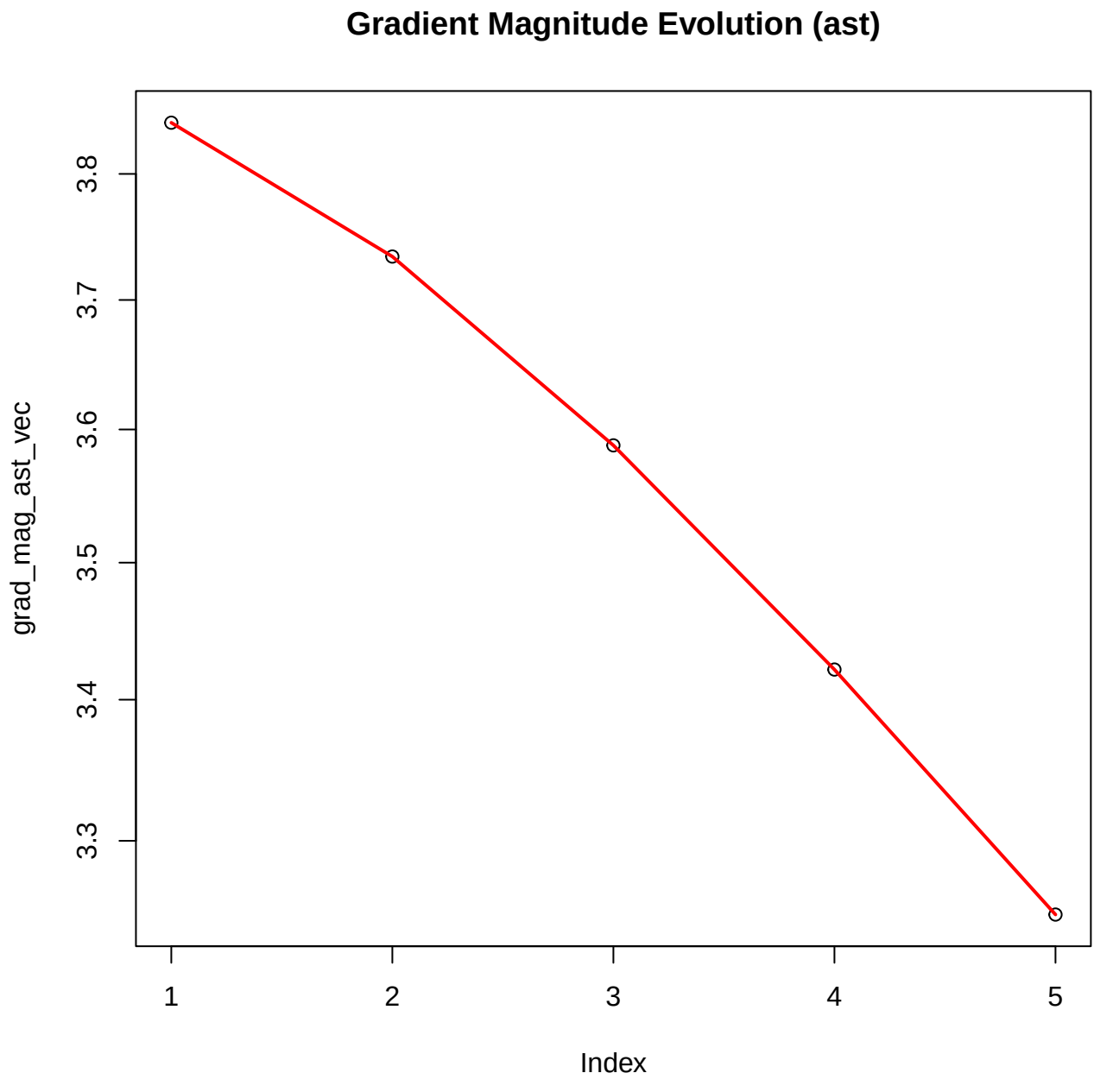

Value (ast)
loss_ast_vec
0.5
1.0
2.0
1
2
3
4
5
Index

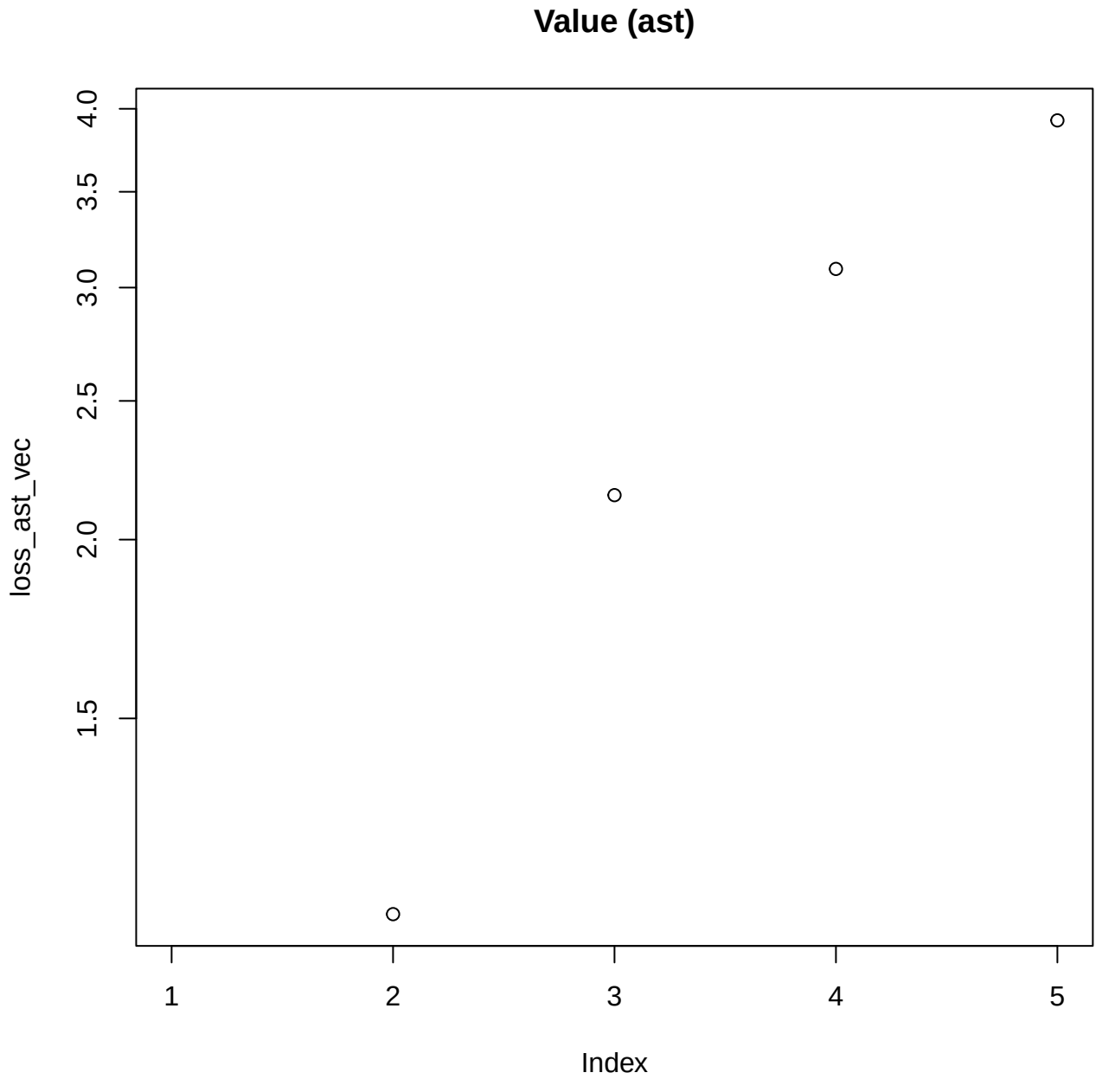

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

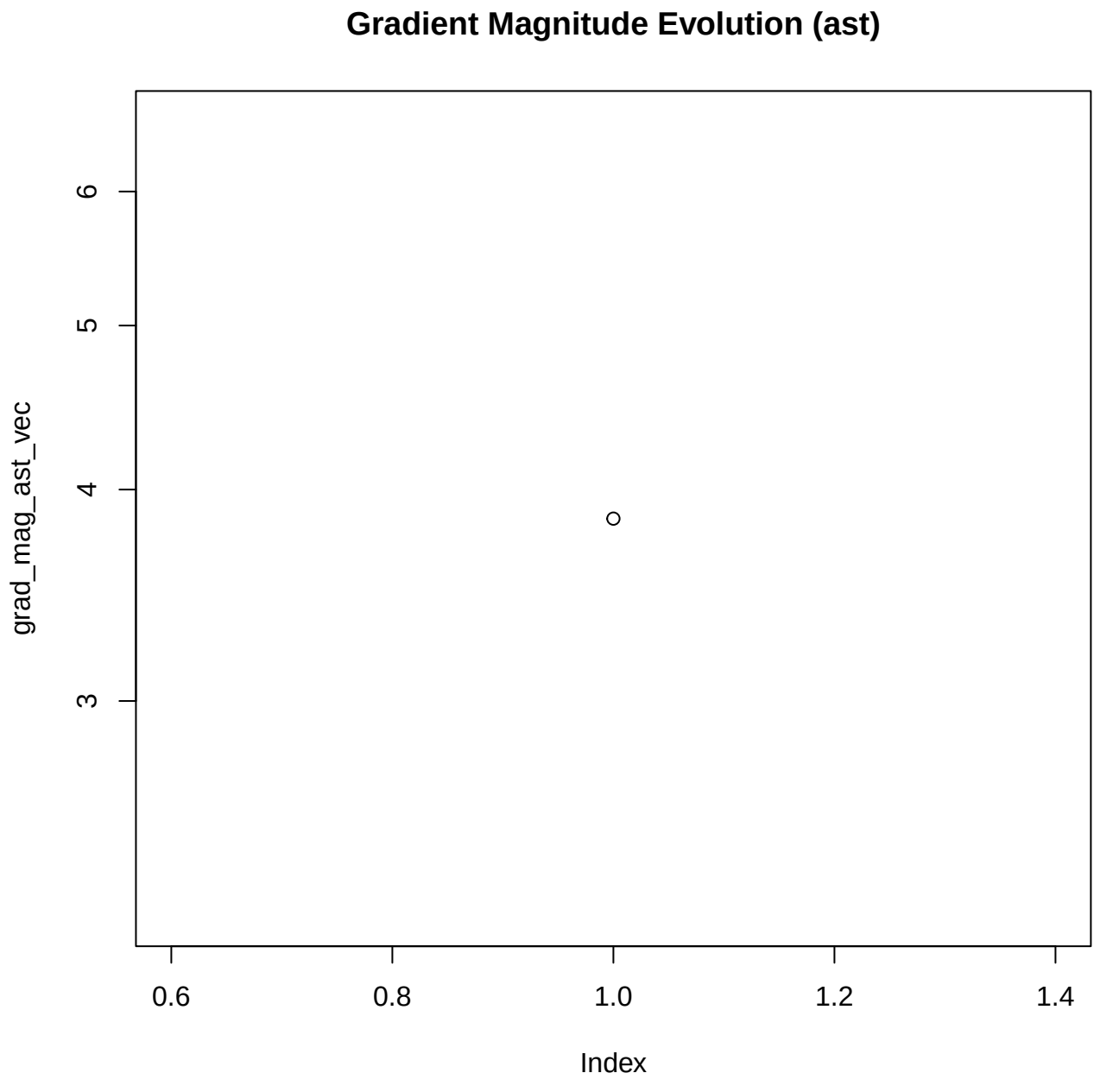

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

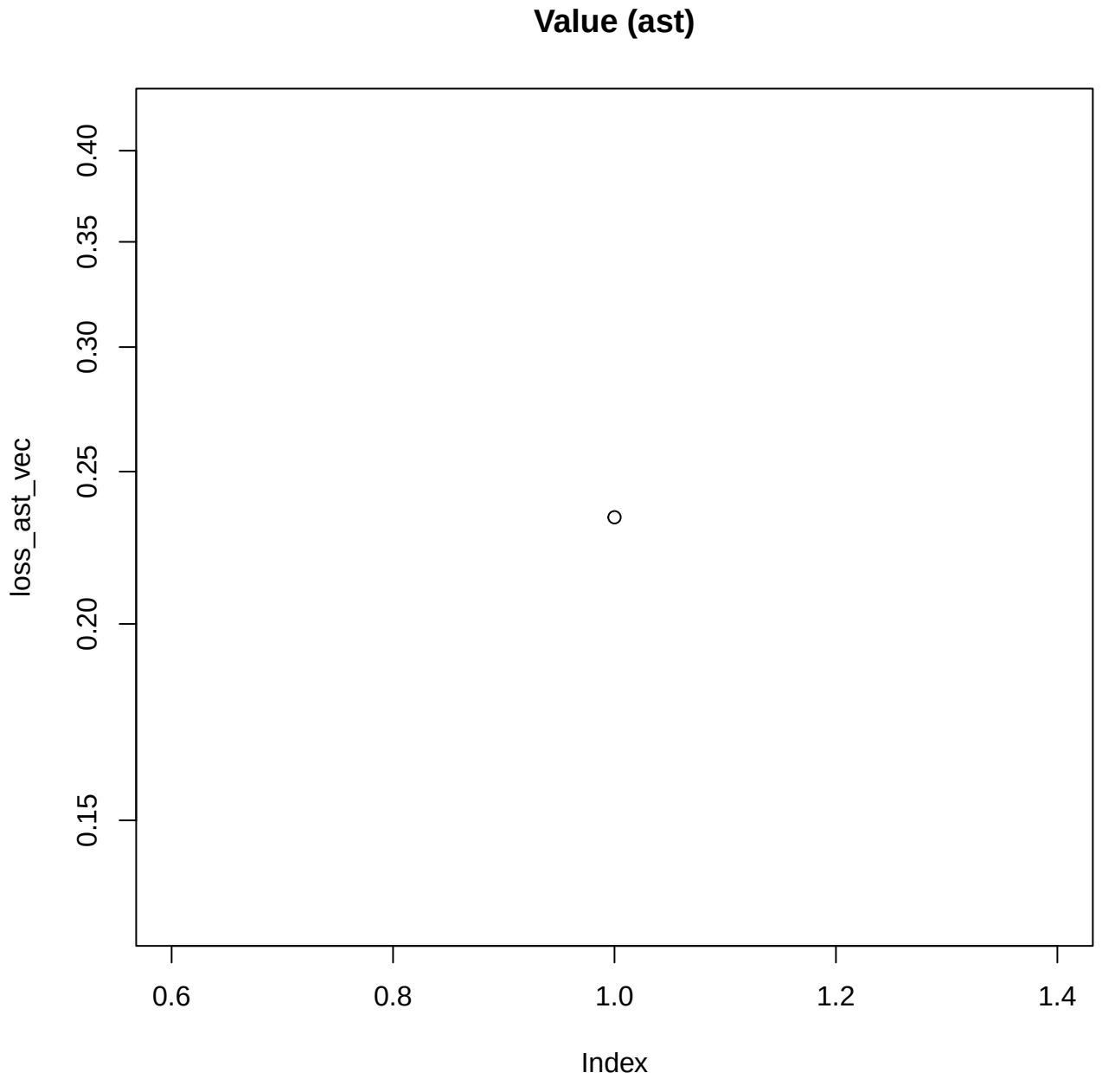

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

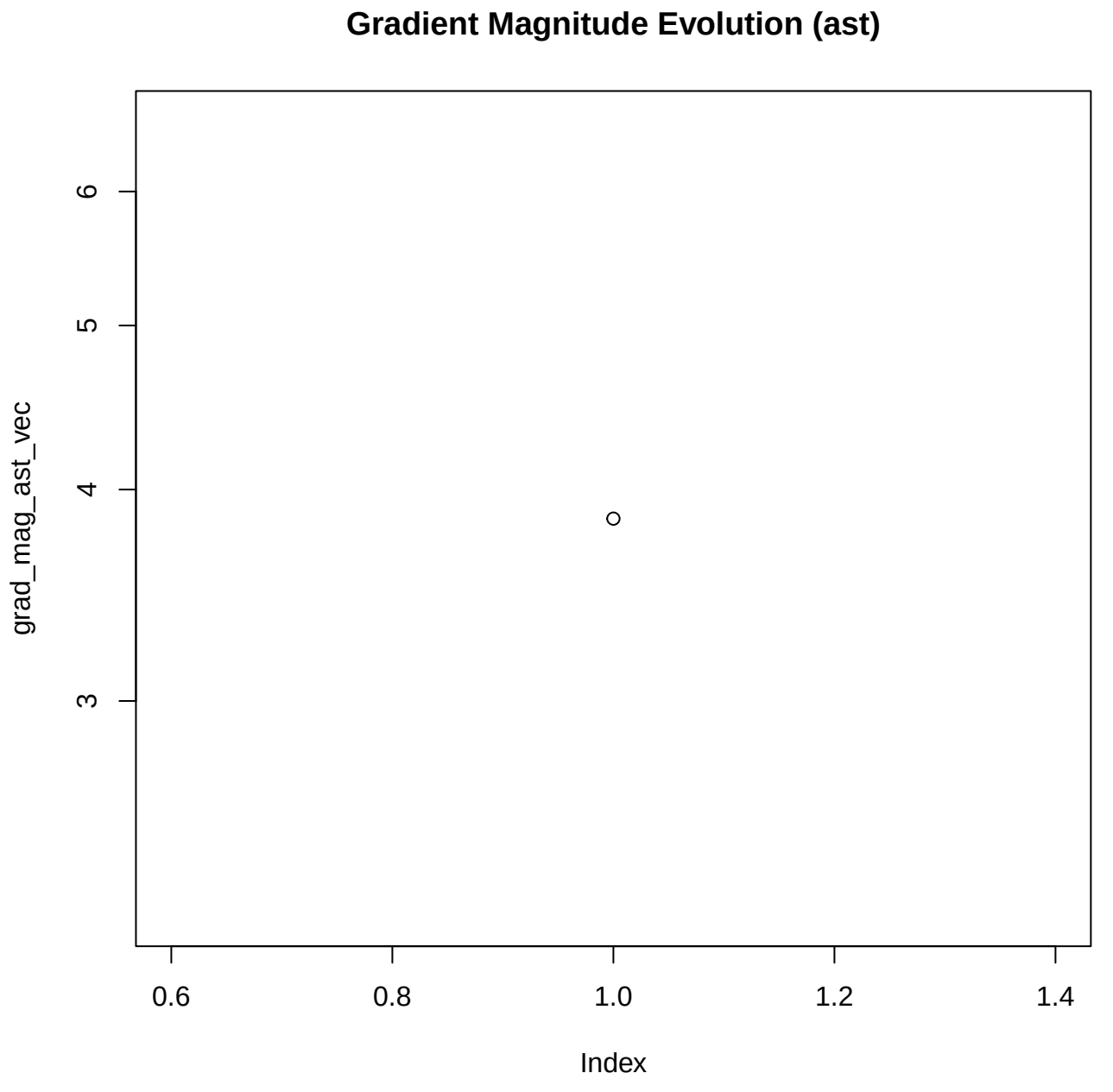

Value (ast)
loss_ast_vec
4.0
3.5
3.0
2.5
2.0
1.5
1
2
3
4
5
Index

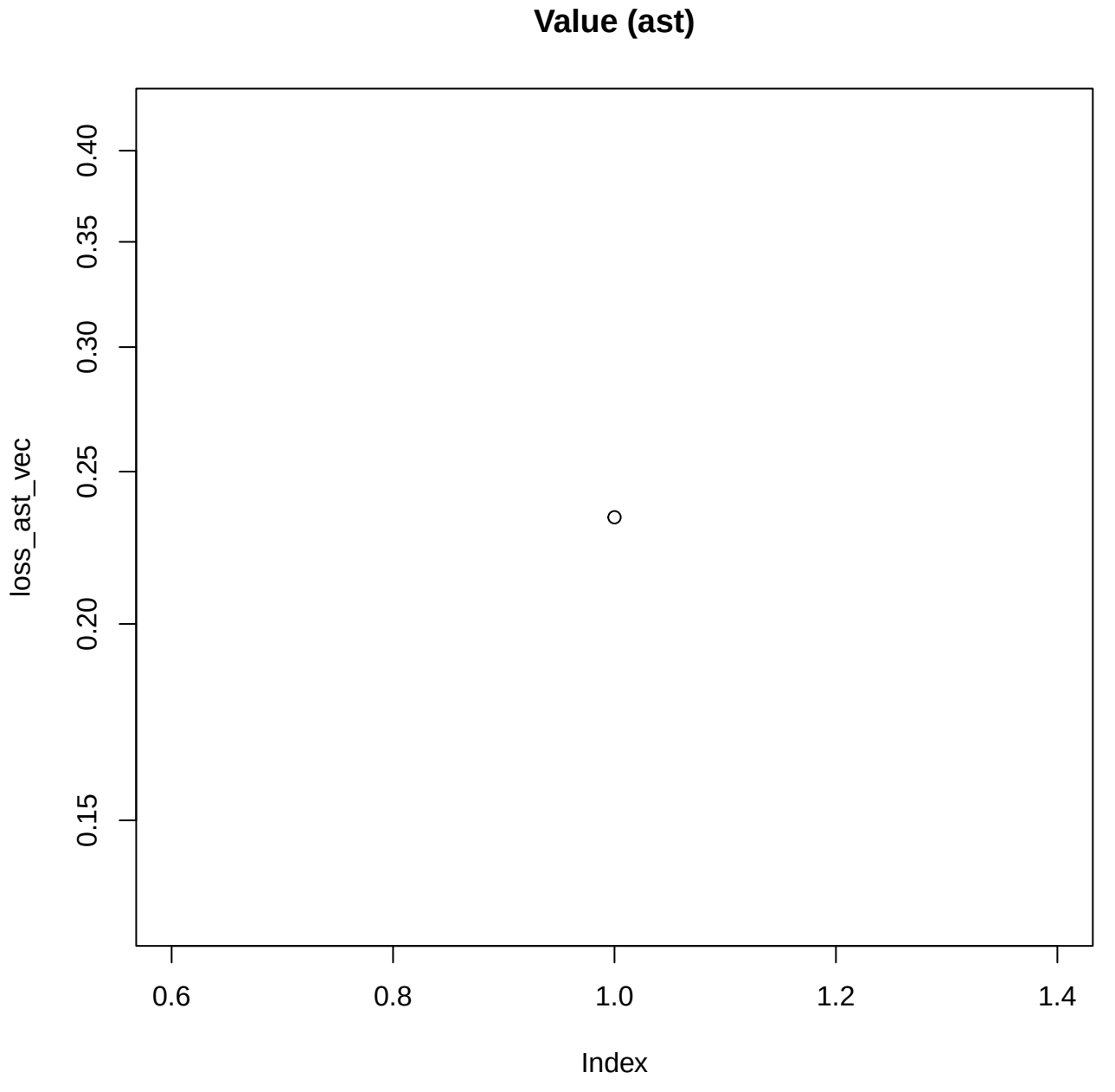

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

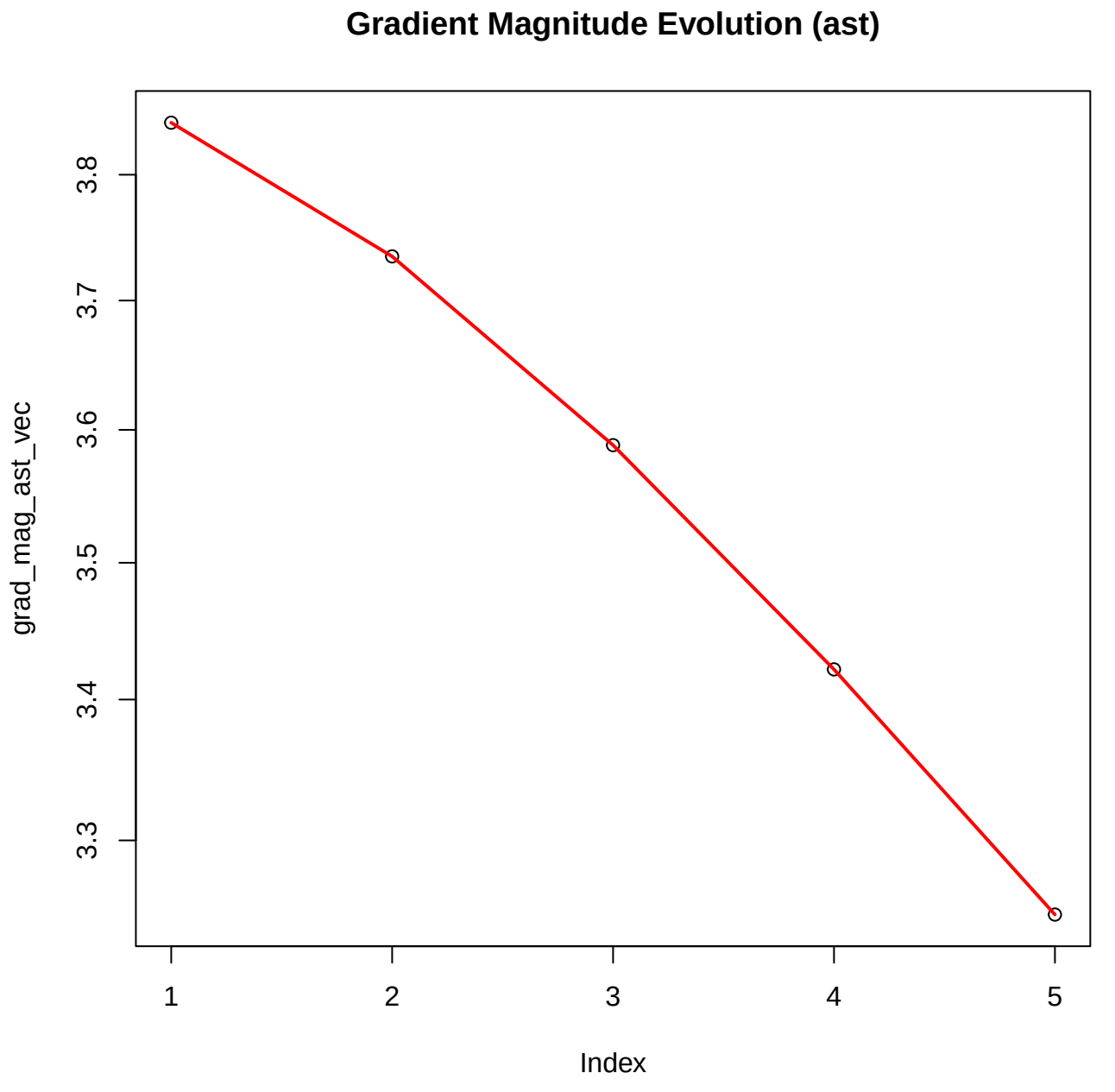

Value (ast)
loss_ast_vec
4.0
3.5
3.0
2.5
2.0
1.5
1
2
3
4
5
Index

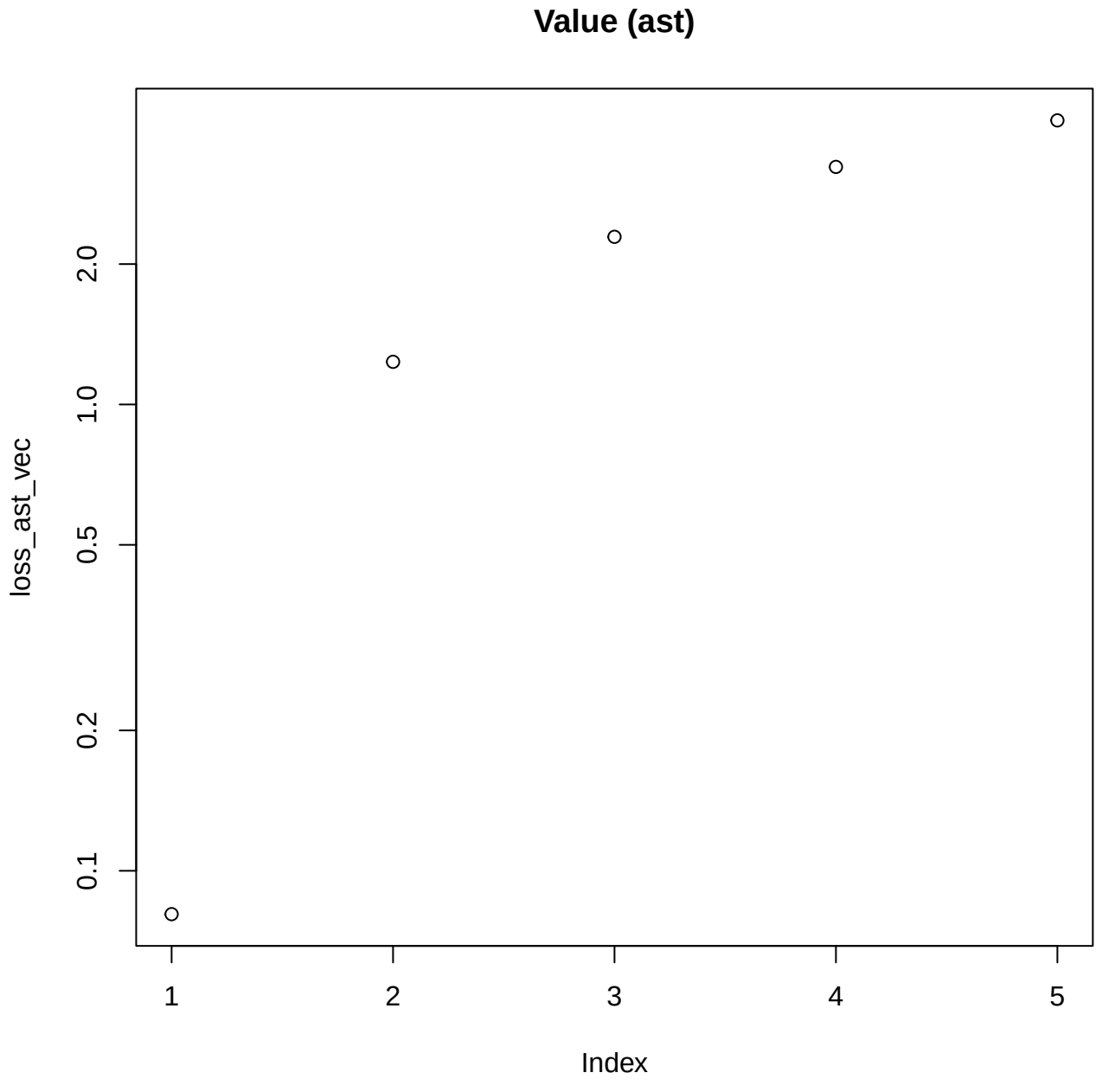

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

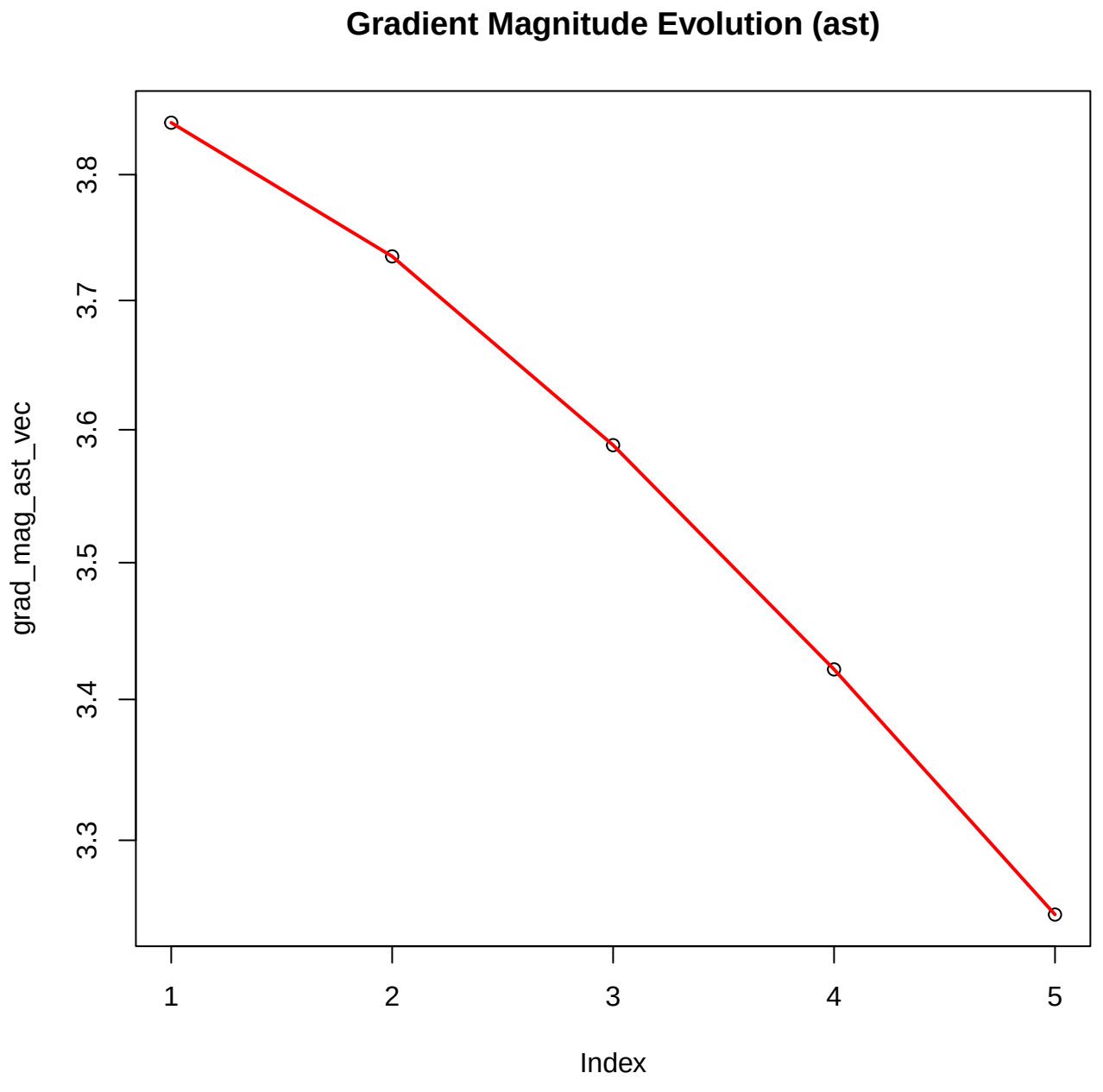

Value (ast)
loss_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.6
0.8
1.0
1.2
1.4
Index

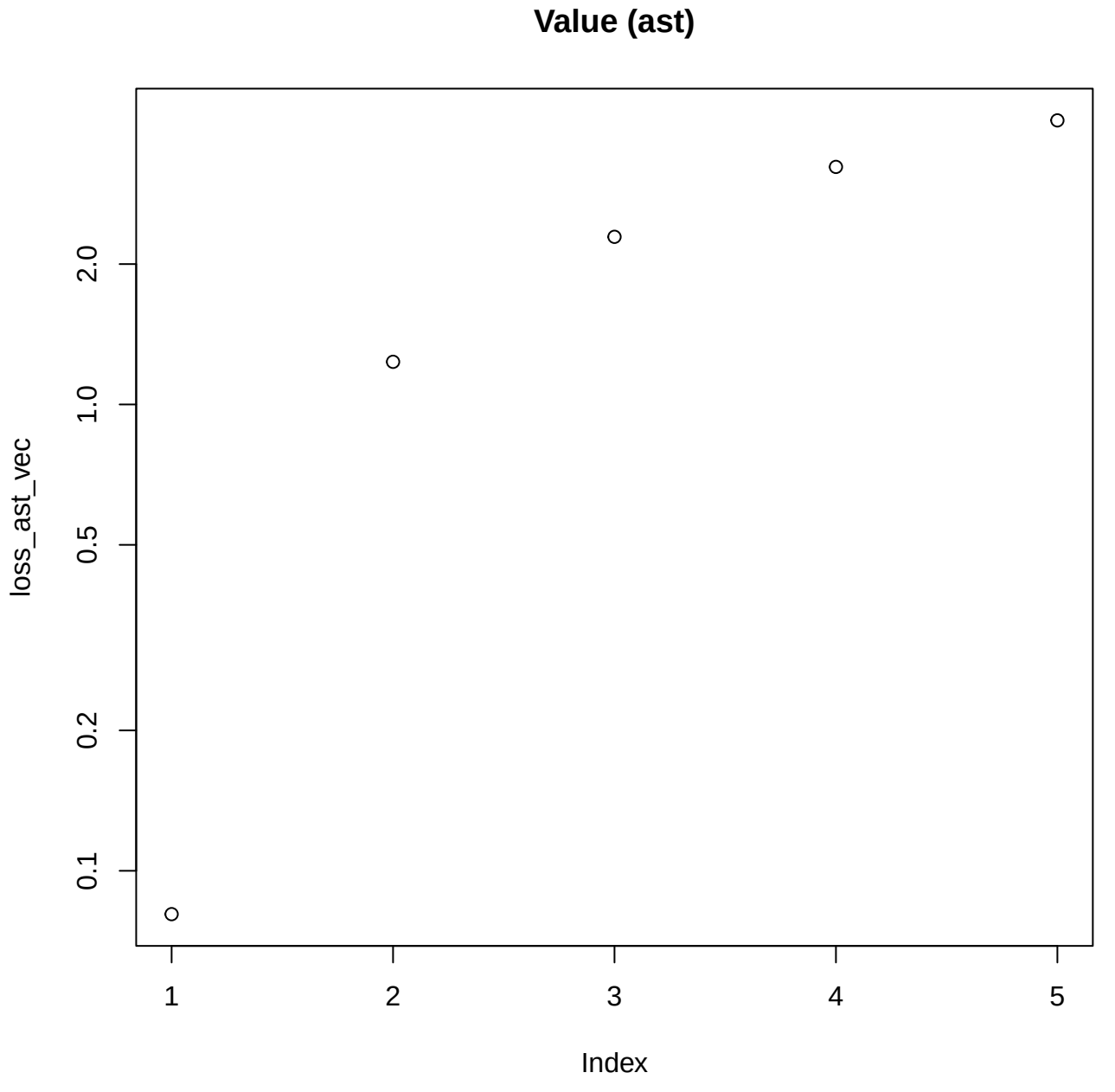

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3
4
5
6
0.6
0.8
1.0
1.2
1.4
Index

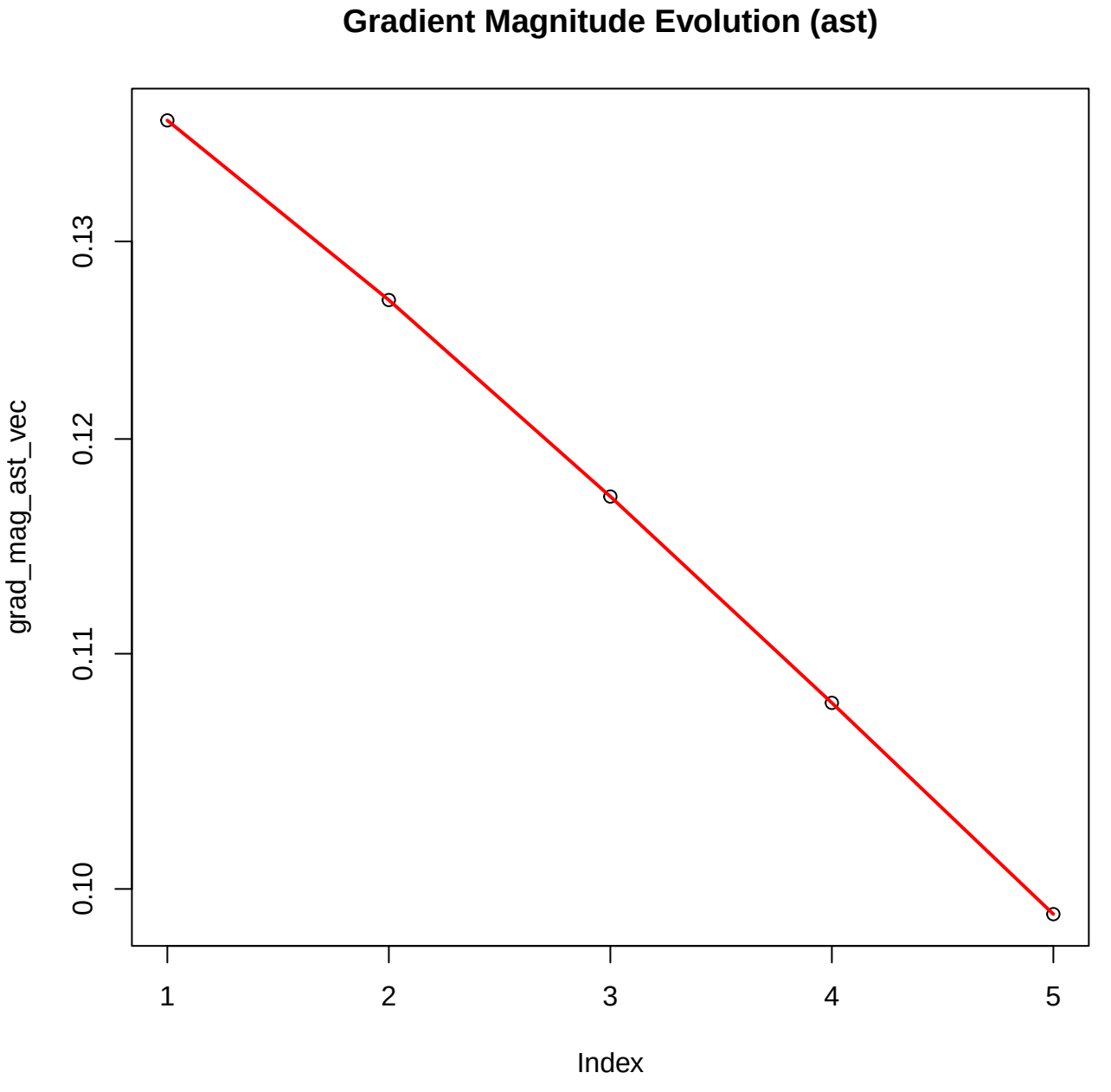

Value (ast)
loss_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
0.6
0.8
1.0
1.2
1.4
Index

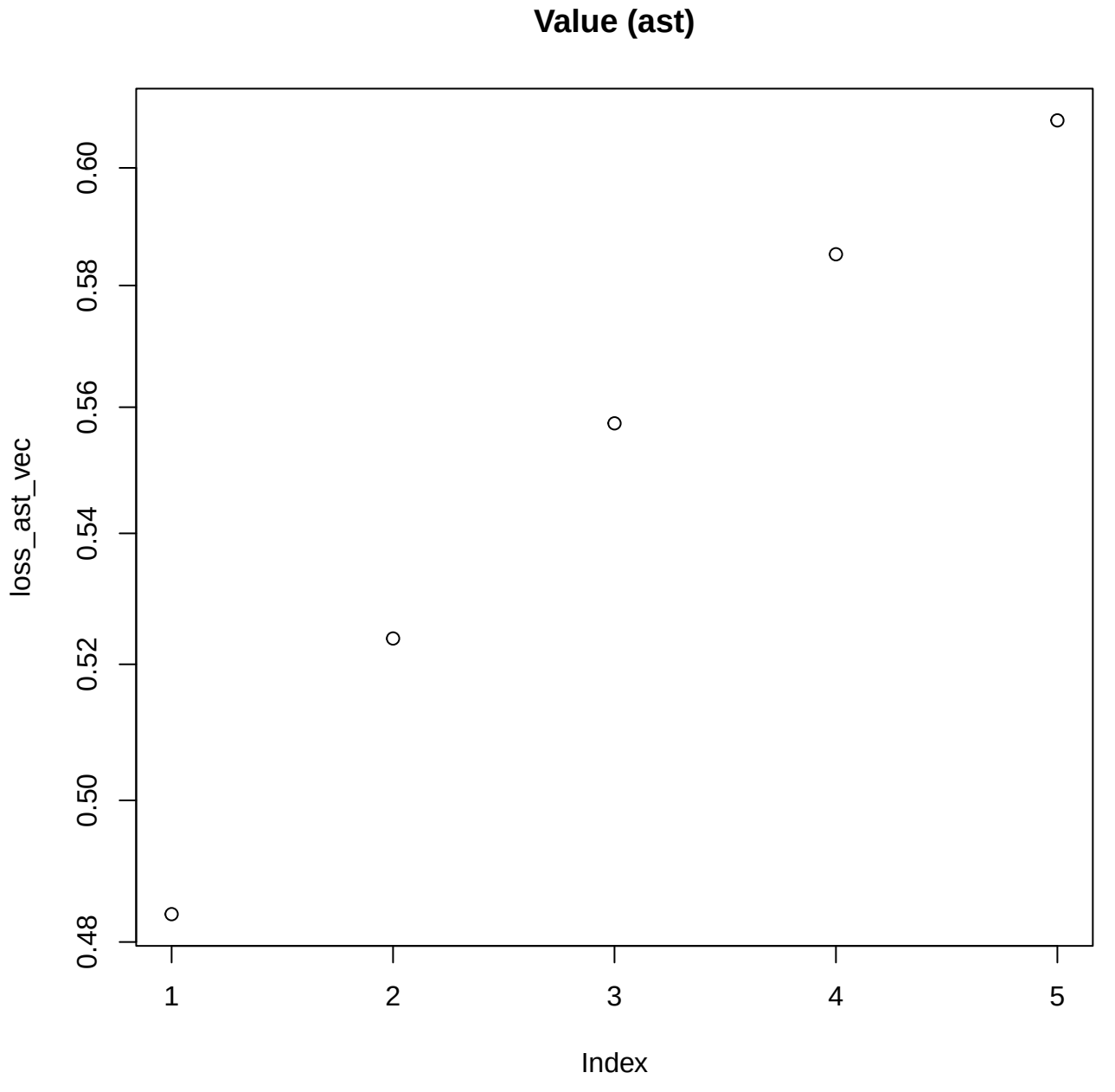

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

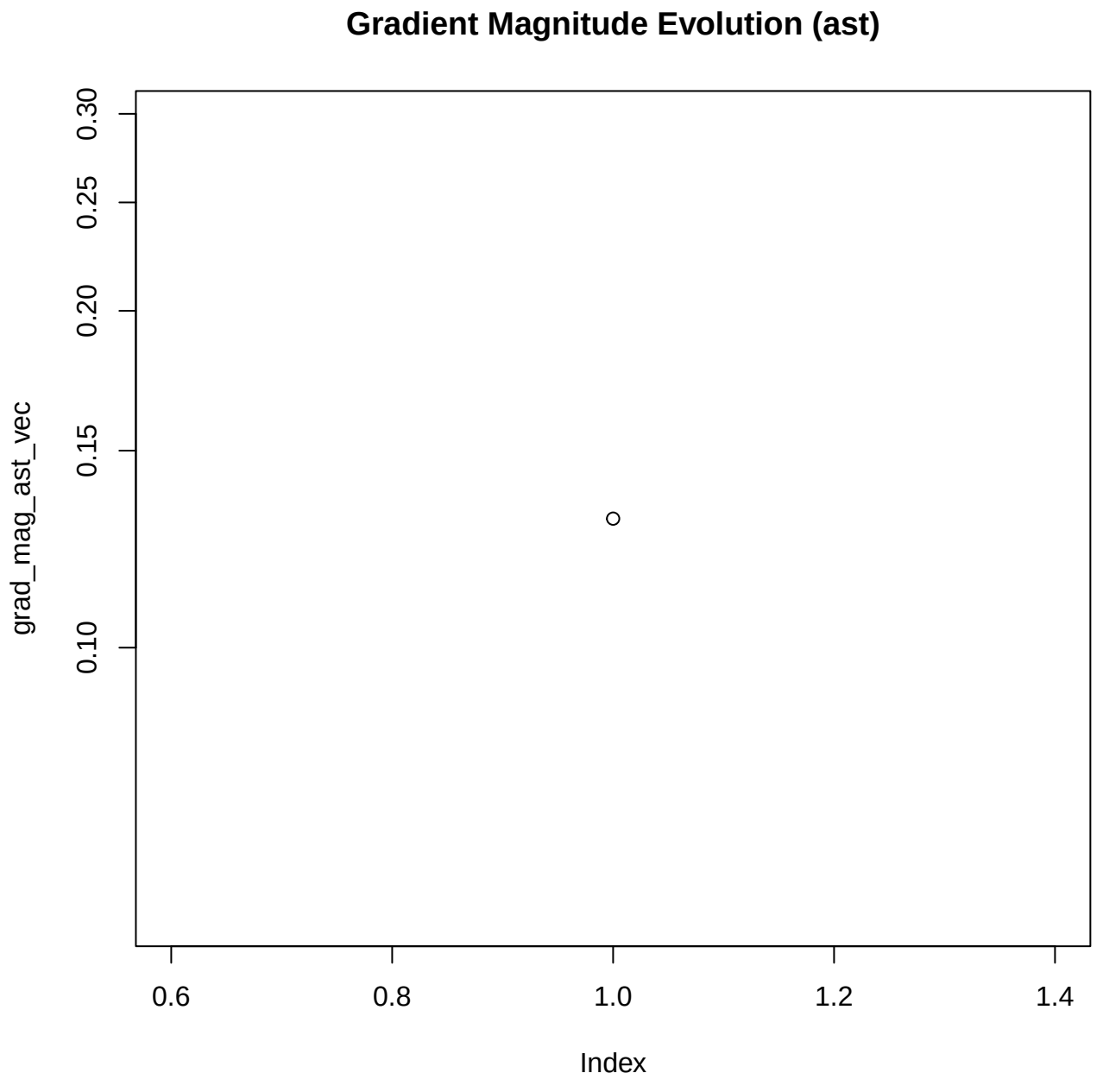

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

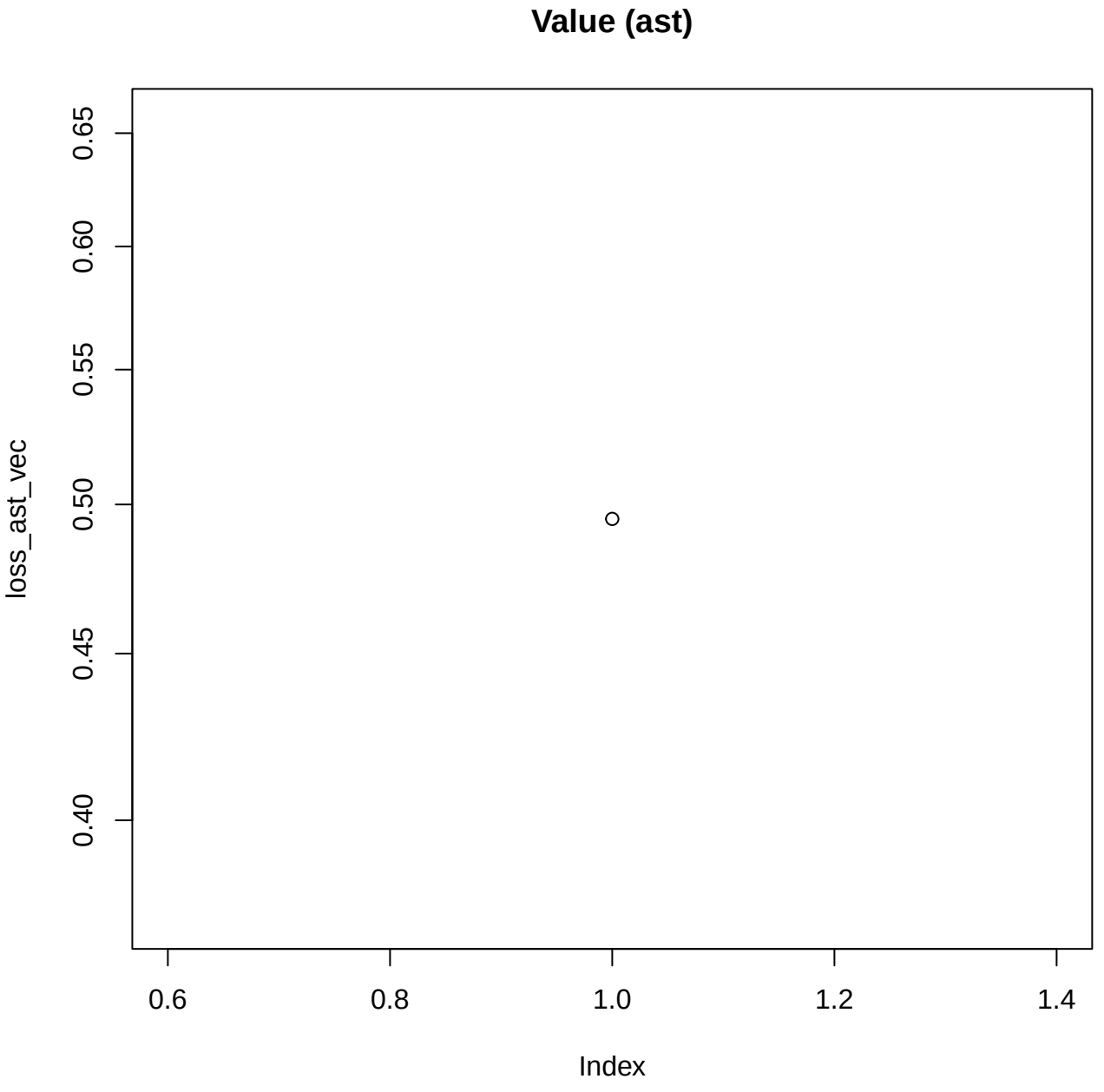

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

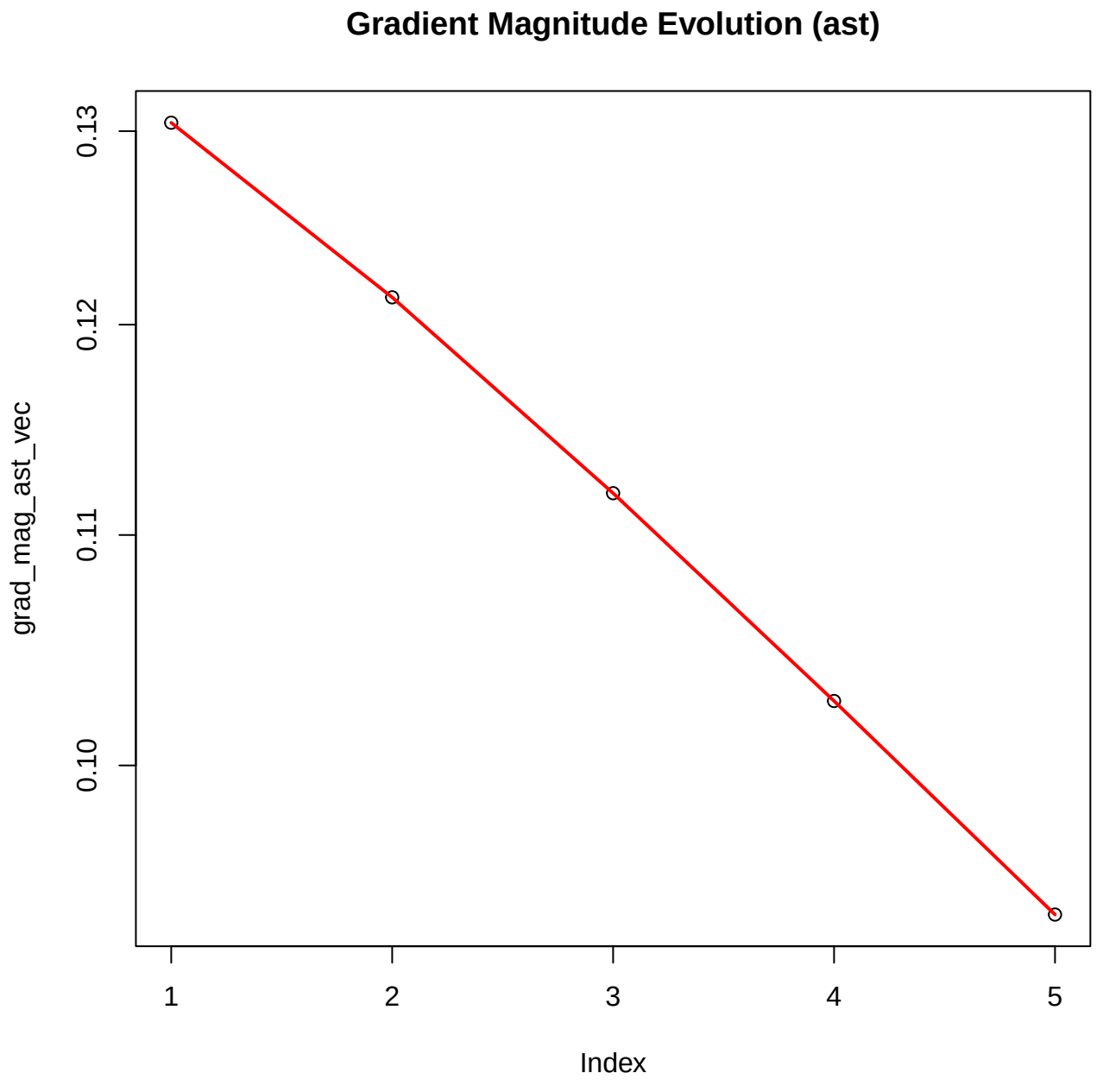

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

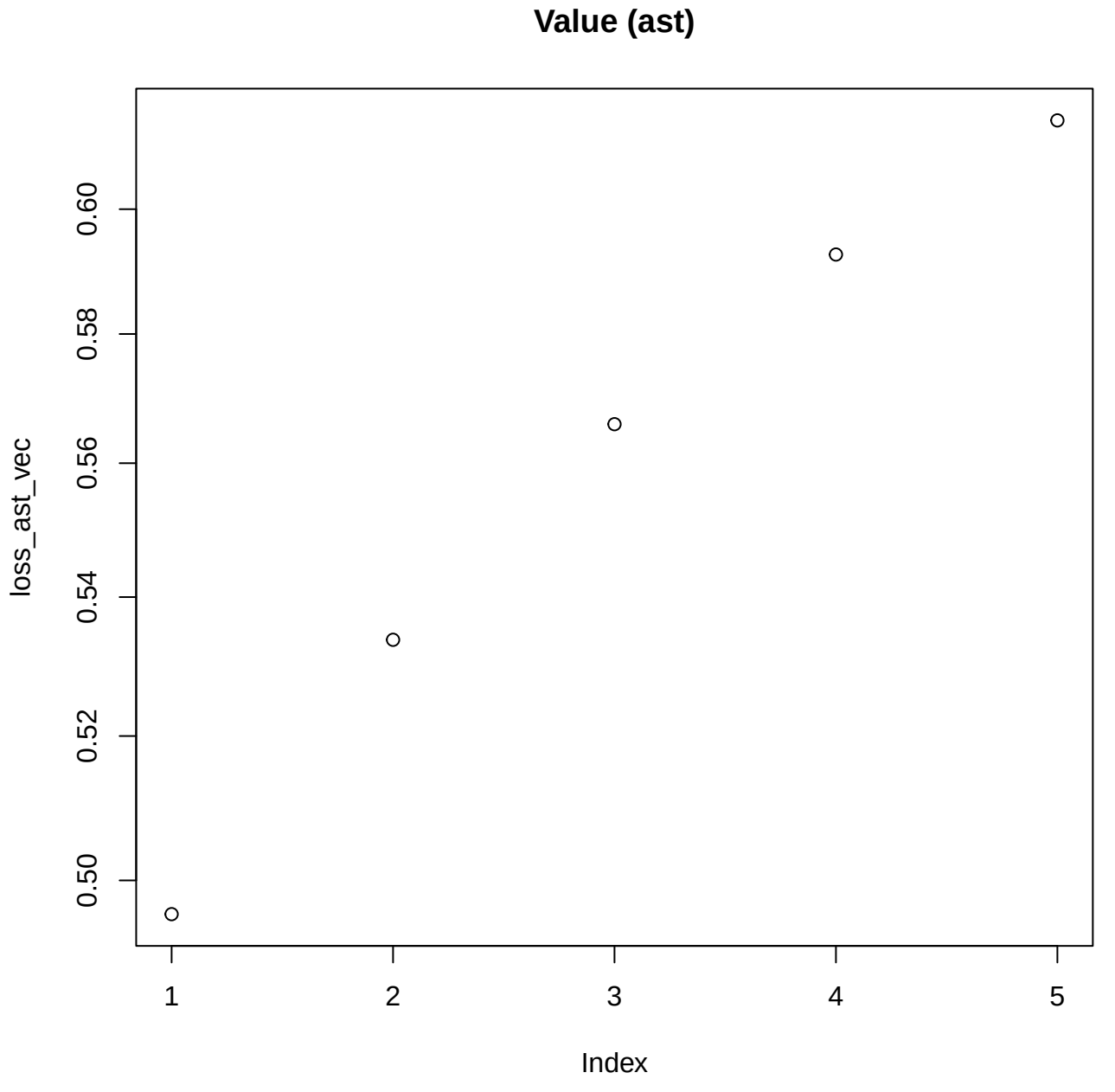

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

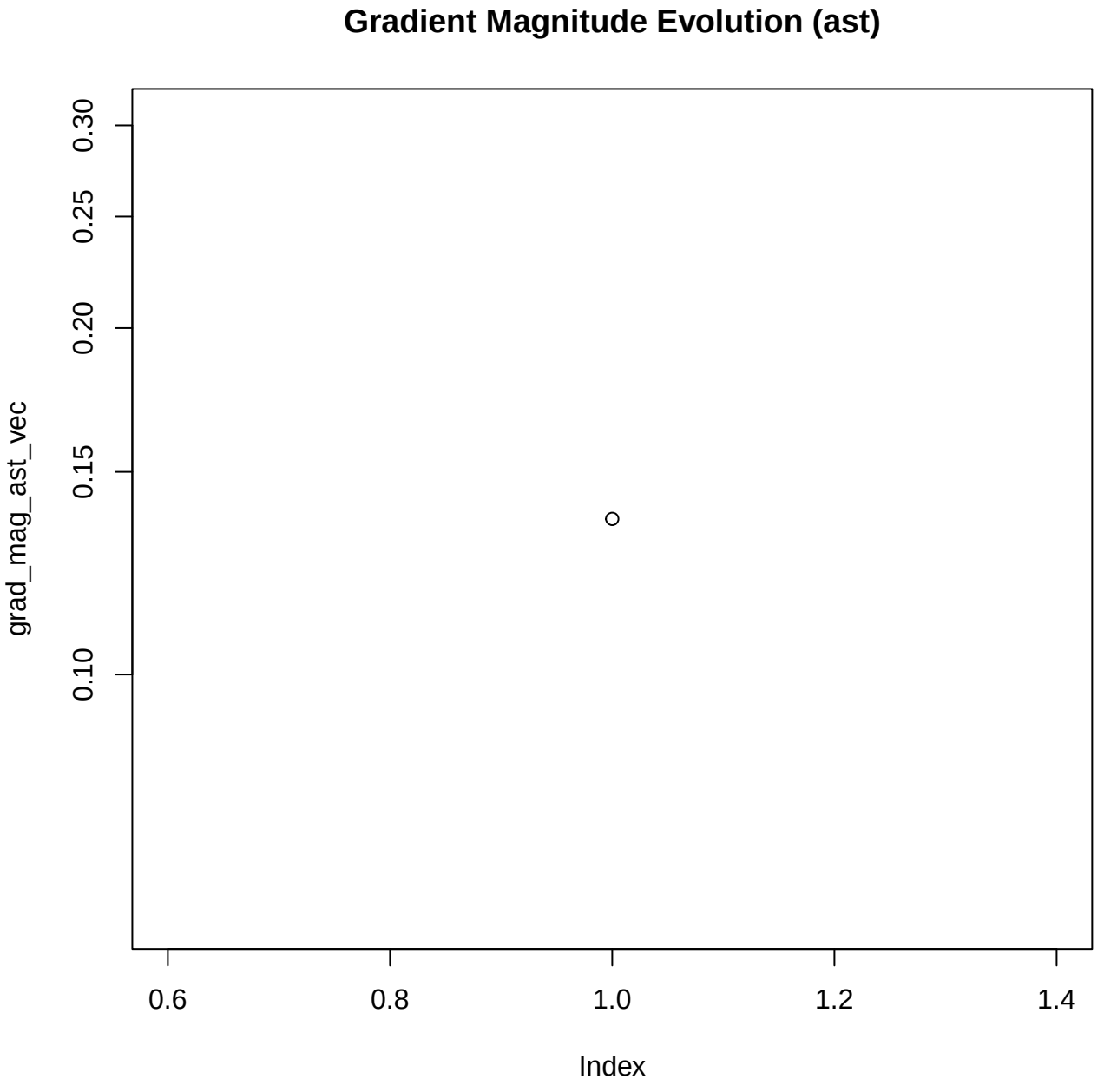

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

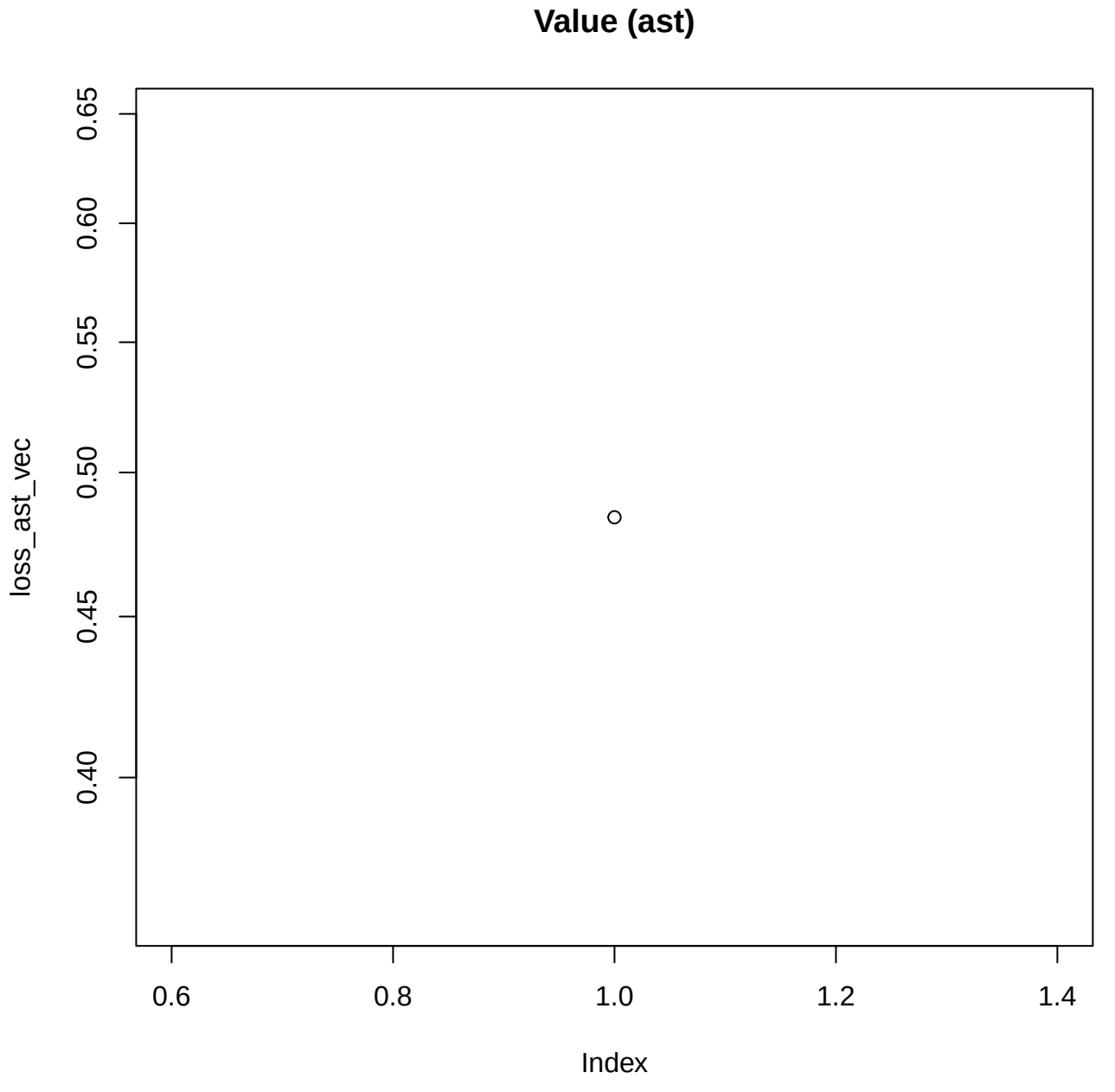

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

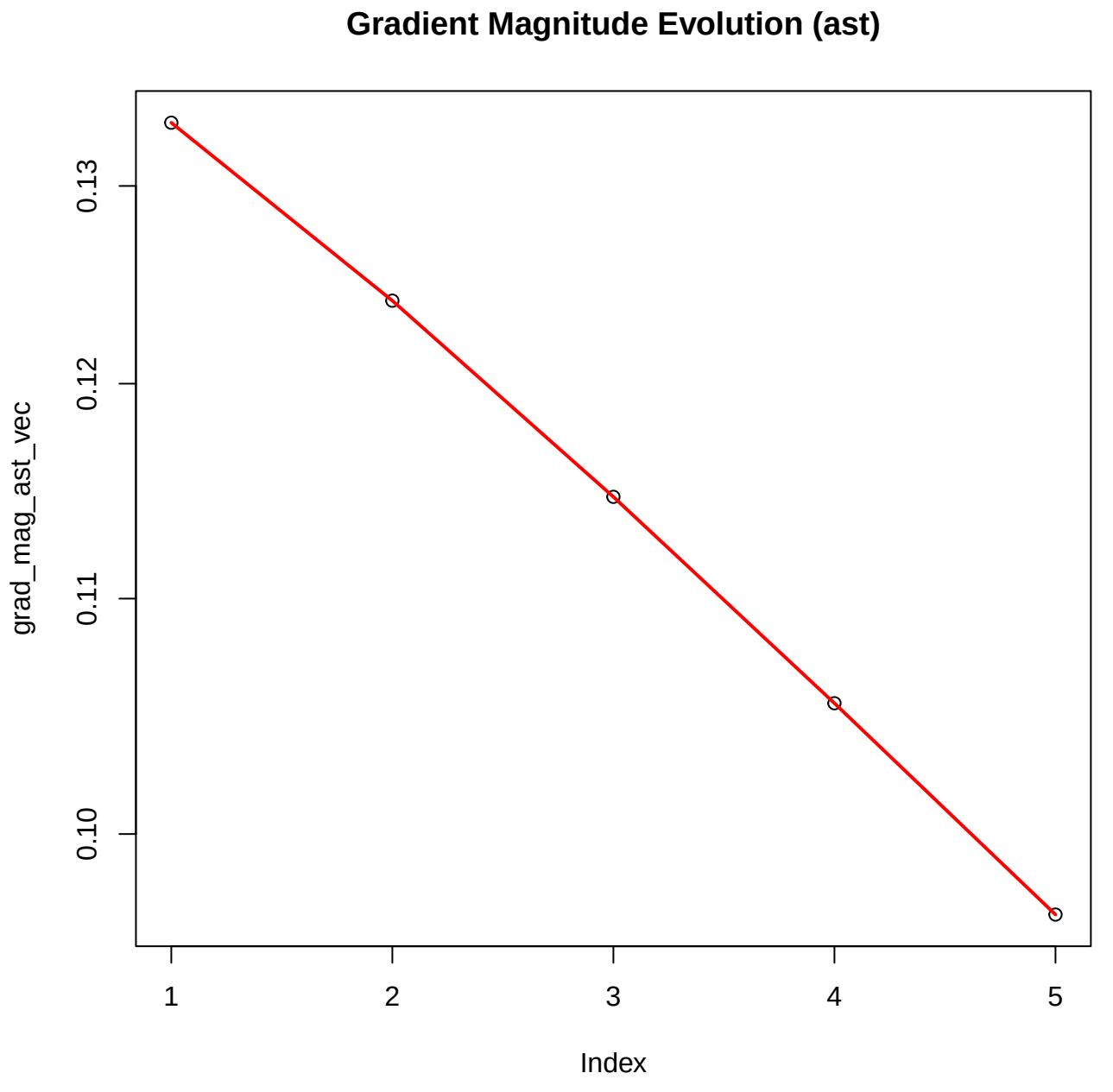

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

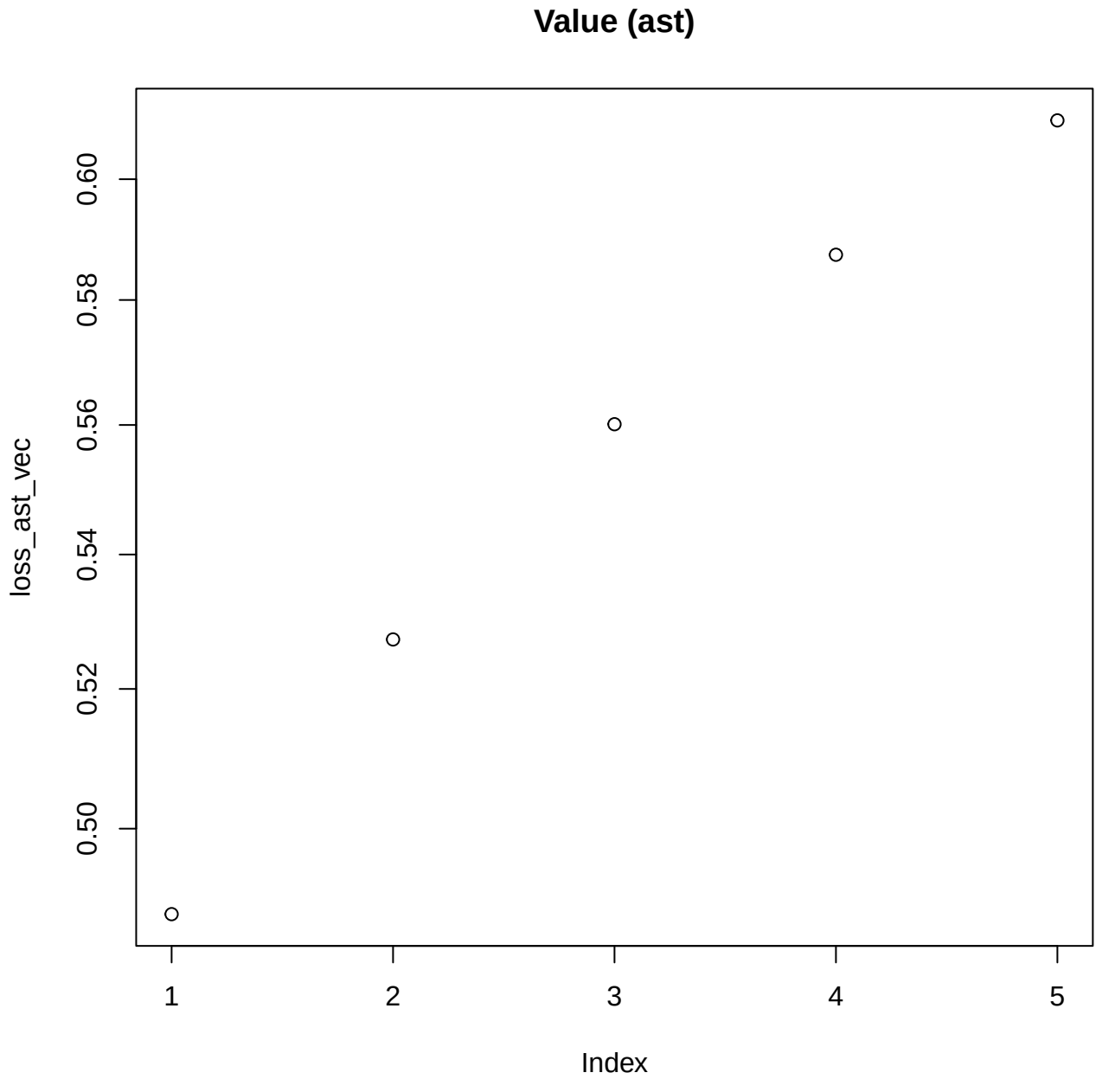

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

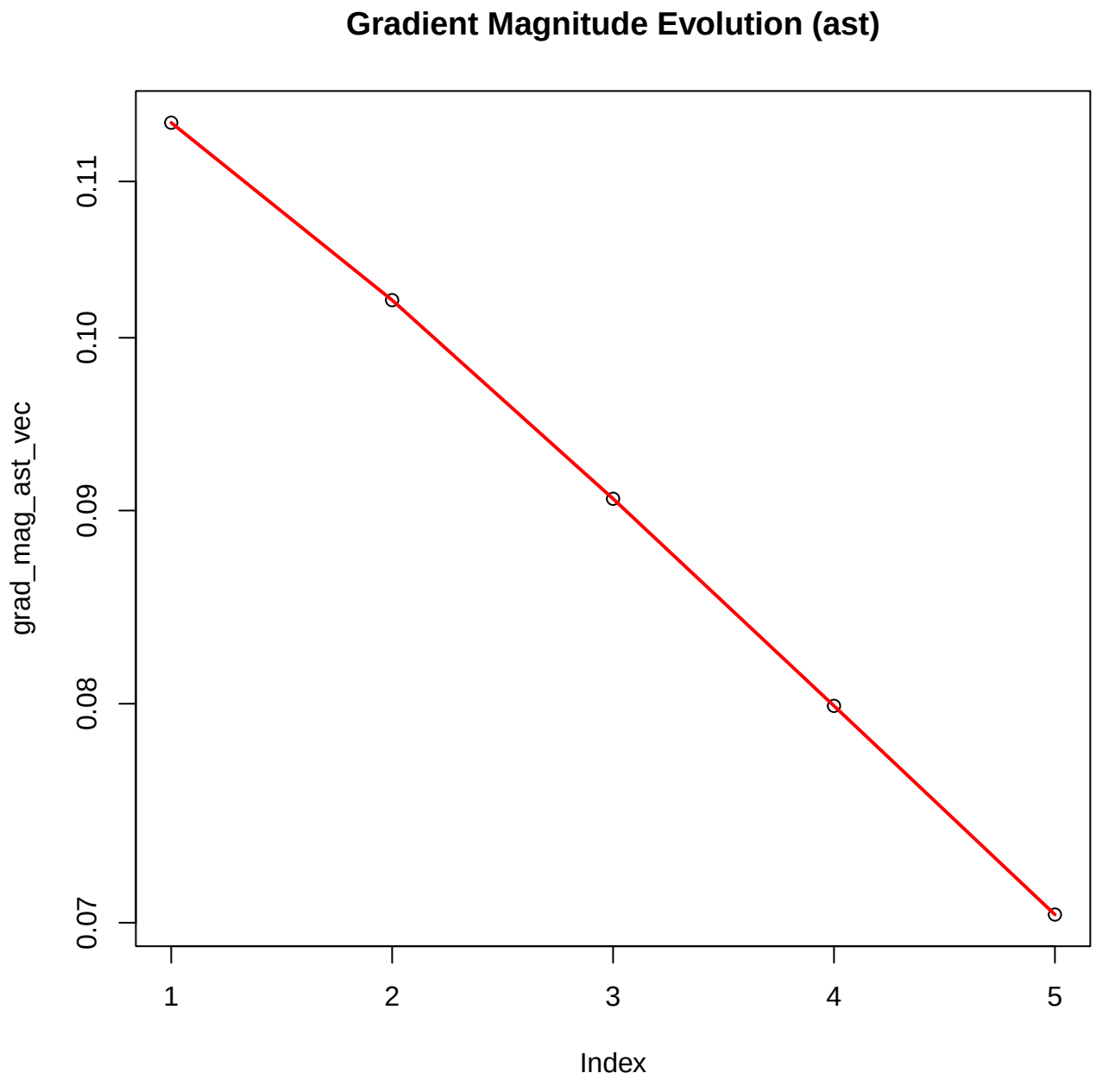

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

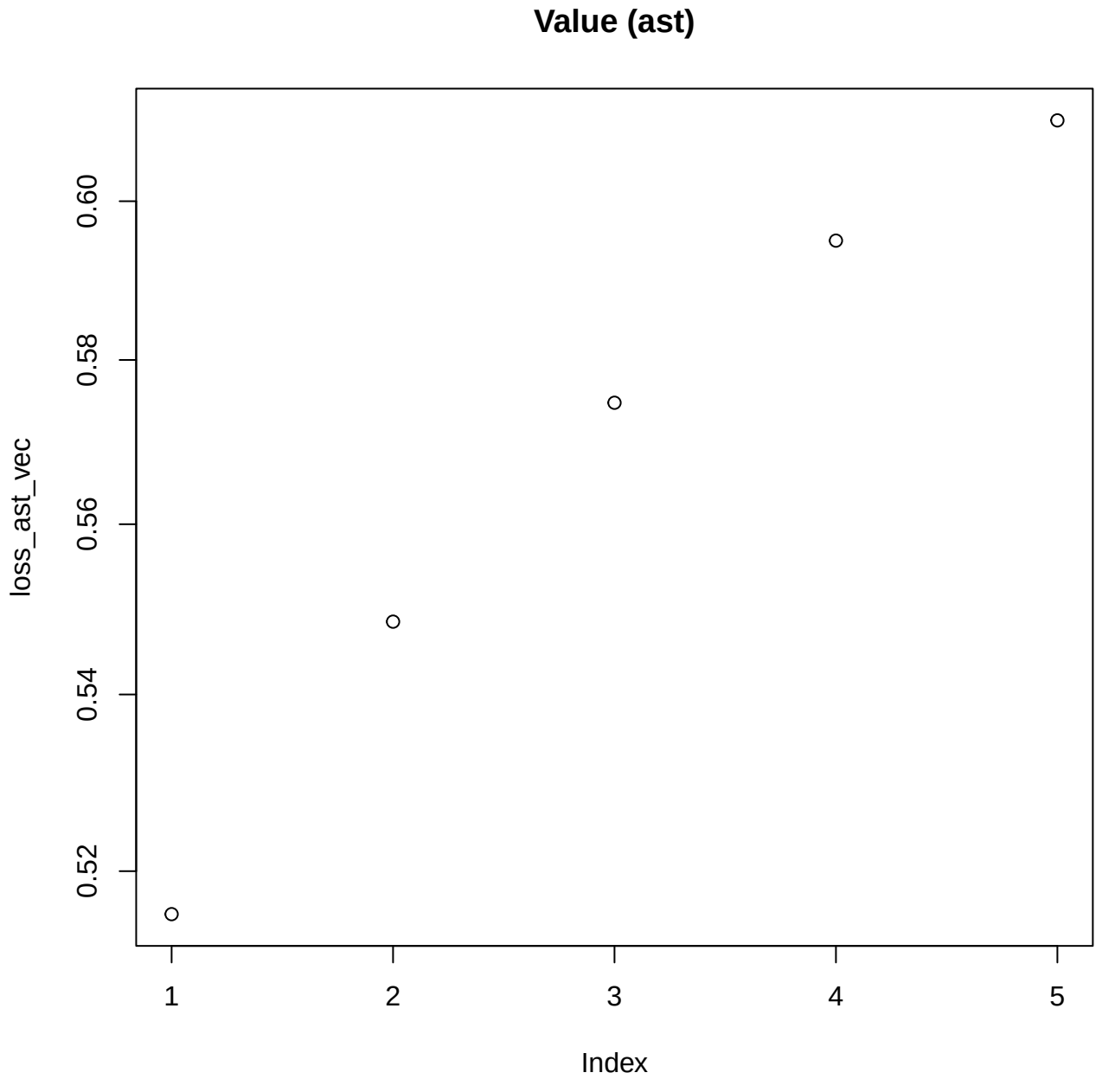

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index

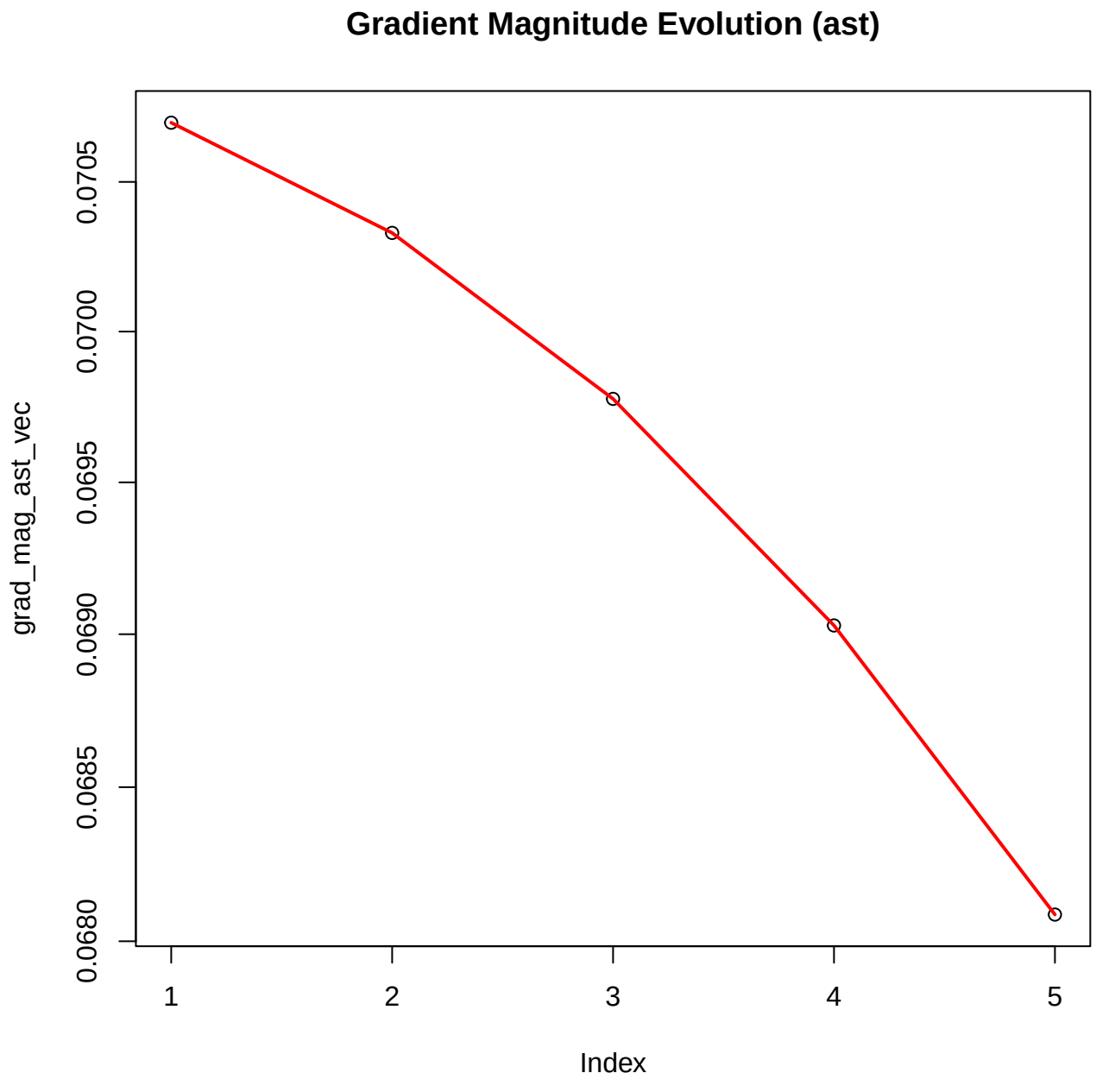

Value (ast)
loss_ast_vec
0.65
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index

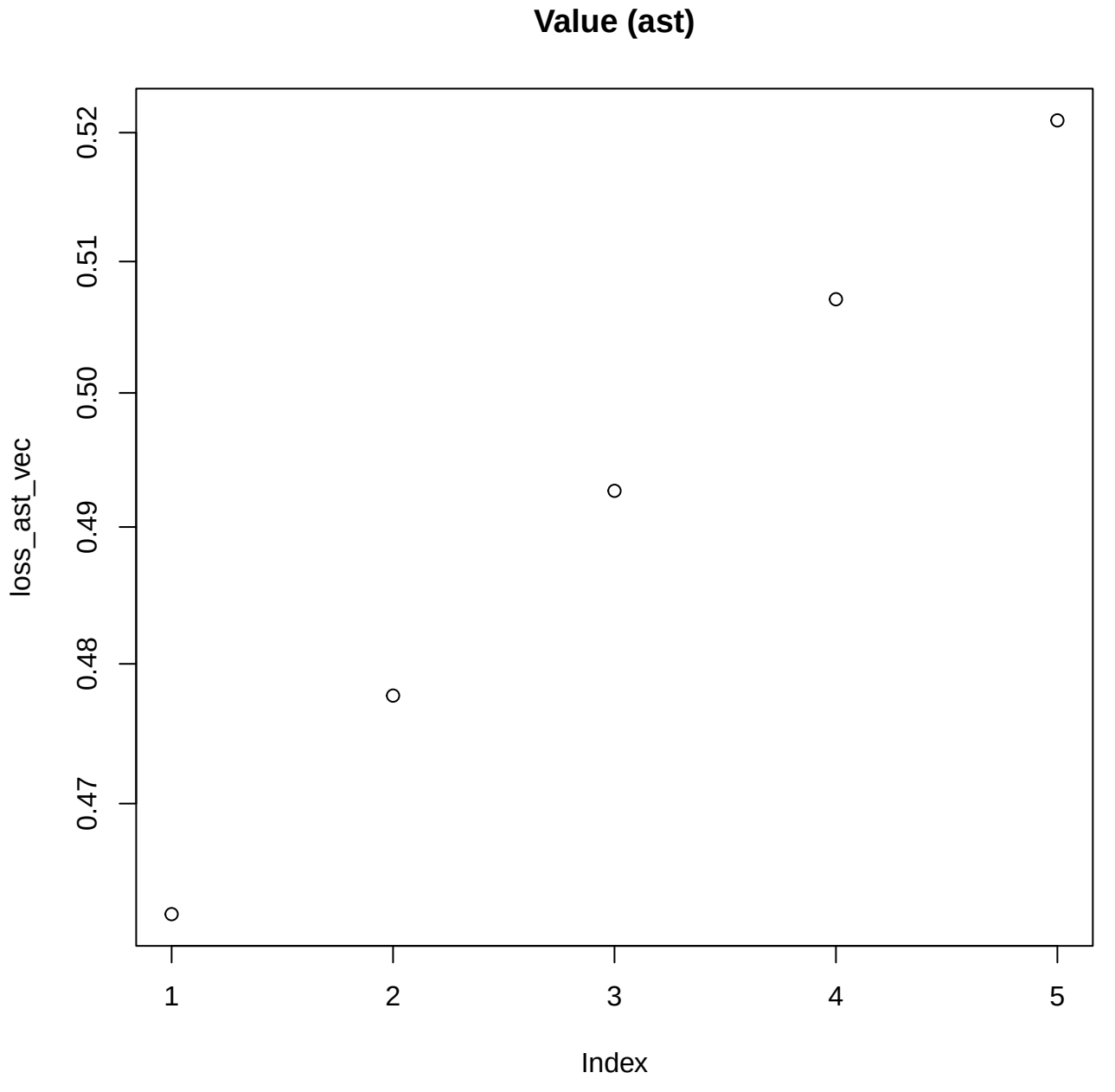

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

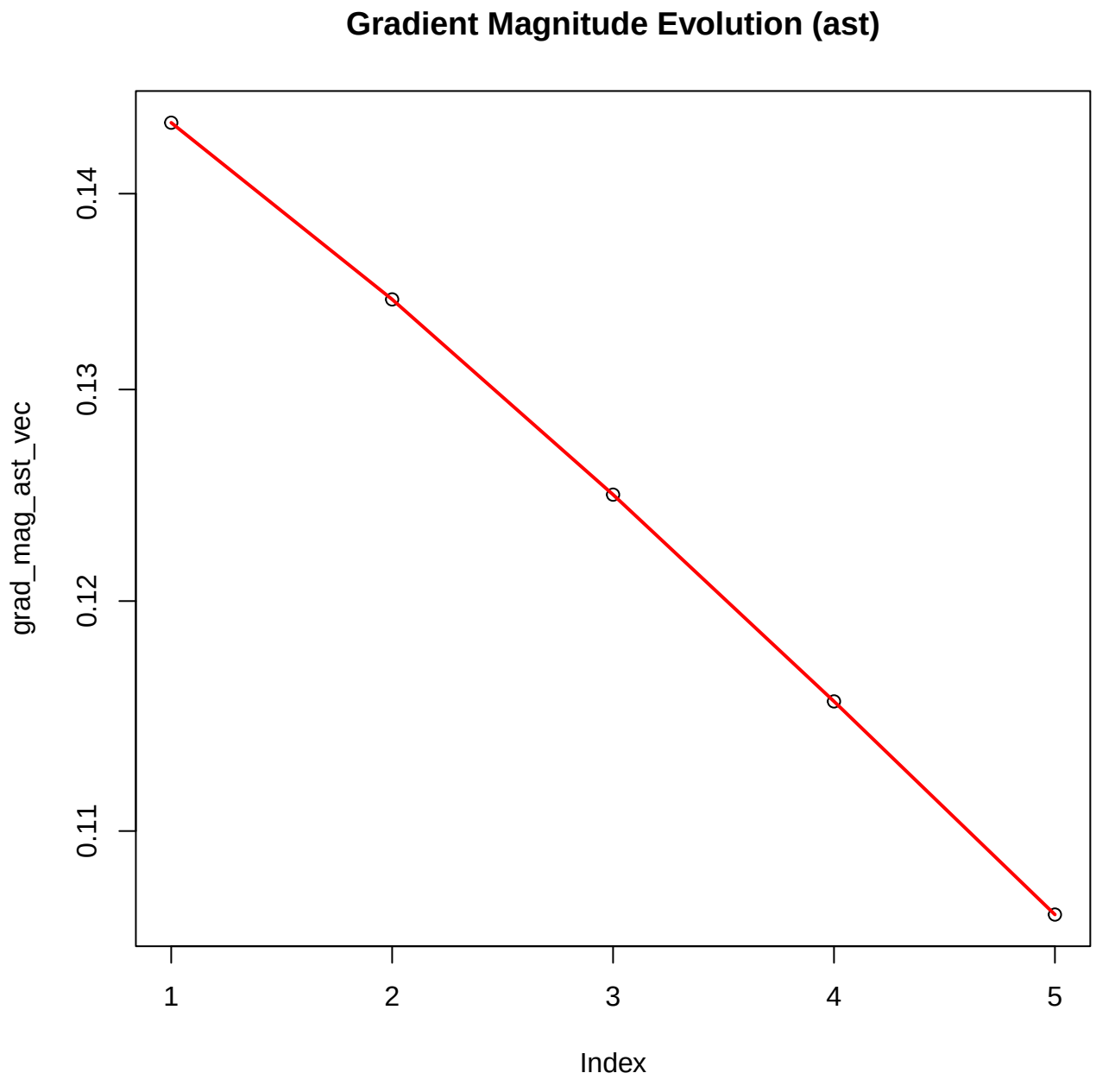

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

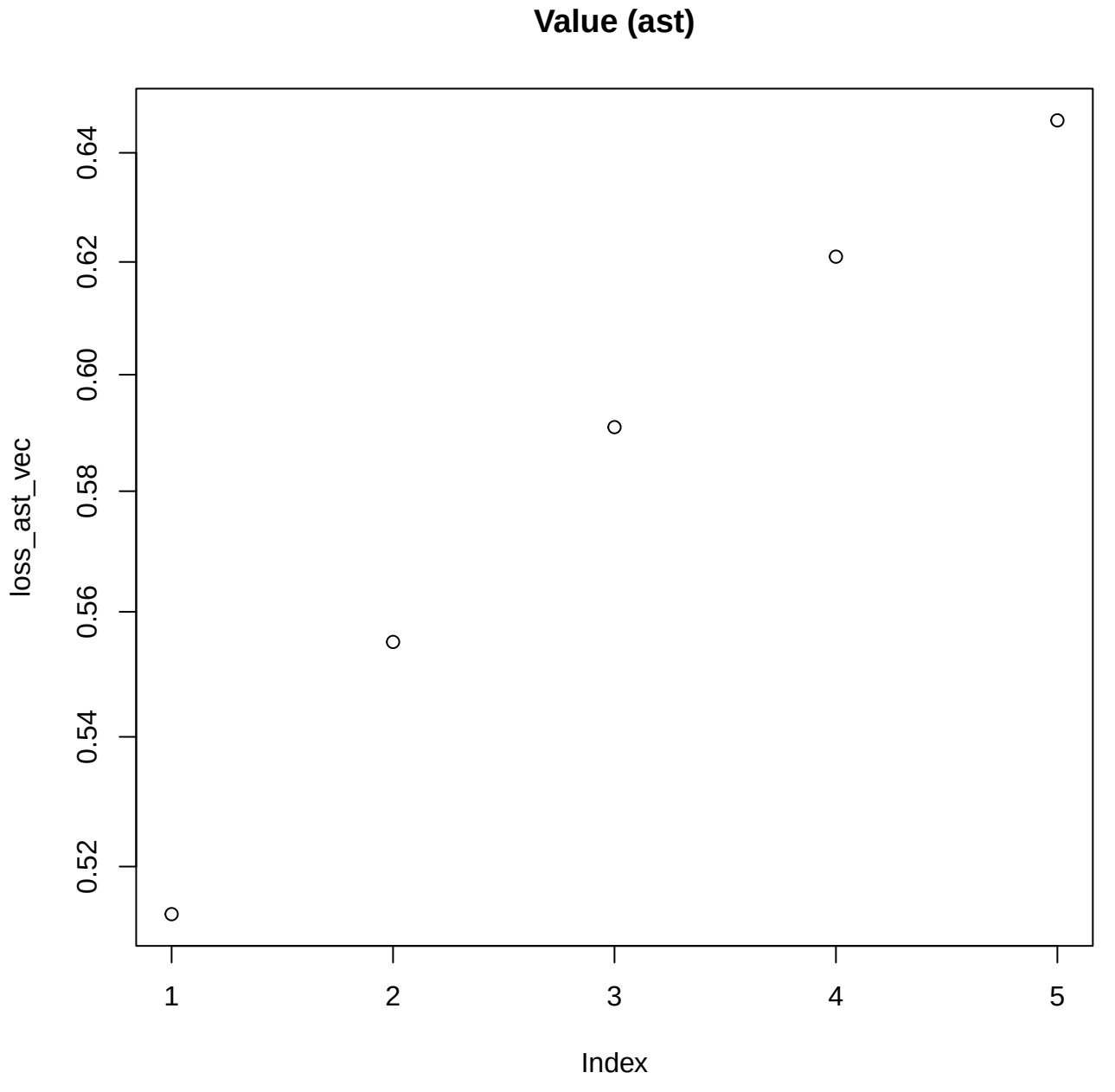

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.07
0.08
0.09
0.10
0.11
1
2
3
4
5
Index

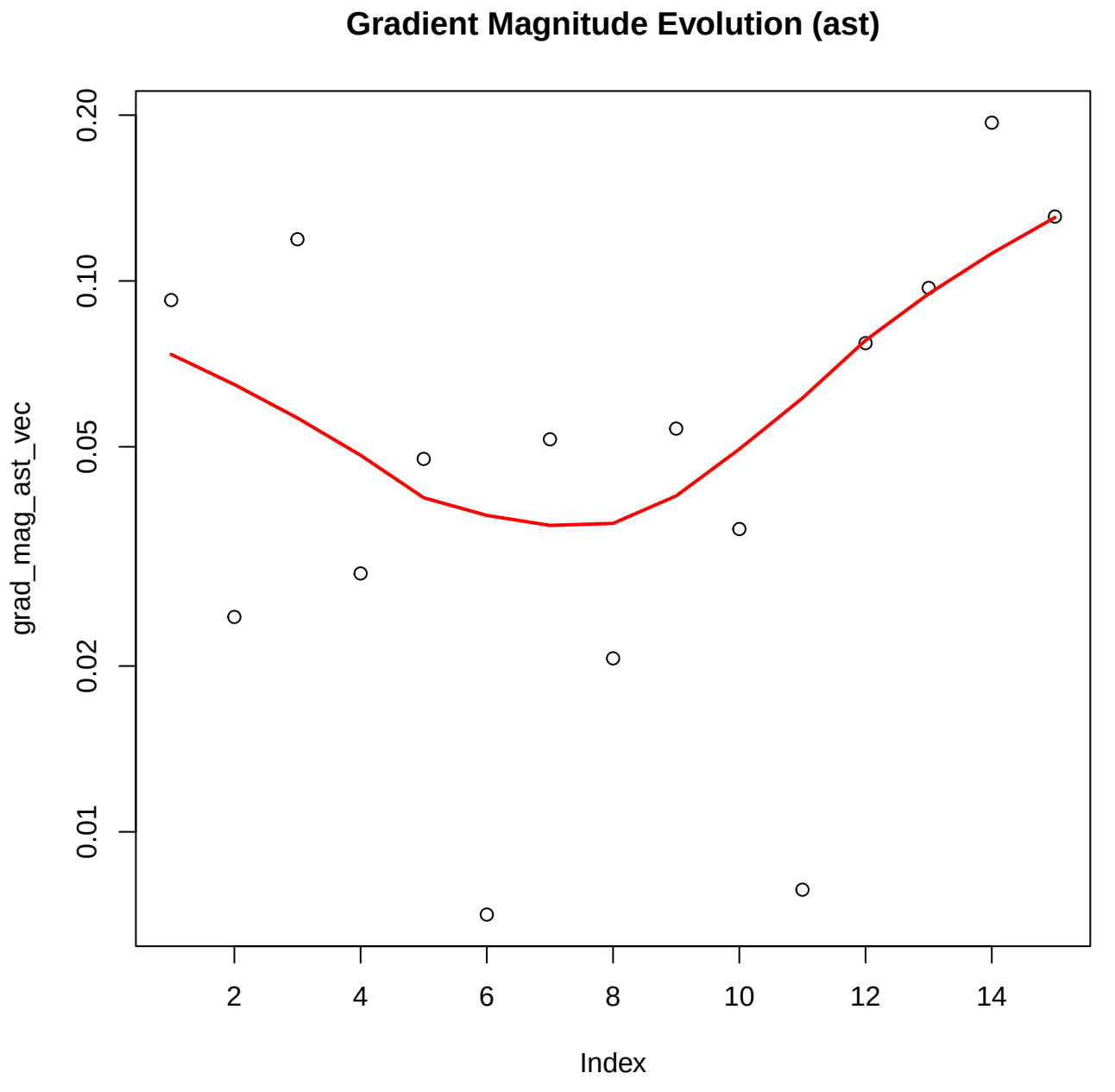

Value (ast)
loss_ast_vec
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

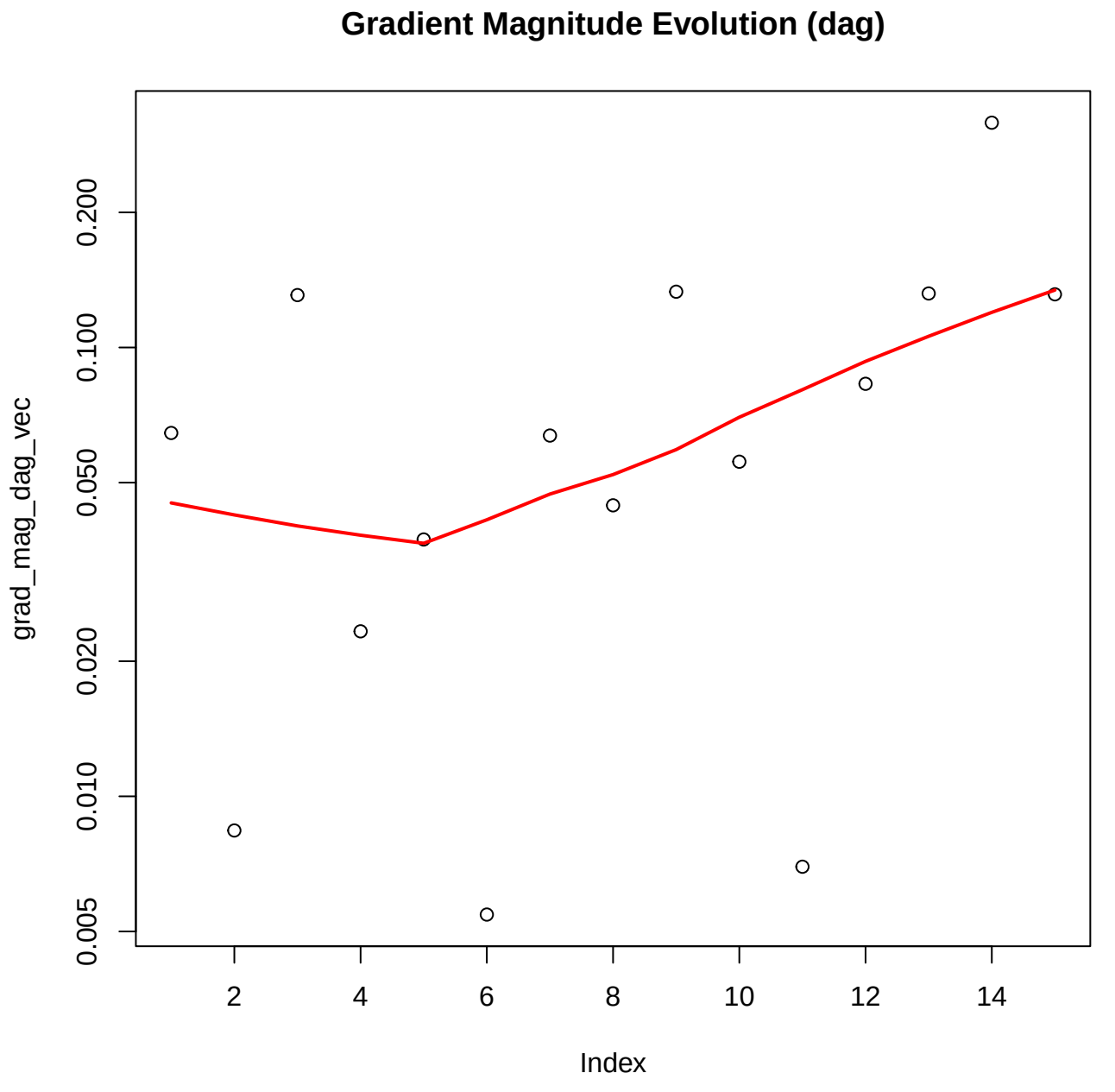

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.0680
0.0685
0.0690
0.0695
0.0700
0.0705
1
2
3
4
5
Index

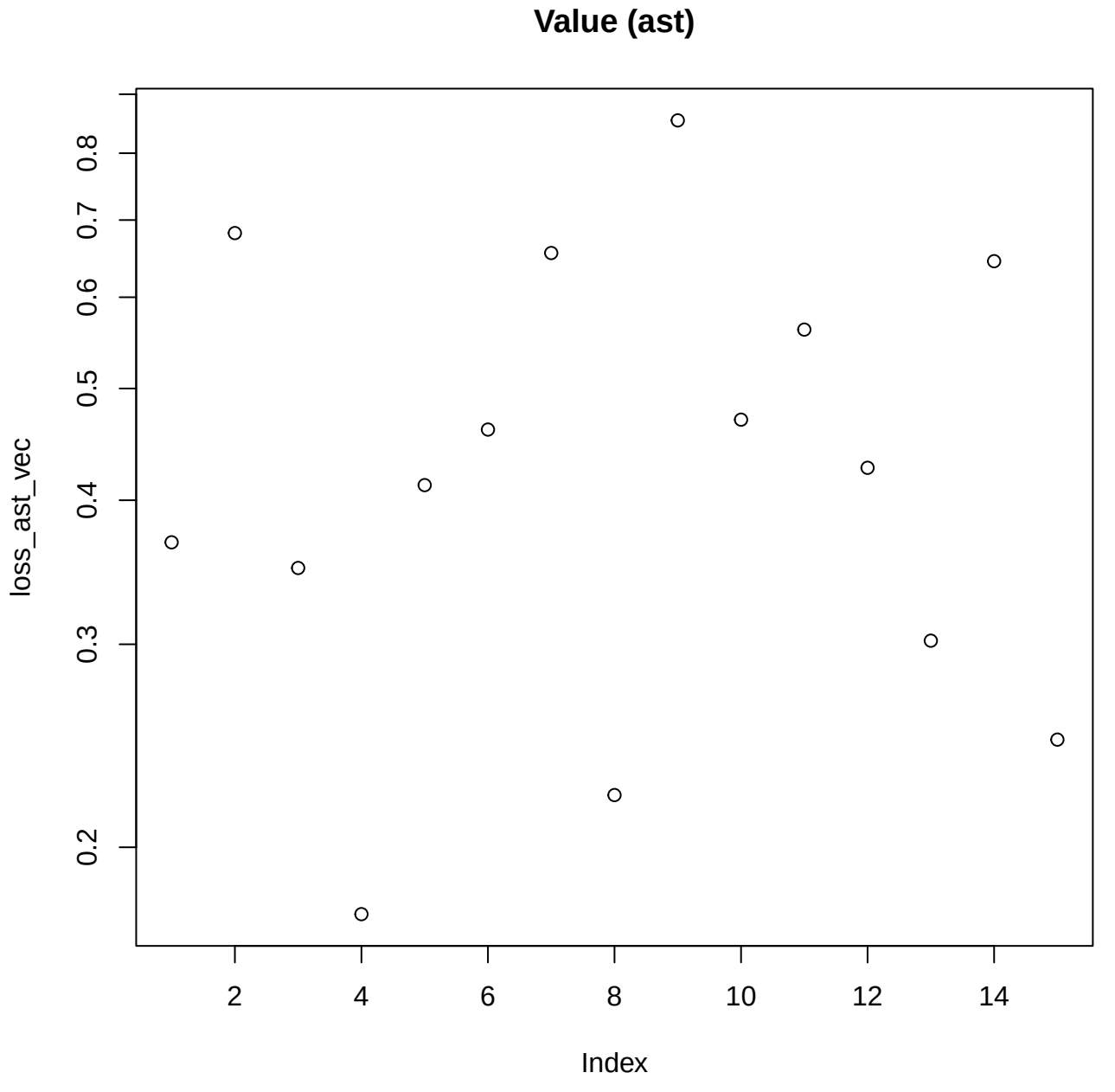

Value (ast)
loss_ast_vec
0.47
0.48
0.49
0.50
0.51
0.52
1
2
3
4
5
Index

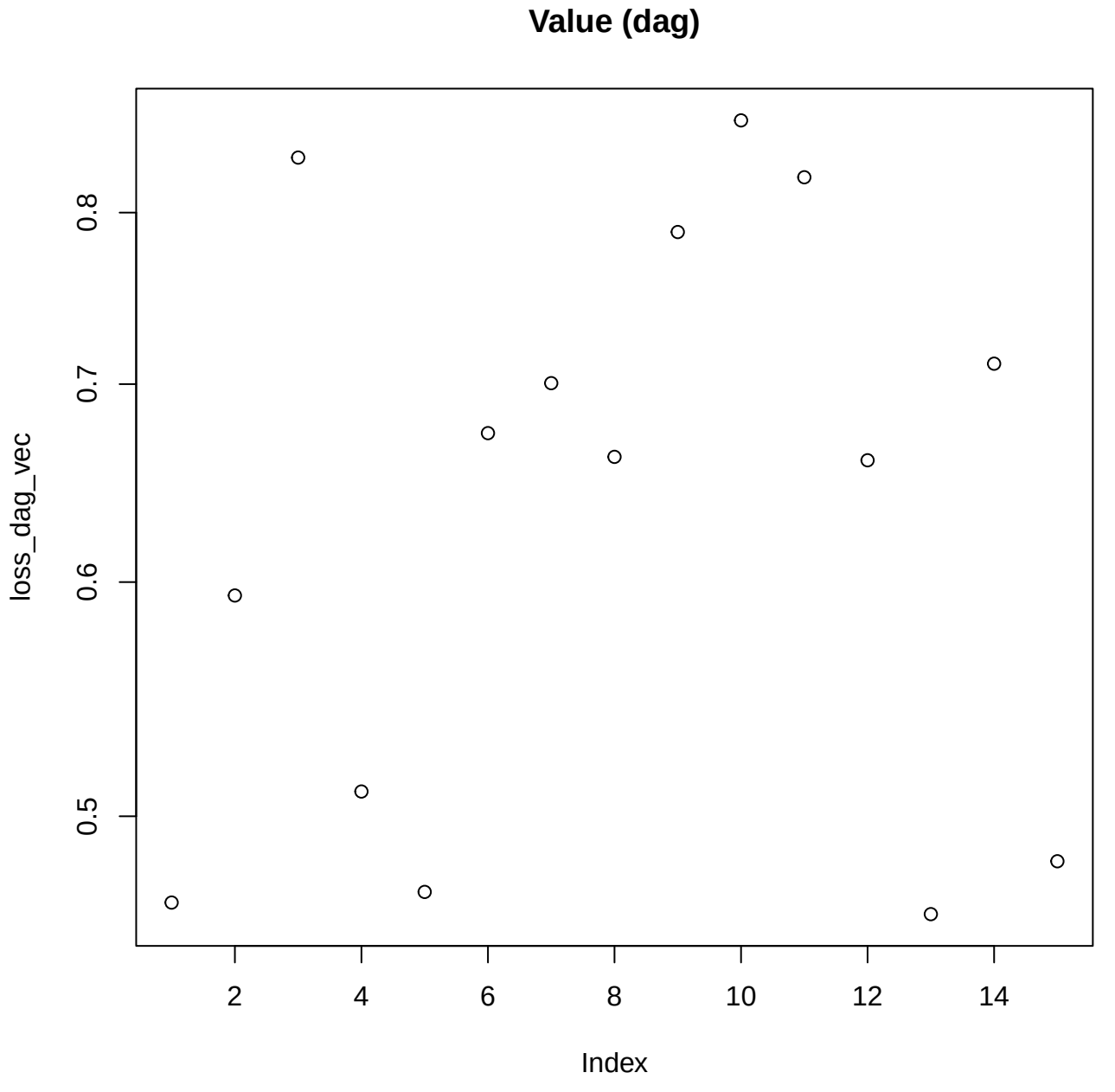

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.11
0.12
0.13
0.14
1
2
3
4
5
Index

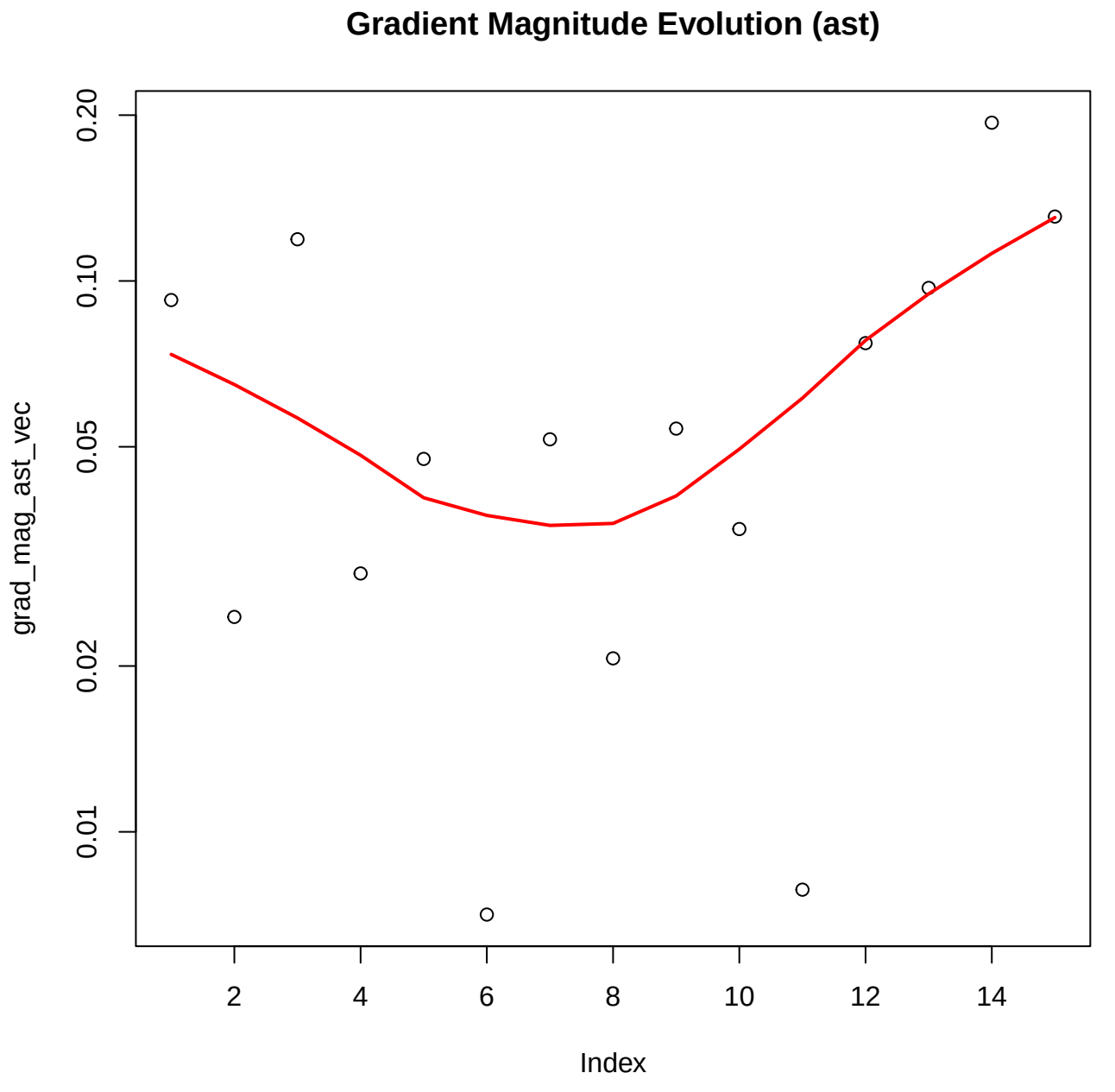

Value (ast)
loss_ast_vec
0.52
0.54
0.56
0.58
0.60
0.62
0.64
1
2
3
4
5
Index

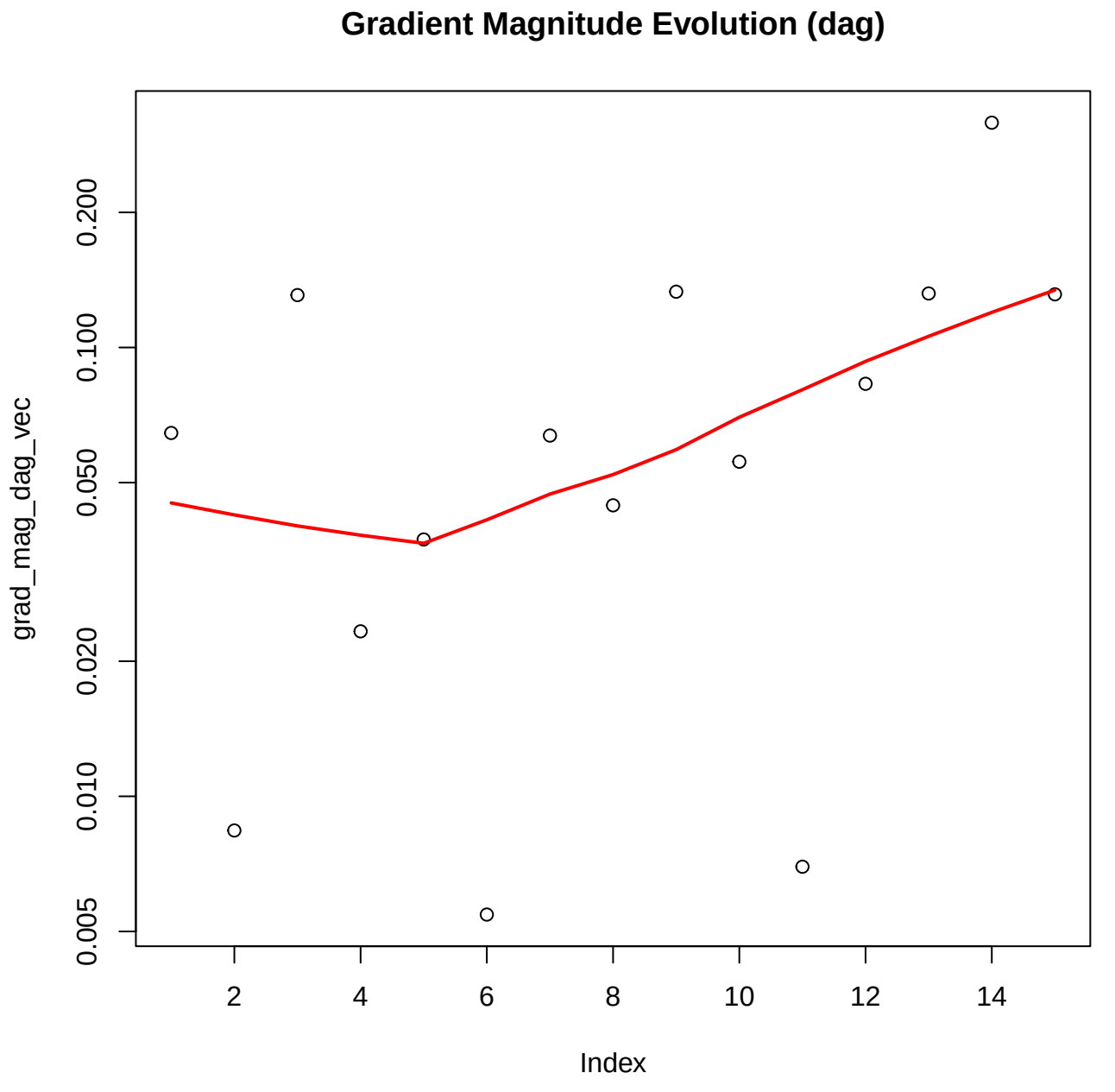

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

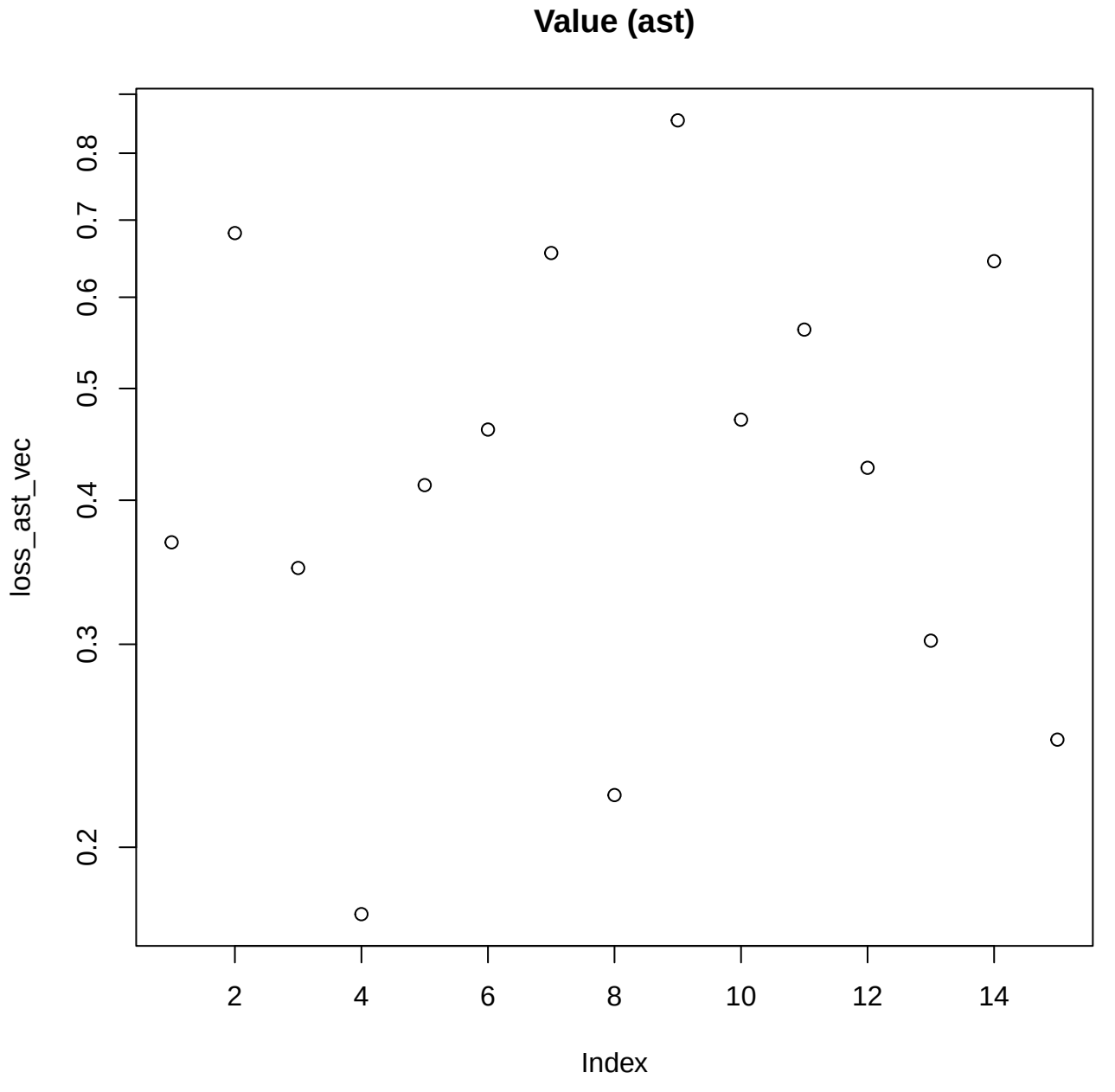

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

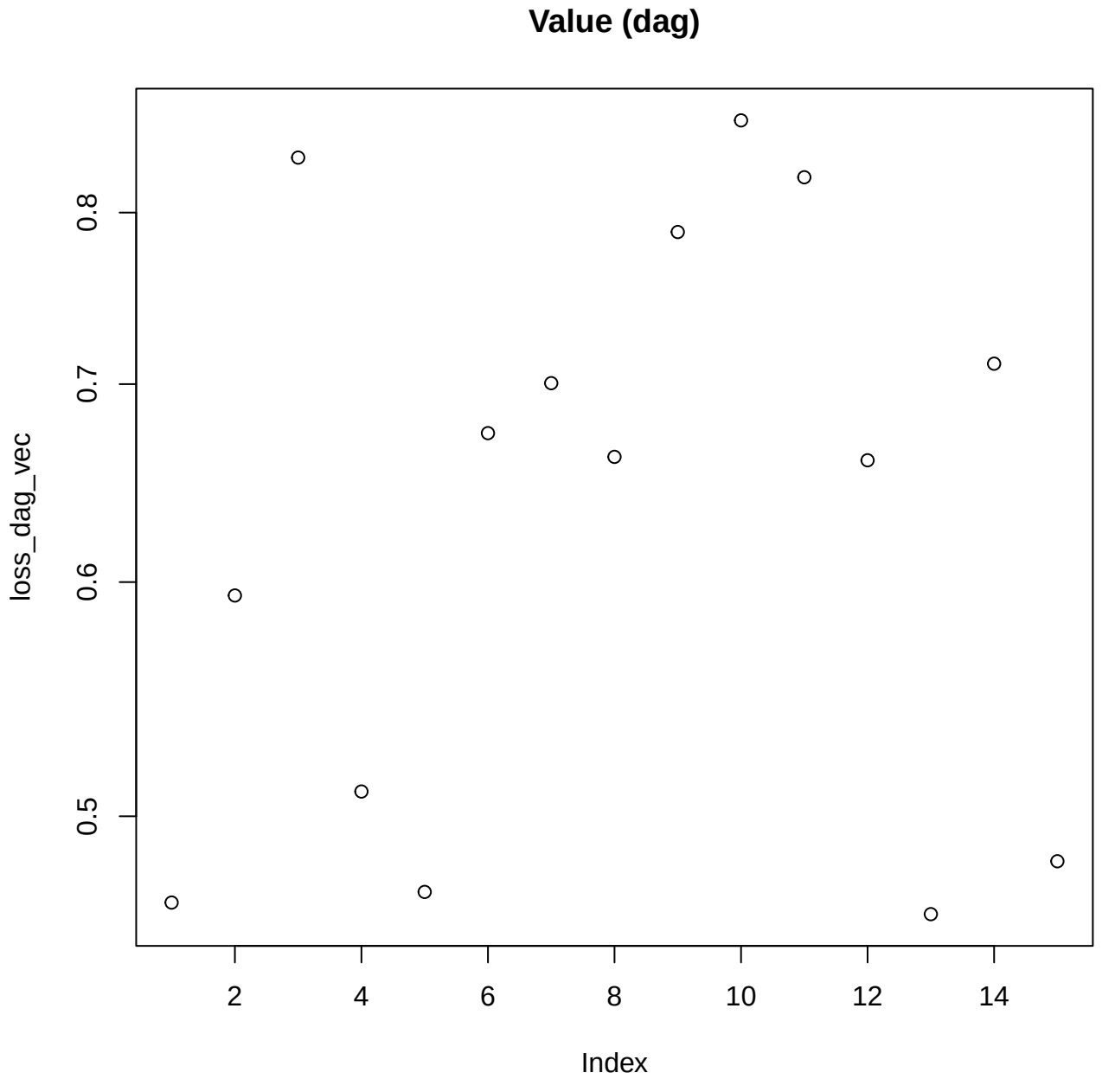

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

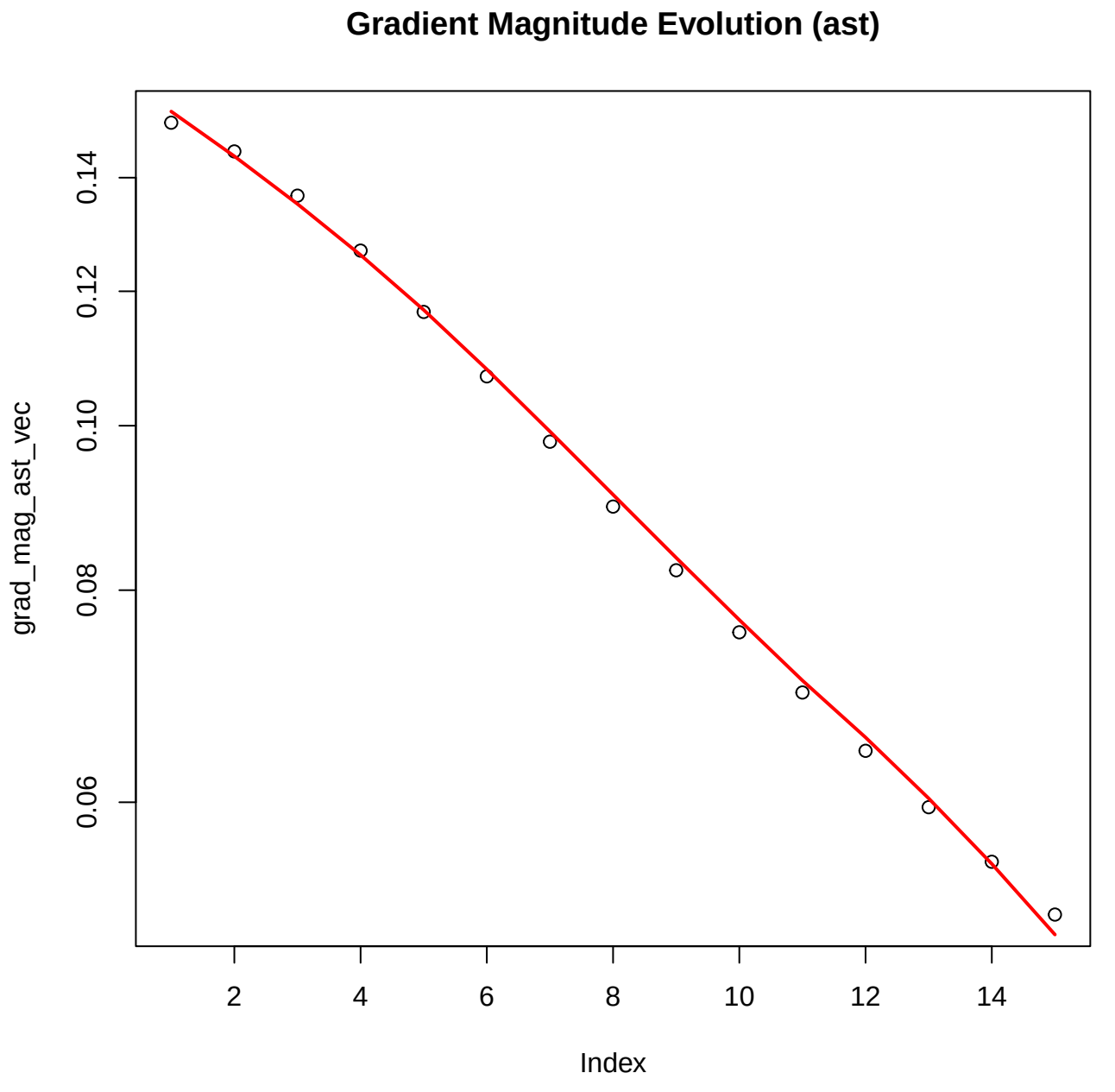

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

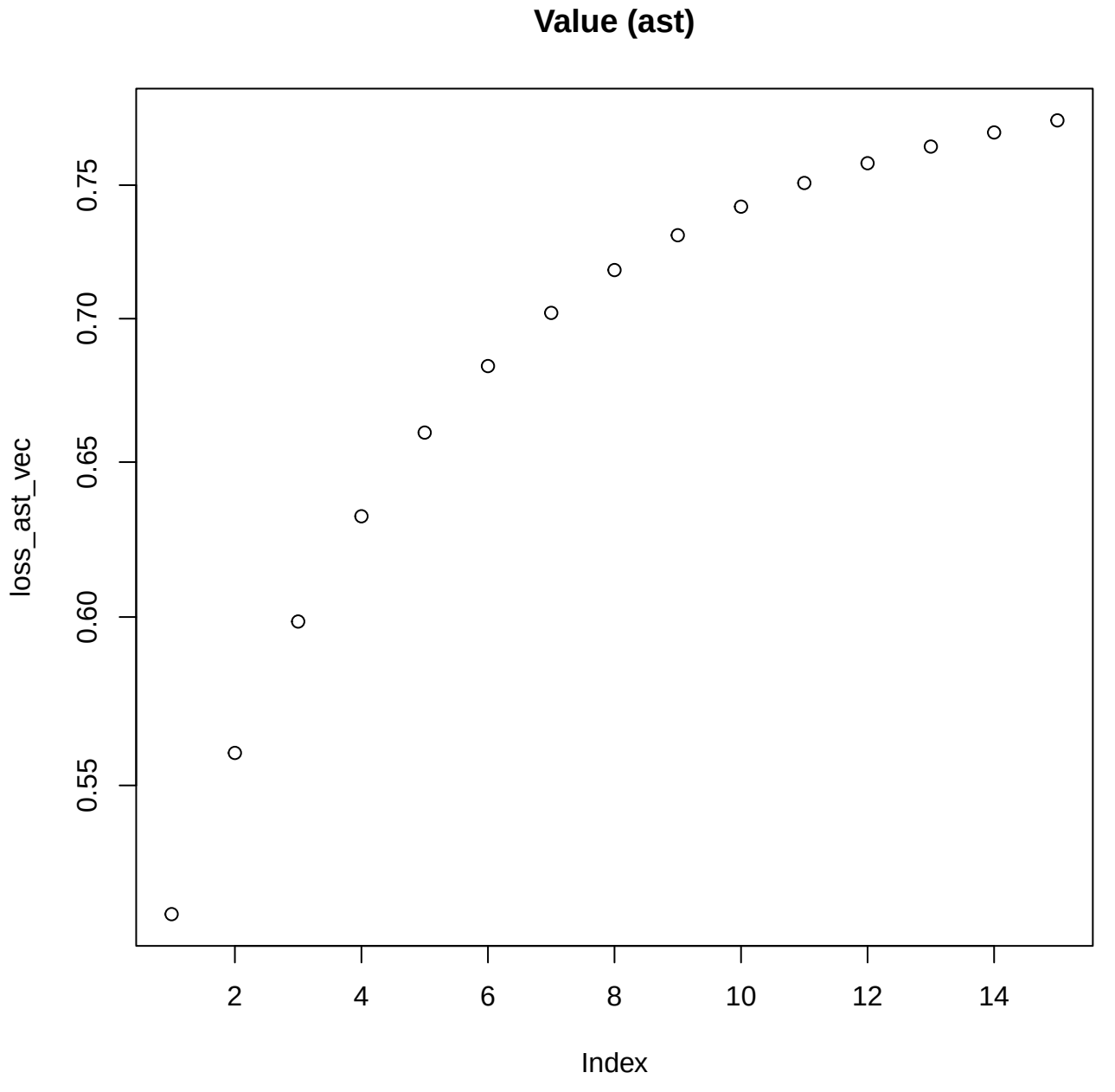

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

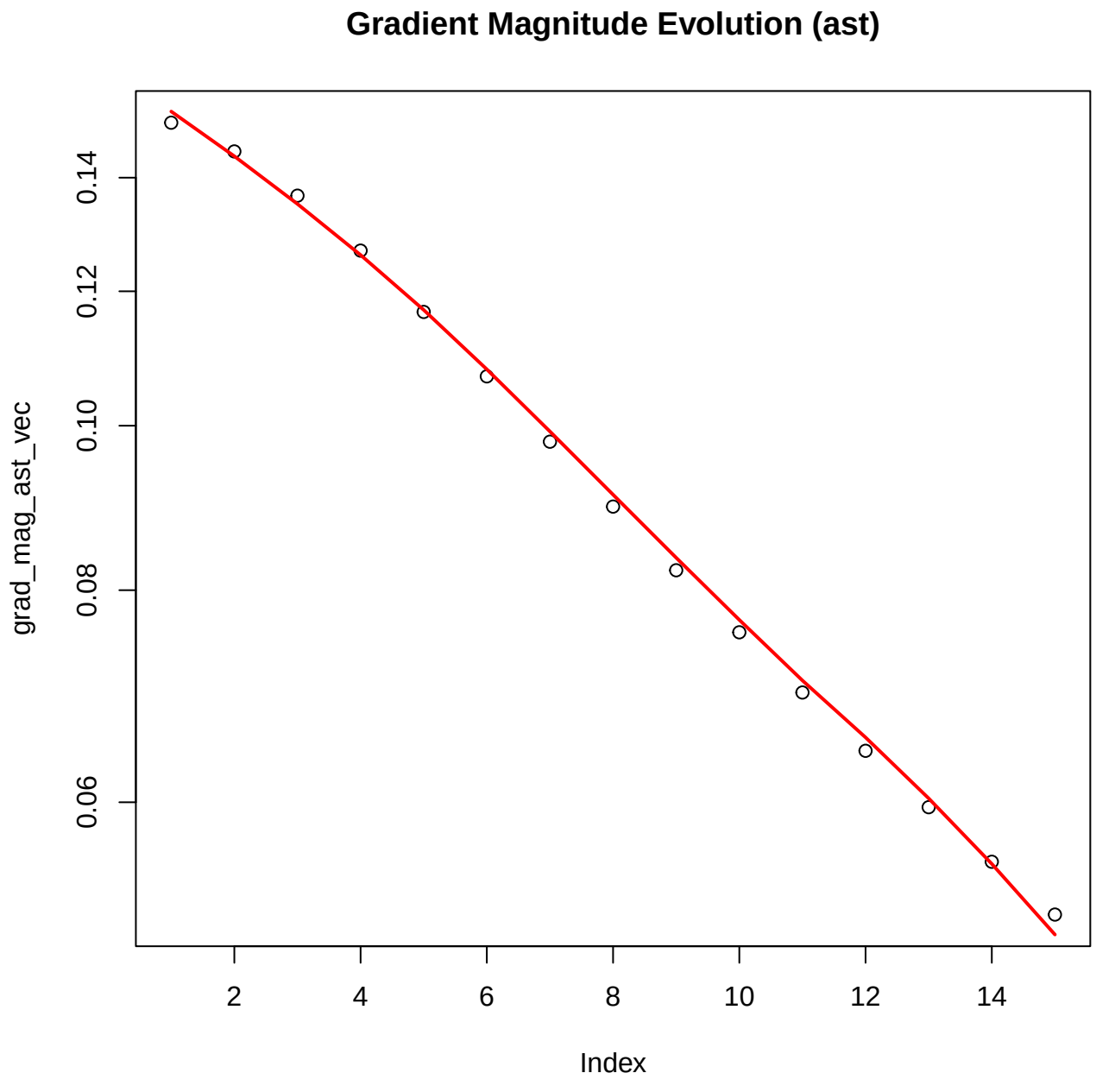

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

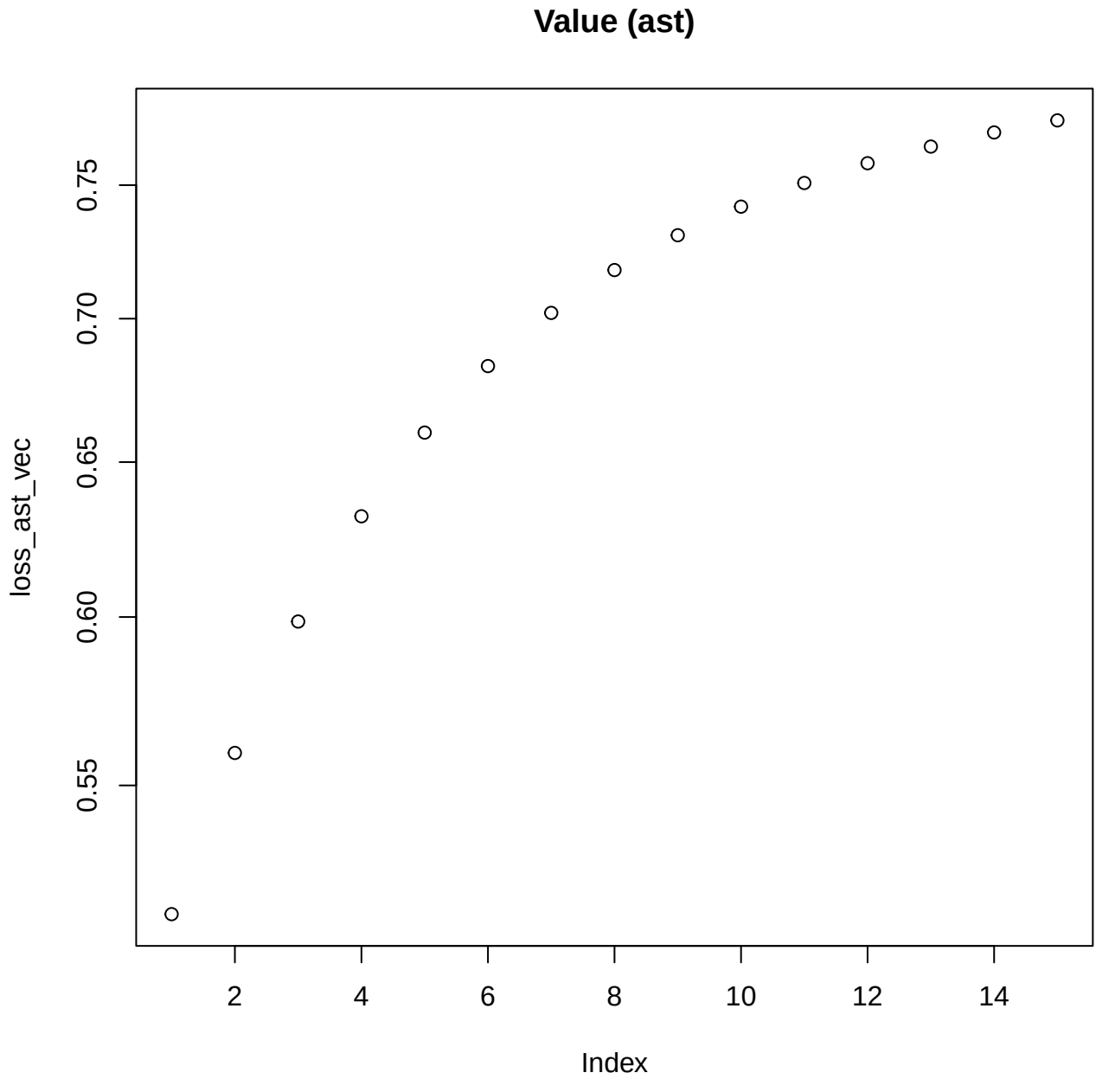

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

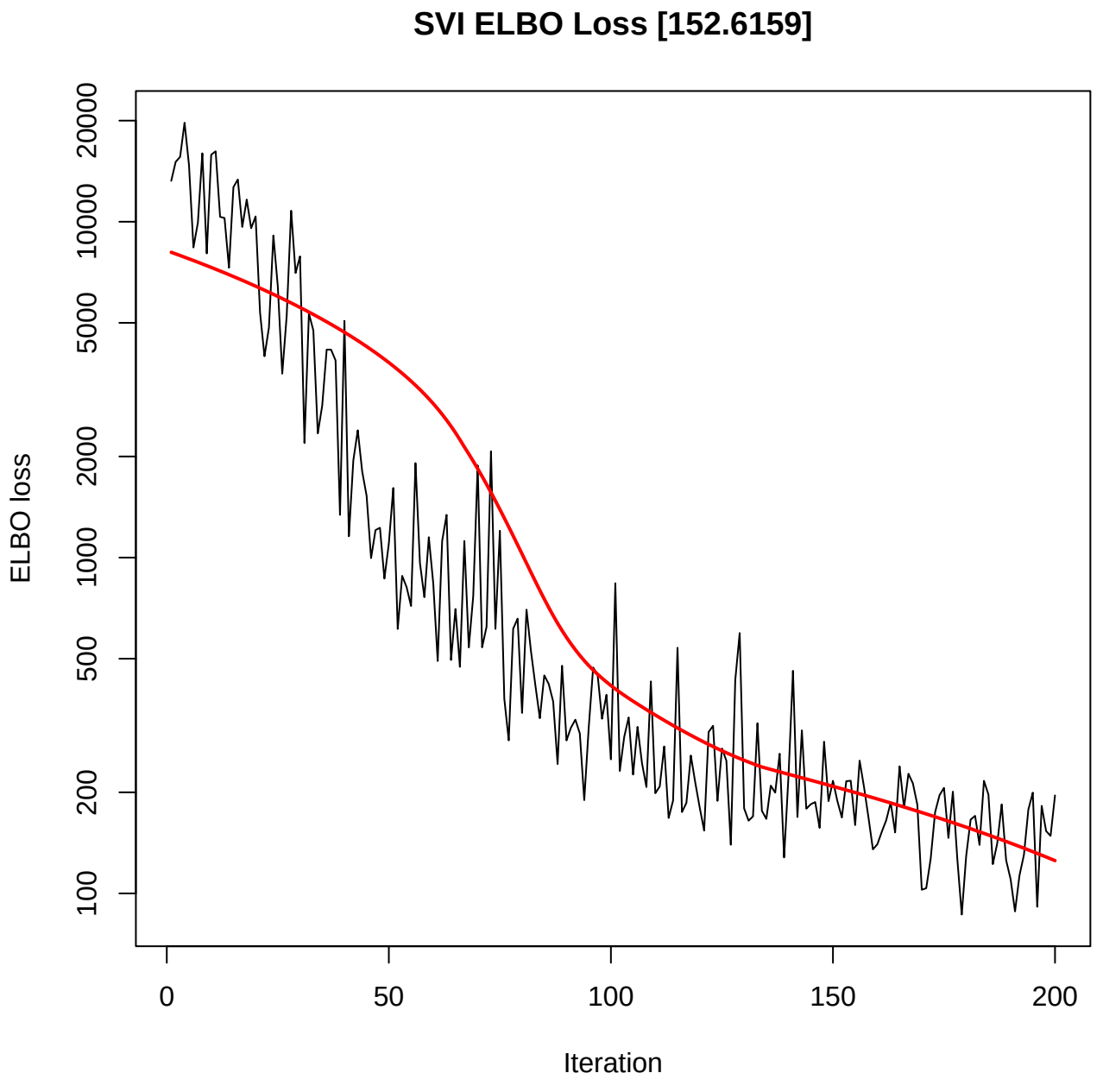

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

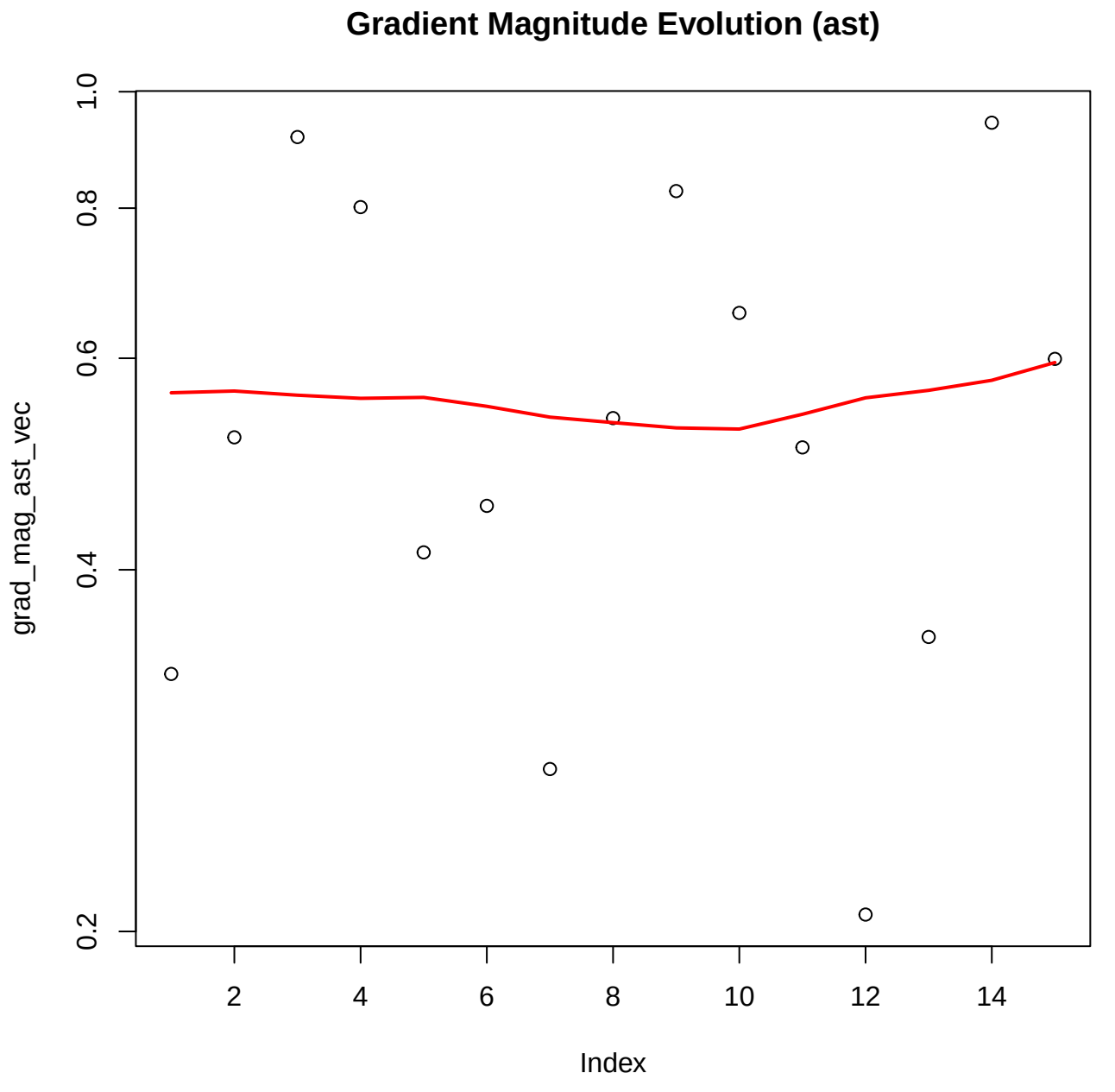

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

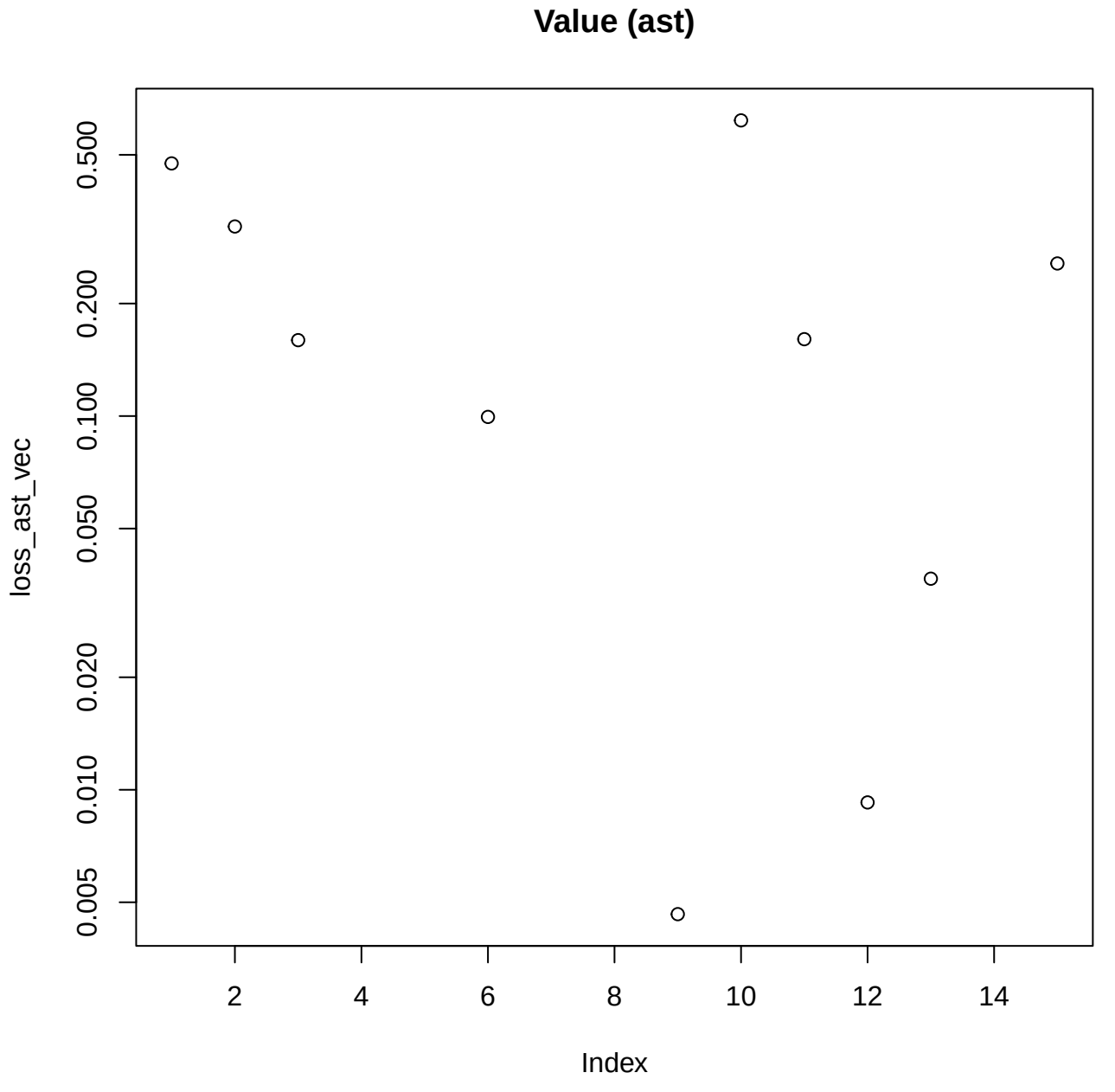

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

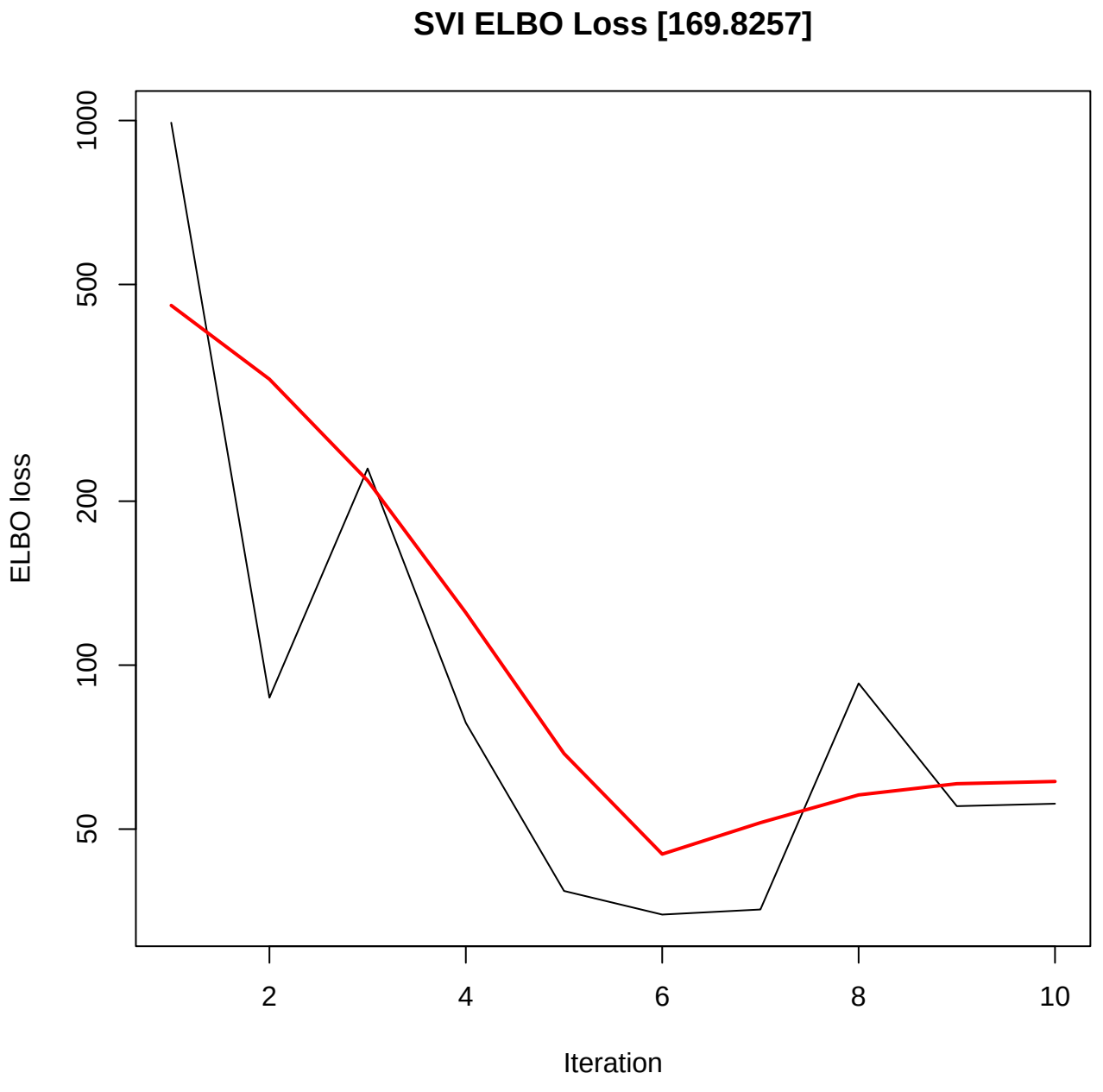

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

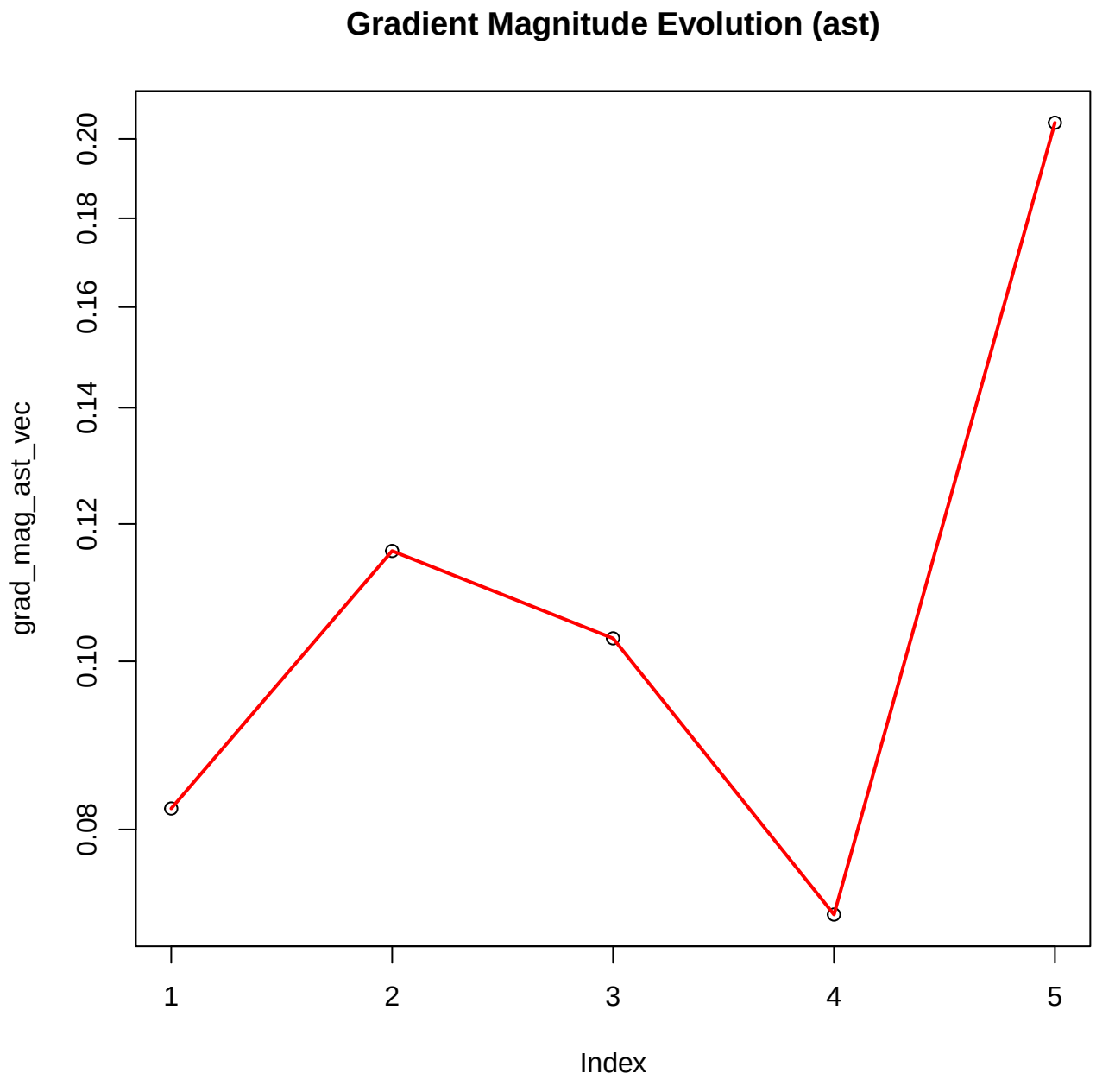

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

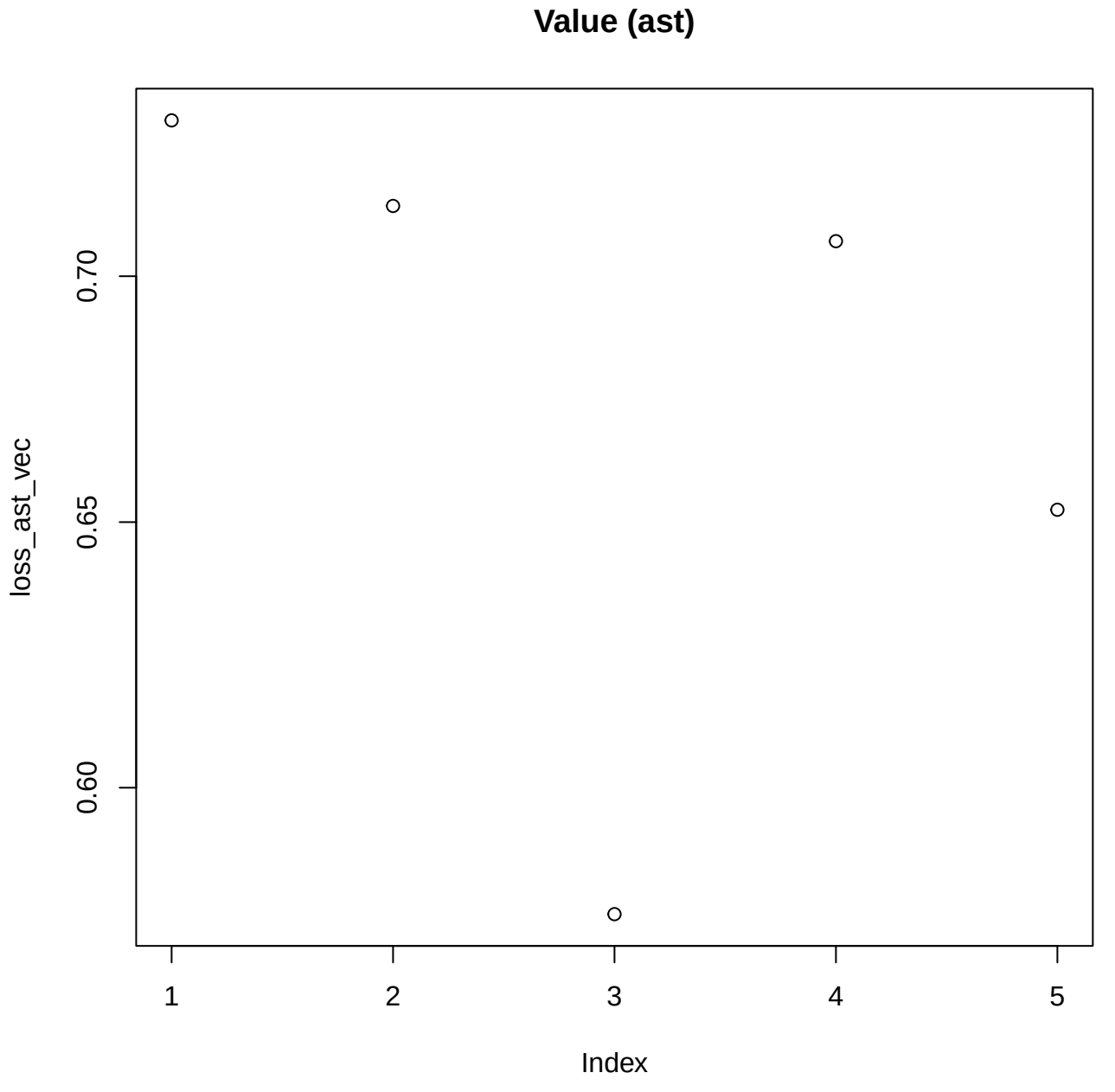

SVI ELBO Loss [152.6159]
ELBO loss
20000
10000
5000
2000
1000
500
200
100
0
50
100
150
200
Iteration

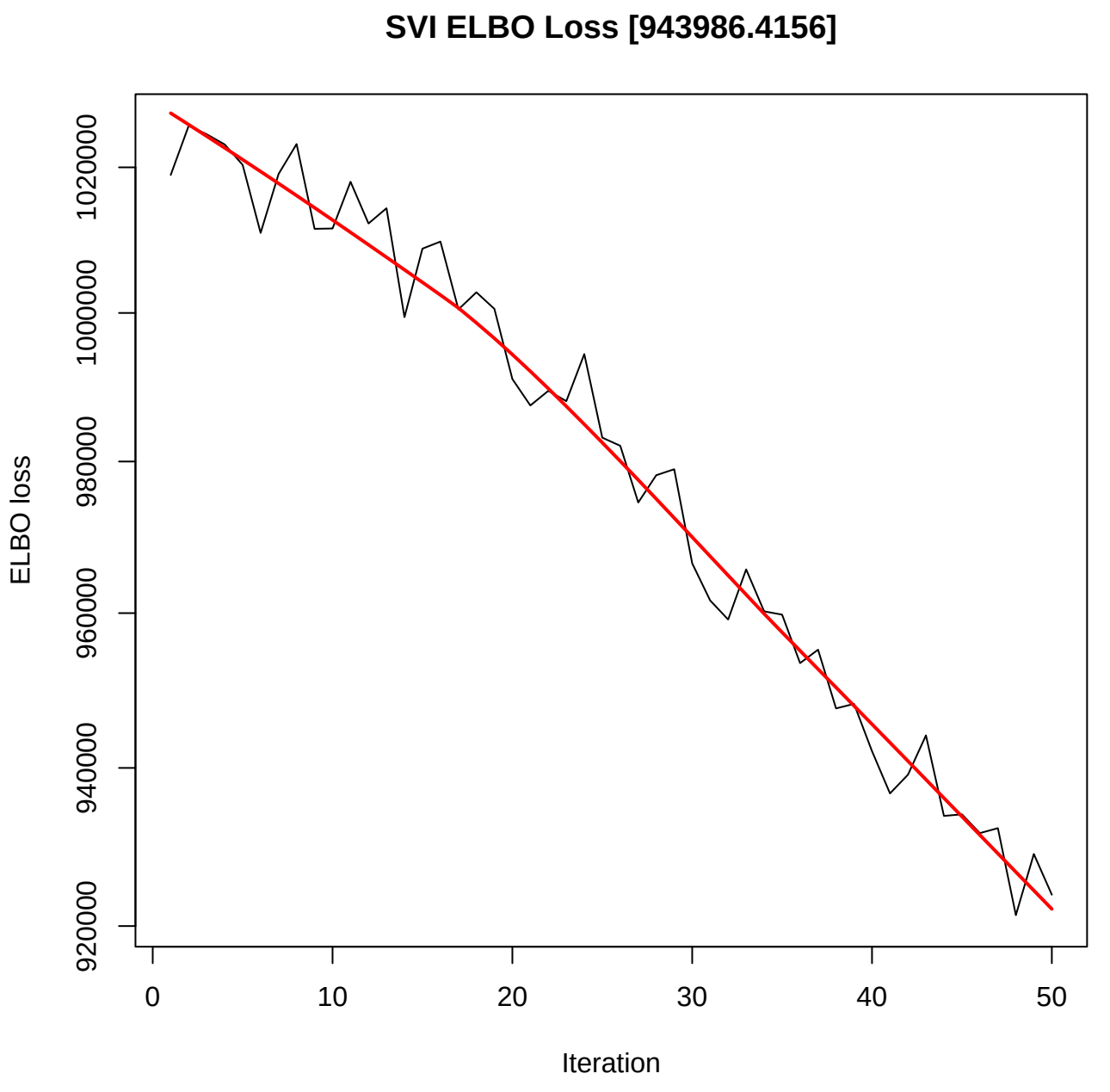

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10
12
14
Index

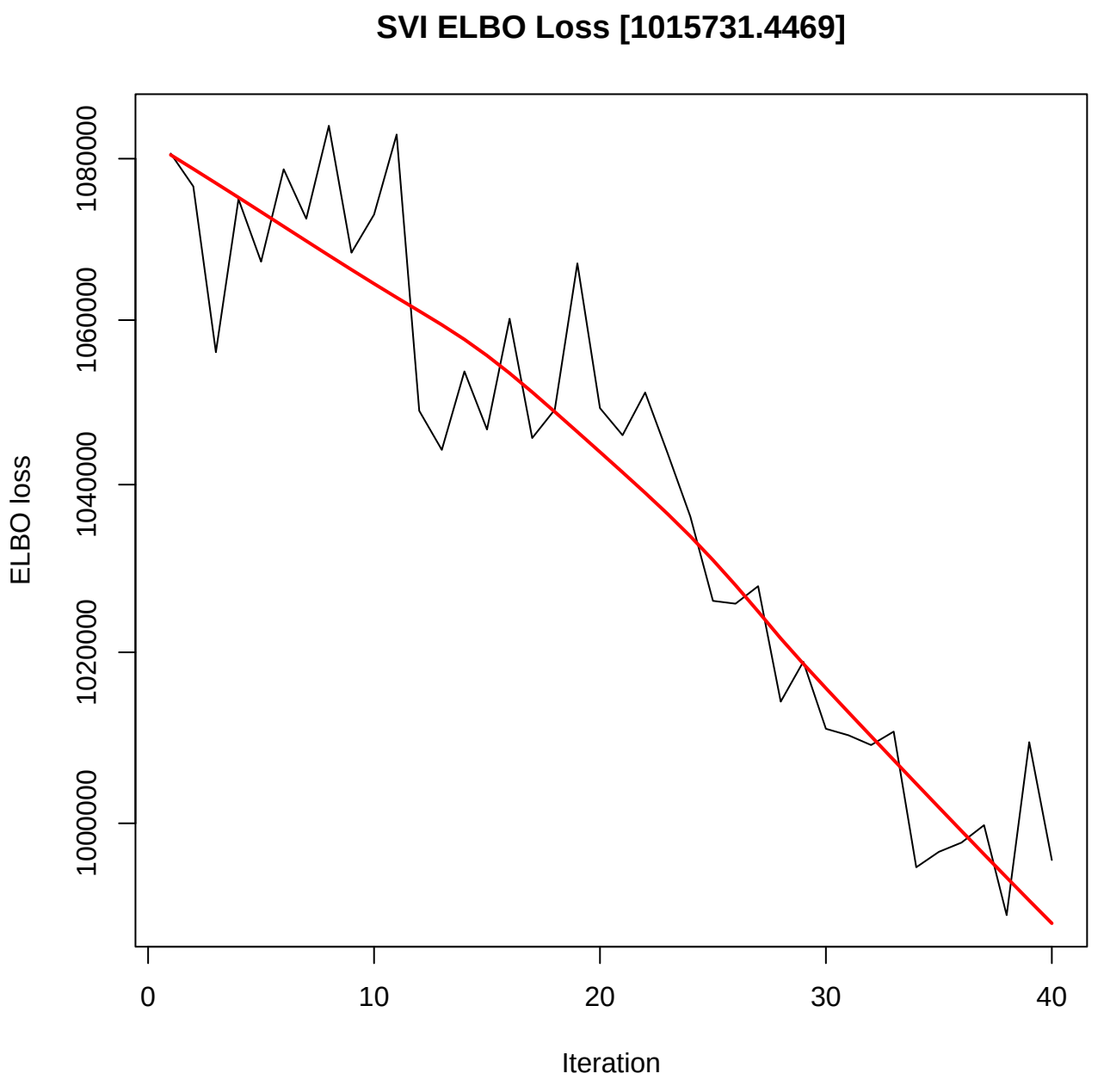

Value (ast)
loss_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
0.500
2
4
6
8
10
12
14
Index

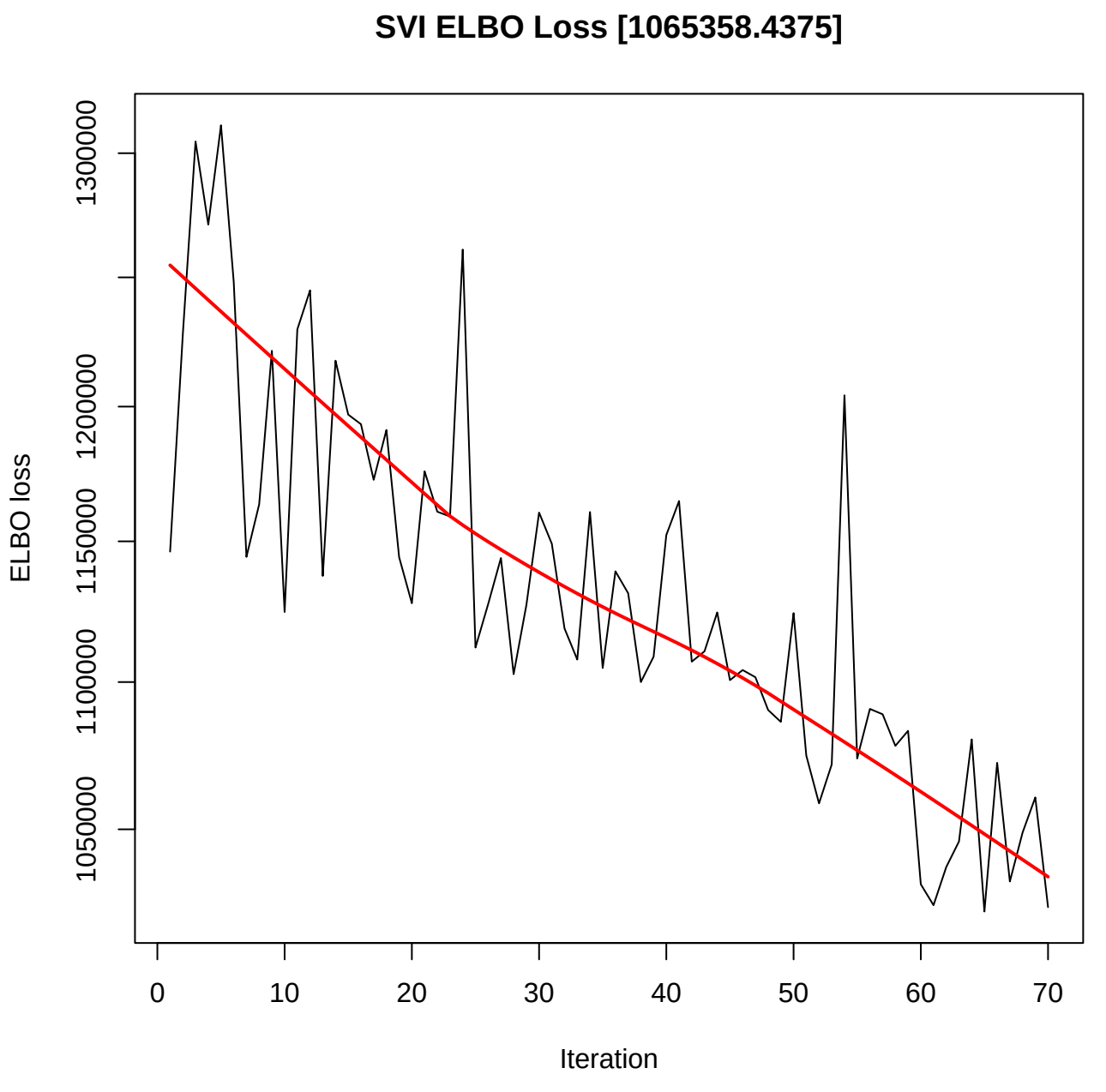

SVI ELBO Loss [169.8257]
ELBO loss
1000
500
200
100
50
2
4
6
8
10
Iteration

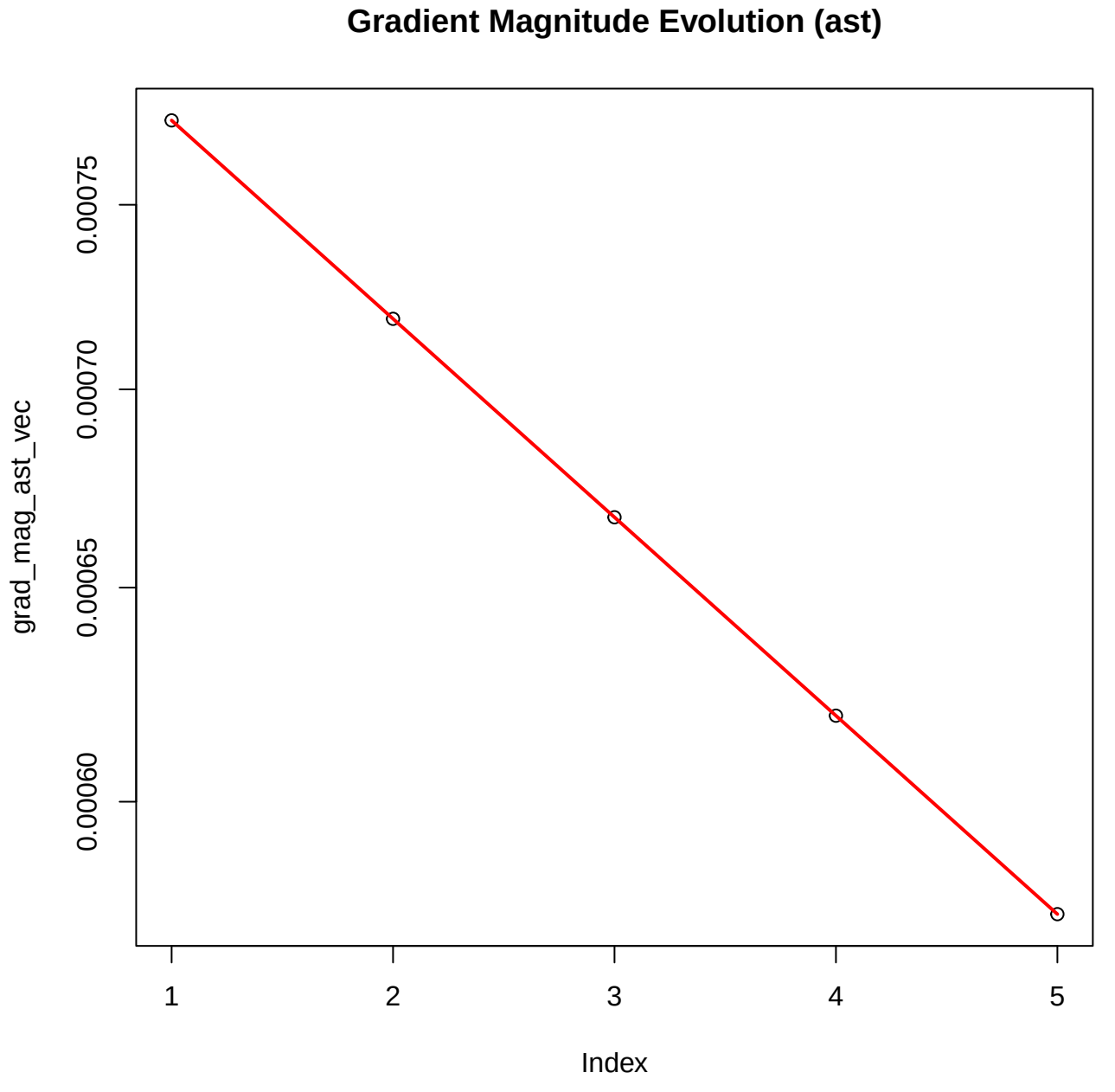

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
0.18
0.20
1
2
3
4
5
Index

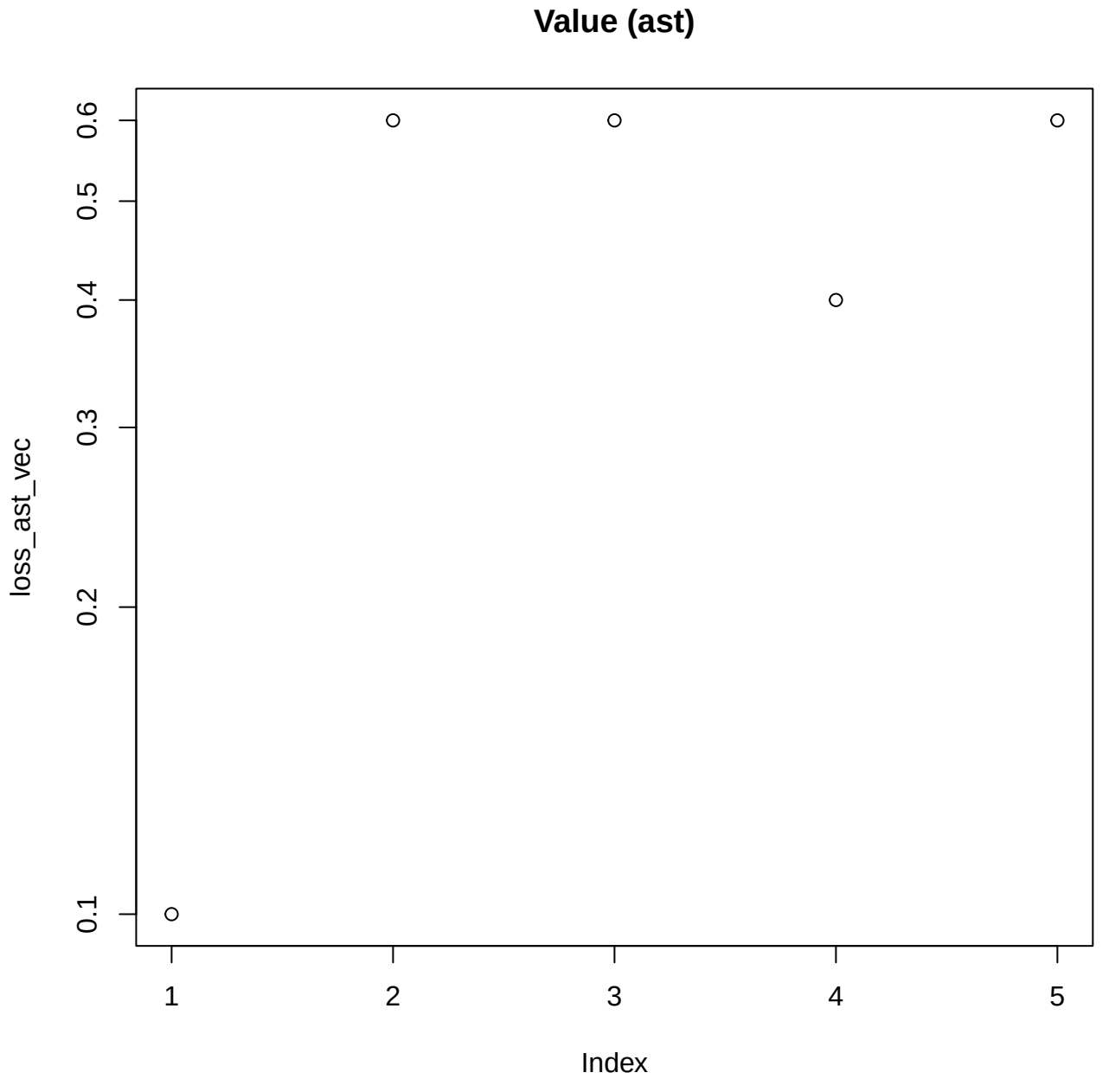

Value (ast)
loss_ast_vec
0.60
0.65
0.70
1
2
3
4
5
Index

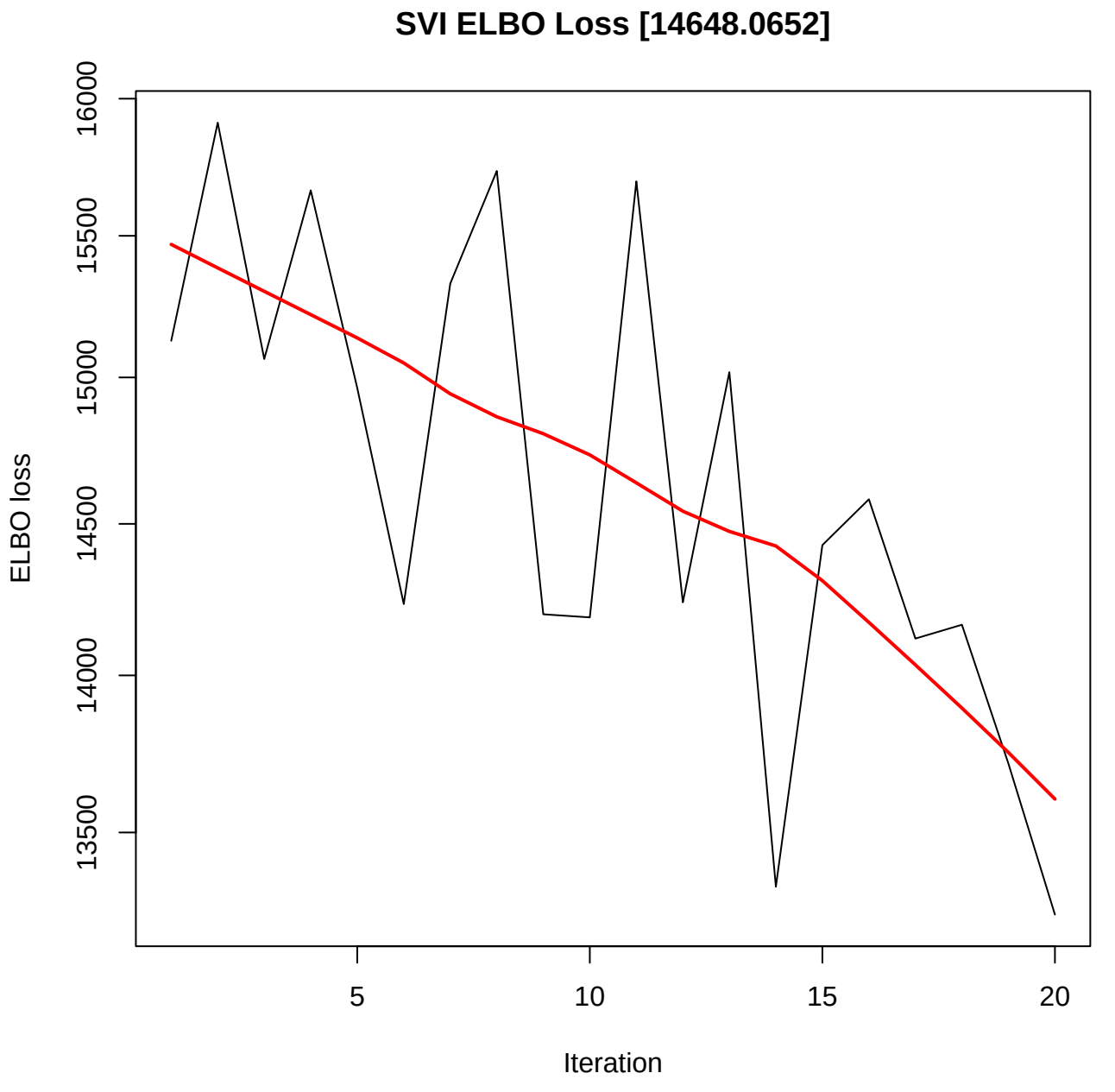

SVI ELBO Loss [943986.4156]
ELBO loss
920000
940000
960000
980000
1000000
1020000
0
10
20
30
40
50
Iteration

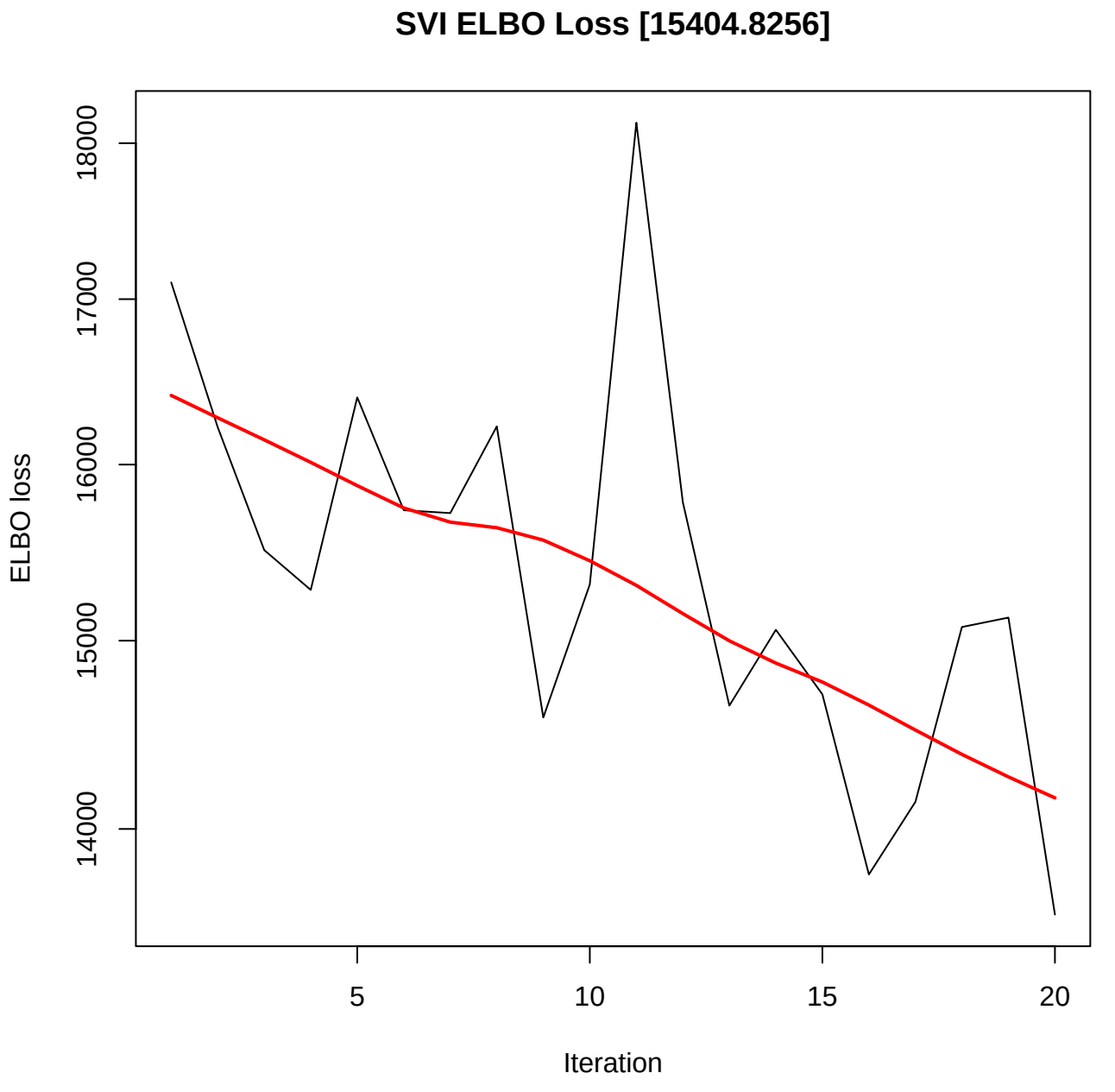

SVI ELBO Loss [1015731.4469]
ELBO loss
1000000
1020000
1040000
1060000
1080000
0
10
20
30
40
Iteration

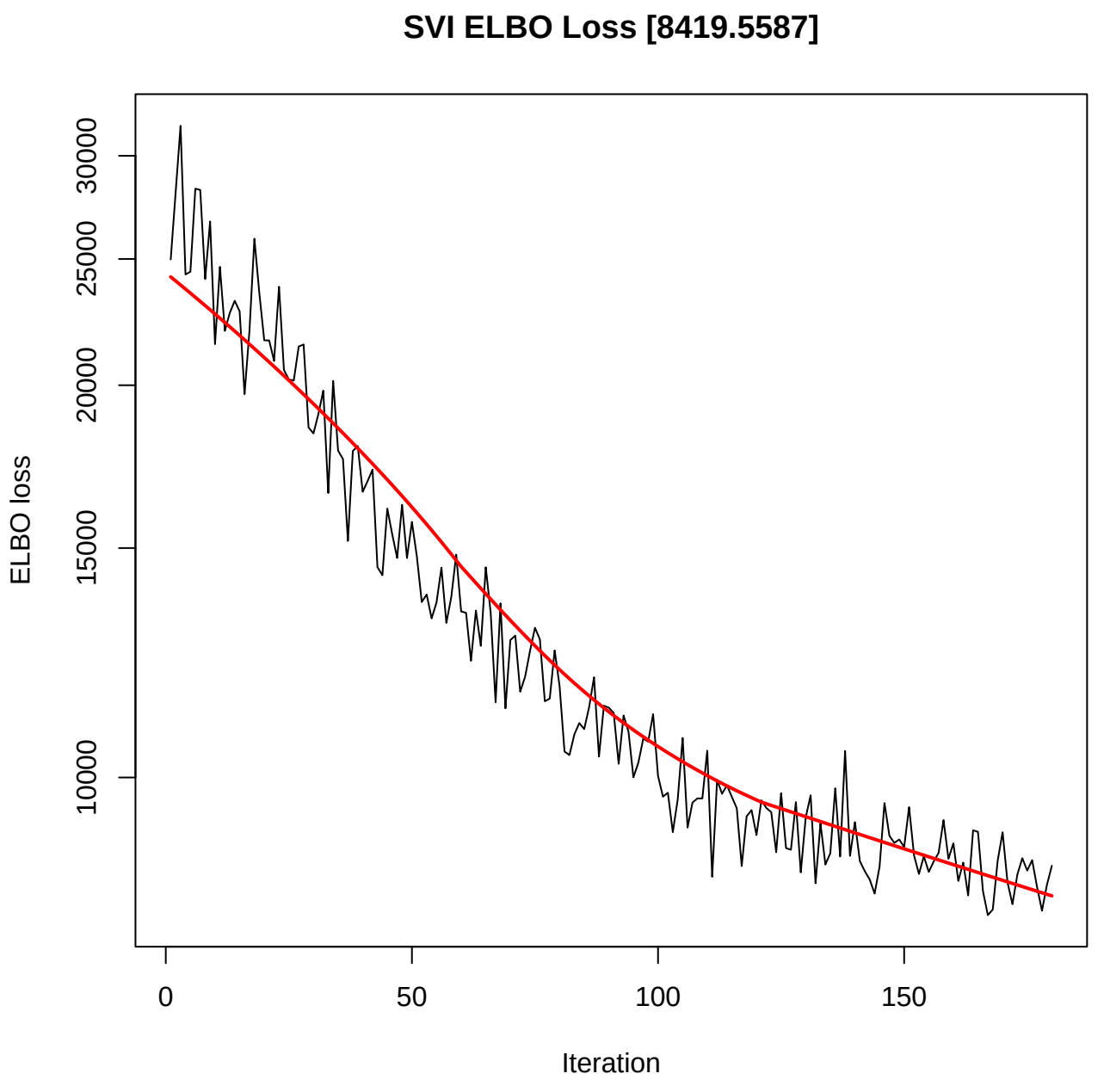

SVI ELBO Loss [1065358.4375]
ELBO loss
1300000
1200000
1150000
1100000
1050000
0
10
20
30
40
50
60
70
Iteration

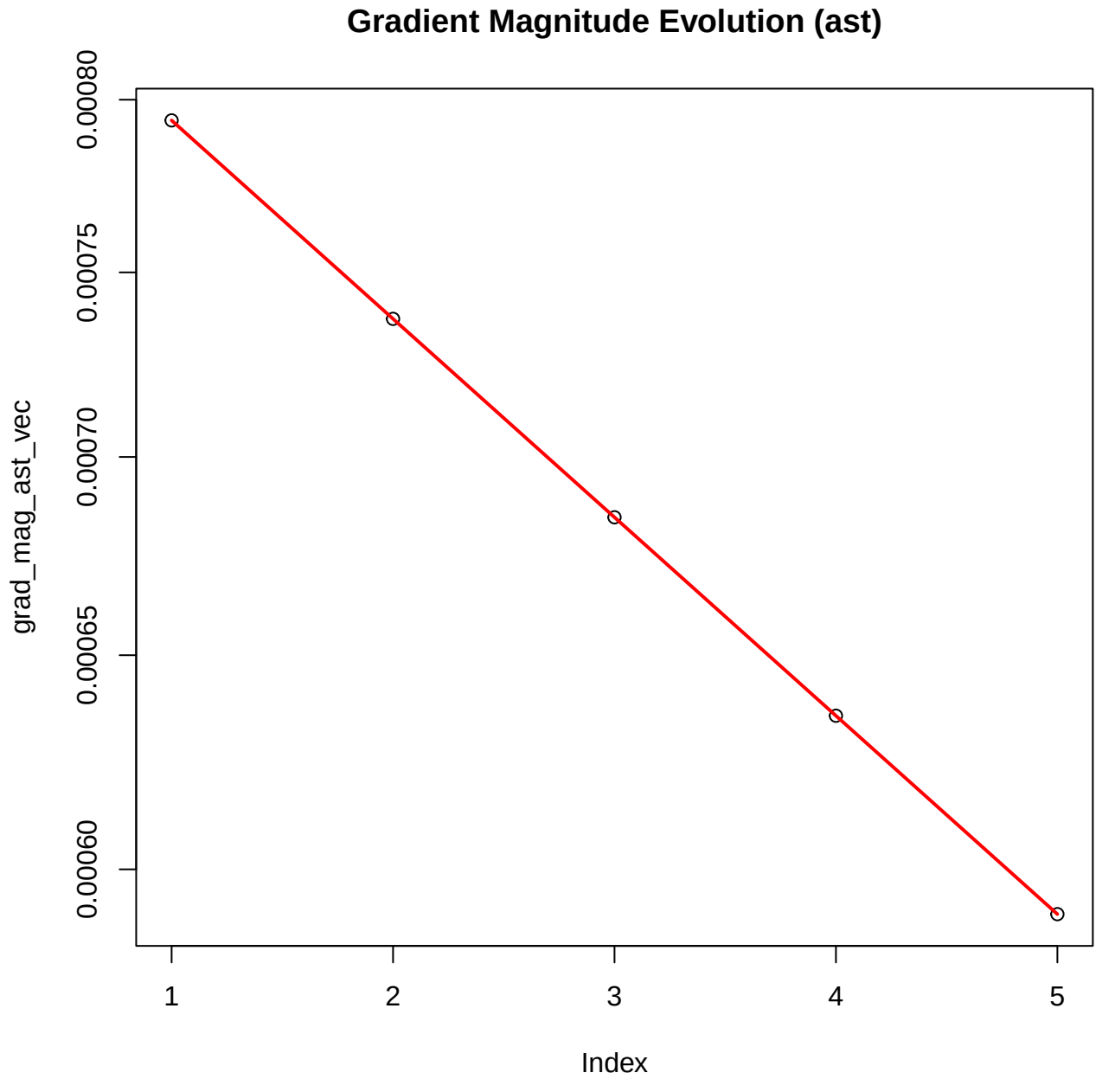

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

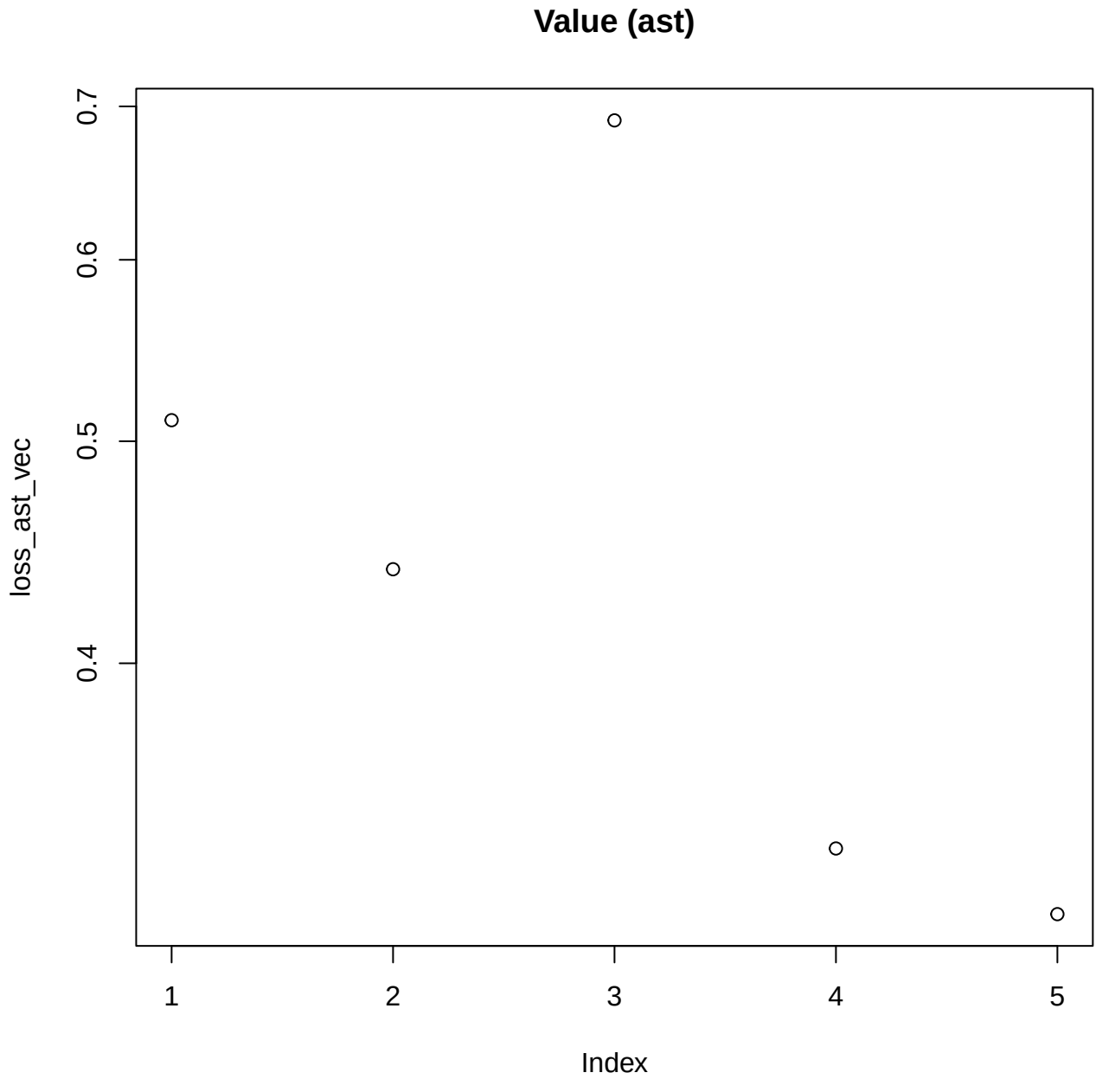

Value (ast)
loss_ast_vec
0.1
0.2
0.3
0.4
0.5
0.6
1
2
3
4
5
Index

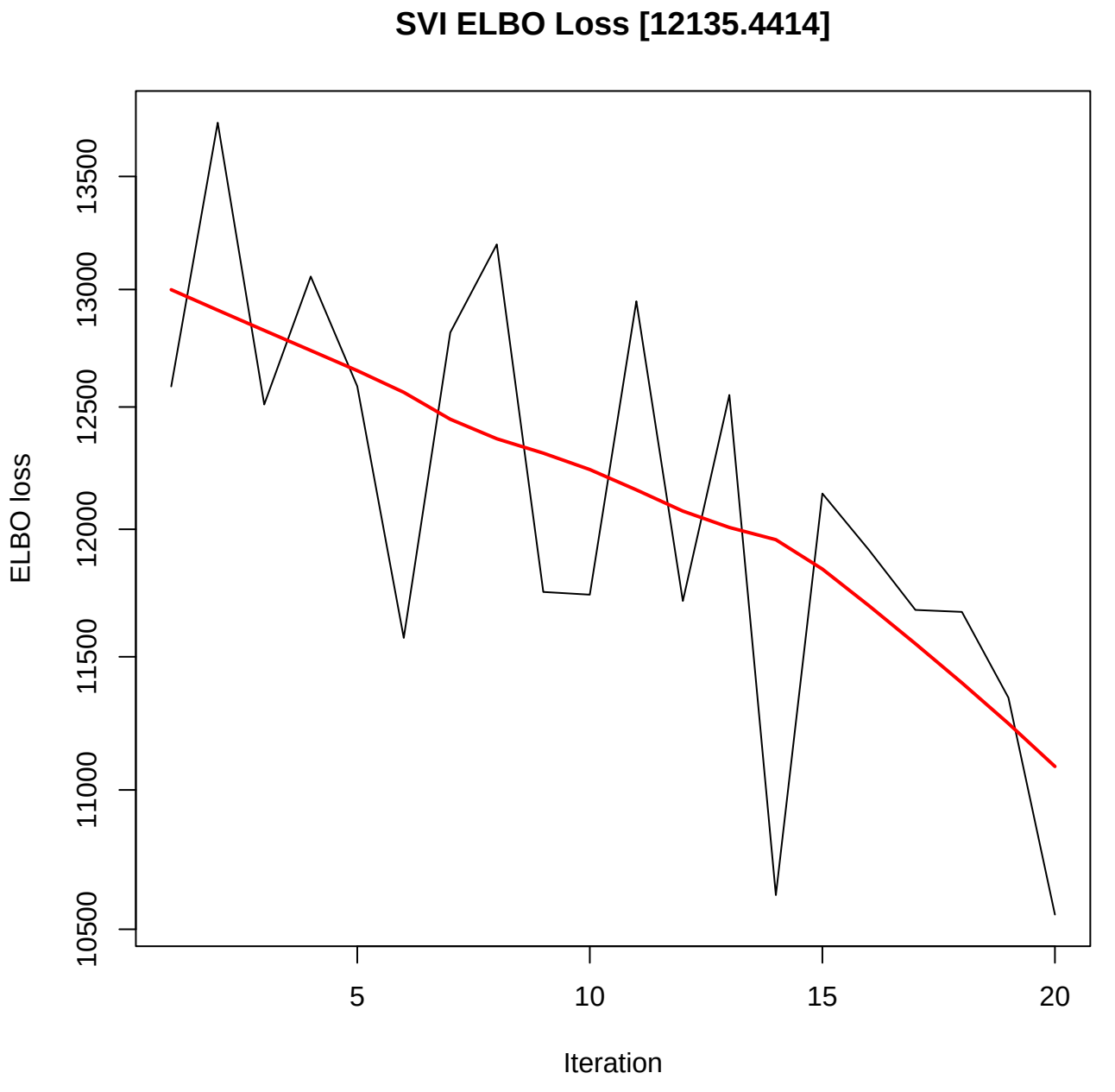

SVI ELBO Loss [14648.0652]
ELBO loss
16000
15500
15000
14500
14000
13500
5
10
15
20
Iteration

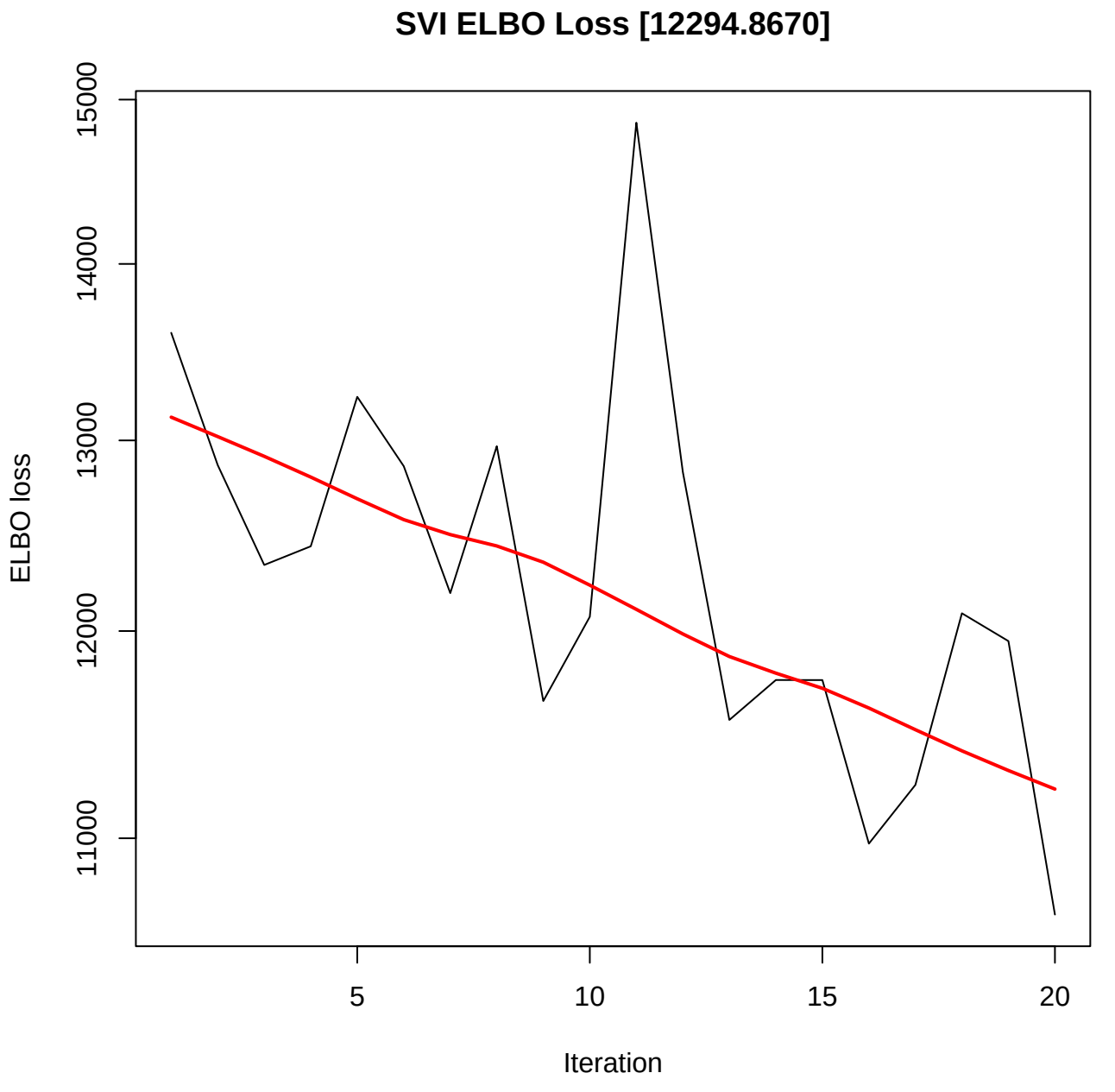

SVI ELBO Loss [15404.8256]
ELBO loss
18000
17000
16000
15000
14000
5
10
15
20
Iteration

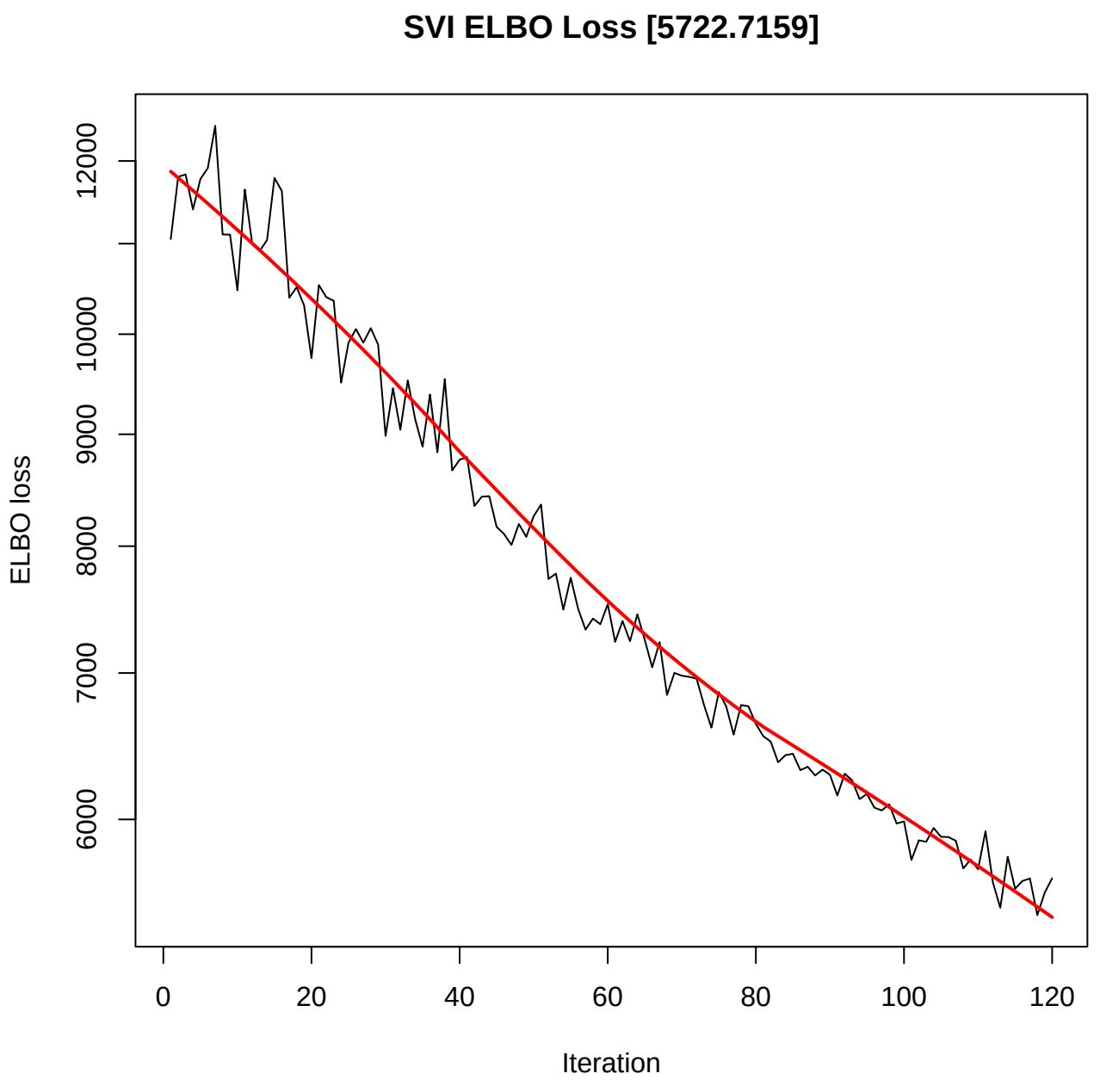

SVI ELBO Loss [8419.5587]
ELBO loss
10000
15000
20000
25000
30000
0
50
100
150
Iteration

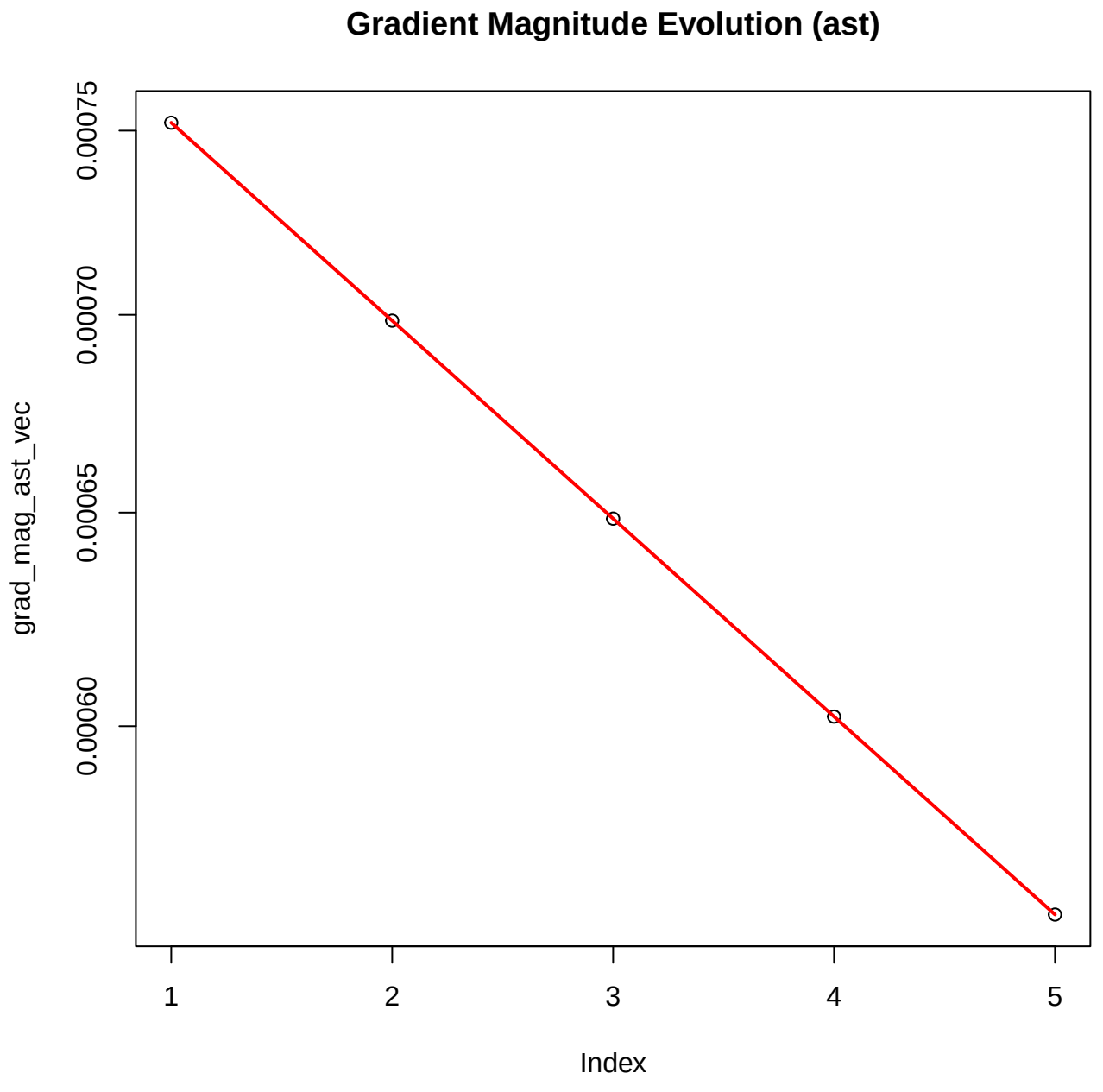

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

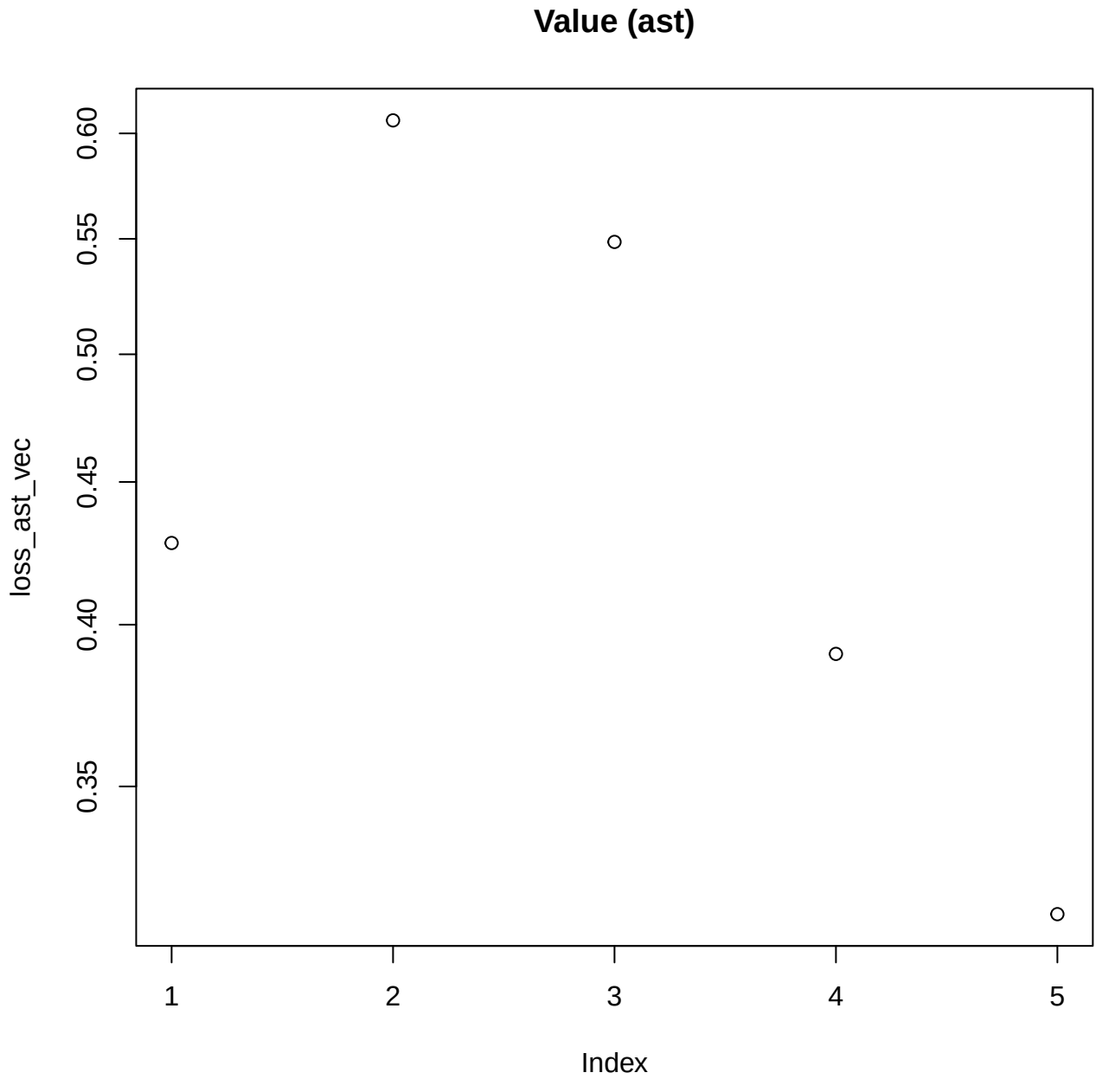

Value (ast)
loss_ast_vec
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

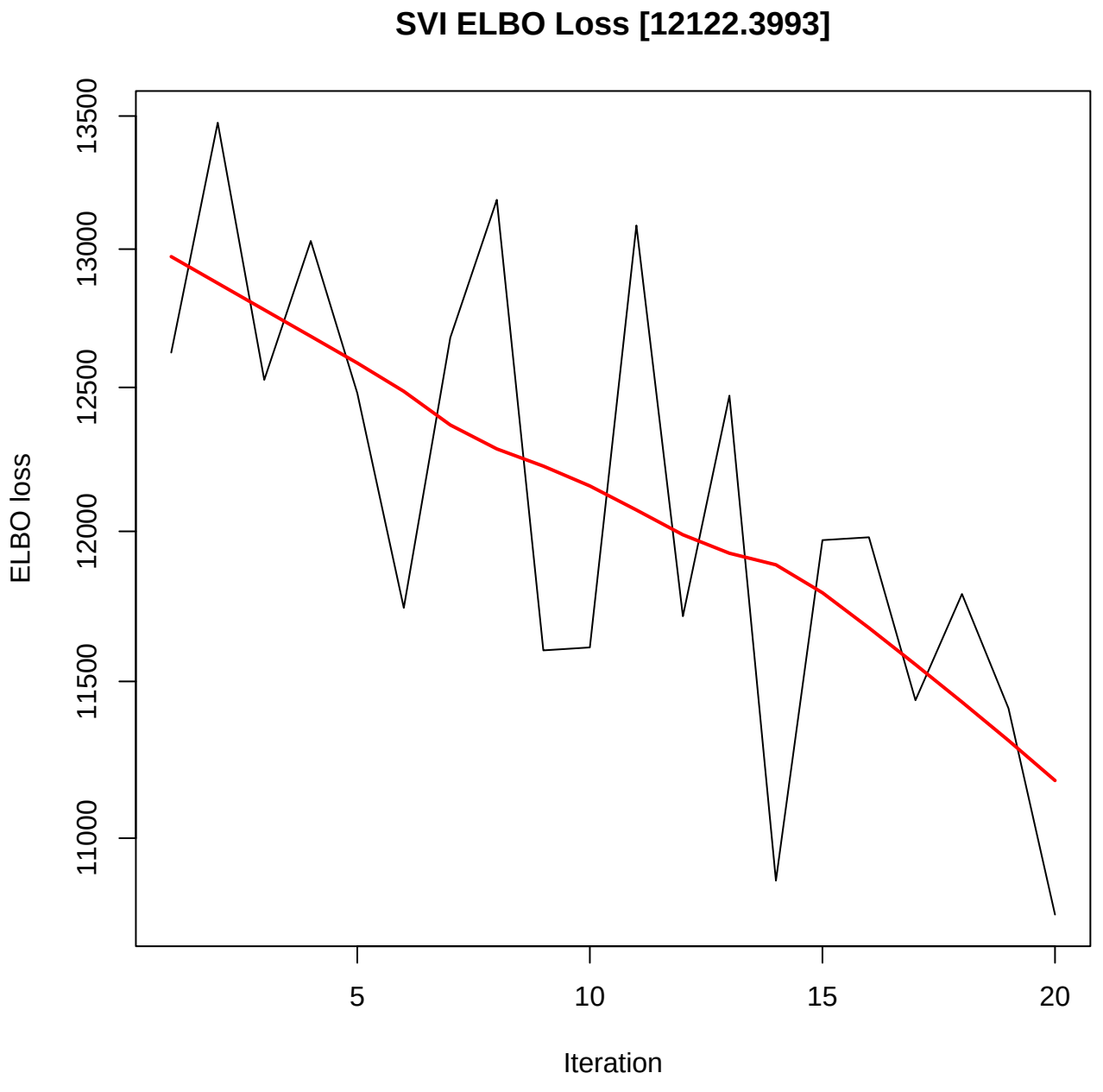

SVI ELBO Loss [12135.4414]
ELBO loss
10500
11000
11500
12000
12500
13000
13500
5
10
15
20
Iteration

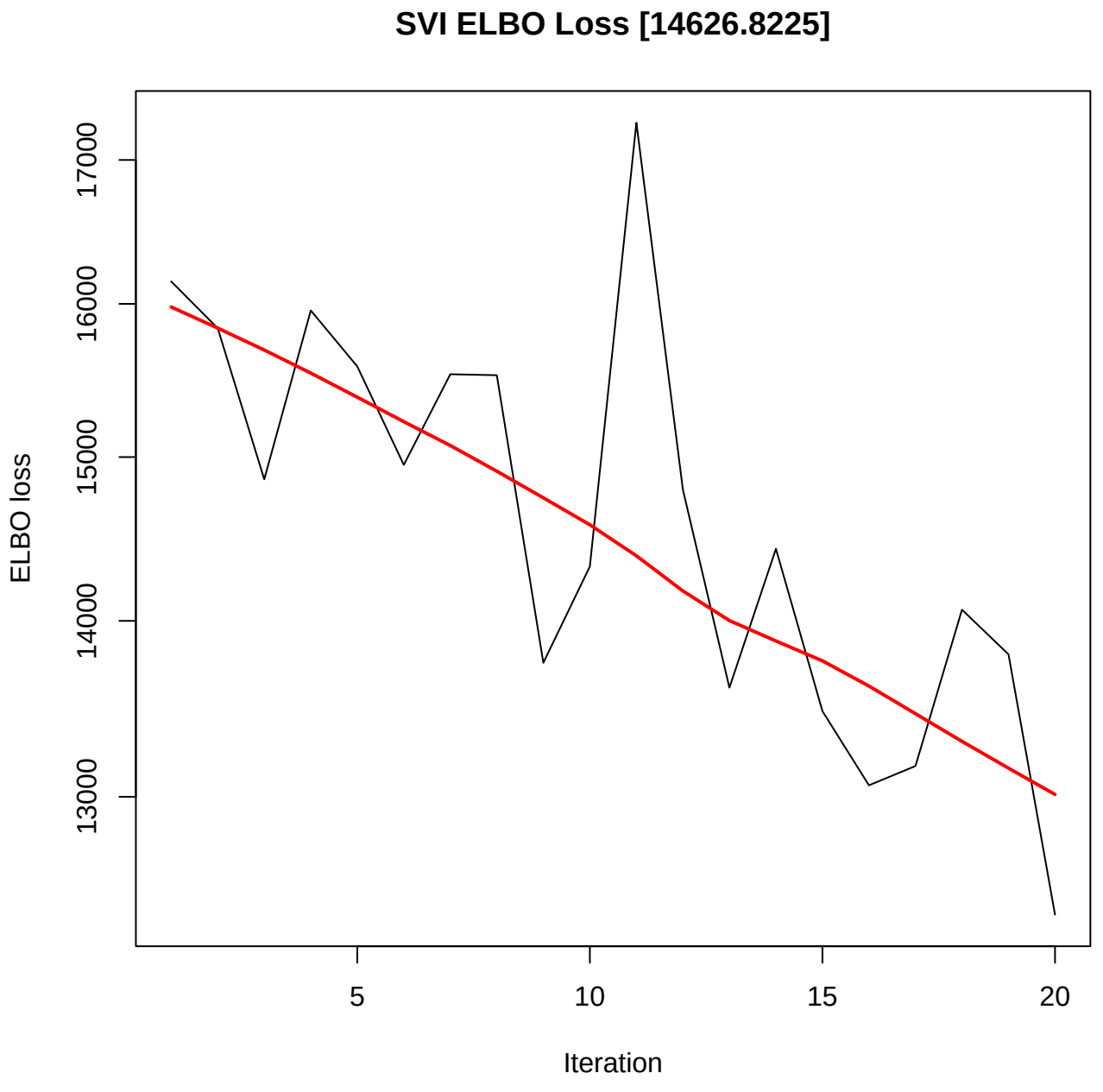

SVI ELBO Loss [12294.8670]
ELBO loss
15000
14000
13000
12000
11000
5
10
15
20
Iteration

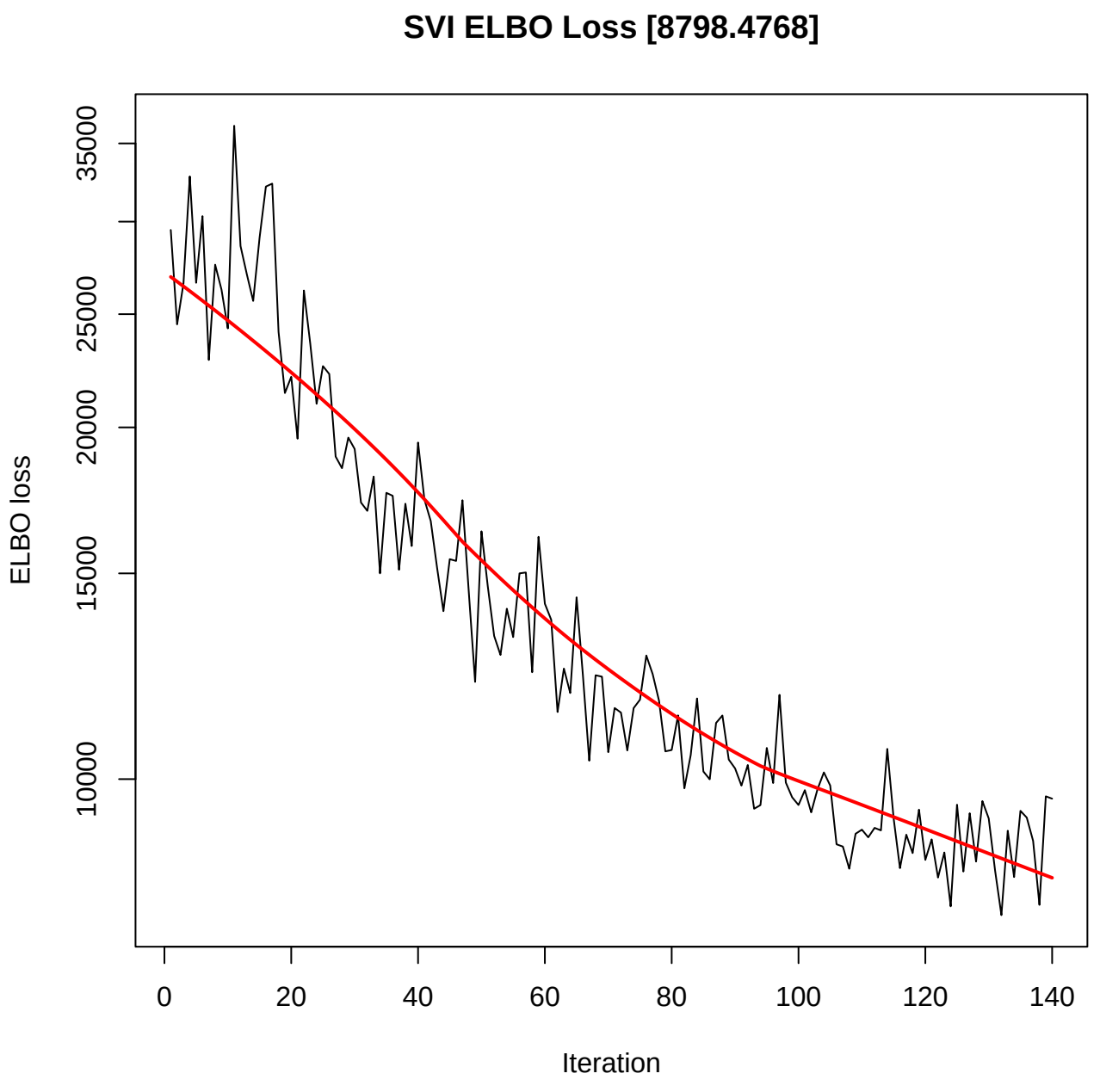

SVI ELBO Loss [5722.7159]
ELBO loss
12000
10000
9000
8000
7000
6000
0
20
40
60
80
100
120
Iteration

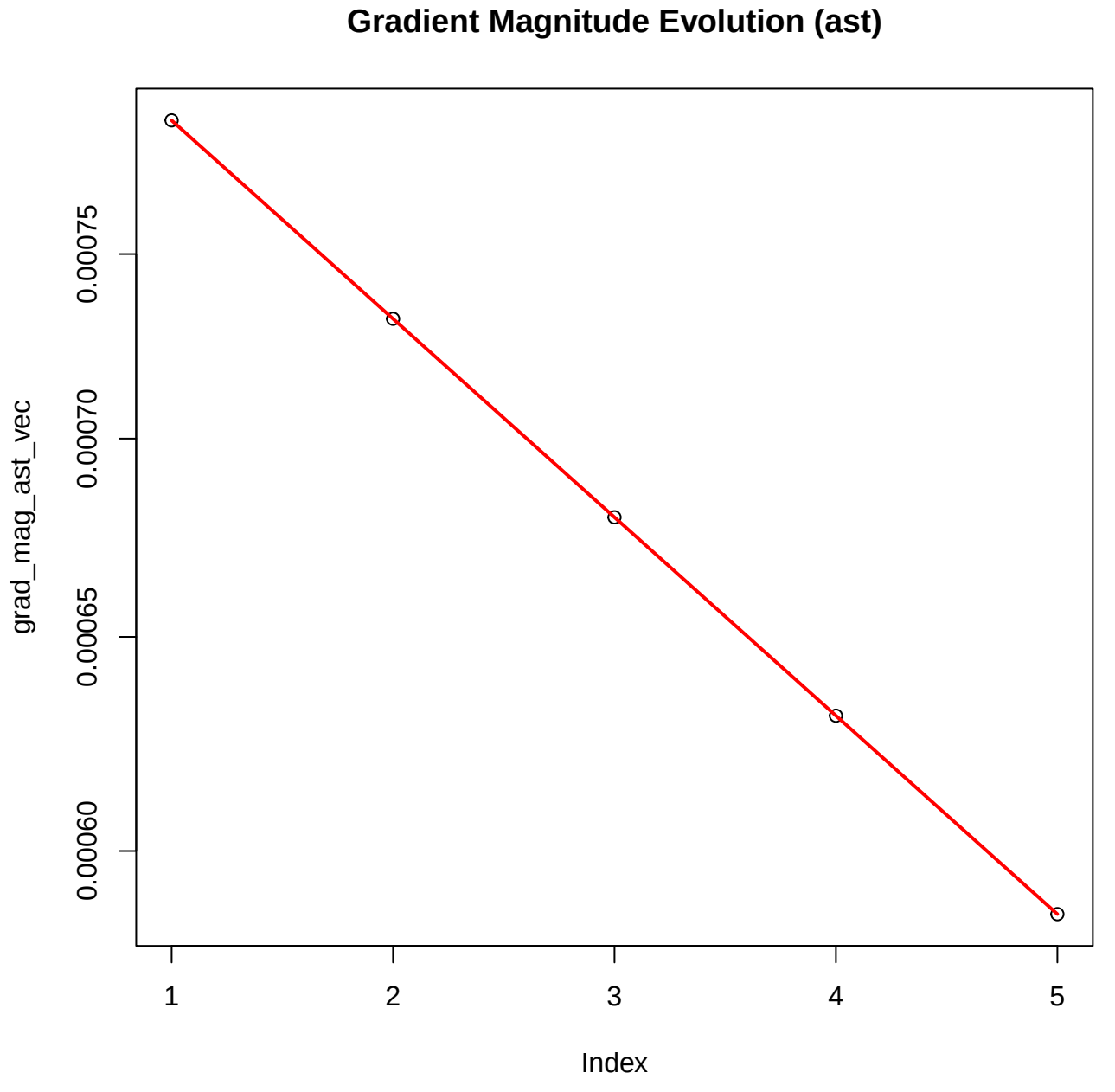

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

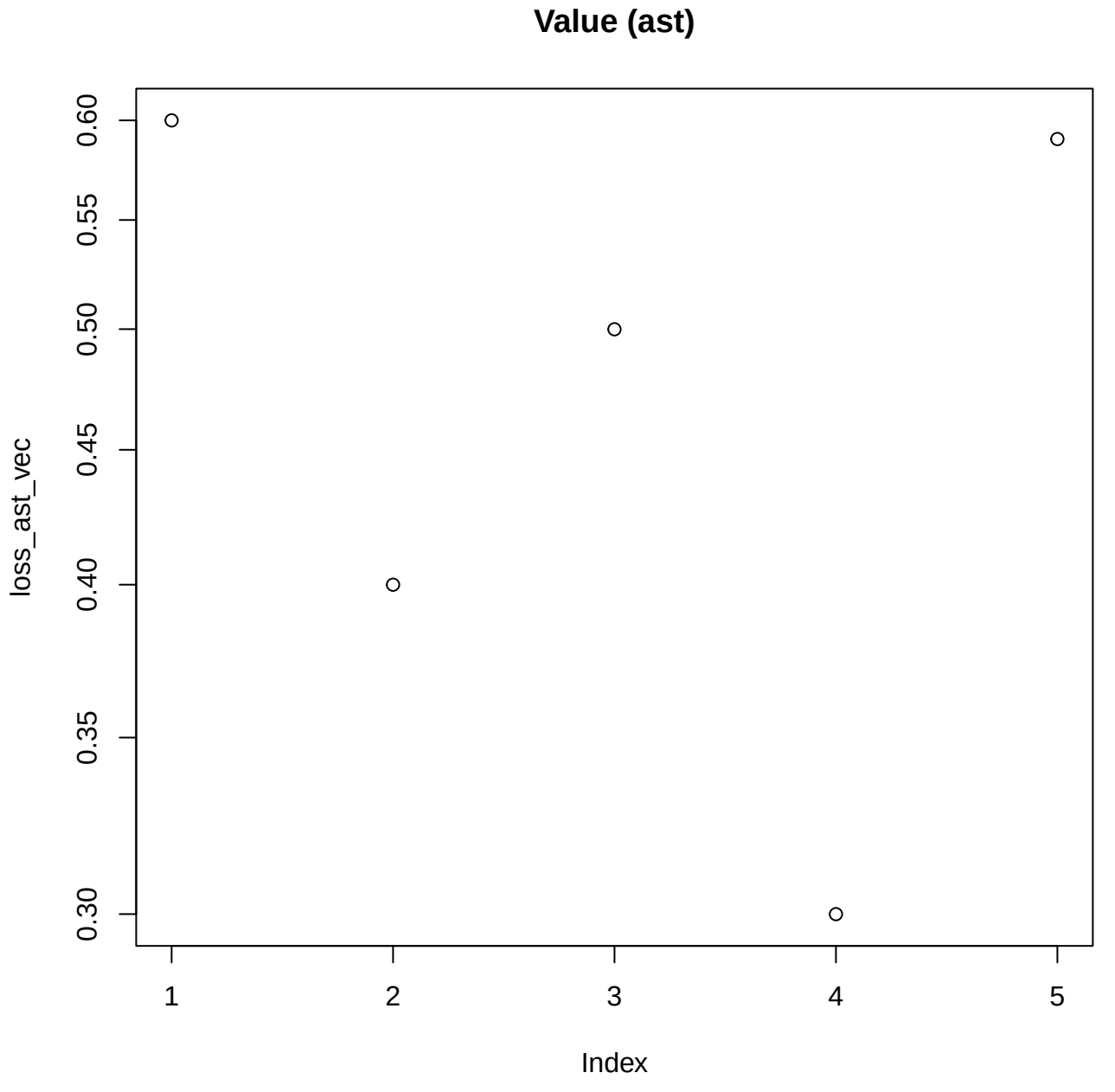

Value (ast)
loss_ast_vec
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

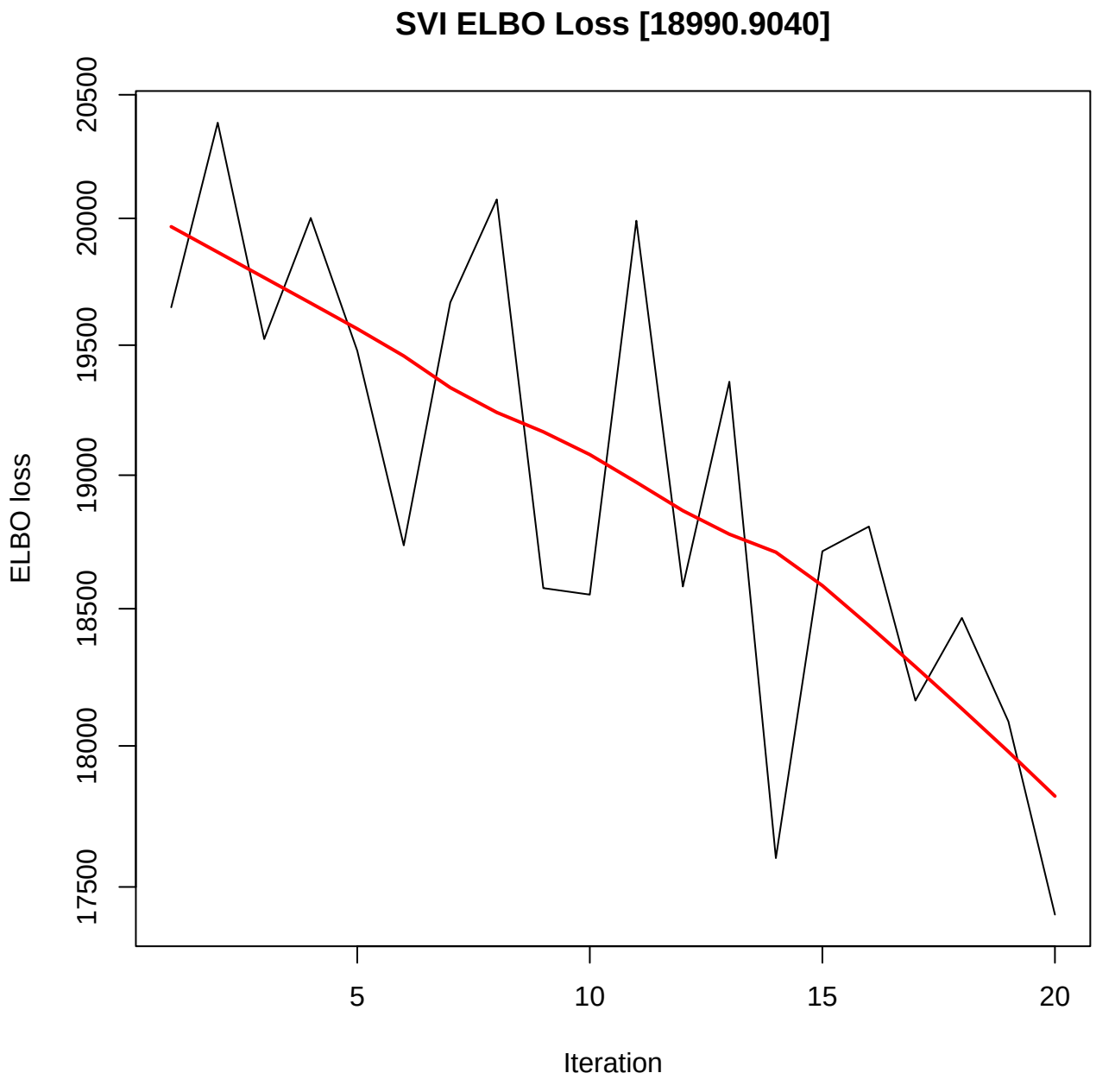

SVI ELBO Loss [12122.3993]
ELBO loss
13500
13000
12500
12000
11500
11000
5
10
15
20
Iteration

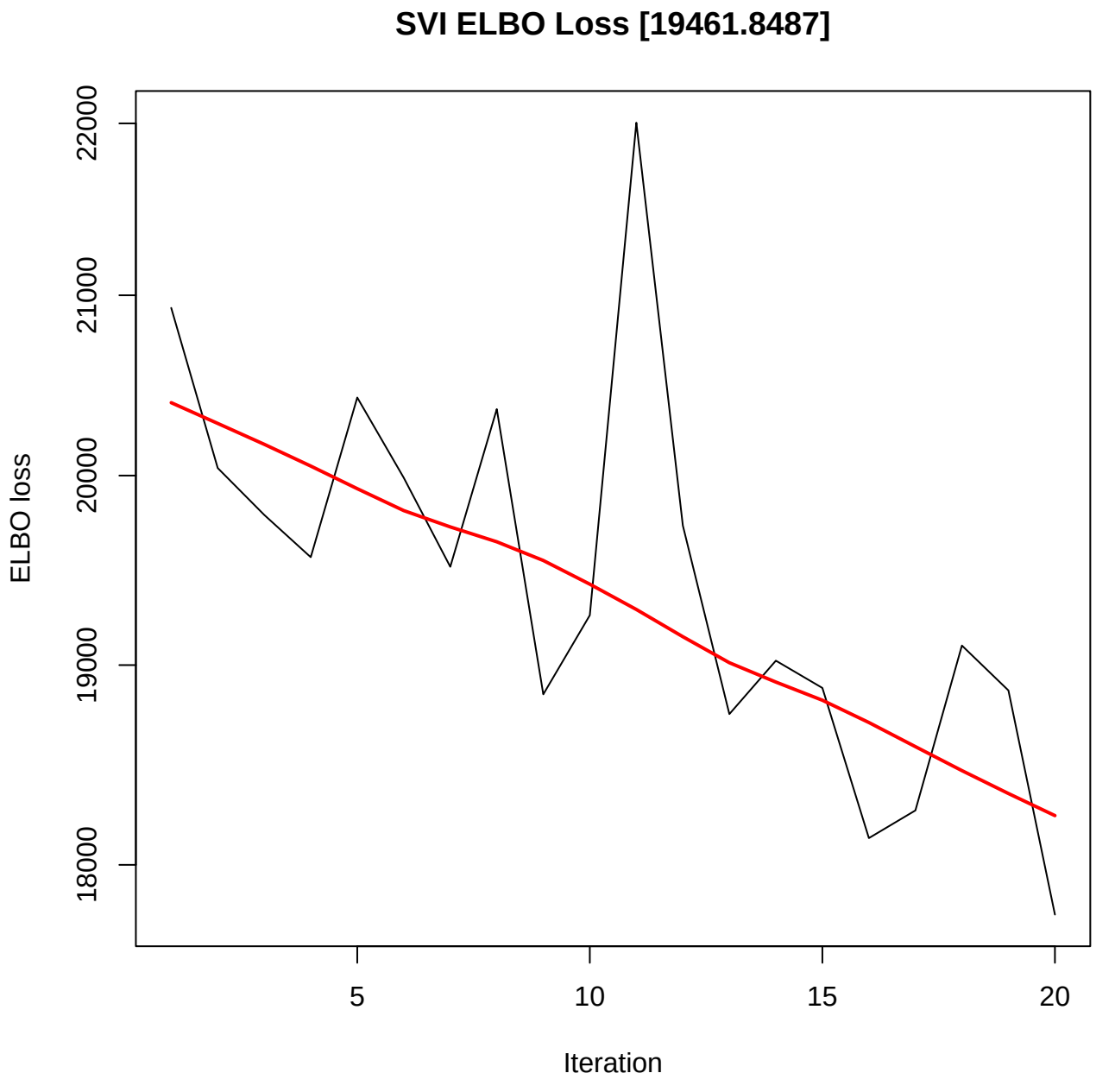

SVI ELBO Loss [14626.8225]
ELBO loss
17000
16000
15000
14000
13000
5
10
15
20
Iteration

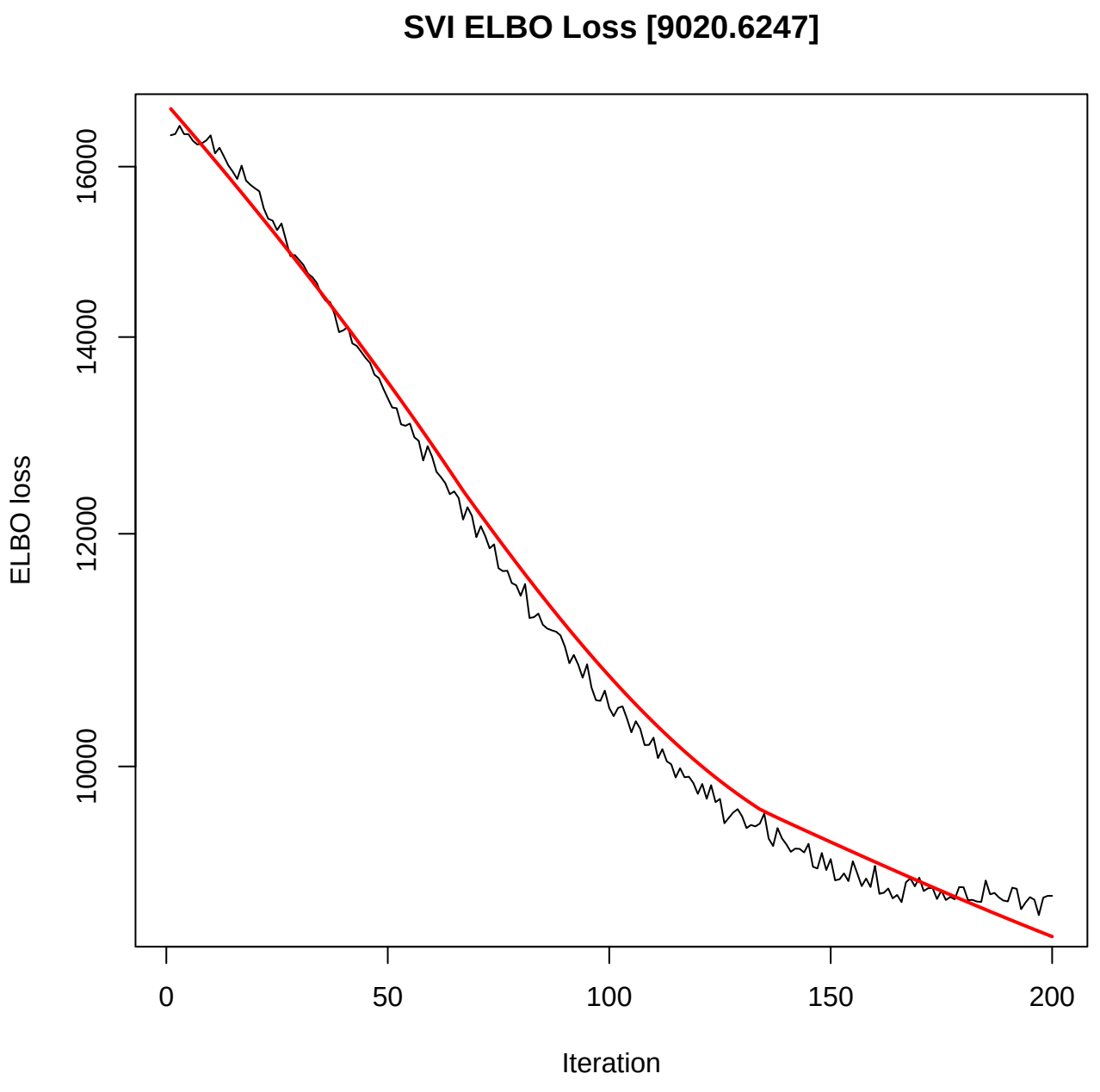

SVI ELBO Loss [8798.4768]
ELBO loss
35000
25000
20000
15000
10000
0
20
40
60
80
100
120
140
Iteration

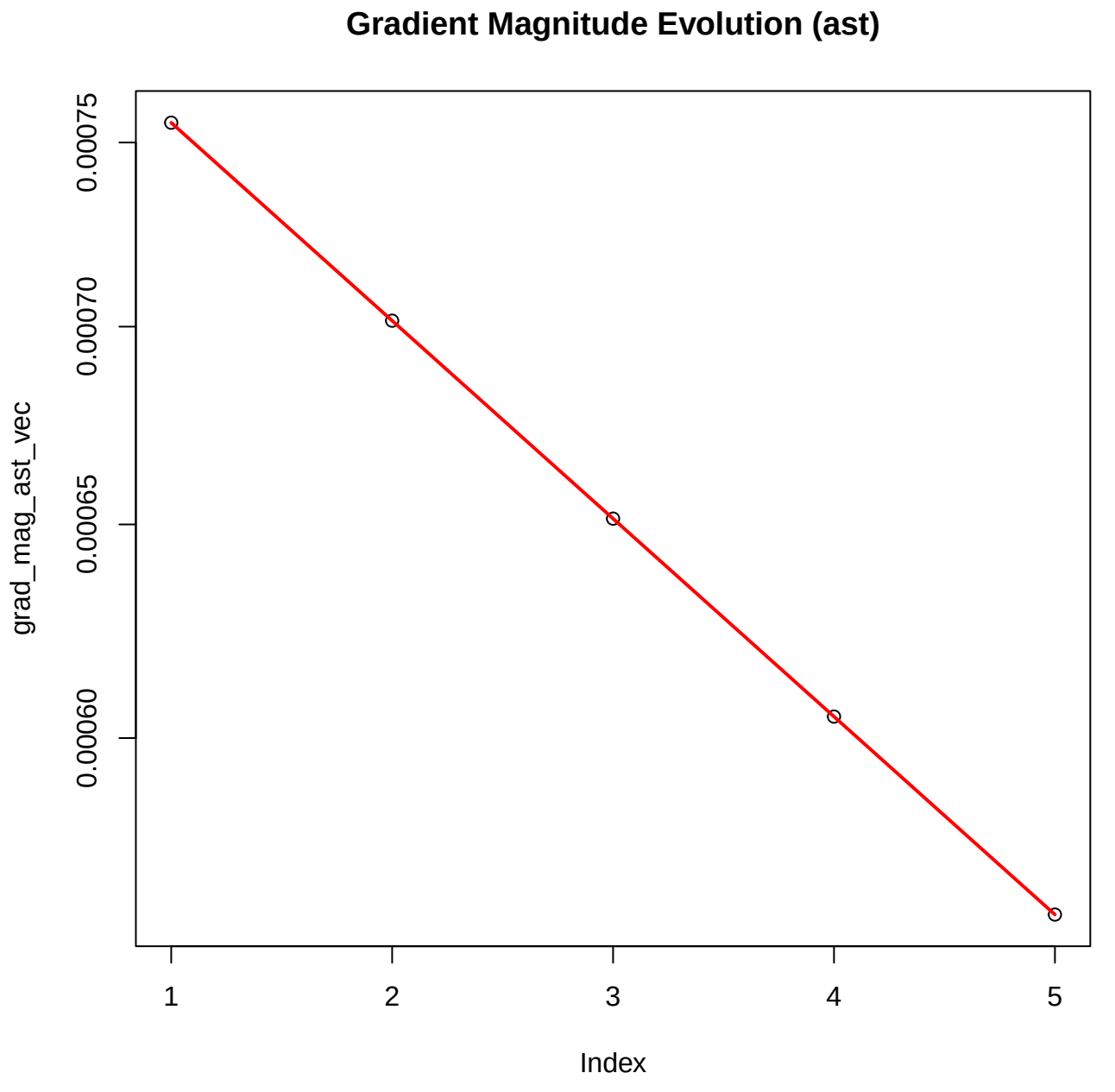

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

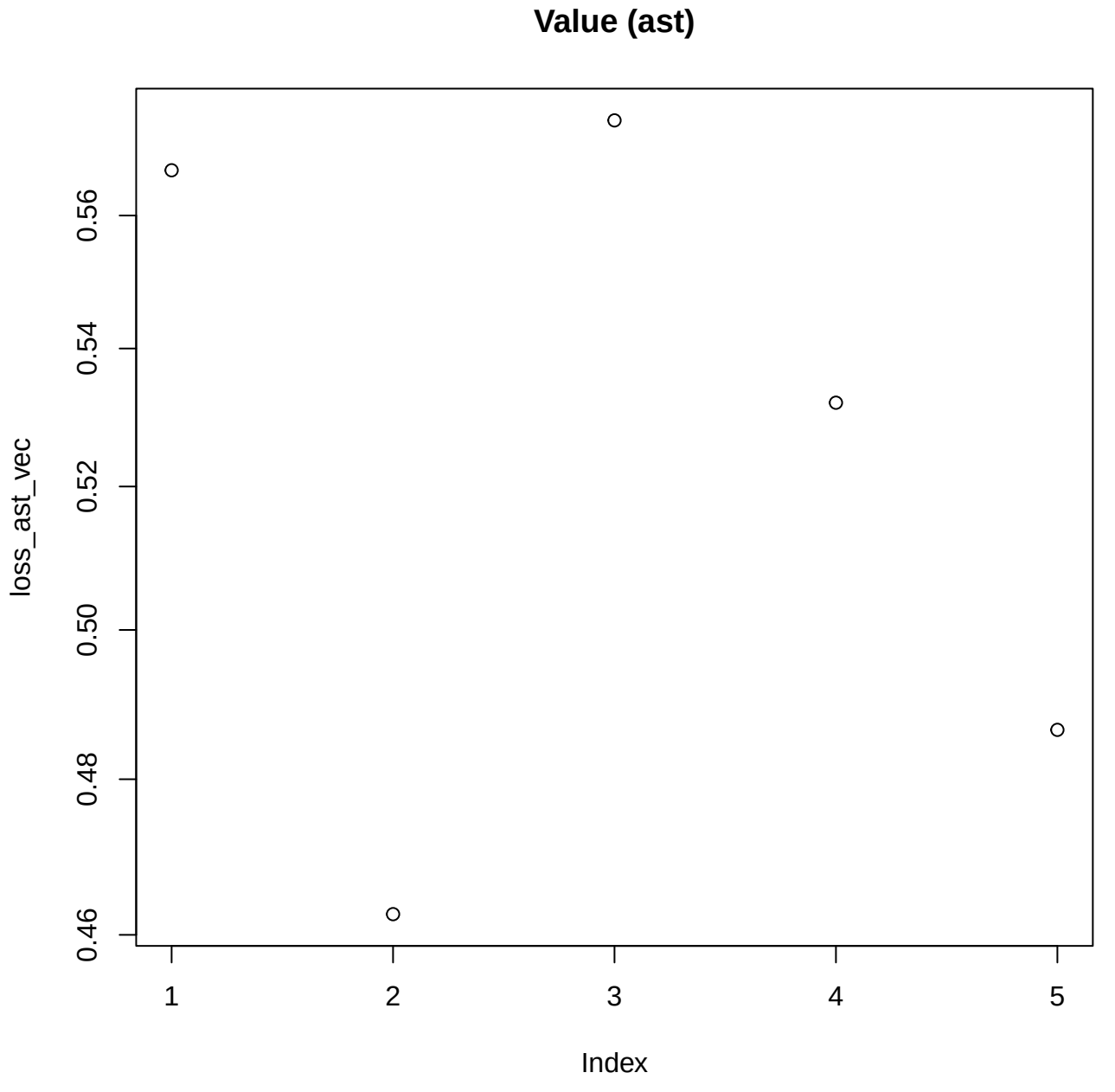

Value (ast)
loss_ast_vec
0.30
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

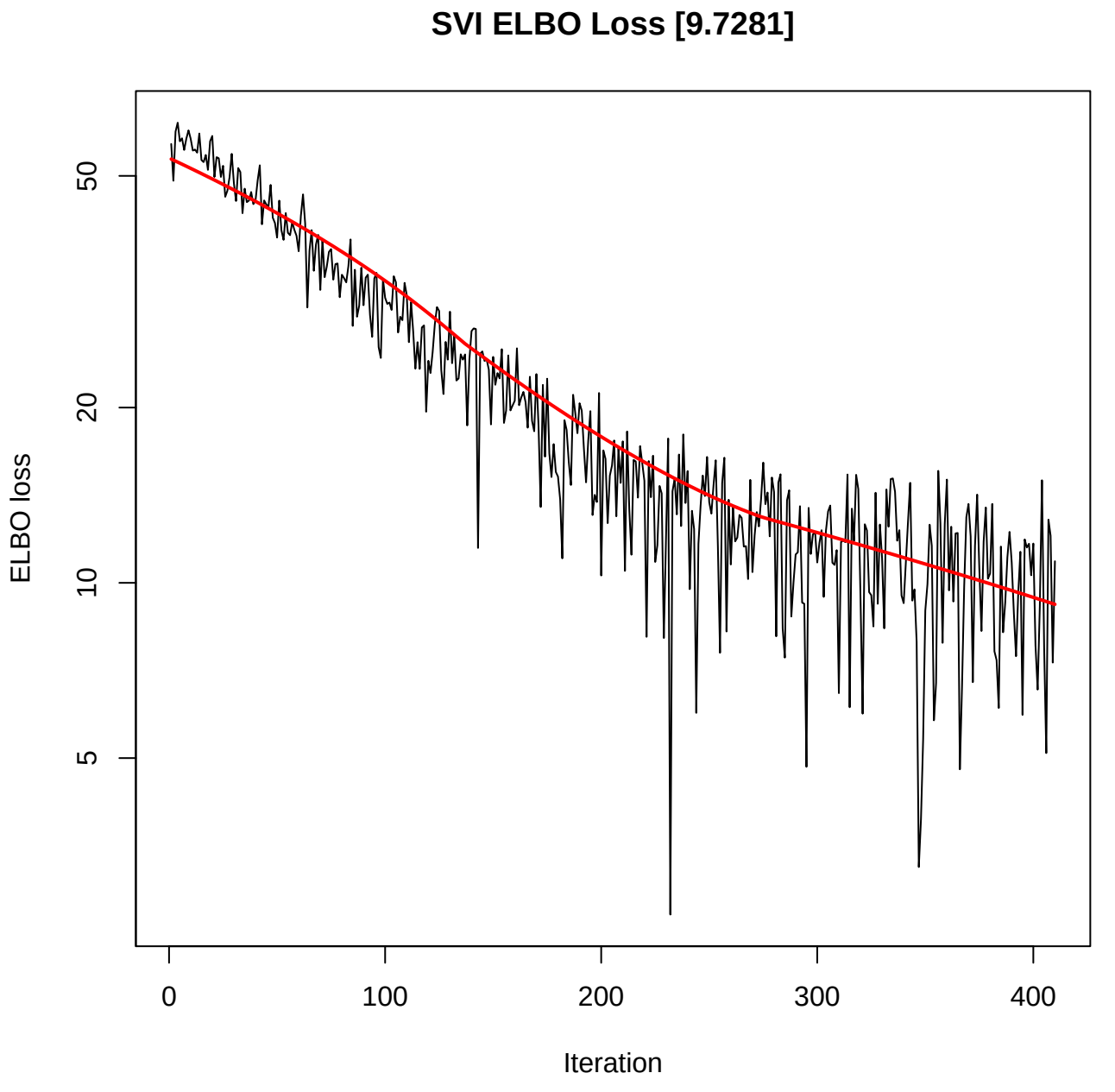

SVI ELBO Loss [18990.9040]
ELBO loss
20500
20000
19500
19000
18500
18000
17500
5
10
15
20
Iteration

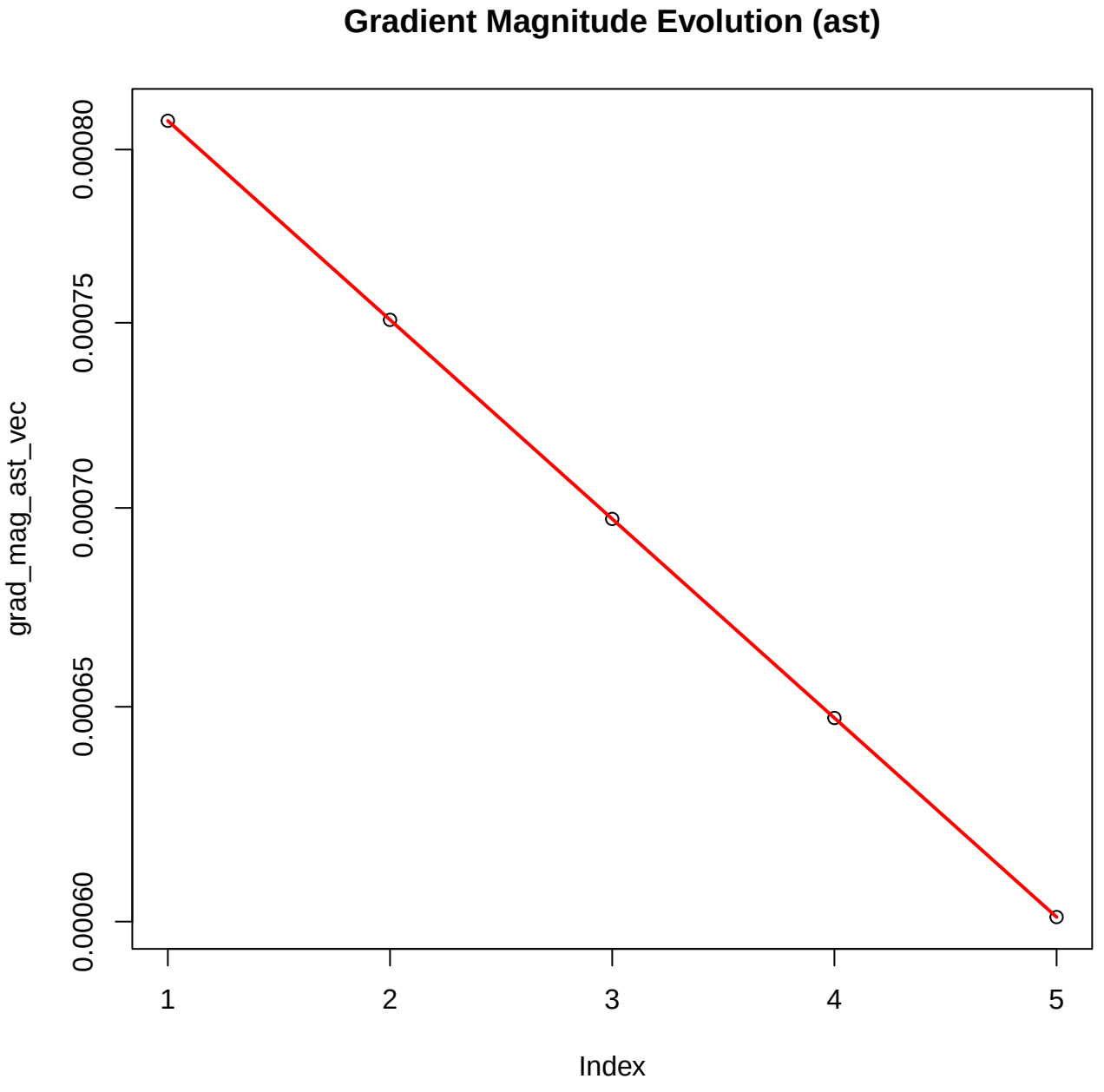

SVI ELBO Loss [19461.8487]
ELBO loss
22000
21000
20000
19000
18000
5
10
15
20
Iteration

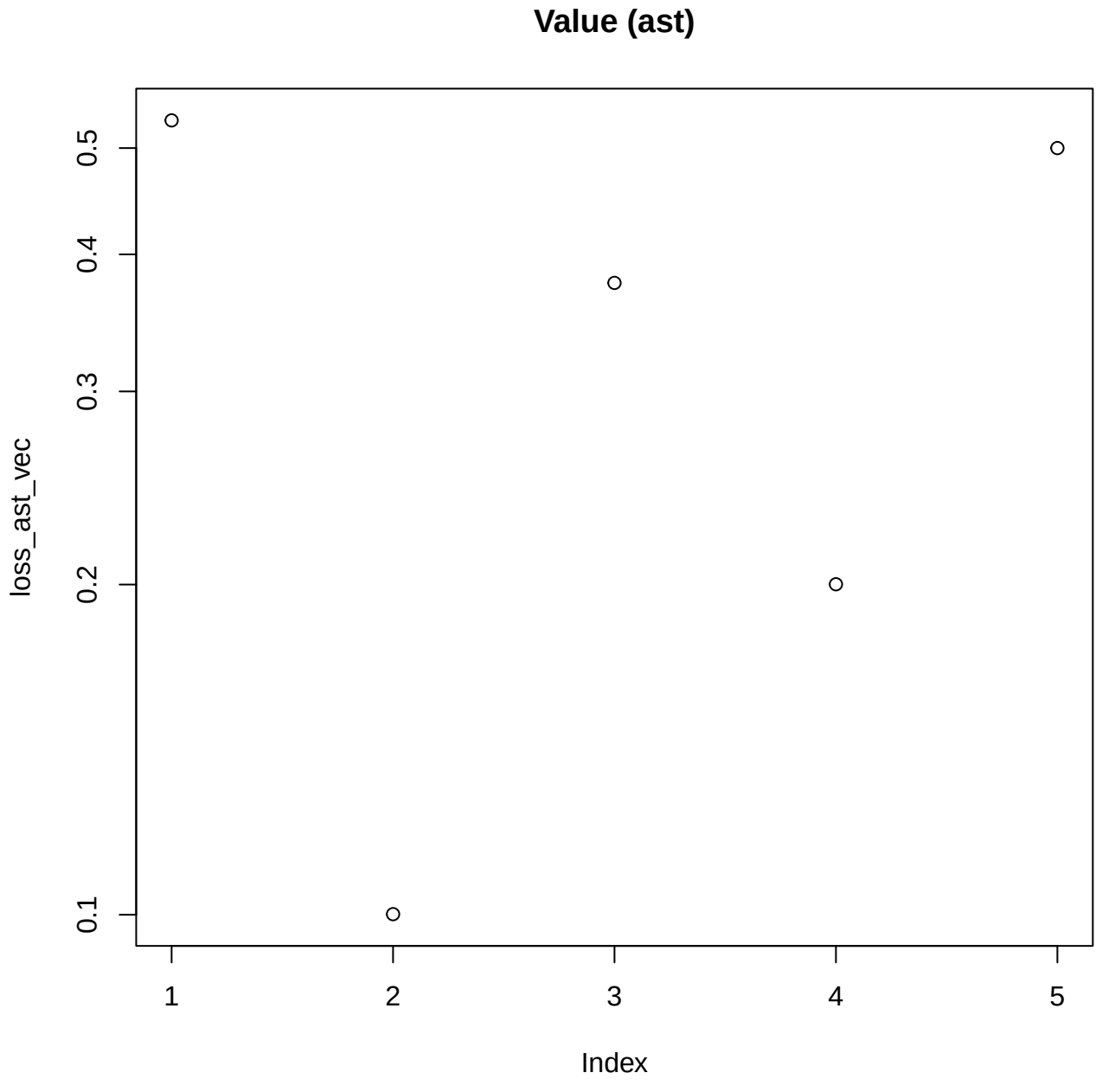

SVI ELBO Loss [9020.6247]
ELBO loss
16000
14000
12000
10000
0
50
100
150
200
Iteration

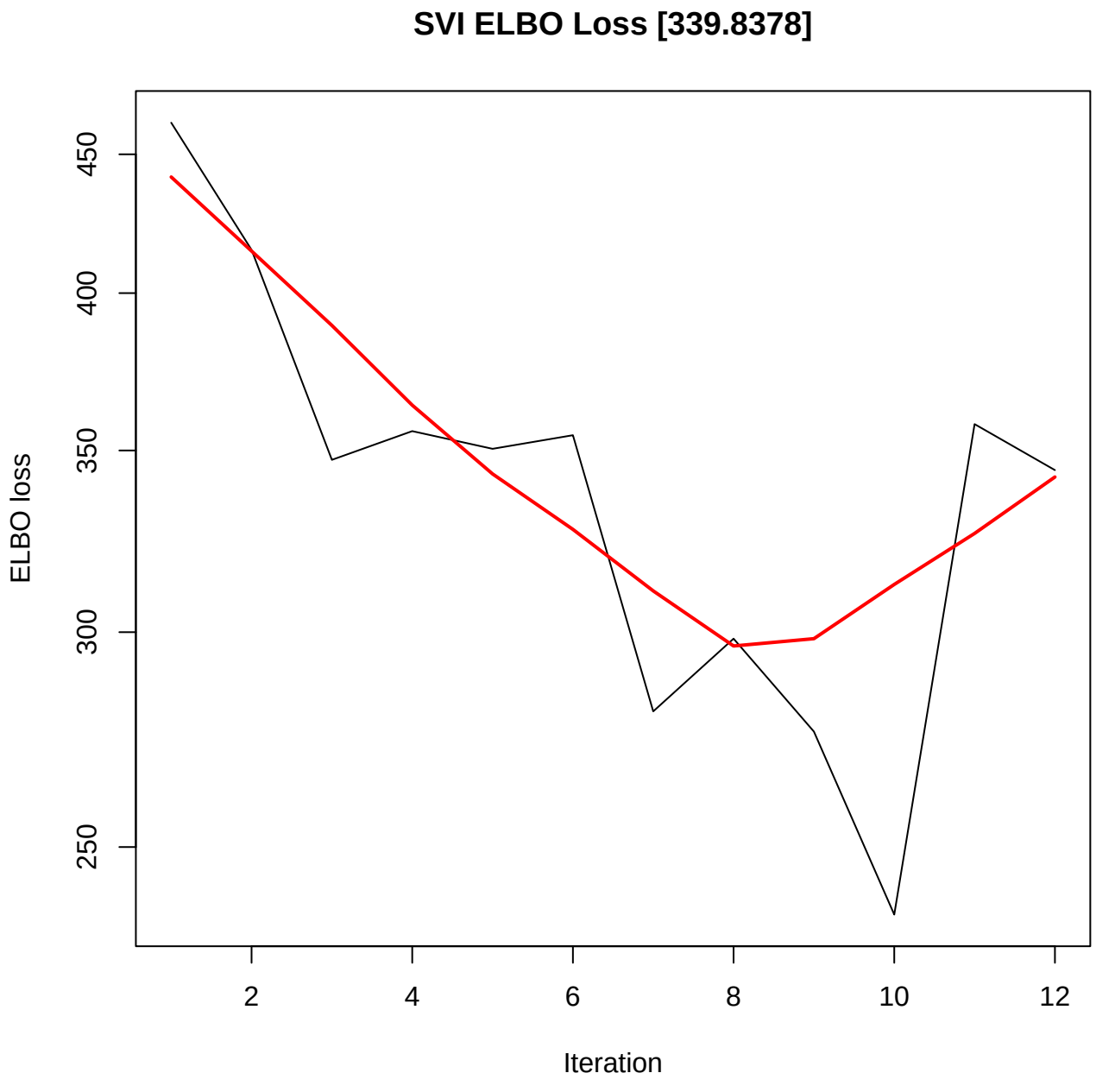

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

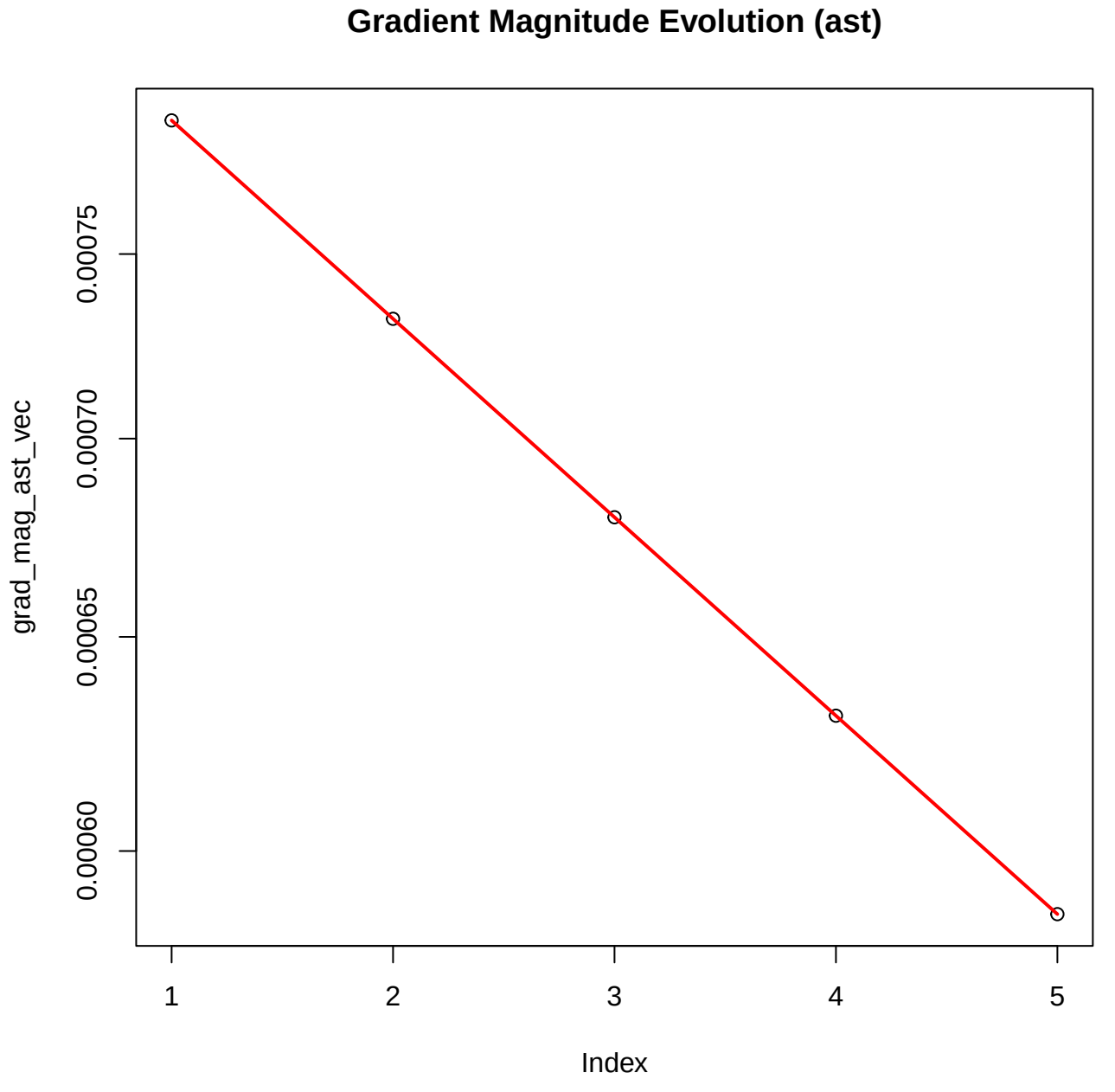

Value (ast)
loss_ast_vec
0.46
0.48
0.50
0.52
0.54
0.56
1
2
3
4
5
Index

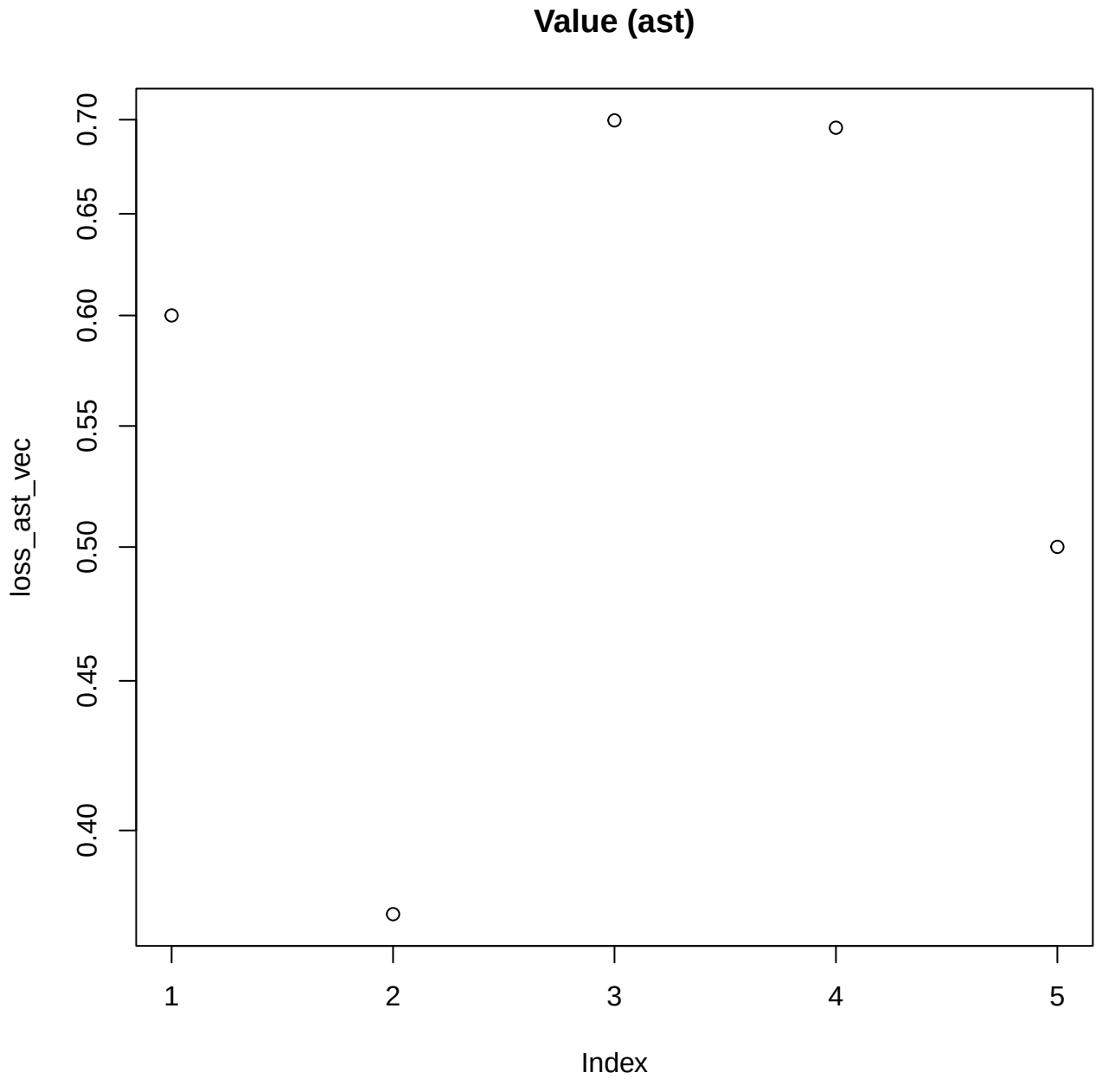

SVI ELBO Loss [9.7281]
ELBO loss
50
20
10
5
0
100
200
300
400
Iteration

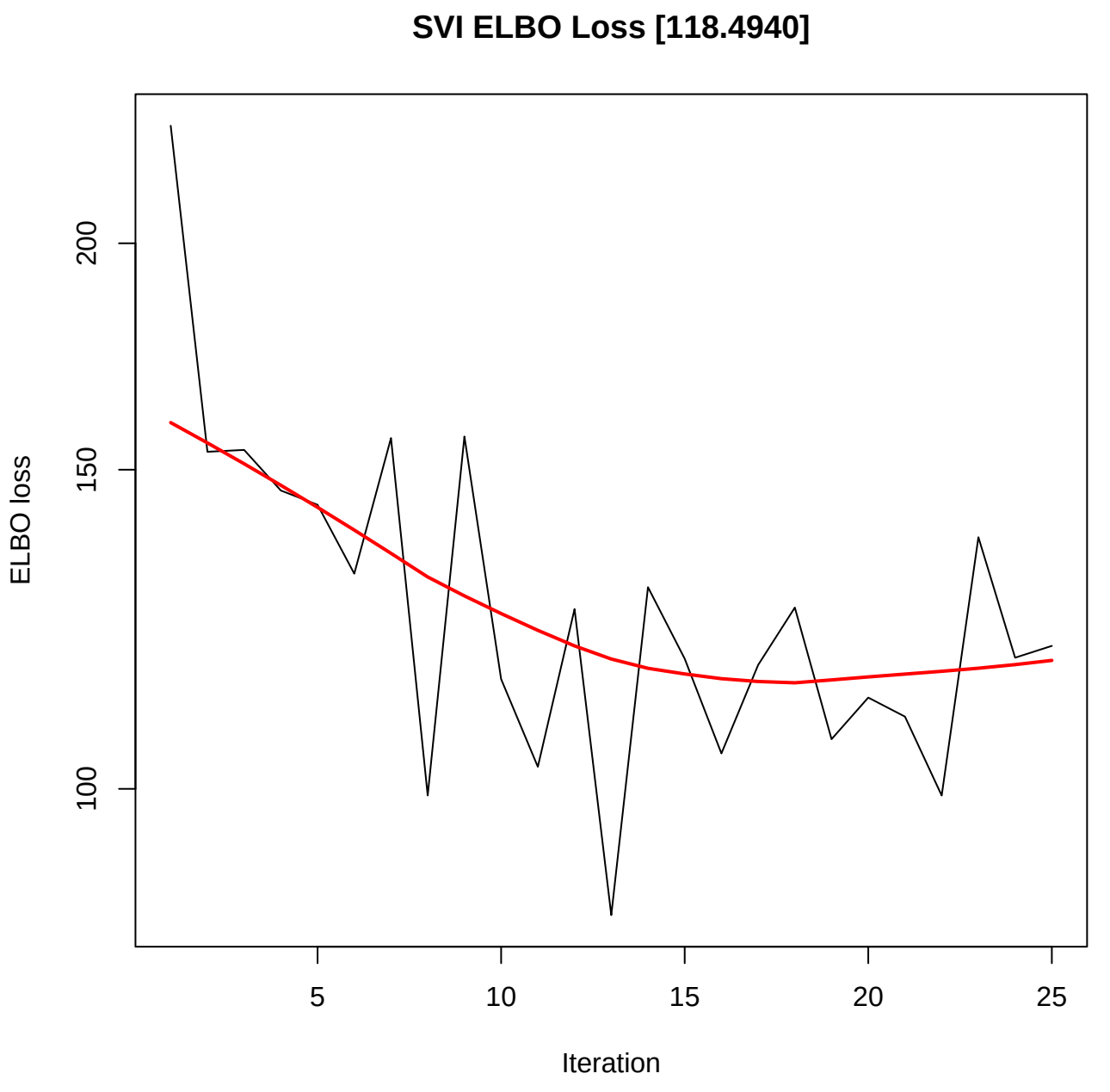

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

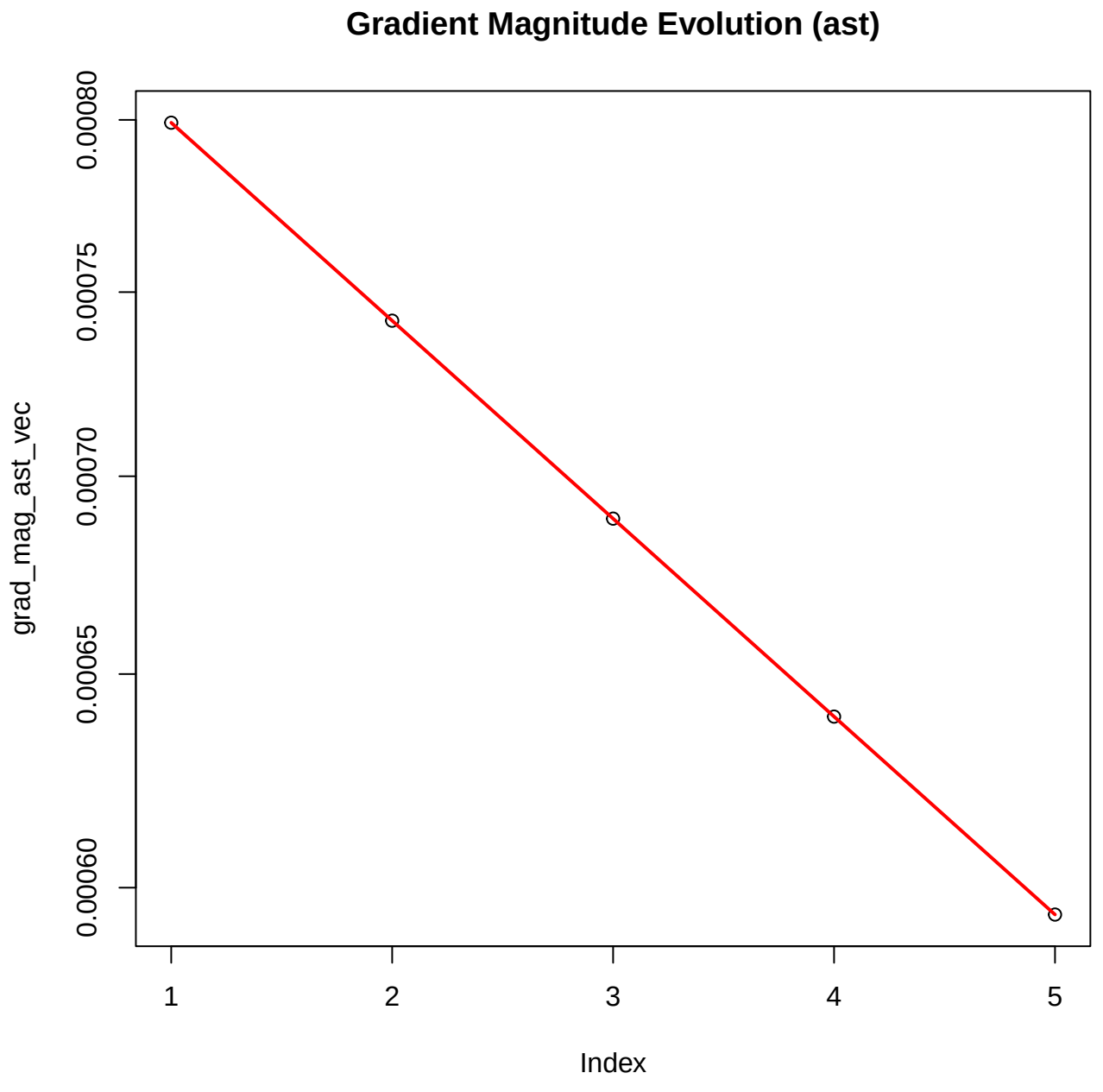

Value (ast)
loss_ast_vec
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
Index

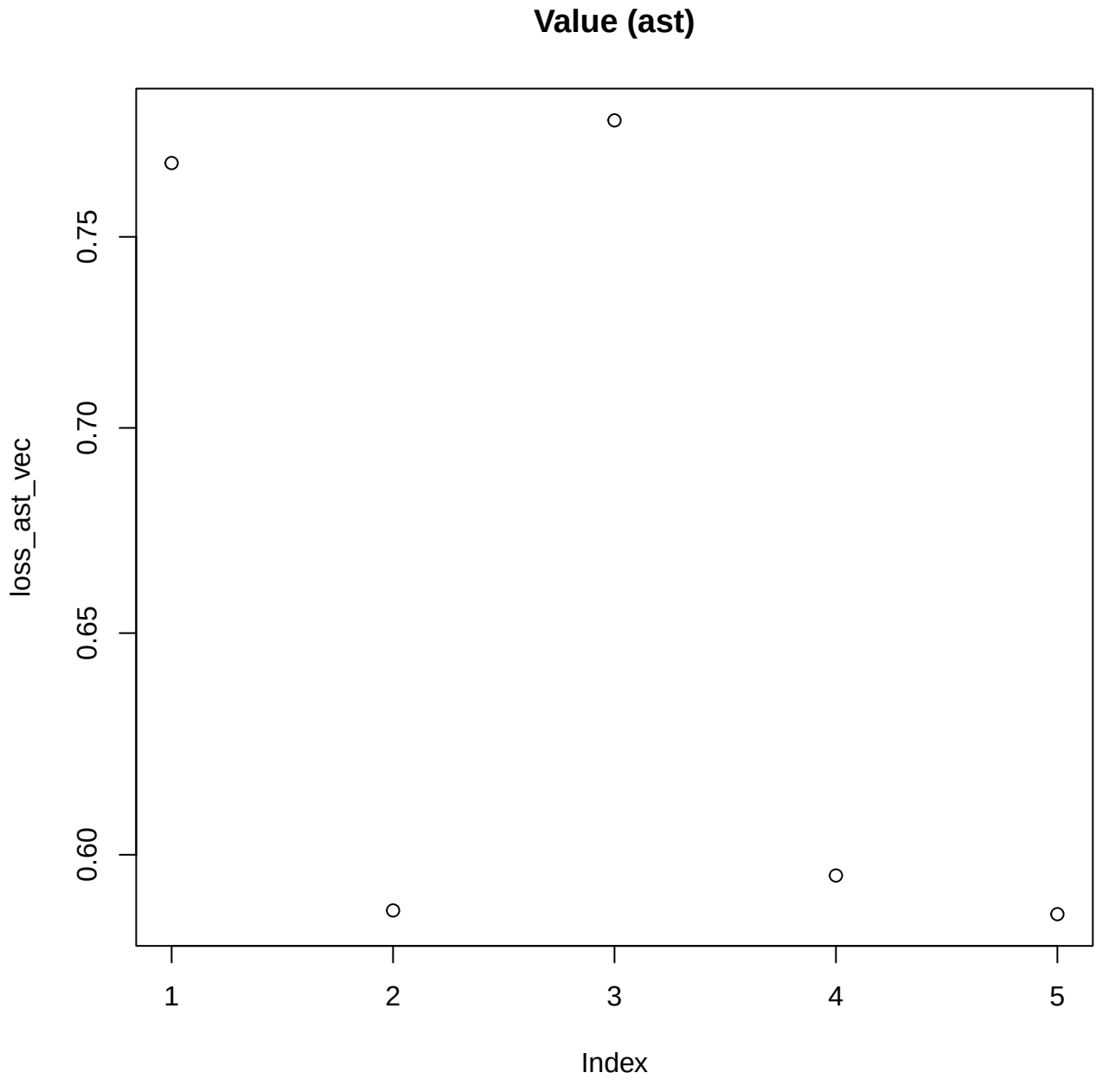

SVI ELBO Loss [339.8378]
ELBO loss
450
400
350
300
250
2
4
6
8
10
12
Iteration

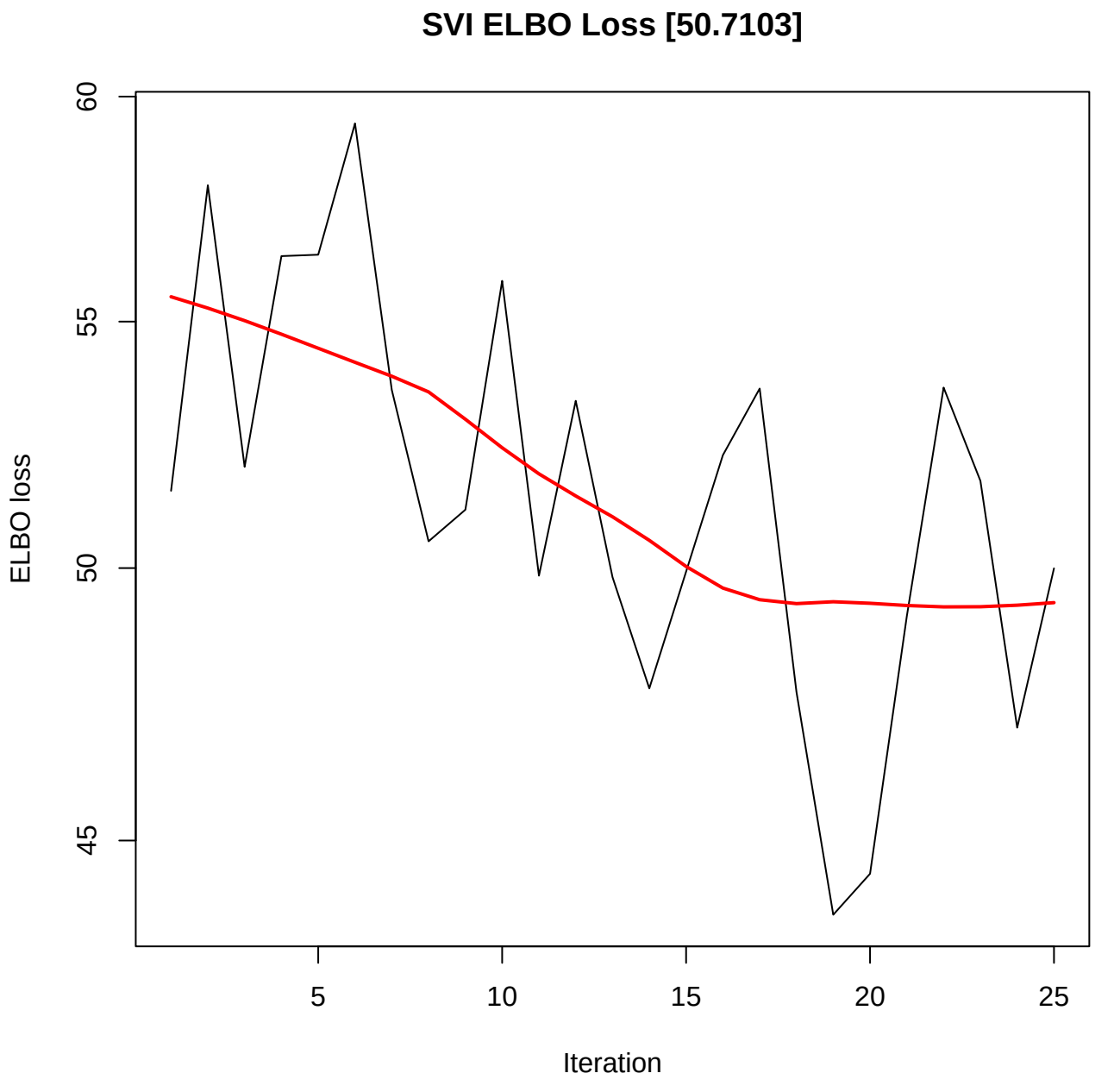

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

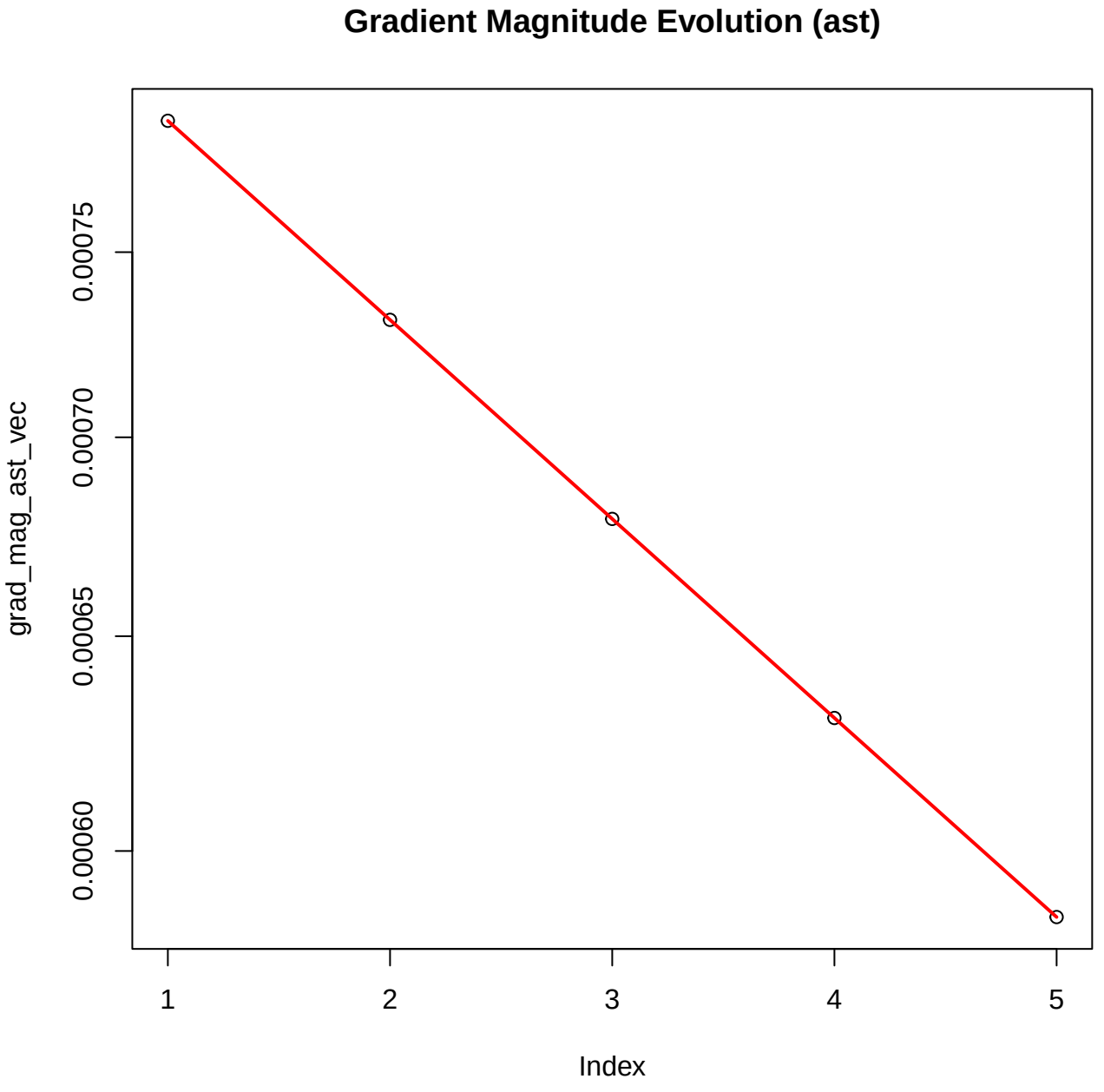

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

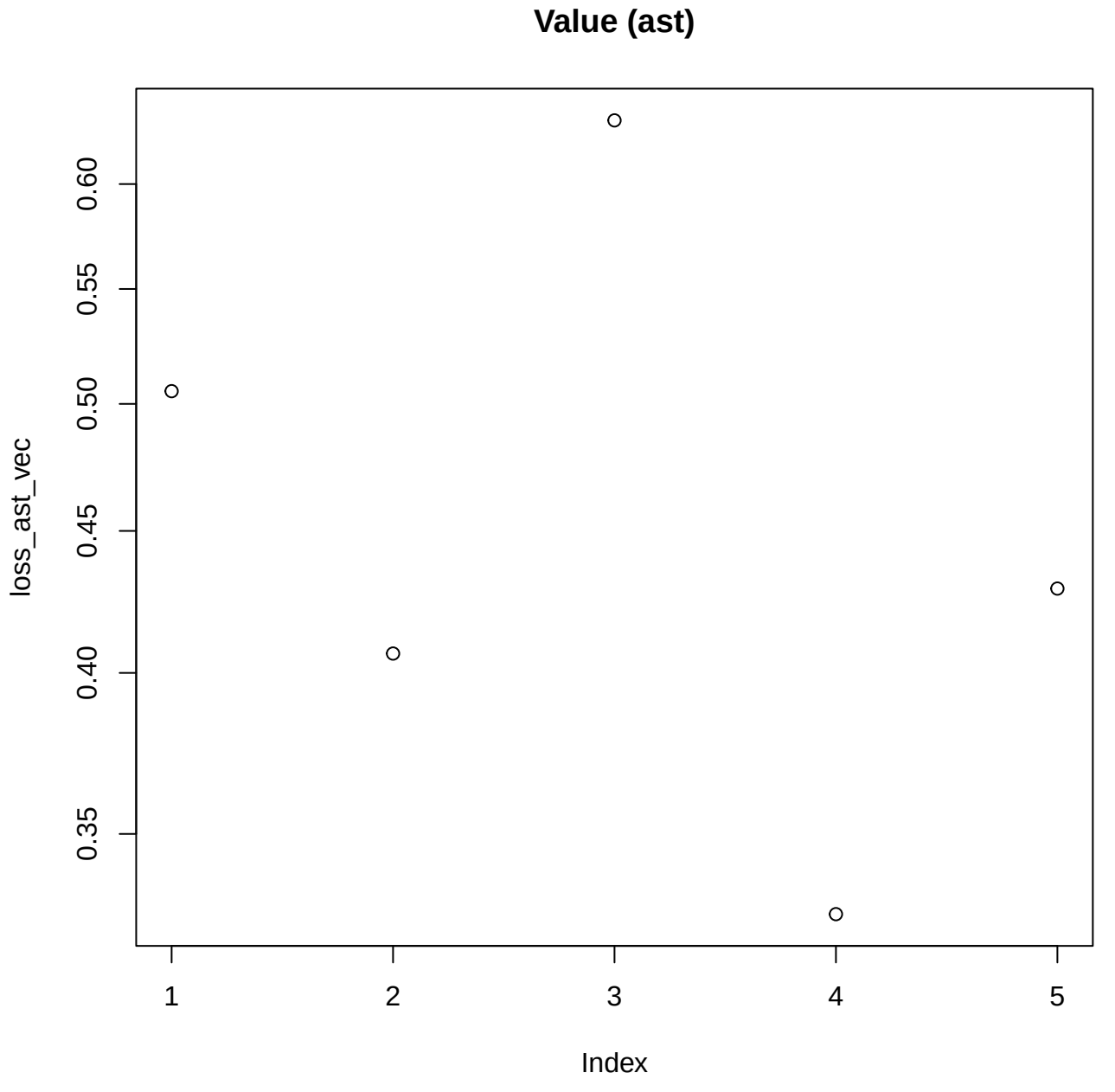

SVI ELBO Loss [118.4940]
ELBO loss
100
150
200
5
10
15
20
25
Iteration

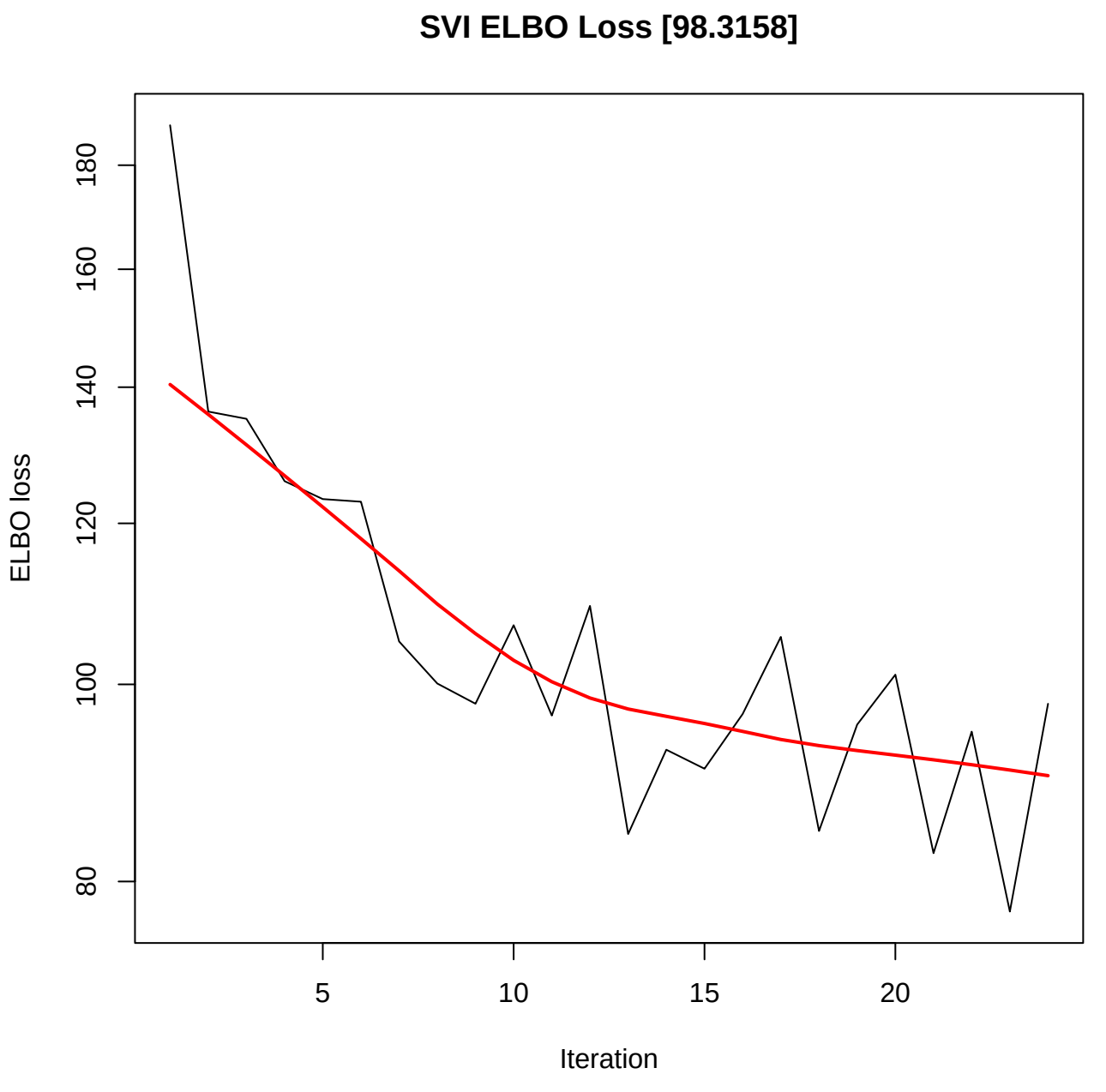

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

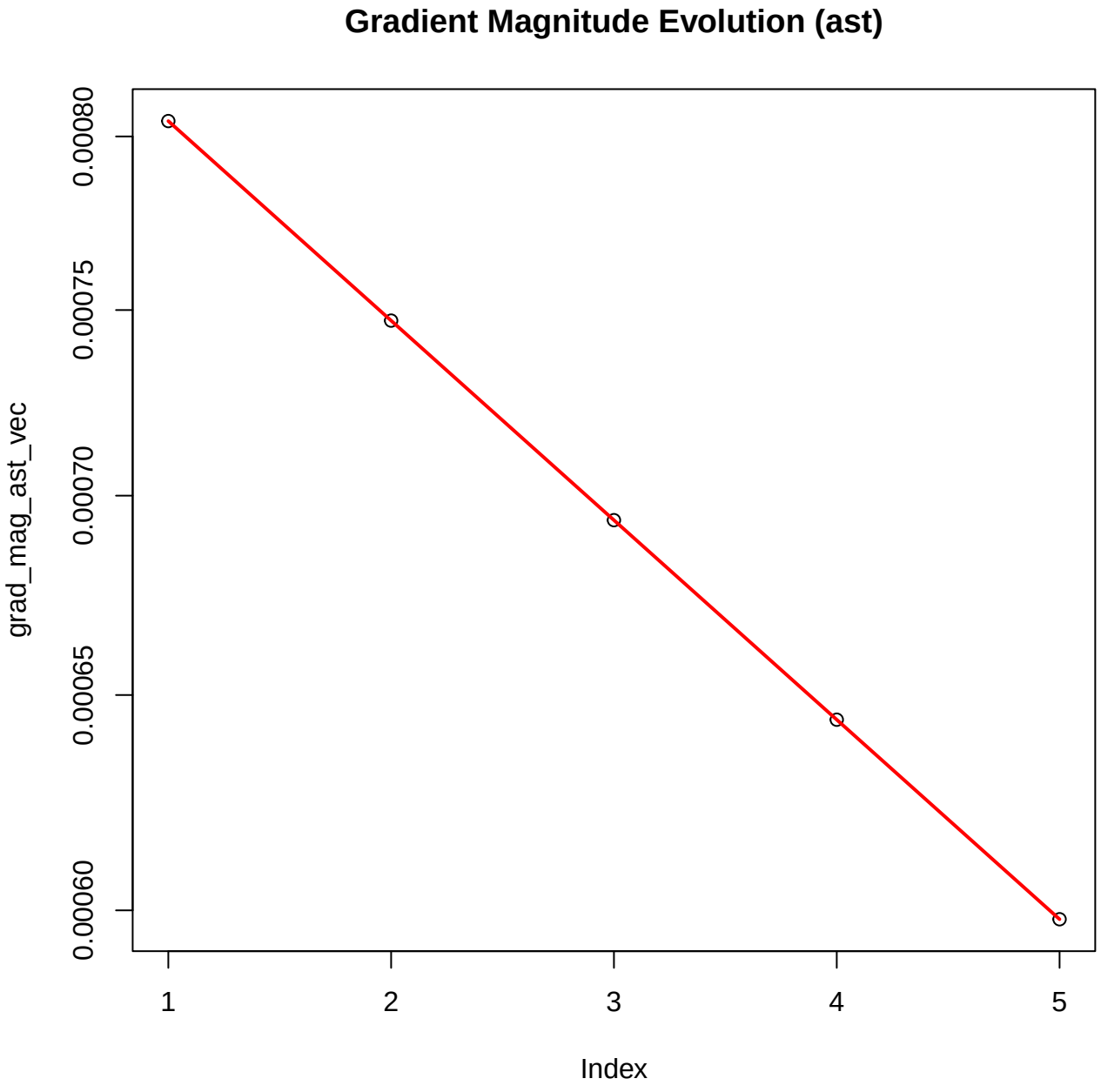

Value (ast)
loss_ast_vec
0.60
0.65
0.70
0.75
1
2
3
4
5
Index

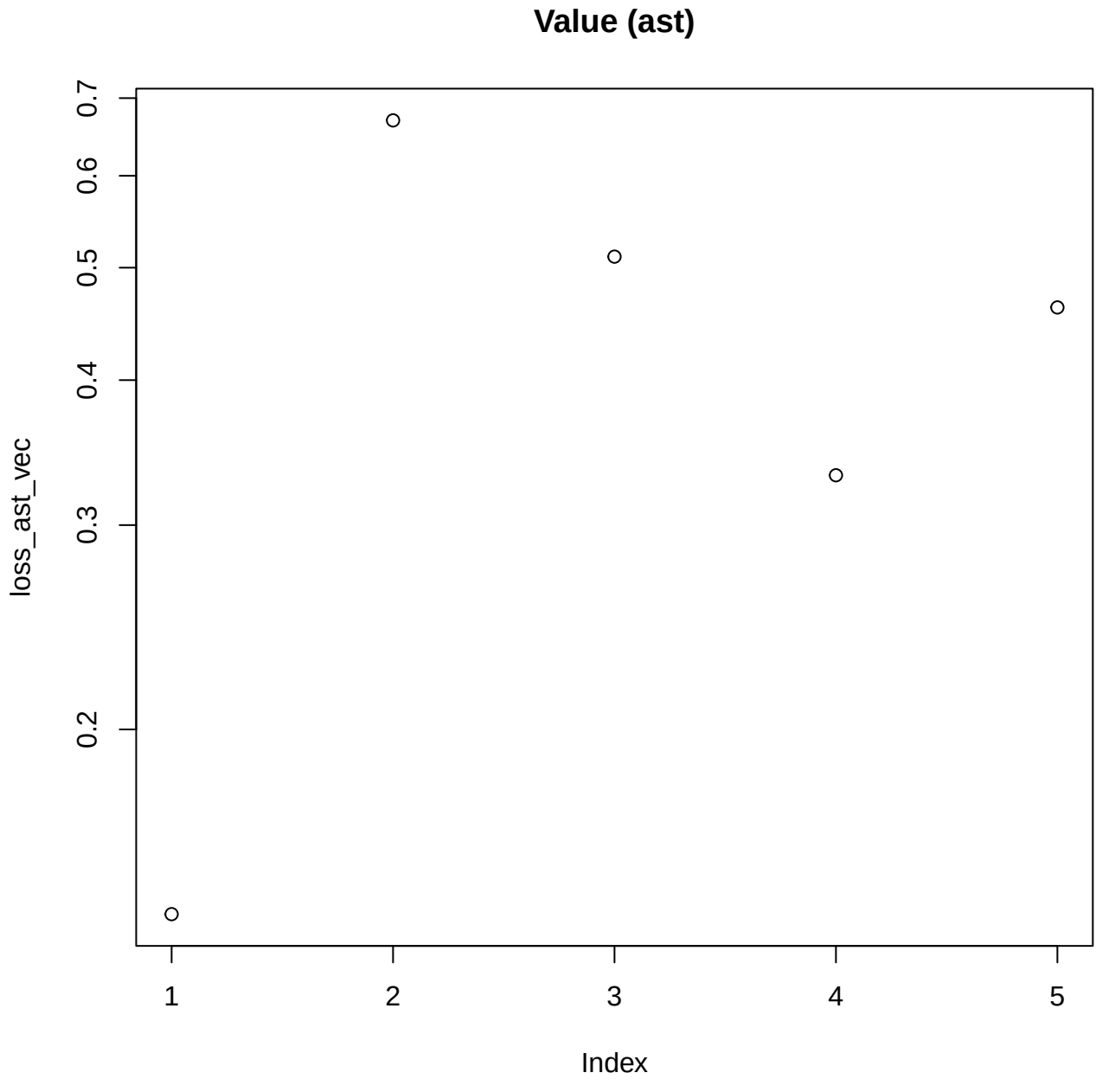

SVI ELBO Loss [50.7103]
ELBO loss
60
55
50
45
5
10
15
20
25
Iteration

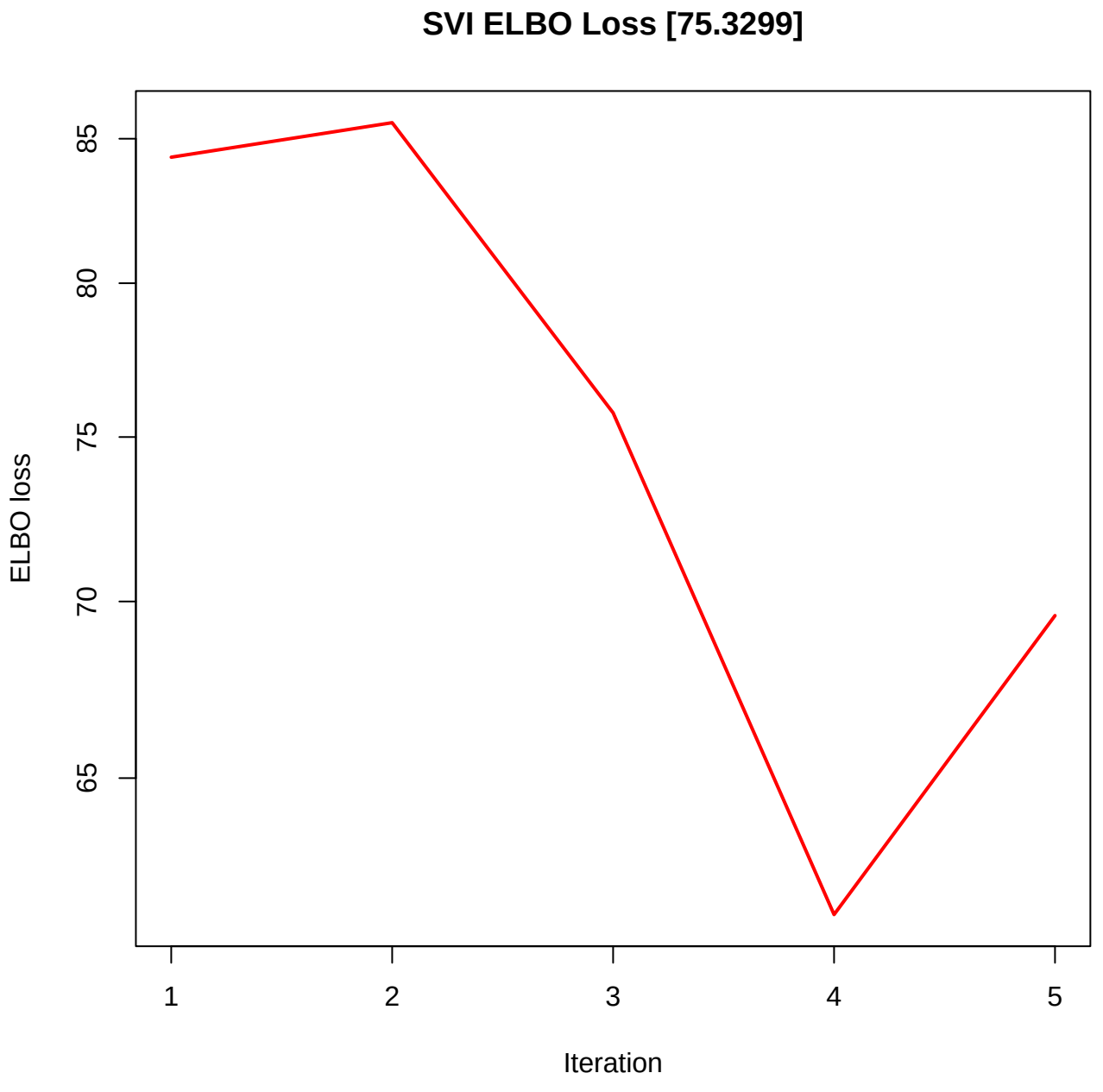

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

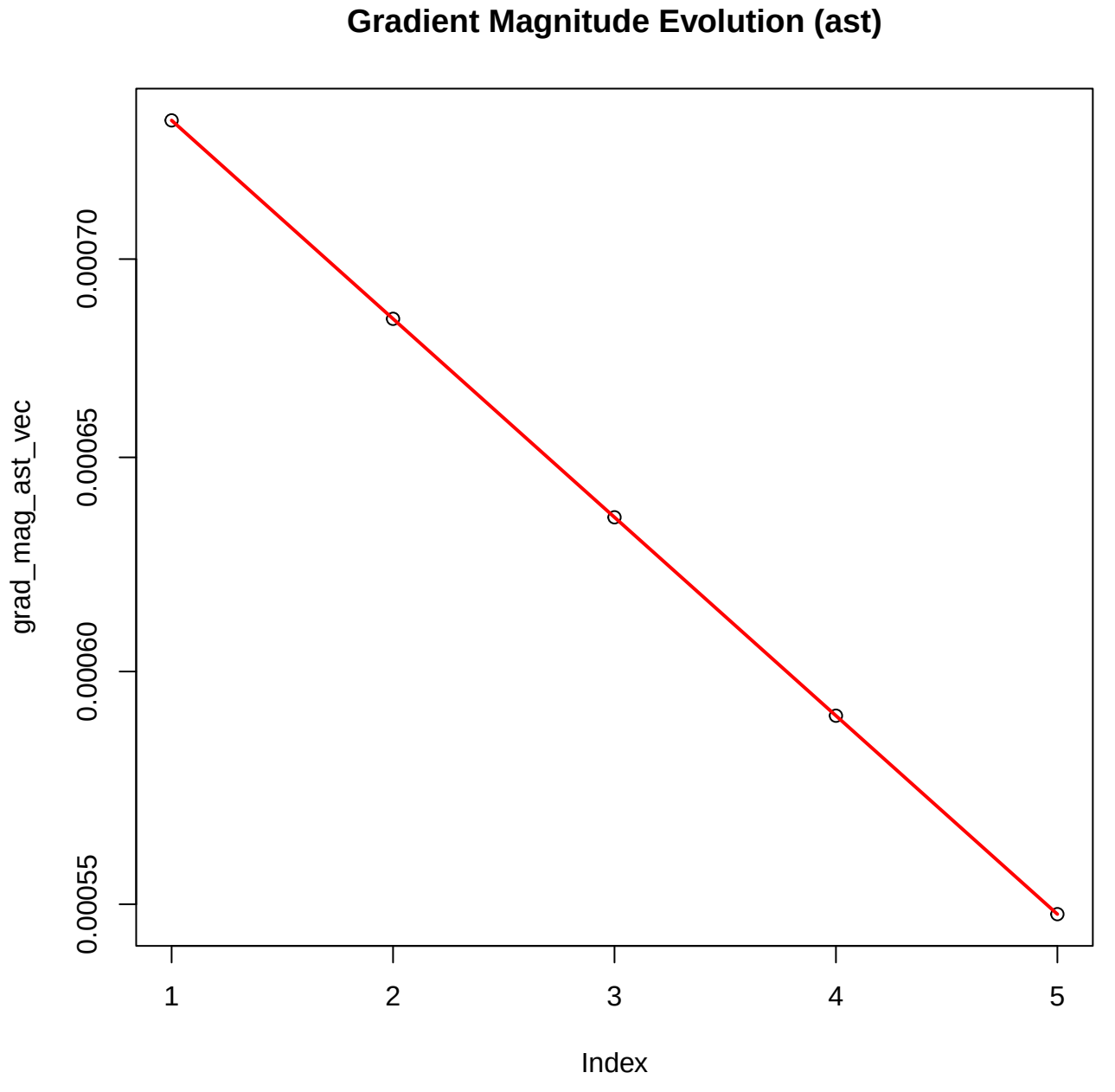

Value (ast)
loss_ast_vec
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

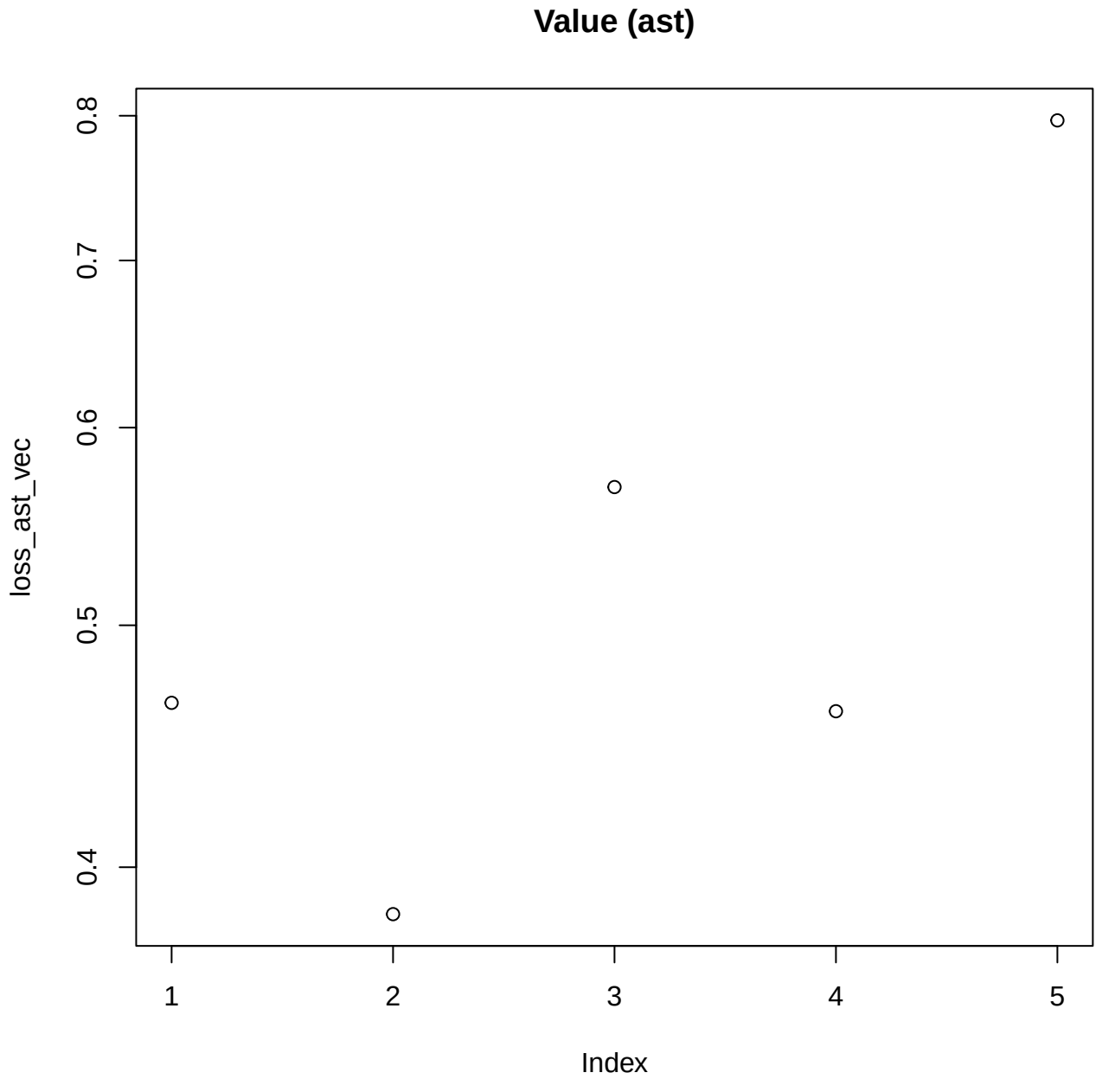

SVI ELBO Loss [98.3158]
ELBO loss
180
160
140
120
100
80
5
10
15
20
Iteration

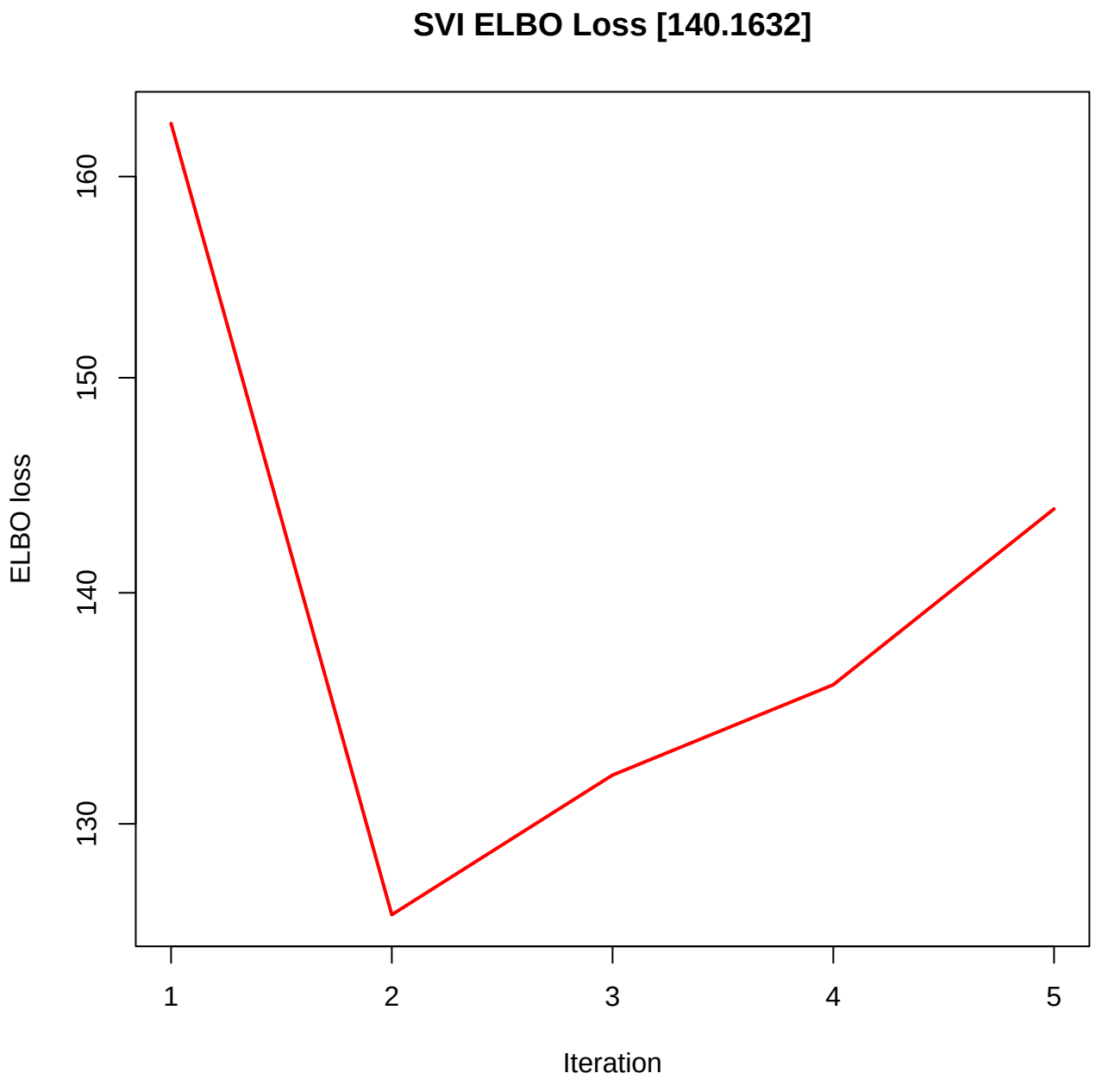

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

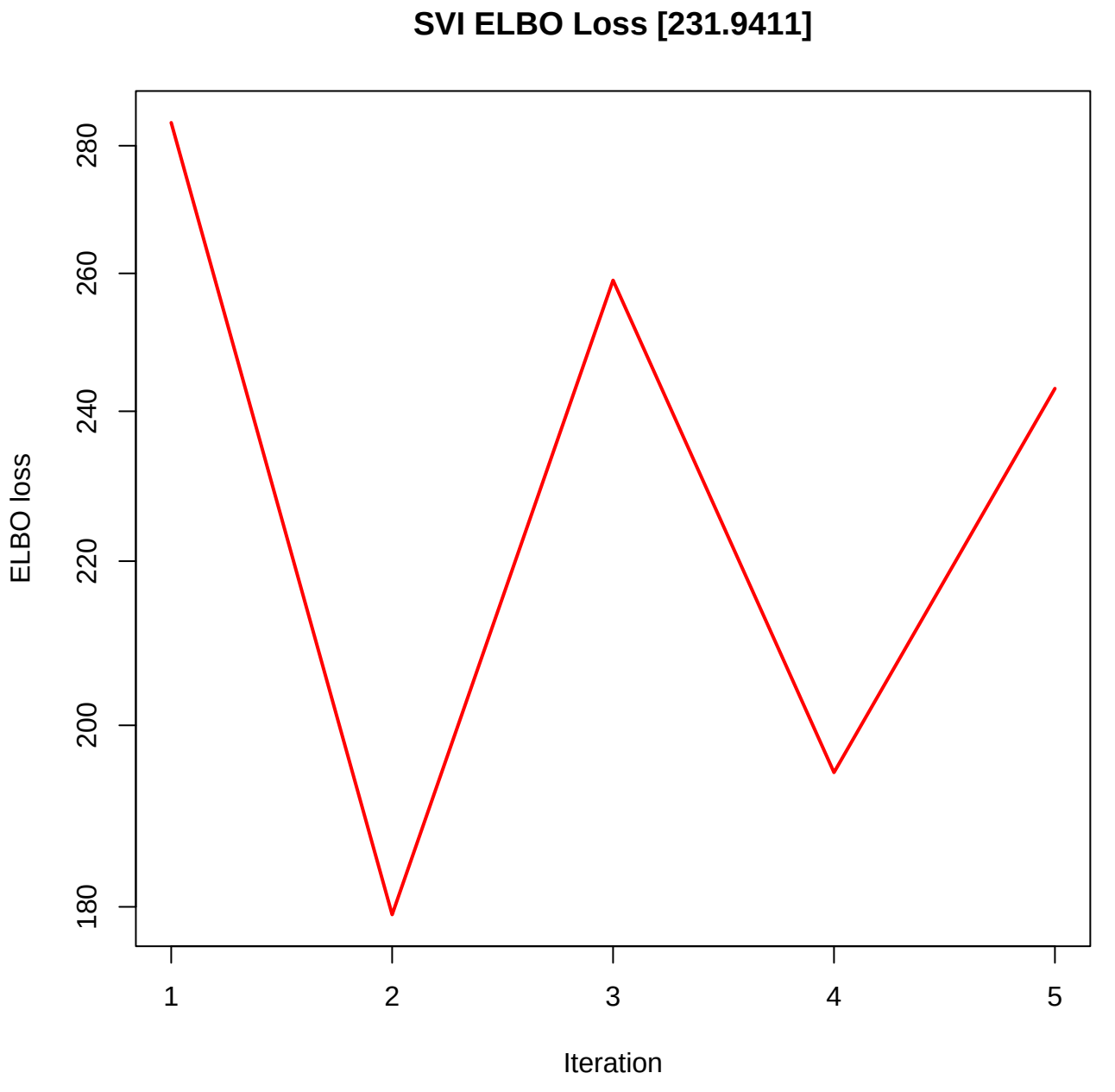

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

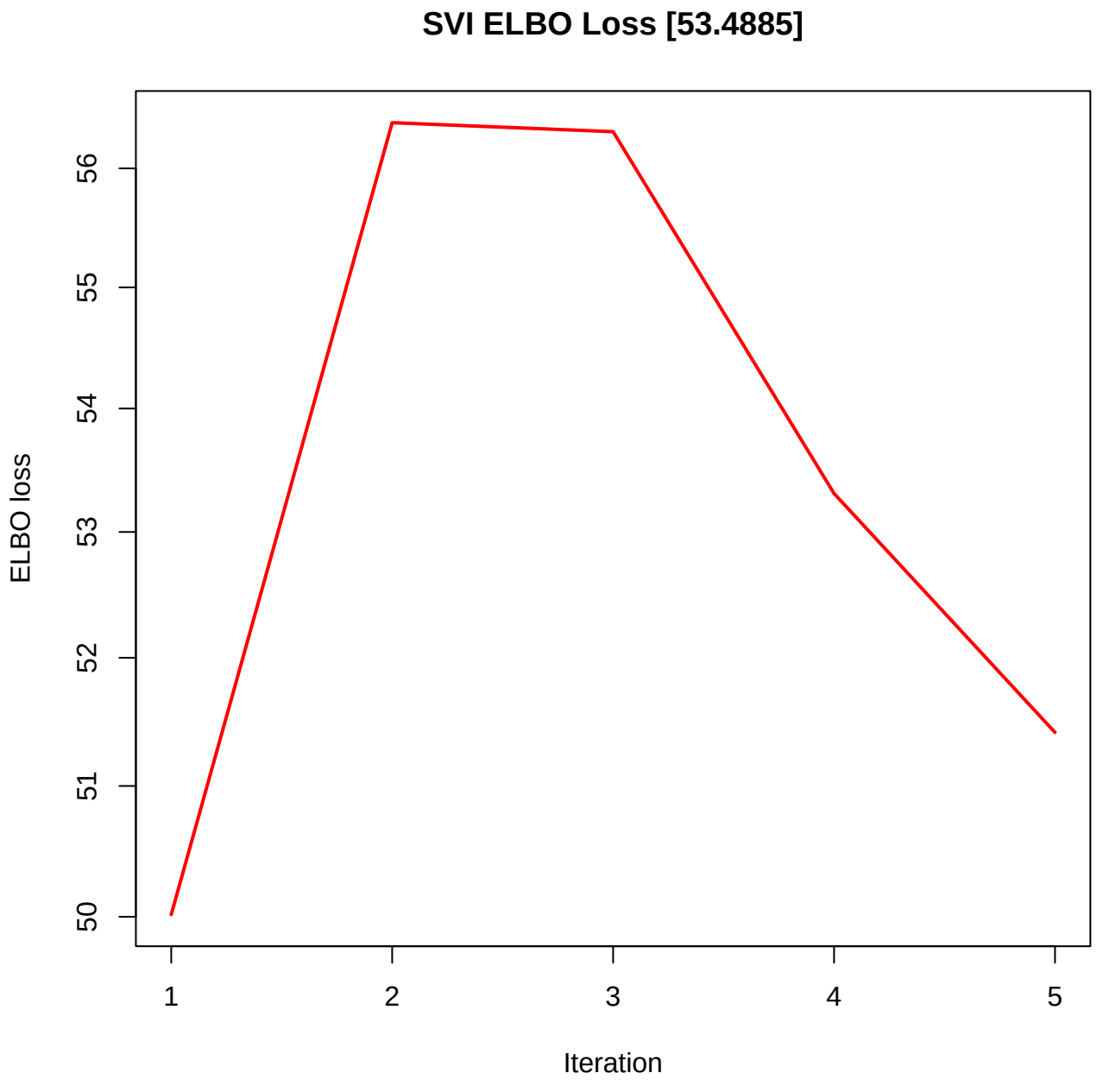

SVI ELBO Loss [75.3299]
ELBO loss
85
80
75
70
65
1
2
3
4
5
Iteration

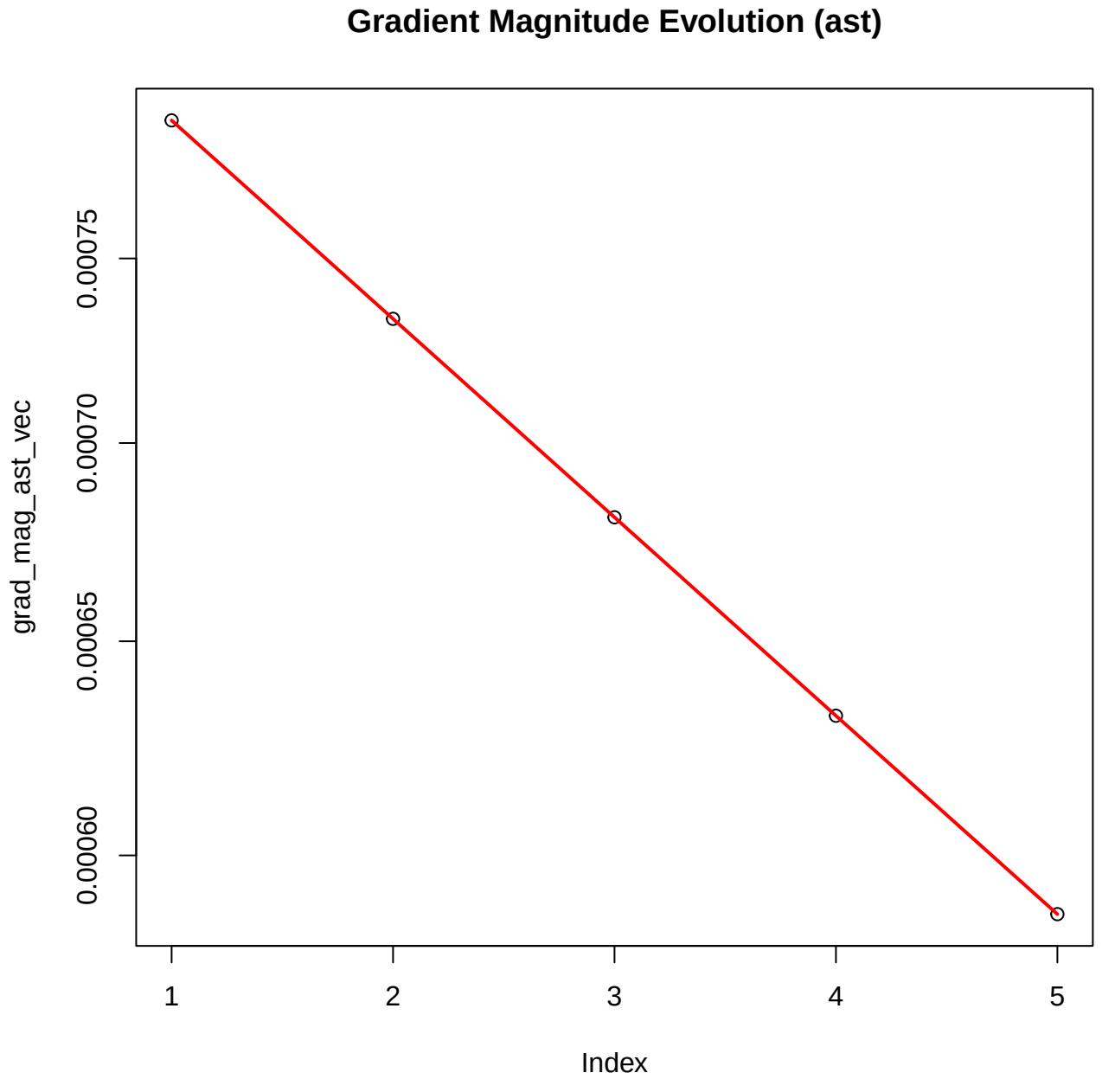

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00055
0.00060
0.00065
0.00070
1
2
3
4
5
Index

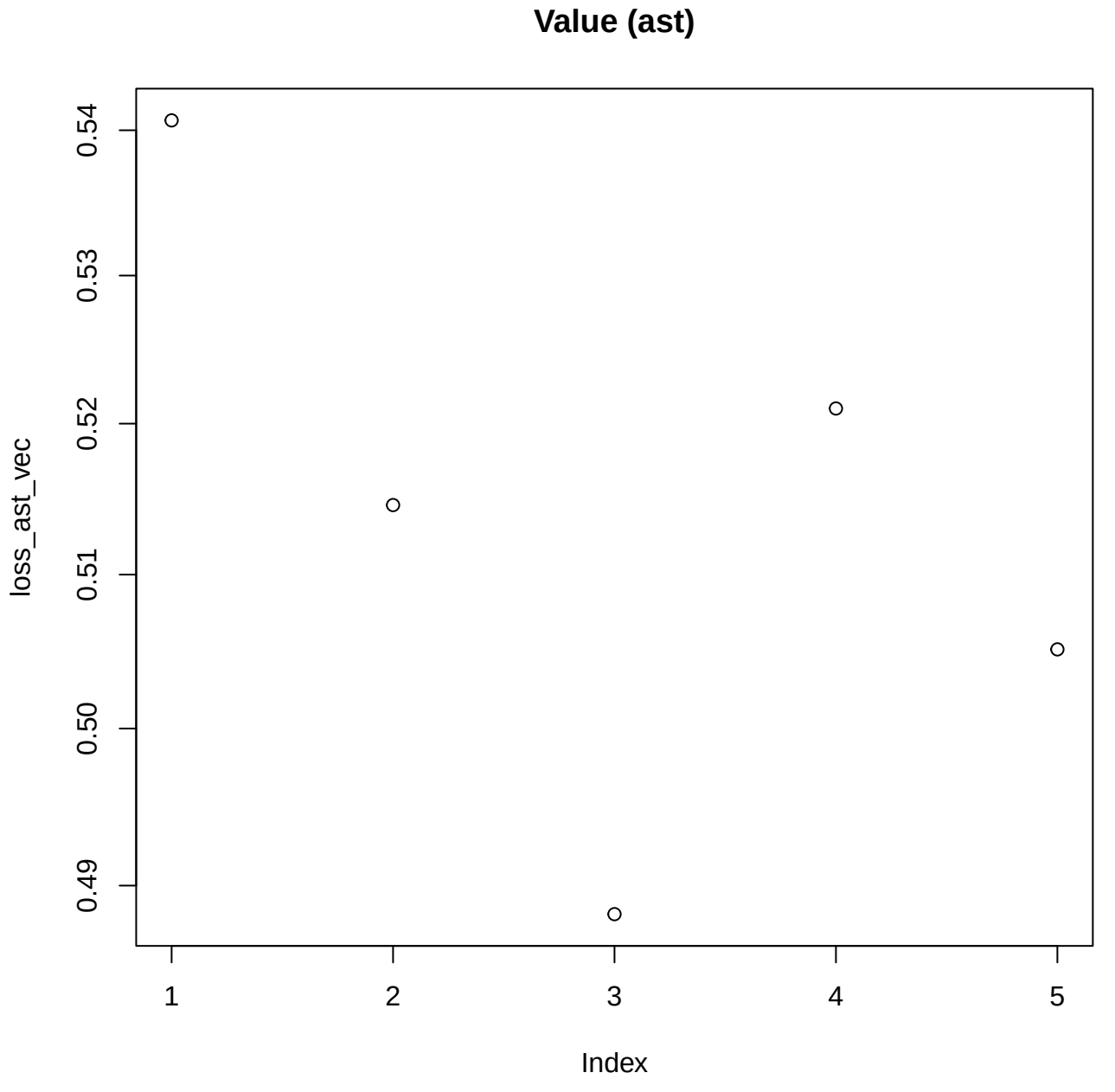

Value (ast)
loss_ast_vec
0.4
0.5
0.6
0.7
0.8
1
2
3
4
5
Index

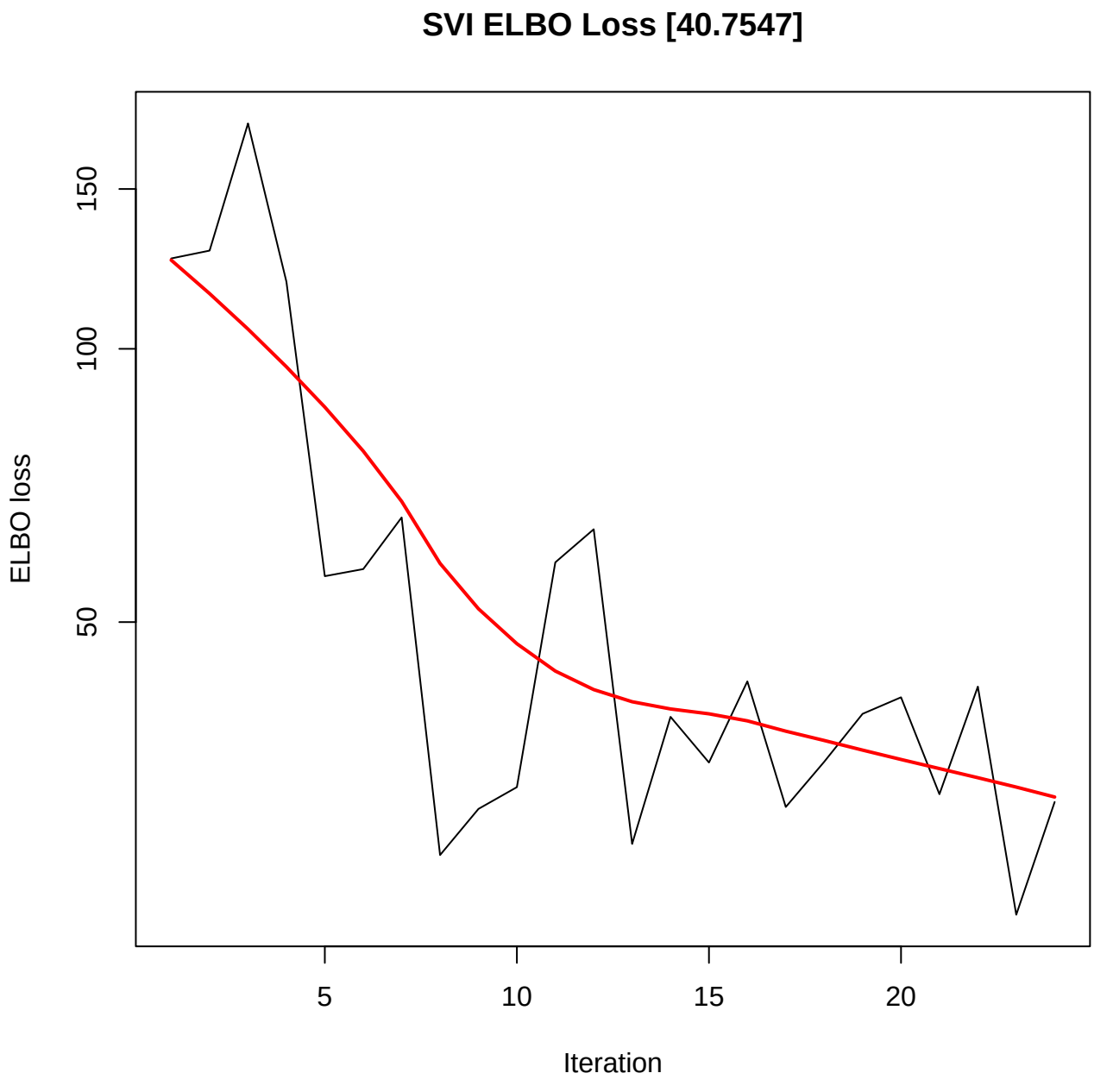

SVI ELBO Loss [140.1632]
ELBO loss
160
150
140
130
1
2
3
4
5
Iteration

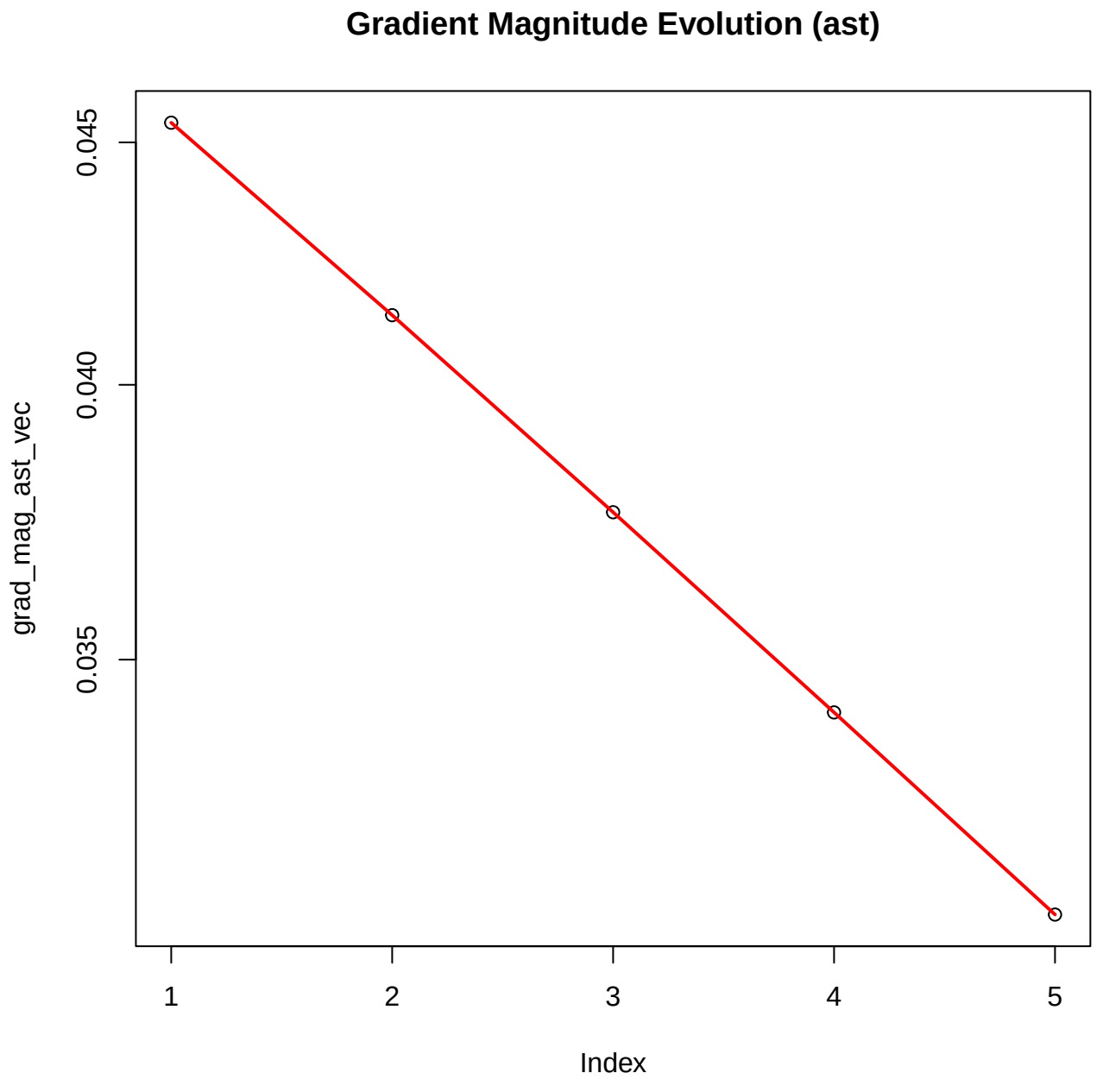

SVI ELBO Loss [231.9411]
ELBO loss
280
260
240
220
200
180
1
2
3
4
5
Iteration

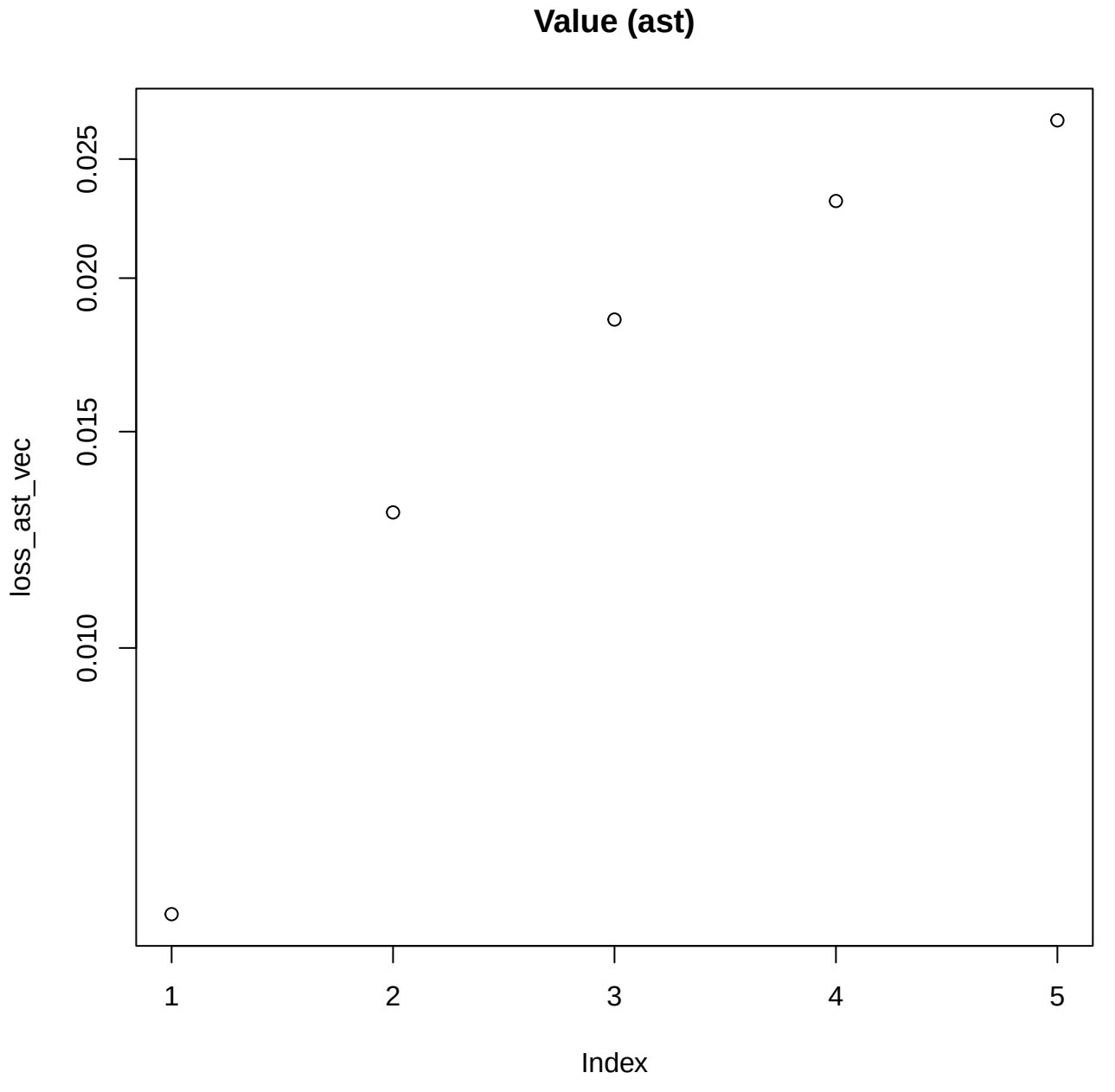

SVI ELBO Loss [53.4885]
ELBO loss
50
51
52
53
54
55
56
1
2
3
4
5
Iteration

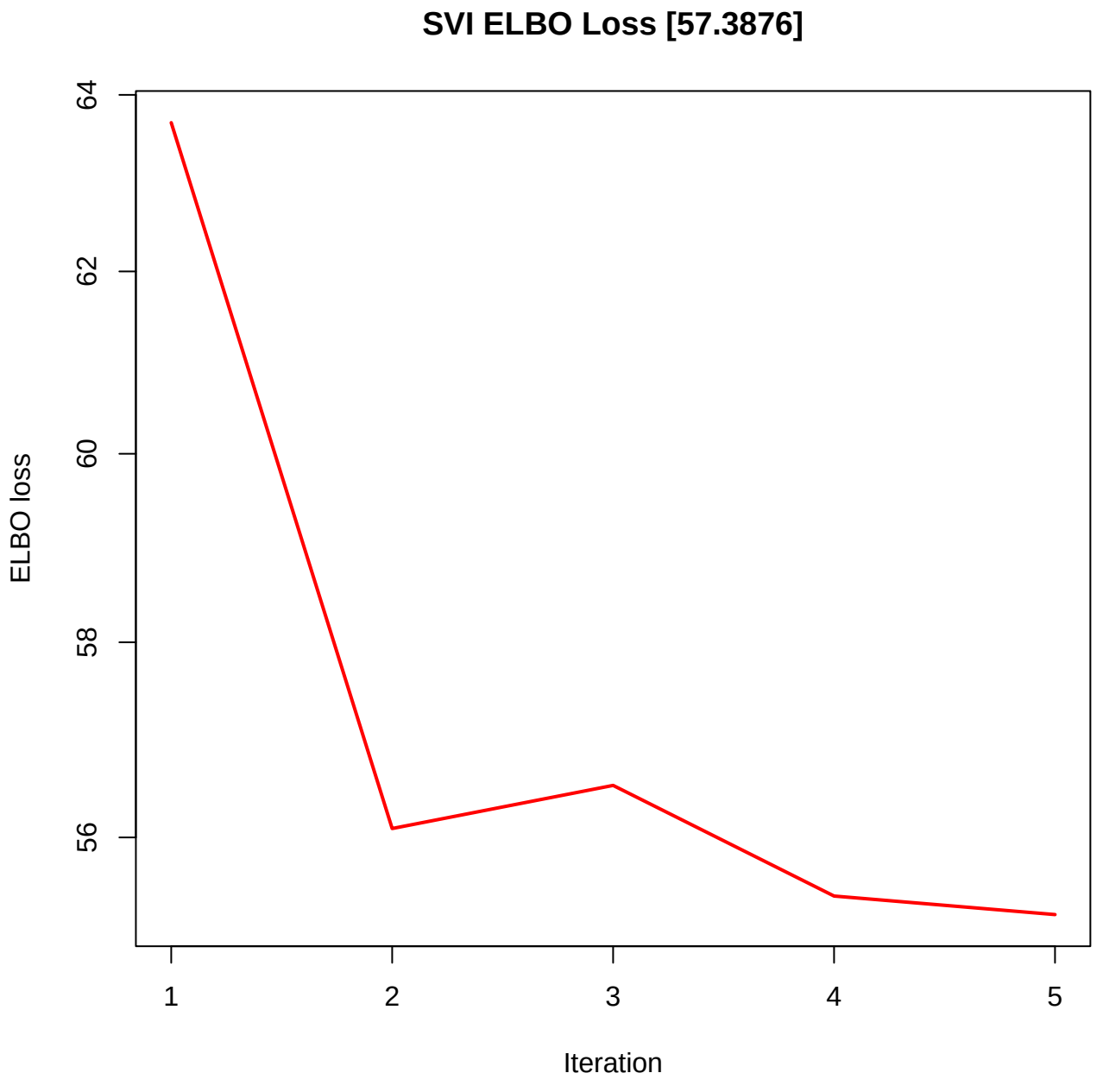

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

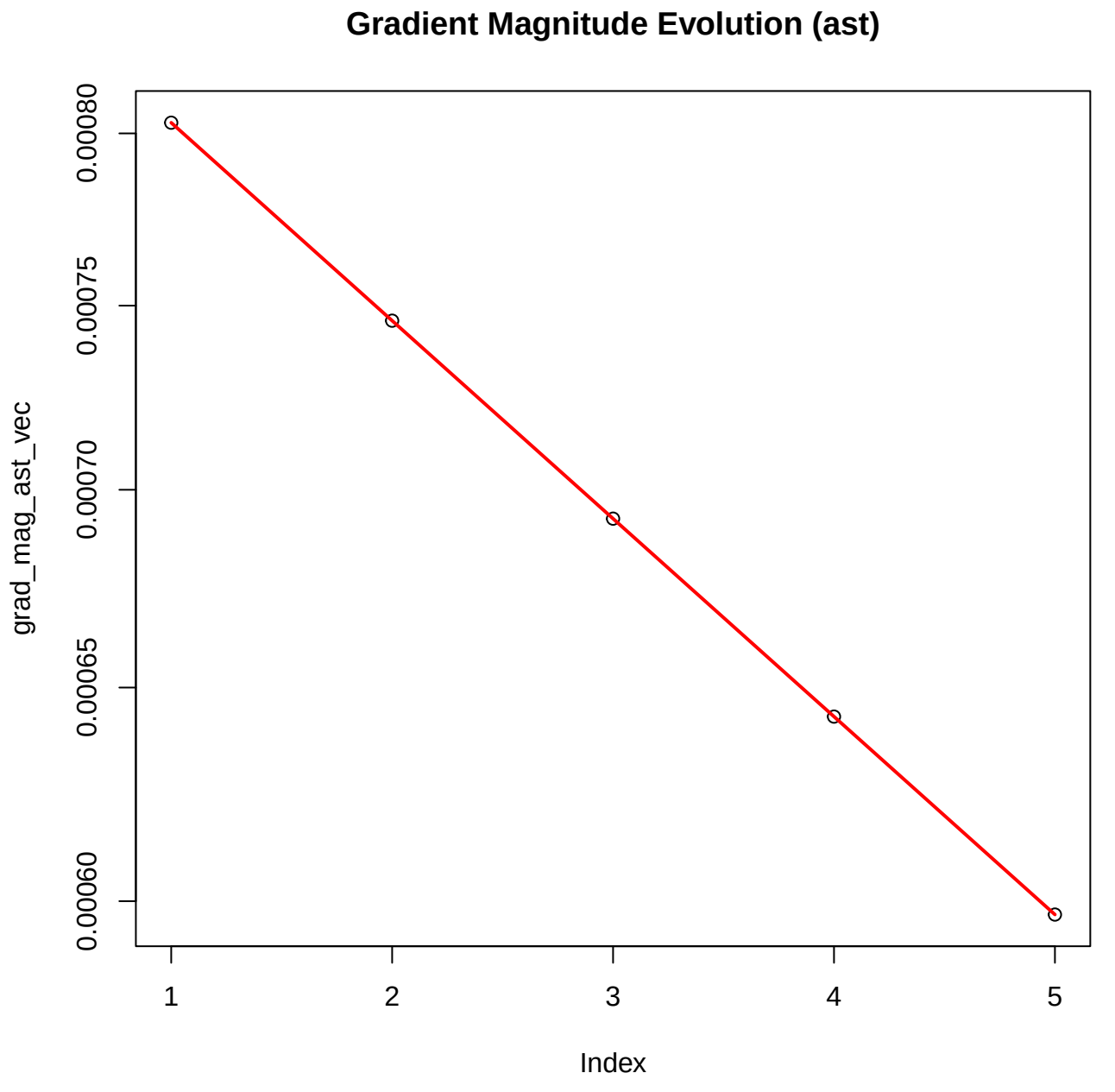

Value (ast)
loss_ast_vec
0.49
0.50
0.51
0.52
0.53
0.54
1
2
3
4
5
Index

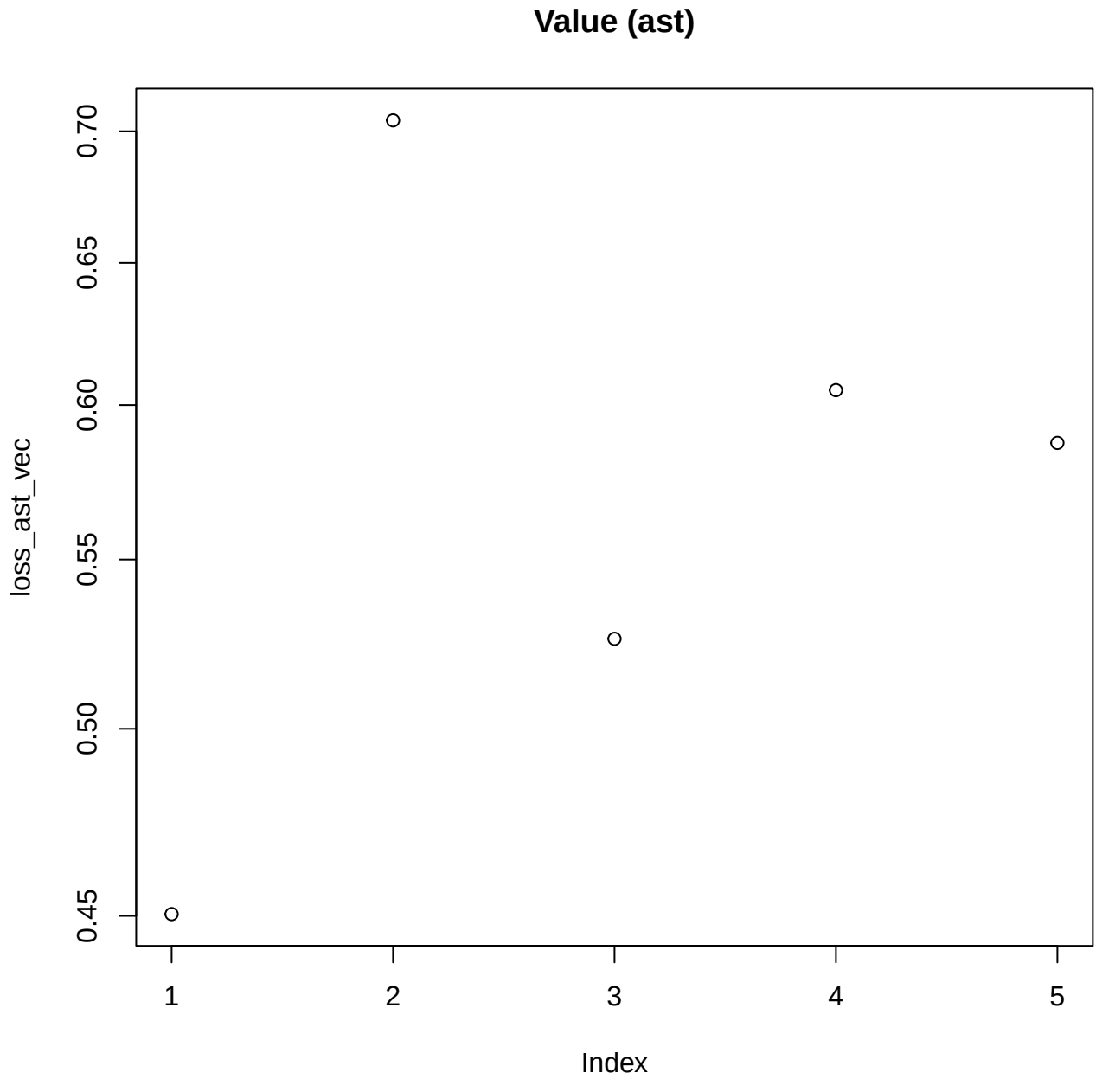

SVI ELBO Loss [40.7547]
ELBO loss
150
100
50
5
10
15
20
Iteration

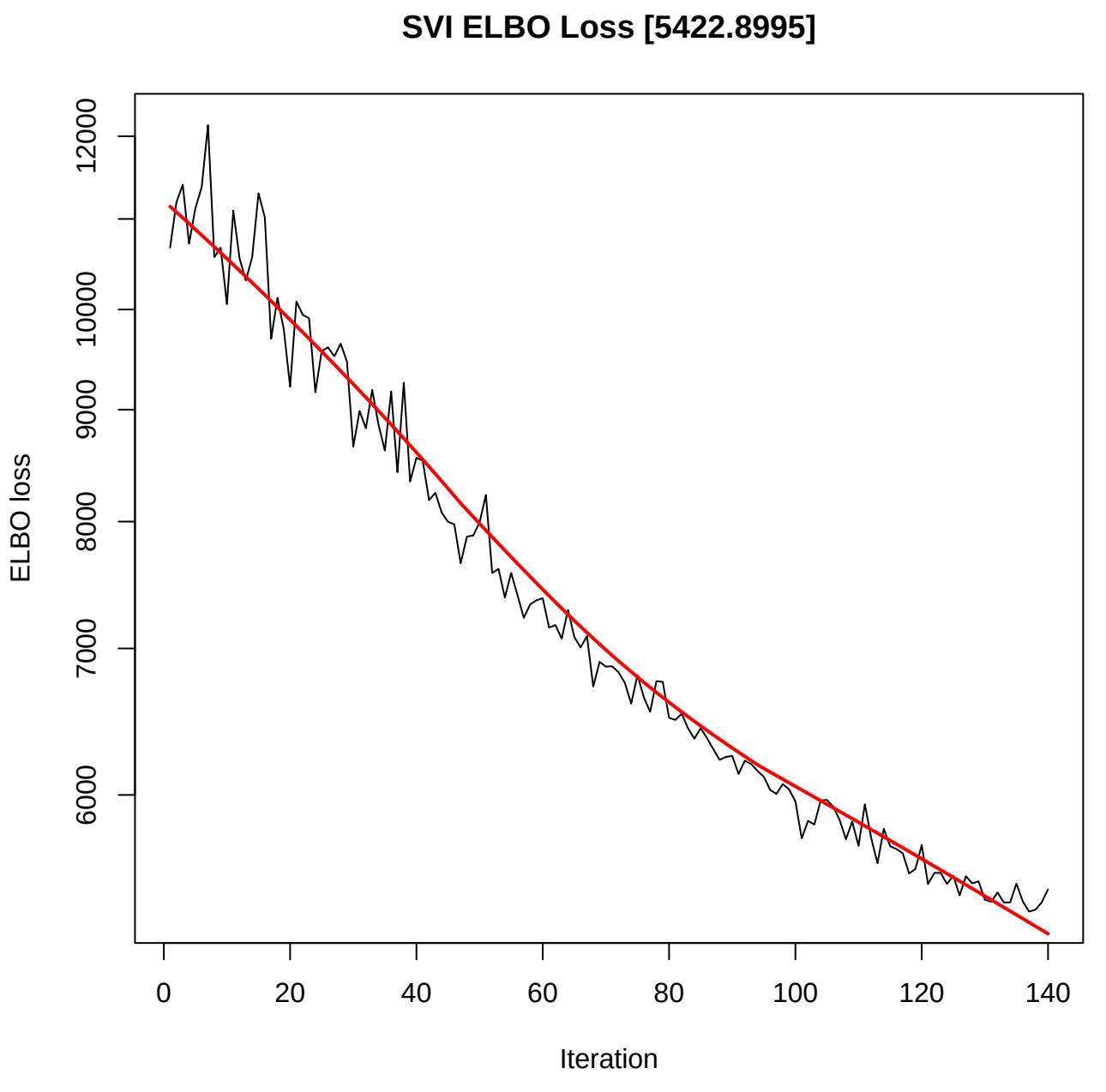

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.035
0.040
0.045
1
2
3
4
5
Index

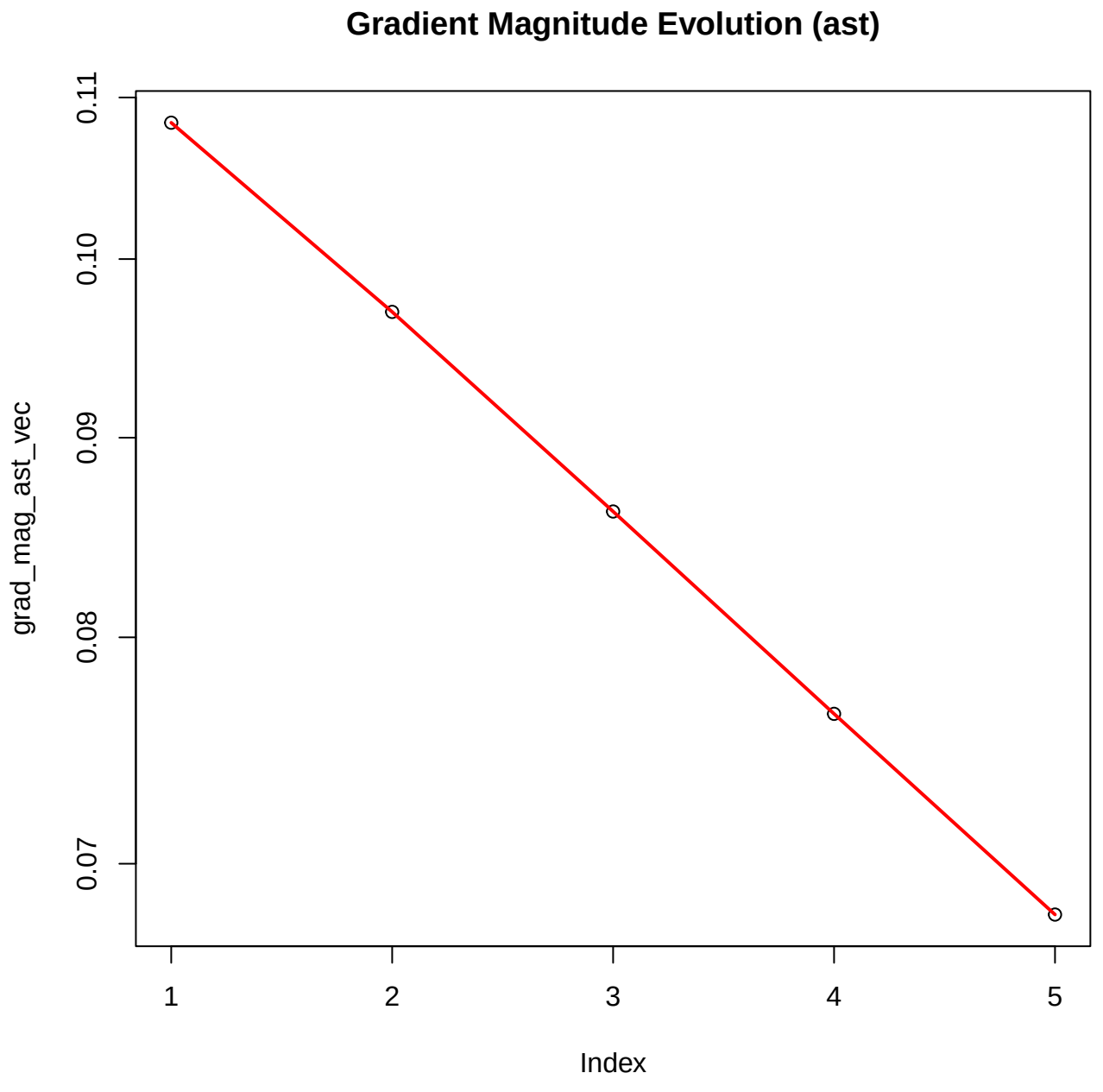

Value (ast)
loss_ast_vec
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

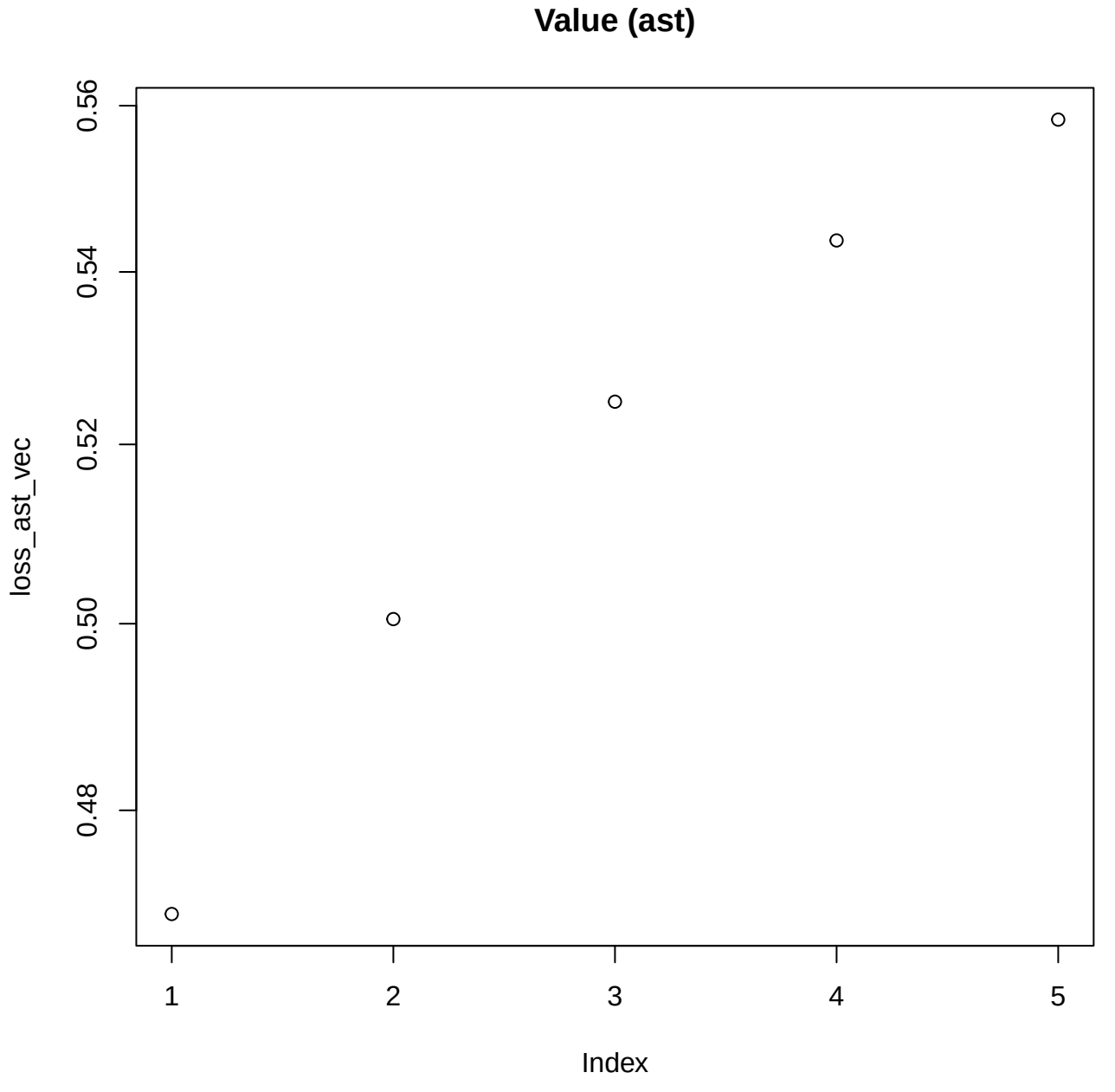

SVI ELBO Loss [57.3876]
ELBO loss
64
62
60
58
56
1
2
3
4
5
Iteration

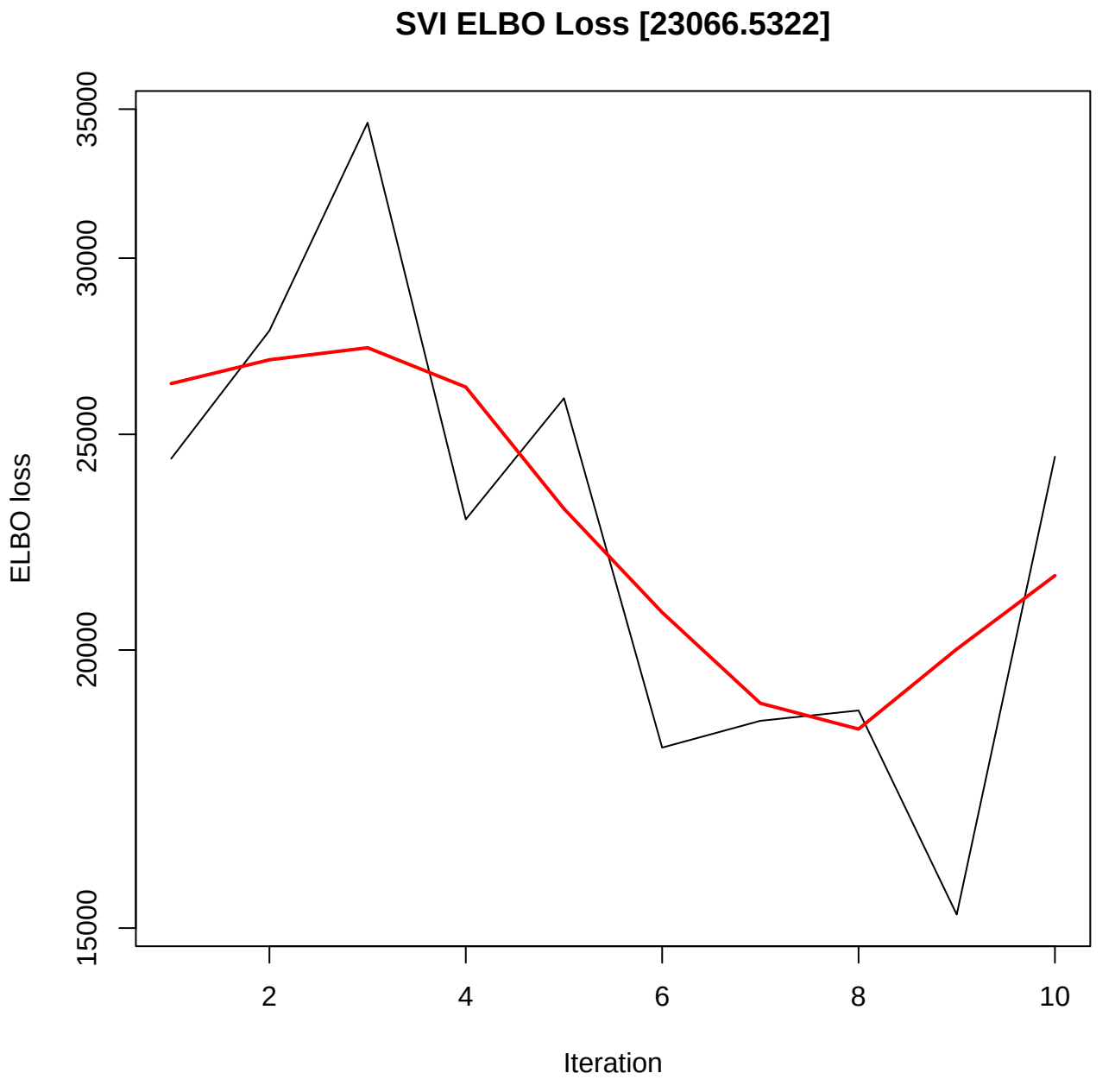

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

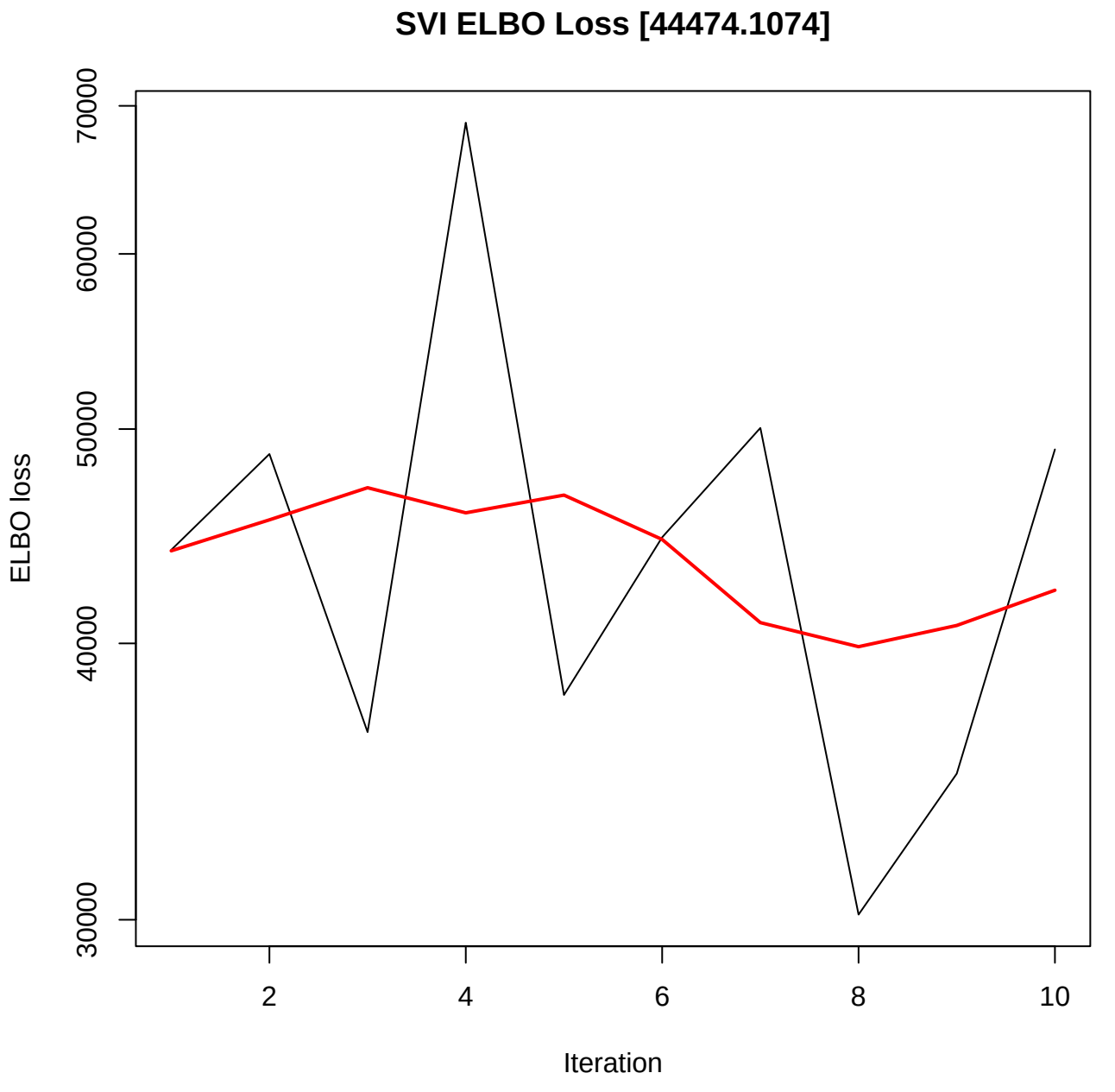

Value (ast)
loss_ast_vec
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

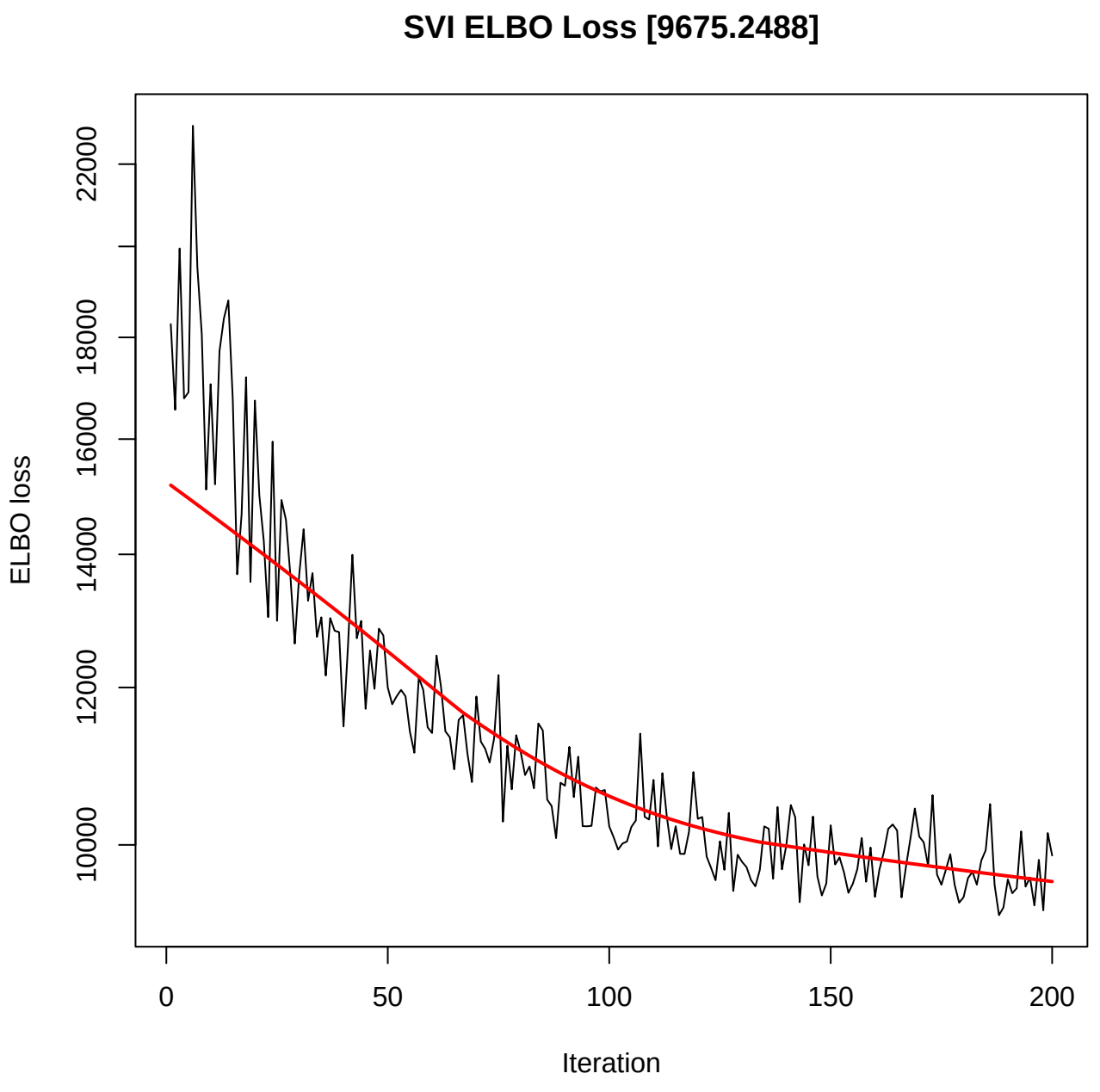

SVI ELBO Loss [5422.8995]
ELBO loss
12000
10000
9000
8000
7000
6000
0
20
40
60
80
100
120
140
Iteration

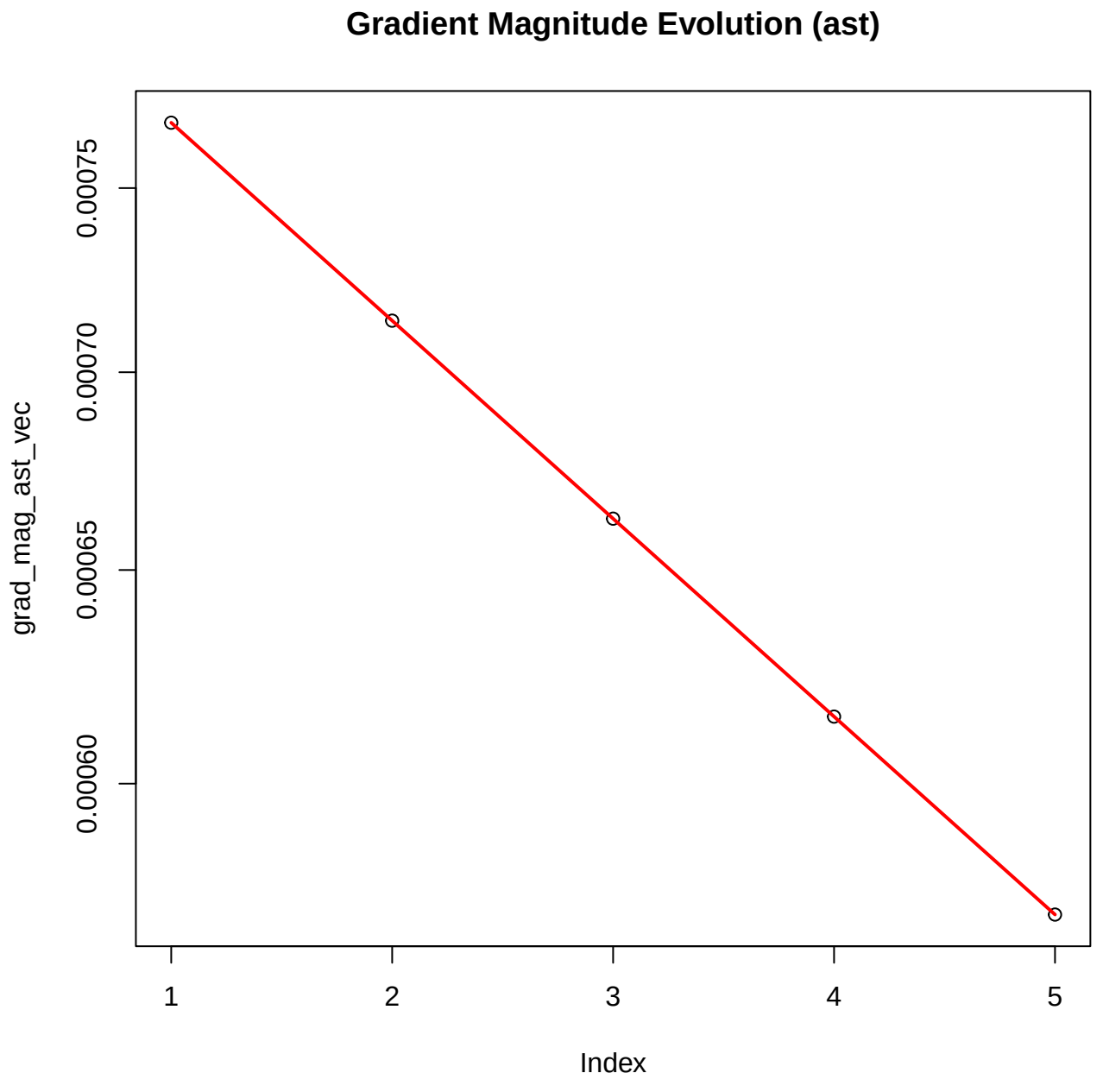

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.07
0.08
0.09
0.10
0.11
1
2
3
4
5
Index

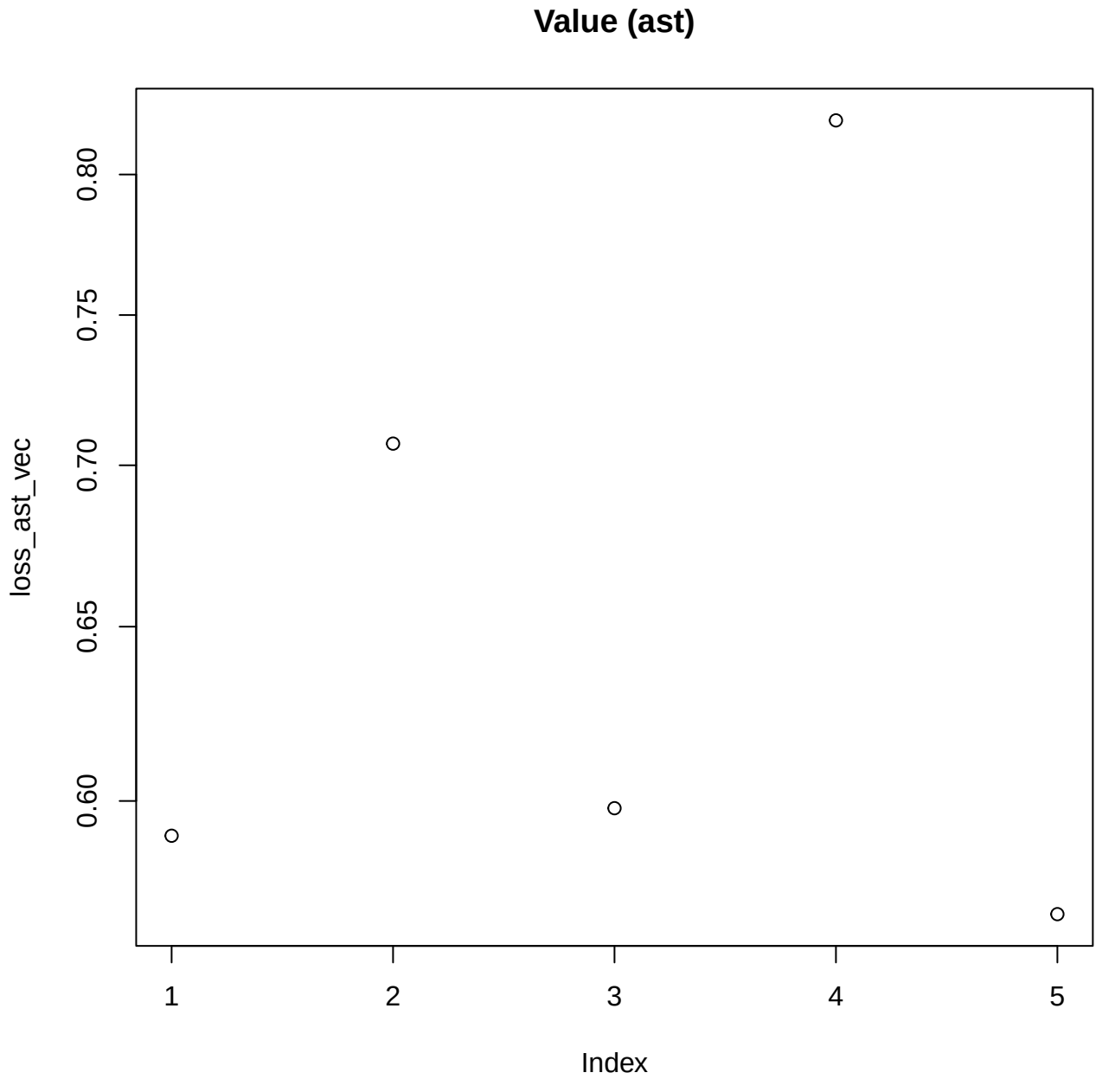

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
1
2
3
4
5
Index

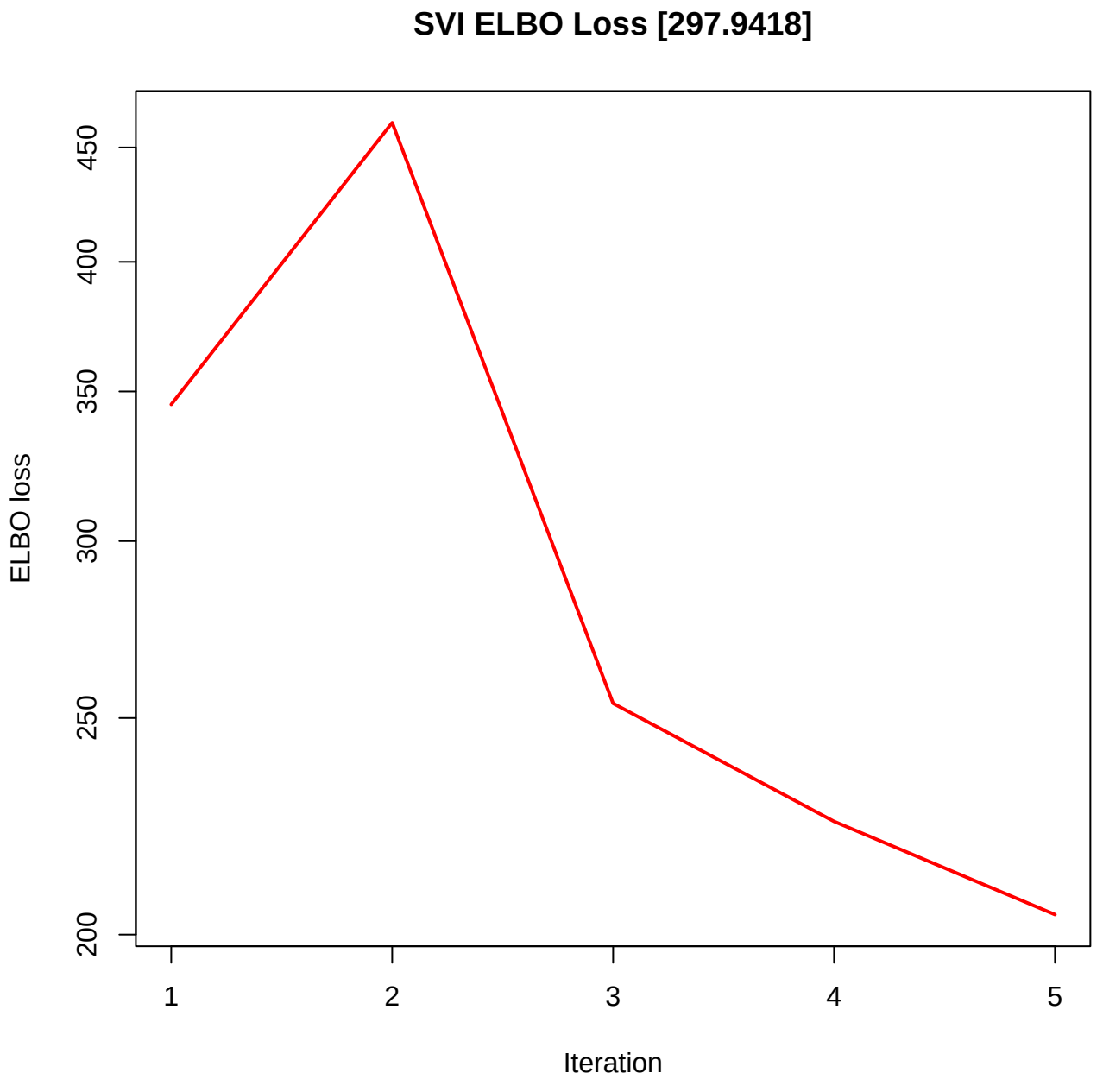

SVI ELBO Loss [23066.5322]
ELBO loss
35000
30000
25000
20000
15000
2
4
6
8
10
Iteration

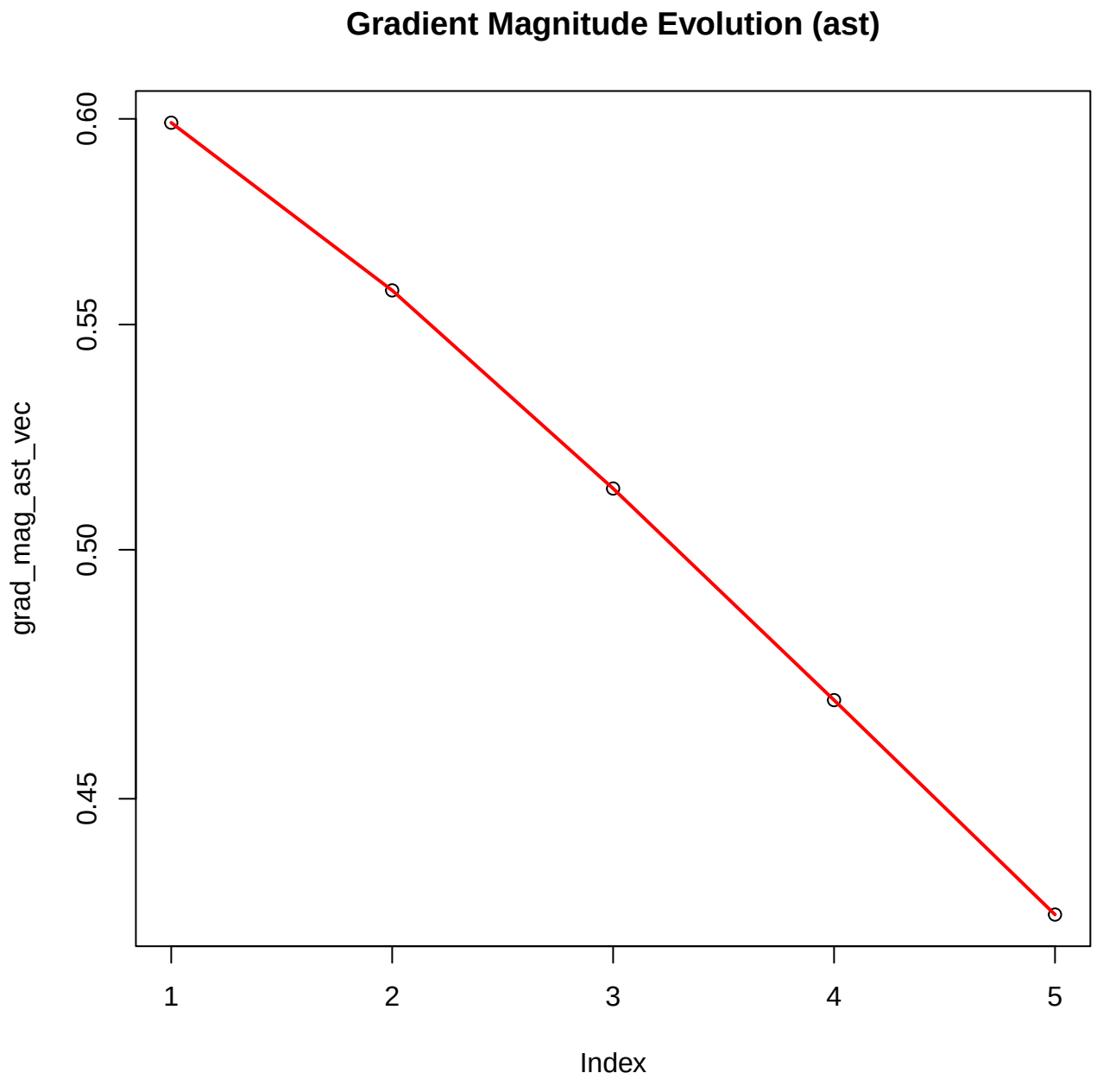

SVI ELBO Loss [44474.1074]
ELBO loss
70000
60000
50000
40000
30000
2
4
6
8
10
Iteration

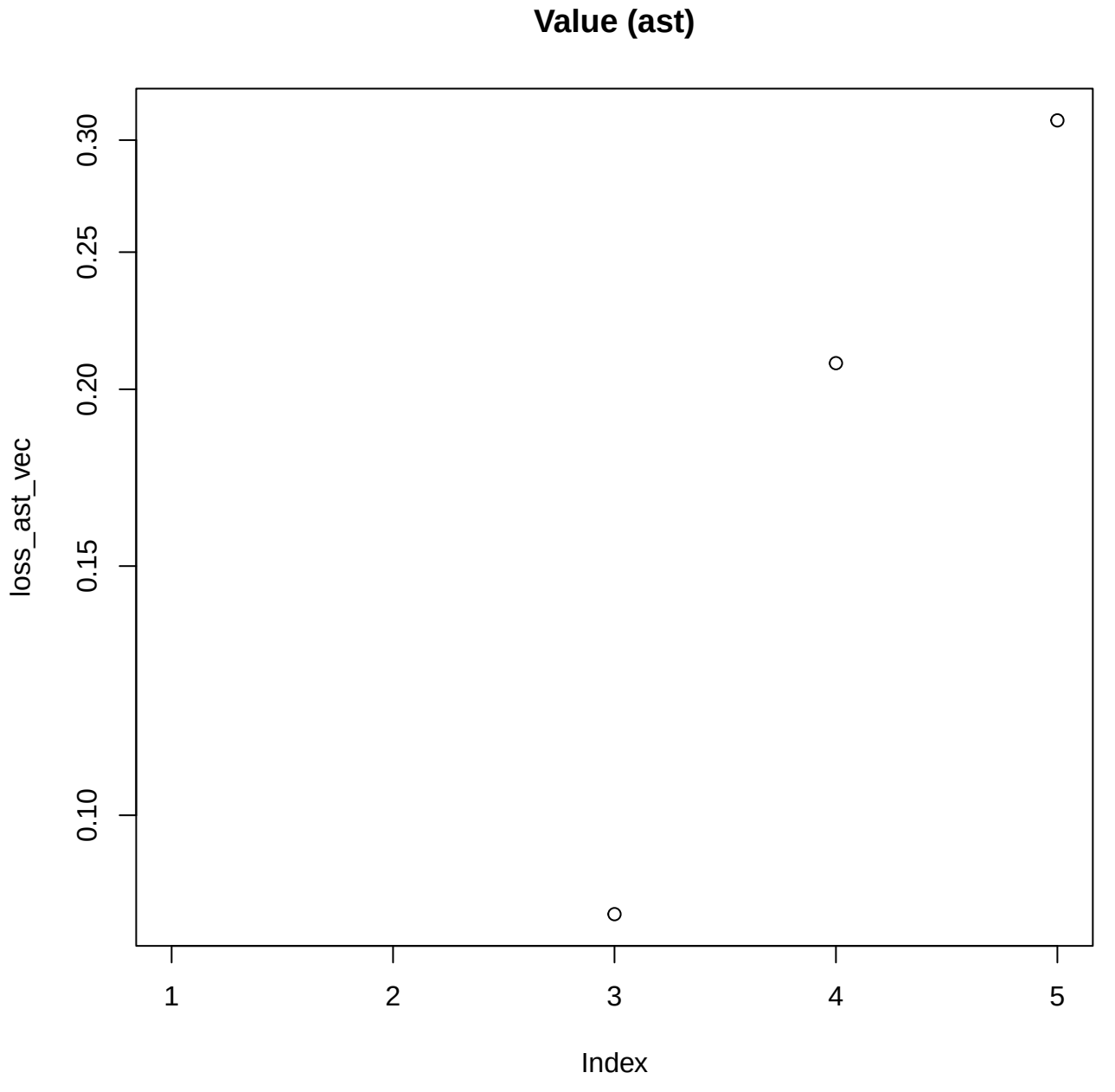

SVI ELBO Loss [9675.2488]
ELBO loss
10000
12000
14000
16000
18000
22000
0
50
100
150
200
Iteration

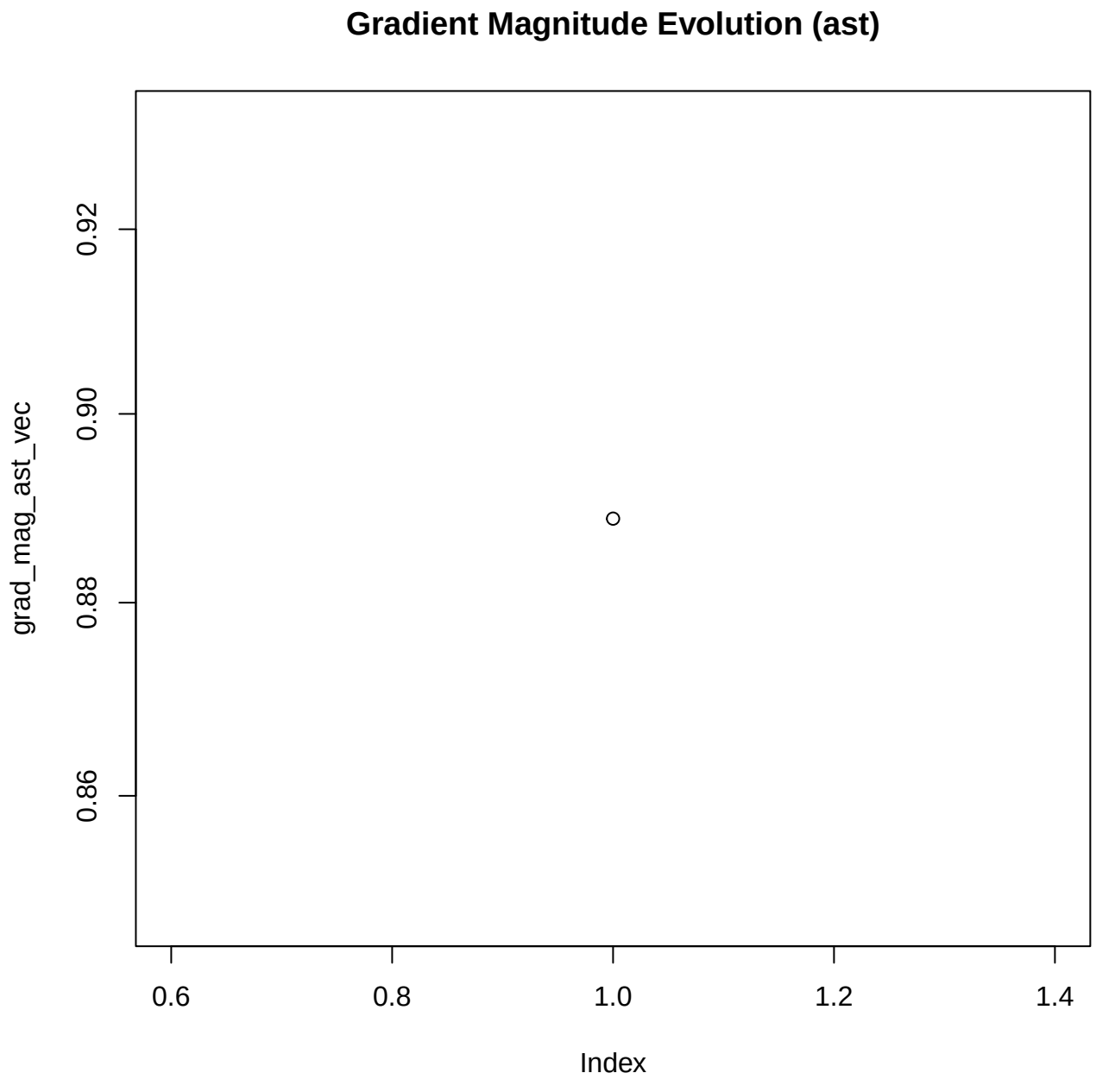

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

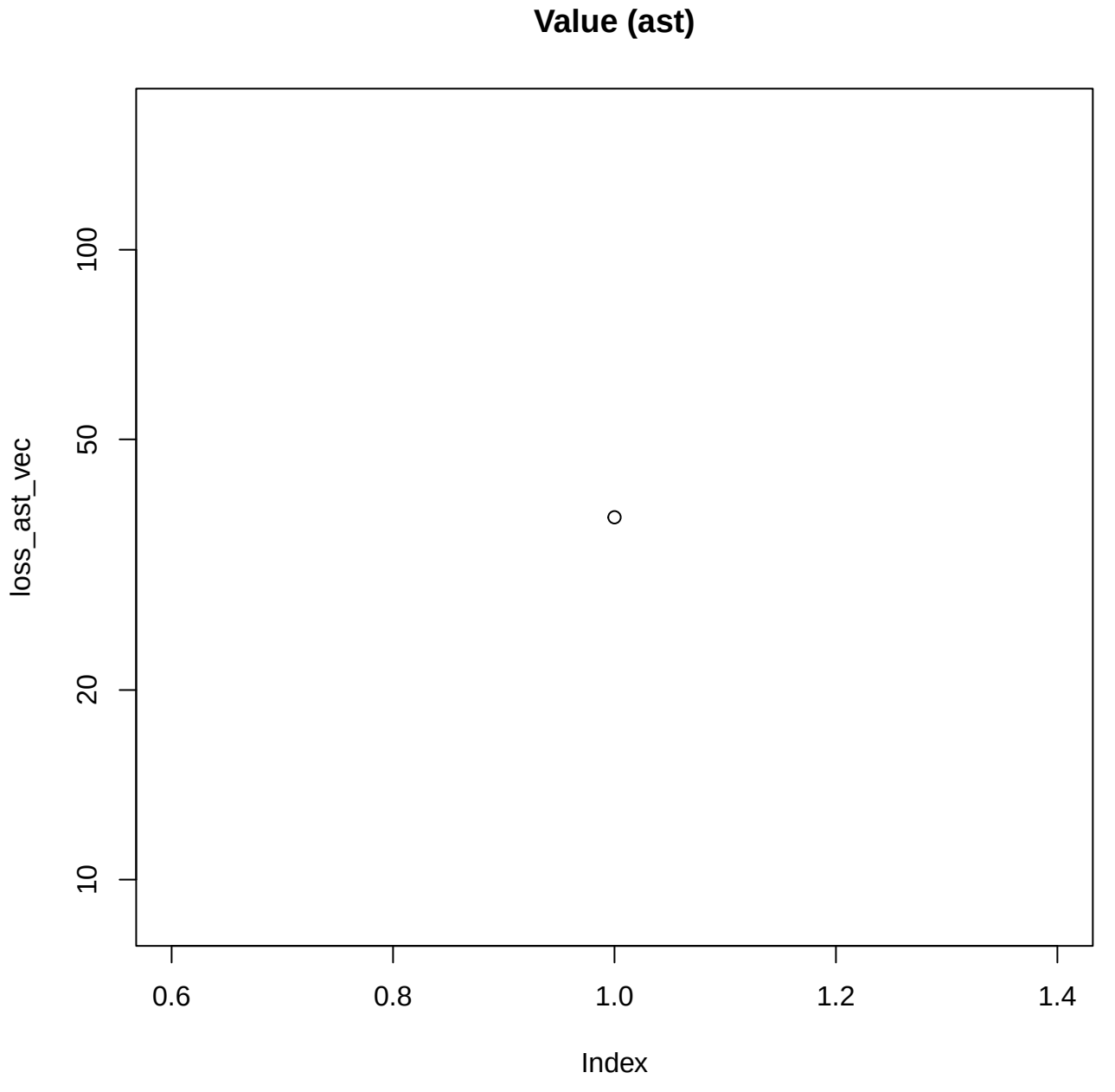

Value (ast)
loss_ast_vec
0.60
0.65
0.70
0.75
0.80
1
2
3
4
5
Index

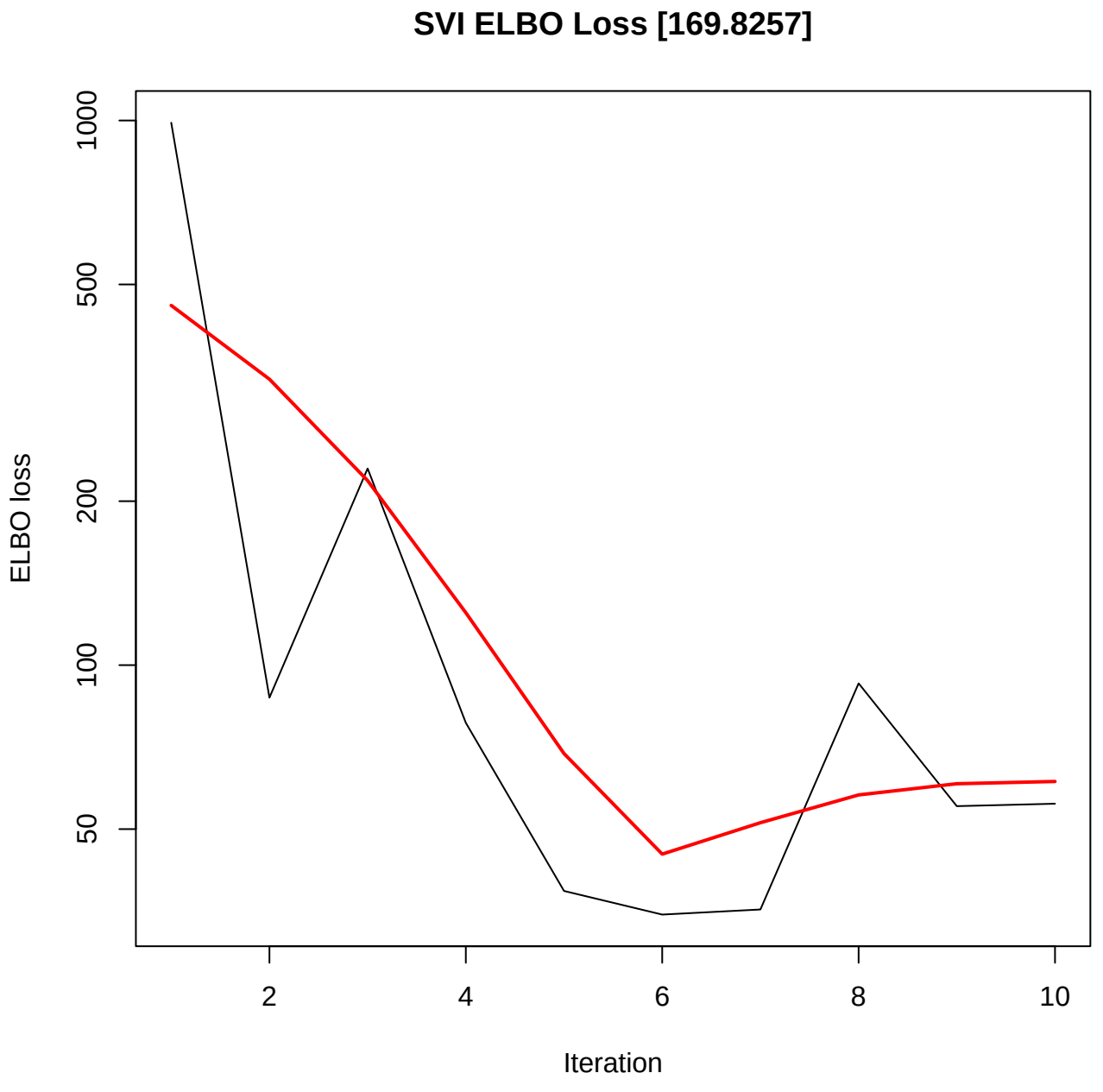

SVI ELBO Loss [297.9418]
ELBO loss
450
400
350
300
250
200
1
2
3
4
5
Iteration

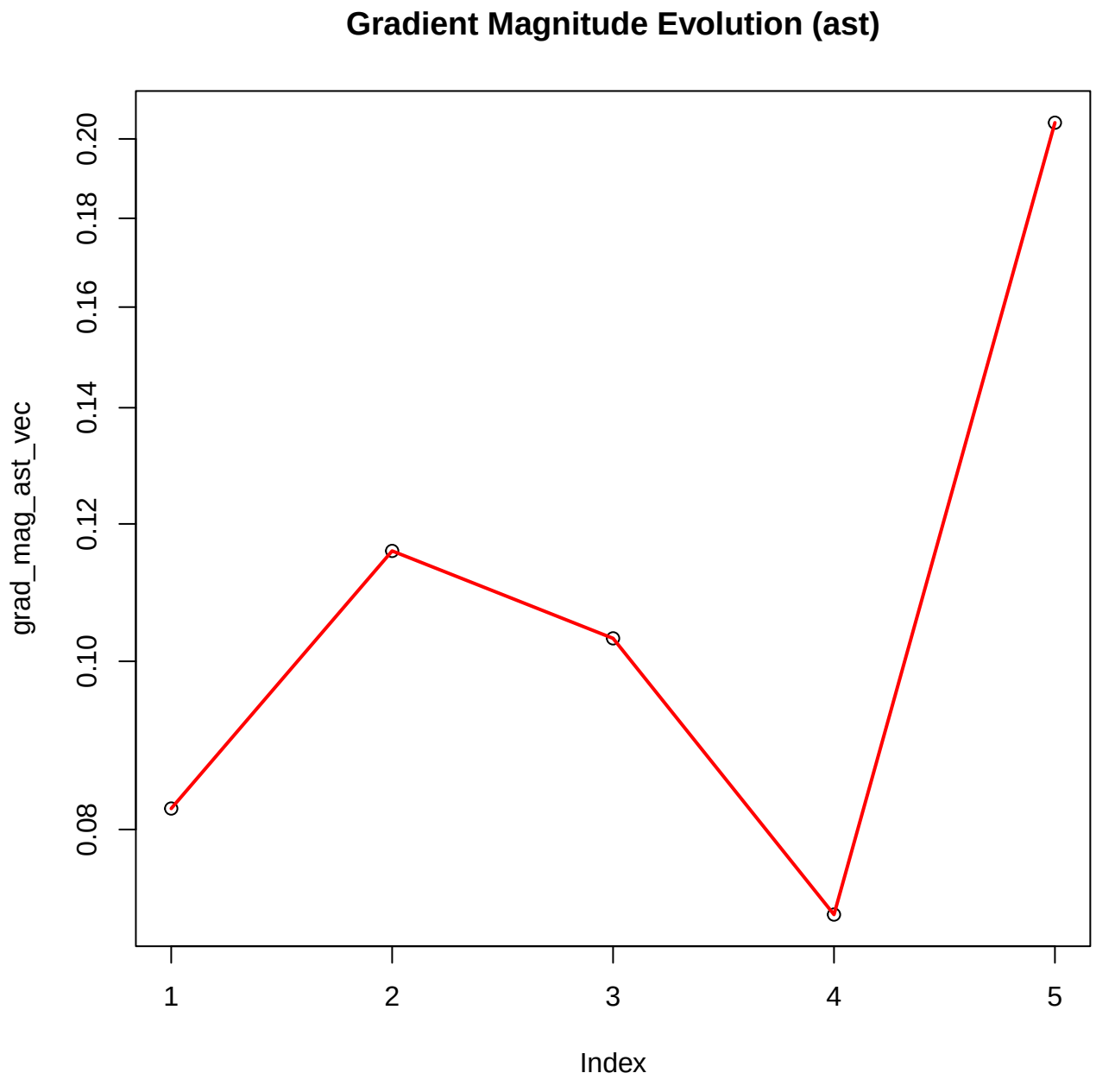

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

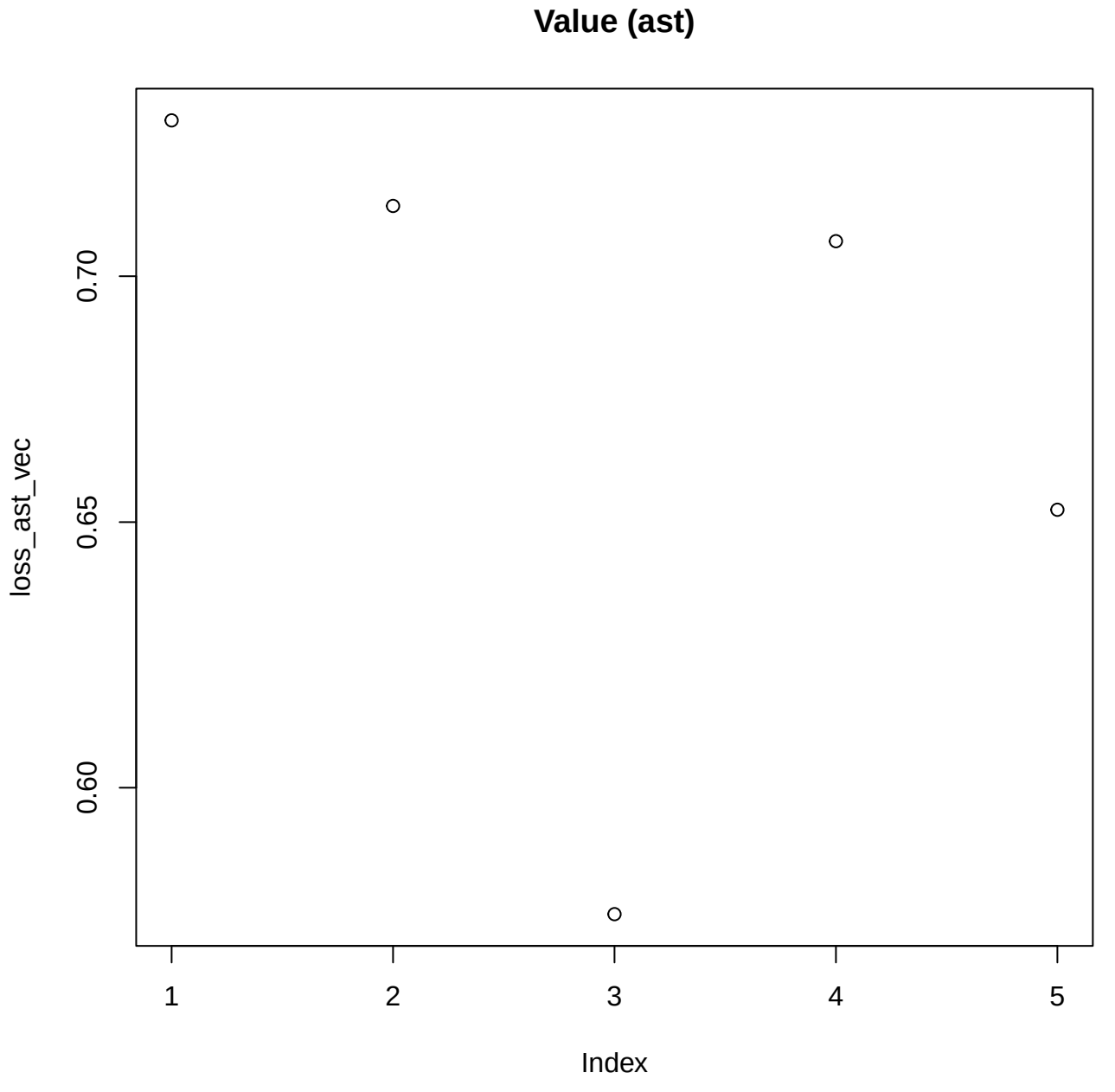

Value (ast)
loss_ast_vec
0.10
0.15
0.20
0.25
0.30
1
2
3
4
5
Index

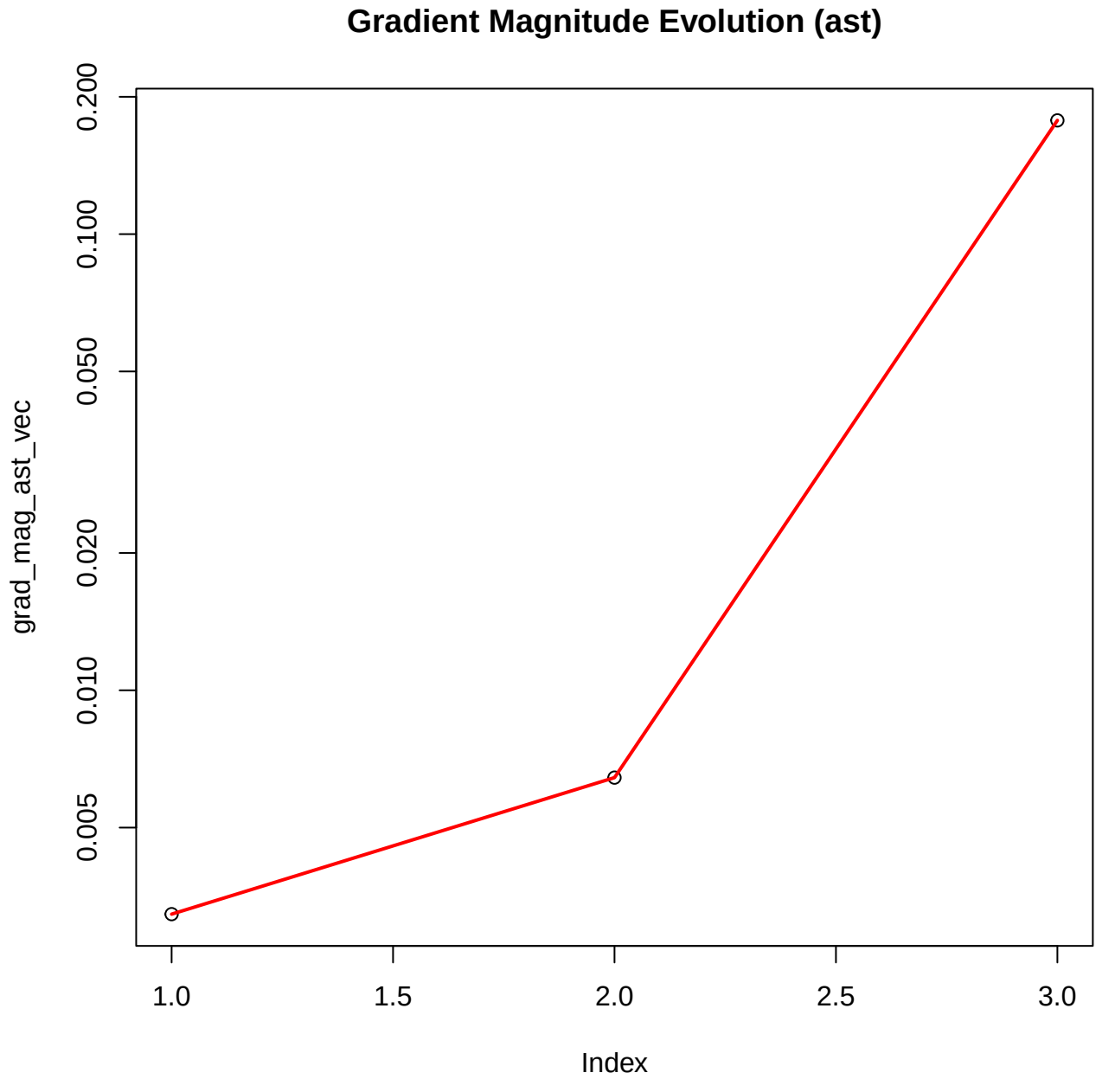

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.86
0.88
0.90
0.92
0.6
0.8
1.0
1.2
1.4
Index

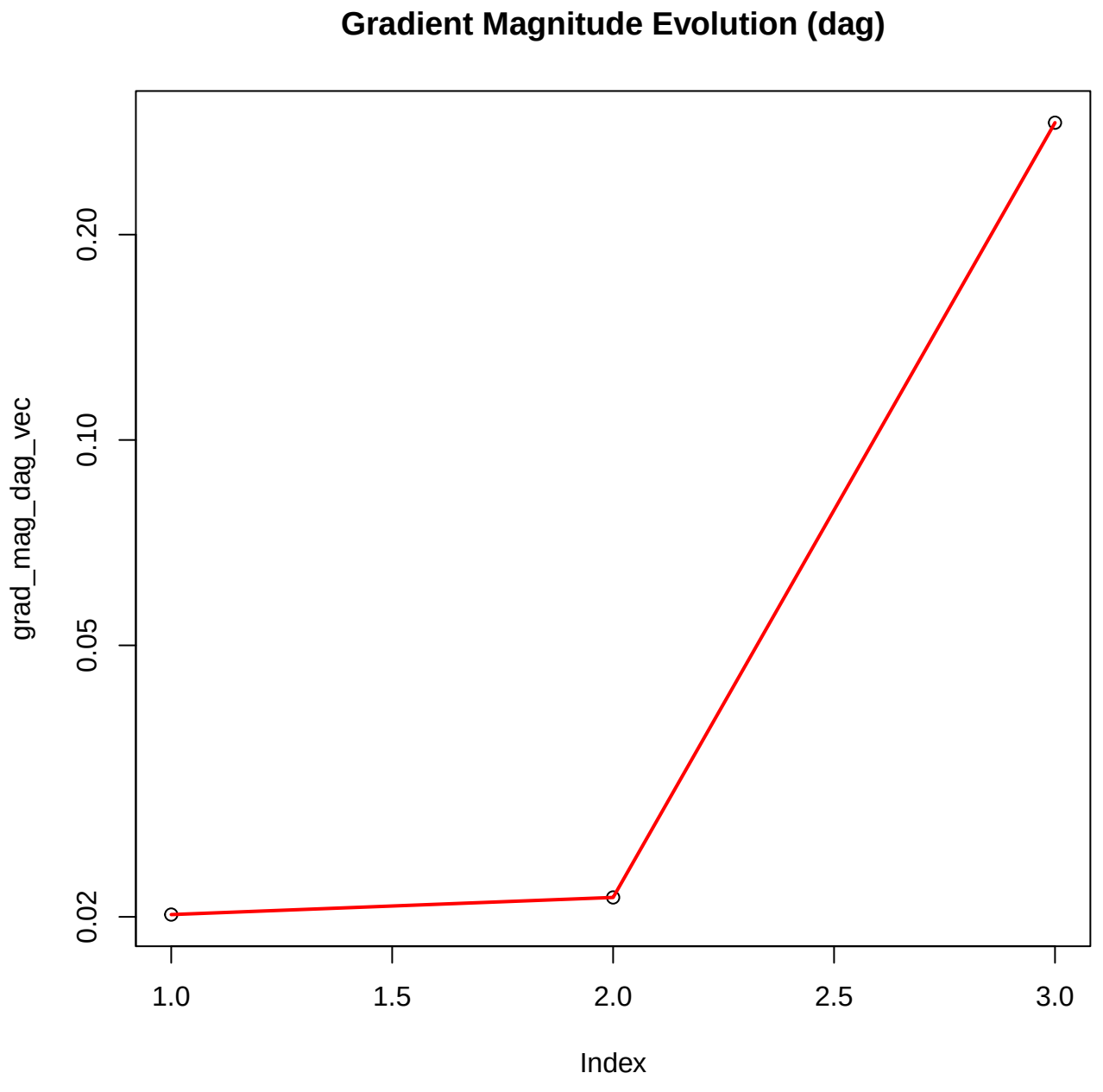

Value (ast)
loss_ast_vec
100
50
20
10
0.6
0.8
1.0
1.2
1.4
Index

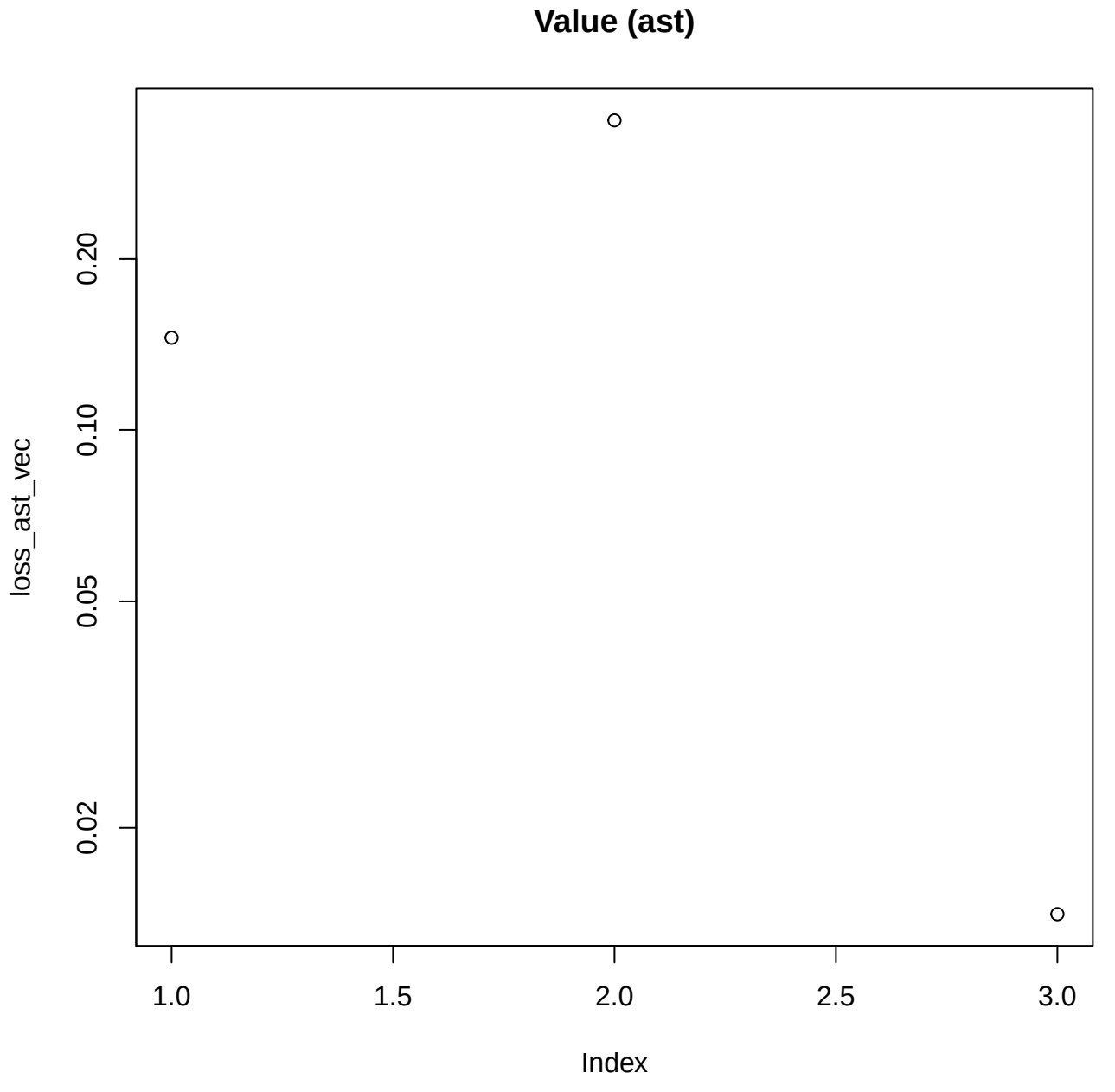

SVI ELBO Loss [169.8257]
ELBO loss
1000
500
200
100
50
2
4
6
8
10
Iteration

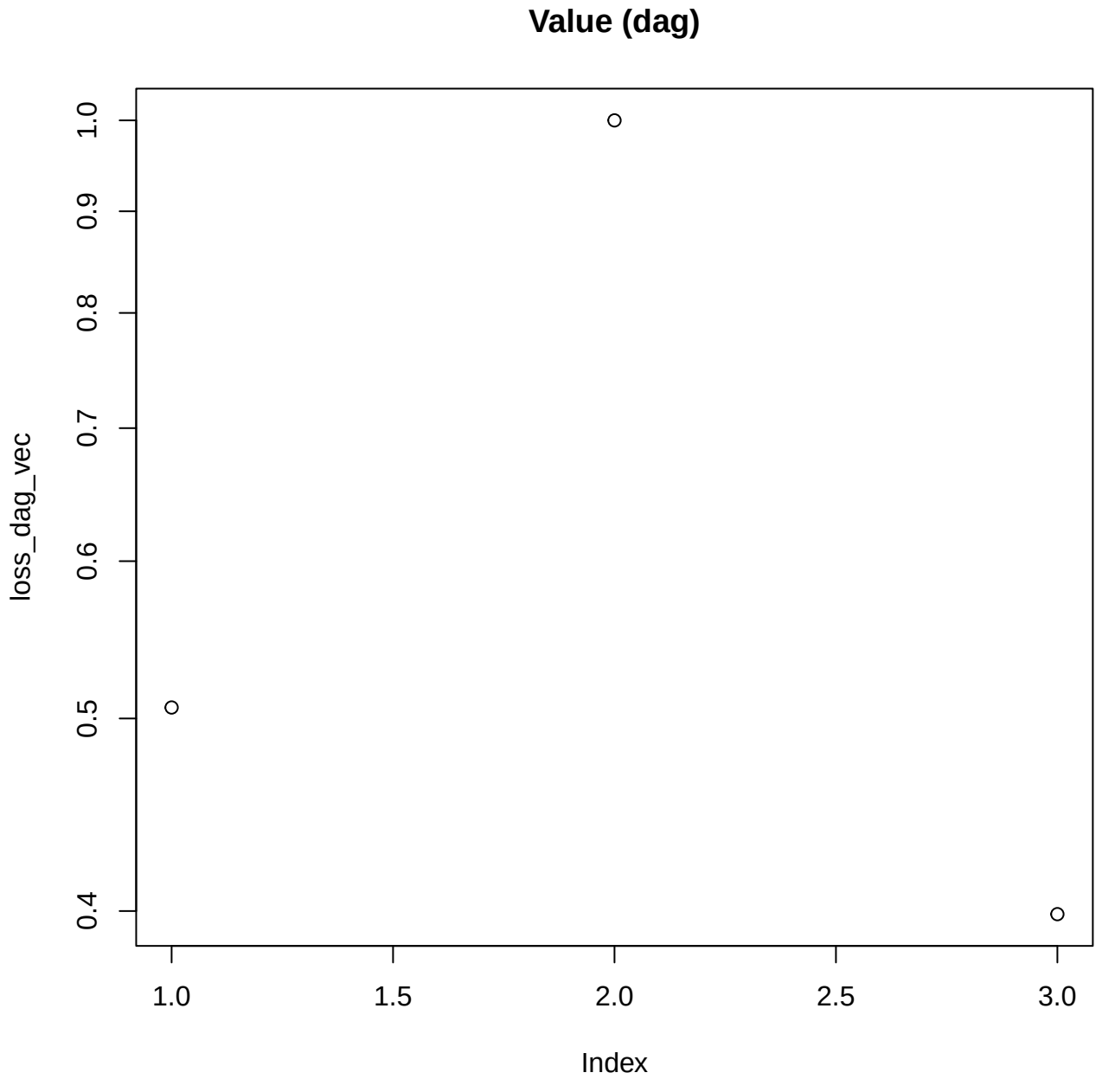

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
0.18
0.20
1
2
3
4
5
Index

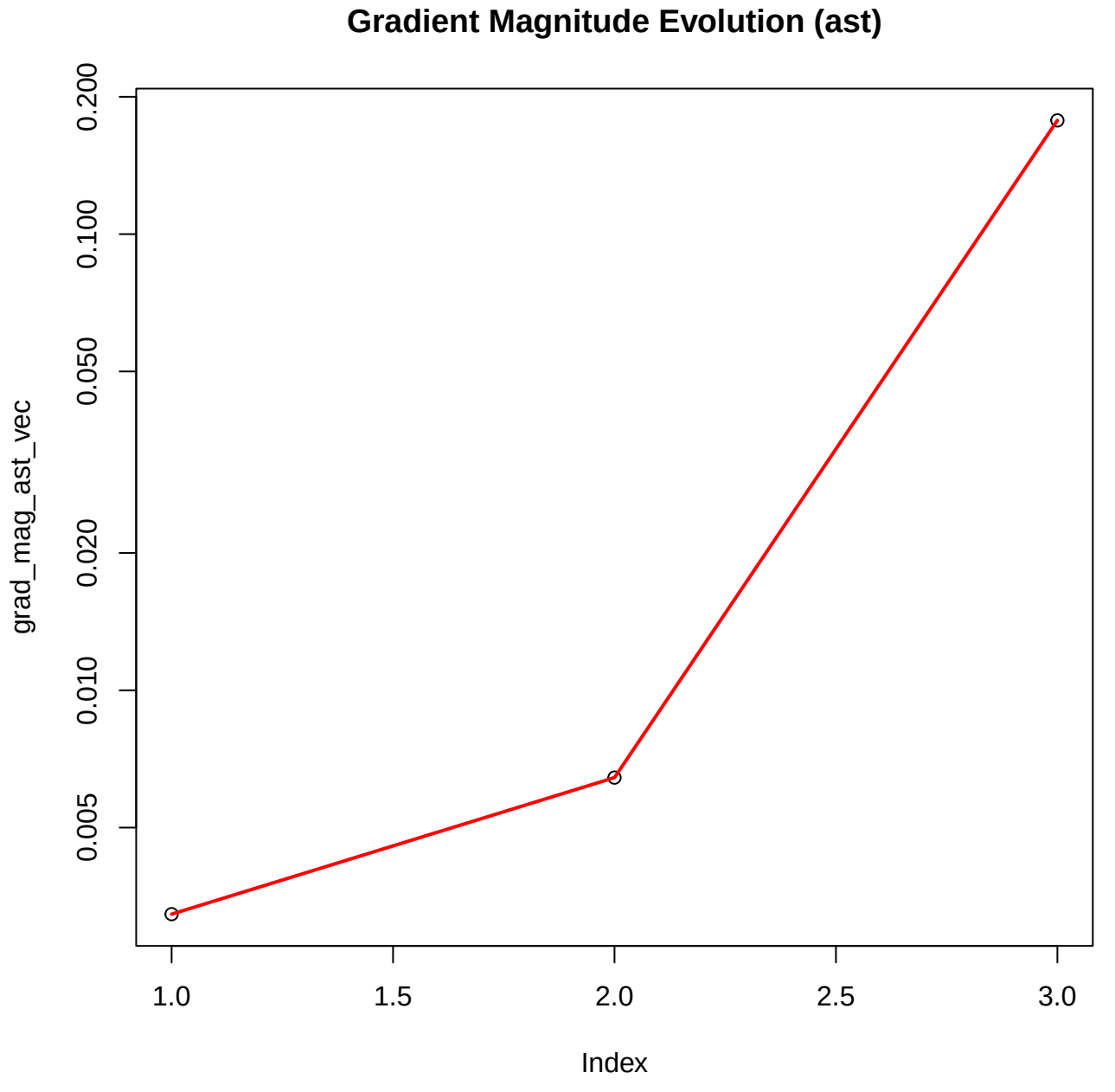

Value (ast)
loss_ast_vec
0.60
0.65
0.70
1
2
3
4
5
Index

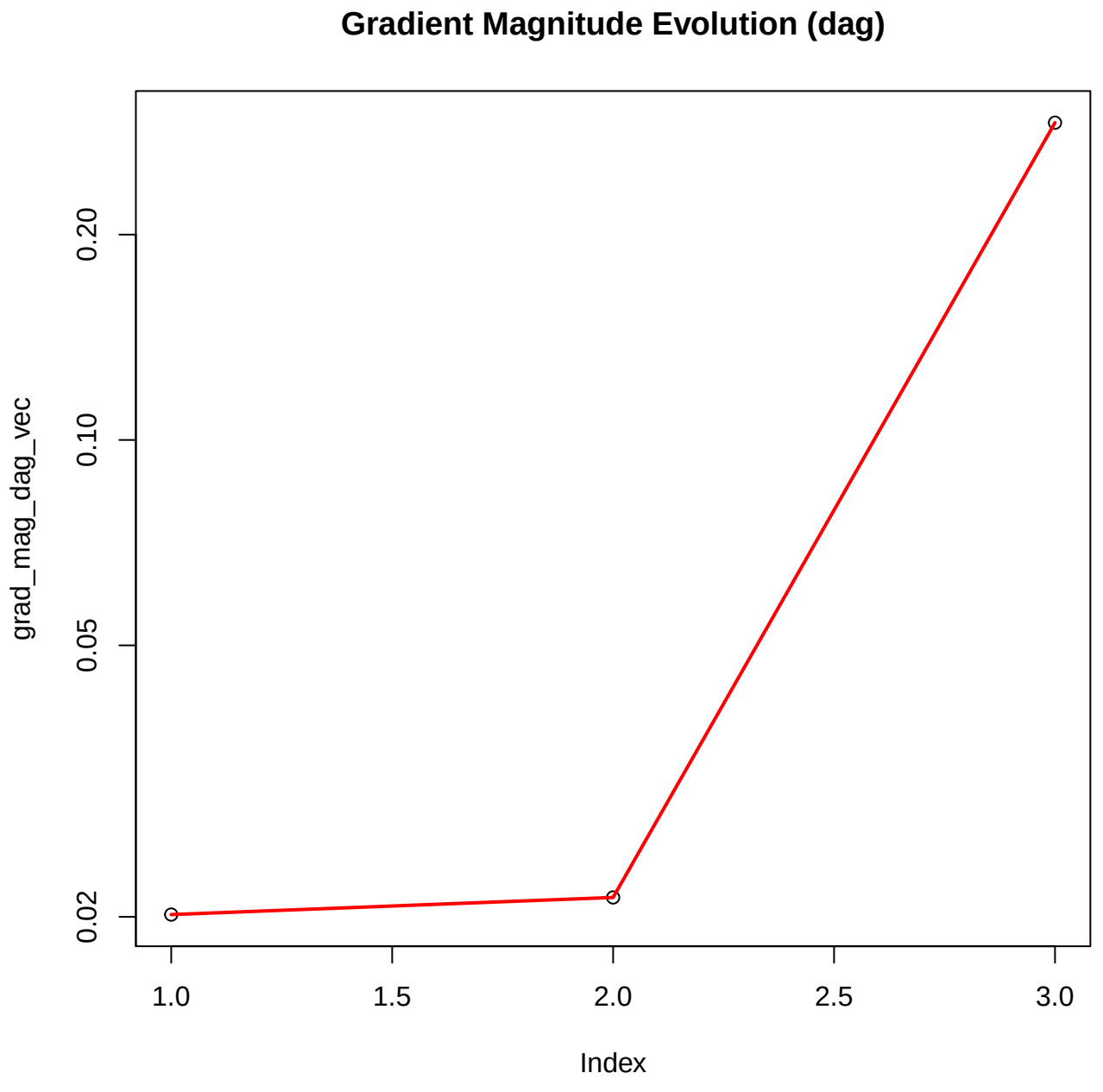

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

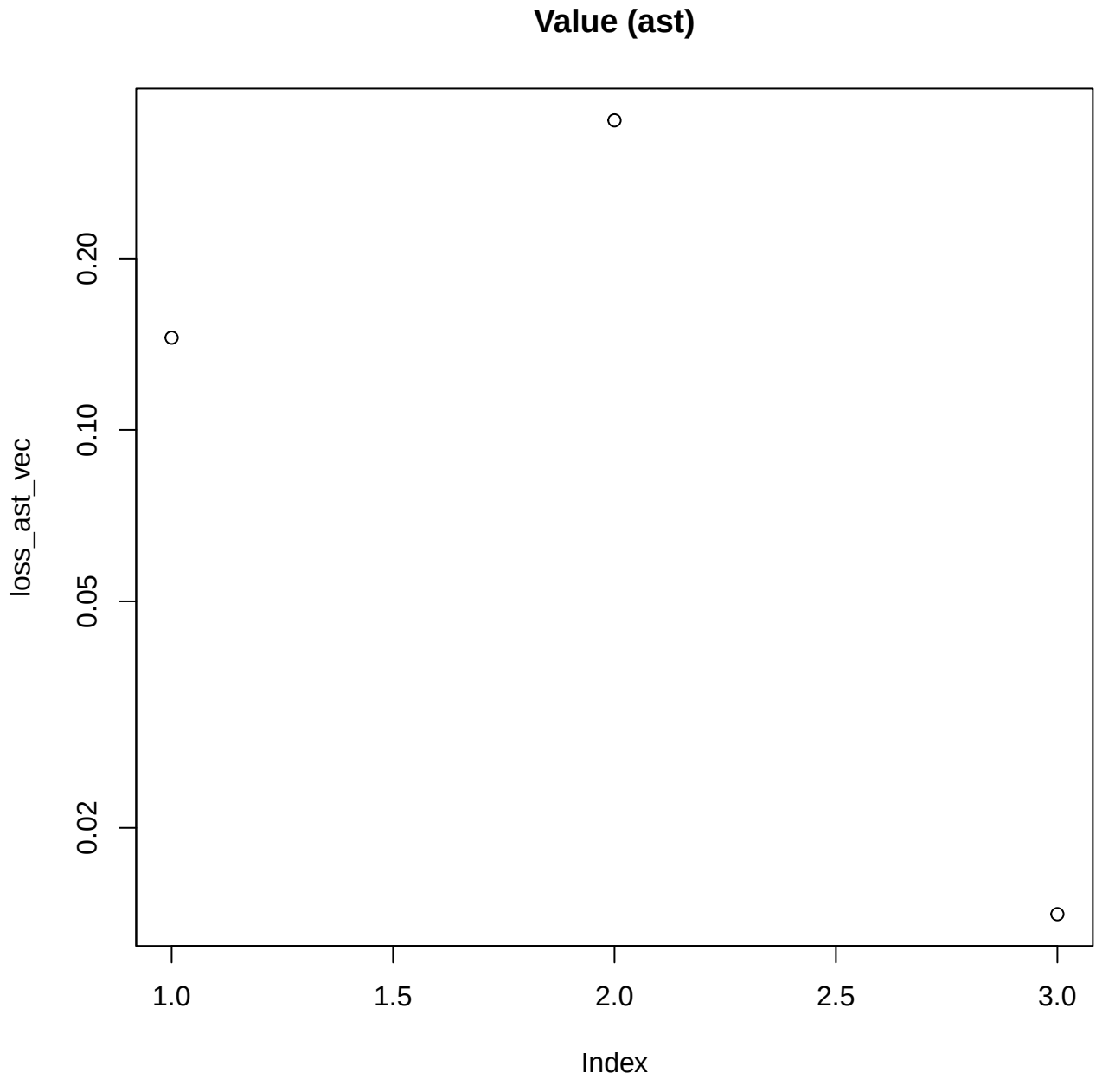

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

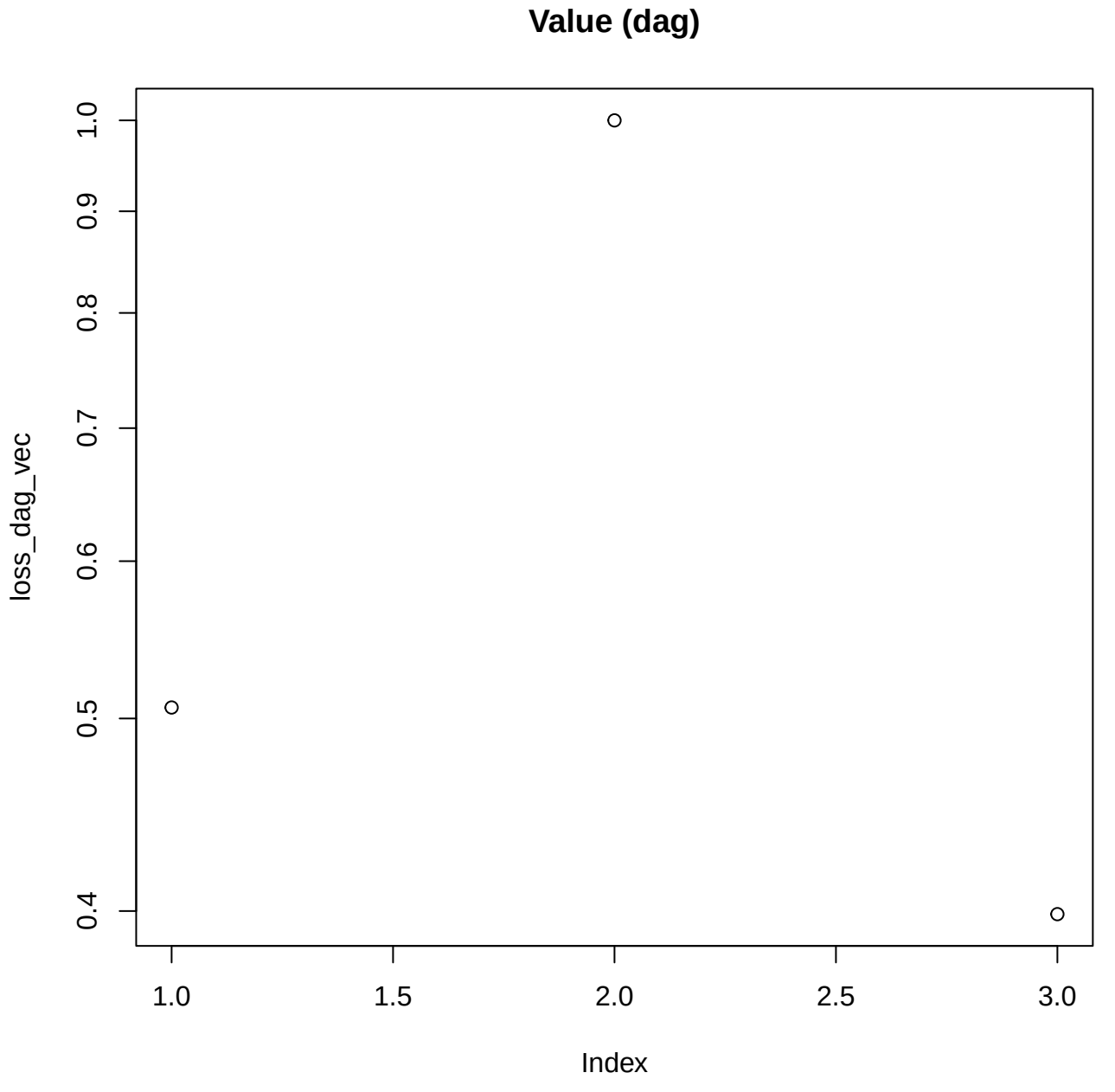

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

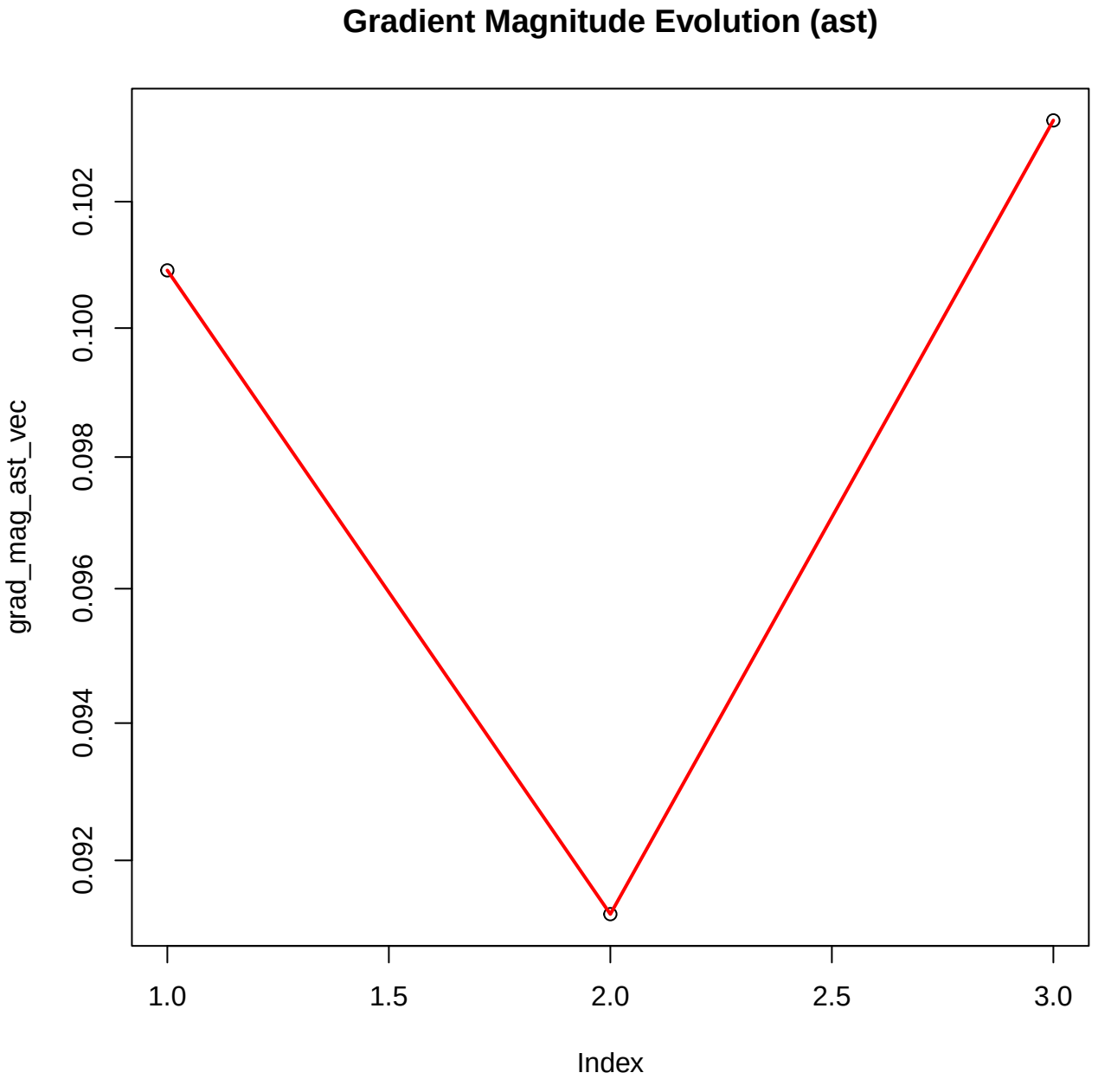

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

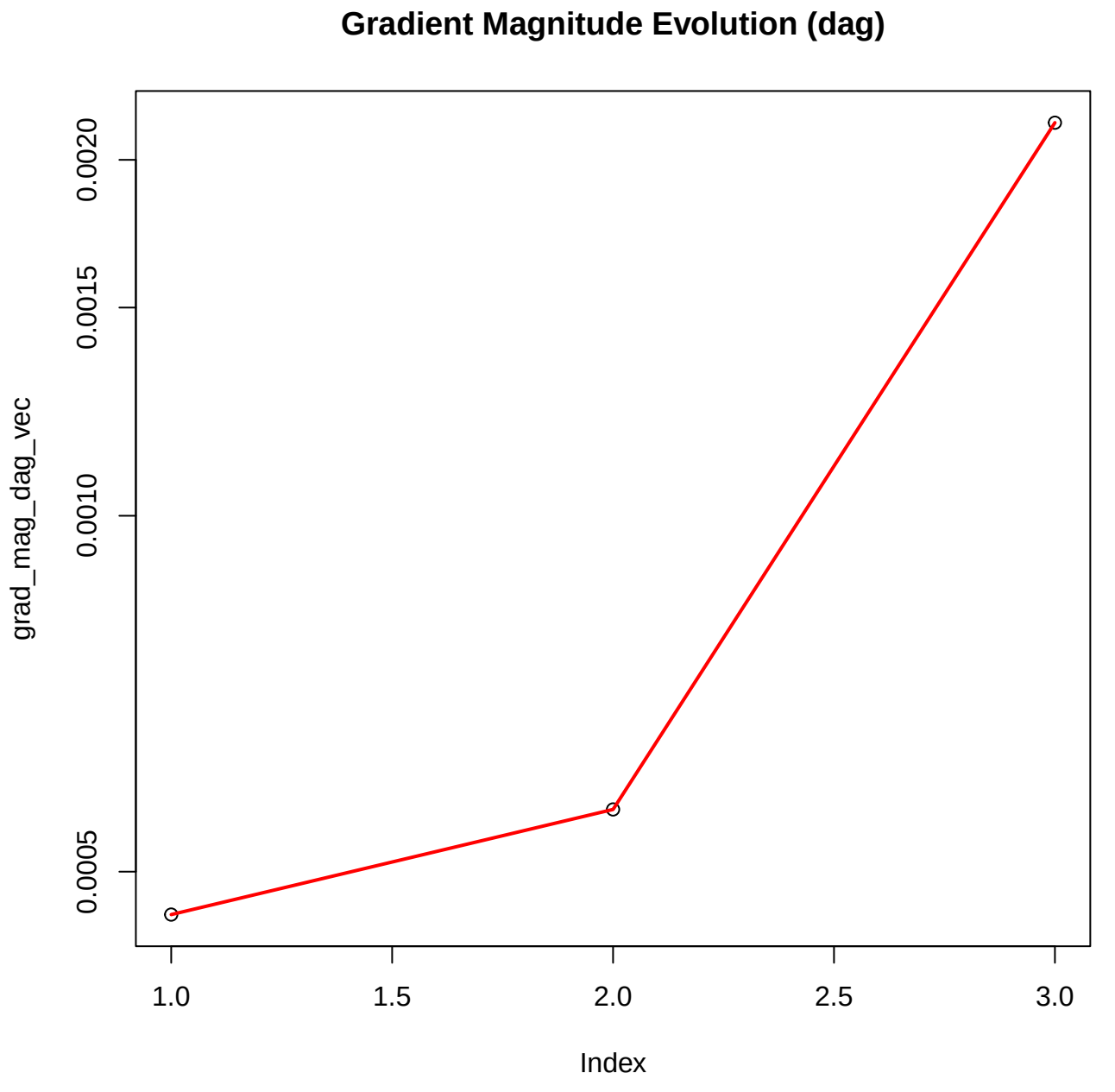

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

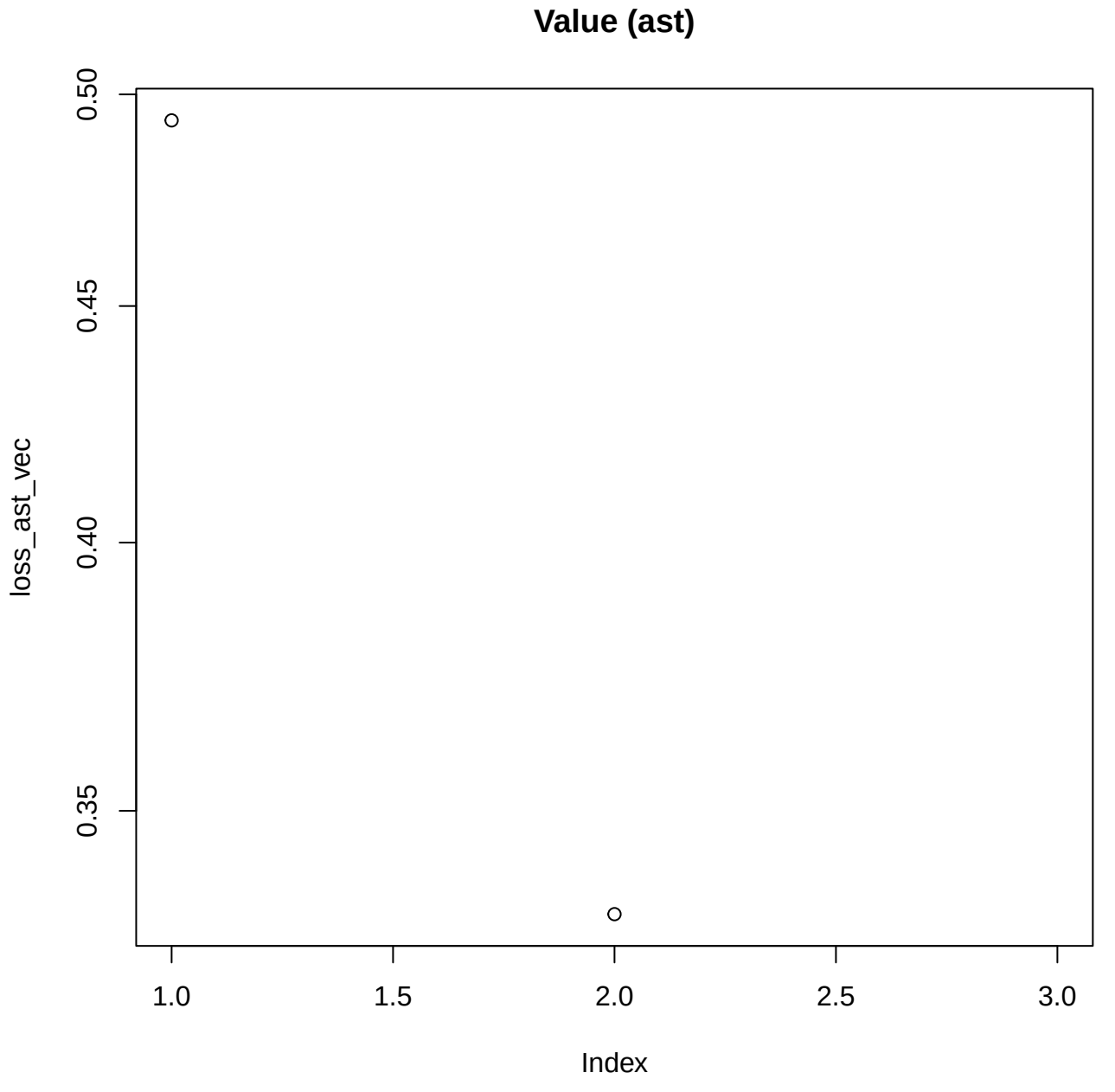

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

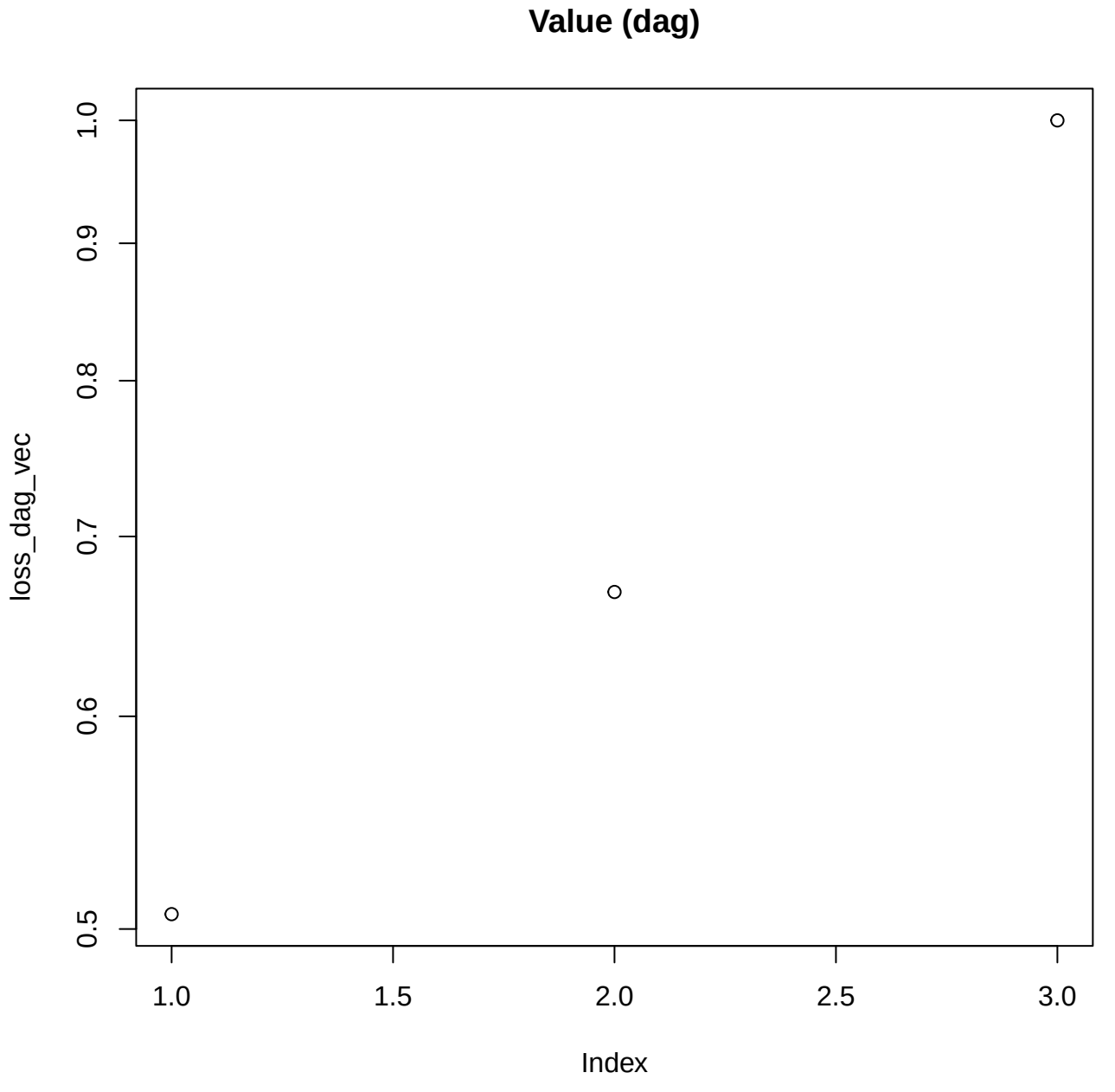

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

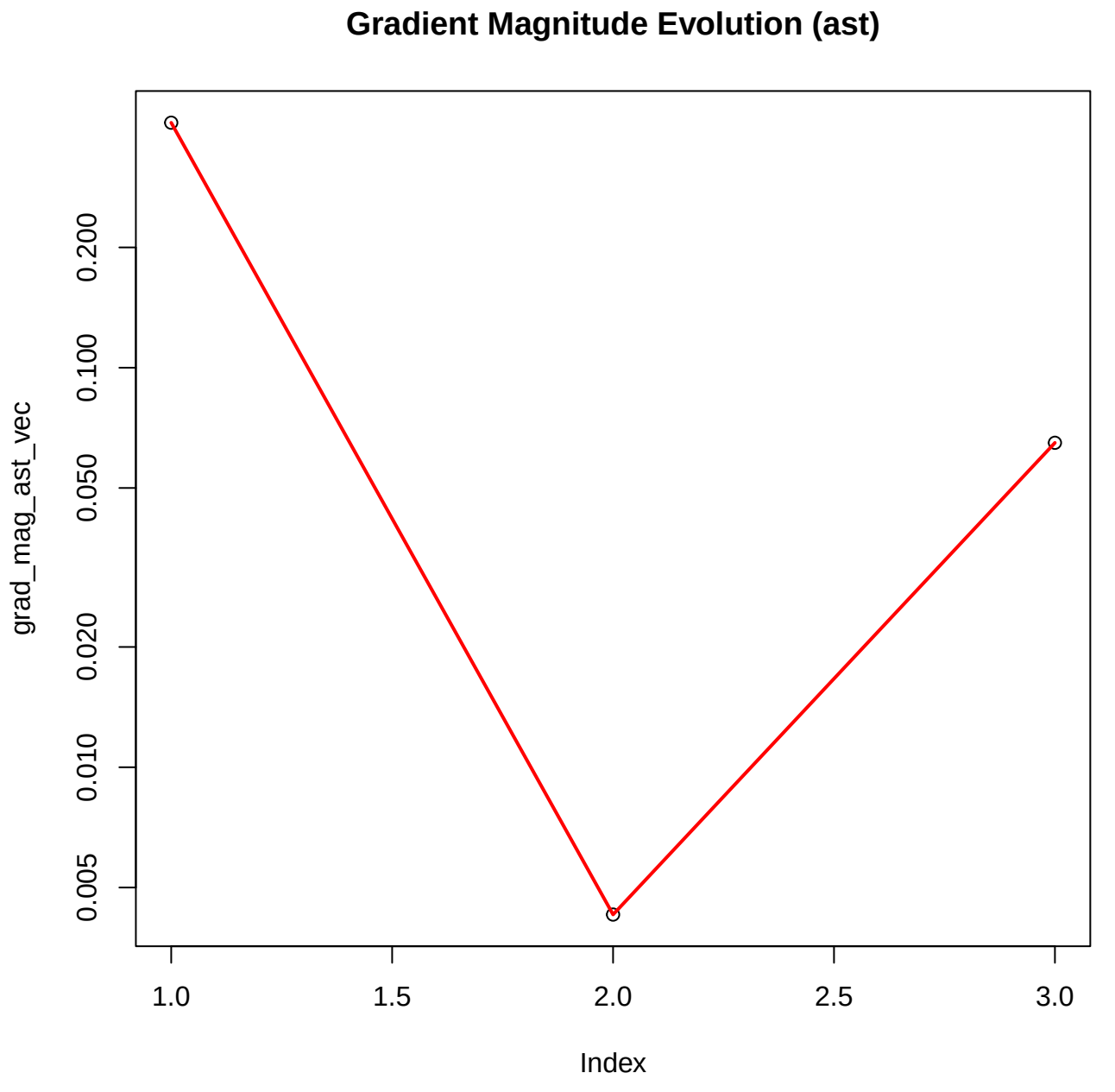

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

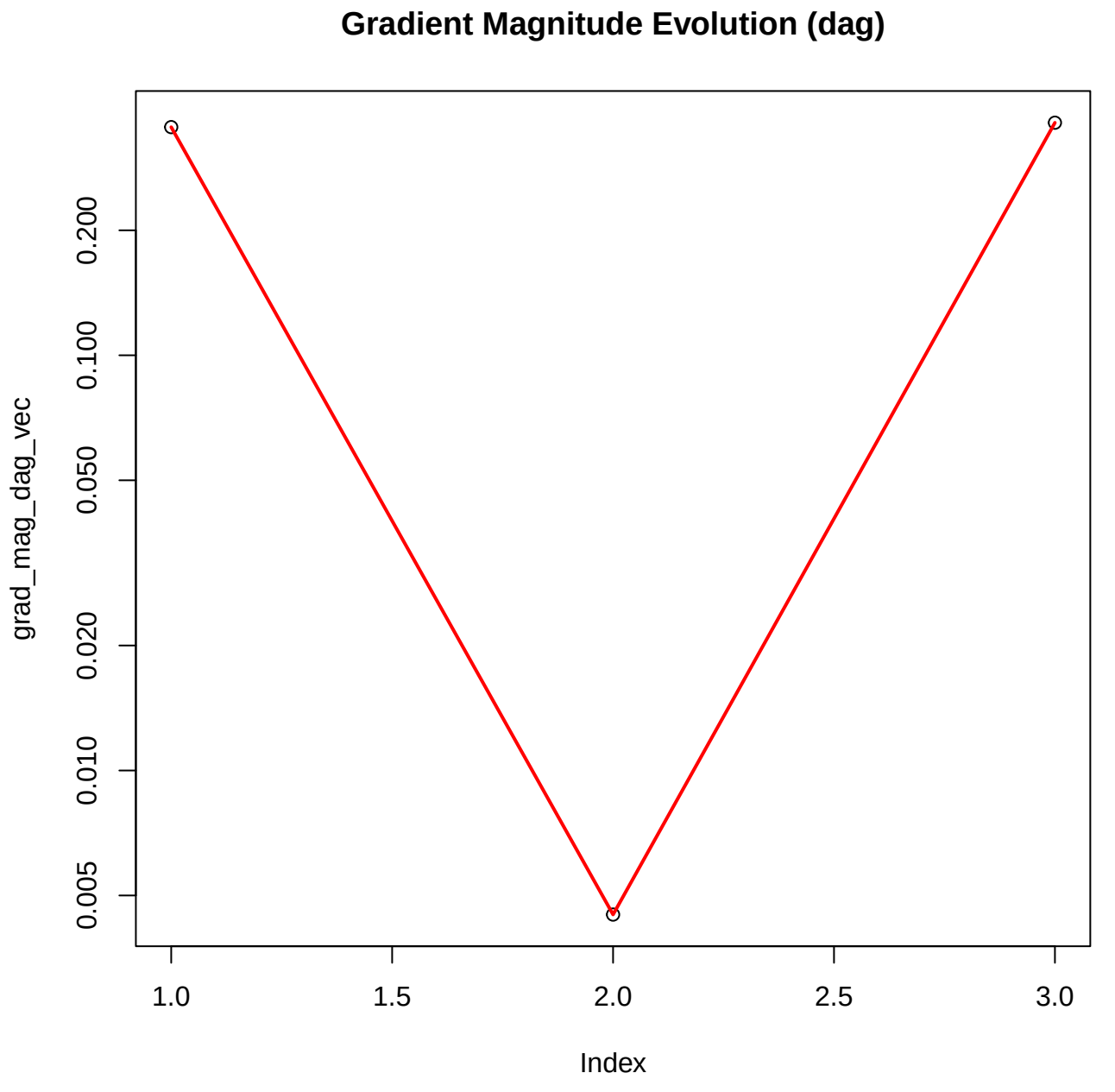

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.092
0.094
0.096
0.098
0.100
0.102
1.0
1.5
2.0
2.5
3.0
Index

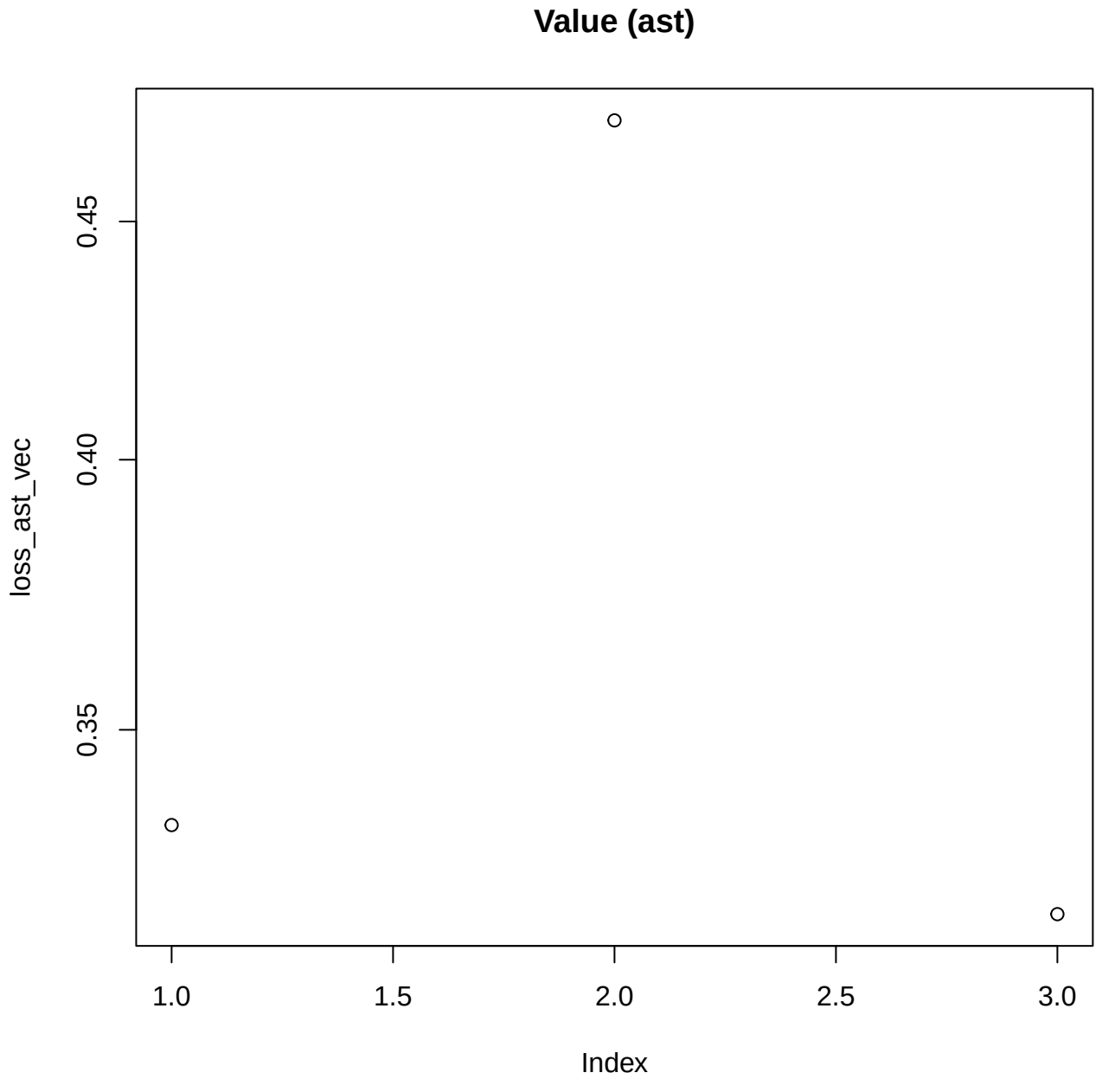

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.0005
0.0010
0.0015
0.0020
1.0
1.5
2.0
2.5
3.0
Index

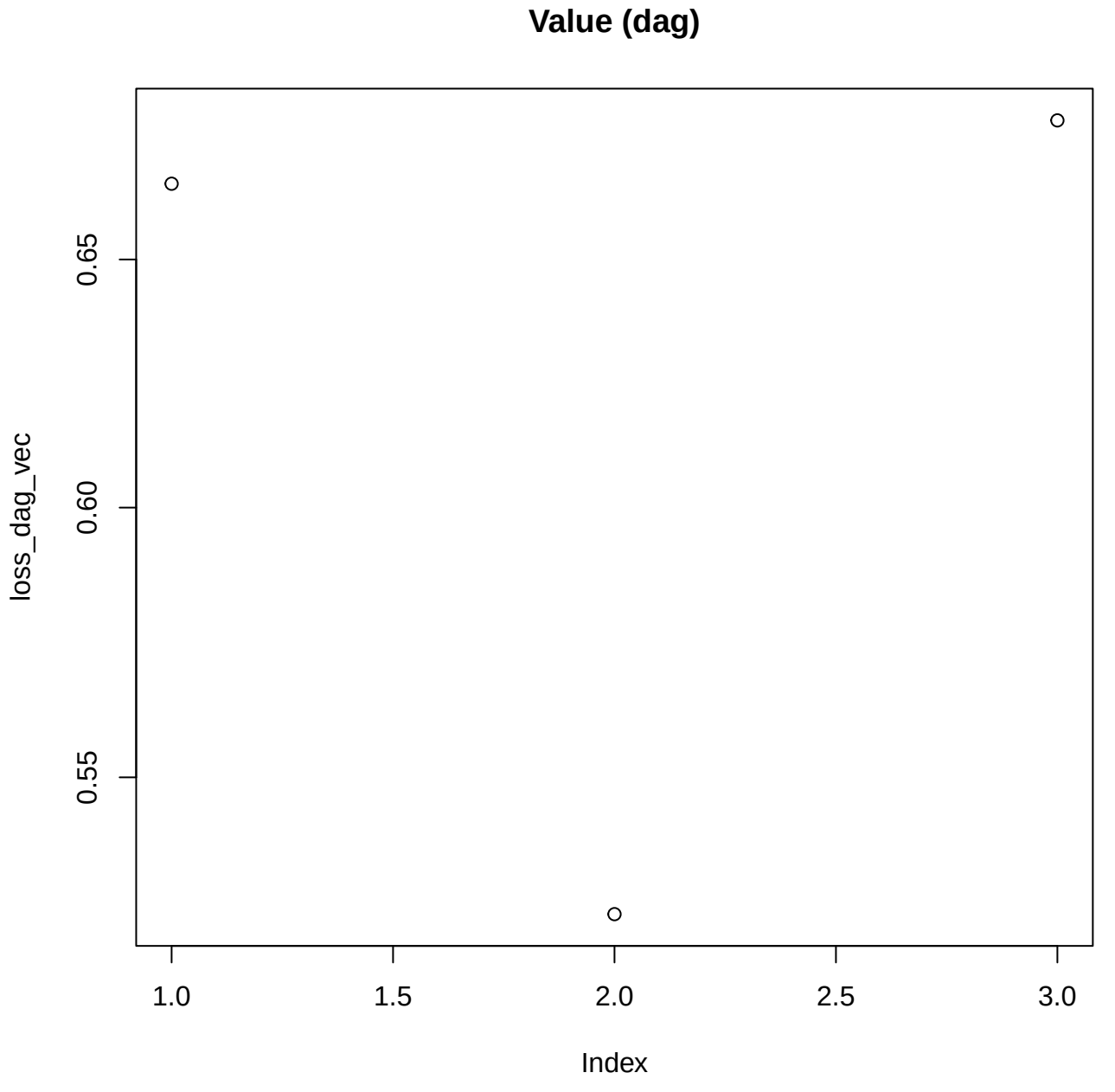

Value (ast)
loss_ast_vec
0.50
0.45
0.40
0.35
1.0
1.5
2.0
2.5
3.0
Index

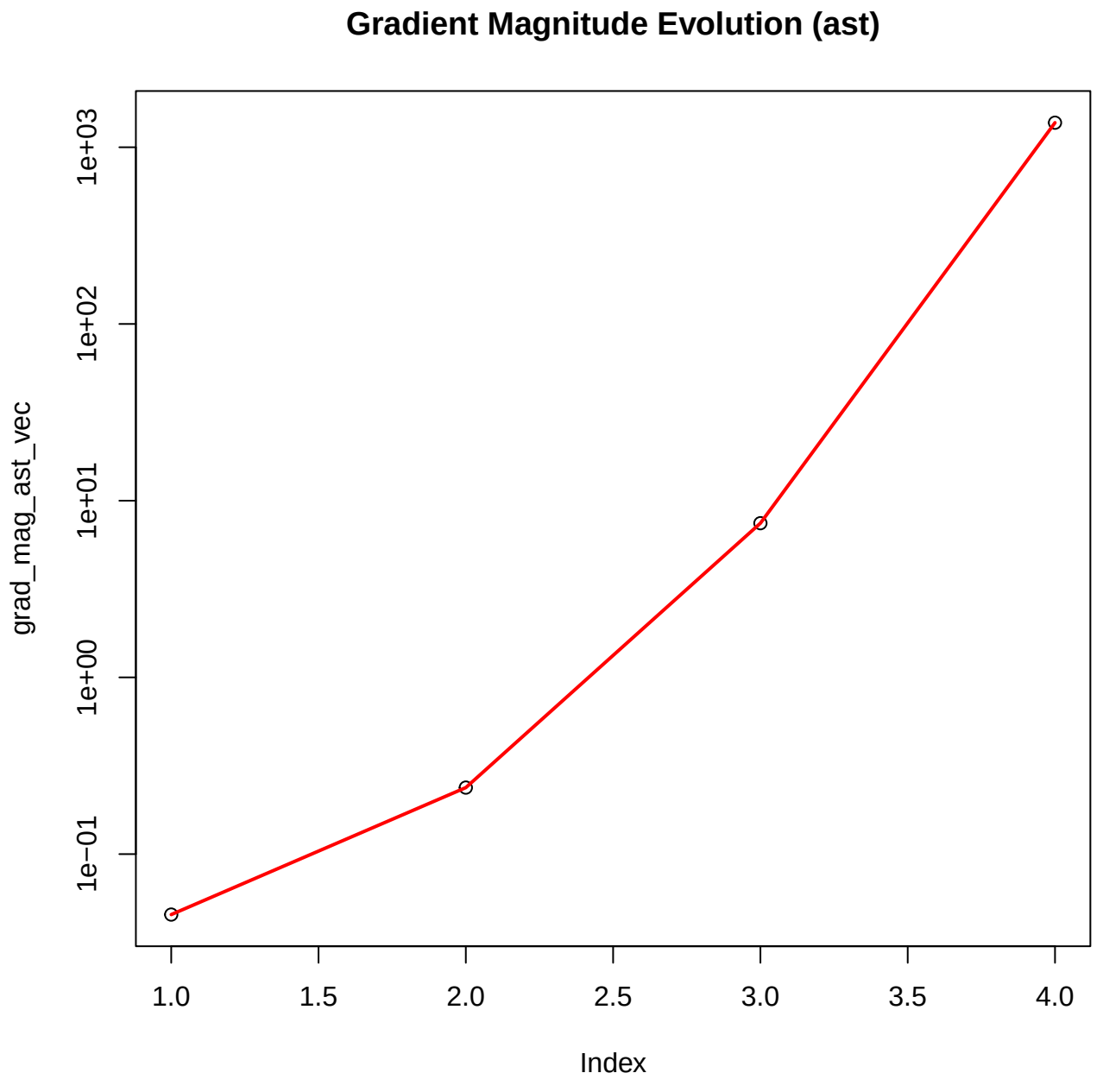

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
0.9
1.0
1.0
1.5
2.0
2.5
3.0
Index

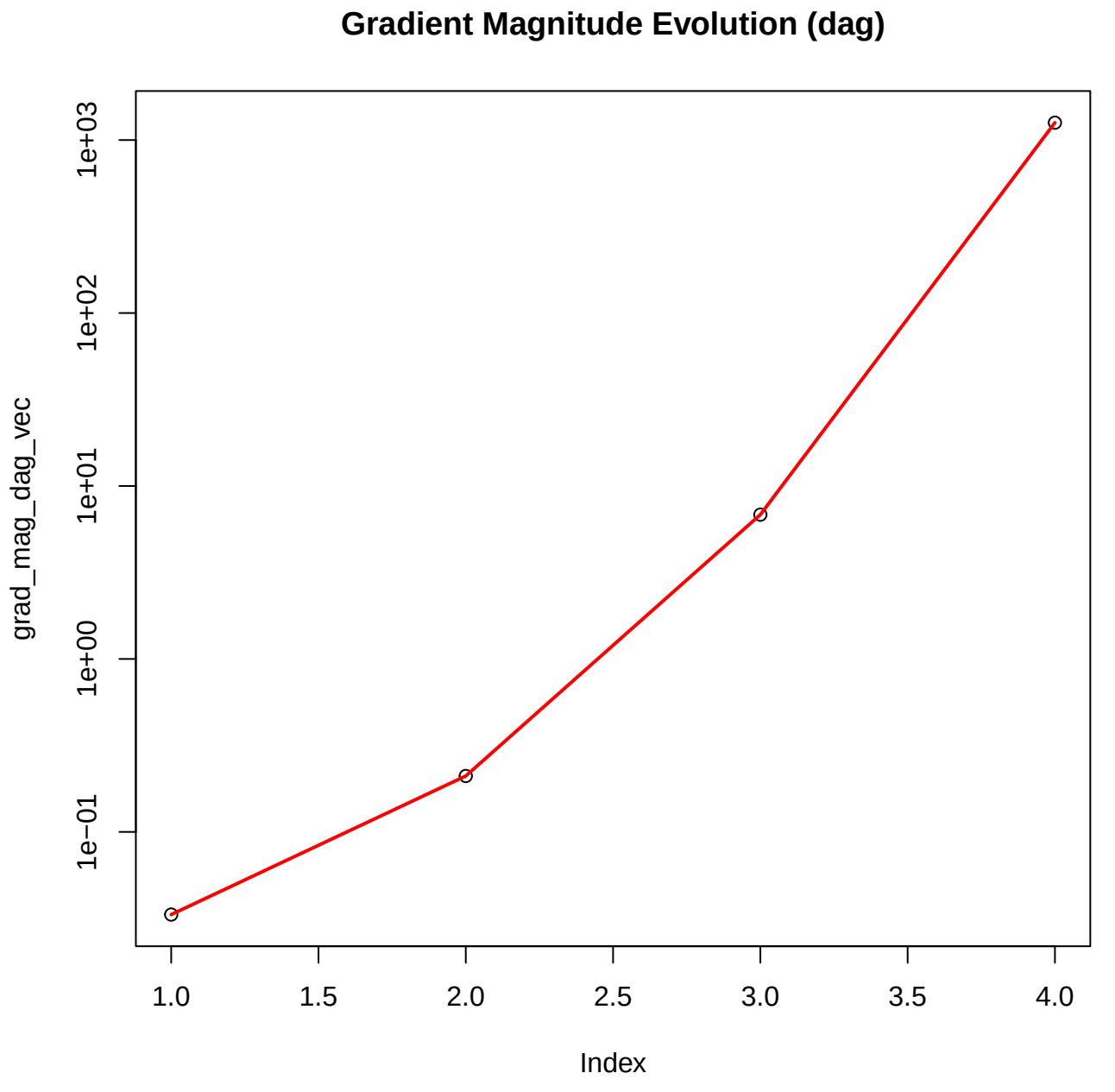

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

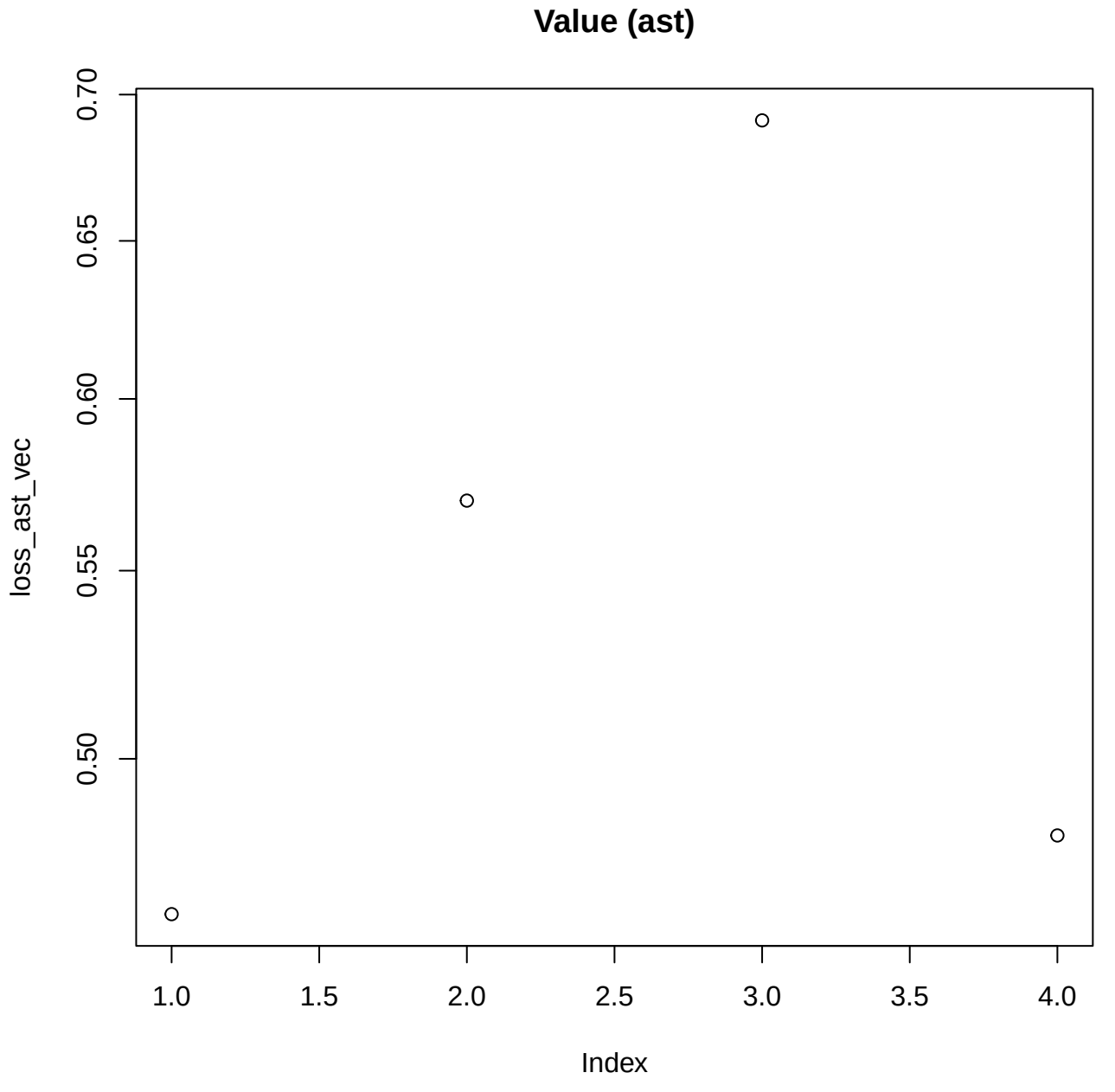

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

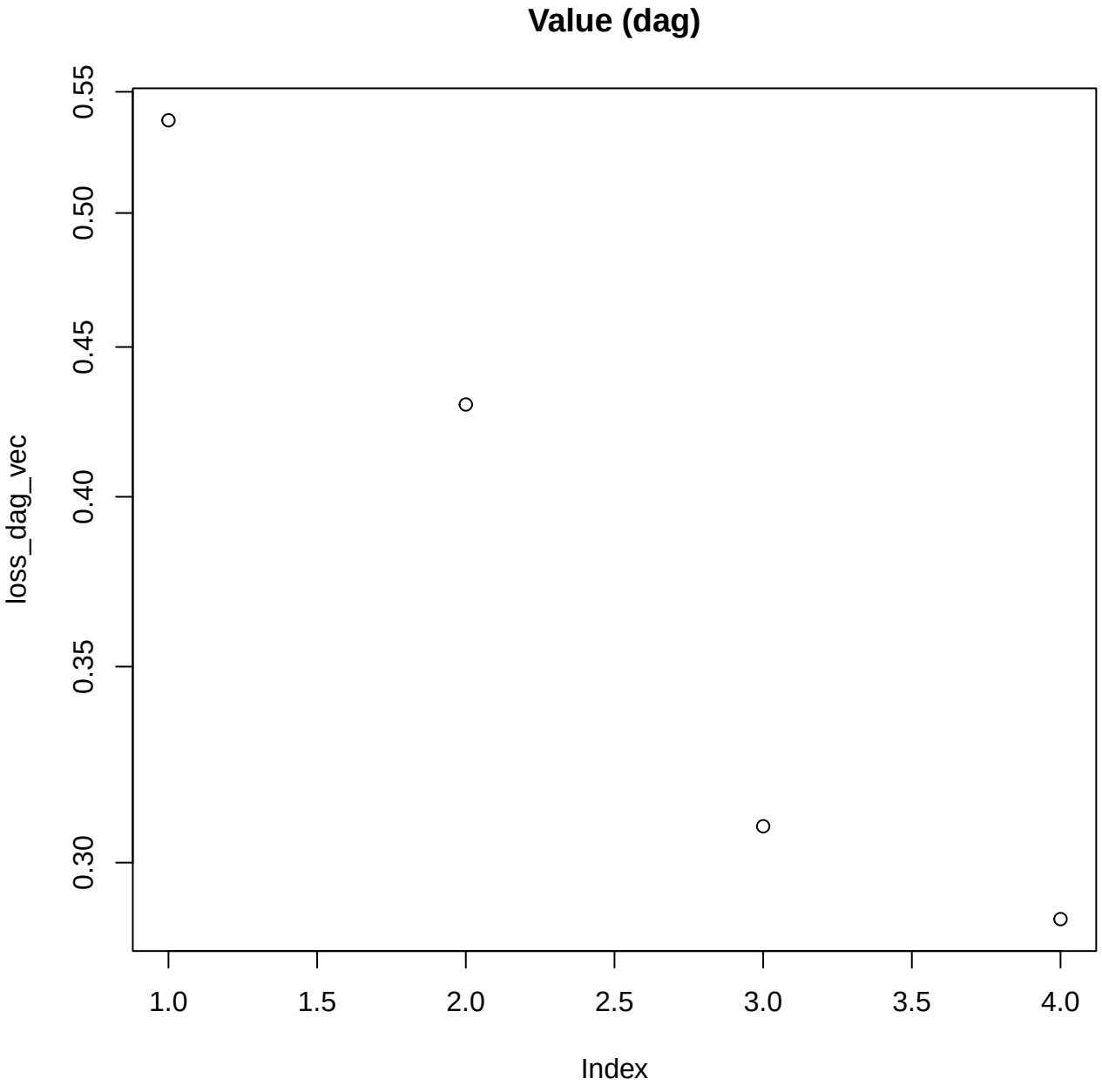

Value (ast)
loss_ast_vec
0.35
0.40
0.45
1.0
1.5
2.0
2.5
3.0
Index

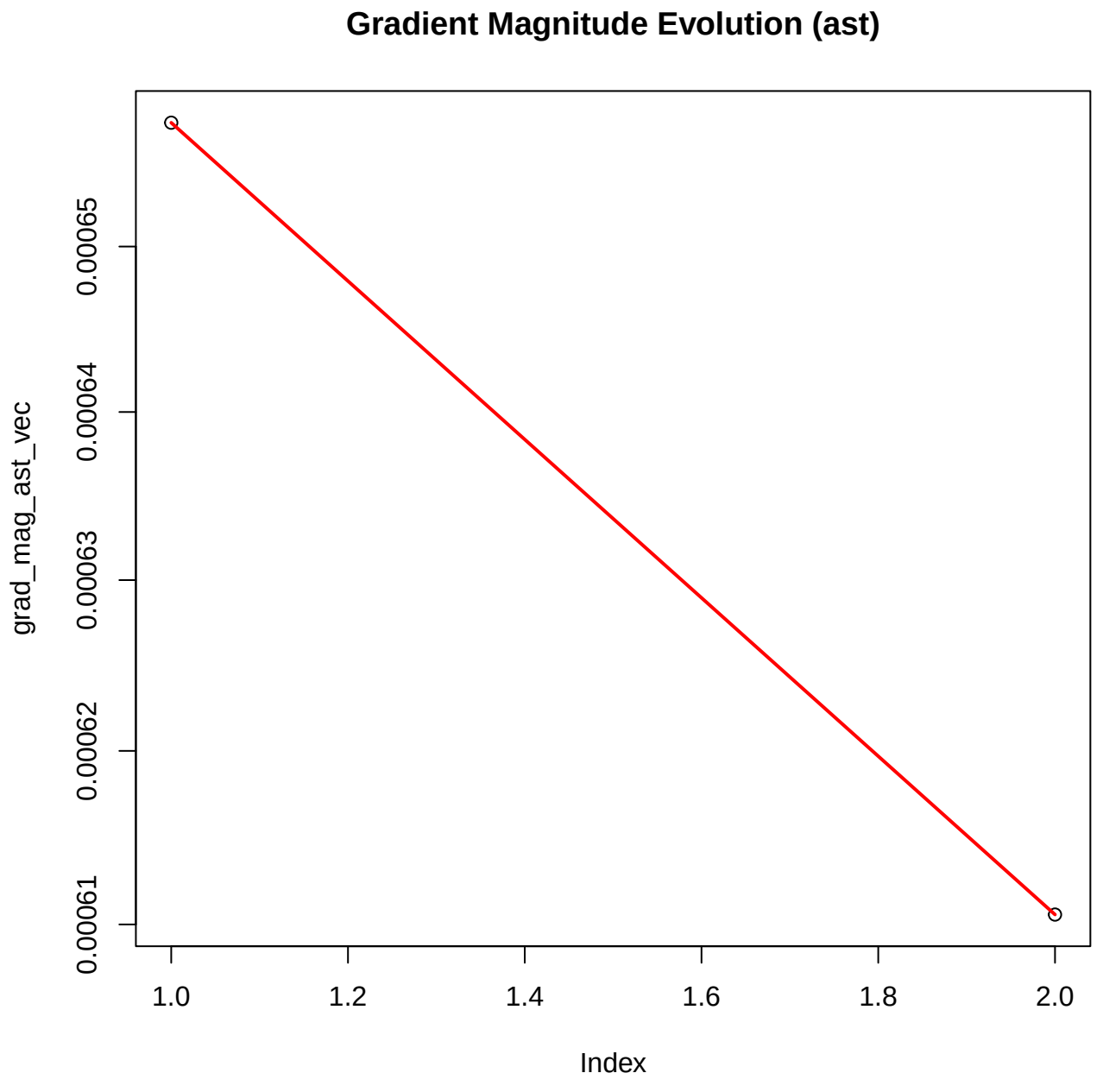

Value (dag)
loss_dag_vec
0.55
0.60
0.65
1.0
1.5
2.0
2.5
3.0
Index

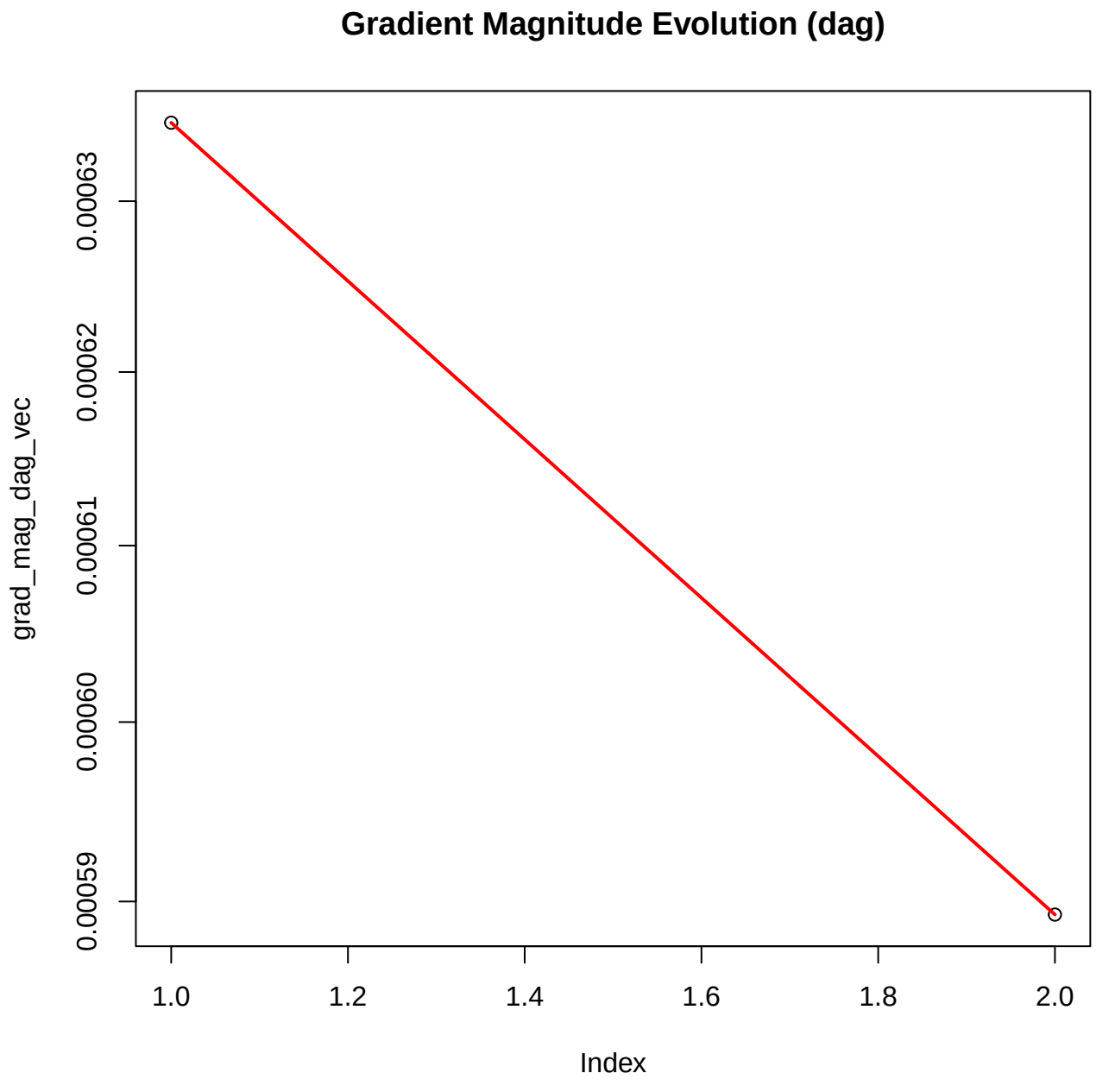

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

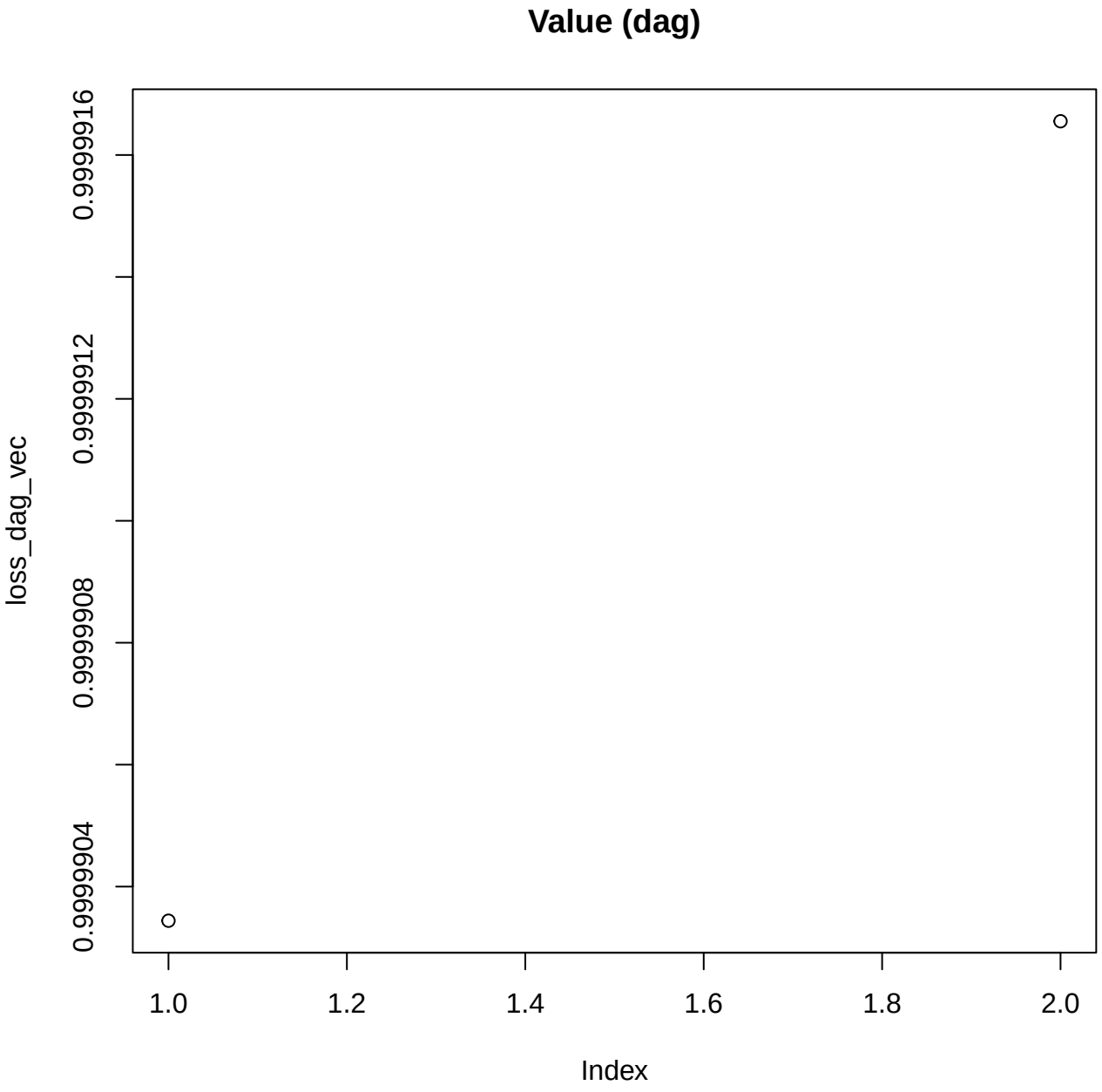

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

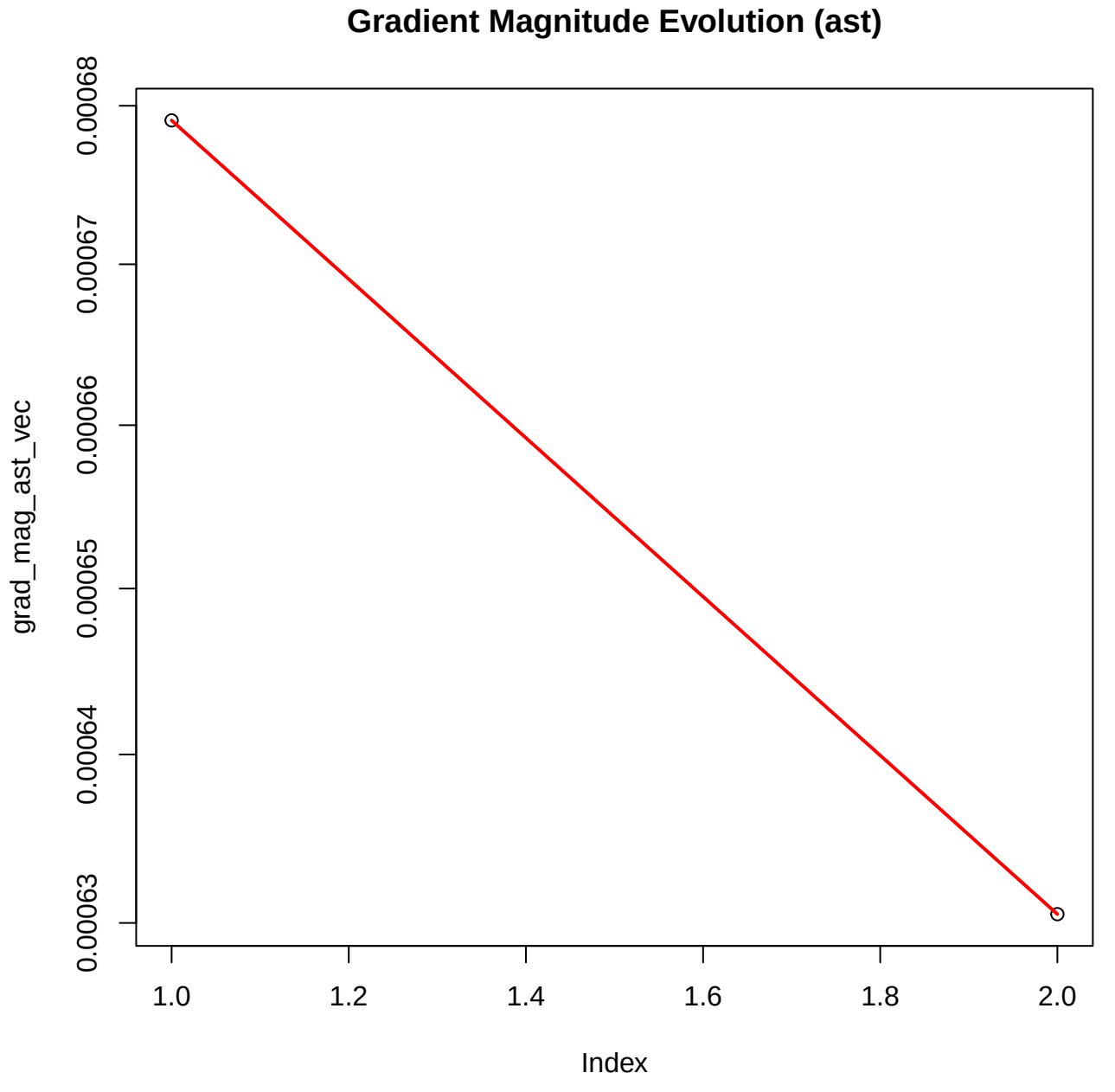

Value (ast)
loss_ast_vec
0.50
0.55
0.60
0.65
0.70
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

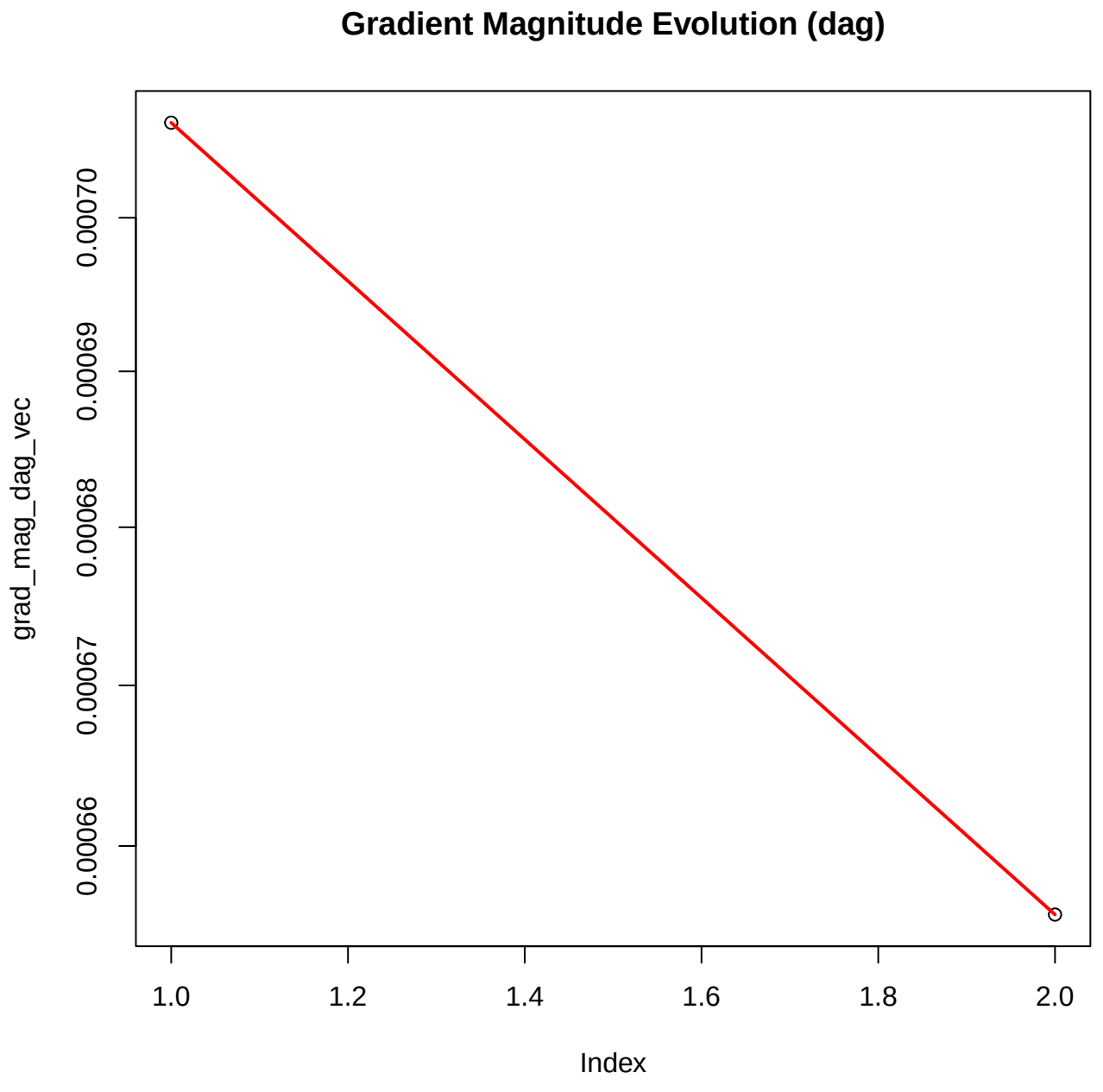

Value (dag)
loss_dag_vec
0.55
0.50
0.45
0.40
0.35
0.30
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

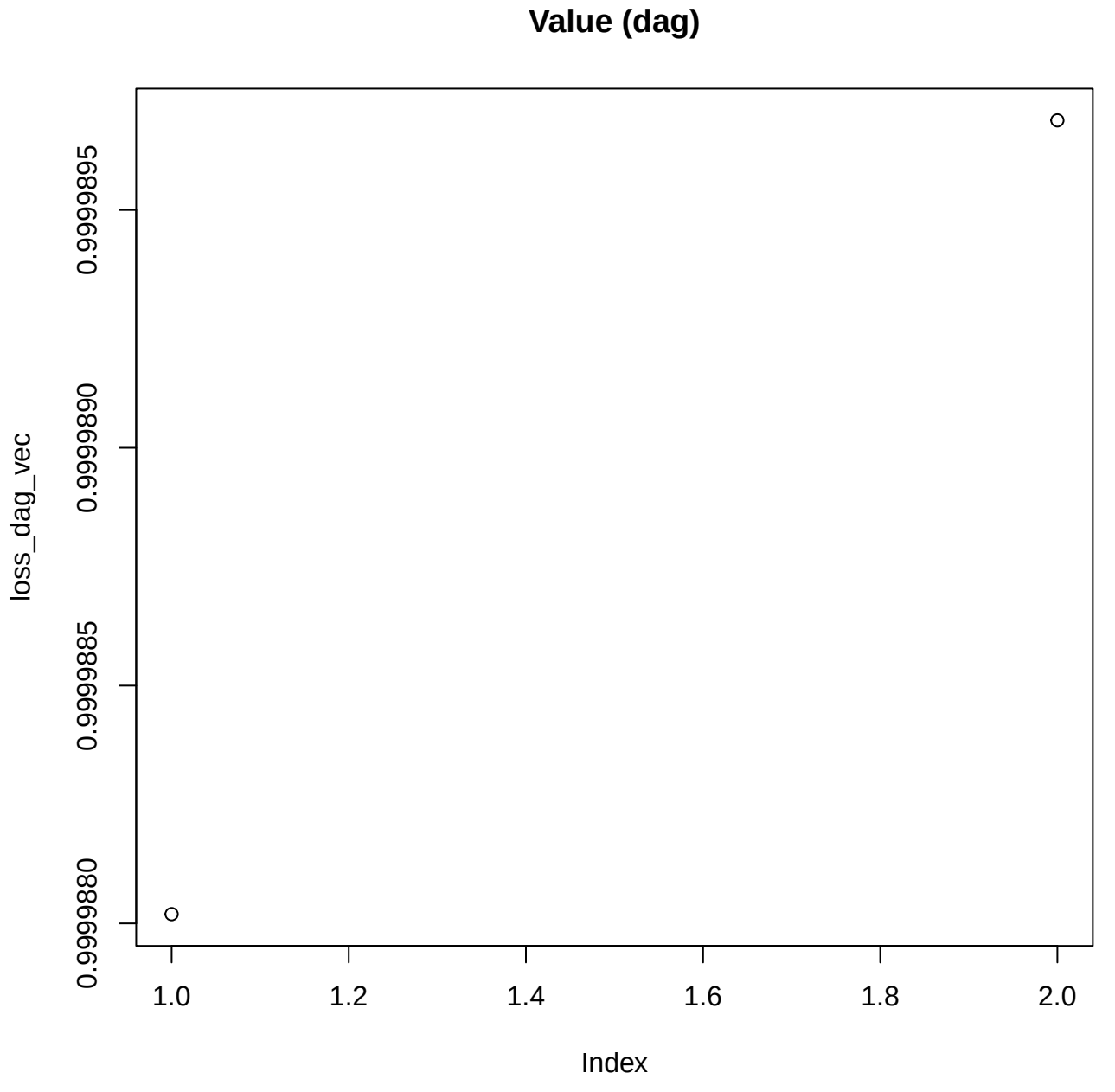

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

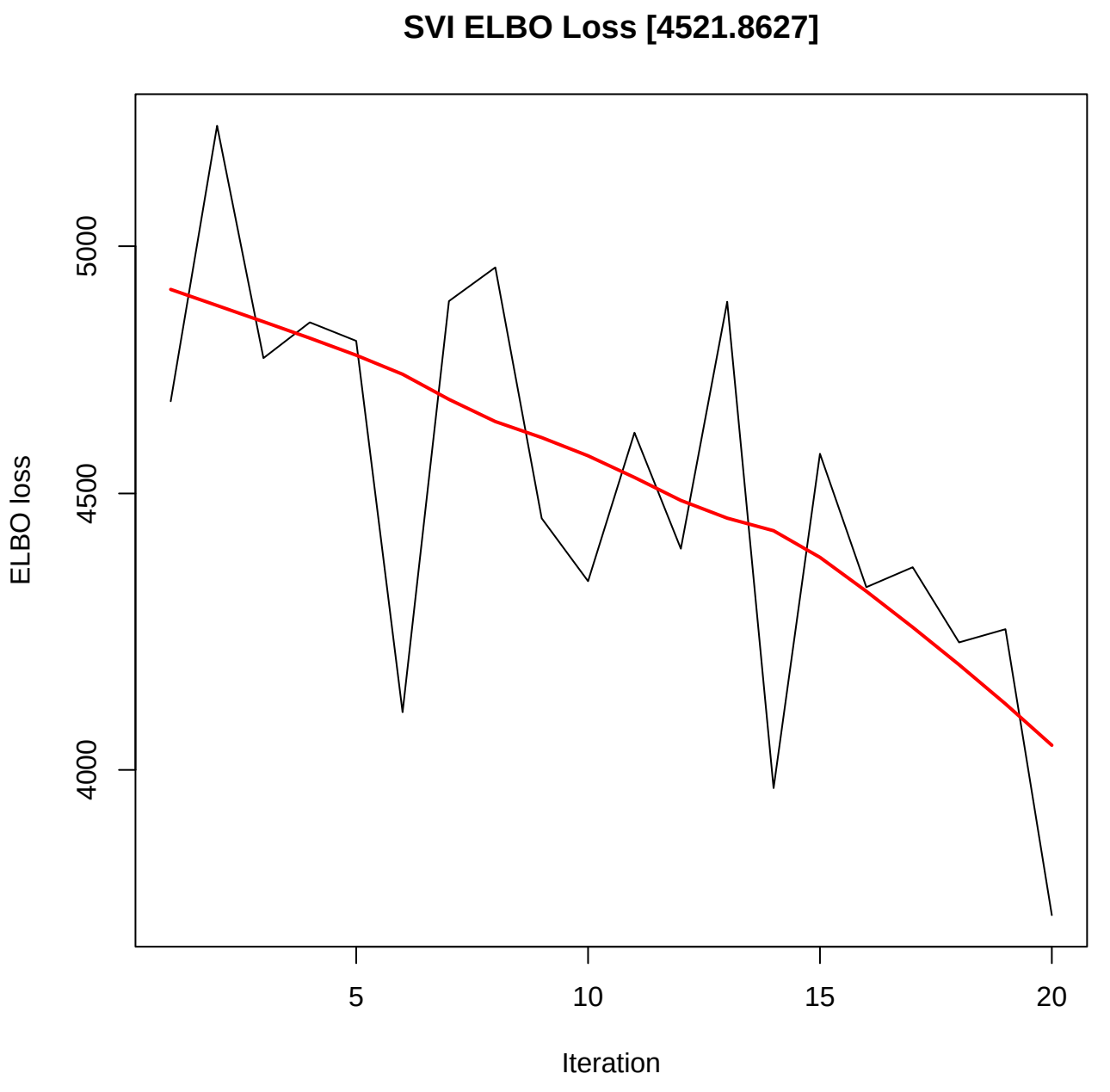

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00059
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index

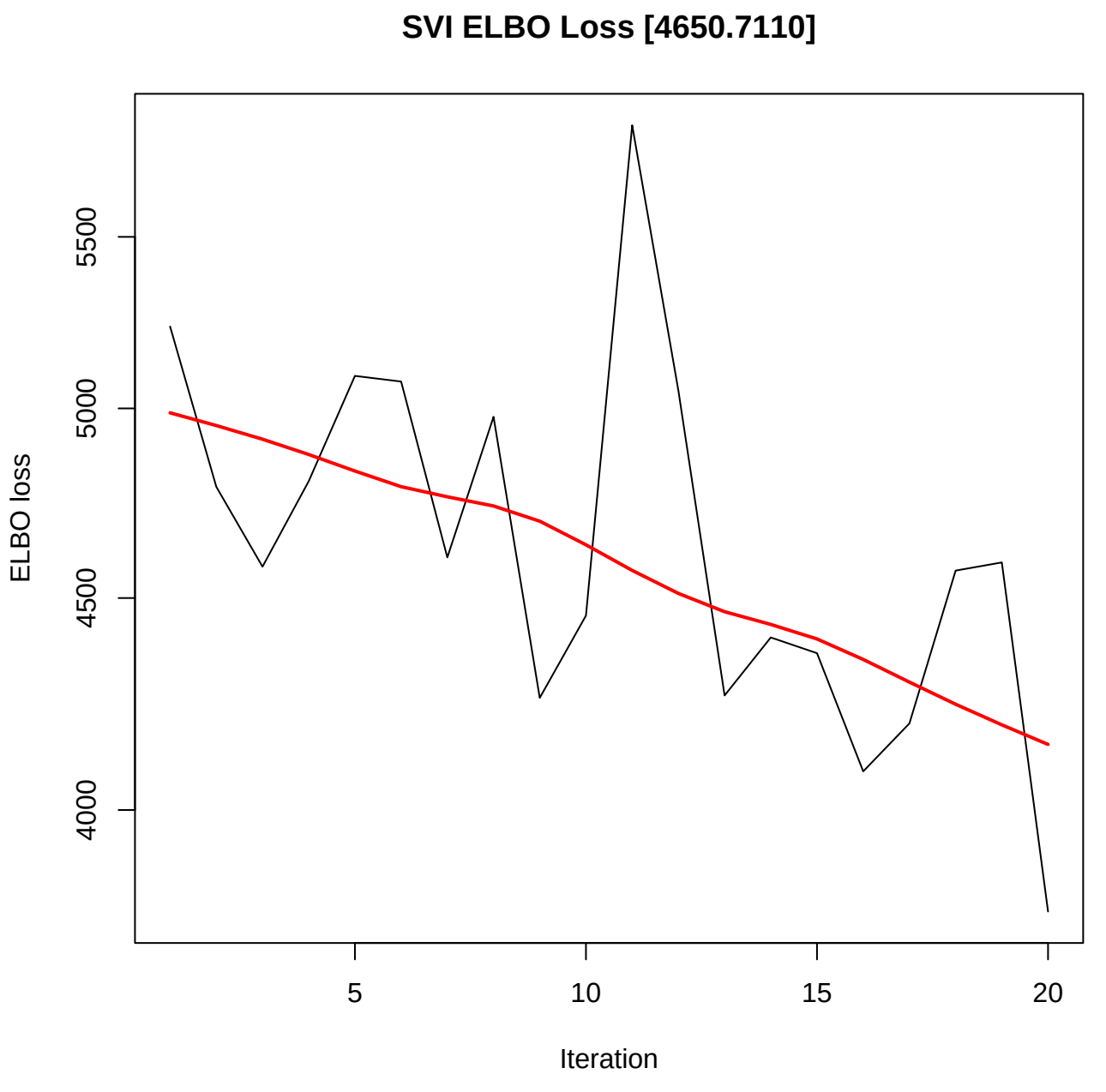

Value (dag)
loss_dag_vec
0.9999904
0.9999908
0.9999912
0.9999916
1.0
1.2
1.4
1.6
1.8
2.0
Index

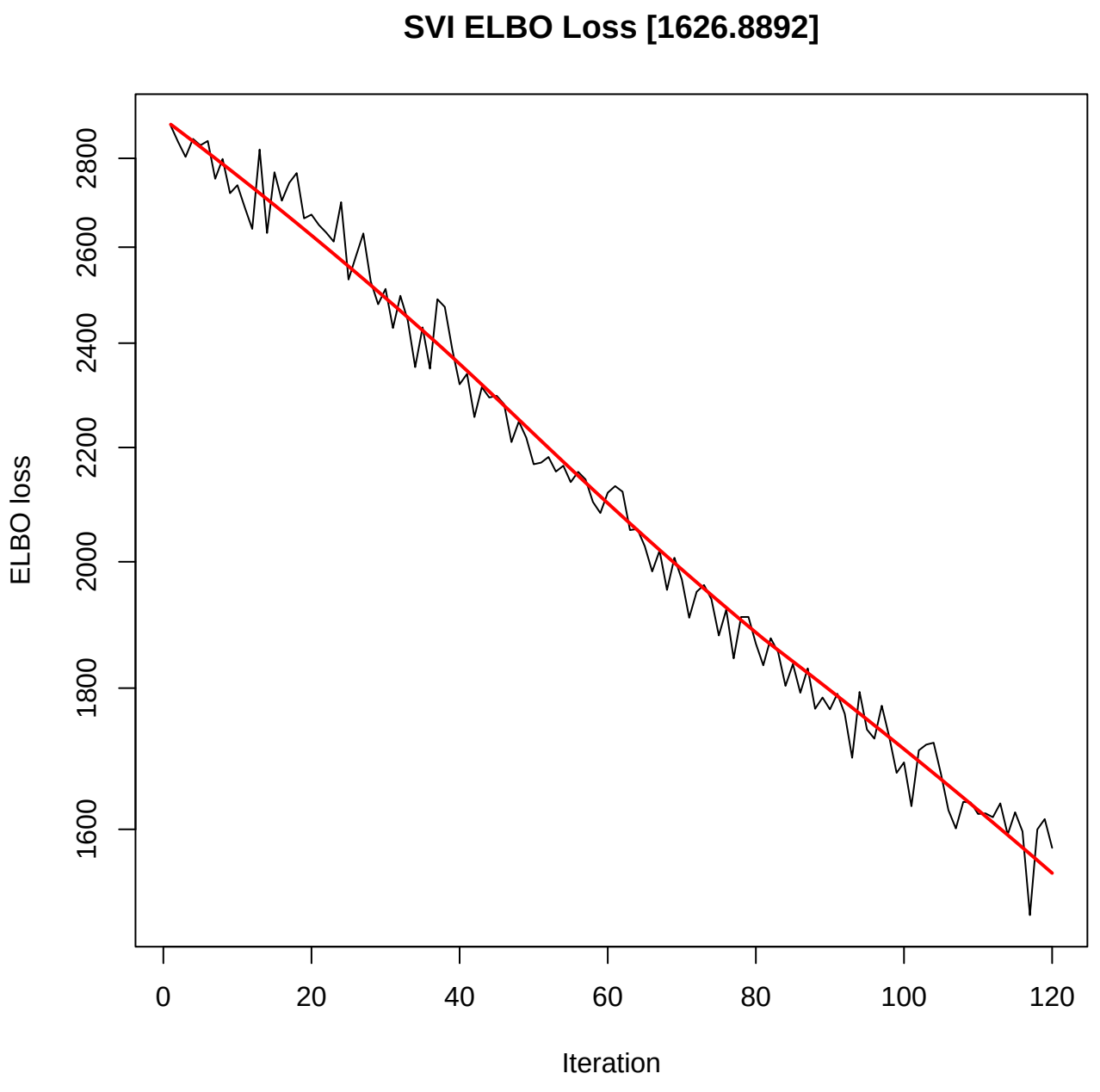

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00063
0.00064
0.00065
0.00066
0.00067
0.00068
1.0
1.2
1.4
1.6
1.8
2.0
Index

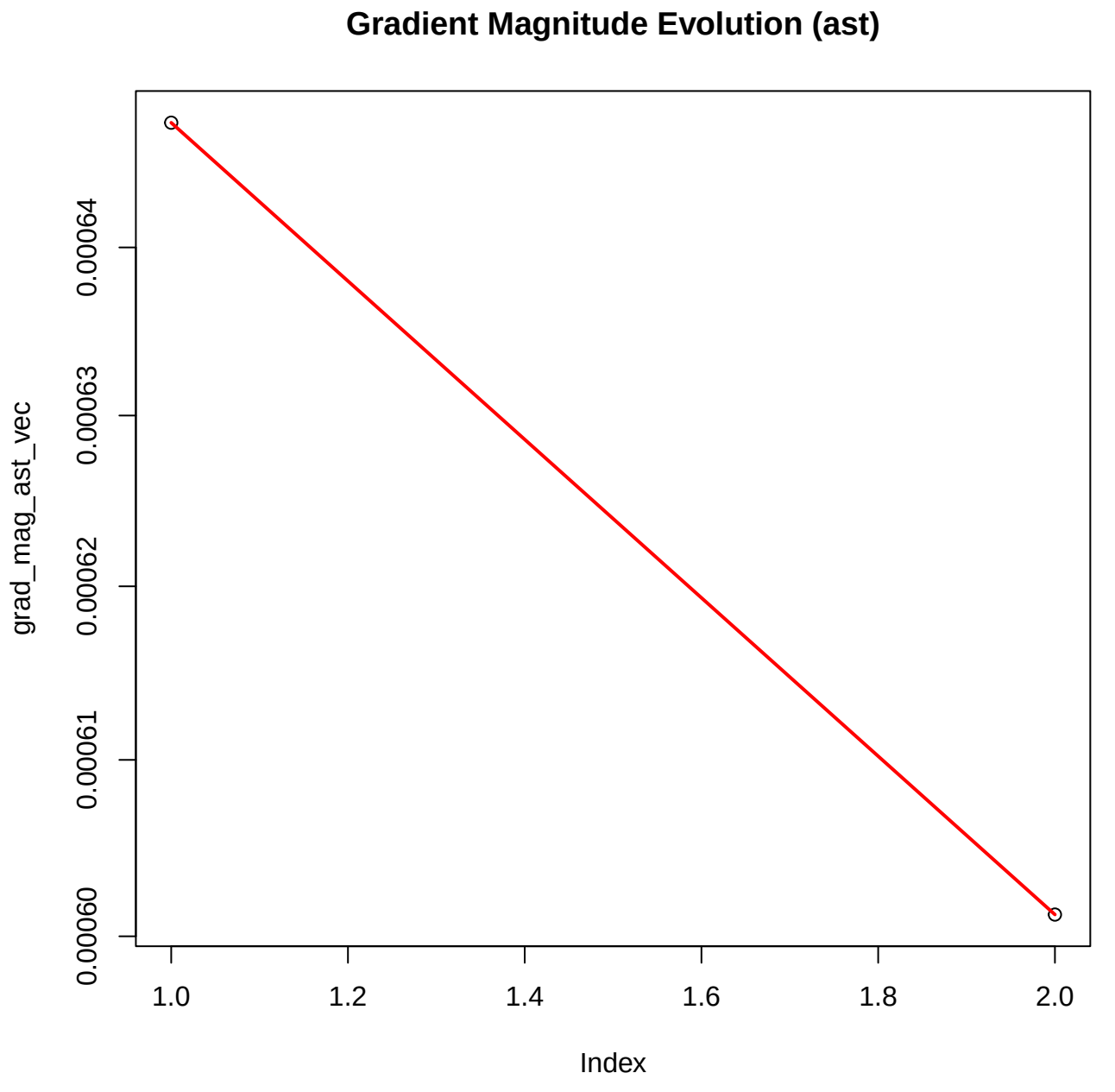

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00066
0.00067
0.00068
0.00069
0.00070
1.0
1.2
1.4
1.6
1.8
2.0
Index

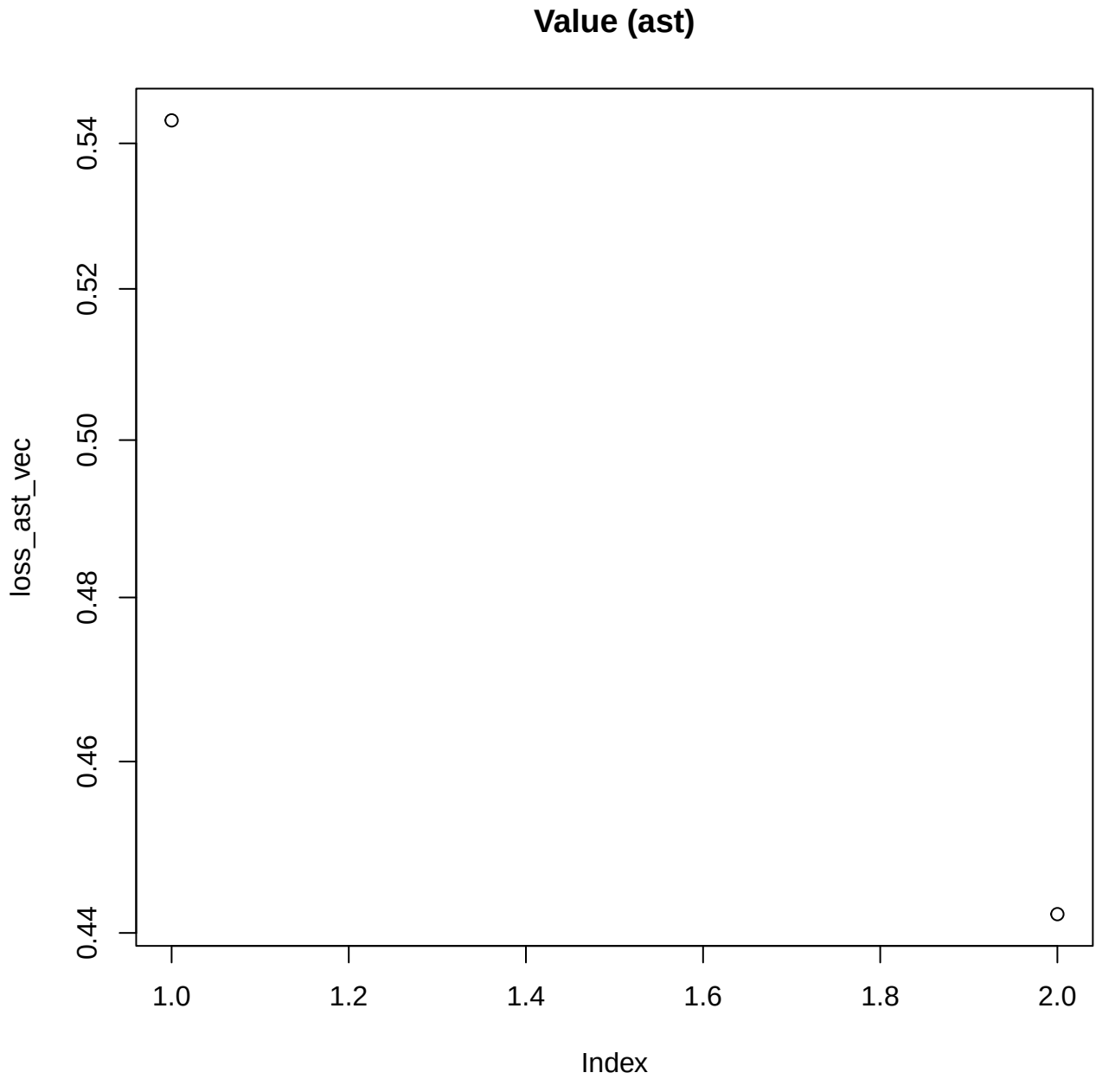

Value (dag)
loss_dag_vec
0.9999880
0.9999885
0.9999890
0.9999895
1.0
1.2
1.4
1.6
1.8
2.0
Index

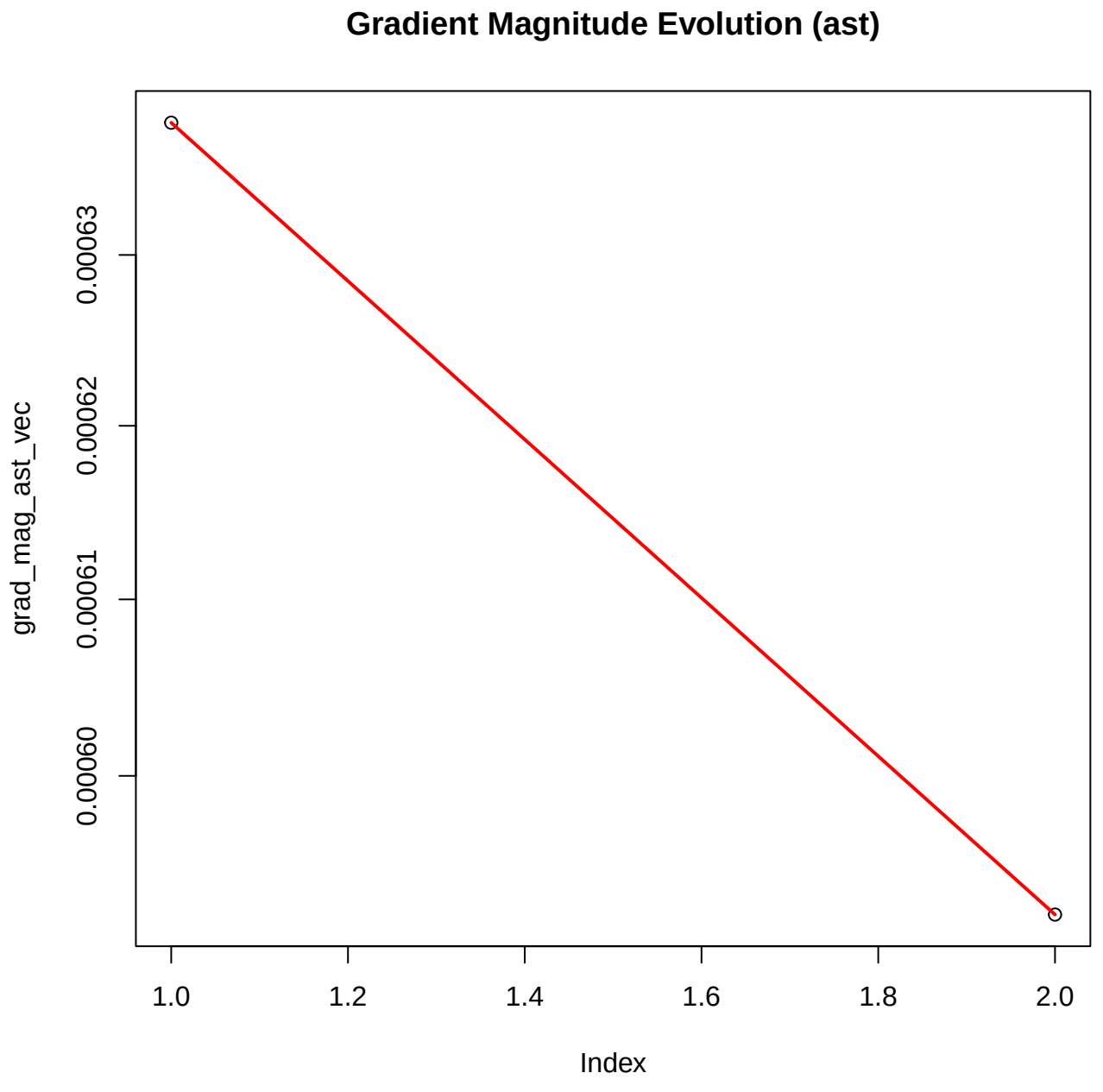

SVI ELBO Loss [4521.8627]
ELBO loss
5000
4500
4000
5
10
15
20
Iteration

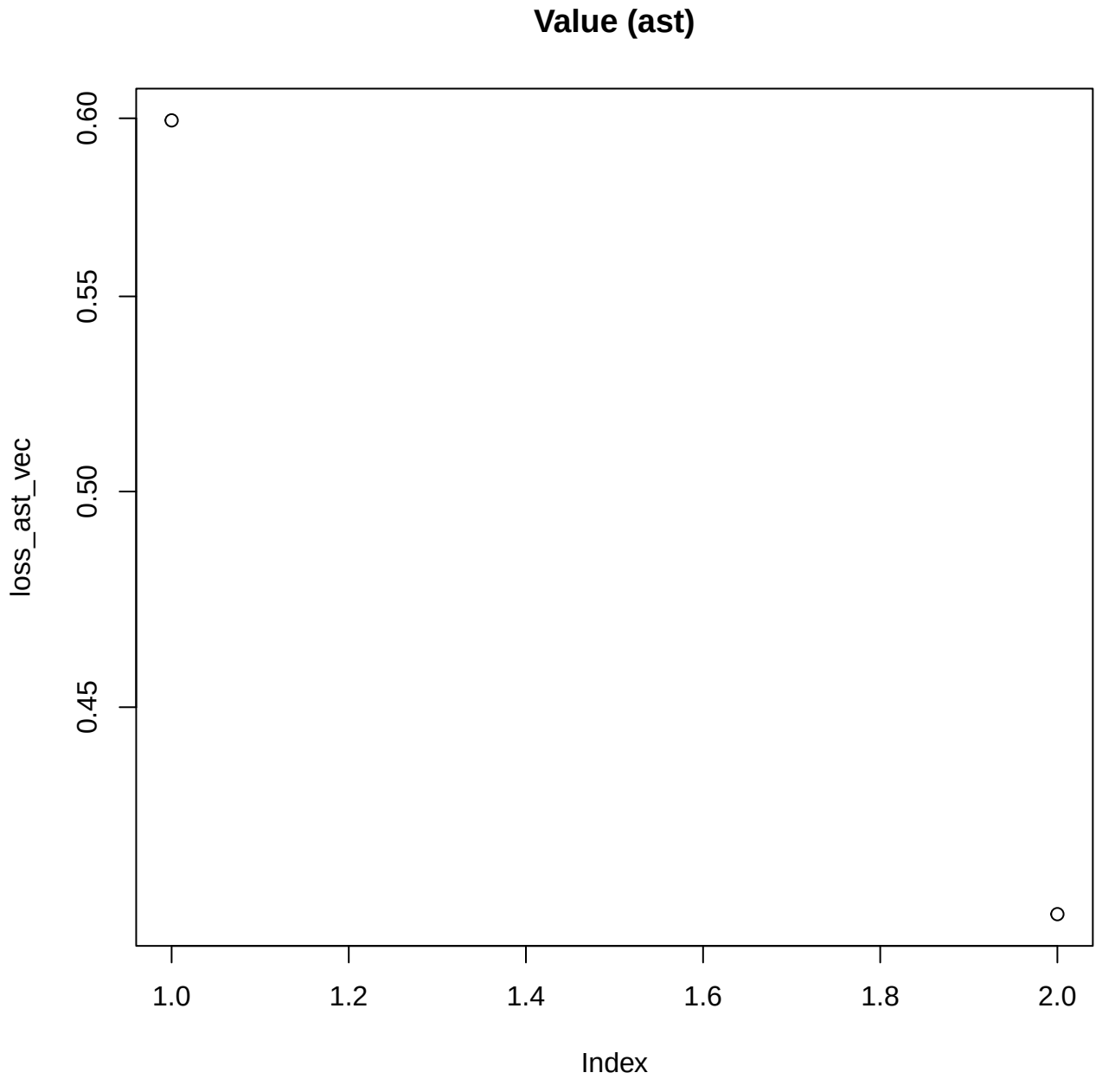

SVI ELBO Loss [4650.7110]
ELBO loss
4000
4500
5000
5500
5
10
15
20
Iteration

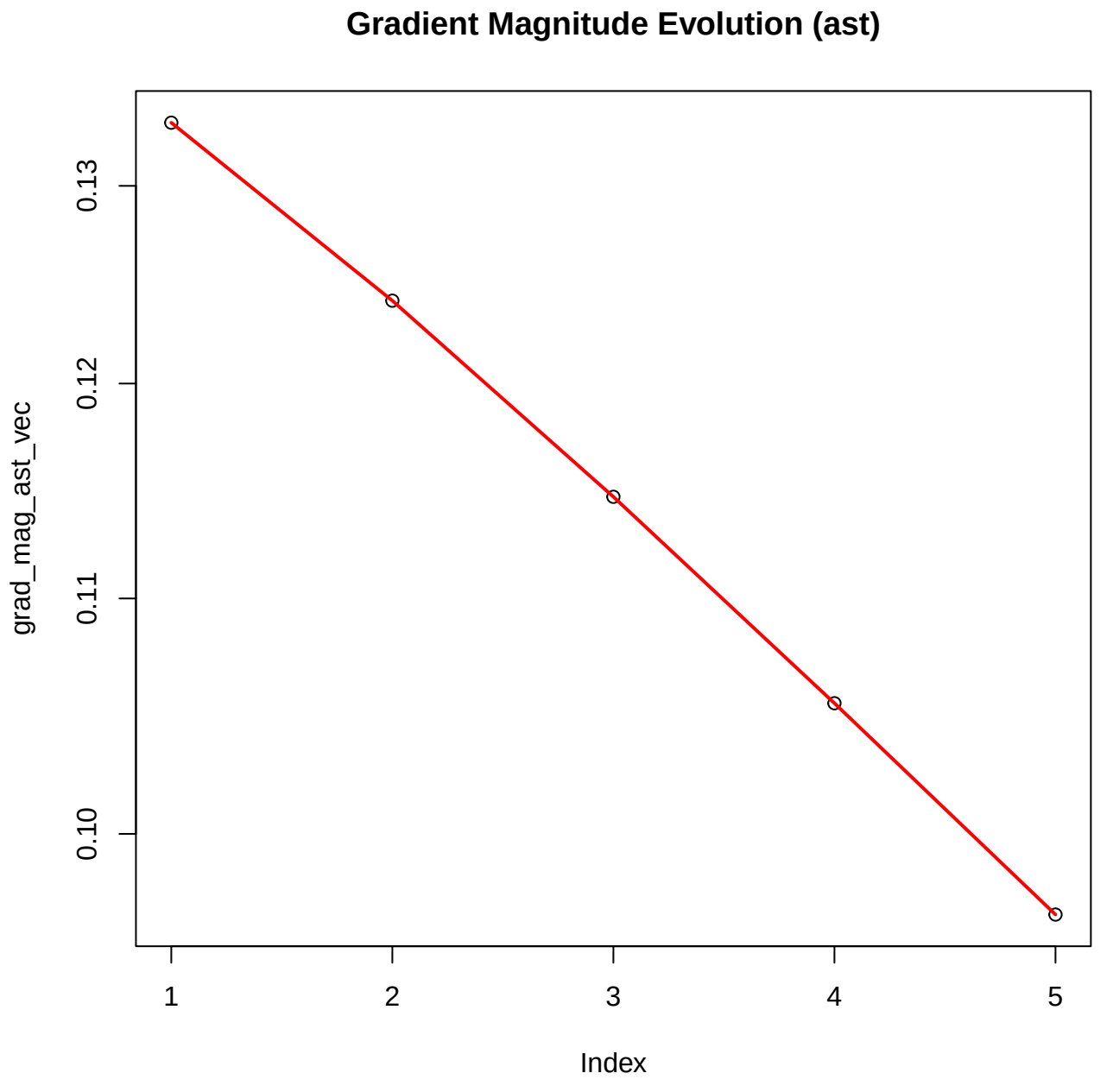

SVI ELBO Loss [1626.8892]
ELBO loss
1600
1800
2000
2200
2400
2600
2800
0
20
40
60
80
100
120
Iteration

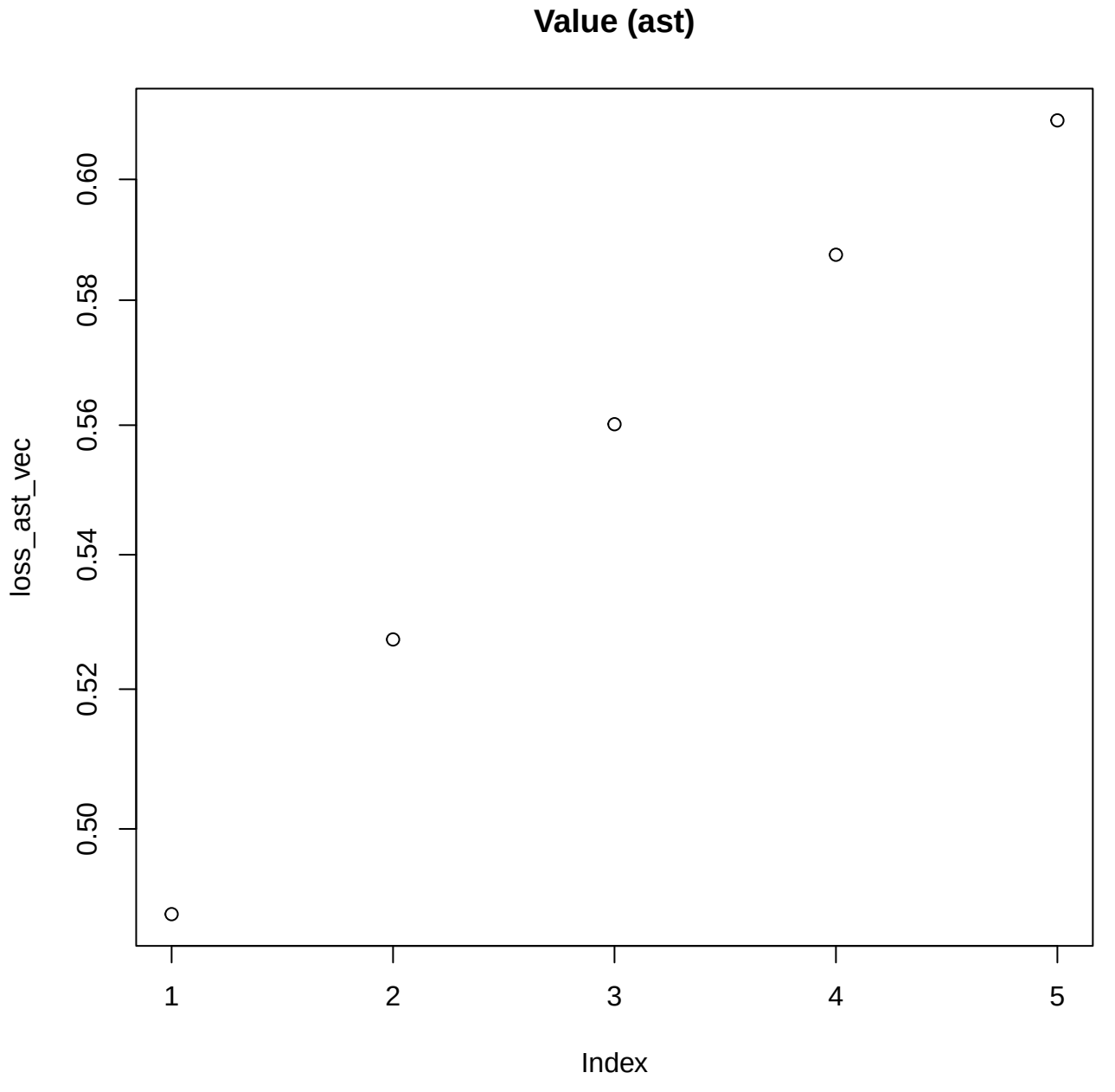

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00061
0.00062
0.00063
0.00064
1.0
1.2
1.4
1.6
1.8
2.0
Index

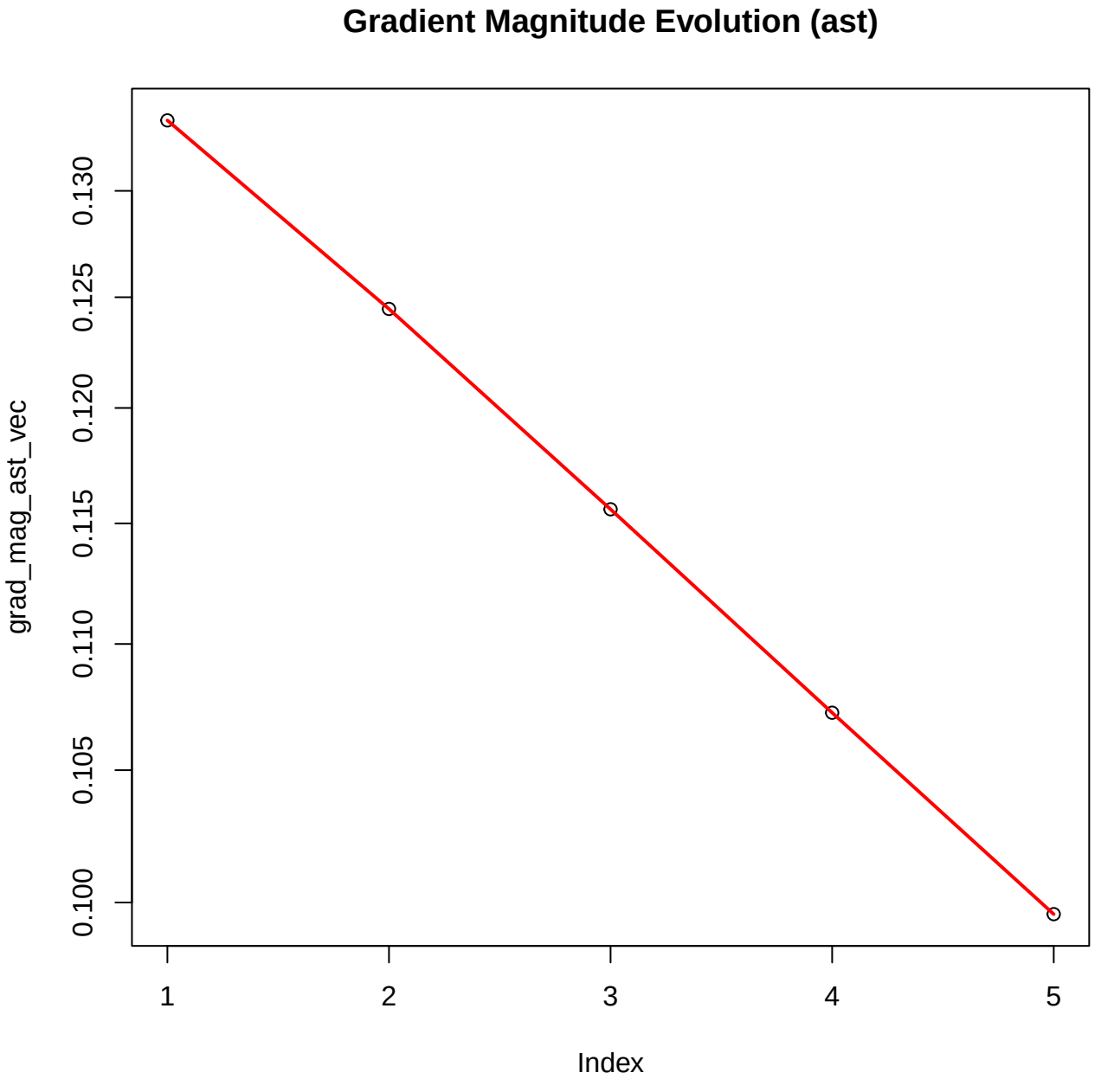

Value (ast)
loss_ast_vec
0.44
0.46
0.48
0.50
0.52
0.54
1.0
1.2
1.4
1.6
1.8
2.0
Index

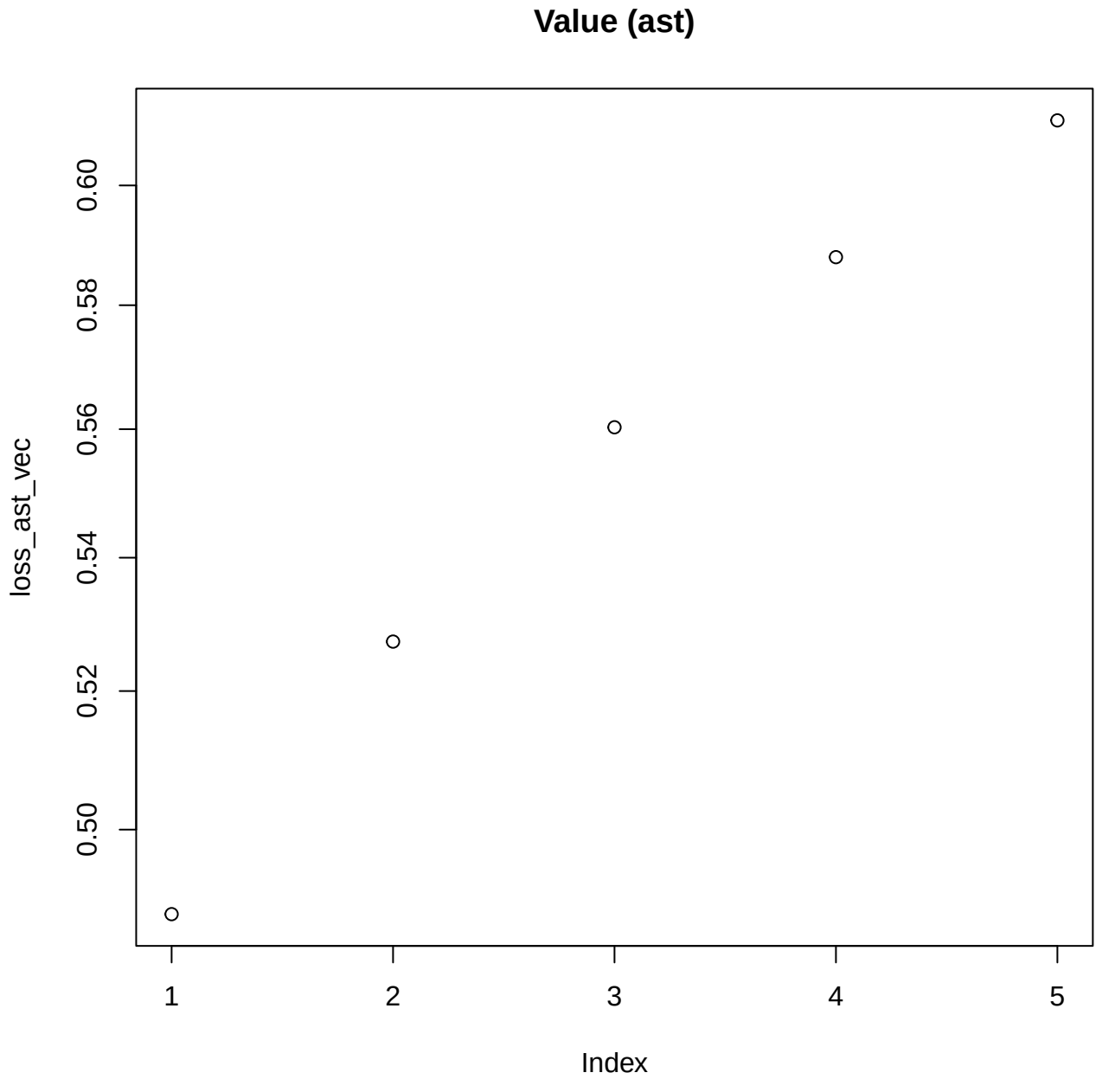

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index

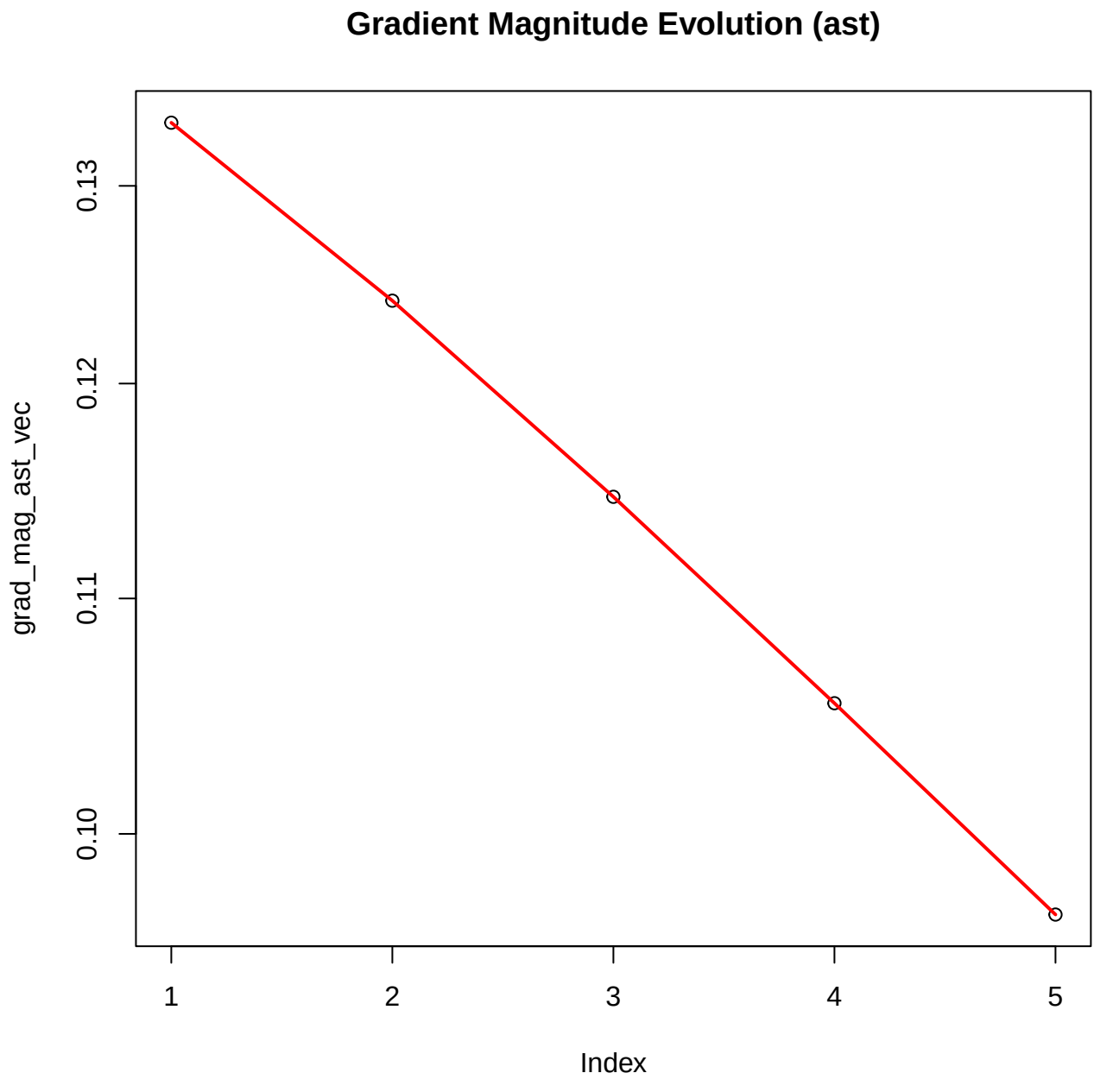

Value (ast)
loss_ast_vec
0.60
0.55
0.50
0.45
1.0
1.2
1.4
1.6
1.8
2.0
Index

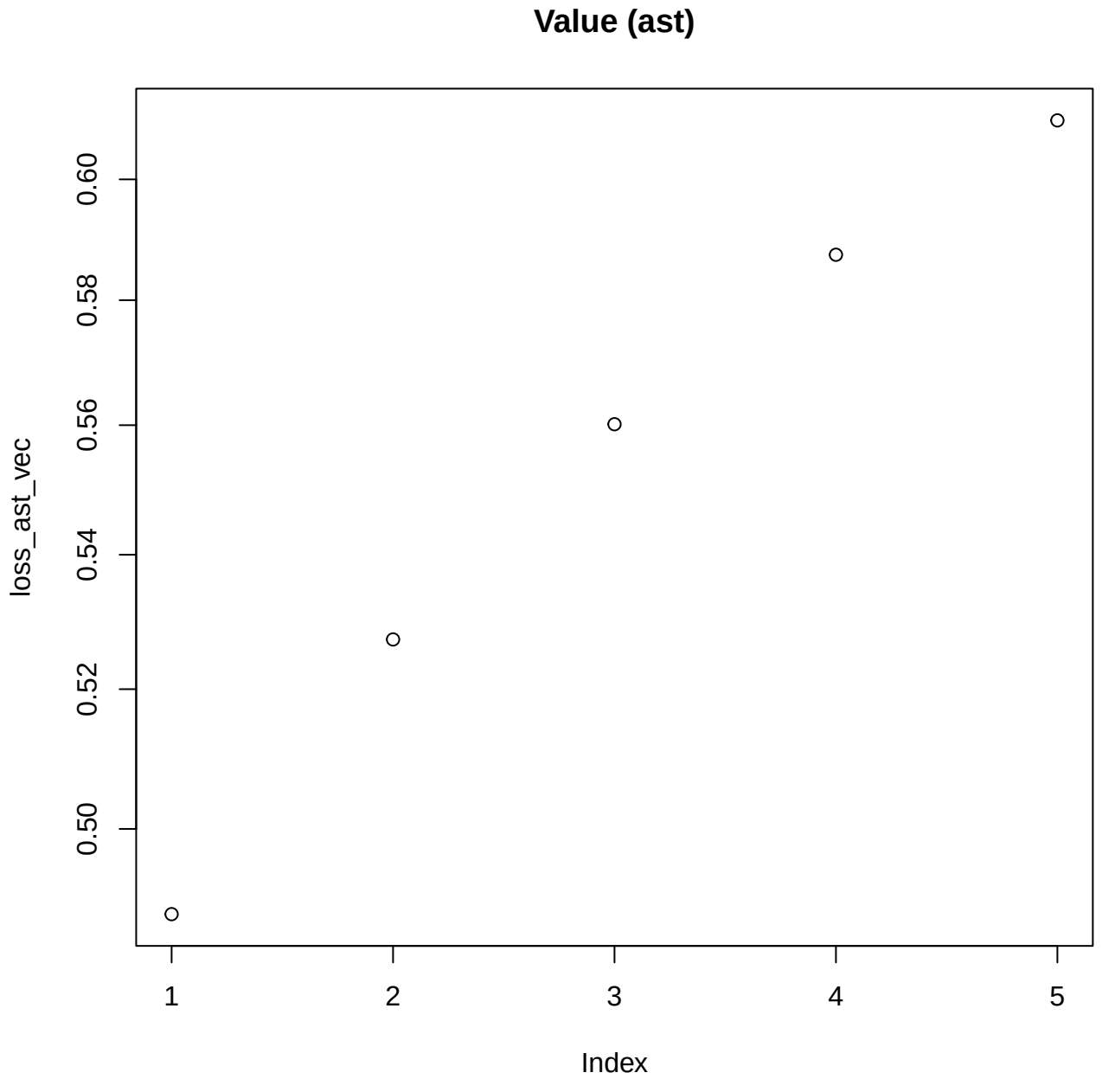

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

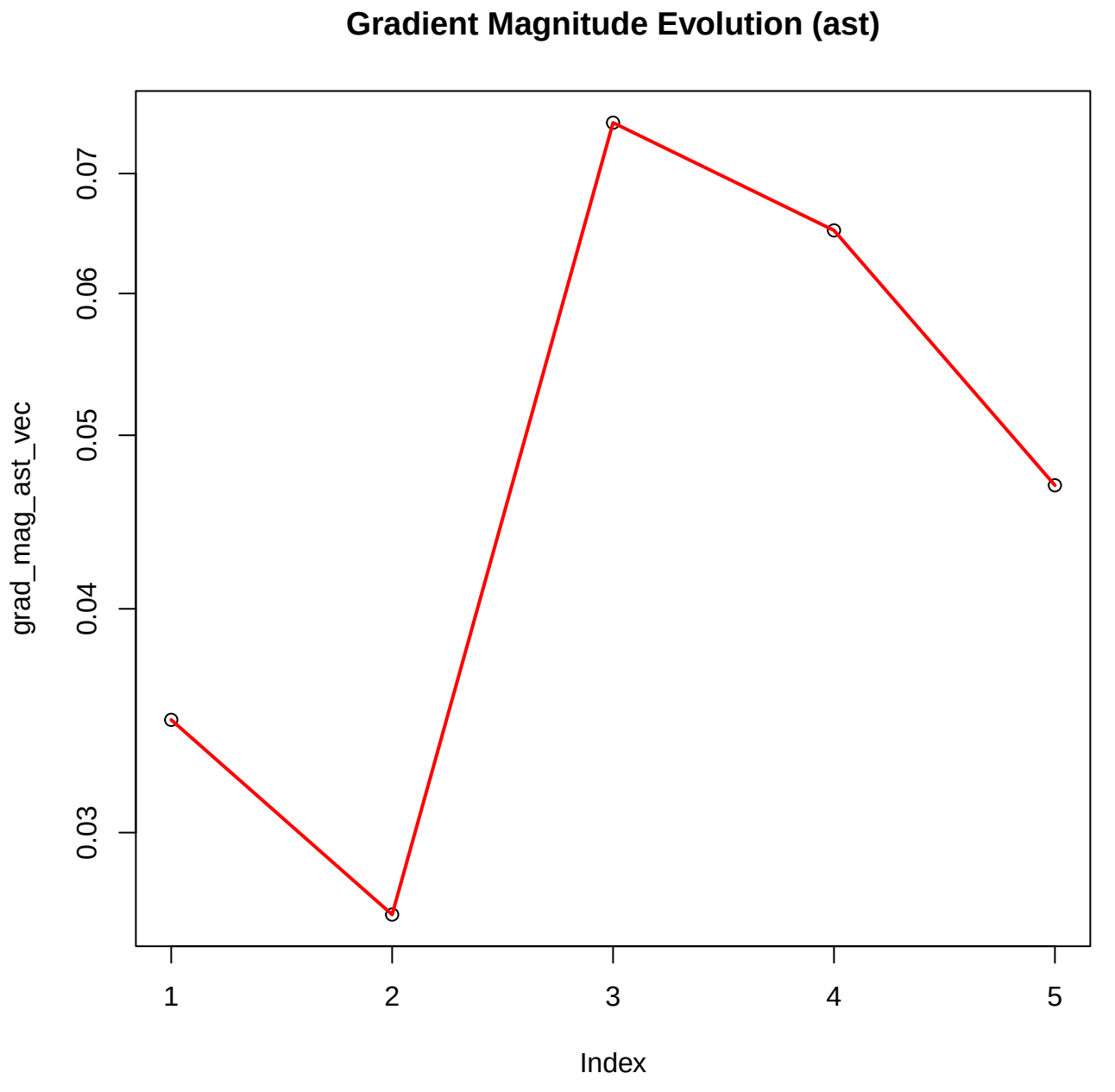

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

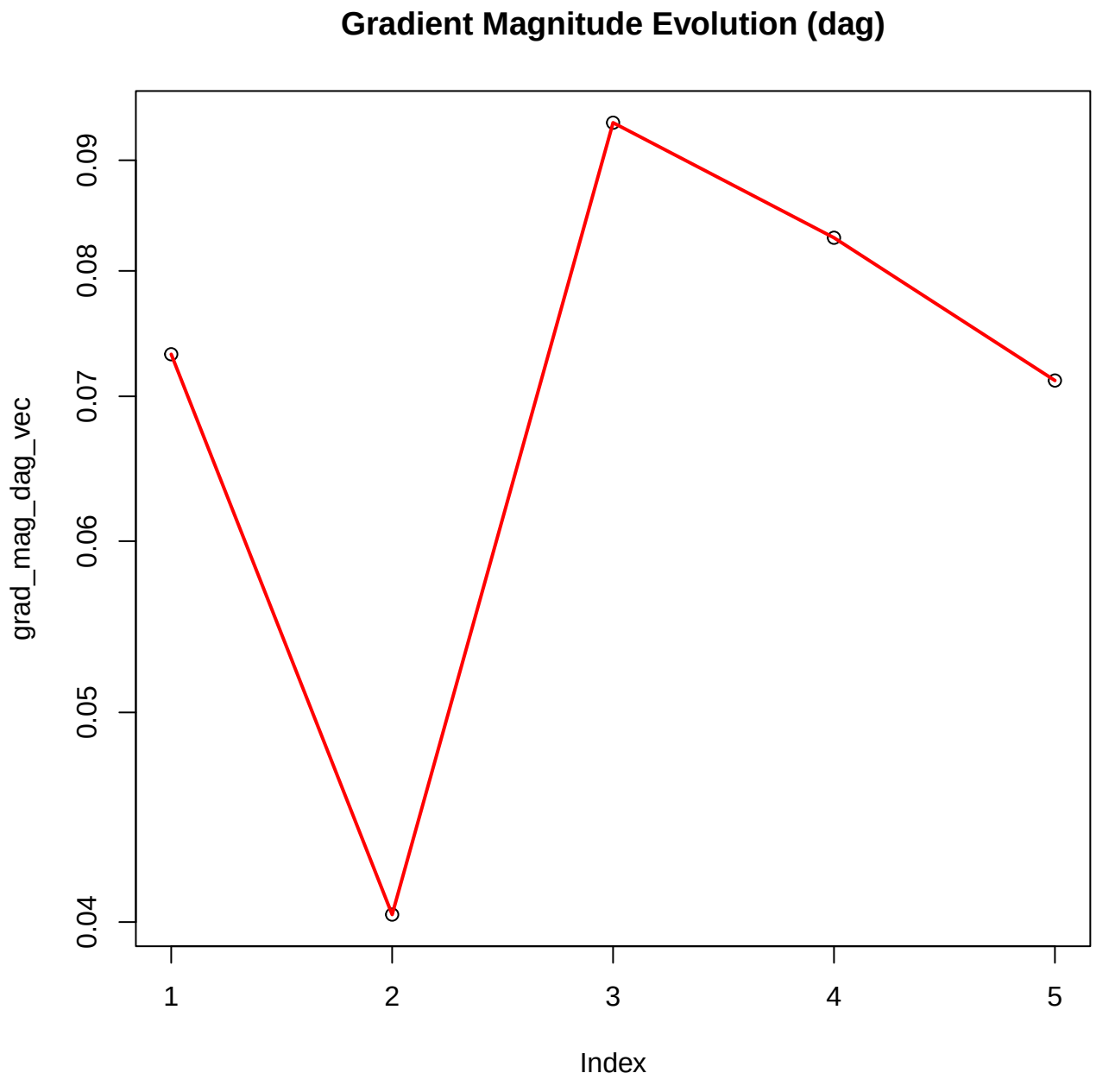

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.100
0.105
0.110
0.115
0.120
0.125
0.130
1
2
3
4
5
Index

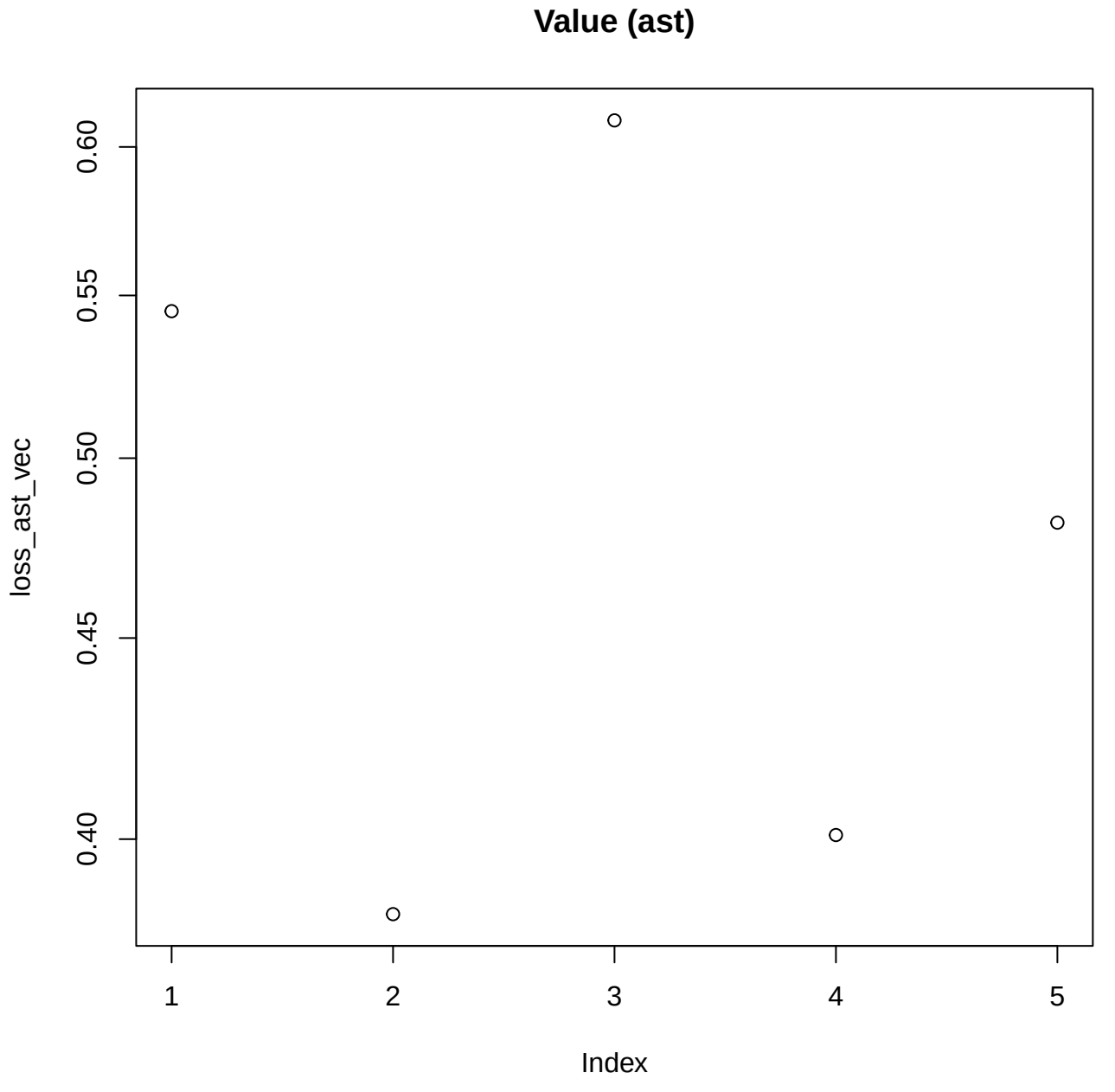

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

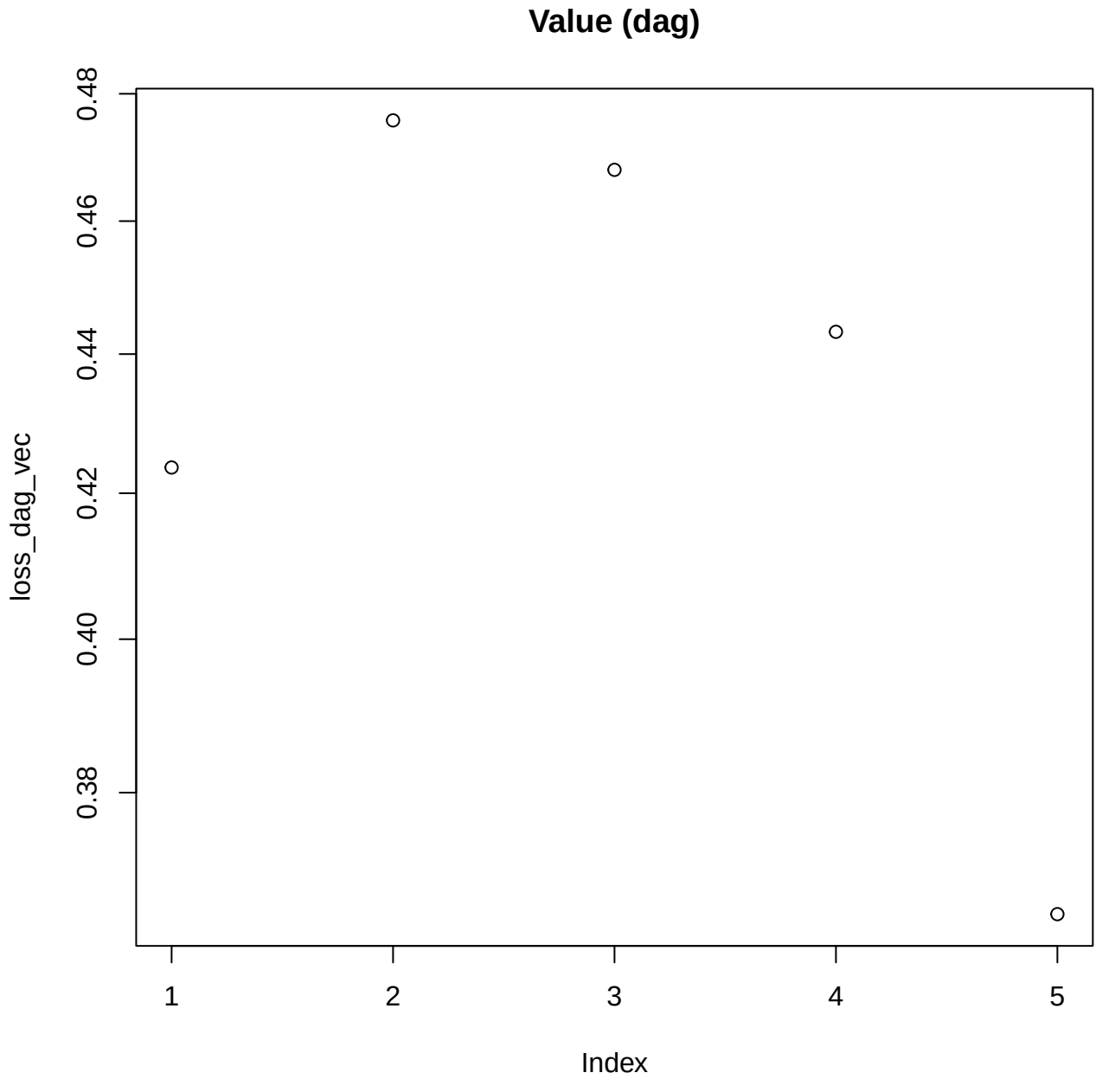

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

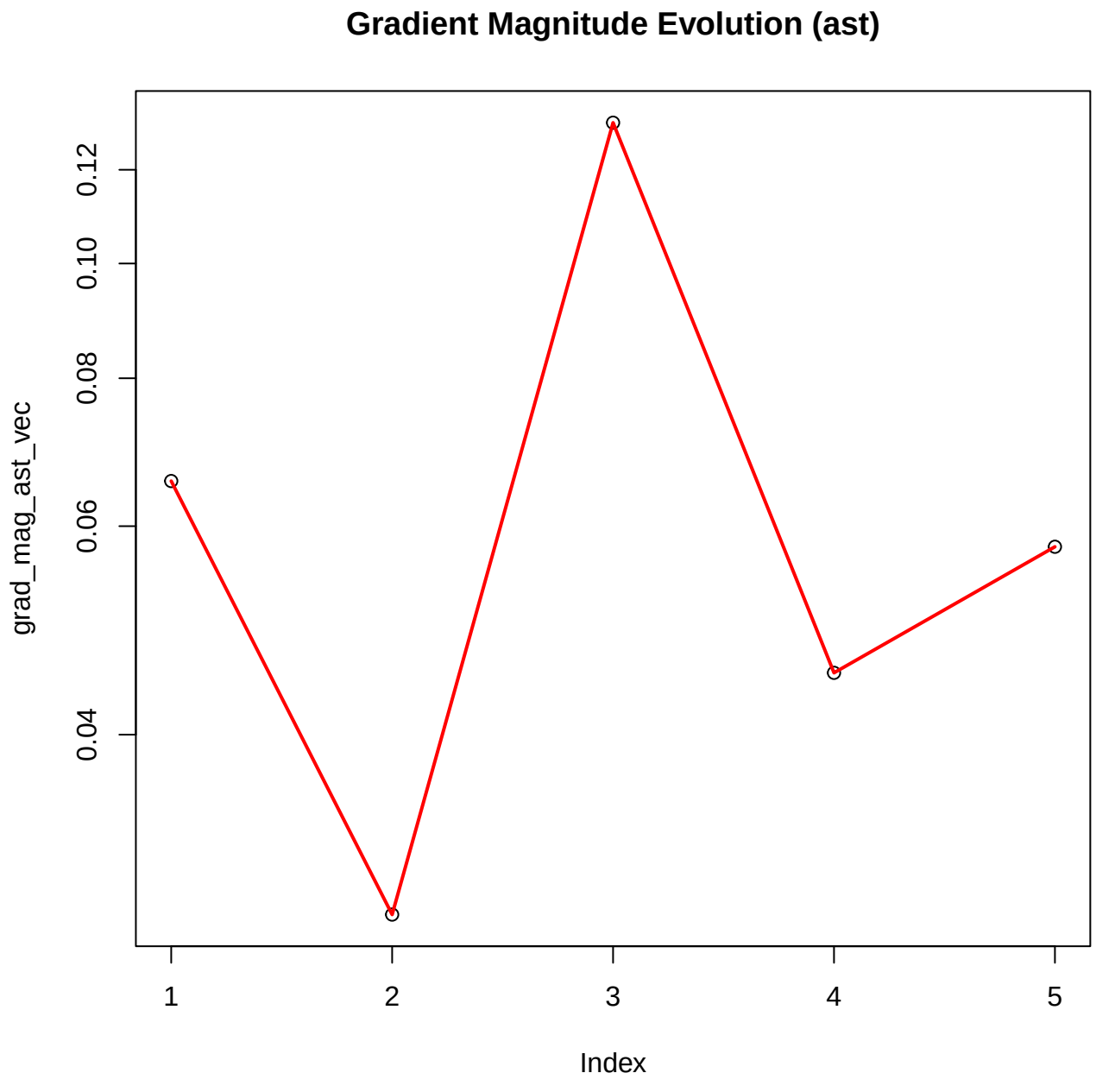

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

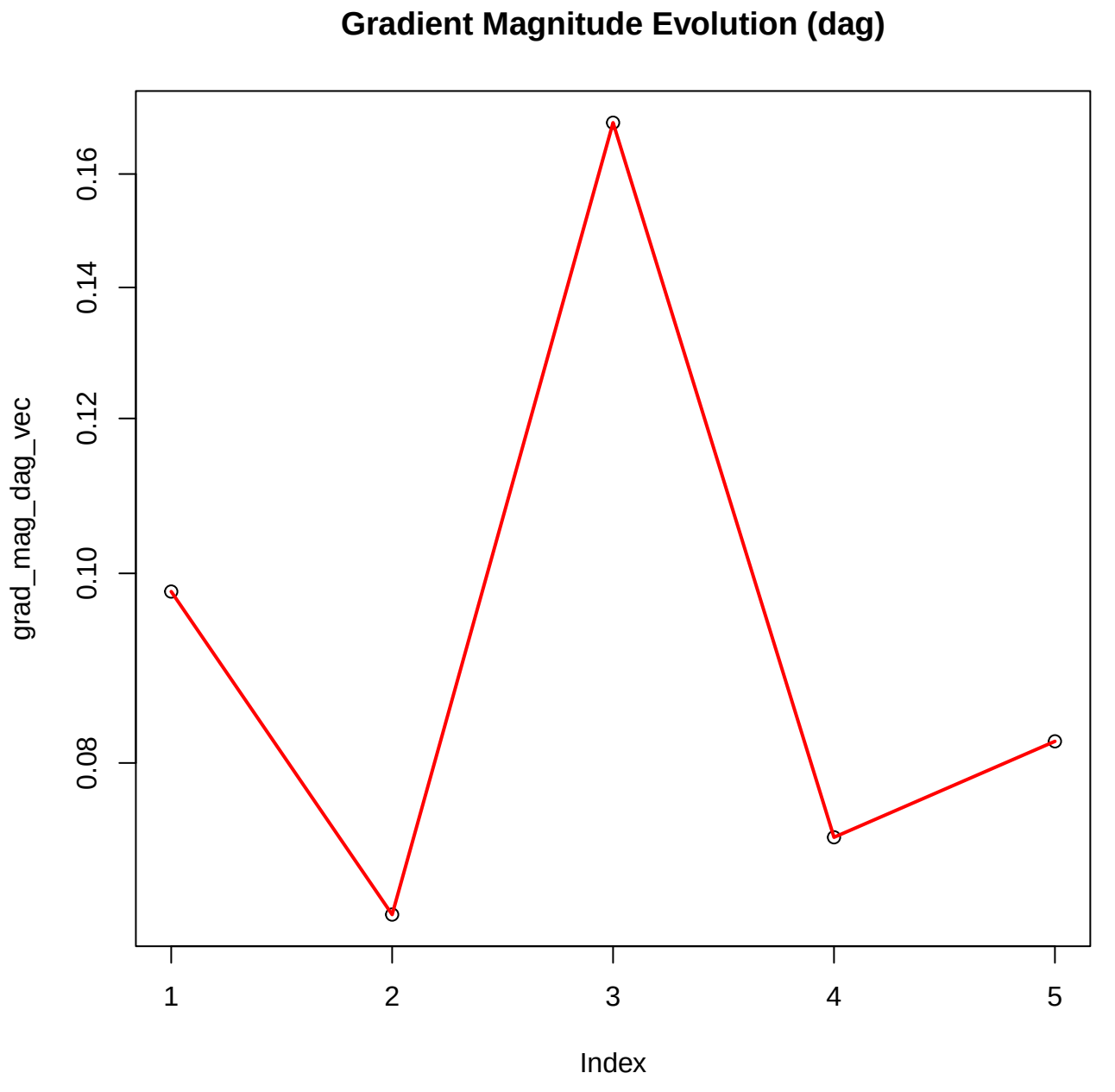

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index

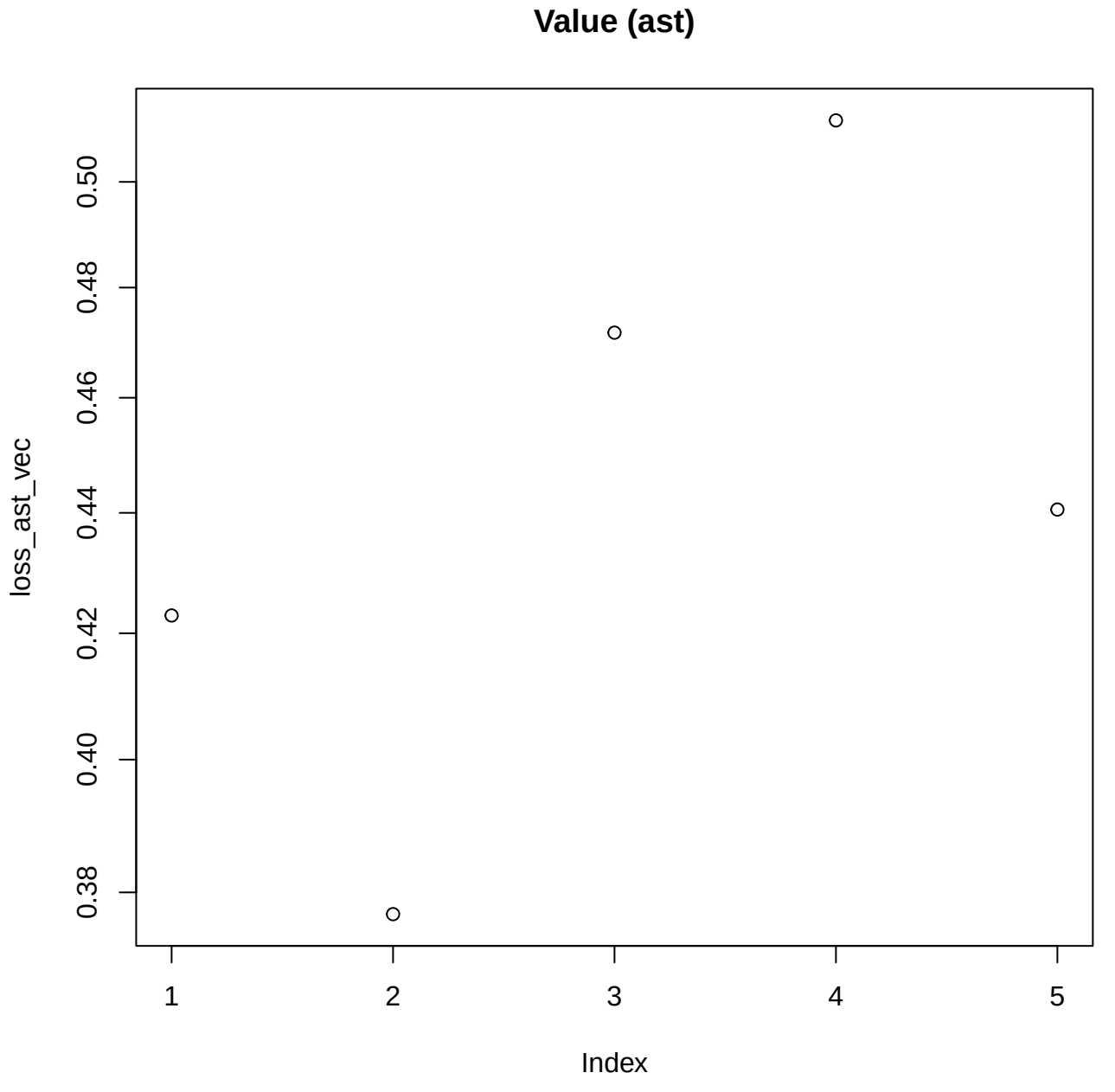

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.04
0.05
0.06
0.07
0.08
0.09
1
2
3
4
5
Index

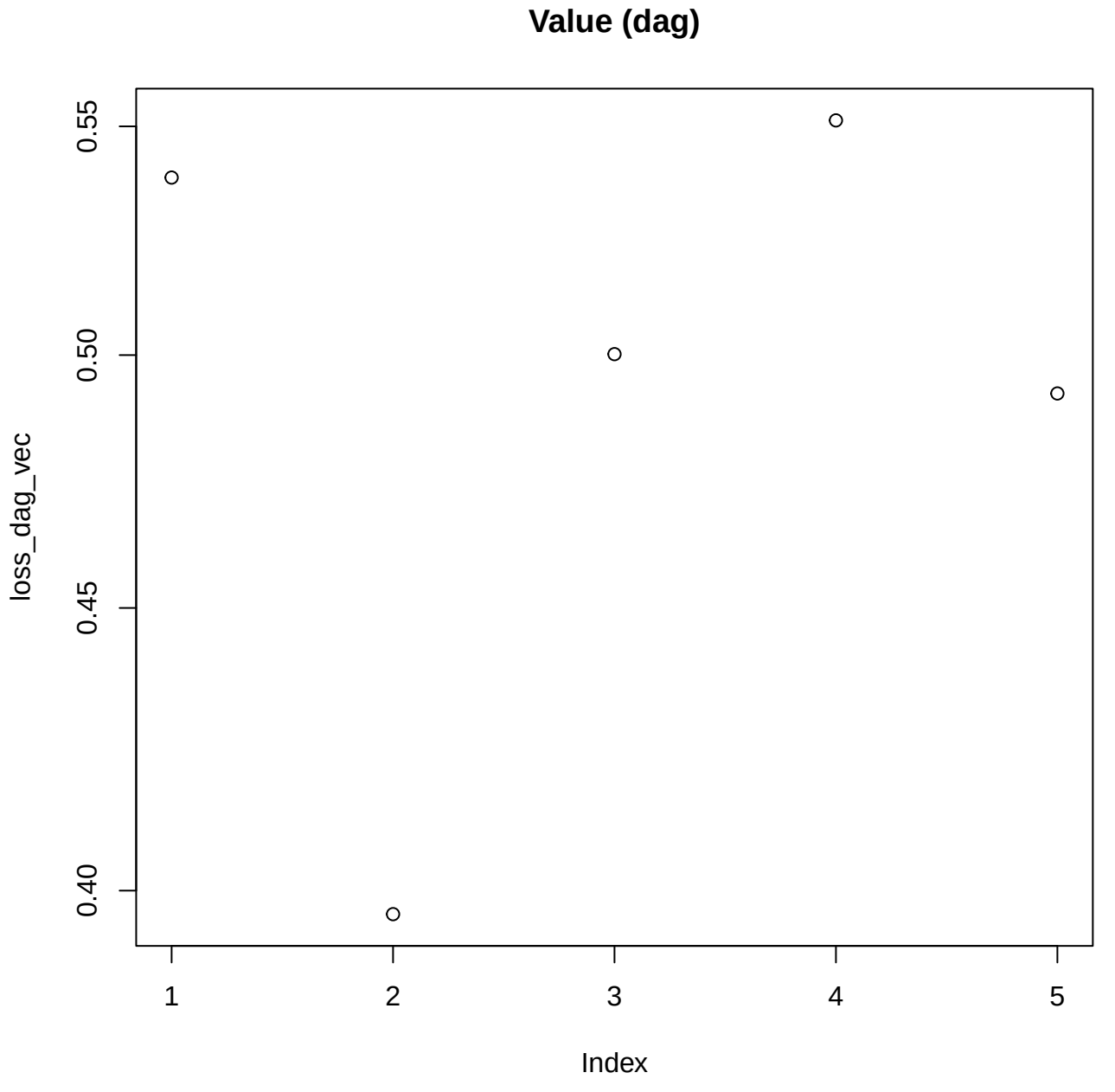

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

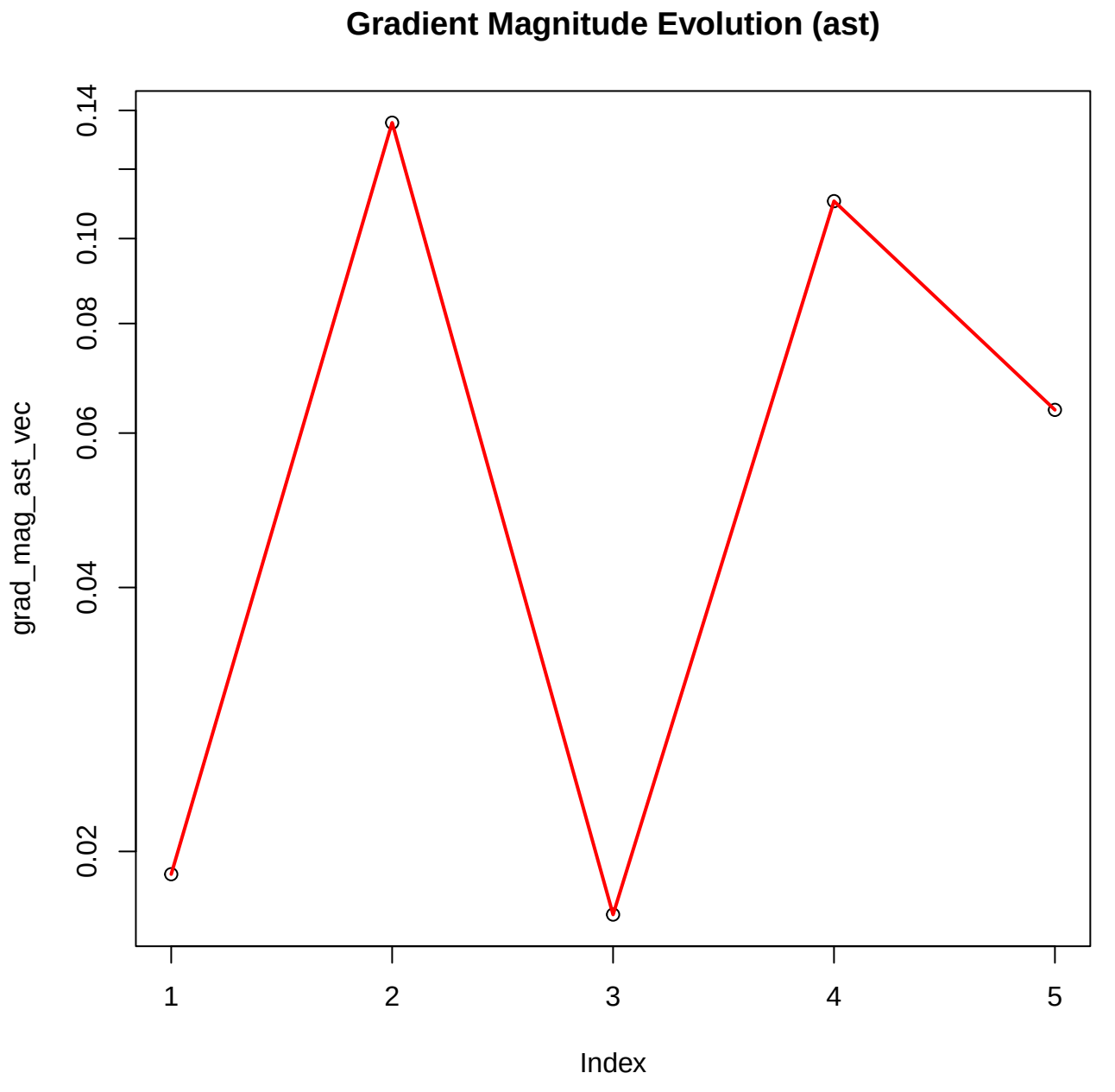

Value (dag)
loss_dag_vec
0.38
0.40
0.42
0.44
0.46
0.48
1
2
3
4
5
Index

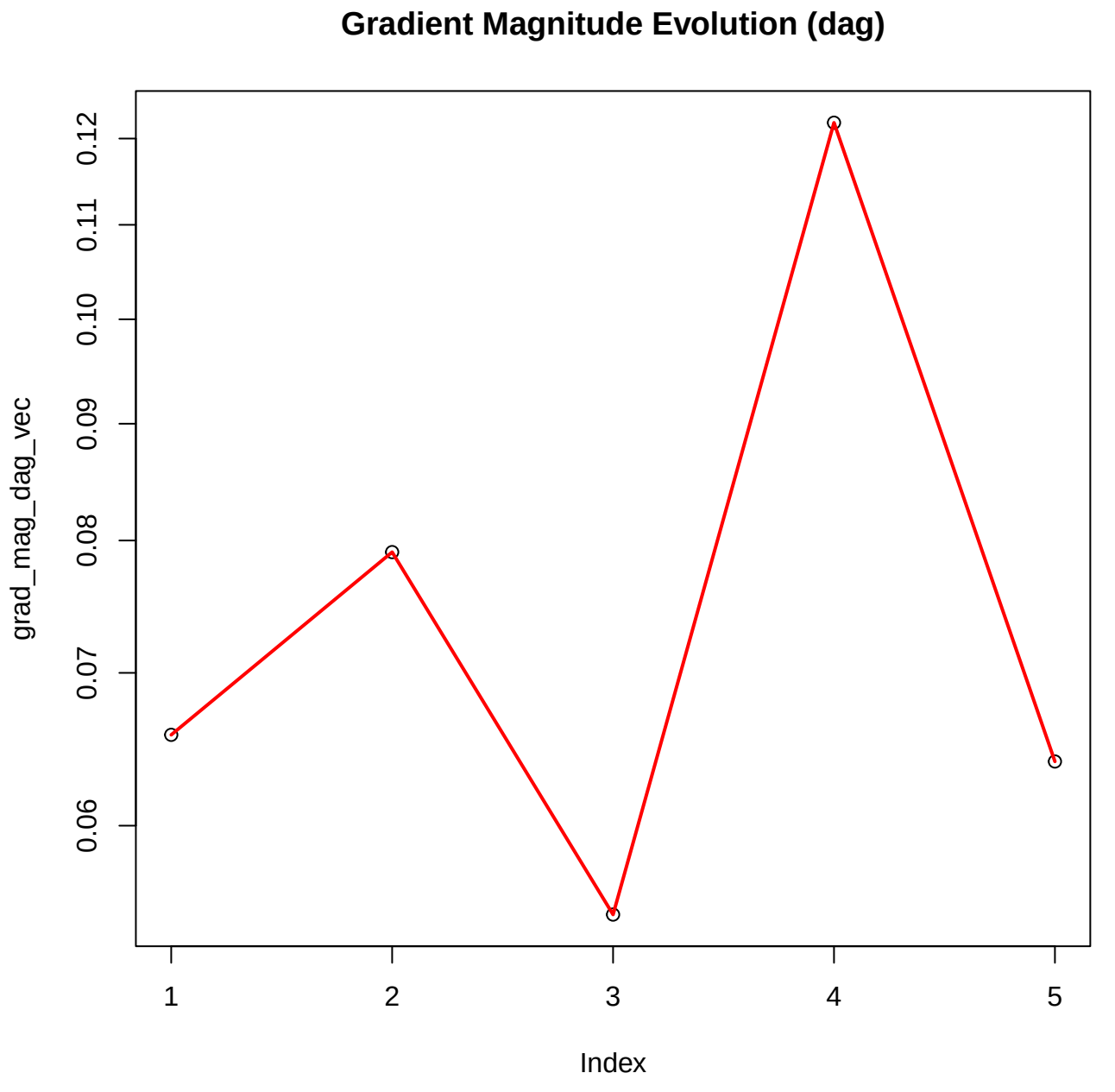

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.04
0.06
0.08
0.10
0.12
1
2
3
4
5
Index

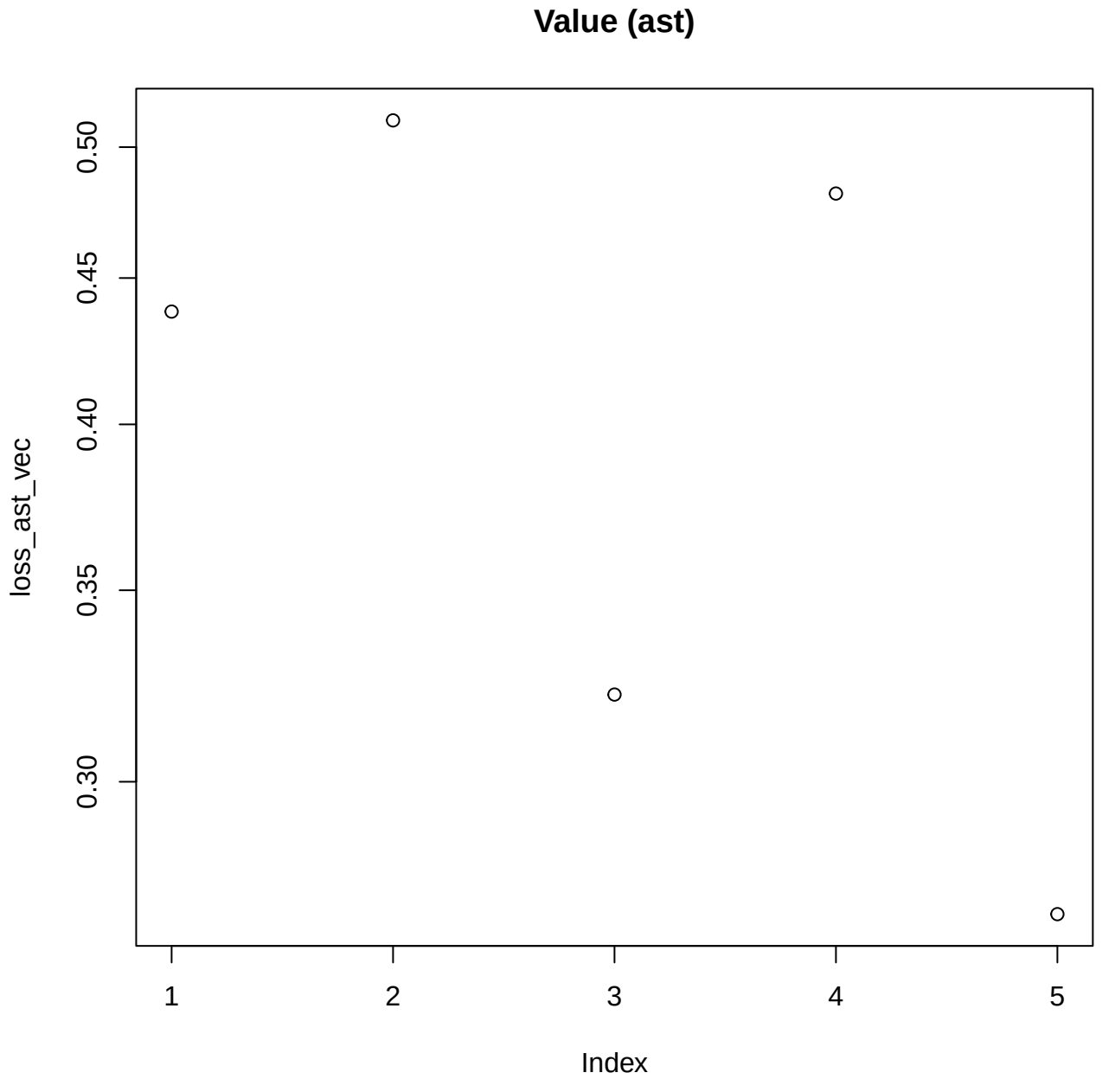

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.08
0.10
0.12
0.14
0.16
1
2
3
4
5
Index

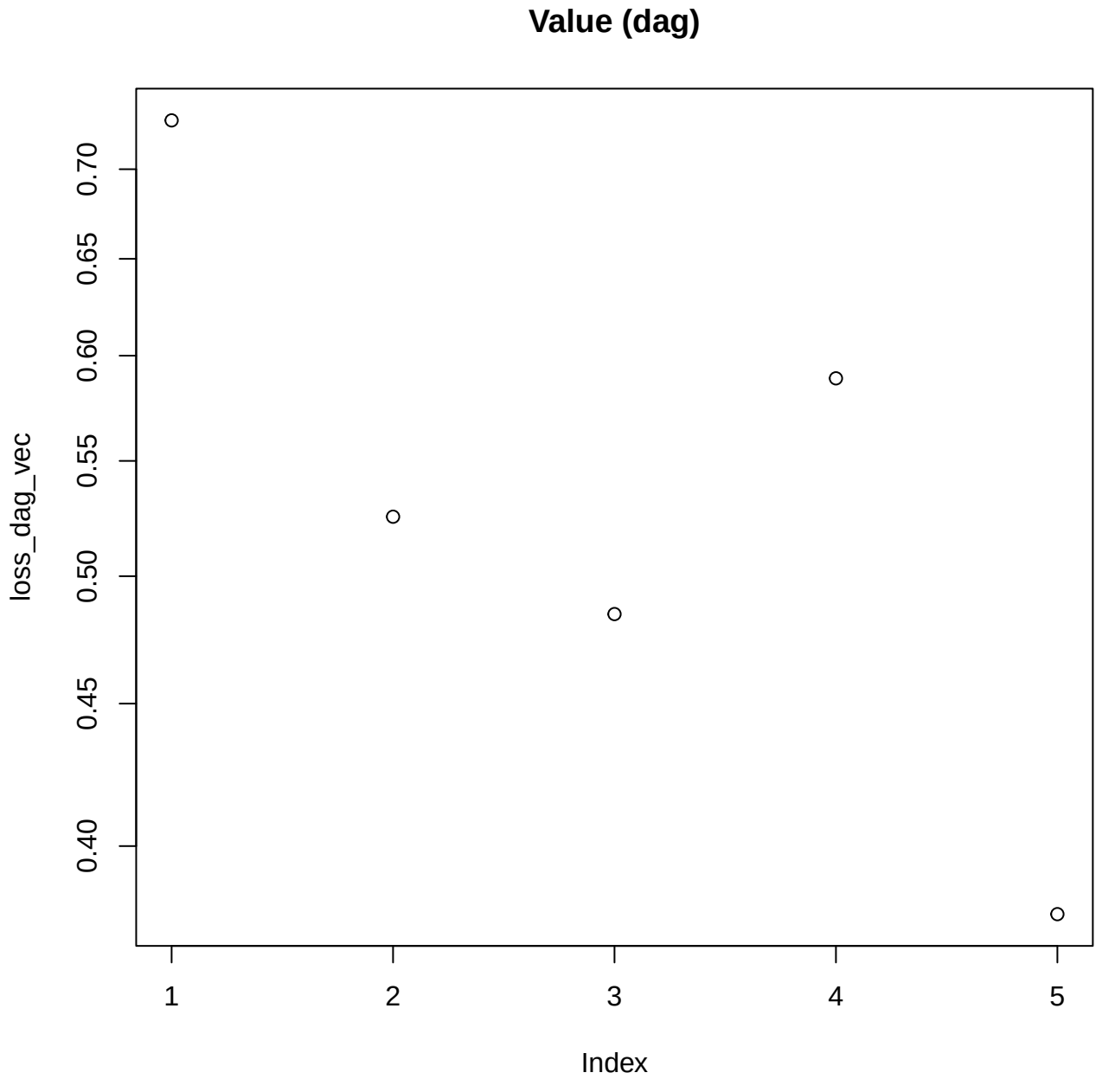

Value (ast)
loss_ast_vec
0.38
0.40
0.42
0.44
0.46
0.48
0.50
1
2
3
4
5
Index

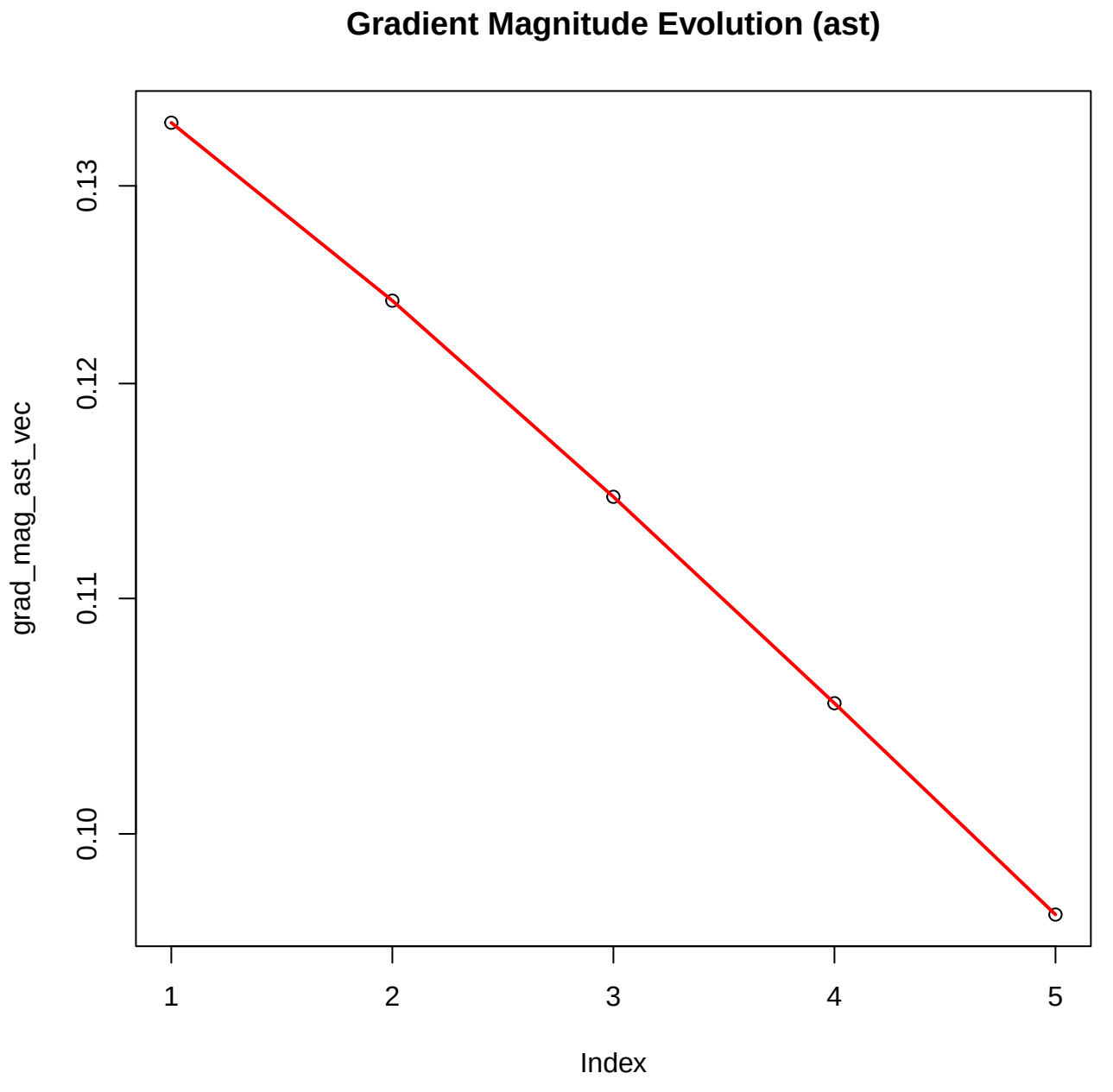

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

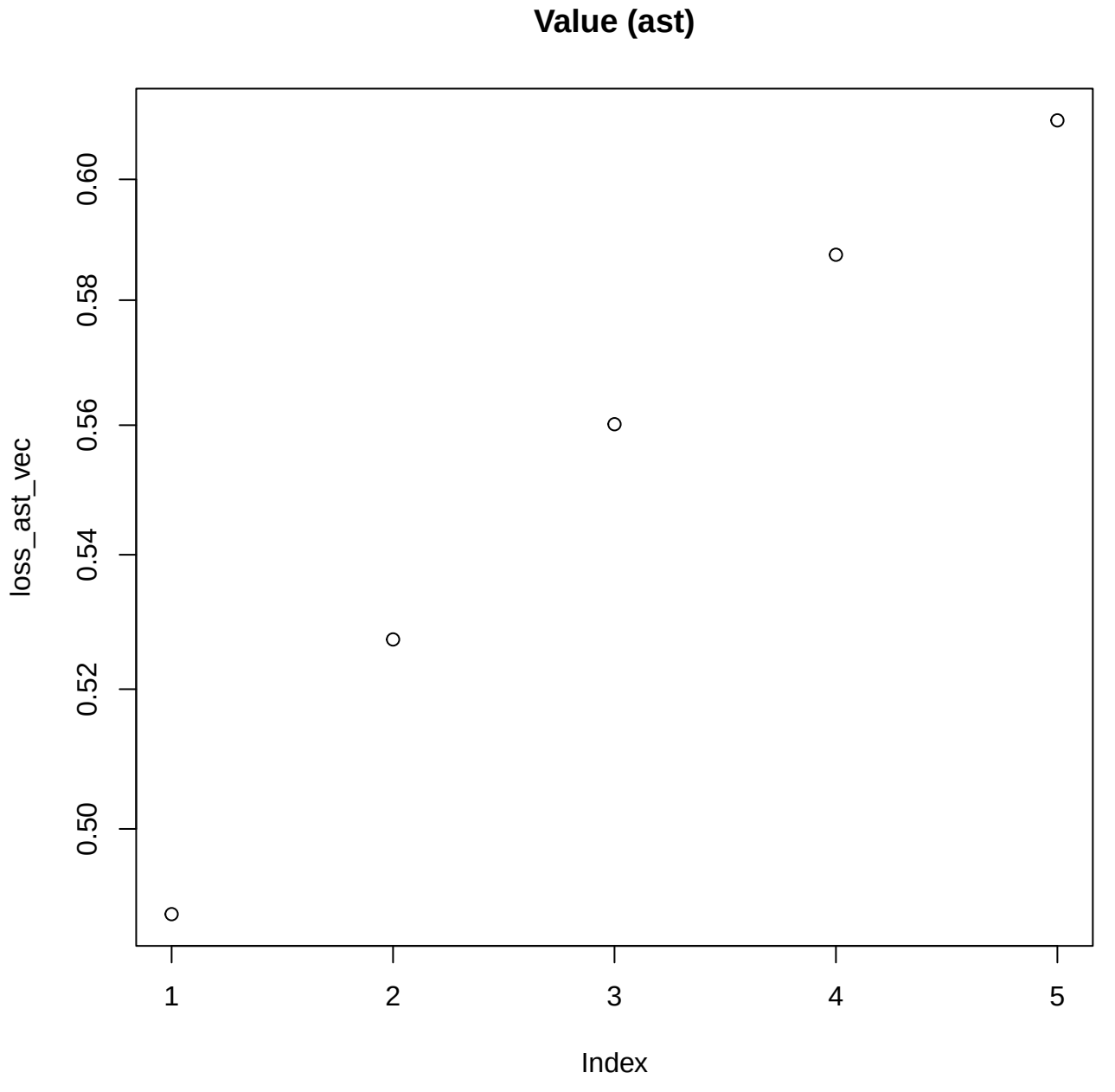

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.04
0.06
0.08
0.10
0.14
1
2
3
4
5
Index

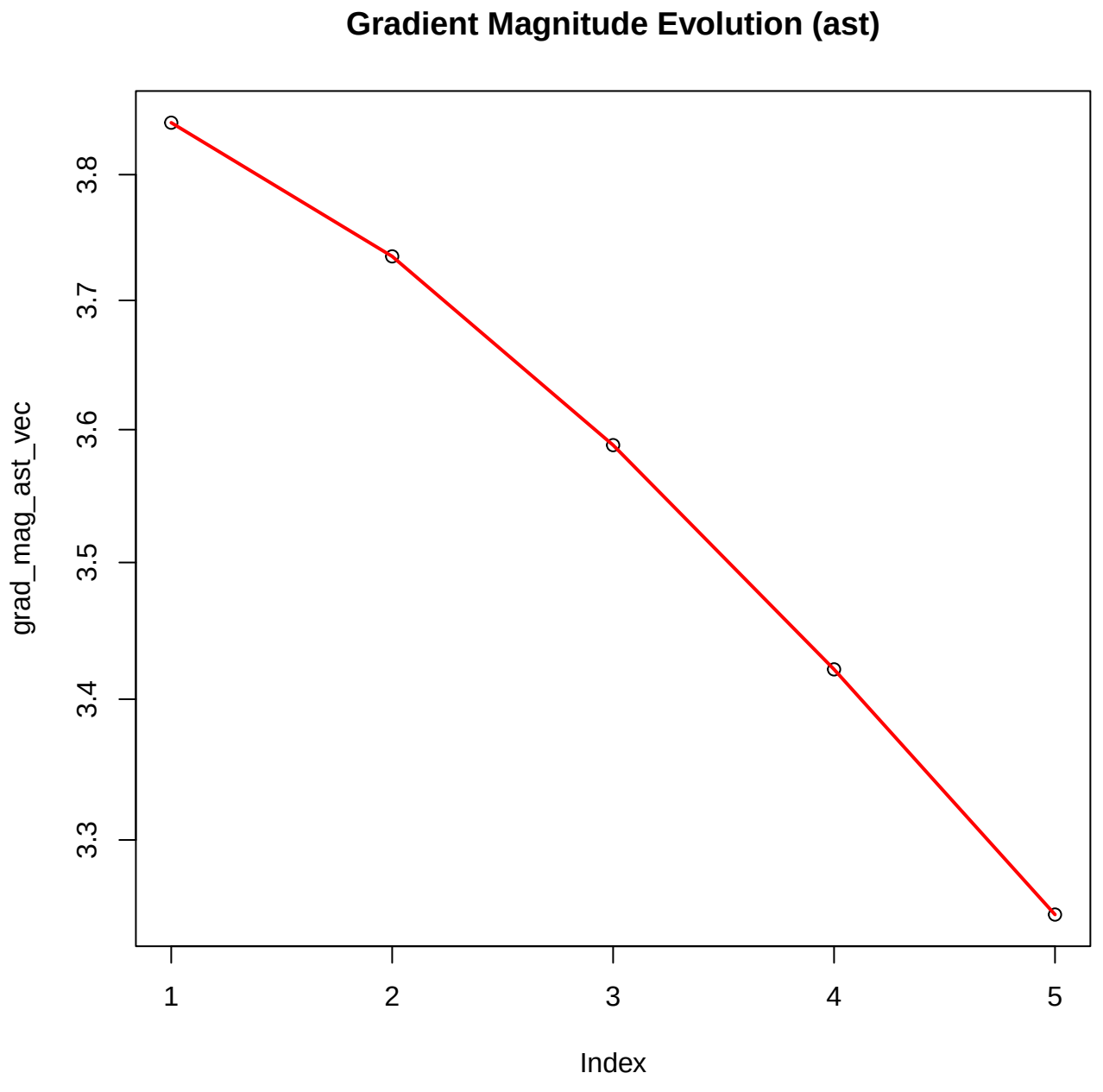

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.06
0.07
0.08
0.09
0.10
0.11
0.12
1
2
3
4
5
Index

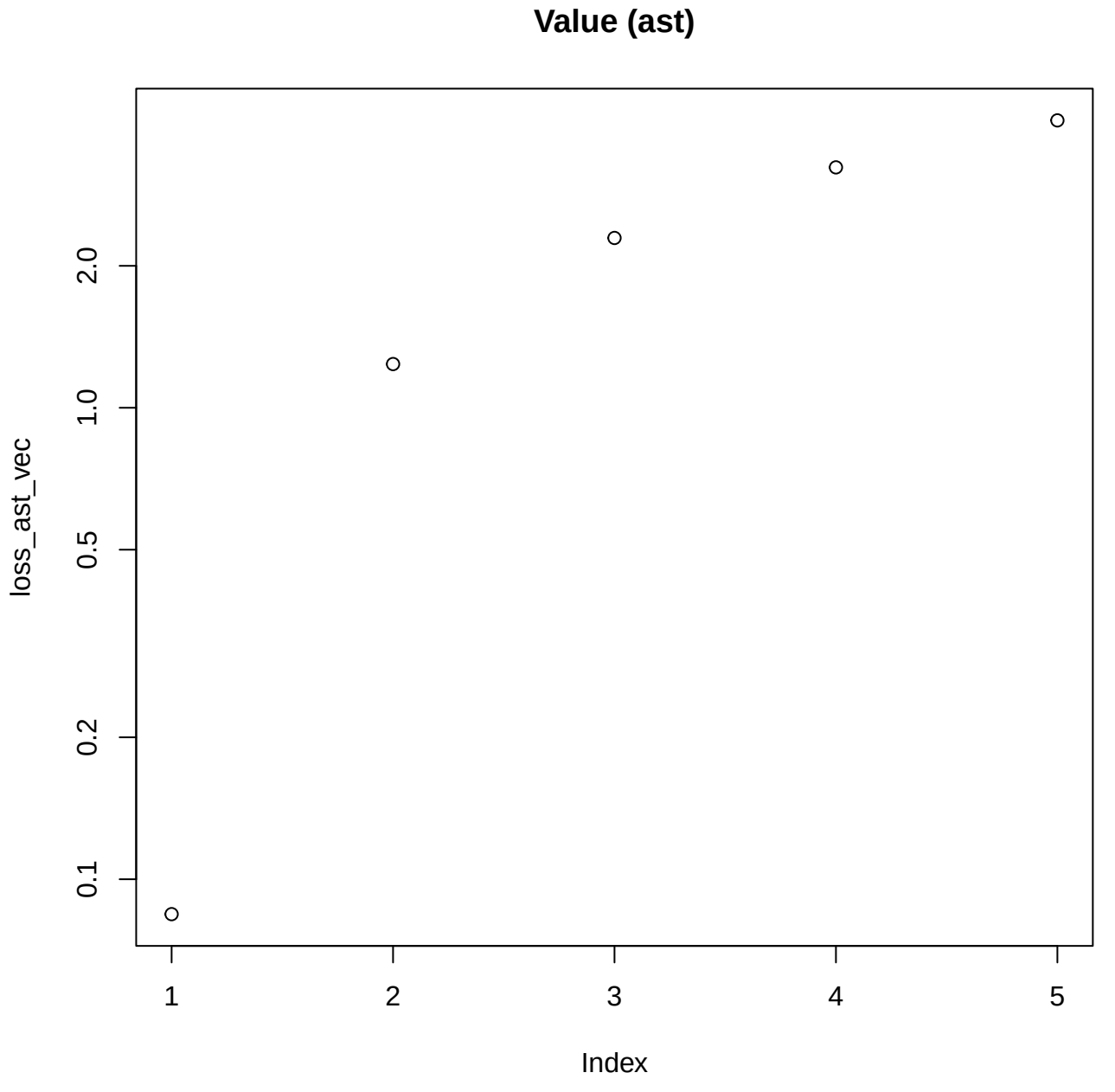

Value (ast)
loss_ast_vec
0.30
0.35
0.40
0.45
0.50
1
2
3
4
5
Index

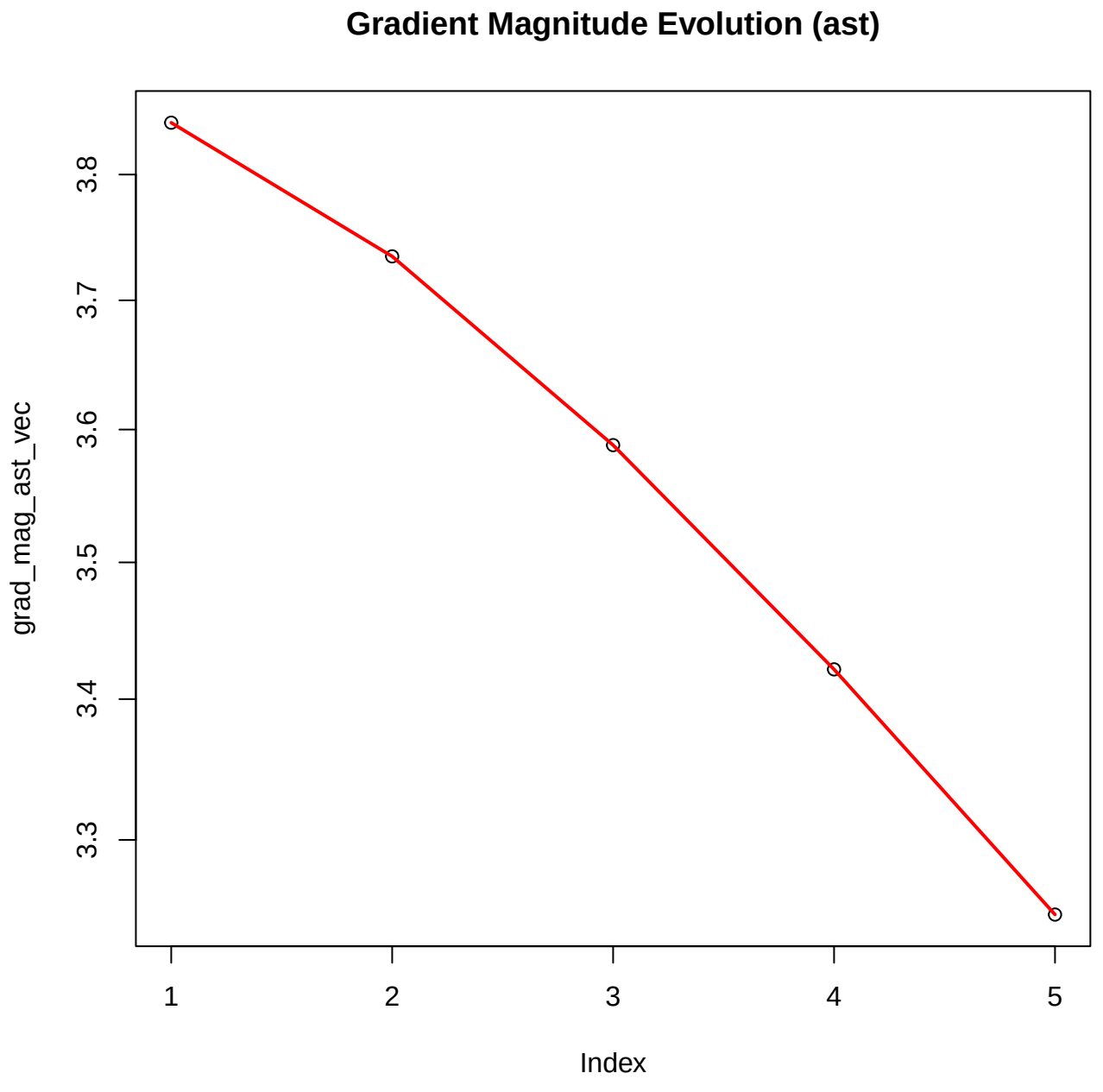

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

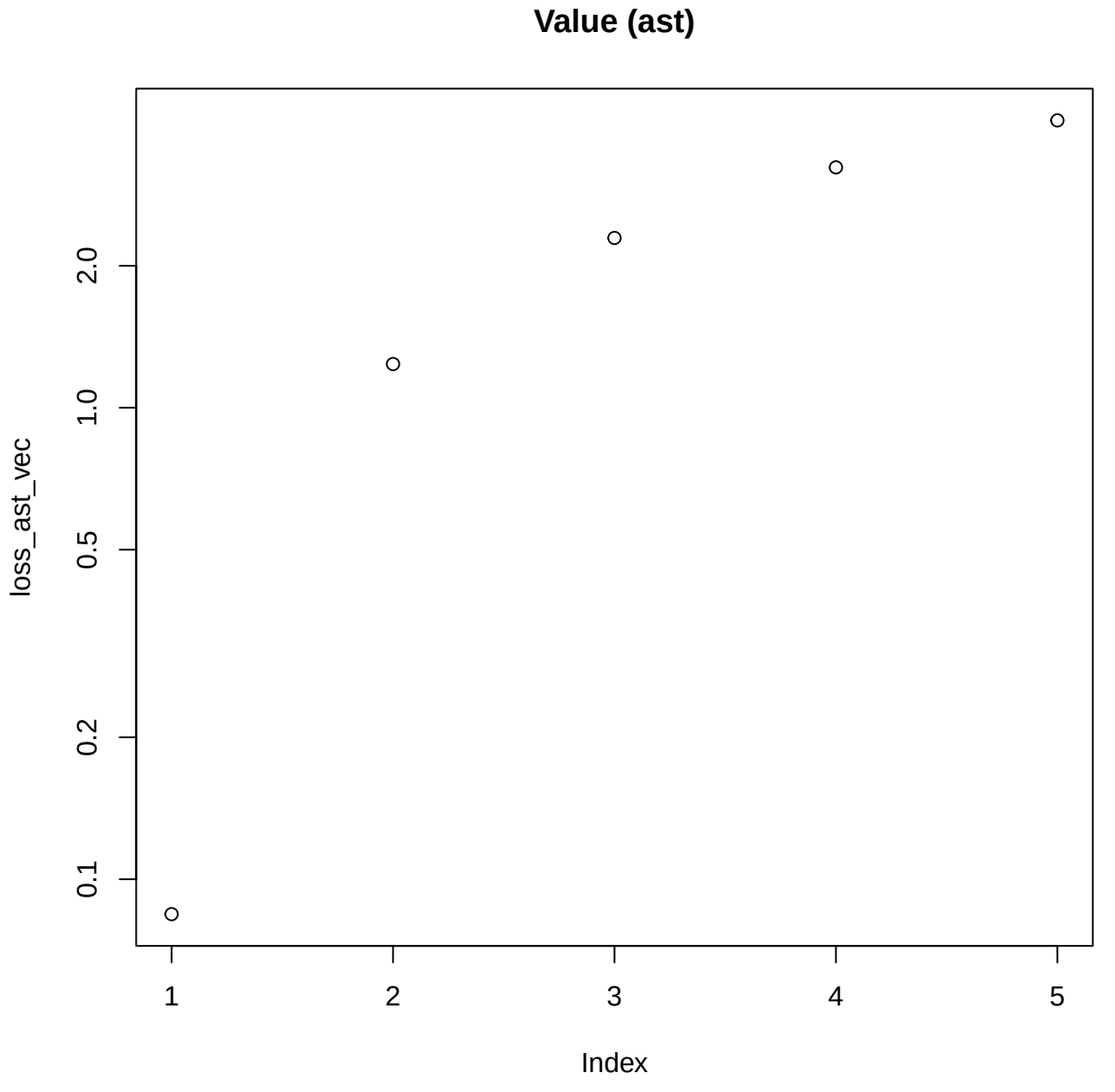

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

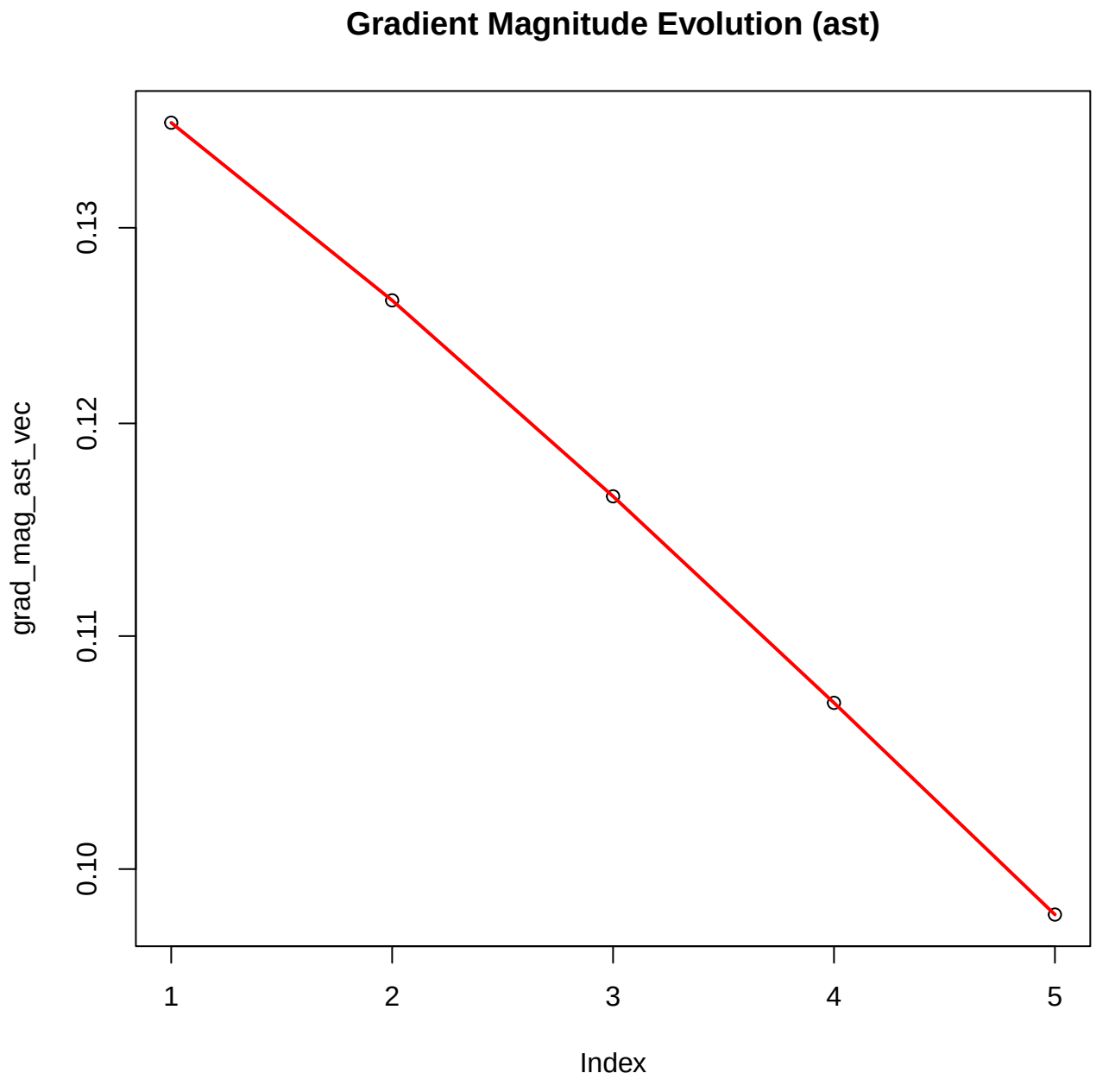

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

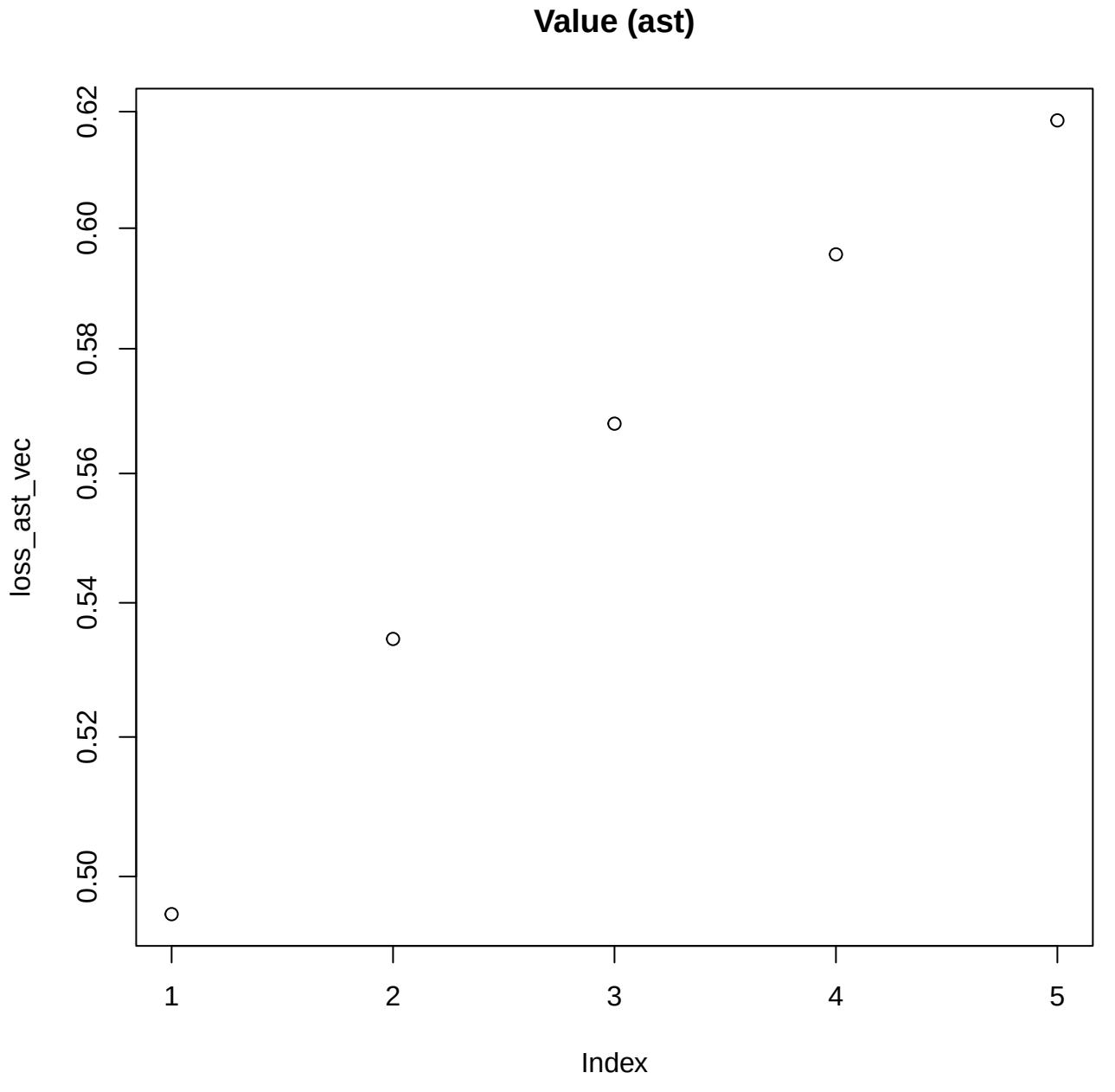

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

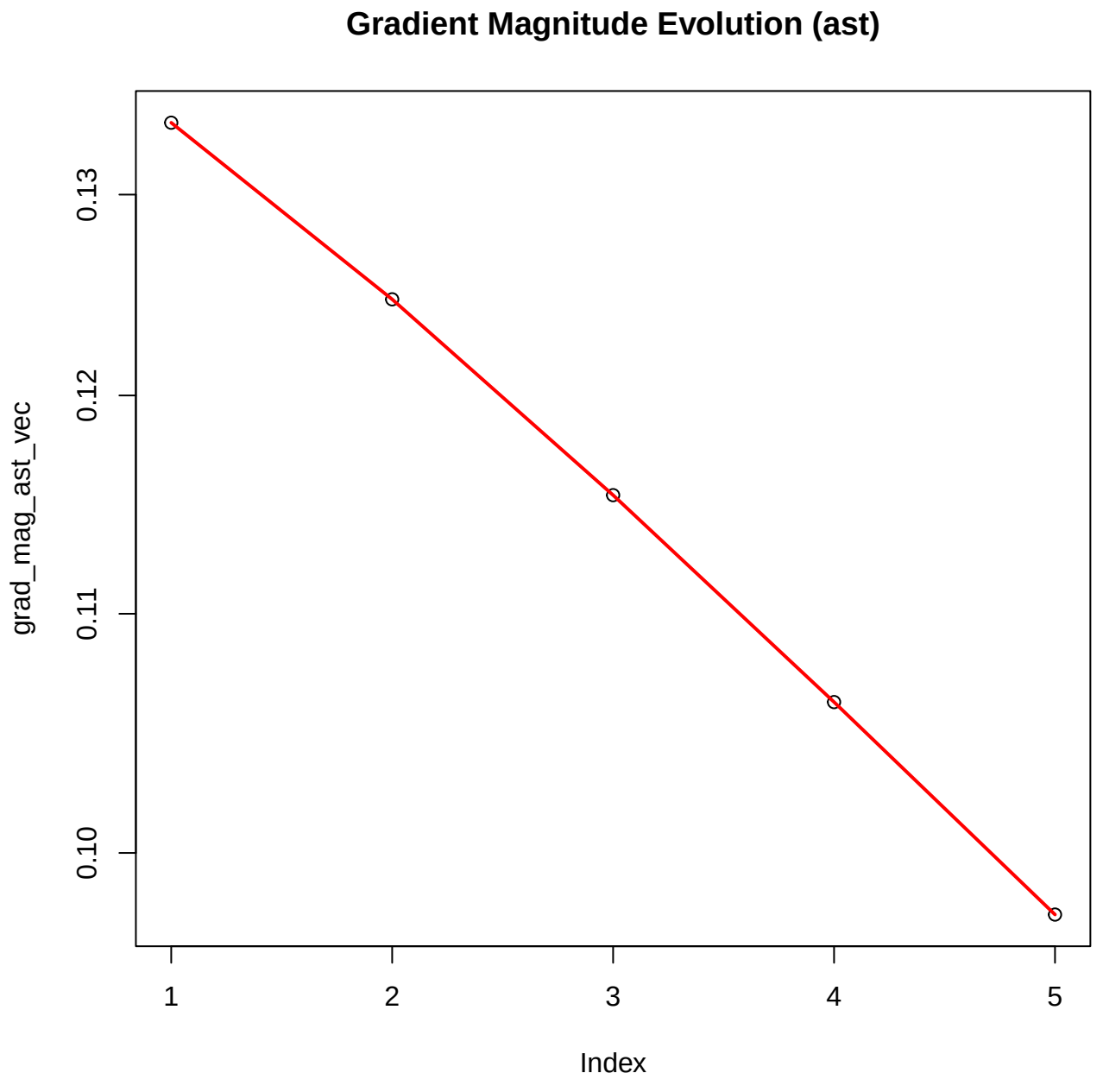

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

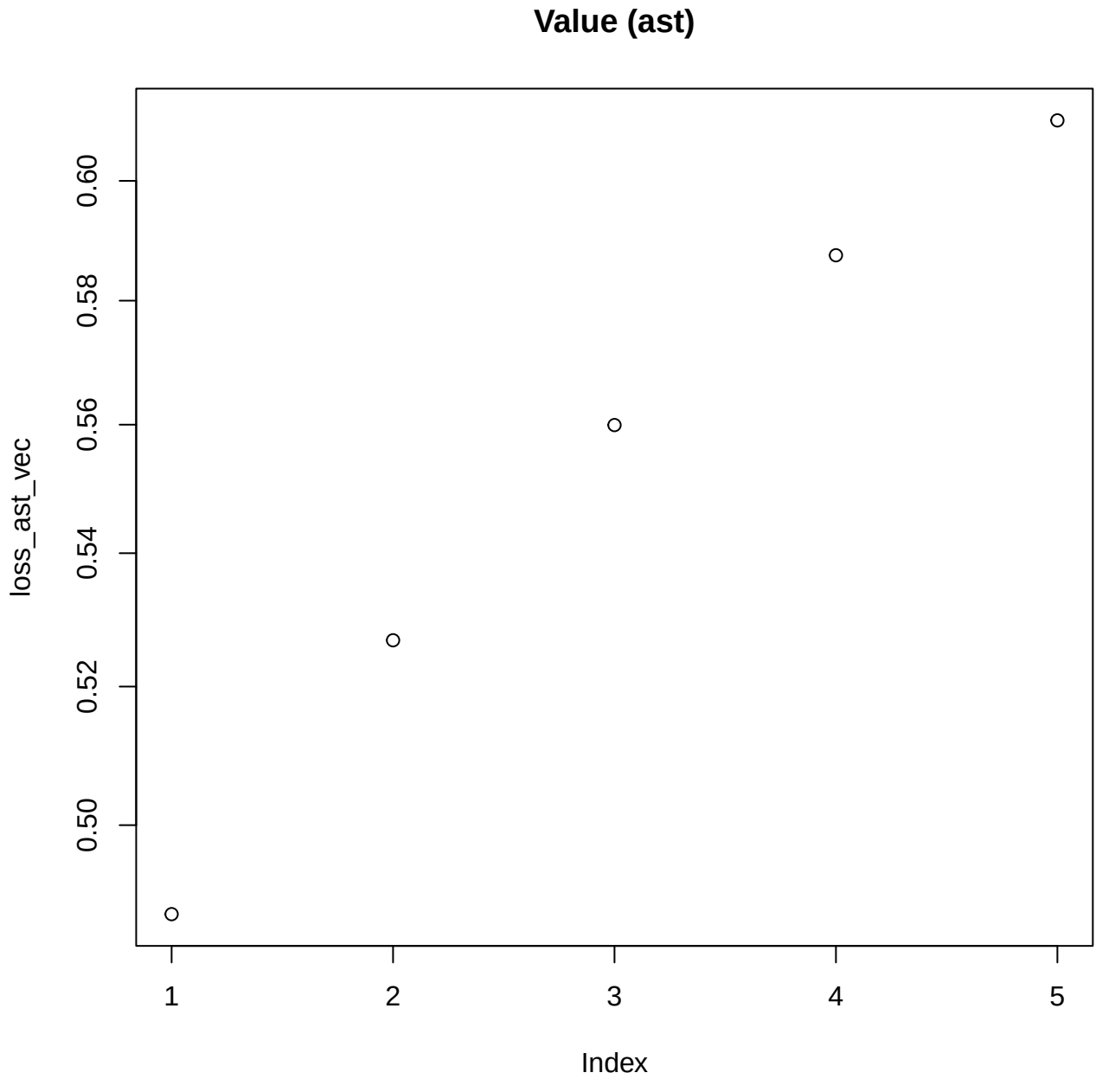

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index

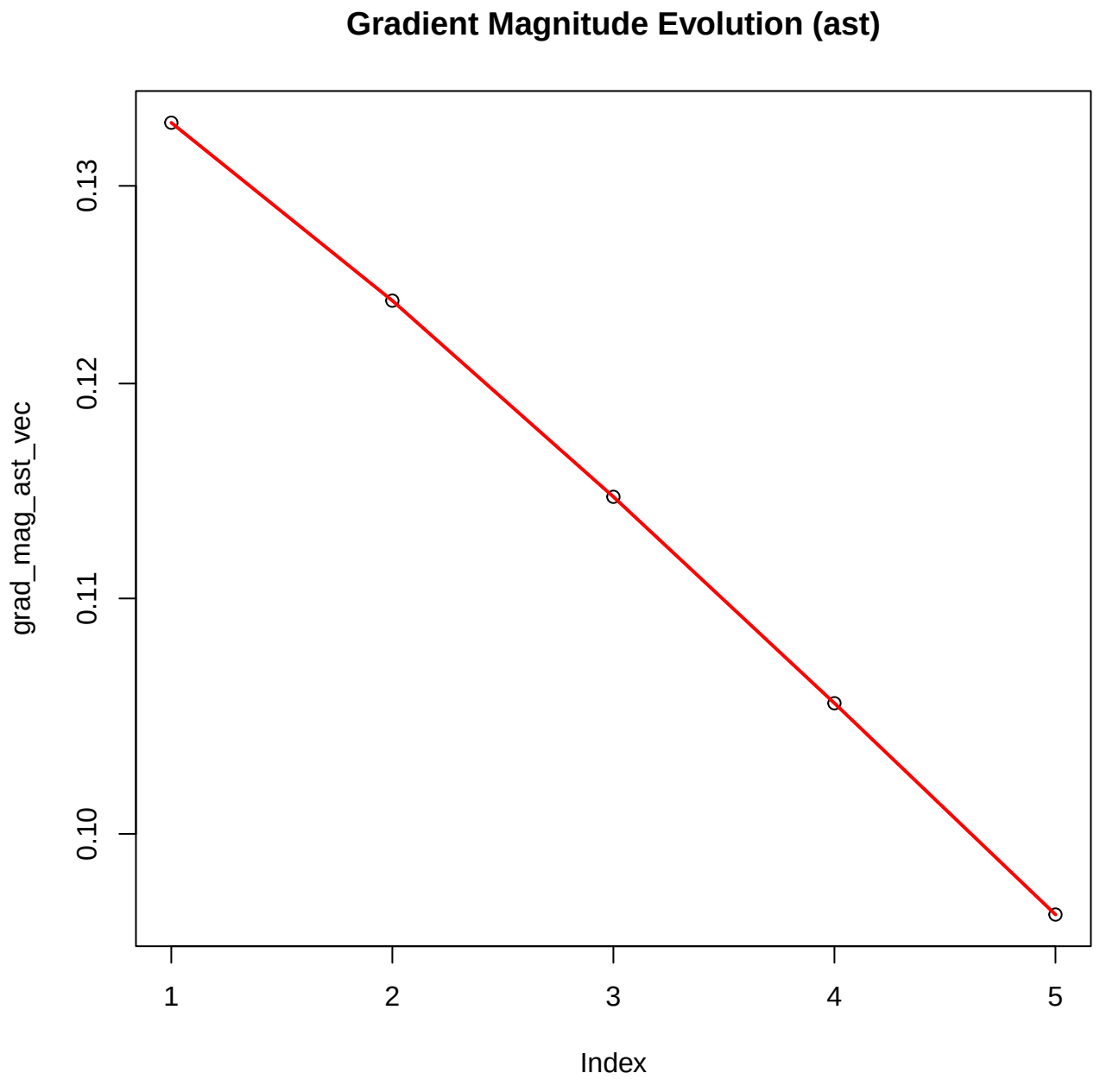

Value (ast)
loss_ast_vec
0.1
0.2
0.5
1.0
2.0
1
2
3
4
5
Index

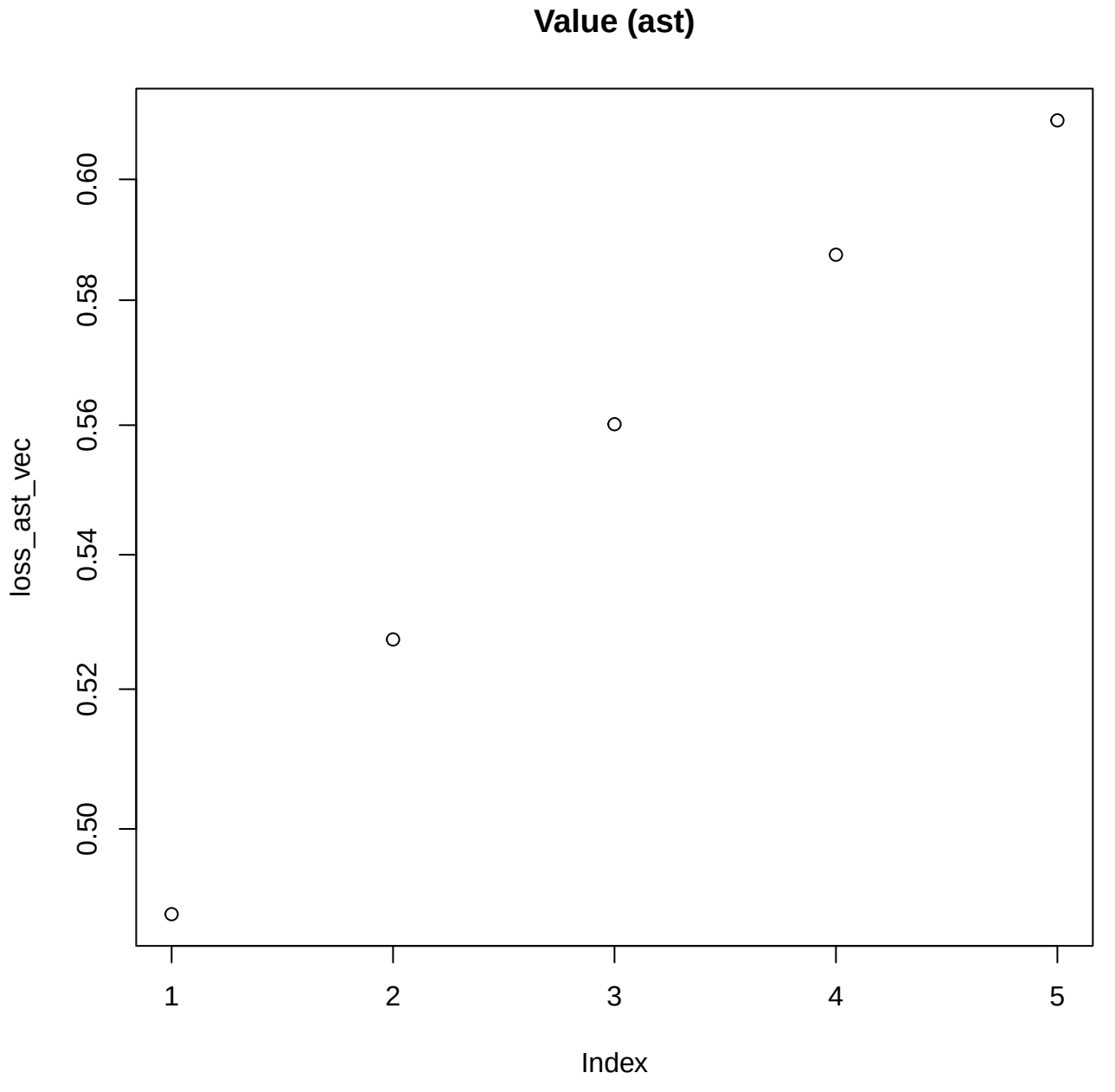

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

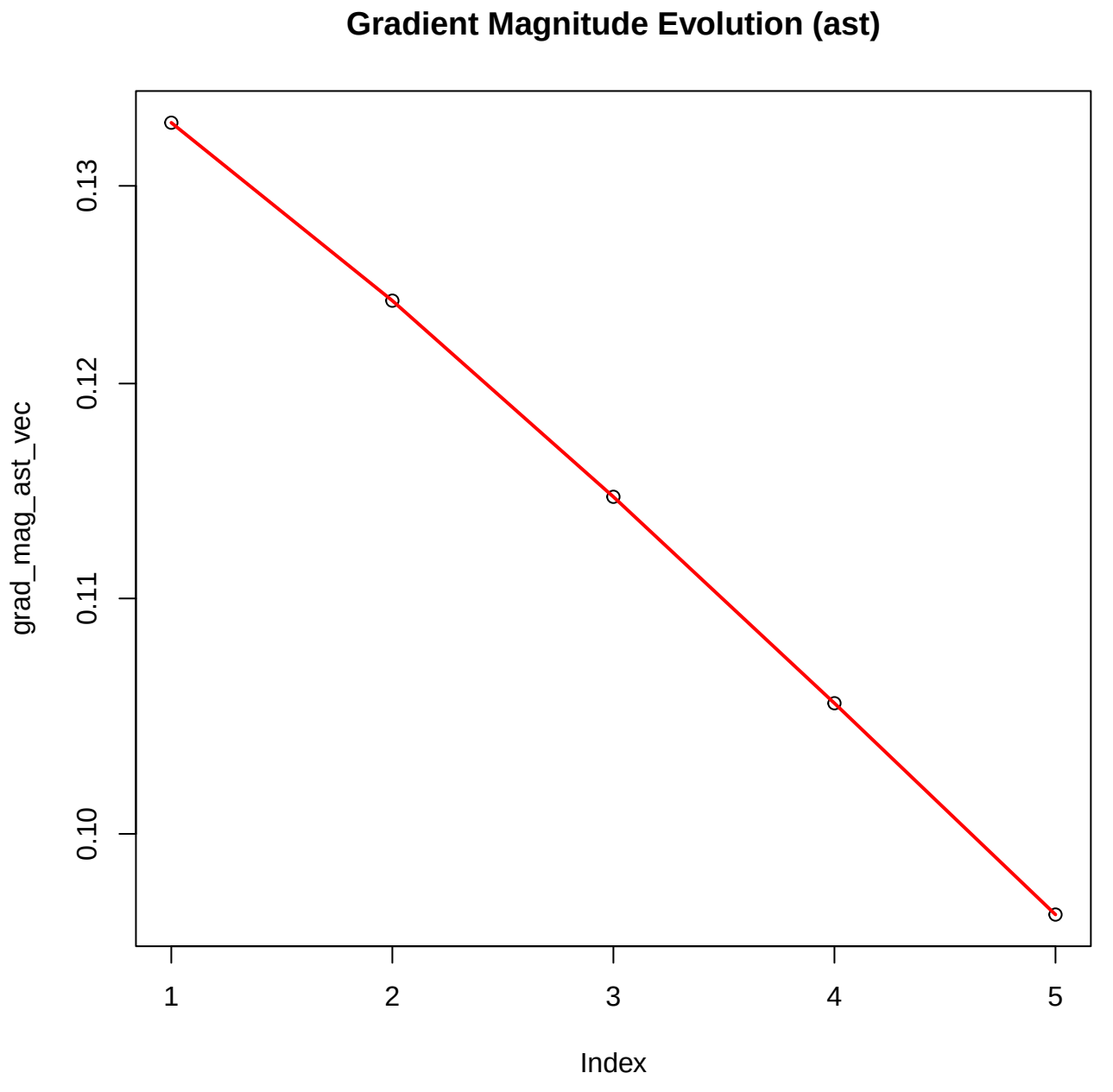

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
0.62
1
2
3
4
5
Index

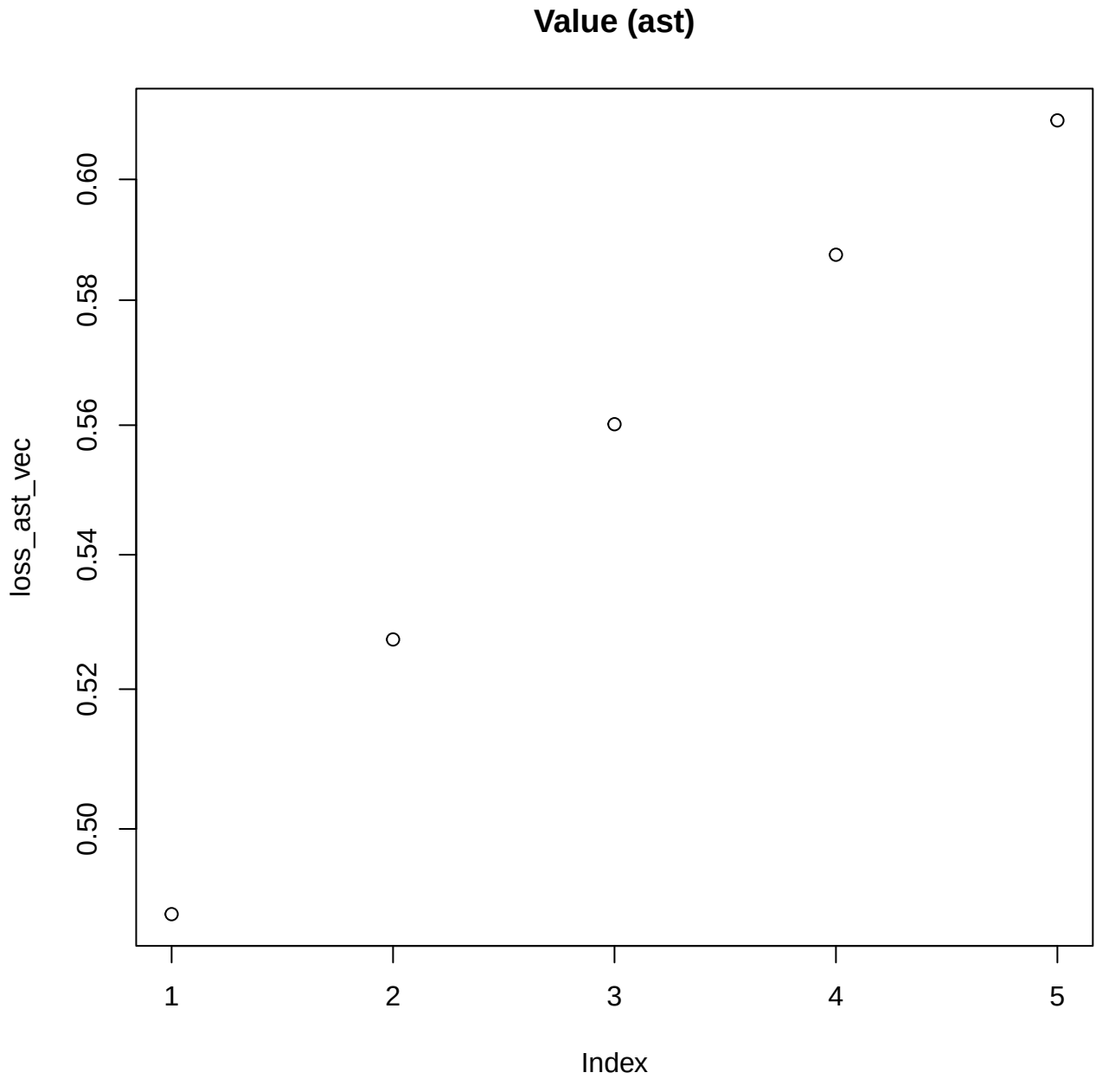

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

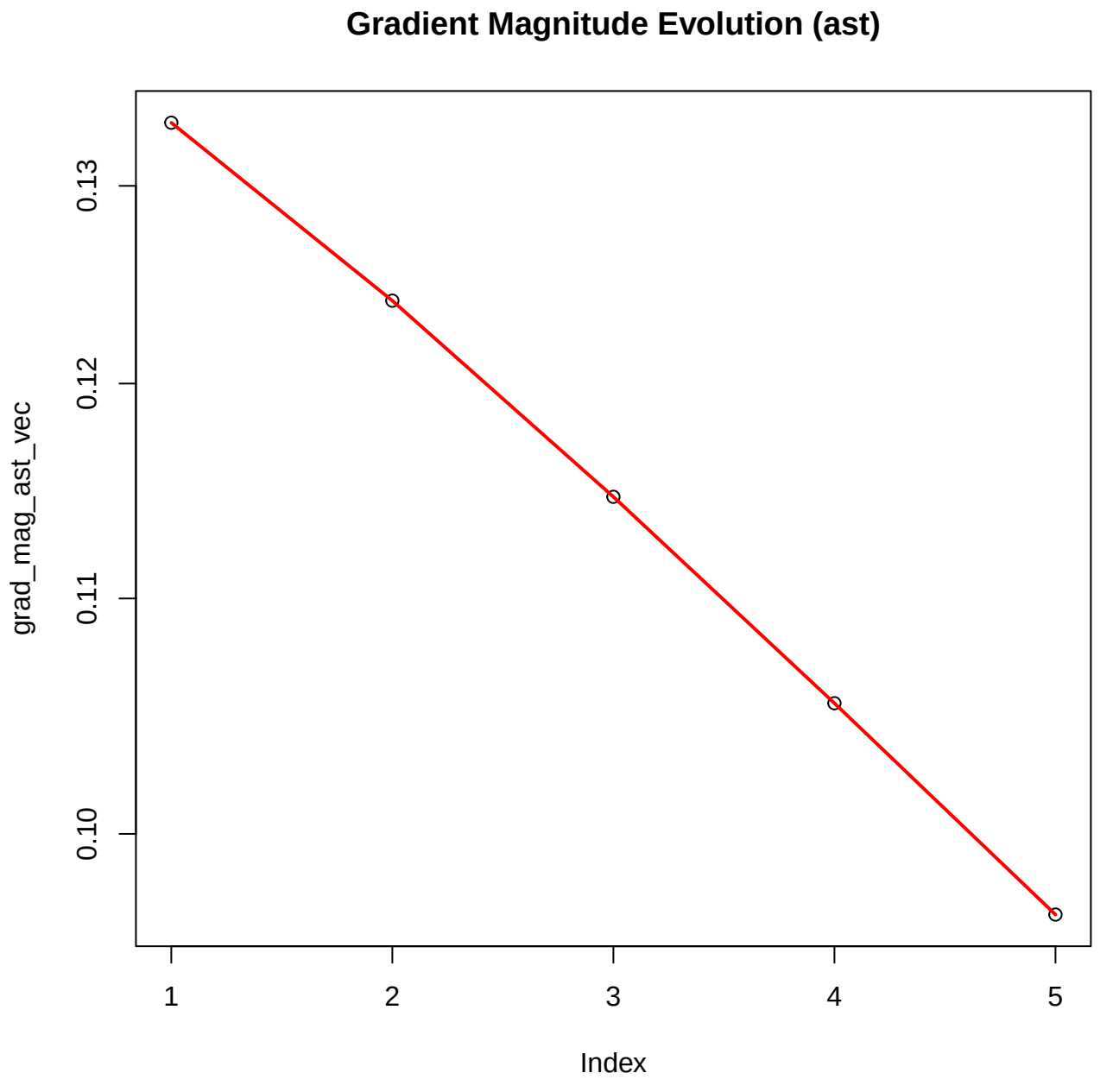

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

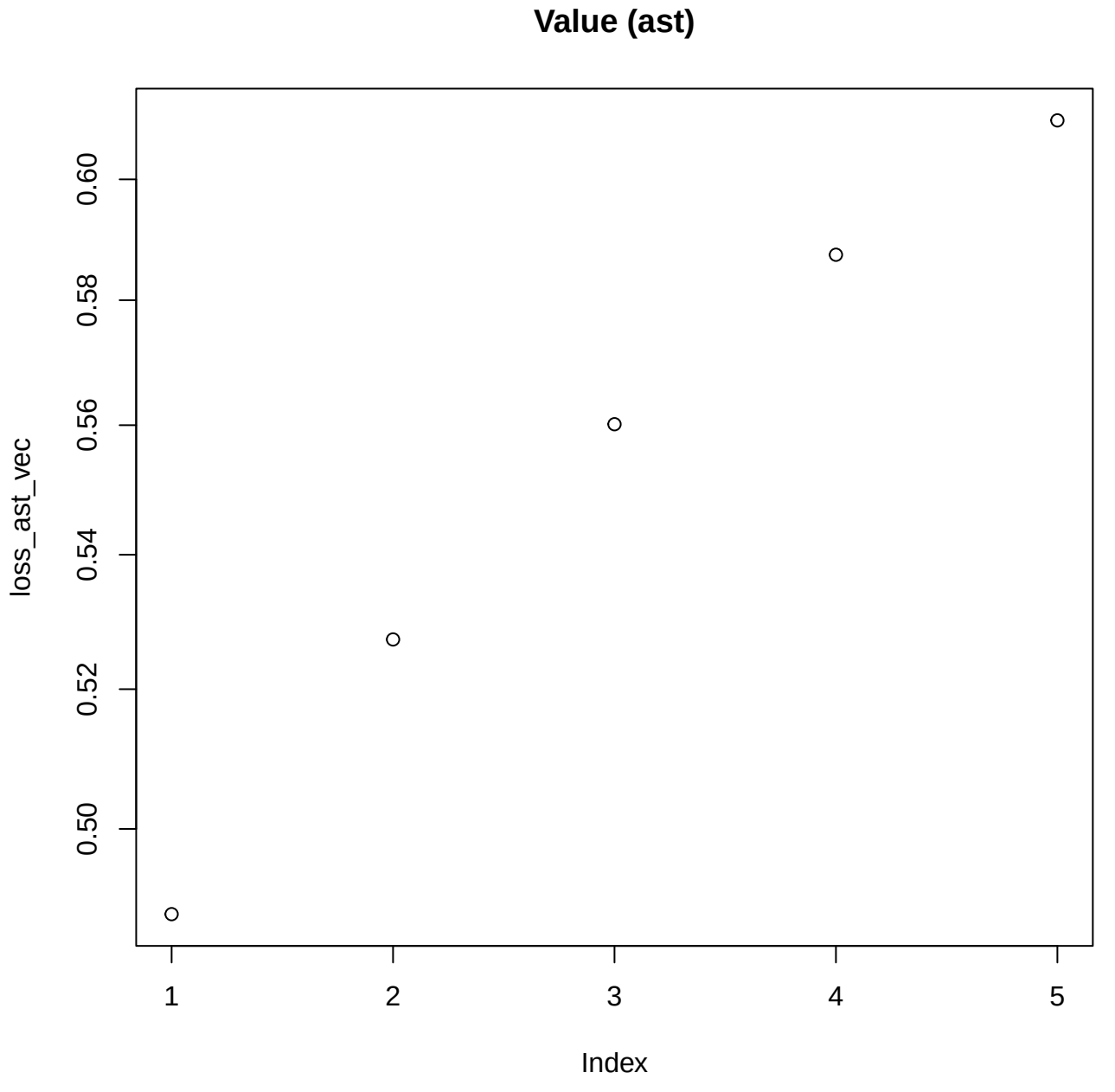

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

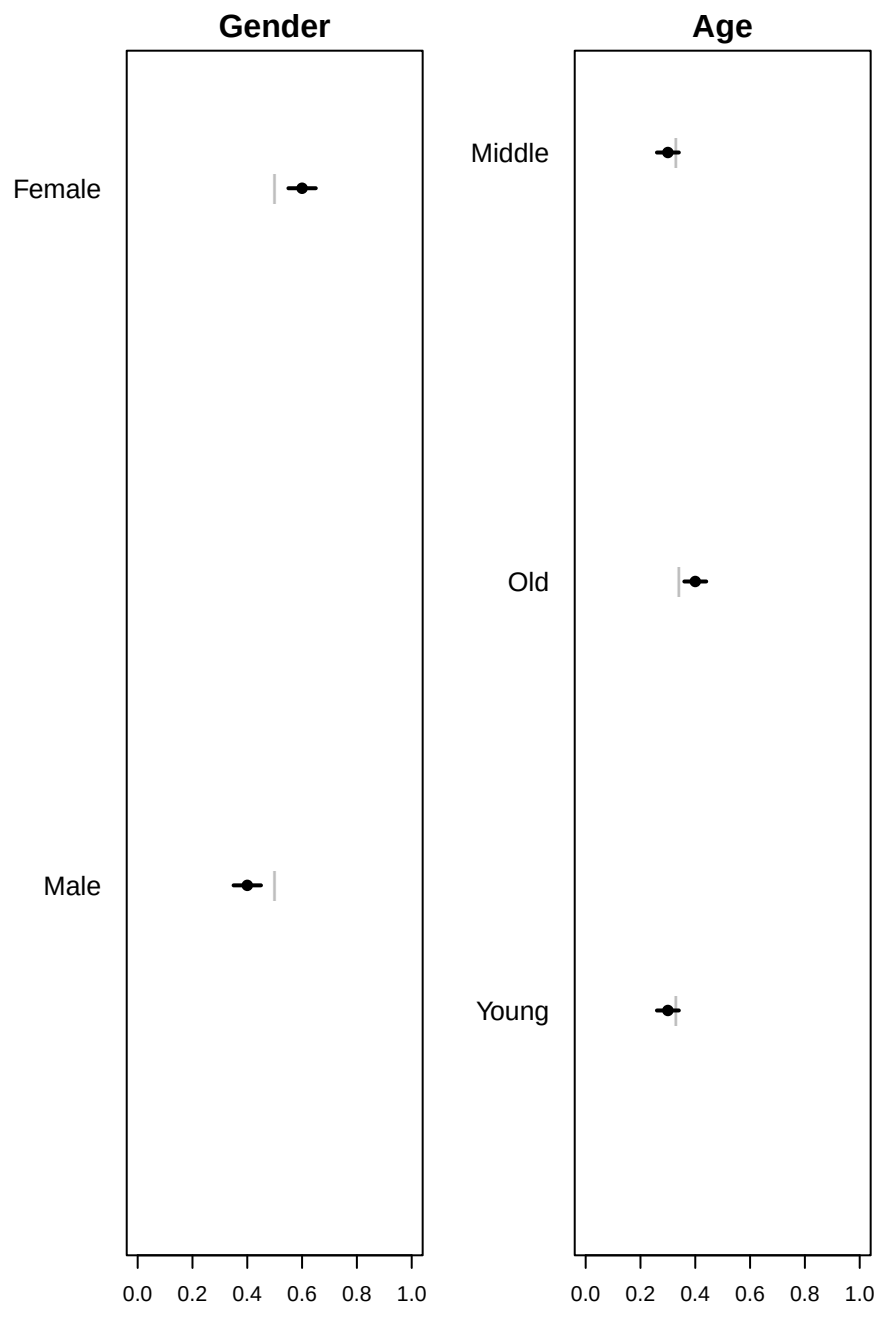

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

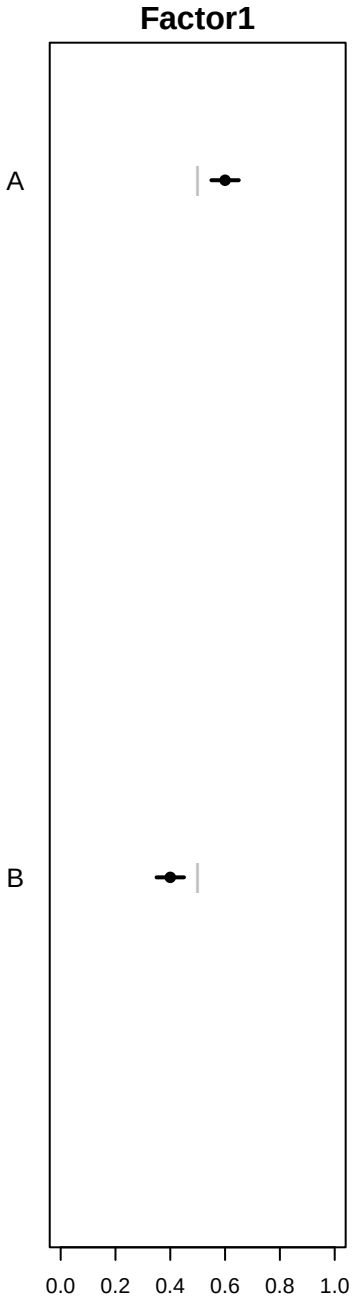

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

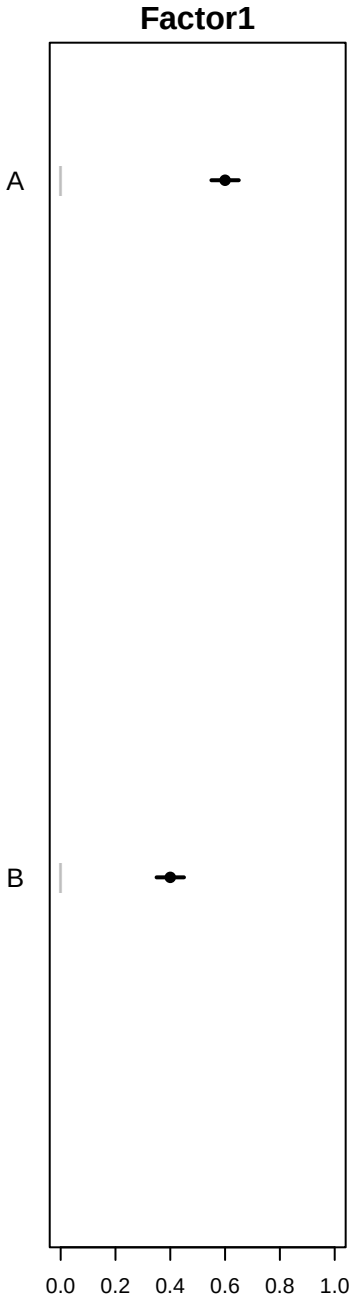

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

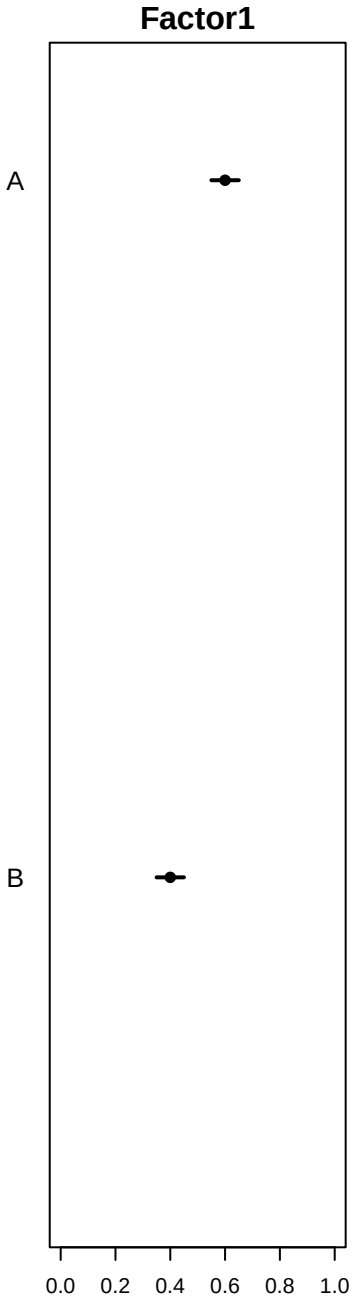

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

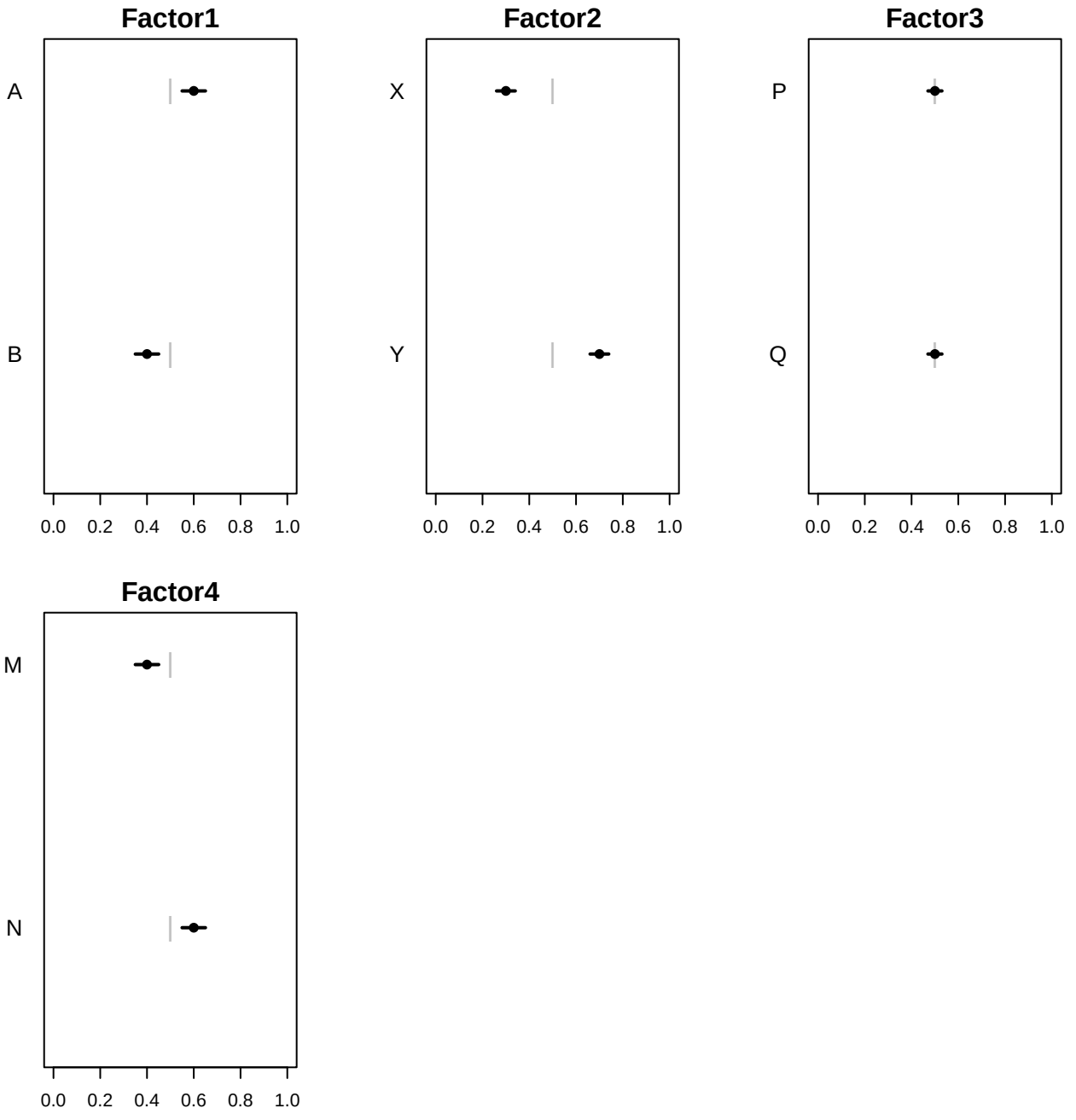

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

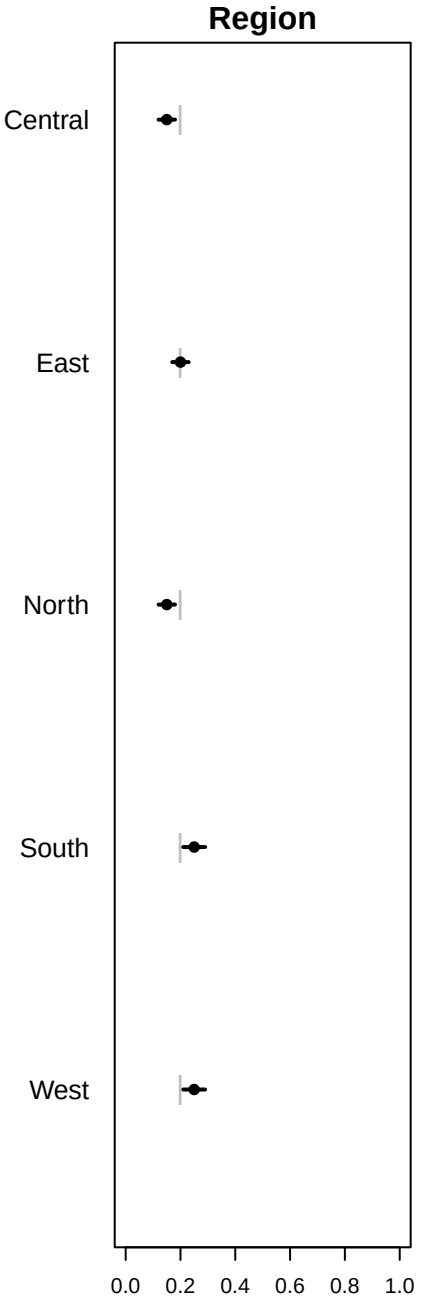

Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
0.0
0.2
0.4
0.6
0.8
1.0

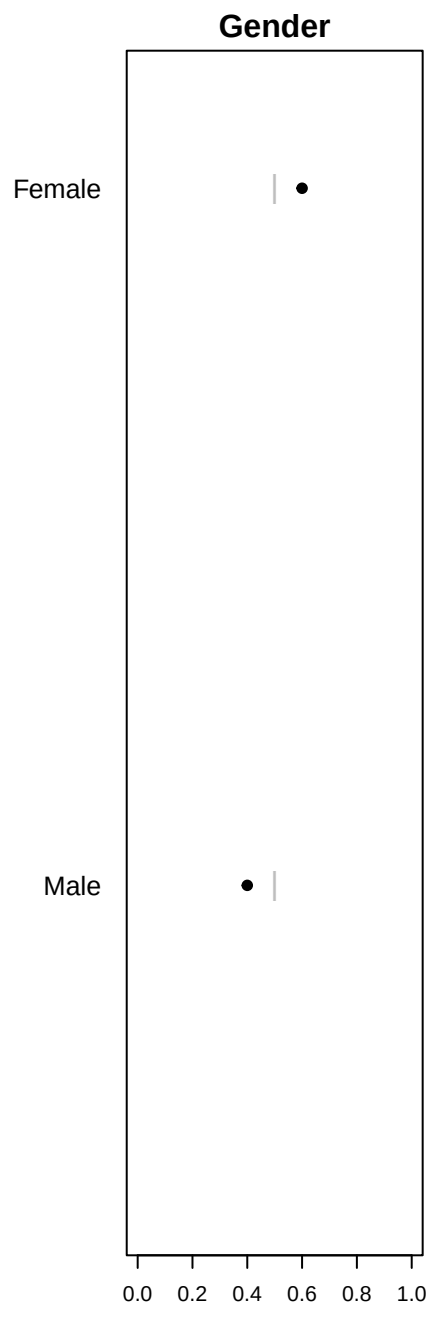

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

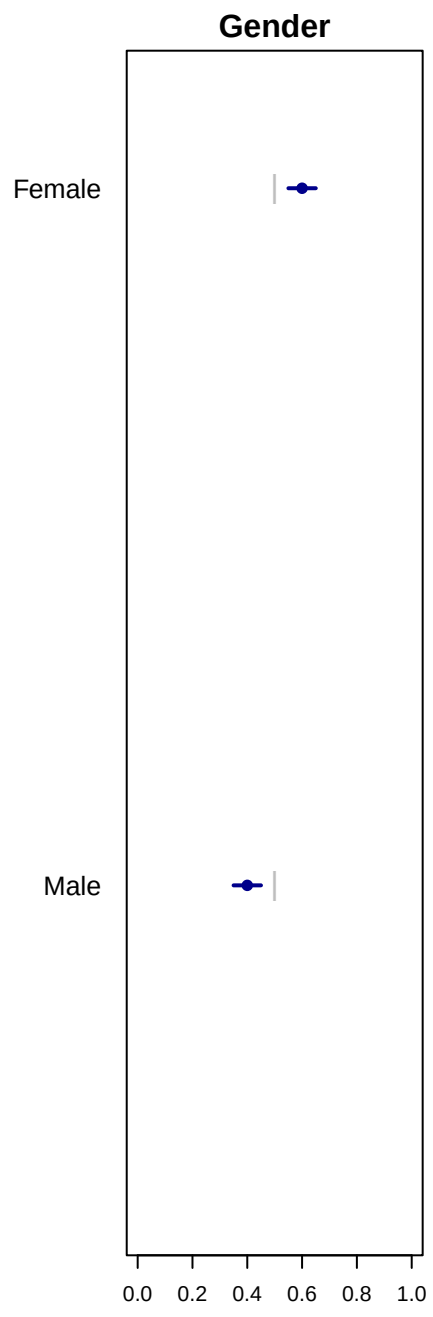

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

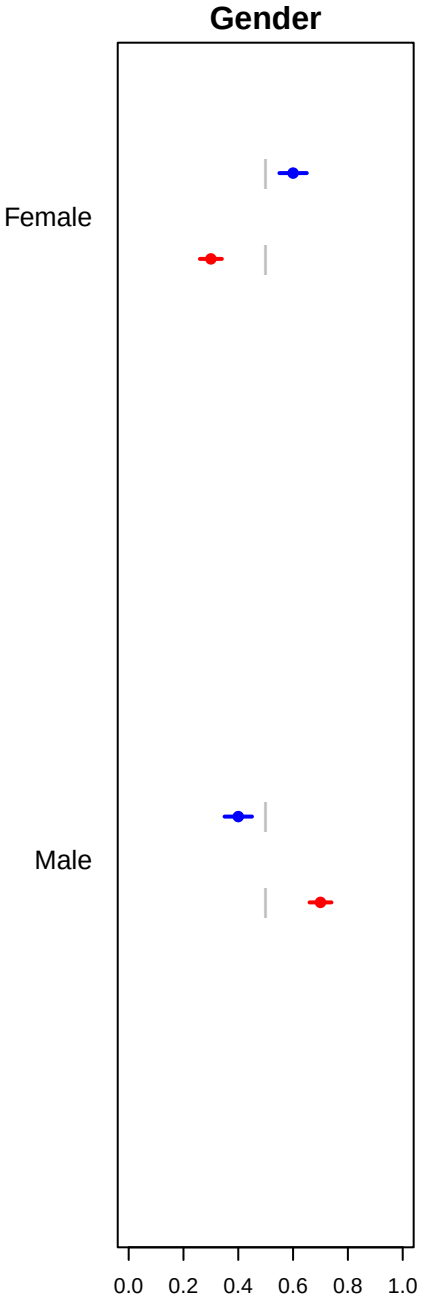

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0

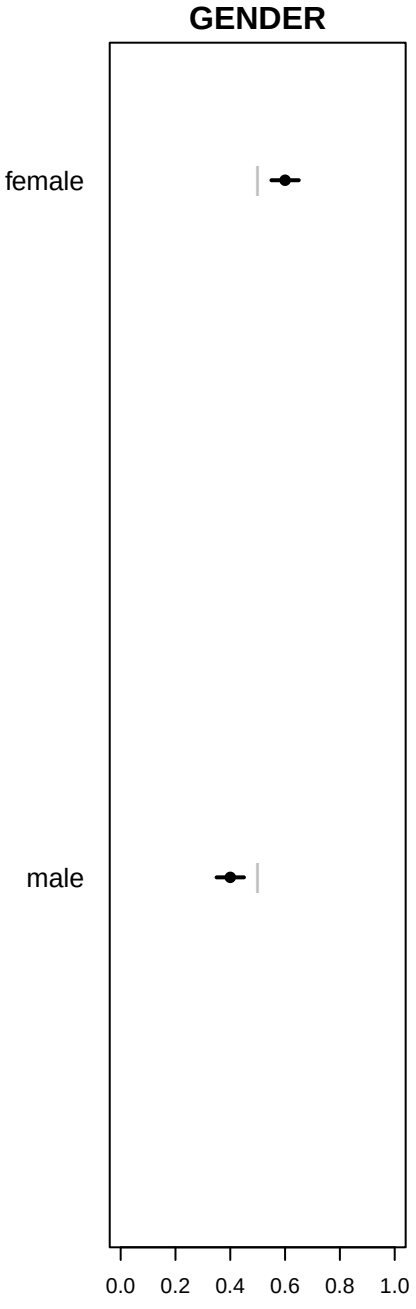

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
0.0
0.2
0.4
0.6
0.8
1.0

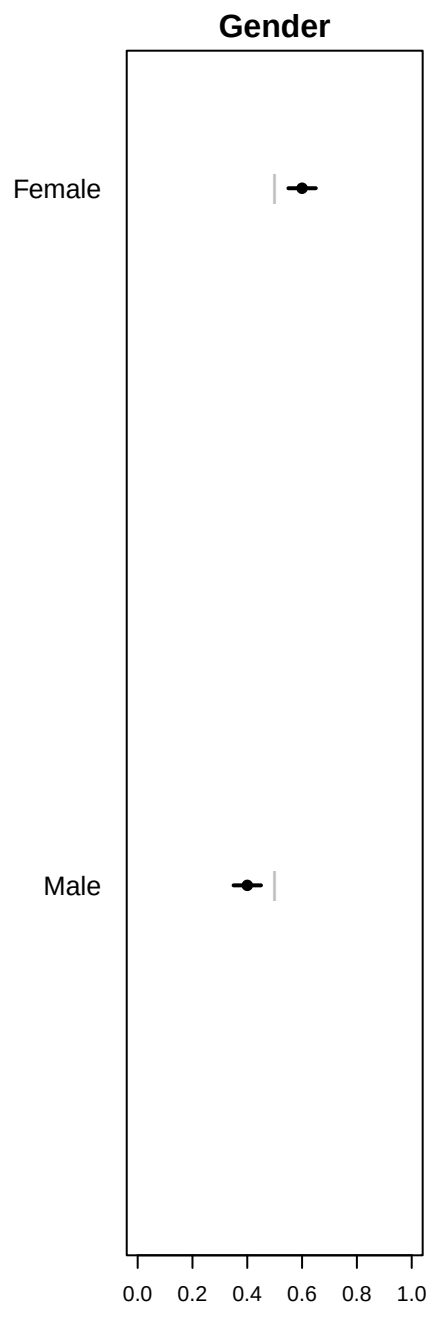

Region
Central
East
North
South
West
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0